FRM Part II 2026

Market Risk Measurement and Management

Gold-source question bank

890 questions, sorted by chapter and learning objective

Scope and LO wording: SchweserNotes 2026 (final authority). Every question solved and independently verified.

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Reading 1. Estimating Market Risk Measures: An Introduction and Overview

Kevin Dowd, Measuring Market Risk, 2nd Edition (West Sussex, UK: John Wiley & Sons, 2005). Chapter 3.

LO	Learning objective	Qs
1.a	estimate VaR using a historical simulation approach.	12
1.b	estimate VaR using a parametric approach for both normal and lognormal asset price distributions.	25
1.c	estimate the expected shortfall given profit and loss (P&L) or return data.	20
1.d	estimate risk measures by estimating quantiles.	14
1.e	evaluate estimators of risk measures by estimating their standard errors.	4
1.f	interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution.	8

LO 1.a

Estimate VaR using a historical simulation approach.

Q1

AP Mock 1 Q1

REBUILT — Original options mixed signs and included a figure (−1.4%) no realistic mistake produces; options replaced with positive-loss figures from common errors and the HS convention stated.

You are assigned to calculate the VaR for the stock of Fooda Inc. The ten worst daily returns of the stock over the last 100 trading days are (unordered):

−1.2%	−0.7%	−3.2%	−2.6%	−2.4%	−2.0%	−1.9%	−1.7%	−1.5%	−1.5%

Use the Dowd/Schweser convention that the HS VaR is the $(\alpha n + 1)$th worst observation. Assuming i.i.d. returns and 20 trading days in a month, which of the following is closest to the 95% monthly VaR of Fooda Inc. (expressed as a loss)?

A. 1.70%
B. 7.60%
C. 8.50%
D. 34.00%

Answer: B

Step by step

1. Order the returns from worst: −3.2, −2.6, −2.4, −2.0, −1.9, −1.7, −1.5, −1.5, −1.2, −0.7.
2. Cut-off observation: $\alpha n + 1 = 0.05 \times 100 + 1 = 6$, so the 6th worst return is used.
3. Daily 95% VaR = 1.70% (6th worst return −1.7%).
4. Scale by the square root of time: $1.70\% \times \sqrt{20} = 1.70\% \times 4.4721 = 7.60\%$.

Why B is correct: With 100 observations, 5 lie in the 5% tail and the 6th worst separates tail from body, giving a 1.70% daily VaR; under i.i.d. returns VaR scales with $\sqrt{T}$, so the 20-day VaR is 7.60%.

Why the others are wrong

- **A:** The daily VaR (6th worst, 1.70%) without scaling to a 20-day horizon.
- **C:** Used the 5th worst return (αn convention, 1.9%) and scaled it: $1.9\% \times \sqrt{20} = 8.50\%$.
- **D:** Scaled linearly with time (1.70% × 20) instead of by $\sqrt{20}$.

Q2

AnalystPrep Q.2628

REBUILT — Option D (−14.5%) matched no realistic error and the HS convention was unstated; replaced the distractors with the 5th-worst and 5th/6th-average figures and stated the $(\alpha n + 1)$ convention.

You are assigned to calculate the monthly VaR for the stock of Apex Inc. The ten worst monthly returns of the stock over the last 100 months are (unordered):

−12%	−7%	−32%	−26%	−24%	−20%	−19%	−17%	−15%	−14%

Use the Dowd/Schweser convention that the HS VaR is the $(\alpha n + 1)$th worst observation. Which of the following is closest to the 95% monthly VaR for Apex (expressed as a loss)?

A. 32%
B. 18%
C. 17%
D. 19%

Answer: C

Step by step

1. Order from worst: −32, −26, −24, −20, −19, −17, −15, −14, −12, −7.
2. Cut-off observation: $\alpha n + 1 = 0.05 \times 100 + 1 = 6$.
3. 6th worst monthly return = −17%, so the 95% monthly VaR = 17%. The data are already monthly, so no time scaling is needed.

Why C is correct: Five observations (5% of 100) form the tail; the 6th worst return, −17%, separates the tail from the body of the distribution.

Why the others are wrong

- **A:** Took the single worst observation (−32%) as the VaR; that is a maximum loss, not the 95% quantile.
- **B:** Interpolated between the 5th (−19%) and 6th (−17%) worst observations rather than using the stated ($\alpha n +$ 1) rule.
- **D:** Used the 5th worst return ($\alpha n = 5$) instead of the 6th.

Q3

AnalystPrep Q.1490

REBUILT — Options B and D ('number of observations' vs 'size of the data' decreasing as holding period increases) were both defensible; distractors rewritten so only one option is correct.

Extending historical simulation (HS) estimates of VaR and expected shortfall to longer holding periods poses no theoretical difficulty, but it does create a practical problem. Which of the following best describes that problem?

A. As the holding period shortens, the number of available P/L observations falls, so daily HS estimates are the least reliable.

B. As the holding period lengthens, the number of non-overlapping P/L observations falls sharply, so longer-horizon HS estimates rest on very few data points.

C. HS requires a parametric distribution to be fitted to P/L, and its parameters must be re-estimated for each new holding period.

D. As the holding period lengthens, HS can no longer produce an expected shortfall estimate because ES requires daily data.

Answer: B

Step by step

1. HS for a longer horizon builds h-day P/L by summing daily P/L over non-overlapping blocks.
2. With 1,000 daily observations (about 4 years at 250 days), 5-day sums give 1,000/5 = 200 weekly observations and 20-day sums give 1,000/20 = 50 monthly observations.
3. With so few observations, especially in the tail, longer-horizon HS VaR and ES become imprecise; this limits how long the holding period can practically be.

Why B is correct: The number of usable observations is inversely proportional to the holding period, so the data constraint, not theory, limits long-horizon HS estimates.

Why the others are wrong

- **A:** Reverses the relationship: shorter holding periods give more observations, not fewer.
- **C:** Confuses HS with parametric approaches; HS is non-parametric and does not fit a distribution.
- **D:** ES can be computed from h-day P/L just like VaR (average of tail losses); the issue is the small number of observations, not the data frequency.

Q4

AnalystPrep Q.2822

REBUILT — Without a stated convention both 17% ($\alpha n + 1$) and 18% (αn) were defensible; stated the ($\alpha n + 1$) convention in the stem.

A dataset contains 30 return observations. The worst 10 returns (in %), ordered, are:

−20	−18	−18	−17	−15	−14	−12	−10	−7	−3

Use the Dowd/Schweser convention that the HS VaR is the (αn+1)th worst observation. What is the 90% historical-simulation VaR?

A. 17%

B. 18%

C. 16%

D. 15%

Answer: A

Step by step

1. Cut-off observation: $\alpha n + 1 = 0.10 \times 30 + 1 = 4$.
2. Ordered worst returns: −20, −18, −18, −17, ...; the 4th worst is −17%.
3. 90% VaR = 17%.

Why A is correct: Three observations (10% of 30) lie in the tail; the 4th worst return separates tail from body, so VaR = 17%.

Why the others are wrong

- **B:** Used the 3rd worst observation ($\alpha n = 3$), the convention sometimes seen on past exams but not the one specified.
- **C:** Averaged the 4th (−17%) and 5th (−15%) worst returns instead of picking the 4th.
- **D:** Took the 5th worst return, counting one observation too far into the body.

Q5

AnalystPrep Q.3010

A trader holds a single long position in stock KKJL; yesterday's closing price was USD 100. You use a 95% historical-simulation VaR based on a 260-day moving window, taking the VaR as the $(\alpha n + 1)$th worst return. As of yesterday, the 16 worst 1-day returns (%) in the window were:

−9.5	−8	−7.6	−7.4	−7.2	−7.18	−7.1	−6.9	−6.57	−6.56	−6.45	−6.4	−6.25	−6.05	−5.99	−5.85

Today the stock fell 10% after an adverse news story. The return that drops out of the window today is −3%. What is today's 95% HS VaR in USD?

A. USD 6.25
B. USD 5.445
C. USD 5.625
D. USD 6.05

Answer: C

Key correction: Source key (AnalystPrep) said A; correct is C because the −6.25% cut-off return must be applied to today's position value of USD 90, not yesterday's USD 100.

Step by step

1. Cut-off observation: $\alpha n + 1 = 0.05 \times 260 + 1 = 14$.
2. Yesterday the 14th worst return was −6.05%.
3. Today's −10% return enters as the new worst; the −3% return leaving the window is not in the tail, so every listed return moves down one rank and the new 14th worst is −6.25% (yesterday's 13th).
4. Today's position value = 100 × (1 − 0.10) = USD 90.
5. VaR = 6.25% × 90 = USD 5.625.

Why C is correct: The new extreme loss pushes the cut-off one rank deeper (to −6.25%), while the position value shrinks to USD 90; HS VaR applies the cut-off return to the current value.

Why the others are wrong

- **A:** Used the correct new cut-off (−6.25%) but applied it to yesterday's USD 100 instead of today's USD 90.
- **B:** Kept yesterday's cut-off (−6.05%), ignoring that the −10% return enters the tail, and applied it to USD 90.
- **D:** Yesterday's VaR (6.05% × 100), ignoring both the new return and the change in position value.

Q6

AnalystPrep Q.3011

Stock A's past 260 daily returns display strong negative skewness. Portfolio P1 is long stock A and portfolio P2 is short the same amount of stock A. Using historical simulation, which statement about the 99% VaR is most likely accurate?

A. |VaR(P1)| > |VaR(P2)|
B. |VaR(P1)| < |VaR(P2)|
C. VaR(P1) = VaR(P2)
D. Cannot be concluded from the given information

Answer: A

Step by step

1. Negative skew means a long left tail: large negative returns are more extreme than large positive returns.
2. P1 (long) loses when returns are negative, so its 99% VaR is read from the long left tail.

3. P2 (short) loses when returns are positive, so its 99% VaR is read from the shorter right tail of A's returns.
4. Hence the long position's VaR is larger in magnitude: |VaR(P1)| > |VaR(P2)|.

Why A is correct: HS VaR uses the empirical tail relevant to each position; the long position faces the longer left tail, so its loss quantile is larger.

Why the others are wrong

- **B:** Reverses the skew logic, treating the short position as exposed to the long tail.
- **C:** Only true for a symmetric distribution (e.g., normal), where long and short VaRs have equal magnitude.
- **D:** The direction of skew is enough to rank the tail quantiles; no further information is needed.

Q7

AnalystPrep Q.3037

REBUILT — Option C (USD 10) was a giveaway and the HS convention was unstated; replaced C with the αn-convention figure and stated the $(\alpha n + 1)$ rule.

A trader holds a long position in stock A worth 100 as of yesterday. You use a 95% historical-simulation VaR based on a 260-day moving window, taking the VaR as the $(\alpha n + 1)$th worst return. As of yesterday the 16 worst 1-day returns (%) were:

−9.5	−8	−7.6	−7.4	−7.2	−7.18	−7.1	−6.9	−6.57	−6.56	−6.45	−6.4	−6.25	−6.05	−5.99	−5.85

Today the stock price rose 10%. The oldest return dropping out of the window is not among those listed. What is today's 95% HS VaR (in absolute value)?

A. USD 6.25
B. USD 6.655
C. USD 6.875
D. USD 6.05

Answer: B

Step by step

1. Cut-off observation: $\alpha n + 1 = 0.05 \times 260 + 1 = 14$.
2. Today's +10% return does not enter the left tail and the dropped return is not in the tail, so the 14th worst return is still −6.05%.
3. Today's position value = 100 × 1.10 = 110.
4. VaR = 6.05% × 110 = USD 6.655.

Why B is correct: The tail ranking is unchanged, so the cut-off return stays −6.05%, but HS VaR is applied to the larger current position value of 110.

Why the others are wrong

- **A:** Used the 13th worst return (−6.25%) on the old value of 100, i.e., the αn convention and a stale position value.
- **C:** Used the 13th worst return ($\alpha n = 13$) on the correct value of 110: 6.25% × 110.
- **D:** Applied the correct cut-off return to yesterday's value of 100, ignoring the 10% price rise.

Q8

BT answers L2.T5.68.1 · BT notes P2.T5.709.3

REBUILT — OCR damaged options C/D (USD 72/USD 73); reconstructed as USD 12.00 and USD 13.00 (undiscounted P/L without and with the dividend), consistent with the question.

An asset is worth USD 100.00 at the beginning of the year and USD 112.00 at the end of the year, and it pays an interim dividend of USD 1.00 during the year. What is the present value of the profit/loss if the discount rate is 3.0%?

A. USD 8.74
B. USD 9.71
C. USD 12.00
D. USD 13.00

Answer: B

Step by step

1. Dowd's discounted P/L: $P/L = \frac{P_t + D_t}{1+d} - P_{t-1}$.
2. End-of-period value plus dividend = 112 + 1 = 113.
3. Discount: 113 / 1.03 = 109.7087.
4. Subtract the starting value: 109.7087 − 100 = USD 9.71.

Why B is correct: P/L is the end-of-period value plus interim payments, discounted to the start date, minus the starting value.

Why the others are wrong

- **A:** Left out the USD 1.00 dividend: 112/1.03 − 100 = USD 8.74.
- **C:** Undiscounted price change without the dividend: 112 − 100.
- **D:** Undiscounted P/L with the dividend (113 − 100), ignoring the 3% discount rate.

Q9

BT answers L2.T5.68.3 · BT notes P2.T5.709.3

REBUILT — Stem did not fix the order-statistic convention and the options (14.01, 14.50) invited a midpoint or other reading; stated the Dowd convention and replaced options with realistic off-by-one and interpolation errors.

An empirical loss/profit (L/P) dataset (positive values are losses, negative values are gains) contains 100 observations. The five highest losses are \$11, \$12, \$13, \$14 and \$15 million. Using the convention in the reading (the VaR observation at confidence level 1 − α among n ordered observations is the (α × n + 1)th largest loss), what is the 99% historical simulation VaR?

A. \$13 million
B. \$14 million
C. \$14.5 million
D. \$15 million

Answer: B

Step by step

1. α = 1 − 0.99 = 0.01 and n = 100, so the VaR observation is the $\alpha n + 1 = 0.01 \times 100 + 1 =$ 2nd largest loss.
2. Order losses from largest: \$15, \$14, \$13, \$12, \$11.
3. The 2nd largest loss is \$14 million, so $\text{VaR}_{99\%} =$ \$14 million.
4. Check: the single worst loss (\$15) alone fills the 1% tail; VaR is the first observation just beyond it.

Why B is correct: Under the reading's convention the 1% tail holds α × n = 1 observation, and VaR is the next one, the 2nd largest loss.

Why the others are wrong

- **A:** \$13 million is the 3rd largest loss; it results from adding one too many to α × n + 1.
- **C:** \$14.5 million averages the 1st and 2nd largest losses, an interpolation not used by the reading's convention.
- **D:** \$15 million is the largest loss, i.e. the α × n = 1st observation, forgetting the +1.

Q10

BT answers P2.T5.22.1.1

REBUILT — Original 250-day sample gave non-integer α × n + 1 (3.5 and 7.25) so the key depended on an unstated weighting convention and the keyed answer 14.0 ignored the sign of X − Y; changed to 200 days and Y − X so the (α × n + 1) convention gives unique integer ranks.

A risk analyst has 200 daily loss/profit observations (millions of dollars, losses positive) for a fund. The ten largest losses, in descending order, are: 39, 38, 33, 31, 26, 23, 19, 15, 12, 9. Using basic (equally weighted) historical simulation and the convention that the VaR at confidence level 1 − α is the (α × n + 1)th largest loss, let X be the one-day 97.5% VaR and Y the one-day 99.0% VaR. What is Y − X?

A. 10
B. 12
C. -10
D. 14

Answer: A

Step by step

1. 99.0% VaR: $\alpha n + 1 = 0.01 \times 200 + 1 = 3$, so Y = 3rd largest loss = 33.
2. 97.5% VaR: $\alpha n + 1 = 0.025 \times 200 + 1 = 6$, so X = 6th largest loss = 23.
3. $Y - X = 33 - 23 = 10$.
4. The higher confidence level must give the larger VaR, so Y − X is positive.

Why A is correct: Each VaR is a specific order statistic of the losses, the 3rd (99%) and 6th (97.5%) largest; their difference is 10.

Why the others are wrong

- **B:** 12 uses the (α × n)th observation (2nd = 38 and 5th = 26) and omits the +1: 38 − 26.
- **C:** -10 computes X − Y, the reverse of what was asked.
- **D:** 14 uses the 7th largest loss (19) for the 97.5% VaR, over-counting by one: 33 − 19.

Q11 BT notes P2.T5.707.1 · BT answers P2.T5.707.1

REBUILT — Histogram was unnecessary and missing in one version; made the stem self-contained with the sorted losses, stated the (α × n + 1) convention, and replaced giveaway options with adjacent-rank errors.

A mutual fund has 300 daily returns. The ten worst daily returns, in dollars per unit, are:

Rank	Return
1	-1.250
2	-0.900
3	-0.870
4	-0.840
5	-0.820
6	-0.800
7	-0.770
8	-0.730
9	-0.710
10	-0.700

Using basic historical simulation and the convention that the VaR at confidence level 1 − α is the (α × n + 1)th worst outcome, what is the one-day 99.0% VaR?

A. 0.800
B. 0.840
C. 0.870
D. 0.900

Answer: B

Step by step

1. α = 0.01, n = 300, so $\alpha n + 1 = 0.01 \times 300 + 1 = 4$.
2. The 4th worst return is -0.840.
3. VaR is quoted as a positive loss: 0.840.

Why B is correct: The 1% tail contains α × n = 3 observations; VaR is the 4th worst, which is the first observation outside the tail.

Why the others are wrong

- **A:** 0.800 is the 6th worst return, the αn observation for a 2% tail (98% confidence), not the 1% tail.
- **C:** 0.870 is the 3rd worst return, the αn observation, omitting the +1.
- **D:** 0.900 is the 2nd worst return, which would result from using n = 200 instead of 300 ($\alpha n = 2$).

Q12 Schweser Module Quiz 1.1 Q1

REBUILT — Stem lacked the reference distribution; specified that returns are standardized (mean 0, σ 1) so the comparison with the normal quantile 1.645 is well defined.

Profit/loss data are standardized to mean 0 and standard deviation 1. The 95% VaR estimated by historical simulation from 1,000 observations is 1.56. Which statement is most likely true?

A. The parametric assumption of normal returns is correct.
B. The parametric assumption of lognormal returns is correct.
C. The historical distribution has fatter tails than a normal distribution.
D. The historical distribution has thinner tails than a normal distribution.

Answer: D

Step by step

1. For a standard normal distribution, the 5% one-tailed critical value is z = 1.645, so normal VaR = 1.645.
2. Historical VaR = 1.56 < 1.645: the 5% loss threshold lies closer to the center than the normal predicts.
3. Hence less probability mass sits far in the tail: the historical distribution has thinner tails than the normal.

Why D is correct: A smaller empirical quantile than the normal quantile at the same confidence level indicates thinner tails.

Why the others are wrong

- **A:** A normal fit would give 1.645, not 1.56, so the normal assumption is not supported.
- **B:** Nothing in a single quantile supports lognormality; geometric-return VaR would also not be 1.56 for these standardized data.
- **C:** Fat tails would give an empirical VaR above 1.645; the direction is reversed.

LO 1.b

Estimate VaR using a parametric approach for both normal and lognormal asset price distributions.

Q13

AnalystPrep Q.2434

REBUILT — Original volatilities (0.25 monthly vs 0.15 weekly) were unrealistic and the stem did not say returns are iid or ask for an annualized comparison; rebuilt with realistic percentages and an explicit annualized-volatility question.

A portfolio manager compares two funds whose volatilities were estimated at different return frequencies. Returns for both funds are assumed to be independent and identically distributed.

Metric	Fund X	Fund Y
Return frequency	Monthly	Weekly
Volatility (per period)	5.0%	2.2%

Using 12 months and 52 weeks per year, which of the following statements is correct?

A. Fund X has the higher annualized volatility, about 17.3% versus 15.9% for Fund Y.
B. Fund Y has the higher annualized volatility, about 114.4% versus 60.0% for Fund X.
C. The funds have approximately equal annualized volatility, so they are equally risky.
D. Annualized volatilities cannot be compared because the funds' return frequencies differ.

Answer: A

Key correction: Source key (AnalystPrep) said B using the original numbers (0.25 monthly vs 0.15 weekly), where sqrt-time scaling gives 0.866 vs 1.082; the item was rebuilt with new data, so the answer is now A.

Step by step

1. Under iid returns, volatility scales with the square root of time: $\sigma_{annual} = \sigma_{period}\sqrt{T}$, where T is the number of periods per year.
2. Fund X: $5.0\% \times \sqrt{12} = 5.0\% \times 3.4641 = 17.32\%$.
3. Fund Y: $2.2\% \times \sqrt{52} = 2.2\% \times 7.2111 = 15.87\%$.
4. Compare on the same horizon: 17.32% > 15.87%, so Fund X is riskier.

Why A is correct: Volatilities quoted at different frequencies must be put on a common horizon before comparing; with iid returns the square-root-of-time rule does this, and Fund X's annualized volatility is higher.

Why the others are wrong

- **B:** Scales volatility linearly by the number of periods ($5.0\% \times 12 = 60.0\%$, $2.2\% \times 52 = 114.4\%$) instead of by the square root, which overstates the weekly fund.
- **C:** Rounds the two annualized figures (17.3% vs 15.9%) into 'about equal'; the gap is material and the question asks which is higher.
- **D:** Volatilities at different frequencies are comparable once converted to a common horizon; the iid assumption makes the square-root-of-time conversion valid.

Q14

AP Mock 1 Q2

An analyst has gathered the following information about a portfolio which has normally distributed geometric returns:

Item	Value
Mean	12%
Standard deviation	32%
Portfolio value	USD 85 million

What is the 95% lognormal VaR for this portfolio?

A. USD 34.54 million.
B. USD 27.20 million.
C. USD 56.61 million.
D. USD 28.39 million.

Answer: D

Step by step

1. Normally distributed geometric returns imply lognormal prices: $\text{VaR} = P\,(1 - e^{\mu - z\sigma})$.
2. Exponent: $0.12 - 1.645 \times 0.32 = -0.4064$.
3. $e^{-0.4064} = 0.6661$, so the loss fraction is $1 - 0.6661 = 0.3339$.

4. VaR = 85 × 0.3339 = USD 28.39 million.

Why D is correct: With lognormal prices, the 5th-percentile price is $Pe^{\mu - z\sigma}$; VaR is the starting value minus that price.

Why the others are wrong

- **A:** This is the normal (arithmetic) VaR, $85 \times (1.645 \times 0.32 - 0.12) = 34.54$; the normal formula was used instead of the lognormal one.
- **B:** 85 × 0.32 = 27.20: used only the volatility, ignoring both z and the drift.
- **C:** 56.61 = 85 × 0.6661 is the 5th-percentile portfolio value, not the loss; forgot to subtract it from P.

Q15

AP Mock 2 Q22

REBUILT — Original options A/B were arbitrary round numbers and the z-value was not stated; added z = 2.326 and replaced distractors with realistic-error results.

A risk analyst wishes to establish the VaR of a portfolio under his management. The portfolio has a value of GBP 10 million. The annual mean and volatility of the portfolio returns are 15% and 30% respectively. Use z = 2.326 for 99%.

By how much does the 1-year 99% VaR calculated under the normal distribution assumption exceed the 1-year 99% VaR calculated under the lognormal distribution assumption?

A. GBP 1,954,800.
B. GBP 5,478,000.
C. GBP 527,800.
D. GBP 1,260,200.

Answer: D

Step by step

1. Normal VaR = $P(z\sigma - \mu)$ = 10,000,000 × (2.326×0.30 − 0.15) = 10,000,000 × 0.5478 = GBP 5,478,000.
2. Lognormal VaR = $P(1 - e^{\mu - z\sigma})$ = 10,000,000 × $(1 - e^{-0.5478})$ = 10,000,000 × (1 − 0.57822) = GBP 4,217,795.
3. Difference = 5,478,000 − 4,217,795 = GBP 1,260,205 ≈ GBP 1,260,200.

Why D is correct: Lognormal VaR is lower than normal VaR because lognormal losses are bounded by the position value; the gap here is about GBP 1.26 million.

Why the others are wrong

- **A:** Ignored the 15% drift in both VaRs (normal 6,978,000 minus lognormal 5,023,200 = 1,954,800).
- **B:** This is the normal VaR itself, not the difference between the two VaRs.
- **C:** Used the 95% z of 1.645 instead of 2.326 for both VaRs (about 3,435,000 − 2,907,200 = 527,800).

Q16

AP Mock 2 Q33

A risk manager wishes to estimate market risk under both the normal and lognormal distribution assumptions:

Item	Value
Daily mean	0.00079365
Daily volatility	0.025198
Trading days in a year	252
Current portfolio value	USD 1,000,000

Which of the following statements is true?

A. Normal 95% VaR exceeds lognormal 95% VaR at the 1-day holding period by USD 36,745.
B. Normal 95% VaR exceeds lognormal 95% VaR at the 1-year holding period by USD 800.
C. Normal 99% VaR exceeds lognormal 99% VaR at the 1-day holding period by USD 1,640.
D. Normal 99% VaR exceeds lognormal 99% VaR at the 1-year holding period by USD 421,255.

Answer: C

Step by step

1. 1-day, 99% (z = 2.326): normal VaR = 1,000,000 × (2.326 × 0.025198 − 0.00079365) = USD 57,817.
2. Lognormal VaR = 1,000,000 × $(1 - e^{-0.057817})$ = USD 56,177.
3. Difference = USD 1,640, so C is true.
4. Checks: the 1-day 95% gap is about 815 (not 36,745); the 1-year ($\mu = 0.2$, $\sigma = 0.4$) 95% gap is about 90,551 (not 800); the 1-year 99% gap is about 212,123 (not 421,255).

Why C is correct: Over short horizons the two VaRs nearly coincide because e^x ≈ 1 + x; the gap widens with horizon and confidence level.

Why the others are wrong

- **A:** The 1-day 95% gap is only about USD 815, so 36,745 is far too large.
- **B:** USD 800 is the order of the 1-day 95% gap (about 815), wrongly attached to the 1-year horizon, where the gap is about USD 90,551.
- **D:** The 1-year 99% gap is about USD 212,123, roughly half the stated figure.

Q17

AP Mock 3 Q28

REBUILT — Original asked a vague 'advantage' question whose key (compounding) is arguable; rebuilt to test why geometric returns underpin lognormal VaR (prices cannot be negative).

A risk analyst is choosing between geometric (continuously compounded) and arithmetic returns when estimating parametric VaR for a long horizon. Which statement best explains why geometric returns underpin the lognormal VaR approach?

A. Geometric returns assume interim payments are not reinvested, which keeps the calculation simple.

B. Geometric returns always have lower volatility than arithmetic returns, so VaR is smaller.

C. Geometric returns assume interim payments are continuously reinvested, and since the price is the starting price times e raised to the return, the price can never be negative.

D. Geometric returns are normally distributed only when prices are themselves normally distributed.

Answer: C

Step by step

1. Geometric return: $R_t = \ln\left((P_t + D_t)/P_{t-1}\right)$, which assumes interim payments are continuously reinvested.
2. Hence $P_t = P_{t-1}e^{R_t} > 0$ for any R_t, so prices cannot be negative.
3. If R is normal, the price is lognormal, giving the lognormal VaR $P(1 - e^{\mu - z\sigma})$.

Why C is correct: Geometric returns reinvest interim payments continuously and map to a strictly positive price, so normal geometric returns imply lognormal prices.

Why the others are wrong

- **A:** This describes arithmetic returns (no reinvestment), which are unsuitable for long horizons.
- **B:** Geometric returns are not inherently less volatile; this is not the basis of lognormal VaR.
- **D:** Reversed: normally distributed geometric returns imply lognormal (not normal) prices.

Q18

AnalystPrep Q.1475

REBUILT — Original distractors (9.87, 13.956, -13.956) did not correspond to realistic mistakes; replaced with sign, missing-mean and wrong-z errors.

Profit/losses (P/L) are normally distributed with a mean of 12 and a standard deviation of 18. What is the corresponding VaR at a 95% confidence level? (Use z = 1.645.)

A. 41.61.

B. 17.61.

C. 29.61.

D. 23.28.

Answer: B

Step by step

1. For normal P/L: VaR $= -\mu + z_\alpha\sigma$.
2. 95% one-tailed z = 1.645.
3. VaR = −12 + 1.645 × 18 = −12 + 29.61 = 17.61.

Why B is correct: VaR is the loss quantile, 1.645σ below the mean, i.e. −μ + 1.645σ in the positive-loss convention.

Why the others are wrong

- **A:** Added the mean (12 + 29.61) instead of subtracting it; a positive mean P/L reduces VaR.
- **C:** Forgot to subtract the mean: 1.645 × 18 = 29.61.
- **D:** Used the two-tailed 1.96 instead of 1.645: −12 + 1.96×18 = 23.28.

Q19

AnalystPrep Q.1482

REBUILT — Original relied on unstated rounded z-values (1.65, 2.33); added them to the stem so the key is unambiguous.

A portfolio has a beginning period value of $200. The arithmetic returns follow a normal distribution with a mean of 5% and a standard deviation of 10%. Using z = 1.65 and z = 2.33, calculate VaR at the 95% and 99% confidence levels, respectively:

A. $23, $36.6.
B. $43, $56.6.
C. $1.65, $2.33.
D. $23, $43.

Answer: A
Step by step
1. VaR = $(-\mu_r + z\sigma_r)P_{t-1}$.
2. 95%: (−0.05 + 1.65×0.10) × 200 = 0.115 × 200 = $23.
3. 99%: (−0.05 + 2.33×0.10) × 200 = 0.183 × 200 = $36.6.

Why A is correct: Parametric normal VaR is the return quantile −μ + zσ multiplied by the starting value.
Why the others are wrong
- **B:** Added the mean instead of subtracting it: (0.05+0.165)×200 = 43 and (0.05+0.233)×200 = 56.6.
- **C:** Reported the z-scores themselves, with no σ, μ or portfolio value.
- **D:** The 95% figure is right, but 43 for the 99% VaR is the 95% added-mean value (0.215×200); the 99% VaR is 36.6.

Q20

AnalystPrep Q.2632

REBUILT — z-value stated explicitly, and an unexplainable distractor replaced with the normal-VaR result (55.8).

An analyst has gathered the following information about a portfolio that has normally distributed geometric returns:

Item	Value
Mean	10%
Standard deviation	40%
Portfolio value	USD 100 million

What is the 95% lognormal VaR for this portfolio? (Use z = 1.645.)

A. USD 74.7 million.
B. USD 55.8 million.
C. USD 42.8 million.
D. USD 113.4 million.

Answer: C
Step by step
1. Lognormal VaR = $P(1 - e^{\mu - z\sigma})$.
2. Exponent: 0.10 − 1.645×0.40 = −0.558.
3. $e^{-0.558} = 0.5724$.
4. VaR = 100 × (1 − 0.5724) = USD 42.8 million.

Why C is correct: The 95% worst-case value is P·e^(μ−zσ) = 57.2 million, so the loss is 42.8 million.
Why the others are wrong
- **A:** 100×(e^{0.558} − 1) = 74.7: sign of the exponent reversed.
- **B:** Normal VaR: 100×(1.645×0.40 − 0.10) = 55.8; used the normal rather than lognormal formula.
- **D:** 100×(e^{0.758} − 1) = 113.4: used e^{μ+zσ} − 1, adding the mean and flipping the exponent sign.

Q21

AnalystPrep Q.2636

REBUILT — Original distractors were arbitrary and the key relied on rounded z = 2.33; fixed z = 2.326, recomputed, and rebuilt distractors from realistic mistakes.

Jacob Watson is a risk manager for a large bank estimating VaR for the equities portfolio using normal and lognormal assumptions. The portfolio value is USD 1 million, the annual mean return is 15% and the annual volatility is 25%. Use z = 2.326 for 99%. What are the 1-year 99% VaRs under the two assumptions?

A. Normal: USD 431,500; Lognormal: USD 350,500.
B. Normal: USD 350,500; Lognormal: USD 431,500.
C. Normal: USD 731,500; Lognormal: USD 518,800.
D. Normal: USD 581,500; Lognormal: USD 440,900.

Answer: A

Step by step

1. Normal: $P(z\sigma - \mu)$ = 1,000,000 × (2.326×0.25 − 0.15) = 1,000,000 × 0.4315 = USD 431,500.
2. Lognormal: $P(1 - e^{\mu - z\sigma})$ = 1,000,000 × $(1 - e^{-0.4315})$ = 1,000,000 × (1 − 0.6495) = USD 350,500.
3. Lognormal VaR is below normal VaR.

Why A is correct: Lognormal VaR is bounded by the portfolio value and is smaller than the normal VaR for the same inputs.

Why the others are wrong

- **B:** Values swapped; lognormal VaR cannot exceed normal VaR here.
- **C:** Added the mean ($z\sigma + \mu = 0.7315$) rather than subtracting it; $1 - e^{-0.7315} = 51.9\%$.
- **D:** Ignored the drift entirely ($z\sigma = 0.5815$; $1 - e^{-0.5815} = 44.1\%$).

Q22

AnalystPrep Q.2817

REBUILT — Original distractors were arbitrary numbers (one option even mixed up the 95% and 99% figures); replaced with options that each reflect a realistic formula error.

Assume that the P/L over a specified period is normally distributed with a mean of 14.1 and a standard deviation of 28.2. What are the 95% VaR and the 99% VaR?

A. 95% VaR = 46.389; 99% VaR = 65.593
B. 95% VaR = 41.172; 99% VaR = 58.543
C. 95% VaR = 32.289; 99% VaR = 51.493
D. 95% VaR = 60.489; 99% VaR = 79.693

Answer: C

Step by step

1. For normally distributed P/L, $\text{VaR}_\alpha = -\mu + \sigma z_\alpha$.
2. 95%: $z = 1.645$, so $-14.1 + 28.2 \times 1.645 = -14.1 + 46.389 = 32.289$.
3. 99%: $z = 2.326$, so $-14.1 + 28.2 \times 2.326 = -14.1 + 65.593 = 51.493$.

Why C is correct: Because the mean P/L is a positive expected profit, it offsets part of the loss quantile, so VaR is the quantile term σz_α minus μ.

Why the others are wrong

- **A:** Omits the mean (σz_α only: 46.389 and 65.593), ignoring that expected profit reduces VaR.
- **B:** Uses two-tailed z-values (1.96 and 2.576) instead of the one-tailed 1.645 and 2.326.
- **D:** Adds the mean instead of subtracting it ($+\mu + \sigma z_\alpha$), treating a profit as if it were an extra loss.

Q23

AnalystPrep Q.2819

REBUILT — Original distractors were arbitrary (two shared the same bad 95% figure); replaced with options reflecting realistic formula errors.

Assume geometric returns R_t are normally distributed with a mean of 0.079 and a standard deviation of 0.312, and the portfolio is currently worth 1 unit. What are the 95% and 99% lognormal VaR?

A. 95% VaR = 0.4469; 99% VaR = 0.5528
B. 95% VaR = 0.4342; 99% VaR = 0.6467
C. 95% VaR = 0.3522; 99% VaR = 0.4762
D. 95% VaR = 0.4014; 99% VaR = 0.5160

Answer: C

Step by step

1. Lognormal VaR = $P_{t-1}\,(1 - e^{\mu_R - \sigma_R z_\alpha})$.
2. 95%: $1 - e^{0.079 - 0.312 \times 1.645} = 1 - e^{-0.4342} = 1 - 0.6478 = 0.3522$.
3. 99%: $1 - e^{0.079 - 0.312 \times 2.326} = 1 - e^{-0.6467} = 1 - 0.5238 = 0.4762$.

Why C is correct: Lognormal VaR is the loss from the price falling to the percentile $e^{\mu_R - \sigma_R z_\alpha}$; the positive mean raises that percentile and lowers VaR.

Why the others are wrong

- **A:** Flips the sign of the mean inside the exponent ($e^{-\mu - \sigma z}$), which treats the drift as a loss.
- **B:** Applies the normal-VaR shortcut $-\mu + \sigma z$ (0.4342, 0.6467) to geometric returns without the exponential transformation.
- **D:** Drops the mean: $1 - e^{-\sigma z}$ (0.4014, 0.5160).

Q24

AnalystPrep Q.2818

REBUILT — Original used absurd units (mean return of 155% on a 1-unit portfolio) and arbitrary distractors; restated as percentages on USD 100 million with error-based options.

Over a one-month horizon, arithmetic returns are normally distributed with a mean of 1.55% and a standard deviation of 1.07%. The portfolio is currently worth USD 100 million. What are the 95% and 99% VaR?

A. 95% VaR = USD 1.760 million; 99% VaR = USD 2.489 million
B. 95% VaR = USD 3.310 million; 99% VaR = USD 4.039 million
C. 95% VaR = USD 0.547 million; 99% VaR = USD 1.206 million
D. 95% VaR = USD 0.210 million; 99% VaR = USD 0.939 million

Answer: D

Step by step

1. Normal VaR for arithmetic returns: $\text{VaR}_\alpha = (-\mu_r + \sigma_r z_\alpha)\, P_{t-1}$.
2. 95%: $(-0.0155 + 0.0107 \times 1.645) \times 100 = 0.00210 \times 100 =$ USD 0.210 million.
3. 99%: $(-0.0155 + 0.0107 \times 2.326) \times 100 = 0.00939 \times 100 =$ USD 0.939 million.

Why D is correct: VaR is the loss quantile σz_α reduced by the expected return, then scaled by portfolio value.

Why the others are wrong

- **A:** Leaves out the mean return ($\sigma z_\alpha \times 100$ only).
- **B:** Adds the mean return ($+\mu + \sigma z$) instead of subtracting it.
- **C:** Uses two-tailed z-values 1.96 and 2.576 instead of 1.645 and 2.326.

Q25

BT Mock Q66 · BT answers P2.T5.22.1.2 · BT notes P2.T5.22.1.2

REBUILT — Merged three identical copies; kept the stem but replaced options A-C, which could not be reproduced by any natural calculation, with options generated by realistic errors (swapped parameters, mean-sign error, ignoring means).

Bertha is a risk manager who estimates the market risk of one of her firm's portfolios. She measures performance with both arithmetic and geometric returns. Assumptions:
- The portfolio value is USD 100.0 million; there are 250 trading days per year.
- Geometric returns: average +12.0% per annum, standard deviation 28.0%.
- Arithmetic returns: average +14.0% per annum, standard deviation 32.0%.

She measures the one-day 95.0% VaR assuming arithmetic returns are normal (normal VaR) and assuming geometric returns are normal (lognormal VaR), with serially independent returns. Which is nearest to the difference between the two?

A. Normal VaR is approximately USD 355,200 less than lognormal VaR.
B. Normal VaR is approximately USD 402,200 greater than lognormal VaR.
C. Normal VaR is approximately USD 458,200 greater than lognormal VaR.
D. Normal VaR is approximately USD 448,800 greater than lognormal VaR.

Answer: D

Step by step

1. Daily arithmetic parameters: $\mu = 0.14/250 = 0.00056$, $\sigma = 0.32/\sqrt{250} = 0.020239$.
2. Normal VaR $= 100\text{m}\,(1.645 \times 0.020239 - 0.00056) = 100\text{m} \times 0.032732 \approx$ USD 3,273,246.
3. Daily geometric parameters: $\mu = 0.12/250 = 0.00048$, $\sigma = 0.28/\sqrt{250} = 0.017709$.
4. Lognormal VaR $= 100\text{m}\,(1 - e^{0.00048 - 1.645 \times 0.017709}) = 100\text{m} \times 0.028244 \approx$ USD 2,824,436.
5. Difference $\approx$ 3,273,246 − 2,824,436 = USD 448,810 (normal is larger).

Why D is correct: Each model must use its own return convention (arithmetic parameters for normal VaR, geometric for lognormal VaR); the gap arises mainly from the different volatility inputs.

Why the others are wrong

- **A:** Swaps the parameter sets (normal VaR on geometric inputs, lognormal VaR on arithmetic inputs), which reverses the sign of the gap.
- **B:** Drops the drift in the lognormal calculation only ($1 - e^{-\sigma z}$ with the mean omitted), which inflates lognormal VaR and shrinks the gap to about USD 402,200.
- **C:** Ignores the means in both calculations (σz vs $1 - e^{-\sigma z}$), overstating the gap.

Q26 BT answers L2.T5.68.2 · BT notes P2.T5.709.3

According to Dowd, each of the following is TRUE except:

A. Geometric returns may be economically more meaningful because an asset price can never become negative.
B. The geometric return over multiple periods is the sum of the one-period geometric returns.
C. The difference between geometric and arithmetic returns is negligible over short periods but significant over long periods.
D. The arithmetic return implicitly assumes that interim payments are continuously reinvested.

Answer: D

Step by step

1. Arithmetic return $r_t = [P_t + D_t - P_{t-1}]/P_{t-1}$ treats the interim payment D_t as earning no return (no reinvestment).
2. Geometric return $R_t = \ln[(P_t + D_t)/P_{t-1}]$ implicitly assumes continuous reinvestment of interim payments.
3. So D reverses the two definitions and is the false statement; A, B and C are true.

Why D is correct: The reinvestment assumption belongs to geometric returns, not arithmetic returns.

Why the others are wrong

- **A:** True: $P_t = P_{t-1}e^{R_t} > 0$, so geometric returns can never imply a negative price.
- **B:** True: log returns are time-additive, so the multi-period geometric return is the sum of one-period returns.
- **C:** True: $R \approx r$ for small returns, with the gap growing as the horizon (and compounding) lengthens.

Q27 BT answers L2.T5.69.1

REBUILT — The 'L/P' wording conflicts with subtracting the mean (a mean loss would be added); restated as P/L and replaced weak distractors (16.00, 17.32) with error-based options.

If the profit/loss (P/L) over some period is normally distributed with mean USD 8 and standard deviation USD 16, what is the 95% VaR?

A. USD 26.32
B. USD 18.32
C. USD 34.32
D. USD 12.51

Answer: B

Step by step

1. For normally distributed P/L, $\text{VaR}_{95\%} = -\mu + \sigma z_{95\%}$.
2. $= -8 + 16 \times 1.645 = -8 + 26.32 =$ USD 18.32.

Why B is correct: The expected profit of USD 8 cushions the loss quantile σz, reducing VaR from 26.32 to 18.32.

Why the others are wrong

- **A:** Ignores the mean ($\sigma z = 26.32$), i.e. relative VaR rather than absolute VaR.
- **C:** Adds the mean ($8 + 26.32$), treating expected profit as an expected loss.
- **D:** Uses z = 1.282 (the 90% quantile): $-8 + 16 \times 1.282 = 12.51$.

Q28 BT answers L2.T5.69.2

REBUILT — Original distractors (0.120, 0.168) were not tied to any error; replaced with options from realistic mistakes, same numbers.

Suppose geometric returns R_t are normally distributed with mean 0.08 and standard deviation 0.10, and the portfolio is currently worth USD 1.00. What is the 99% lognormal VaR?

A. USD 0.081
B. USD 0.142
C. USD 0.153
D. USD 0.208

Answer: B

Step by step

1. Lognormal VaR $= P_{t-1}\,(1 - e^{\mu_R - \sigma_R z_\alpha})$, with $z_{99\%} = 2.326$.
2. Exponent: $0.08 - 0.10 \times 2.326 = -0.1526$.
3. $e^{-0.1526} = 0.8585$, so VaR $= 1 - 0.8585 =$ USD 0.142.

Why B is correct: Lognormal VaR converts the normal percentile of the geometric return into a price percentile,

so the loss is $1 - e^{\mu - \sigma z}$.

Why the others are wrong

- **A:** Uses z = 1.645 (95%) instead of 2.326: $1 - e^{0.08 - 0.1645} = 0.081$.
- **C:** Applies the normal VaR formula $-\mu + \sigma z = 0.1526$ without exponentiating.
- **D:** Ignores the mean: $1 - e^{-0.2326} = 0.208$.

Q29

BT answers P2.T5.707.2 · BT notes P2.T5.707.2

Consider the following asset over three periods that starts with an initial price of $20.00 and pays a $2.00 dividend each period. In addition to the asset's price and dividends over three periods, we also show its per-period arithmetic (aka, simple) return:

Period	Initial	1	2	3
Price	$20.00	$23.00	$19.00	$25.00
Dividend	| $2.00	$2.00	$2.00	
Arithmetic returns	| 25.00%	-8.70%	42.11%	

About this asset's performance, each of the following statements is true EXCEPT which is not accurate?

A. Geometric returns cannot be greater than arithmetic returns

B. The geometric returns in each period are approximately +22.31%, -9.10% and 35.14%

C. If the asset instead paid zero dividends, then the geometric return would equal the arithmetic return

D. The three-period geometric return is conveniently the sum of per-period geometric returns which, in this case, is about +22.31% -9.10% +35.14% = 48.36%

Answer: C

Step by step

1. Geometric return $R_t = \ln[(P_t + D_t)/P_{t-1}]$.
2. Period 1: $\ln(25/20) = 22.31\%$; period 2: $\ln(21/23) = -9.10\%$; period 3: $\ln(27/19) = 35.14\%$. So B is true.
3. Log returns add over time: 22.31% - 9.10% + 35.14% = 48.36%. So D is true.
4. Since $\ln(1 + r) \leq r$ for every r > -1 (equality only at r = 0), geometric never exceeds arithmetic. So A is true.
5. Without dividends: arithmetic 15.00%, -17.39%, 31.58% vs geometric $\ln(23/20) = 13.98\%$, $\ln(19/23) = -19.11\%$, $\ln(25/19) = 27.44\%$: still different, so C is false.

Why C is correct: The gap between geometric and arithmetic returns comes from the concavity of the log function, not from dividends; removing dividends changes both returns but the geometric return stays below the arithmetic one for any nonzero return.

Why the others are wrong

- **A:** True statement: $\ln(1 + r) \leq r$ always, so geometric returns can at most equal arithmetic returns (at r = 0); choosing it confuses the direction of the inequality.
- **B:** True statement: the values are $\ln(25/20)$, $\ln(21/23)$, $\ln(27/19)$; choosing it usually comes from forgetting to add the dividend to the ending price when checking.
- **D:** True statement: additivity over time is the key convenience of log returns; choosing it confuses log returns with simple returns, which compound multiplicatively rather than add.

Q30

BT answers P2.T5.707.3 · BT notes P2.T5.707.3

A risk manager is estimating the market risk of a portfolio using both the normal distribution and the lognormal distribution assumptions. The manager gathers the following data on the portfolio:
- Annual mean: 12.0%
- Annual volatility: 37.0%
- Current portfolio value: USD $1.0 million due to 12,500 shares at a current price of $80.00 per share
- Trading days in a year: 250

Which of the following statements is correct?

A. Lognormal 95% VaR is greater than normal 95% VaR at the one-day holding period by about 2.43%

B. Lognormal 95% VaR is less than normal 95% VaR at the one-year (250 days) holding period by about 3.90%

C. Lognormal 99% VaR is greater than normal 99% VaR at the one-day holding period by about 4.04%

D. Lognormal 99% VaR is less than normal 99% VaR at the one-year (250 days) holding period by about 21.75%

Answer: D

Step by step

1. Normal VaR (as % of value) = $-\mu_T + z\sigma_T$; lognormal VaR = $1 - e^{\mu_T - z\sigma_T}$, with $\mu_T = 12\% \cdot T$, $\sigma_T = 37\%\sqrt{T}$.
2. One-year 99% (z = 2.326): normal = $-12\% + 2.326(37\%) = 74.07\%$ (USD 740,749); lognormal = $1 - e^{0.12-0.8607} = 1 - e^{-0.7407} = 52.32\%$ (USD 523,243); difference 21.75%.
3. One-year 95% (z = 1.645): normal 48.86%, lognormal 38.65%; difference 10.21%, not 3.90%.
4. One-day (T = 1/250): 95% normal 3.80% vs lognormal 3.73% (diff 0.07%); 99% normal 5.40% vs lognormal 5.25% (diff 0.14%).
5. With identical μ and σ, lognormal VaR is always smaller because $1 - e^{-x} < x$ for x > 0.

Why D is correct: At the same mean and volatility, $1 - e^{-x} < x$, so lognormal VaR is below normal VaR, and the gap grows with the horizon; at one year and 99% it is 74.07% - 52.32% = 21.75%.

Why the others are wrong

- **A:** Wrong direction and size: lognormal VaR is lower than normal VaR (3.73% vs 3.80% at one day, 95%); the gap is only about 0.07%.
- **B:** Right direction but wrong size: the one-year 95% gap is 48.86% - 38.65% = 10.21%, not 3.90%.
- **C:** Wrong direction: one-day 99% lognormal VaR (5.25%) is below normal VaR (5.40%), a gap of only about 0.14%.

Q31

GARP PE1 Q2 · Mock 3 Q78

REBUILT — Source options were rounded with z = 1.65 and the distractors could not be traced to clear mistakes; options were recomputed exactly with z = 1.645, and each distractor now comes from a specific error.

A risk manager is estimating the market risk of a portfolio under two assumptions: normally distributed arithmetic returns (normal VaR) and normally distributed geometric returns (lognormal VaR). The manager gathers the following data on the portfolio's daily returns:
Annualized average of arithmetic returns: 16%
Annualized standard deviation of arithmetic returns: 27%
Annualized average of geometric returns: 13%
Annualized standard deviation of geometric returns: 29%
Current portfolio value: EUR 5,200,000
Trading days in a year: 252
Assuming both daily arithmetic returns and daily geometric returns are serially independent, which of the following statements is correct?

A. The 1-day 95% normal VaR is 2.12% and the 1-day 95% lognormal VaR is 2.26%.
B. The 1-day 95% normal VaR is 2.80% and the 1-day 95% lognormal VaR is 2.96%.
C. The 1-day 95% normal VaR is 2.73% and the 1-day 95% lognormal VaR is 2.91%.
D. The 1-day 95% normal VaR is 3.27% and the 1-day 95% lognormal VaR is 3.47%.

Answer: C

Step by step

1. Scale the arithmetic inputs to one day: $\mu_a = 0.16/252 = 0.000635$, $\sigma_a = 0.27/\sqrt{252} = 0.017008$.
2. Scale the geometric inputs to one day: $\mu_g = 0.13/252 = 0.000516$, $\sigma_g = 0.29/\sqrt{252} = 0.018268$.
3. Normal VaR = $-\mu_a + 1.645\sigma_a = -0.000635 + 0.027978 = 2.73\%$.
4. Lognormal VaR = $1 - e^{\mu_g - 1.645\sigma_g} = 1 - e^{-0.029534} = 2.91\%$.
5. The portfolio value is not needed for percentage VaR (in currency: about EUR 142,190 and EUR 151,340).

Why C is correct: Each VaR uses its own return series: arithmetic parameters for normal VaR and geometric parameters for lognormal VaR, both scaled to one day (mean by T, volatility by √T) and evaluated at the one-tailed 95% z of 1.645. Here lognormal VaR is higher only because geometric volatility is higher.

Why the others are wrong

- **A:** Used the 90% quantile (z = 1.282) instead of the 95% quantile of 1.645.
- **B:** Ignored the daily mean: $1.645\sigma_a = 2.80\%$ and $1 - e^{-1.645\sigma_g} = 2.96\%$.
- **D:** Used the two-tailed 1.96 instead of the one-tailed 1.645 multiplier.

Q32

Mock 1 Q47

REBUILT — The source key USD 1,180,920 matched neither z = 1.645 (USD 1,175,735) nor z = 1.65 (USD 1,180,496); the stem now fixes z = 1.65 and all options were recomputed, with the distractors tied to specific errors.

A diversified portfolio exhibits a normally distributed geometric return with a mean and standard deviation of 12.2% and 23%, respectively. What is the 5% lognormal VaR, assuming the beginning period portfolio value is USD 5.2 million? Use z = 1.65.

A. USD 1,180,500
B. USD 1,339,000
C. USD 4,019,500
D. USD 1,642,100

Answer: A
Step by step
1. Lognormal VaR = $P[1 - e^{\mu - z\sigma}]$.
2. Exponent: $0.122 - 1.65(0.23) = 0.122 - 0.3795 = -0.2575$.
3. $e^{-0.2575} = 0.77298$.
4. VaR = 5,200,000 × (1 - 0.77298) = USD 1,180,496 ≈ USD 1,180,500.

Why A is correct: If geometric returns are normal, the 5% quantile of the end-of-period value is $Pe^{\mu - z\sigma}$; the loss relative to the starting value is $P[1 - e^{\mu - z\sigma}]$.

Why the others are wrong
- **B:** Normal (arithmetic) VaR: 5.2m × (0.3795 - 0.122) = 1,339,000; it does not exponentiate the geometric quantile.
- **C:** Reports the 5th-percentile portfolio value $Pe^{\mu - z\sigma}$ (5.2m × 0.77298) instead of the loss $P[1 - e^{\mu - z\sigma}]$.
- **D:** Ignored the mean: 5.2m × $[1 - e^{-0.3795}]$ = 1,642,142.

Q33

Mock 2 Q73

REBUILT — Source distractors (USD 1,800 either way) did not come from any identifiable mistake; they were replaced with values produced by realistic errors (flipped direction, 1.96 multiplier, ignored mean).

A diversified portfolio has a current value of USD 100,000, a mean return of 8%, and a standard deviation of 15%. What is the approximate difference between the value at risk (VaR) measures computed using a normal distribution assumption and a lognormal distribution assumption (same mean and standard deviation), assuming a 95% confidence level?

A. Lognormal VaR is less than normal VaR by about USD 2,100.
B. Lognormal VaR is greater than normal VaR by about USD 1,300.
C. Lognormal VaR is less than normal VaR by about USD 1,300.
D. Lognormal VaR is less than normal VaR by about USD 2,800.

Answer: C
Step by step
1. Normal VaR = $P(-\mu + z\sigma) = 100{,}000(-0.08 + 1.645 \times 0.15)$ = USD 16,673.
2. Lognormal VaR = $P[1 - e^{\mu - z\sigma}] = 100{,}000[1 - e^{-0.16673}] = 100{,}000(1 - 0.84643)$ = USD 15,357.
3. Difference = 16,673 - 15,357 = USD 1,316 ≈ USD 1,300, with lognormal VaR the smaller one.

Why C is correct: With the same μ and σ, the lognormal VaR $1 - e^{-x}$ is always below the normal VaR x (here x = 16.67%), so lognormal is lower by about USD 1,300.

Why the others are wrong
- **A:** Used the two-tailed 1.96: normal 21,400 vs lognormal 19,265, a gap of about 2,135.
- **B:** Right magnitude, wrong direction: $1 - e^{-x} < x$, so lognormal VaR cannot exceed normal VaR at the same parameters.
- **D:** Ignored the mean: normal 24,673 vs lognormal 21,865, a gap of about 2,808.

Q34

Schweser Module Quiz 1.1 Q2

Assume the profit/loss distribution for XYZ is normally distributed with an annual mean of $20 million and a standard deviation of $10 million. The 5% VaR is calculated and interpreted as which of the following statements?

A. 5% probability of losses of at least $3.50 million.
B. 5% probability of earnings of at least $3.50 million.
C. 95% probability of losses of at least $3.50 million.
D. 95% probability of earnings of at least $3.50 million.

Answer: D
Step by step
1. Normal VaR (P/L in $) = $-\mu_{P/L} + z\sigma_{P/L} = -20 + 1.65(10) = -3.50$ million.

2. Negative VaR means the 5th-percentile P/L is a profit: 20 - 16.5 = +$3.50 million.
3. So P/L falls below +$3.50 million with 5% probability; equivalently, earnings are at least $3.50 million with 95% probability.

Why D is correct: The 5% quantile of the P/L distribution is +$3.5 million, so with 95% confidence the firm earns at least $3.5 million; the negative VaR signals that even the bad tail is profitable.

Why the others are wrong

- **A:** Treats the -$3.5 million VaR as a $3.5 million loss, ignoring the sign: the 5th-percentile outcome is a $3.5 million profit, not a loss.
- **B:** Wrong tail: earnings of at least $3.5 million happen with 95% probability; only 5% of outcomes fall below $3.5 million.
- **C:** Mixes up both the sign and the tail: losses never occur at the 95% level here, because 95% of outcomes are profits of $3.5 million or more.

Q35

AnalystPrep Q.2820

REBUILT — The stem was clumsy and the distractors (11.27%, 11.32%, 36.72%) were giveaways tied to no realistic error; the stem was cleaned up and the distractors replaced with the results of common mistakes.

A portfolio's returns are normally distributed with a mean of 0.04 and a standard deviation of 0.32 over the horizon. Using z = 1.28 for the 10th percentile of the standard normal distribution, what is the 90% VaR as a percentage of portfolio value?

A. 36.96%
B. 40.96%
C. 44.96%
D. 48.64%

Answer: A

Step by step

1. Normal VaR $= -\mu + z\sigma$.
2. $z\sigma = 1.28 \times 0.32 = 0.4096$.
3. VaR = -0.04 + 0.4096 = 0.3696 = 36.96%; equivalently, the 10th-percentile return is 0.04 - 0.4096 = -36.96%.

Why A is correct: The 10th-percentile return is $\mu - 1.28\sigma = -36.96\%$, so with 90% confidence the loss will not exceed 36.96% of portfolio value.

Why the others are wrong

- **B:** Ignored the mean: reports only $z\sigma = 40.96\%$.
- **C:** Added the mean to the loss instead of subtracting it: 0.4096 + 0.04 = 44.96%.
- **D:** Used the 95% one-tailed z of 1.645 instead of 1.28: 1.645 × 0.32 - 0.04 = 48.64%.

Q36

AnalystPrep Q.1502

An interviewer asks a candidate to interpret a $20 million 15-day VaR figure at a 95% confidence level. Which of the following is the CORRECT interpretation?

A. There is a 5% chance of a gain greater than $20 million over a 15-day period.
B. The one-day VaR is $20 million at the 95% confidence level.
C. The minimum loss over the next 15 days is at least $20 million with 95% confidence.
D. The loss over the next 15 days is expected to be less than $20 million in 95% of scenarios.

Answer: D

Step by step

1. VaR at confidence level 95% is the loss threshold that is exceeded with probability 5% over the stated horizon: $P(L > \text{VaR}_{95\%}) = 5\%$.
2. Here the horizon is 15 days and the threshold is $20 million, so $P(L_{15d} > \$20\text{m}) = 5\%$.
3. Equivalently, $P(L_{15d} \leq \$20\text{m}) = 95\%$: in 95% of scenarios the 15-day loss is below $20 million, which is statement D.

Why D is correct: VaR is a quantile of the loss distribution over a specific horizon: at 95% confidence, losses stay below the VaR figure 95% of the time and exceed it 5% of the time. Statement D restates exactly that for the 15-day horizon.

Why the others are wrong

- **A:** VaR describes the left (loss) tail, not gains; this confuses the loss quantile with the upper tail of the P/L

distribution.
- **B:** The figure is a 15-day VaR; the one-day VaR would be smaller (roughly $\$20\text{m}/\sqrt{15} \approx \$5.16\text{m}$ under i.i.d. square-root-of-time scaling), so the horizon cannot simply be relabelled.
- **C:** VaR is not a minimum loss; it is the threshold that losses exceed only 5% of the time. Losses are below $20 million in 95% of cases, so 'at least $20 million with 95% confidence' reverses the meaning.

Q37 AnalystPrep Q.3036

REBUILT — Source option A was printed as a truncated '56' with no derivable meaning; it was replaced with 6.58 (the VaR that ignores the mean) so every option maps to a specific error, and the stem now states that the second parameter is the variance.

A portfolio P = A + B consists of two independent, normally distributed positions with A ~ N(0.5, 1) and B ~ N(3, 15), where the parameters are the mean and the variance of profit/loss. What is the 95% parametric (normal) VaR of the portfolio?

A. 6.58
B. 4.87
C. 3.08
D. 1.54

Answer: C

Step by step
1. Sum of independent normals is normal: $\mu_P = 0.5 + 3 = 3.5$.
2. Independence means zero covariance, so the variances add: $\sigma_P^2 = 1 + 15 = 16$, hence $\sigma_P = 4$.
3. The 95% one-tailed standard normal quantile is $z_{0.95} = 1.645$.
4. $\text{VaR}_{95\%} = -\mu_P + z\,\sigma_P = -3.5 + 1.645 \times 4 = -3.5 + 6.58 = 3.08$.

Why C is correct: For normal P/L, VaR is the negative of the 5% quantile, $-\mu + z\sigma$. Independence lets the variances (not the standard deviations) add, giving σ = 4, and the positive expected profit of 3.5 offsets part of the 6.58 tail term, leaving 3.08.

Why the others are wrong
- **A:** 6.58 is 1.645 × 4 alone; it ignores the expected profit $\mu_P = 3.5$, which must be subtracted from the tail term.
- **B:** 4.87 is $\sqrt{1} + \sqrt{15} = 1 + 3.873$, the sum of the standard deviations (the perfect-correlation case) reported as if it were the VaR, with no z-multiplier or mean adjustment and wrongly ignoring independence.
- **D:** 1.54 treats P as the equally weighted average $(A + B)/2$: mean 1.75, variance 4, σ = 2, so $-1.75 + 1.645 \times 2 = 1.54$; but the portfolio is the sum A + B, not the average.

LO 1.c

Estimate the expected shortfall given profit and loss (P&L) or return data.

Q38 GARP book Chapter 18 end-of-chapter Q18.2

REBUILT — Original referred to an unreproduced footnote; restated as a self-contained derivation request with the expected-shortfall formula for a normal loss.

Losses are normally distributed with mean μ and standard deviation σ. Using the fact that the expected shortfall at confidence level α of a normal loss is $\text{ES}_\alpha = \mu + \sigma\,\phi(z_\alpha)/(1-\alpha)$, where ϕ is the standard normal density and z_α the α-quantile, verify that the 97.5% expected shortfall equals $\mu + 2.338\sigma$.

Answer: see model answer — For $\alpha = 97.5\%$, $z_\alpha = 1.96$ and $\phi(1.96) = \frac{1}{\sqrt{2\pi}} e^{-1.96^2/2} = 0.05845$. Dividing by the tail probability $1 - \alpha = 0.025$ gives $0.05845/0.025 = 2.338$. Hence $\text{ES}_{97.5\%} = \mu + 2.338\sigma$. For comparison, the 97.5% VaR is only $\mu + 1.96\sigma$, so ES exceeds VaR as it must.

Step by step
1. Find $z_{97.5\%} = 1.96$.
2. Compute $\phi(1.96) = 0.3989 \times e^{-1.9208} = 0.05845$.
3. Tail probability $1 - \alpha = 0.025$, so $\phi/(1-\alpha) = 2.3378 \approx 2.338$.
4. $\text{ES} = \mu + 2.338\sigma$, as required.

Why this is correct: ES is the conditional mean of the loss beyond VaR; for a normal this adds the term $\sigma\phi(z)/(1-\alpha)$ to the mean, which at 97.5% is 2.338σ.

Q39 AnalystPrep Mock 2 Q73

An investment bank's CRO computes daily VaR at various confidence levels:

Confidence level	VaR
90.0%	125,555,000
95.0%	180,250,500
95.5%	185,000,500
96.0%	195,500,000
96.5%	203,205,500
97.0%	211,425,000
97.5%	222,233,500
98.0%	233,562,500
98.5%	241,523,000
99.0%	256,300,000
99.5%	280,500,000

Determine the closest estimate of the daily expected shortfall at the 96.5% confidence level.

A. USD 203,205,500.
B. USD 240,924,000.
C. USD 280,500,000.
D. USD 235,535,643.

Answer: B

Step by step

1. The tail beyond 96.5% has mass 3.5%; the table slices it into seven 0.5% slices, so there are six interior VaRs at 97.0%, 97.5%, 98.0%, 98.5%, 99.0%, 99.5%.
2. Sum = 211,425,000 + 222,233,500 + 233,562,500 + 241,523,000 + 256,300,000 + 280,500,000 = 1,445,544,000.
3. $ES_{96.5\%} \approx 1{,}445{,}544{,}000/6 = 240{,}924{,}000$.

Why B is correct: Expected shortfall is estimated as the average of the equally spaced tail VaRs strictly beyond the confidence level, which gives USD 240.924 million.

Why the others are wrong

- **A:** This is the 96.5% VaR itself; ES must exceed VaR because it averages the losses beyond it.
- **C:** This is the largest (99.5%) VaR, the extreme of the tail rather than its average.
- **D:** Averages seven values by including the 96.5% VaR (203,205,500) in the tail, although it is not beyond the threshold: 1,648,749,500/7 = 235,535,643.

Q40 AnalystPrep Mock 3 Q10

REBUILT — Stem did not state the tail-averaging convention and distractor A had no clear origin; added the convention and replaced distractors with realistic mistakes.

Jimmy Ray, a risk analyst at Alcoa Bank, has performed a historical simulation using the last 500 trading days of returns for the bank's fixed-income portfolio. The 10 worst one-day losses (USD) are:

9,111	8,669	8,127	7,098	6,712	6,698	5,743	5,189	4,811	4,775

Which of the following is the 99% one-day expected shortfall, taken as the average of the losses in the worst 1% tail of observations?

A. 7,943.
B. 8,251.
C. 6,712.
D. 9,111.

Answer: A

Step by step

1. Tail size = 500 × (1 − 0.99) = 5 observations.
2. The five worst losses are 9,111, 8,669, 8,127, 7,098, 6,712.
3. Sum = 39,717.
4. $ES_{99\%} = 39{,}717/5 = 7{,}943.4 \approx 7{,}943$.

Why A is correct: ES averages all losses in the 1% tail (the five worst of 500), giving 7,943, which lies between the 99% VaR (6,712) and the worst loss (9,111).

Why the others are wrong

- **B:** Averaged only the four worst losses (9,111, 8,669, 8,127, 7,098 = 8,251), dropping the fifth tail observation.
- **C:** This is the 99% VaR (5th worst loss), not the average of the tail.
- **D:** This is the single worst loss, not an average.

Q41

BT Mock Q1 · BT notes P2.T5.23.8.3

REBUILT — Original distractors (6.60, 9.38) did not correspond to identifiable errors; replaced with options generated by realistic mistakes (VaR, drift sign, ignoring drift).

An exposure has a value of USD 30.0 million. In L(+)/P(−) format its expected loss is 4.0% with volatility 15.0%, i.e. $\mu_{L/P} = +4.0\%$ and $\sigma = 15.0\%$. The exposure is normally distributed. Which is nearest to the single-period 97.5% expected shortfall (ES)?

A. USD 9.32 million.
B. USD 10.02 million.
C. USD 10.52 million.
D. USD 11.72 million.

Answer: D

Step by step

1. Tail multiplier for a normal: $\phi(z)/(1-\alpha)$ with $z_{97.5\%} = 1.96$: $\phi(1.96) = 0.05845$, so $0.05845/0.025 = 2.3378$.
2. In loss-positive format ES $= \mu + \sigma \times 2.3378 = 4.0\% + 15.0\% \times 2.3378 = 39.07\%$.
3. Dollar ES = 0.3907 × 30.0 = USD 11.72 million.

Why D is correct: Under normality, ES = μ + σφ(z)/(1−α); because losses are positive, the +4% expected loss adds to the tail loss.

Why the others are wrong

- **A:** Subtracted the expected loss instead of adding it: $(15\% \times 2.3378 - 4\%) \times 30 = 9.32$.
- **B:** Computed the 97.5% VaR, $(4\% + 15\% \times 1.96) \times 30 = 10.02$, which is smaller than ES.
- **C:** Ignored the 4% expected loss: $15\% \times 2.3378 \times 30 = 10.52$.

Q42

BT answers L2.T5.70.2

REBUILT — Distractors A (0.77) and B (3.29) were unexplained giveaways; replaced with options from realistic mistakes (unconditional mean, severity VaR, severity ES).

A process has a 5% probability of producing a material loss and otherwise produces no loss (Bernoulli). If the loss occurs, its severity is normally distributed with mean USD 4 million and standard deviation USD 2 million. What is the 95% expected shortfall of the loss?

A. USD 0.20 million.
B. USD 7.29 million.
C. USD 4.00 million.
D. USD 8.13 million.

Answer: C

Step by step

1. The worst 5% of outcomes has probability mass exactly equal to the probability of a loss event (5%).
2. ES = average loss over that 5% tail = expected severity given the event occurs = USD 4.00 million.
3. Check: ES $= \frac{1}{0.05}\int_{0.95}^{1} q_p\, dp$; for $p > 0.95$ the quantile is the severity quantile at $(p - 0.95)/0.05$, which is uniform over (0,1), so the average is the severity mean, 4.

Why C is correct: The 5% tail coincides with the loss event, so the average tail loss is the mean of the severity distribution, USD 4 million.

Why the others are wrong

- **A:** Unconditional expected loss, $5\% \times 4 = 0.20$; ES conditions on being in the tail.
- **B:** Severity 95% VaR, $4 + 1.645 \times 2 = 7.29$: applies the 95% quantile within the severity distribution, forgetting the tail is already the event itself.
- **D:** Severity 95% ES, $4 + 2 \times 2.063 = 8.13$: applies a second tail cut inside the severity distribution.

Q43 BT answers L2.T5.72.1

REBUILT — Option values were OCR-uncertain; options rebuilt around exactly computed values and plausible errors.

To estimate the 95% expected shortfall, the 5% tail is cut into five equal-probability slices, producing four (n − 1) tail VaRs that are averaged. The loss distribution is standard normal (mean 0, variance 1). Which is closest to the estimated 95% ES?

A. 1.65
B. 1.93
C. 2.00
D. 2.06

Answer: C

Step by step

1. Slices of 1% give interior VaRs at 96%, 97%, 98%, 99%.
2. Quantiles: 1.7507, 1.8808, 2.0537, 2.3263.
3. Average = (1.7507 + 1.8808 + 2.0537 + 2.3263)/4 = 8.0115/4 = 2.0029 ≈ 2.00.

Why C is correct: The average of the four interior tail VaRs approximates the tail mean; five slices give 2.00, below the theoretical value 2.063.

Why the others are wrong

- **A:** This is the 95% VaR (1.645), a single quantile rather than a tail average.
- **B:** Included the 95% VaR in the average: (1.645 + 1.7507 + 1.8808 + 2.0537 + 2.3263)/5 = 1.93.
- **D:** This is the true ES (2.063) obtained with infinitely many slices; the five-slice estimate is 2.00.

Q44 BT answers P2.T5.22.2.1

REBUILT — Option C was OCR-garbled and the options mixed up sum/average and slice counts; rewrote the options so exactly one (the Schweser n-1 tail-VaR average) is correct.

Peter has fitted a parametric distribution to his portfolio's returns and can compute any quantile (the inverse CDF), but has no closed-form formula for the expected shortfall. Let C be the confidence level, $S = 1 - C$ the significance level, N the number of equal-probability slices of the tail, and $L = S/N$ the size of each slice. How can Peter approximate the C expected shortfall?

A. Average the N tail VaRs at confidence levels $C + L,\ C + 2L,\ \ldots,\ C + NL$.
B. Average the $N - 1$ tail VaRs at confidence levels $C + L,\ C + 2L,\ \ldots,\ C + (N - 1)L$.
C. Average the N tail VaRs at confidence levels $C,\ C + L,\ \ldots,\ C + (N - 1)L$.
D. It is not possible to approximate ES from a set of tail VaRs; ES must be derived from the density directly.

Answer: B

Step by step

1. Divide the tail (probability mass S) into N equal slices of size $L = S/N$.
2. The slice boundaries inside the tail are at confidence levels $C + L, C + 2L, \ldots, C + (N-1)L$, giving $N - 1$ quantiles (the last boundary, $C + NL = 1$, is the 100% quantile, which is unbounded for most distributions, and C itself is the start of the tail).
3. ES is approximated by the simple average of these $N - 1$ tail VaRs; it converges to the true ES as N grows. Example: $N = 10$, $C = 95\%$ gives the average of the VaRs at 95.5%, 96%, ..., 99.5%.

Why B is correct: ES is the average of VaR over the tail, so it is approximated by averaging the VaRs at the $N - 1$ interior slice boundaries.

Why the others are wrong

- **A:** Includes the N-th level $C + NL = 100\%$, where the VaR is unbounded (or the worst possible loss) and not an interior slice boundary; the count is $N - 1$, not N.
- **C:** Starts the sequence at C itself, so the VaR at the tail threshold is included; this pulls the average down and uses the wrong set of levels.
- **D:** False: Schweser's approach explicitly estimates ES by averaging VaRs at increasing confidence levels in the tail.

Q45 BT answers P2.T5.708.1 · BT notes P2.T5.708.1

A hedge fund has 300 daily P/L observations. Its worst 20 daily losses are sorted below (losses in dollars):

Rank	Loss	Rank	Loss
1	1.350	11	0.740
2	1.000	12	0.730
3	0.950	13	0.730
4	0.920	14	0.710
5	0.850	15	0.700
6	0.830	16	0.700
7	0.830	17	0.680
8	0.790	18	0.680
9	0.760	19	0.670
10	0.740	20	0.670

Which is nearest to the 99.0% expected shortfall (ES)?

A. 0.920
B. 0.950
C. 1.100
D. 1.350

Answer: C

Step by step

1. Tail probability $1 - 0.99 = 1\%$; number of tail observations $= 1\% \times 300 = 3$.
2. The three worst losses are 1.350, 1.000 and 0.950.
3. $\text{ES}_{99\%} = (1.350 + 1.000 + 0.950)/3 = 3.300/3 = 1.100$.

Why C is correct: Historical-simulation ES is the average of the losses in the worst $(1 - c)n$ observations, here 3.

Why the others are wrong

- **A:** 0.920 is the 4th-worst loss, a single order statistic; ES requires averaging the tail losses, not picking one.
- **B:** 0.950 is the 3rd-worst loss, i.e. a VaR-type cutoff at the edge of the 3-observation tail; it ignores the larger losses beyond it.
- **D:** 1.350 is only the single worst loss; ES averages all three tail losses.

Q46 BT answers P2.T5.708.2 · BT notes P2.T5.708.2

The daily profit/loss (P/L) follows the discrete distribution below, where a is a constant:

P/L = X	f(X)
($3.00)	1a
($2.00)	4a
($1.00)	7a
$0.00	13a
$1.00	11a
$2.00	9a
$3.00	5a

What is the 95.0% expected shortfall, measured relative to zero (not relative to the mean)?

A. $1.50
B. $2.40
C. $2.67
D. $3.00

Answer: B

Step by step

1. Probabilities sum to 1: $(1 + 4 + 7 + 13 + 11 + 9 + 5)a = 50a = 1$, so $a = 2\%$.
2. $P(X = -3) = 2\%$; $P(X = -2) = 8\%$. The 5% tail therefore consists of the 2% at a loss of $3.00 plus 3% (of the 8%) at a loss of $2.00.
3. $\text{ES} = (2\% \times 3.00 + 3\% \times 2.00)/5\% = 0.12/0.05 = \$2.40$.

Why B is correct: ES is the probability-weighted average loss in the worst 5% of the distribution, splitting the $2.00 outcome so that exactly 5% of probability is used.

Why the others are wrong

- **A:** \$1.50 averages the three worst outcomes (3, 2, 1) weighted by probability (3 + 8 + 7 = 18, divided by 12), which is a 24% tail rather than 5%.
- **C:** \$2.67 weights the \$3.00 and \$2.00 losses 2:1 instead of using the correct 2% and 3% tail probabilities, overweighting the worst loss.
- **D:** \$3.00 is only the worst loss (the 2% tail); the 5% tail also includes outcomes at \$2.00.

Q47 BT answers P2.T5.708.3 · BT notes P2.T5.708.3

REBUILT — Distractor D (5.95) did not correspond to any identifiable mistake and B was rounded loosely (3.80 vs 3.82); replaced with distractors that each reflect a specific error. Plot dropped (not needed).

At the end of an investment horizon the final value X of a risky investment lies between 0 and 10 and has density $f(x) = 3x^2/1000$ for $0 \le x \le 10$. Define the 95.0% expected shortfall as the expected loss in the worst 5% tail measured relative to the expected final value; for example, if the expected final value is 7.50 and the 5% tail conditional expectation (TCE) is 1.0, ES would be 6.50. Which is nearest to the 95.0% expected shortfall?

A. 2.76
B. 3.82
C. 4.74
D. 5.66

Answer: C

Step by step

1. Mean: $E[X] = \int_0^{10} x \cdot 3x^2/1000\, dx = 3 \cdot 10^4/4000 = 7.500$.
2. CDF: $F(x) = x^3/1000$. The 5% quantile solves $x^3 = 50$: $q = 50^{1/3} = 3.6840$.
3. Tail integral: $\int_0^q xf(x)\, dx = 3q^4/4000 = 0.13815$; TCE $= 0.13815/0.05 = 2.7630$.
4. ES $= 7.500 - 2.763 = 4.737 \approx 4.74$.

Why C is correct: ES here is the gap between the mean and the average of values in the worst 5% tail, $7.500 - 2.763$.

Why the others are wrong

- **A:** 2.76 is the tail conditional expectation itself; it forgets to measure the loss relative to the expected value of 7.50.
- **B:** 3.82 is $7.50 - 3.684$, the VaR relative to the mean; it uses the 5% quantile rather than the average of the tail values.
- **D:** 5.66 assumes the tail values are uniform on $[0, q]$, so TCE $= q/2 = 1.842$ and ES $= 7.50 - 1.842$; the density is increasing, so the tail average is well above $q/2$.

Q48 BT answers P2.T5.709.1 · BT notes P2.T5.709.1

REBUILT — OCR digits in the quantile table were uncertain (reconstructed to match the printed total 186.380); removed the pre-printed sum/average rows, which handed a distractor and the arithmetic to the candidate.

Sally wants the 95.0% expected shortfall (ES) but can only retrieve quantiles of the loss distribution. She slices the 5% tail into ten equal-probability slices and obtains the following tail VaRs:

Confidence Level	Tail VaR
95.00%	6.330
95.50%	7.050
96.00%	7.940
96.50%	9.080
97.00%	10.600
97.50%	12.730
98.00%	15.910
98.50%	21.220
99.00%	31.840
99.50%	63.680

Following the standard approach of averaging the VaRs at the interior slice boundaries, which is the BEST estimate of the 95.0% ES?

A. 7.05
B. 12.73
C. 18.64
D. 20.01

Answer: D

Step by step

1. With $n = 10$ slices there are $n - 1 = 9$ interior boundaries, at 95.5%, 96.0%, ..., 99.5%. The 95.00% VaR is the tail threshold, not a slice boundary.
2. Sum of all ten VaRs = 186.380; remove the 95% VaR: $186.380 - 6.330 = 180.050$.
3. ES $\approx 180.050/9 = 20.006 \approx 20.01$.

Why D is correct: ES is estimated as the average of the $n - 1$ tail VaRs strictly inside the tail, excluding the VaR at the 95% threshold.

Why the others are wrong

- **A:** 7.05 is just the first slice VaR (95.5%); a single quantile close to the threshold is not an average across the tail.
- **B:** 12.73 is the 97.5% VaR, the middle level of the tail; it is a single quantile and ignores the heavy extreme quantiles.
- **C:** 18.64 averages all ten rows (186.380/10), wrongly including the 95% VaR and dividing by 10 instead of 9.

Q49

BT answers P2.T7.608.3 · BT notes P2.T5.608.3

Dowd shows that, for normally distributed returns, expected shortfall is ES $= -\mu + \sigma\,\phi(z_\alpha)/(1-\alpha)$, where ϕ is the standard normal pdf. Hull gives an equivalent form that avoids a pdf lookup: ES $= -\mu + \sigma\,\dfrac{e^{-Y^2/2}}{\sqrt{2\pi}\,(1-X)}$, where X is the confidence level and Y is the standard normal point with probability $1 - X$ of being exceeded (e.g., X = 99% gives Y = 2.326). Daily returns are normally distributed with mean +1.0% and standard deviation 5.0%. Which of the following is closest to the daily 97.5% expected shortfall (loss expressed as a positive number)?

A. 4.00%
B. 7.22%
C. 8.80%
D. 10.69%

Answer: D

Step by step

1. X = 97.5%, so Y = 1.960 (2.5% of the distribution lies above it).
2. $e^{-Y^2/2} = e^{-1.9208} = 0.1465$; $\sqrt{2\pi} = 2.5066$.
3. Tail factor: $0.1465/(2.5066 \times 0.025) = 2.338$ (this equals $\phi(1.96)/0.025$).
4. ES $= -1.0\% + 5.0\% \times 2.338 = -1.0\% + 11.69\% = 10.69\%$.

Why D is correct: ES is the mean loss beyond VaR; for a normal it equals -μ plus σ times the pdf at the quantile divided by the tail probability, which at 97.5% is 2.338σ (slightly above the 97.5% VaR multiple of 1.96).

Why the others are wrong

- **A:** 4.00% = σ − μ = 5% − 1%: it sets the multiple to 1 and ignores the tail formula entirely.
- **B:** 7.22% = −1% + 5% × 1.645: this is the 95% VaR, using the wrong confidence level.
- **C:** 8.80% = −1% + 5% × 1.96: this is the 97.5% VaR, not ES; it omits the extra tail severity (2.338 vs 1.96).

Q50

BT notes P2.T5.22.1.2 · BT answers P2.T5.22.1.3

Consider a portfolio of three independent, identical junk bonds. Each has a default probability of 10.0% over a single period, a face value of $100.00 and is priced at par; the loss given default is 100% (no recovery). The portfolio value is $300.00. What is the portfolio's 95.0% expected shortfall (ES)?

A. $100.00
B. $300.00
C. $158.00
D. Ambiguous

Answer: C

Step by step

1. Number of defaults ~ Binomial(3, 0.10): P(0)=72.9%, P(1)=24.3%, P(2)=2.7%, P(3)=0.1%.
2. Losses: 0, $100, $200, $300. The worst 5% tail mass is built from the worst outcomes first.
3. Take 0.1% at $300 and 2.7% at $200; that is 2.8%, so 2.2% more is needed from the $100 outcome.
4. ES = (0.1% × 300 + 2.7% × 200 + 2.2% × 100)/5% = (0.3 + 5.4 + 2.2)/5% = 7.9/0.05 = $158.00.

Why C is correct: Discrete distributions have no outcome sitting exactly at the 5% boundary, so ES splits the $100

atom and averages exactly 5% of probability mass of the worst losses.

Why the others are wrong

- **A:** $100.00 is the 95% VaR (the loss level not exceeded with 95% confidence), not the average tail loss.
- **B:** $300.00 is the maximum loss (all three default), which has only 0.1% probability; it ignores the rest of the tail.
- **D:** ES is not ambiguous here: although VaR is a single value ($100), ES is well defined by splitting the $100 outcome. A naive E(L | L > VaR) = (2.7%×200 + 0.1%×300)/2.8% = $203.57 is the mistake that makes it look odd.

Q51

GARP PE1 Q52 · Mock 3 Q28

REBUILT — Original distractors included JPY 497m, above the largest VaR in the table (a giveaway) and JPY 400m with no realistic error path; replaced with options from realistic averaging mistakes.

A wealth management firm has JPY 72 billion in assets under management. The portfolio manager estimates the daily VaR at various confidence levels as follows:

Confidence level	VaR (JPY)
95.0%	332,760,000
95.5%	336,292,500
96.0%	340,095,000
96.5%	350,332,500
97.0%	359,107,500
97.5%	367,882,500
98.0%	378,412,500
98.5%	392,452,500
99.0%	410,880,000
99.5%	439,252,500

Using the approach of dividing the tail into equal-probability slices and averaging the VaRs, what is the closest estimate of the daily expected shortfall at the 97.5% confidence level?

A. JPY 398 million
B. JPY 394 million
C. JPY 405 million
D. JPY 414 million

Answer: C

Step by step

1. The 2.5% tail beyond 97.5% is cut into 5 slices of 0.5%, giving 4 interior VaRs: 98.0%, 98.5%, 99.0%, 99.5%.
2. Sum = 378,412,500 + 392,452,500 + 410,880,000 + 439,252,500 = 1,620,997,500.
3. ES ≈ 1,620,997,500 / 4 = 405,249,375, about JPY 405 million.

Why C is correct: ES is approximated by averaging the VaRs at the n − 1 quantiles inside the tail; the 97.5% VaR itself is the tail boundary and is excluded.

Why the others are wrong

- **A:** JPY 398m averages five VaRs including the 97.5% boundary row (1,988.88m/5 = 397.8m): an off-by-one error.
- **B:** JPY 394m drops the 99.5% VaR and averages only 98.0%, 98.5%, 99.0% (393.9m).
- **D:** JPY 414m drops the 98.0% VaR and averages 98.5%, 99.0%, 99.5% (414.2m).

Q52

GARP PE2 Q21 · Mock 4 Q22

REBUILT — Original options A (JPY 398m) and B (JPY 400m) were below the 97.5% VaR of 439m and D was copied from another table, so they were giveaways; replaced with options from realistic averaging mistakes.

A wealth management firm has JPY 86 billion in assets under management. The portfolio manager estimates the daily VaR at various confidence levels as follows:

Confidence level	VaR (JPY)
95.0%	397,463,000
95.5%	401,682,500
96.0%	406,224,500
96.5%	418,453,000
97.0%	428,934,000
97.5%	439,415,500
98.0%	451,993,000
98.5%	468,763,000
99.0%	490,773,000
99.5%	524,663,000

Using the approach of dividing the tail into equal-probability slices and averaging the VaRs, what is the closest estimate of the daily expected shortfall at the 97.5% confidence level?

A. JPY 484 million
B. JPY 475 million
C. JPY 471 million
D. JPY 495 million

Answer: A

Step by step

1. The 2.5% tail beyond 97.5% is cut into 5 slices of 0.5%, giving 4 interior VaRs: 98.0%, 98.5%, 99.0%, 99.5%.
2. Sum = 451,993,000 + 468,763,000 + 490,773,000 + 524,663,000 = 1,936,192,000.
3. ES ≈ 1,936,192,000 / 4 = 484,048,000, about JPY 484 million.

Why A is correct: ES is approximated by averaging the VaRs at the n − 1 quantiles inside the tail; the 97.5% VaR itself is the tail boundary and is excluded.

Why the others are wrong

- **B:** JPY 475m averages five VaRs including the 97.5% boundary row (2,375.6m/5 = 475.1m): an off-by-one error.
- **C:** JPY 471m drops the 99.5% VaR and averages 98.0%, 98.5%, 99.0% (470.5m).
- **D:** JPY 495m drops the 98.0% VaR and averages 98.5%, 99.0%, 99.5% (494.7m).

Q53

BT answers L2.T5.70.1

Assume an empirical loss/profit distribution consists of 100 equally likely observations in which the seven highest losses are $94, $95, $96, $97, $98, $99 and $100. What is the 95% expected shortfall (ES)?

A. $95.00
B. $97.00
C. $97.50
D. $98.00

Answer: D

Step by step

1. With 100 observations, the 5% tail contains the 5 worst losses: $96, $97, $98, $99, $100.
2. The 95% VaR is the next loss, $95 (the 6th worst), the boundary of the tail.
3. ES = (96 + 97 + 98 + 99 + 100)/5 = 490/5 = $98.00.

Why D is correct: ES is the average of the losses beyond VaR; with equal probabilities it is the simple mean of the worst 5% of observations.

Why the others are wrong

- **A:** $95.00 is the 95% VaR (6th worst loss), not the average tail loss.
- **B:** $97.00 averages the seven worst losses ($94 to $100), i.e., a 7% tail instead of 5%.
- **C:** $97.50 averages the six worst losses ($95 to $100), including the VaR observation itself.

Q54

Mock 2 Q63

REBUILT — Original distractors (minimum 12, maximum 51) were giveaways; replaced them with results of realistic averaging mistakes and clarified that exactly nine losses exceed VaR.

A risk manager at Bank ABC uses profit/loss data from the previous 1,000 trading days and estimates a 1-day 99% VaR of USD 10 million. Exactly nine historical losses exceed this VaR; in USD millions they are: 12, 15, 17, 18, 21,

22, 24, 27 and 51. Estimating expected shortfall as the average of the losses beyond VaR, what is the portfolio's 99% expected shortfall?

A. USD 19.5 million
B. USD 20.7 million
C. USD 21.7 million
D. USD 23.0 million

Answer: D

Step by step

1. ES is the average of the tail losses that exceed the VaR threshold.
2. Sum of the nine tail losses: 12 + 15 + 17 + 18 + 21 + 22 + 24 + 27 + 51 = 207.
3. $ES_{99\%} = 207/9 = 23.0$ (USD millions).
4. Check: ES (23.0) exceeds VaR (10), as it must.

Why D is correct: ES measures the expected loss given that the loss exceeds VaR, so every one of the nine exceedances (including the 51 outlier) is averaged with equal weight.

Why the others are wrong

- **A:** Drops the USD 51 million loss as an outlier and averages the remaining eight: 156/8 = 19.5. ES must include all tail losses; capturing extreme losses is its purpose.
- **B:** Divides the sum by 10 (1% of 1,000 days) instead of by the nine observed exceedances: 207/10 = 20.7.
- **C:** Includes the VaR itself (10) as a tenth tail observation: (207 + 10)/10 = 21.7.

Q55

BT answers L2.T5.70.3

REBUILT — OCR face value '$700' restored to $100 (consistent with the source solution), and the weak $10 distractor replaced by results of realistic errors (unconditional expected loss; conditioning on default outcomes only).

A portfolio holds two bonds, each with face value $100 and zero recovery. Each bond defaults over the horizon with probability 2%, independently of the other. What is the portfolio's 95% expected shortfall (expected loss in the worst 5% of outcomes)?

A. $0.00
B. $4.00
C. $80.00
D. $101.01

Answer: C

Step by step

1. Binomial loss distribution: $P(0) = 0.98^2 = 0.9604$ (loss $0); $P(1) = 2 \times 0.02 \times 0.98 = 0.0392$ (loss $100); $P(2) = 0.02^2 = 0.0004$ (loss $200).
2. The worst 5% tail contains 0.04% at $200, 3.92% at $100 and the remaining 5% − 3.96% = 1.04% at $0. (The 95% VaR is therefore $0.)
3. $ES_{95\%} = (0.0004 \times 200 + 0.0392 \times 100 + 0.0104 \times 0)/0.05 = 4.00/0.05 = \$80.00$.

Why C is correct: ES averages over exactly 5% of probability mass in the tail; because the tail must be filled out with 1.04% of zero-loss outcomes, the probability-weighted average is $80, even though the 95% VaR is zero.

Why the others are wrong

- **A:** $0 is the 95% VaR (the cumulative probability of any default, 3.96%, is below 5%), not the expected shortfall; it ignores the losses beyond the quantile.
- **B:** $4.00 is the unconditional expected loss (2 × 2% × $100), averaging over the whole distribution rather than the 5% tail.
- **D:** Divides the tail loss 4.00 by 3.96% (only the default outcomes) instead of the full 5% tail: 4.00/0.0396 = $101.01; ES must condition on the full 5% tail mass.

Q56

BT answers L2.T5.71.4 · BT notes P2.T5.71.4

REBUILT — Original asked only to name the coherence axiom violated by R(X+Y) > R(X)+R(Y), which R01/LO 1.c does not cover; rebuilt as an ES-estimation problem on a two-bond loss distribution in which the subadditivity comparison must be computed for both VaR and ES.

Two bonds each have face value $100, zero recovery and a 4% default probability over the horizon, and they default independently. Using the 95% confidence level, compare the risk measures of the two-bond portfolio with the sum of the two stand-alone measures. Which statement is correct?

A. Portfolio VaR is $100 versus a stand-alone VaR sum of $0, while portfolio ES is $103.20 versus a stand-alone

ES sum of $160; VaR violates subadditivity but ES does not.

B. Portfolio VaR is $0 and portfolio ES is $160, both equal to the stand-alone sums; neither measure benefits from or penalizes diversification.

C. Portfolio VaR is $100 versus $0, and portfolio ES is $200 versus $160; both VaR and ES violate subadditivity.

D. Portfolio VaR is $100 versus $0, and portfolio ES is $100 versus $160; VaR violates subadditivity but ES does not, and ES equals VaR for the portfolio.

Answer: A

Step by step

1. Stand-alone bond: P(default) = 4% < 5%, so $\text{VaR}_{95\%} = 0$; the 5% tail holds 4% at $100 and 1% at $0, so $\text{ES}_{95\%} = (0.04 \times 100)/0.05 = 80$. Sums: VaR 0, ES 160.
2. Portfolio distribution: $P(0) = 0.96^2 = 0.9216$, $P(1) = 2 \times 0.04 \times 0.96 = 0.0768$ (loss $100), $P(2) = 0.04^2 = 0.0016$ (loss $200).
3. Portfolio VaR: $P(L \geq 100) = 0.0784 > 0.05$ while $P(L = 200) = 0.0016 < 0.05$, so $\text{VaR}_{95\%} = 100 > 0$ = sum of stand-alone VaRs (subadditivity violated).
4. Portfolio ES: the 5% tail holds 0.16% at $200 and 4.84% at $100, so $\text{ES}_{95\%} = (0.0016 \times 200 + 0.0484 \times 100)/0.05 = 5.16/0.05 = 103.20$.
5. $103.20 \leq 160$, so ES is subadditive here while VaR is not.

Why A is correct: Because each stand-alone default probability (4%) sits below the 5% tail, each VaR is zero, but combining the bonds pushes the one-default probability above 5% and VaR jumps to $100. ES averages the whole 5% tail, so it registers the default risk in each bond and the portfolio ES ($103.20) is below the $160 sum.

Why the others are wrong

- **B:** Treats both measures as simply additive and ignores that the portfolio's probability of at least one default (7.84%) exceeds 5%, which lifts portfolio VaR to $100 and reduces ES to $103.20.
- **C:** Takes the worst outcome ($200) as the ES instead of the probability-weighted average loss across the full 5% tail.
- **D:** Fills the whole 5% tail with $100 losses and ignores the 0.16% probability of losing $200, so it understates portfolio ES (true value $103.20).

Q57

Mock 2 Q27

REBUILT — Original relied on VaR being subadditive (which VaR does not guarantee) and tested a coherence axiom outside the 2026 MR LOs; replaced with a data-based expected shortfall estimate from P&L data (LO 1.c).

A risk manager has 100 daily P&L observations for a portfolio. The six largest daily losses (USD million) are 14, 11, 9, 8, 7 and 6; every other observation is a smaller loss or a profit. Using the historical simulation convention that expected shortfall is the average of the losses in the tail beyond the VaR cut-off, the 95% one-day expected shortfall is closest to:

A. USD 7.0 million

B. USD 9.2 million

C. USD 9.8 million

D. USD 10.5 million

Answer: C

Step by step

1. With n = 100 and 95% confidence, the tail contains (1 - 0.95) x 100 = 5 observations.
2. The five largest losses are 14, 11, 9, 8 and 7 (USD million).
3. $\text{ES}_{95\%} = (14 + 11 + 9 + 8 + 7)/5 = 49/5 = 9.8$.
4. The 6th loss (6) lies at the VaR cut-off boundary and is not averaged in.

Why C is correct: Expected shortfall is the average of the losses in the tail beyond the VaR level; with 5% of 100 observations that is the average of the 5 worst losses.

Why the others are wrong

- **A:** This is the 5th worst loss, i.e. a VaR-type quantile, not an average of tail losses.
- **B:** Averages the 6 worst losses (49 + 6 = 55, 55/6 = 9.17), including the cut-off observation and using the wrong tail size.
- **D:** Averages only the 4 worst losses (42/4 = 10.5), using too few tail observations.

LO 1.d

Estimate risk measures by estimating quantiles.

Q58

BT answers L2.T5.72.2

Expected shortfall can be expressed as a weighted average of loss quantiles. For a 95% expected shortfall, what weight is applied to the quantiles in the 5% loss tail (all other quantiles receive zero weight)?

A. 5%
B. 1.0
C. 5.0
D. 20.0

Answer: D

Step by step

1. ES weighting function: 0 for $p < 95\%$ and $1/(1 - \text{confidence level})$ for $p \geq 95\%$.
2. $1/(1 - 0.95) = 1/0.05 = 20$.
3. Check: the weights must integrate to 1: $0.05 \times 20 = 1$.

Why D is correct: Spreading a total weight of 1 evenly over a tail of probability 5% requires a weight of $1/0.05 = 20$ on each tail quantile.

Why the others are wrong

- **A:** This is the tail probability itself, not the weight; the weights would then sum to far less than one.
- **B:** A weight of 1 on tail quantiles would give a total weight of only 0.05, not 1; 1.0 is the total weight, not the per-quantile weight.
- **C:** Reads the 5% as the number 5 rather than taking its reciprocal; 5.0 is not $1/0.05$.

Q59

BT answers P2.T5.72.3

REBUILT — Options were loosely worded (D garbled, 'decreasingly less' phrasing) so more than one reading was arguable; reworded so exactly one option matches the Schweser description of ES as a coherent (weighted-quantile) measure.

Which of the following statements about expected shortfall (ES) viewed as a weighted average of loss quantiles is correct?

A. ES is a special case of a coherent risk measure: it gives equal weight, $1/(1 - c)$, to every quantile in the tail and zero weight to all other quantiles.
B. ES is a coherent risk measure that gives progressively greater weight to the higher (more extreme) tail quantiles.
C. ES is a coherent risk measure that gives progressively smaller weight to the higher (more extreme) tail quantiles.
D. ES places all of its weight on the single quantile at the confidence level c and zero weight on every other quantile.

Answer: A

Step by step

1. A coherent risk measure is a weighted average of the quantiles of the loss distribution, with user-specific weights reflecting risk aversion.
2. For ES the weighting function is set to $1/(1 - c)$ for every tail quantile (equal weights) and to zero for all quantiles outside the tail.
3. Hence ES is a special case: equal tail weights, zero elsewhere. Choice A states exactly this.

Why A is correct: ES averages the tail quantiles with identical weights $1/(1 - c)$ and ignores the rest of the distribution, which is the special case of the general weighted-quantile (coherent) measure.

Why the others are wrong

- **B:** Increasing weights on the extreme quantiles describe a more risk-averse weighting function, which is a different (more general) coherent measure, not ES; ES weights are flat across the tail.
- **C:** Decreasing weights on extreme tail quantiles would downplay the worst losses, the opposite of a risk-averse weighting, and ES does not do this; its weights are equal.
- **D:** Confusing ES with VaR: putting all weight on the single quantile at c is VaR, which ignores losses beyond the threshold, whereas ES averages over the whole tail.

Q60

Schweser Module Quiz 1.2 Q1

REBUILT — Original options were vague/self-inconsistent (source explanation contradicted the keyed option); rewritten to test the precise ES versus general coherent-risk-measure weighting distinction.

Which of the following statements about expected shortfall (ES) and coherent risk measures, each estimated as a weighted average of quantiles, is correct?

A. ES and a general coherent risk measure both place non-zero weight only on quantiles in the tail region.

B. ES averages VaR quantiles over the tail only (zero weight elsewhere), whereas a general coherent risk measure weights quantiles across the entire distribution using a risk-aversion function; ES is a special case.

C. ES weights quantiles across the entire distribution, whereas a coherent risk measure uses only tail quantiles.

D. A coherent risk measure always places equal weights on all quantiles, whereas ES uses a risk-aversion weighting function.

Answer: B

Step by step

1. A coherent risk measure is a weighted average of the quantiles of the loss distribution, with weights set by the user's risk aversion.
2. ES sets the weight to 1/(1 − confidence level) for tail quantiles and zero for all other quantiles, so it only uses tail quantiles.
3. A general coherent measure divides the whole distribution into equal-probability slices and applies a risk-aversion weighting function to each quantile.
4. Hence ES is a special case: B.

Why B is correct: ES is the special case of a weighted-quantile risk measure with equal weights in the tail and zero elsewhere; the general coherent measure may weight every quantile.

Why the others are wrong

- **A:** Wrong for the general coherent measure: it uses quantiles over the entire distribution, not just the tail.
- **C:** Reversed: it is ES that is confined to the tail, and the general coherent measure that spans the whole distribution.
- **D:** Reversed and wrong: equal weights on tail quantiles characterise ES, while coherent measures use user-specific risk-aversion weights.

Q61

AnalystPrep Q.1479

A coherent risk measure is a generalization of expected shortfall (ES), which can be viewed as the average of tail VaRs (quantiles) with equal weights. Which of the following is the appropriate replacement for those equal weights when estimating any coherent risk measure?

A. Average weights.

B. Exponential weights.

C. Weights appropriate to the risk measure being estimated.

D. Fixed weights regardless of the risk measure being estimated.

Answer: C

Step by step

1. ES = average of the tail quantiles (VaRs), each given the same weight 1/(nâˆ'1) (zero weight outside the tail).
2. A coherent risk measure is a weighted average of quantiles where the weighting function reflects the user's risk aversion.
3. So the equal weights are replaced by a weighting function suited to the specific risk measure being estimated.

Why C is correct: A coherent risk measure keeps the same 'average of quantiles' machinery as ES but lets the weighting function vary with the measure (and the user's risk aversion).

Why the others are wrong

- **A:** Equal ('average') weights simply reproduce ES; they are not a replacement.
- **B:** Exponential weights are one possible scheme (e.g. age-weighting), but they are not the general replacement for every risk measure.
- **D:** Fixed weights regardless of the measure would make every measure identical, defeating the purpose of a user-specific weighting function.

Q62

AnalystPrep Q.1480

Coherent risk measures can be expressed as weighted averages of the quantiles of the loss distribution. The key to estimating a coherent risk measure therefore lies in the:

A. ability to assign weights to the assets in a portfolio accurately.
B. ability to calculate exponential values accurately.
C. ability to estimate quantiles.
D. ability to approximate risk exposure.

Answer: C

Step by step

1. A coherent risk measure = weighted average of loss quantiles (VaRs at different confidence levels).
2. The weights are chosen by the user (risk aversion), so the estimation challenge is obtaining the quantiles themselves.
3. Hence the key ingredient is the ability to estimate quantiles.

Why C is correct: Every coherent measure (including ES) is built from quantiles, so quantile estimation is the core task.

Why the others are wrong

- **A:** Portfolio asset weights are inputs to the P&L distribution, not the weights of the quantile average.
- **B:** Exponentials are irrelevant to the estimation; this confuses the weighting function with exponential weighting schemes.
- **D:** Exposure approximation is not what defines the estimator; the quantile of the loss distribution is.

Q63

AnalystPrep Q.2969

Which of the following risk measures is the least commonly used in the practice of risk management?

A. Standard deviation
B. Spectral risk measures
C. Value at risk
D. Expected shortfall

Answer: B

Step by step

1. Standard deviation, VaR and ES are standard, widely implemented measures.
2. Spectral (general coherent) risk measures require choosing a user-specific risk-aversion weighting function over all quantiles, which is complex and rarely done in practice.
3. So spectral risk measures are the least commonly used.

Why B is correct: Spectral/coherent measures are theoretically attractive but need a subjective weighting function and are seldom implemented.

Why the others are wrong

- **A:** Standard deviation is the oldest and most common dispersion measure.
- **C:** VaR is the industry and regulatory standard.
- **D:** ES is widely used; it is the equal-weight special case of a spectral measure.

Q64

BT answers P2.T5.22.2.2 · BT notes P2.T5.22.2

REBUILT — Merged two identical items; restored the garbled formula as the weighted-average-of-quantiles definition.

A general risk measure is defined as a weighted average of loss quantiles: $M_\phi = \int_0^1 \phi(p)\, q_p\, dp$, where $\phi(p)$ is the weighting (risk aversion) function with non-negative weights summing to one. Which of the following is true?

A. All cases of this general risk measure are coherent.
B. If the weights applied to the tail-loss quantiles are unequal, this measure is not coherent.
C. If the weights applied to the tail-loss quantiles are weakly increasing, this measure is coherent.
D. The weakness of this measure is that it cannot represent different degrees of risk aversion.

Answer: C

Step by step

1. M_ϕ is coherent when $\phi(p)$ is weakly increasing in p, so larger losses get at least as much weight (a risk-averse

user).
2. ES (equal weights on the tail, zero elsewhere) satisfies this and is coherent; VaR (all weight on one quantile, zero on worse losses) does not.
3. Thus only C is true.

Why C is correct: Coherence requires non-decreasing weights on higher loss quantiles, i.e. risk aversion that does not fall as losses get worse.

Why the others are wrong

- **A:** Too broad: weight functions that decrease in the tail (e.g. VaR) give non-coherent measures.
- **B:** Unequal weights are fine if they are weakly increasing; equal weights (ES) are only one coherent case.
- **D:** The weighting function is exactly what represents risk aversion, so this is a strength, not a weakness.

Q65

BT answers P2.T5.405.1

REBUILT — Source relied on a Basel working paper outside the reading and had a vague EXCEPT framing; reworded as a general statement about VaR as a quantile measure, answer unchanged.

Regarding value at risk (VaR) as a quantile-based risk measure, each of the following is accurate EXCEPT:

A. VaR measures only a loss quantile; by disregarding losses beyond the VaR level, a manager relying only on VaR may be tempted to avoid losses within the confidence level while increasing losses beyond it.

B. Because VaR is only a quantile, backtesting is easier, since empirical quantiles are robust to extreme outliers, unlike typical estimators of expected shortfall.

C. Subadditivity reflects diversification; using non-subadditive measures in portfolio optimization may lead to very concentrated portfolios that would be judged risky.

D. Although VaR is not subadditive (and so not coherent), the literature has produced no case where this lack of coherence causes a practical problem for market risk measurement.

Answer: D

Step by step

1. A is true: VaR ignores the size of losses beyond the quantile (the tail), which ES addresses.
2. B is true: a quantile is not driven by extreme observations, which helps backtesting.
3. C is true: without subadditivity, splitting or concentrating positions can lower measured risk, so optimization can favor concentration.
4. D is false: failure of subadditivity can be a real practical problem (e.g. incentives to split portfolios, misleading diversification), so it is the exception.

Why D is correct: VaR's non-subadditivity is not merely academic; it can mislead risk measurement and optimization, so the 'no practical problem' claim is wrong.

Why the others are wrong

- **A:** This is accurate: it is the standard tail-blindness criticism of VaR.
- **B:** This is accurate: the 'positive flipside' of VaR being only a quantile.
- **C:** This is accurate: it describes the link between subadditivity and diversification.

Q66

BT answers P2.T5.405.3

REBUILT — Source cited Basel-specific advantages of spectral measures that are not in the reading; rewrote the false option as a clear misstatement consistent with the reading.

A spectral risk measure is defined as $M_\phi = \int_0^1 \phi(p)\, q_p\, dp$, a weighted average of loss quantiles with non-negative weights summing to one. Each of the following statements is true EXCEPT:

A. An advantage of spectral measures over VaR and ES is that they are not tied to a single confidence level.

B. A spectral risk measure places weight only on the quantile at the chosen confidence level and ignores worse losses.

C. For the measure to be coherent, the weighting function must be weakly increasing over [0,1].

D. A weakly increasing weighting function ensures the spectral risk measure is coherent.

Answer: B

Step by step

1. A spectral measure averages many quantiles, so it is not tied to one confidence level (A true).
2. Coherence requires weights that are weakly increasing in p, reflecting risk aversion (C and D true).
3. B describes VaR, which puts all weight on one quantile; a spectral measure weights many quantiles, including

worse tail losses. B is false.

Why B is correct: Placing all weight on a single quantile is VaR, not a spectral measure, which spreads weights across quantiles.

Why the others are wrong

- **A:** True: spreading weights across quantiles removes dependence on one confidence level.
- **C:** True: weakly increasing weights are the requirement for coherence.
- **D:** True: this is the sufficient condition for coherence.

Q67

BT answers P2.T5.709.2 · BT notes P2.T5.709.2

REBUILT — Merged two identical items; original tested the stand-alone coherence axioms (outside the 2026 LOs), so rewrote it to test ES as the equal-weight special case of a coherent risk measure.

Which of the following statements about expected shortfall (ES) and coherent risk measures is most accurate?

A. ES is a coherent risk measure whose weighting function places equal weight on all tail quantiles and zero weight on the rest.

B. VaR is a coherent risk measure that applies equal weights to all quantiles in the tail.

C. Coherent risk measures have smaller standard errors than VaR because they average many quantiles.

D. ES is the best coherent risk measure for every user regardless of risk aversion.

Answer: A

Step by step

1. A coherent risk measure is a weighted average of loss quantiles with user-specific weights.
2. ES sets the weight to 1/(1â˜Î±) on each tail quantile and zero elsewhere, i.e. equal weighting of the tail, so it is the special equal-weight case.
3. VaR puts all weight on one quantile; standard errors of coherent measures are typically larger, and the best weighting depends on each user's risk aversion.

Why A is correct: ES is the equal-tail-weight special case of the general coherent (weighted-quantile) risk measure.

Why the others are wrong

- **B:** Equal tail weights describe ES; VaR uses a single quantile.
- **C:** Averaging tail quantiles that rely on sparse data typically gives larger, not smaller, standard errors; coherent measures are more sensitive to n.
- **D:** The optimal weighting depends on each user's risk aversion; ES is just one choice.

Q68

BT notes P2.T5.72.1

REBUILT — Distractors were arbitrary; replaced with options reflecting realistic errors (95% VaR, including the 95% quantile, 99% VaR). Answer unchanged.

You estimate the 95% expected shortfall of a standard normal loss distribution (mean 0, standard deviation 1) by slicing the tail into five equal-probability slices, producing four tail VaRs. What is the estimated 95% ES?

A. 1.65

B. 1.93

C. 2.00

D. 2.33

Answer: C

Step by step

1. Slicing the 5% tail into 5 slices gives 4 breakpoints at confidence levels 96%, 97%, 98%, 99%.
2. Quantiles: 1.7507, 1.8808, 2.0537, 2.3263.
3. Average = (1.7507 + 1.8808 + 2.0537 + 2.3263)/4 = 8.0115/4 = 2.0029, about 2.00.

Why C is correct: ES is estimated as the average of the nâ˜1 equally spaced tail VaRs, which gives about 2.00.

Why the others are wrong

- **A:** 1.65 is the 95% VaR itself, ignoring the tail beyond it.
- **B:** 1.93 averages five quantiles including the 95% VaR (1.645), which is not one of the four tail VaRs.
- **D:** 2.33 is only the 99% VaR, the deepest slice, not the average of all four.

Q69 BT notes P2.T5.72.1 (72.2)

For a 95% expected shortfall (ES), what weight is applied to each quantile in the 5% loss tail (with zero weight on all quantiles outside the tail)?

A. 5%
B. 1.0
C. 5.0
D. 20.0

Answer: D

Step by step

1. ES is a coherent risk measure: a weighted average of loss quantiles, with weights that sum (integrate) to 1.
2. For ES, the weight is a constant $1/(1-\alpha)$ on every quantile in the tail and 0 elsewhere.
3. $1/(1-0.95) = 1/0.05 = 20.0$.
4. Check: tail probability mass 0.05 x weight 20 = 1, so the weights sum to 1.

Why D is correct: The tail has probability mass 5%, so for the weights to sum to one each tail quantile carries weight 1/0.05 = 20, i.e., ES is the simple average of the tail quantiles.

Why the others are wrong

- **A:** 5% is the tail probability itself (1 - confidence level), not its reciprocal; tail weights of 0.05 would sum to only 0.0025.
- **B:** 1.0 is the total weight over the whole distribution, not the per-quantile weight; it would give tail weights summing to only 0.05.
- **C:** 5.0 comes from mishandling the inversion, e.g. treating 1/(1 - 0.95) as 0.05 x 100 (reading the tail as a percent) instead of 1/0.05.

Q70 BT notes P2.T5.72.1 (72.3)

REBUILT — Source had no key and confusing options (C worded 'decreasingly less', D mixed in VaR coherence); rewrote options so exactly one clearly describes ES's weighting within the coherent-measure framework.

Within the framework of coherent risk measures as weighted averages of loss quantiles, which of the following statements about expected shortfall (ES) at confidence level α is correct?

A. ES gives equal weight, $1/(1-\alpha)$, to every quantile in the tail and zero weight to quantiles outside the tail.
B. ES gives progressively larger weights to quantiles deeper in the tail, reflecting increasing risk aversion to extreme losses.
C. ES gives progressively smaller weights to quantiles deeper in the tail, because extreme losses are less likely.
D. ES gives equal weight to every quantile of the entire loss distribution, which makes it a simple average of all quantiles.

Answer: A

Step by step

1. A coherent risk measure is a weighted average of quantiles, with a user-specific (risk-aversion) weighting function.
2. ES is the special case in which the weighting function is flat in the tail: weight $1/(1-\alpha)$ for each tail quantile.
3. All quantiles outside the tail receive weight zero.
4. So ES is the equal-weighted average of the tail quantiles; statement A matches.

Why A is correct: ES is the special case of a coherent measure whose weights are constant at 1/(1 - confidence level) across the tail and zero elsewhere, i.e., an equal-weighted average of tail quantiles.

Why the others are wrong

- **B:** Increasing weights deeper in the tail describe a more general risk-averse coherent measure (e.g., exponential weighting), not ES, which weights all tail quantiles equally.
- **C:** Decreasing weights confuse the falling tail probability density with the weighting function; ES does not discount deeper tail quantiles.
- **D:** Equal weight on every quantile of the whole distribution gives the mean of the distribution, not a tail measure; ES puts zero weight outside the tail.

Q71 GARP PE2 Q61

REBUILT — Source key D claimed the standard error falls as the number of slices rises, but standard error depends on sample size, not slices (only the halving/discretization error does), so D was not defensible; rewrote the options around the slicing and halving procedure.

A risk analyst estimates a coherent risk measure by dividing the loss distribution into n - 1 equal-probability slices, applying a risk-aversion weighting function to each loss quantile, and averaging. Which of the following statements about this procedure is correct?

A. A coherent risk measure must assign the same weight to every quantile of the loss distribution.
B. The analyst can start with a small n and keep doubling it until the estimate stabilizes, i.e., the halving error becomes close to zero.
C. The halving error is the standard error of the estimator and falls as more historical observations are collected.
D. QQ plots are the standard tool for judging how precisely the quantile-based estimator approximates the true risk measure.

Answer: B

Key correction: Source key (GARP PE2 Q61) said D (halving error and standard error both fall as the number of quantiles rises); correct is B because the standard error depends on the sample size, not the number of slices, and the halving error is a discretization error, not a standard error.

Step by step

1. The estimate is a discretized integral of weighted quantiles, so accuracy depends on how many slices n are used.
2. Procedure: start with small n, double n (halving the slice width), recompute; the difference between successive estimates is the halving error.
3. When the halving error is near zero the estimate has converged, so n is large enough: statement B.
4. A is false: weights are user-specific (risk aversion); equal weighting in the tail is only the ES special case.
5. C is false: halving error is a discretization difference; standard error is a sampling concept tied to sample size.
6. D is false: QQ plots compare a sample with a reference distribution, not estimator precision.

Why B is correct: The slicing approximation converges as n increases, and the standard practice is to double n until successive estimates differ by almost nothing (halving error near zero).

Why the others are wrong

- **A:** Confuses coherent measures with one special case; weights are user-specific and need not be equal (equal weights in the tail give ES).
- **C:** Mislabels the halving error as a standard error; it measures discretization (number of slices), while standard error reflects sampling variability and falls with sample size.
- **D:** QQ plots assess whether a sample's distribution matches a theoretical one (LO 1.f); they do not quantify the precision of an estimator.

LO 1.e

Evaluate estimators of risk measures by estimating their standard errors.

Q72 AnalystPrep Q.1481

REBUILT — Original distractors were nonsensical (e.g., 'square root of the error', 'standard error s'); replaced with plausible alternative input sets.

An analyst constructs a confidence interval around a VaR estimate (a quantile of the loss distribution) using the standard error of the quantile estimator. Which set of inputs determines this standard error?

A. The left-tail probability p, the sample size n, and the probability mass f(q) around the quantile
B. The sample size n and the volatility of returns only
C. The left-tail probability p and the probability mass f(q) only, independent of sample size
D. The sample size n, the probability mass f(q), and the expected shortfall at the quantile

Answer: A

Step by step

1. The variance of the quantile estimator is approximated by $p(1-p)/(n\,[f(q)]^2)$, where f(q) is the probability mass in a bin of width h around q.
2. Standard error = square root of this variance: $\text{se}(q) = \sqrt{p(1-p)/n}\,/\,f(q)$.
3. The inputs are therefore p, n and f(q); nothing else enters.

Why A is correct: The standard error of a quantile estimator is $\sqrt{p(1-p)/n}/f(q)$, so it depends only on the tail

probability p, the sample size n and the density (probability mass) f(q) at the quantile.

Why the others are wrong

- **B:** Omits p and f(q); return volatility is not an input of the quantile standard-error formula (it affects the quantile itself, not its sampling precision in this approximation).
- **C:** Drops n; but the standard error shrinks with the square root of n, so larger samples give narrower confidence intervals.
- **D:** Includes expected shortfall, which is a different risk measure and is not an input; it also omits p.

Q73

BT answers L2.T5.73.1

REBUILT — Original option B (SE rises further into the tail) conflicts with Schweser's convention (smaller p implies smaller SE, f(q) held fixed) and option D was outside the Schweser text; rebuilt with statements drawn from Schweser's comparative statics.

Each of the following is TRUE of the standard error of a quantile (VaR) estimator EXCEPT, holding all other factors constant:

A. The standard error falls as the sample size n rises
B. The standard error rises as the bin width h around the quantile is widened
C. The standard error rises as the left-tail probability p rises toward 0.5
D. The standard error depends on the probability mass f(q) around the quantile

Answer: B

Step by step

1. Standard error: $\text{se}(q) = \sqrt{p(1-p)/n} \,/\, f(q)$.
2. A: se is proportional to $1/\sqrt{n}$, so it falls as n rises: true.
3. B: a wider bin h raises the probability mass f(q) (and lowers p), which lowers se: the statement is false.
4. C: p(1-p) increases until p = 0.5, so se rises: true.
5. D: f(q) is in the denominator: true.

Why B is correct: Widening the bin increases f(q) and reduces p, both of which reduce the standard error, so the statement that it rises is false.

Why the others are wrong

- **A:** True statement: larger samples give more precise estimates, so choosing it as the exception reverses the $1/\sqrt{n}$ effect.
- **C:** True statement: p(1−p) is maximized at p = 0.5, so tail probabilities that are larger give less precise estimates; choosing it confuses VaR confidence level with p.
- **D:** True statement: f(q) is in the denominator of the formula; picking it ignores the density term.

Q74

BT answers P2.T5.22.2.3 · BT notes P2.T5.22.3

REBUILT — Merged two copies; replaced option B (asymptotic normality of coherent estimators, not in Schweser) with a Schweser-supported statement and made the false option unambiguous.

Dowd notes that the standard errors of risk-measure estimators are almost as important as the estimates themselves, and that bootstrapping can be used to estimate them. Each of the following statements about standard errors of risk measures is true EXCEPT:

A. The standard error is a measure of the precision of an estimate
B. A larger standard error implies a wider confidence interval around the risk-measure estimate
C. The standard error decreases as the sample size increases
D. The standard error of VaR can be bootstrapped, but this requires a parametric distribution to be assumed

Answer: D

Step by step

1. A: standard error measures how precisely an estimator is determined: true.
2. B: a confidence interval is the estimate ± a multiple of the standard error, so a larger SE gives a wider interval: true.
3. C: se is proportional to $1/\sqrt{n}$: true.
4. D: bootstrapping resamples with replacement from the observed (empirical) data, so no parametric distribution is required: false.

Why D is correct: Bootstrapping is a resampling technique that works directly with the empirical data, so it does not require a parametric distribution; statement D is the false one.

Why the others are wrong

- **A:** True: precision, not bias or accuracy, is what the standard error describes.
- **B:** True: the interval width is proportional to the standard error; rejecting it confuses precision with level of confidence.
- **C:** True: the standard error falls with the square root of n.

Q75 Schweser Module Quiz 1.2 Q2

Which of the following statements most likely increases standard errors from coherent risk measures?

A. Increasing sample size and increasing the left tail probability.
B. Increasing sample size and decreasing the left tail probability.
C. Decreasing sample size and increasing the left tail probability.
D. Decreasing sample size and decreasing the left tail probability.

Answer: C

Step by step

1. Standard error: se$(q) = \sqrt{p(1-p)/n}\,/\,f(q)$.
2. Smaller n raises se (illustration: p = 0.05, f = 0.01: n = 500 gives 0.975; n = 250 gives 1.378).
3. Larger p (toward 0.5) raises p(1-p) and thus se (n = 500: p = 0.05 gives 0.975; p = 0.10 gives 1.342).
4. Both effects raise se only in option C.

Why C is correct: Standard error rises when the sample is smaller and when the left-tail probability p is larger (up to 0.5), so decreasing n and increasing p raises it most.

Why the others are wrong

- **A:** Raising p increases se but raising n reduces it, so the net effect is ambiguous; the sample-size direction is wrong.
- **B:** Both changes lower the standard error: larger n helps and smaller p shrinks p(1-p).
- **D:** Smaller n raises se, but smaller p lowers it (p = 0.01 gives $0.995/\sqrt{500}/0.01 \approx 0.445$ against 0.975 at p = 0.05), so the net effect is not clearly an increase; the tail-probability direction is wrong.

LO 1.f

Interpret quantile-quantile (QQ) plots to identify the characteristics of a distribution.

Q76 AP Mock 2 Q12

REBUILT — Original referenced a missing exhibit and stated no axis convention; added axes and tightened distractor D.

Richard Wayne, FRM, plots the empirical quantiles of a return sample (vertical axis) against the quantiles of a standard normal distribution (horizontal axis). The points lie close to the 45-degree reference line through the central portion of the distribution, but at both ends they curve away from the line: below it in the lower tail and above it in the upper tail, forming an S-shape.

Based on this QQ plot, which of the following is most likely correct?

A. The returns are normally distributed.
B. The returns have fatter tails than the normal distribution, with a roughly symmetric center.
C. The returns are negatively skewed relative to the normal distribution.
D. The returns are normal, but with a larger standard deviation than the sample estimate.

Answer: B

Step by step

1. Empirical quantiles are on the vertical axis, normal quantiles on the horizontal axis; a perfect normal fit would lie on the reference line.
2. The center matches the line, so the middle of the distribution resembles the normal.
3. Lower tail below the line means empirical losses are more extreme than the normal predicts; upper tail above the line means empirical gains are more extreme too.
4. Extreme observations in both tails beyond normal predictions, with a matching center, is the signature of a symmetric fat-tailed (leptokurtic) distribution.

Why B is correct: Tails deviating away from the line in both directions while the center is linear indicates sym-

metric tails fatter than the normal.

Why the others are wrong

- **A:** A normal sample would plot along the line in the tails too; the visible tail departure rejects this.
- **C:** Skewness would show as an asymmetric departure (one tail heavy, the other light), not symmetric departures in both tails.
- **D:** A different scale alone would give a straight line with slope different from 1 across the whole range, not a linear center with bending tails.

Q77 AnalystPrep Q.5292

A data scientist wants to determine whether his dataset follows a particular distribution and decides to use a QQ plot. Which of the following statements about QQ plots is correct?

A. QQ plots are used to evaluate the precision of a statistical estimator.
B. QQ plots are used to determine the statistical significance of a hypothesis test.
C. QQ plots can only be used when the sample size is greater than 100.
D. QQ plots are useful in visually assessing whether a dataset follows a reference distribution such as the normal.

Answer: D

Step by step

1. A QQ plot graphs sample quantiles against the quantiles of a reference distribution.
2. If the sample comes from the reference distribution (up to location and scale), the points are approximately linear.
3. So it is a visual tool for judging distributional fit; it provides no standard errors, p-values or sample-size requirement.

Why D is correct: The QQ plot is a visual diagnostic of how well an empirical distribution matches a reference distribution.

Why the others are wrong

- **A:** Estimator precision is assessed with standard errors or confidence intervals, not QQ plots.
- **B:** A QQ plot is informal and gives no test statistic or p-value.
- **C:** There is no minimum sample size; with small samples the plot is just noisier.

Q78 BT answers L2.T5.74.1

REBUILT — Original depended on a missing QQ figure; replaced with a verbal description of the plot (linear center, steeper slope in both tails) and re-fit the options so only one is right.

An analyst plots the quantiles of an empirical return sample (vertical axis) against standard normal quantiles (horizontal axis). The plot is approximately a straight line with slope near 1 through the central region, but the slope becomes noticeably steeper than 1 in both the left and right tails, and the plot is symmetric about the origin. Which statement is most likely true?

A. The empirical distribution is normal with a larger variance than the reference.
B. The empirical distribution has positive skewness.
C. The empirical distribution is leptokurtic (kurtosis greater than 3) with symmetric, heavy tails.
D. A linear transformation of location and scale would make the empirical distribution approximately normal.

Answer: C

Step by step

1. Center is linear: the body of the distribution resembles the normal.
2. Slope steeper than 1 in the tails means empirical extreme quantiles are further from the center than the normal's, i.e., heavier tails.
3. The plot is symmetric, so tails are heavy on both sides and skew is not indicated.
4. Symmetric heavy tails mean excess kurtosis: kurtosis greater than 3, as for a Student t with few degrees of freedom.

Why C is correct: A steeper-than-1 slope in the tails of a QQ plot signals tails heavier than the reference; symmetry points to leptokurtosis rather than skew.

Why the others are wrong

- **A:** A pure variance difference would give a straight line with slope above 1 everywhere, not only in the tails.
- **B:** Positive skew would show only a heavy right tail (upward bend on one side), not both.

- **D:** A location/scale change only moves the intercept and slope of a straight line; it cannot straighten tails that bend.

Q79 BT answers L2.T5.74.2

Each of the following statements about the QQ plot is true EXCEPT:

A. Any distribution that produces a linear QQ plot is a good candidate distribution for the data.
B. The QQ plot is ideal for evaluating kurtosis but cannot effectively reveal skew.
C. The intercept and slope of a linear QQ plot give a rough idea of the location and scale parameters of the sample data.
D. A QQ plot can identify outliers even when the other observations are broadly consistent with the reference distribution.

Answer: B

Step by step

1. A: a linear plot means the quantiles match one-to-one up to location and scale, so the reference is a good candidate. True.
2. C: if empirical = a + b x reference, intercept a approximates location and slope b approximates scale. True.
3. D: an outlier appears as an isolated point far from the line. True.
4. B: skew shows up as asymmetric tail behavior (one tail bending away more than the other), so QQ plots do reveal skew. This statement is false, hence the EXCEPT answer.

Why B is correct: QQ plots reveal asymmetric tail departures (skew) as well as heavy or light tails (kurtosis), so B is the false statement.

Why the others are wrong

- **A:** True statement; linearity indicates a good reference distribution, so it is not the exception.
- **C:** True statement; intercept and slope map to location and scale, so it is not the exception.
- **D:** True statement; outliers stand out from the line, so it is not the exception.

Q80 BT answers P2.T5.709.3 · BT notes P2.T5.709.3

REBUILT — The QQ exhibit was missing in both sources; replaced with a table of key plotted points consistent with the original answer and explanation.

Peter, a risk analyst, wants to know whether a dataset is approximately normal. He plots empirical (actual) quantiles on the vertical axis against standard normal (predicted) quantiles on the horizontal axis and records these points:

Predicted quantile	Actual quantile
-3.0	-6.0
-2.0	-3.5
0.0	0.0
2.0	1.0
3.0	1.5

Based on the QQ plot, which statement is most likely true about this distribution?

A. The distribution is approximately normal.
B. It has light tails (left and right) with positive skew.
C. It has a very heavy left tail, a light right tail, and negative (left) skew.
D. It has a very heavy right tail, a light left tail, and positive (right) skew.

Answer: C

Step by step

1. Under a normal fit, actual quantiles equal predicted quantiles (points on the 45-degree line).
2. Left tail: at -3.0 the actual quantile is -6.0, far below the line, so losses are far more extreme than normal: very heavy left tail.
3. Right tail: at 2.0 the actual is 1.0 and at 3.0 it is 1.5, below the line, so the right tail is shorter than normal: light right tail.
4. A long left tail with a short right tail is negative (left) skew.

Why C is correct: Points below the line on the left show a heavier tail; points below the line on the right show a lighter tail; the asymmetry is negative skew.

Why the others are wrong

- **A:** The points depart sharply from the line in the tails, so the normal does not fit.
- **B:** The left tail is not light; actual quantiles at -3 are far more negative than normal, and the skew is negative.
- **D:** Reverses the reading: points below the line on the right mean a lighter (not heavier) right tail, and the extreme points sit on the left.

Q81 BT answers P2.T7.506.1

Standard models of asset prices assume normally distributed returns (geometric Brownian motion). Real-world asset returns, however, tend to exhibit each of the following EXCEPT:

A. Leptokurtosis (fat tails).
B. A mean that differs from the median (skewness).
C. A QQ plot against the normal close to the 45-degree line.
D. Time variation in volatility.

Answer: C

Step by step

1. A QQ plot near the 45-degree line indicates the data match the normal reference.
2. Real returns are leptokurtic, skewed and have time-varying volatility, so their QQ plots against the normal bend away from the line in the tails.
3. Thus a near-45-degree QQ plot is not generally true of real returns; it is the exception.

Why C is correct: Fat tails make normal QQ plots of real returns bend away from the line; only C describes a normal-like pattern.

Why the others are wrong

- **A:** Fat tails are a well-documented feature of returns, so this is true and not the exception.
- **B:** Skewness makes the mean differ from the median in practice, so this is true and not the exception.
- **D:** Volatility clustering and time variation are standard stylized facts, so this is true and not the exception.

Q82 GARP PE2 Q43 · Mock 4 Q44

A risk analyst is studying a series of graphs plotting the empirical distribution of a portfolio's profit and loss quantiles against the quantiles of different specified reference distributions. The analyst plans to use these QQ plots to gain insights into the properties of the empirical distribution. Which of the following statements is correct regarding QQ plots?

A. The tails of the empirical distribution are heavier than those of the reference distribution if the QQ plot has a steeper slope at its tails.
B. QQ plots are used to smooth the empirical distribution by removing outliers through the use of quantiles.
C. A distribution is a good reference distribution if data drawn from it produce a QQ plot that is non-linear in the central mass of the distribution.
D. The primary purpose of a QQ plot is to measure the skewness and kurtosis of the empirical data.

Answer: A

Step by step

1. With the empirical quantiles on the vertical axis, a slope steeper than 1 in a tail means the empirical extreme quantiles lie further out than the reference's.
2. That is the definition of heavier tails than the reference.
3. B, C and D misstate the purpose or the linearity criterion.

Why A is correct: A steeper tail slope means empirical tail quantiles exceed those of the reference, i.e., heavier tails.

Why the others are wrong

- **B:** QQ plots make outliers visible; they do not smooth data or remove them.
- **C:** Reversed: a good reference yields an approximately linear plot, including the central mass.
- **D:** A QQ plot is a visual fit diagnostic; skewness and kurtosis are measured with moment statistics.

Q83 Schweser Module Quiz 1.2 Q3

REBUILT — Original options were ambiguous (two defensible answers, and 'testing' is wrong); replaced with clearly distinct purposes.

Which of the following best describes the main purpose of a quantile-quantile (QQ) plot?

A. To perform a formal statistical test that rejects or accepts a reference distribution at a given confidence level.

B. To visually compare the quantiles of an empirical sample with those of a reference distribution to judge whether the reference is a plausible fit.

C. To estimate the standard error of a quantile-based risk measure.

D. To compute VaR directly from the empirical distribution without assuming a model.

Answer: B

Key correction: Source key (Schweser) said C ('Identifying an empirical distribution from a theoretical distribution'); the wording is ambiguous versus D, so the item was rebuilt around the underlying concept (visual comparison, no formal test).

Step by step

1. A QQ plot maps empirical quantiles one-to-one against quantiles of a hypothesized distribution.
2. If the resulting plot is approximately linear, the reference distribution is a plausible match; departures in the tails show fat or thin tails.
3. This is visual inspection only, with no formal test statistic.

Why B is correct: The QQ plot is an informal visual tool for matching an empirical sample to a reference distribution.

Why the others are wrong

- **A:** A QQ plot is visual only and has no formal test or p-value.
- **C:** Standard errors come from sampling theory or bootstrap, not from QQ plots.
- **D:** Nonparametric VaR comes from empirical quantiles (historical simulation), which is not the purpose of the plot.

Reading 2. Non-Parametric Approaches

Kevin Dowd, Measuring Market Risk, 2nd Edition (West Sussex, UK: John Wiley & Sons, 2005). Chapter 4.

LO	Learning objective	Qs
2.a	apply the bootstrap historical simulation approach to estimate coherent risk measures.	10
2.b	describe historical simulation using non-parametric density estimation.	7
2.c	compare and contrast the age-weighted, the volatility-weighted, the correlation- weighted, and the filtered historical simulation approaches.	21
2.d	identify advantages and disadvantages of non-parametric estimation methods.	10

LO 2.a

Apply the bootstrap historical simulation approach to estimate coherent risk measures.

Q84 BT notes P2.T5.22.3.1 · BT answers P2.T5.22.1

Which of the following is an essential difference between basic historical simulation and bootstrap historical simulation; i.e., one of them involves this feature or element, but the other does not?

A. Random number generator
B. Non-parametric methodology
C. Parametric distributional assumption
D. Actual observed (i.e., historical) loss data

Answer: A

Step by step

1. Basic HS orders the single historical sample and reads off the VaR quantile; no randomness is involved.
2. Bootstrap HS repeatedly draws observations from the sample with replacement, which needs random draws (a random number generator), records each resample's VaR, and averages them.
3. Both methods are non-parametric and both use only actual historical data, so only the random resampling distinguishes them.

Why A is correct: Random resampling with replacement is unique to the bootstrap; everything else is shared with basic HS.

Why the others are wrong

- **B:** Both approaches are non-parametric, so this does not distinguish them.
- **C:** Neither approach assumes a parametric distribution.
- **D:** Both approaches rely entirely on actual historical data.

Q85 AP Mock 3 Q26

A risk manager wishes to use the bootstrap historical simulation approach to estimate VaR. What is the main reason for using historical returns to forecast VaR in this approach?

A. The distribution of returns is expected to change over time.
B. The distribution of returns will remain the same in the past and in the future.
C. The historical returns are always normally distributed.
D. The historical returns have no impact on future returns.

Answer: B

Step by step

1. Bootstrap HS resamples observed returns and treats their empirical distribution as the forecast distribution.
2. That is only valid if the return distribution is assumed to be the same in the past and the future (stationarity).
3. Check each other option against this premise: a changing distribution, forced normality, or no link between past and future would each undermine the method.

Why B is correct: Using past returns as a forecast presupposes that the return distribution is unchanged going forward.

Why the others are wrong

- **A:** A changing distribution is a criticism of historical methods, not their rationale.

- **C:** Bootstrap HS is non-parametric; it does not assume normality.
- **D:** If past returns had no bearing on future returns, historical data would be useless for forecasting.

Q86 AnalystPrep Q.1492

Which of the following statements about a bootstrapping exercise for historical simulation VaR is most likely correct? A bootstrapping exercise:

A. resamples from the existing data set without replacement.
B. assumes that the distribution of returns will remain the same in the past and in the future.
C. assumes that the distribution of returns in the future will be markedly different from past distributions.
D. results in a VaR estimate that is a sum of sample VaRs after repeated sampling.

Answer: B

Step by step

1. Bootstrap draws a sample, records its VaR, returns the data to the pool, and repeats: this is sampling WITH replacement (rules out A).
2. Using historical data to forecast presupposes the return distribution stays the same (B), not that it differs (C).
3. The final estimate is the AVERAGE of the sample VaRs, not the sum (rules out D).

Why B is correct: The bootstrap treats historical returns as representative of the future distribution, with resampling done with replacement and sample VaRs averaged.

Why the others are wrong

- **A:** Data are 'returned' after each draw, so sampling is with replacement.
- **C:** Assuming a markedly different future distribution would contradict using past data.
- **D:** The best estimate is the mean of the sample VaRs; summing would multiply the loss by the number of resamples.

Q87 AnalystPrep Q.5295

A data analyst wishes to calculate the VaR of a credit firm using the bootstrap historical simulation approach. How is the final VaR estimate calculated?

A. By taking the highest VaR from all resamples.
B. By taking the lowest VaR from all resamples.
C. By averaging the VaR from all resamples.
D. By taking the median VaR from all resamples.

Answer: C

Step by step

1. Each resample (drawn with replacement) produces its own VaR.
2. The best VaR estimate for the full dataset is the average of all the sample VaRs.
3. Extremes or the median discard information from the other resamples.

Why C is correct: Averaging across resample VaRs reduces sampling noise and is the prescribed bootstrap estimate.

Why the others are wrong

- **A:** The maximum VaR is an upper extreme, not the estimate; it overstates risk.
- **B:** The minimum VaR is a lower extreme and understates risk.
- **D:** The median is not the prescribed aggregator; the method uses the mean.

Q88 AnalystPrep Q.6431

Compared to basic (raw) historical simulation, which of the following statements correctly characterizes what occurs during bootstrapping?

A. Bootstrapping compresses the original data, leading to repeated observations.
B. It replicates the most extreme historical returns in every sample.
C. It scales each historical return by a constant factor obtained from the sample's mean.
D. It resamples (with replacement) many times, creating multiple "alternative" data sets.

Answer: D

Step by step

1. Basic HS uses the single original sample once.
2. Bootstrapping draws many samples with replacement from that sample, each giving a different 'alternative' data set (some observations repeated, some omitted).
3. No scaling or forced inclusion of extremes occurs.

Why D is correct: The defining feature of bootstrapping is repeated resampling with replacement to create many alternative data sets.

Why the others are wrong

- **A:** Data are not compressed; resamples are the same size as the original, with random repeats as a by-product.
- **B:** Extremes appear only if randomly drawn; they are not forced into every sample.
- **C:** Scaling by the mean is not part of bootstrapping (volatility weighting rescales by volatility ratios, a different method).

Q89

AnalystPrep Q.1495

REBUILT — Original asked for the term 'resampling variability', which is not in the assigned reading; replaced with a computational item on averaging resample ES, which is part of LO 2.a.

A risk manager applies bootstrap historical simulation to estimate 95% expected shortfall (ES). Four resamples, each drawn with replacement from the original loss dataset, give the following ES estimates (in USD millions): 3.0, 3.1, 3.2 and 3.9. What is the bootstrap estimate of ES?

A. 3.00
B. 3.15
C. 3.30
D. 3.90

Answer: C

Step by step

1. Each resample computes its own ES by averaging the tail quantile VaRs of that resample.
2. The best ES estimate for the original dataset is the average of the resample ESs.
3. Mean = (3.0 + 3.1 + 3.2 + 3.9)/4 = 13.2/4 = 3.30.

Why C is correct: Bootstrap ES, like bootstrap VaR, is the average of the ES values from all resamples.

Why the others are wrong

- **A:** Takes the lowest resample ES, ignoring the other resamples.
- **B:** Takes the median of 3.1 and 3.2 = 3.15; the method uses the mean, which is pulled up by the 3.9 outcome.
- **D:** Takes the highest resample ES, an upper extreme rather than the average.

Q90

BT answers L2.T5.75.1

According to Dowd, EACH of the following is TRUE about bootstrapped historical simulation EXCEPT:

A. Bootstrap draws at random without replacement.
B. The bootstrap is very intuitive and easy to apply.
C. A bootstrapped estimate will often be more accurate than a 'raw' sample estimate.
D. Bootstraps are useful for gauging the precision (i.e., standard errors) of our estimates.

Answer: A

Step by step

1. The bootstrap draws a sample, records the VaR, and returns the data, i.e., sampling WITH replacement, so A is false.
2. B is true: the procedure is simple and intuitive.
3. C is true: bootstrapping gives more precise estimates of coherent risk measures than raw HS.
4. D is true: the spread of resample estimates gauges precision.

Why A is correct: The only false statement is A because bootstrap sampling is with replacement.

Why the others are wrong

- **B:** True statement, so it is not the exception.
- **C:** True statement (bootstrap improves precision over raw HS).
- **D:** True statement (resample dispersion measures standard errors).

Q91 BT answers L2.T5.75.2

What is the primary difference between historical simulation and bootstrapped historical simulation?

A. Bootstrapping is non-parametric.
B. Bootstrapping can be used to compute both value at risk (VaR) and expected shortfall (ES).
C. Bootstrapping does not require a variance-covariance matrix.
D. Bootstrapping generates multiple samples.

Answer: D

Step by step

1. Both methods are non-parametric, need no variance-covariance matrix, and can give VaR and ES (rules out A, B, C).
2. Basic HS uses the one historical sample; bootstrap generates many resamples with replacement, computes a risk measure on each, and averages.

Why D is correct: Generating many resamples is what distinguishes the bootstrap from basic historical simulation.

Why the others are wrong

- **A:** Also true of basic HS.
- **B:** Basic HS also yields VaR and ES.
- **C:** Neither method uses a variance-covariance matrix.

Q92 BT answers P2.T5.710.1 · BT notes P2.T5.710.1

REBUILT — Original had two defensible answers (B about ease of communication was also arguably true) and tested peripheral points; rebuilt around the core bootstrap HS mechanics with exactly one correct statement.

Betty has 1,000 daily P&L observations and applies the bootstrap historical simulation approach to estimate the 95% VaR and expected shortfall (ES). Which of the following statements about the procedure is TRUE?

A. Because resampling adds randomness, the bootstrap can produce a VaR or ES estimate larger than the worst loss in the original dataset.
B. The best VaR estimate is the VaR computed once from the original dataset; the resamples are used only to assess its accuracy.
C. Each resample is drawn without replacement so that no observation appears twice within a resample.
D. The best estimates of VaR and ES are the averages of the VaRs and ES values computed from the many resampled datasets.

Answer: D

Key correction: Source key (BT answers/notes) said D for a different version of the question (about avoiding closed-form confidence intervals); the rebuilt question has D correct as well.

Step by step

1. Bootstrap HS draws a sample from the original data, records the VaR (and ES, by averaging tail-slice VaRs) from that sample, and returns the data to the dataset.
2. Because the data are returned each time, sampling is with replacement.
3. The procedure is repeated many times, giving many sample VaRs and ES values.
4. The best estimate for the original dataset is the average of the sample VaRs (and of the sample ES values), which is more precise than raw HS.
5. Every draw uses only observed losses, so no estimate can exceed the worst observed loss.

Why D is correct: Bootstrapping resamples with replacement, computes a risk measure on each resample and averages them; the average is the best estimate of VaR/ES.

Why the others are wrong

- **A:** False: resamples contain only observed values, so no resample can contain a loss worse than the historical worst; the bootstrap cannot exceed it (a common confusion with filtered HS).
- **B:** False: this describes plain HS; in the bootstrap the average of the resample VaRs, not the single original-sample VaR, is the best estimate.
- **C:** False: the data are 'returned' after each draw, i.e., sampling is with replacement, so observations can repeat.

Q93 Schweser Module Quiz 2.1 Q1

Johanna Roberto has collected a dataset of 1,000 daily observations on equity returns. She is concerned about the appropriateness of using parametric techniques as the data appears skewed. Ultimately, she decides to use historical simulation and bootstrapping to estimate the 5% VaR. Which of the following steps is most likely to be part of the estimation procedure?

A. Filter the data to remove the obvious outliers.
B. Repeated sampling with replacement.
C. Identify the tail region from reordering the original data.
D. Apply a weighting procedure to reduce the impact of older data.

Answer: B

Step by step

1. Bootstrap HS draws a sample from the 1,000 observations, records the 5% VaR, and returns the data to the dataset.
2. Returning the data means sampling with replacement.
3. The draw-and-record step is repeated many times and the sample VaRs are averaged for the final estimate.

Why B is correct: Bootstrapping is repeated resampling with replacement, with the risk measure estimated on each resample and averaged.

Why the others are wrong

- **A:** Outliers are real tail observations; removing them would bias VaR, and bootstrapping does no filtering.
- **C:** Ordering data to find the tail is part of computing the VaR within a sample, but it is not the distinguishing step; the defining feature of the bootstrap is repeated sampling with replacement.
- **D:** Age-weighting is a different HS refinement (BRW); bootstrapping gives all observations equal probability of being drawn.

LO 2.b

Describe historical simulation using non-parametric density estimation.

Q94 BT answers L2.T5.76.2

REBUILT — Source option D relied on Dowd's kernel-estimator remark not covered in the Schweser reading; rewrote the EXCEPT statements around the midpoint-connecting method taught in the reading.

Regarding historical simulation using non-parametric density estimation, each of the following statements is true EXCEPT:

A. By smoothing the data, it creates new information, so with 100 observations a 99.9% VaR beyond the largest observed loss can be estimated reliably.
B. It is free from restrictive assumptions about the shape of the underlying distribution.
C. Connecting the midpoints of successive histogram bars allows VaR to be estimated at confidence levels between observed data points.
D. When midpoints are connected, the area gained by the lower bar equals the area lost by the upper bar, so total probability remains one.

Answer: A

Step by step

1. Check B: the method uses the data themselves and imposes no distribution, so it is true.
2. Check C: connecting bar midpoints makes the density continuous, so VaR is available at any confidence level; true.
3. Check D: area is displaced, not created or lost, so the surrogate is still a valid pdf; true.
4. Check A: smoothing interpolates between observed points; it cannot add information about losses worse than the worst observation, so a reliable 99.9% VaR from 100 observations is false.

Why A is correct: Non-parametric density estimation only redistributes area between existing observations to fill gaps; it interpolates and cannot supply information beyond the data.

Why the others are wrong

- **B:** True statement: the approach is free of restrictive distributional assumptions; not the exception.
- **C:** True statement: this is the core mechanism that fills the gaps between ordered observations.

- **D:** True statement: area is displaced from the upper bar to the lower bar, preserving total area of one.

Q95

AnalystPrep Q.1487

REBUILT — Source distractors hinged on an inconsistent order-statistic convention (fourth/seventh/ninth-largest loss); replaced with distinct, unambiguous drawback statements.

Non-parametric density estimation rests on the premise that basic historical simulation (HS) does not extract all the information in the data. Which of the following best demonstrates this drawback?

A. With 100 P/L observations, basic HS gives every observation the same weight regardless of how old it is.

B. With 100 P/L observations, basic HS can estimate VaR only at discrete confidence levels (e.g., 95%, 96%), so a 95.5% VaR cannot be obtained.

C. With 100 P/L observations, basic HS requires the assumption that P/L is normally distributed.

D. With 100 P/L observations, basic HS cannot reflect the fact that current volatility differs from volatility in the sample.

Answer: B

Step by step

1. With n observations, basic HS provides only n distinct confidence levels (steps of 1/100 = 1%).
2. A 95.5% VaR lies between two ordered observations, so basic HS cannot give it.
3. Non-parametric density estimation connects the midpoints of histogram bars to smooth the data and fill this gap.
4. Therefore B is the drawback this method addresses; the other options describe different issues.

Why B is correct: Discreteness of the data limits basic HS to n confidence levels; smoothing the histogram removes that limitation.

Why the others are wrong

- **A:** Equal weighting is a real drawback, but it is addressed by age-weighted HS, not non-parametric density estimation.
- **C:** Basic HS is non-parametric and assumes no distribution; this confuses it with the variance-covariance method.
- **D:** Stale volatility is a real drawback, but it is addressed by volatility-weighted HS, not by smoothing between data points.

Q96

AnalystPrep Q.1496

A drawback of historical simulation is that the discreteness of the data rules out VaR estimates between data points. For example, with 100 historical observations, VaR is straightforward at the 95% or 96% confidence level but impossible at 95.5%. Which of the following methods can solve this problem?

A. Applying brute force

B. Bootstrapping

C. Non-parametric density estimation

D. Use of a large number of re-samples

Answer: C

Step by step

1. With 100 observations, basic HS offers only discrete levels (steps of 1%); 95.5% falls between two ordered losses.
2. Non-parametric density estimation connects the midpoints of successive histogram bars, producing a continuous surrogate density.
3. VaR can then be read at any confidence level, including 95.5%.

Why C is correct: Smoothing the empirical distribution into a continuous density permits VaR at any confidence level, not just the n discrete ones.

Why the others are wrong

- **A:** 'Brute force' is not a remedy for discreteness; it does nothing to fill gaps between observations.
- **B:** Bootstrapping resamples the same data with replacement to improve accuracy, but resampled values still come from the discrete set of observed points.
- **D:** More resamples reduce sampling noise in a bootstrap, but the resamples remain drawn from the same discrete observations.

Q97

AnalystPrep Q.6434

REBUILT — Source distractors were giveaways (Zero outliers, Constant correlation) and 97.5% is exactly attainable with 80 observations; changed the target level to 98% and built plausible distractors.

A small prop trading firm has only 80 daily P/L observations and wants a VaR at the 98% confidence level. With basic historical simulation, available confidence levels are spaced 1/80 = 1.25% apart (e.g., 97.5% and 98.75%), so 98% cannot be obtained. The head quant proposes historical simulation with non-parametric density estimation. What is the key benefit of this approach for the firm?

A. It generates additional independent observations, enlarging the effective sample beyond 80.

B. It fits a normal distribution to the 80 observations so VaR follows from the mean and standard deviation.

C. It allows reliable VaR estimates at levels beyond the worst observed loss, such as 99.9%.

D. It smooths between data points so VaR can be estimated at any confidence level, including 98%.

Answer: D

Step by step

1. Basic HS levels: 100% - k/80: 97.5% (k = 2) and 98.75% (k = 1); 98% lies between them.
2. Non-parametric density estimation connects histogram midpoints to form a continuous density.
3. VaR can be read at 98% (or any level) from this surrogate density.
4. The method adds no new information and assumes no distribution.

Why D is correct: Smoothing the empirical histogram creates a continuum of attainable confidence levels without distributional assumptions.

Why the others are wrong

- **A:** Smoothing reshapes the existing 80 points; it does not create new independent observations.
- **B:** That describes a parametric approach; non-parametric density estimation imposes no distribution.
- **C:** The method interpolates between observations; it cannot add information about losses beyond the worst observation.

Q98

BT answers P2.T5.22.3 · BT notes P2.T5.22.3.3 · BT Mock Q71

REBUILT — Source options relied on confidence-interval/order-statistic claims outside the reading and option A was arguable; rewrote around the 0.4% discreteness of 250 observations and what smoothing can and cannot do.

Peter has 250 daily P/L observations for a new equity portfolio. His board wants VaR at confidence levels 99.0%, 99.1%, ..., 99.9% and prefers a non-parametric approach. With 250 observations, basic historical simulation (BHS) can only provide confidence levels spaced 1/250 = 0.4% apart (e.g., 99.2%, 99.6%). Which of the following statements is TRUE?

A. BHS can directly retrieve a distinct VaR for each of the levels 99.0%, 99.1%, ..., 99.9% because the data are non-parametric.

B. Non-parametric density estimation requires Peter to assume normally distributed P/L in order to interpolate between observations.

C. Non-parametric density estimation smooths between data points, so Peter can obtain VaR at granular levels such as 99.1% without a distributional assumption.

D. Non-parametric density estimation allows reliable VaR at 99.9% even though it lies beyond the worst of the 250 observations.

Answer: C

Step by step

1. Tail spacing with 250 observations: 1/250 = 0.4%, so BHS levels are 98.8%, 99.2%, 99.6%, ...; 99.1% is not attainable.
2. Eliminate A: ten requested levels exceed the number of distinct BHS levels near 99%.
3. Non-parametric density estimation connects histogram midpoints to make the distribution continuous, with no parametric assumption.
4. Thus 99.1% and every level between observations is available; it cannot reliably extend beyond the worst loss, so D fails.

Why C is correct: Smoothing makes the empirical distribution behave like a continuous pdf, giving VaR at granular levels without a distributional assumption.

Why the others are wrong

- **A:** BHS yields only n discrete levels (0.4% spacing here), so it cannot produce a distinct VaR for each 0.1% step.
- **B:** Non-parametric methods impose no distribution; this confuses interpolation with parametric (normal) fitting.

- **D:** Smoothing interpolates between observations; it cannot add information about losses worse than anything observed.

Q99

BT answers L2.T5.76.1

REBUILT — Source relied on Dowd's histogram/naive-estimator terminology and an ambiguous 'sixth worst loss' convention; reframed in Schweser terms.

Which of the following LEAST describes historical simulation using non-parametric density estimation?

A. Fitting a normal distribution with the sample mean and standard deviation, then reading VaR from the fitted curve.

B. Connecting the midpoints of successive histogram bars to create a smooth surrogate density.

C. Shifting area from the upper bar to the lower bar so that total area remains one.

D. Obtaining VaR at confidence levels that fall between ordered observations.

Answer: A

Step by step

1. Non-parametric density estimation starts from the empirical histogram and imposes no distribution.
2. B, C and D are the mechanism and benefit described in the reading: connect midpoints, reallocate area, VaR at any level.
3. A imposes a normal distribution calibrated by mean and standard deviation, which is parametric, so it least describes the method.

Why A is correct: The method smooths the empirical histogram; fitting a distribution with estimated parameters is a parametric approach.

Why the others are wrong

- **B:** Describes the method itself: the surrogate density is formed by connecting bar midpoints.
- **C:** Describes what happens to area when midpoints are connected; total area is preserved.
- **D:** Describes the main benefit: VaR on a continuum of confidence levels.

Q100

BT answers P2.T5.710.3 · BT notes P2.T5.710.3

REBUILT — Reworded the key option to the Schweser statement of the main improvement (VaR on a continuum of confidence levels) and replaced the kernel-superiority distractor, which is outside the reading.

What is the primary advantage of historical simulation using non-parametric density estimation over basic historical simulation?

A. It is simpler to implement because no smoothing step is required.

B. VaR can be calculated at a continuum of confidence levels, not just the n levels implied by the ordered observations.

C. It places more weight on recent observations than on older observations.

D. It calibrates a familiar distribution (e.g., beta or gamma) to the data and reads VaR from it.

Answer: B

Step by step

1. Basic HS with n observations gives only n confidence levels.
2. Non-parametric density estimation connects histogram midpoints, creating a continuous surrogate density.
3. VaR can then be computed for a continuum of points; this is the major improvement cited in the reading.

Why B is correct: Smoothing the empirical distribution lets VaR be read at any confidence level while staying free of restrictive distributional assumptions.

Why the others are wrong

- **A:** The extra smoothing step makes it more, not less, involved; simplicity is the advantage of basic HS.
- **C:** That describes age-weighted HS, a different refinement.
- **D:** That describes parametric density estimation; the non-parametric approach assumes no distribution.

LO 2.c

Compare and contrast the age-weighted, the volatility-weighted, the correlation- weighted, and the filtered historical simulation approaches.

Q101

AP Mock 3 Q55

REBUILT — Original item had two defensible answers (plain HS can over-estimate after a volatility fall as well as under-estimate); fixed the scenario to a volatility rise and replaced the options.

Risk factors exhibit time-varying volatility, and volatility has recently risen well above its average level over the historical window. Relative to the VaR implied by the true current volatility, an unadjusted equal-weighted historical simulation VaR will most likely:

A. Over-estimate risk, because older observations carry more weight than recent ones.
B. Under-estimate risk, because the window is dominated by lower-volatility observations.
C. Be unbiased, because historical simulation makes no distributional assumption.
D. Be unaffected, because VaR is a quantile and quantiles ignore volatility.

Answer: B

Step by step

1. Equal-weighted HS gives every observation in the window weight $1/n$, regardless of age or the volatility regime.
2. If current volatility is above the window's typical volatility, most observations come from calmer periods.
3. The empirical quantile therefore reflects calmer conditions than today's, so VaR is too low.
4. The remedy is the volatility-weighted approach: $r_t^* = r_t \times \sigma_T/\sigma_t$, which scales returns up when current volatility is higher.

Why B is correct: Plain HS implicitly assumes i.i.d. returns with constant volatility, so after a volatility rise it stays anchored to calmer past data and under-estimates current risk.

Why the others are wrong

- **A:** Equal weighting gives old data the same weight as new data, not more; this also confuses the direction of the bias (over-estimation occurs after a volatility fall).
- **C:** Being non-parametric does not remove the i.i.d./constant-volatility assumption; plain HS does not adapt to changes in volatility.
- **D:** The sample quantile depends on the dispersion of the returns in the window, which reflects the volatility regime of that window.

Q102

BT answers L2.T5.78.2

In comparison to equal-weighted and age-weighted approaches, Dowd asserts EACH of the following is TRUE about volatility-weighted historical simulation EXCEPT for:

A. Allows us to obtain VaR and ES estimates that can exceed the maximum loss in our historical data set
B. Produces risk estimates that are naturally sensitive to current volatility estimates, enabling us to incorporate information from GARCH forecasts into HS VaR and ES estimation
C. Empirical evidence indicates it produces superior VaR estimates to the age-weighted approach
D. Only drawback is that it does not naturally combine with age-weighted approach or bootstrap, forcing a choice

Answer: D

Step by step

1. A: scaling by σ_T/σ_t can push adjusted losses beyond any historical loss, so true.
2. B: estimates respond to current volatility, using GARCH/EWMA forecasts, so true.
3. C: Dowd cites empirical evidence that it outperforms age-weighting, so true.
4. D: volatility weighting can be combined with other refinements (e.g., age weighting, bootstrap), so false, and the EXCEPT answer.

Why D is correct: The volatility-weighting adjustment changes the data, not the method, so it can be combined with other HS refinements.

Why the others are wrong

- **A:** True statement (a listed advantage), so not the exception.
- **B:** True statement (volatility sensitivity is the main advantage), so not the exception.
- **C:** True per Dowd's cited evidence, so not the exception.

Q103

AnalystPrep Q.2631

An analyst performing a historical simulation to measure the VaR of a portfolio uses a volatility-weighted approach. One month ago, the actual return of the asset was 5% and its daily volatility estimate was 2%. If the current daily volatility is 1.5%, calculate the volatility-adjusted return.

A. 0.03
B. 0.0167
C. 0.0375
D. 0.0667

Answer: C
Step by step
1. Volatility-adjusted return: $r^* = r_t \times \sigma_T/\sigma_t$.
2. $r^* = 5\% \times 1.5\%/2\% = 5\% \times 0.75 = 3.75\% = 0.0375$.

Why C is correct: Current volatility is lower than the volatility on the historical date, so the historical return is scaled down by 0.75.
Why the others are wrong
- **A:** Subtracted the volatility from the return ($5\% - 2\% = 3\%$) instead of scaling by the volatility ratio.
- **B:** Divided the return by 3 (or otherwise combined the three numbers without the ratio σ_T/σ_t); it is not a valid adjustment.
- **D:** Inverted the ratio: $5\% \times 2\%/1.5\% = 6.67\%$, scaling up although current volatility is lower.

Q104
AnalystPrep Q.2831

REBUILT — Original options included 0.014 (a rounding of the correct value) and an unexplained 0.0205; replaced with distractors that each come from a specific formula error.

In an age-weighted historical simulation, the historical window contains $n = 300$ daily observations and the decay factor is $\lambda = 0.8$. What is the weight on the observation that is 13 days old?

A. 0.00333
B. 0.01100
C. 0.01374
D. 0.06872

Answer: C
Step by step
1. Weight of the observation i days old: $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$.
2. $w(13) = 0.8^{12} \times 0.2/(1 - 0.8^{300})$.
3. $0.8^{12} = 0.068719$, so the numerator is 0.013744; $0.8^{300} \approx 0$, so $w(13) = 0.01374$.

Why C is correct: Weights decay geometrically by λ per day and are scaled by $(1-\lambda)/(1-\lambda^n)$ so they sum to one; day 13 has exponent 12.
Why the others are wrong
- **A:** Used the equal weight $1/300 = 0.00333$, i.e. no decay (ordinary HS).
- **B:** Used exponent $i = 13$ instead of $i - 1 = 12$: $0.8^{13} \times 0.2 = 0.01100$.
- **D:** Dropped the $(1-\lambda)$ factor: $0.8^{12} = 0.0687$.

Q105
AnalystPrep Q.3012

Simon is using the age-weighted historical simulation approach to estimate the VaR of a stock portfolio. Under age-weighted historical simulation,

A. more recent observations are weighted more and distant observations weighted less.
B. all observations are weighted equally.
C. the decay parameter takes a value of 1.
D. the historical window of observations must not exceed 250 days.

Answer: A
Step by step
1. Weights follow $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$ with $0 \le \lambda \le 1$, so weight falls geometrically with age.
2. Hence recent observations get more weight and distant ones less.

Why A is correct: The decay factor makes weights decline with age, reducing ghost effects of old events.
Why the others are wrong
- **B:** Equal weighting describes plain historical simulation, which is the special case $\lambda = 1$.
- **C:** $\lambda = 1$ means no decay, which collapses to equal-weighted HS; the approach uses $\lambda < 1$.

- **D:** There is no 250-day limit; the window length n is a free choice.

Q106

AnalystPrep Q.3035

REBUILT — Original distractors 0.05, 0.0025 and 0.0065 were arbitrary or near-duplicates of the correct value; replaced with options from specific formula errors.

Paul uses age-weighted historical simulation with a window of 100 days and decay factor $\lambda = 0.96$. Over the recent past, the portfolio has registered the following returns:

Return	Periods Ago (Days)
-3.2%	109
-3.3%	75
-2.3%	66
-1.3%	22
-3.0%	45

Determine the weight on the return earned 45 days ago.

A. 0.010000
B. 0.006372
C. 0.006637
D. 0.006751

Answer: D

Step by step

1. $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$ with $i = 45$, $n = 100$.
2. Numerator: $0.96^{44} \times 0.04 = 0.006637$.
3. Denominator: $1 - 0.96^{100} = 1 - 0.01687 = 0.98313$.
4. $w(45) = 0.006637/0.98313 = 0.006751$.

Why D is correct: Dividing by $1-\lambda^n$ rescales the geometric weights over the 100-day window so they sum to one.

Why the others are wrong

- **A:** Used the equal weight 1/100 (ordinary HS), ignoring decay.
- **B:** Used exponent 45 instead of 44 and no denominator: $0.96^{45} \times 0.04 = 0.006372$.
- **C:** Used the correct exponent but forgot the $1/(1-\lambda^{100})$ normalization, giving 0.006637.

Q107

AnalystPrep Q.6432

Which statement correctly captures the essence of a correlation-weighted approach within the historical simulation framework?

A. It forces all pairwise asset correlations to zero by partitioning returns into uncorrelated blocks while leaving individual volatilities unchanged.
B. It imposes a desired set of updated correlations among assets by systematically adjusting historical returns, typically without altering each asset's overall volatility.
C. It re-weights observed co-movements to ensure the sample correlation matrix matches a diagonal matrix (i.e., only variances remain, covariances are nullified).
D. It replicates only those return paths whose correlations exceed a specified threshold, discarding paths that deviate from the target correlation profile.

Answer: B

Step by step

1. Correlation-weighted HS replaces the historical correlation (variance-covariance) matrix with an updated one.
2. Historical returns are transformed (loosely, multiplied by the revised matrix) to give correlation-adjusted returns; VaR/ES are then computed as usual.

Why B is correct: The returns themselves are adjusted so the data reflect the desired current correlation structure, rather than filtering or zeroing co-movements.

Why the others are wrong

- **A:** Zeroing correlations is not the aim; the goal is to impose updated, realistic correlations.
- **C:** Forcing a diagonal matrix would eliminate dependence, the opposite of updating it.
- **D:** This describes selecting paths (filtering), not systematically adjusting returns; observations are not discarded.

Q108 AnalystPrep Q.6433

Which of the following most closely characterizes the role of filtered historical simulation in estimating risk?

A. It fits a time-varying volatility model (e.g., GARCH) and then adjusts historical returns before resampling, reflecting evolving market conditions.

B. It reorders past returns chronologically to match any short-term volatility changes but leaves correlation unaltered.

C. It discards all returns that fall outside a user-defined confidence interval, ensuring that the final distribution is free of extreme tail events.

D. It transforms each asset's average return into a standardized zero mean but leaves the standard deviation fixed at its historical level.

Answer: A

Step by step

1. FHS fits a conditional volatility model (GARCH/asymmetric GARCH) to the history.
2. Returns are standardized by the model's volatility, the standardized residuals are bootstrapped, and the draws are rescaled by current volatility.
3. VaR is read from the simulated distribution.

Why A is correct: FHS combines historical simulation and bootstrapping with a conditional volatility model, so risk responds to current market conditions.

Why the others are wrong

- **B:** Reordering returns is not part of FHS; it resamples standardized returns and rescales them to current volatility.
- **C:** Discarding tail returns would remove exactly the information VaR is meant to capture; FHS can even predict losses beyond the historical range.
- **D:** Demeaning with a fixed standard deviation does not remove time-varying volatility, which is the whole point of FHS.

Q109 AnalystPrep Q.6435

REBUILT — Original distractors (zero-sum weighting, ignoring non-positive returns) were implausible giveaways; replaced with the other HS refinements as distractors.

An analyst implements filtered historical simulation on commodity returns. She fits a GARCH model, divides each day's return by that day's GARCH volatility, resamples the standardized returns, and rescales the draws by today's volatility forecast. Which aspect best defines "filtered" in this context?

A. Exponentially down-weighting older observations

B. Replacing the historical correlation matrix with a current one

C. Removing time-varying volatility using a conditional volatility model

D. Excluding returns beyond the chosen confidence level

Answer: C

Step by step

1. The GARCH step estimates conditional volatility for each past day.
2. Dividing returns by it removes heteroskedasticity, leaving roughly i.i.d. standardized residuals (the "filtering").
3. These are bootstrapped and multiplied by the current volatility forecast to give returns consistent with today's market.

Why C is correct: "Filtered" refers to stripping out conditional (time-varying) volatility via a model such as GARCH before resampling.

Why the others are wrong

- **A:** This describes age-weighted (hybrid) HS, which weights by recency rather than modeling volatility.
- **B:** This describes correlation-weighted HS, which adjusts the correlation structure; the question describes a volatility filter.
- **D:** No returns are excluded; discarding the tail would defeat the purpose of VaR estimation.

Q110 BT answers L2.T5.78.1

If volatility(0) is the current (today's) volatility estimate and volatility(t) is the volatility estimate on a previous day (t), which best describes volatility-weighted historical simulation?

A. First, conduct typical historical simulation (HS) on return series. Then multiply VaR by volatility(0)/volatility(t)

B. First, conduct typical historical simulation (HS) on return series. Then multiply VaR by volatility(t)/volatility(0)

C. Each historical return(t) is replaced by: return(t)*volatility(0)/volatility(t). Then conduct typical historical simulation (HS) on adjusted return series

D. Each historical return(t) is replaced by: return(t)*volatility(t)/volatility(0). Then conduct typical historical simulation (HS) on adjusted return series

Answer: C

Step by step

1. Each past return is rescaled: $r_t^* = r_t \times \sigma_0/\sigma_t$ (current over then-prevailing volatility).
2. Returns rise when current volatility exceeds past volatility and fall when it is lower.
3. Standard equal-weighted HS VaR/ES is then run on the adjusted series. Example: $-2.0\%\times 20\%/10\% = -4.0\%$.

Why C is correct: Adjustment is applied to each return before the HS quantile is taken, using the ratio of current to historical volatility.

Why the others are wrong

- **A:** Adjusting the final VaR by a single ratio ignores that the volatility ratio differs for each observation.
- **B:** Same error as A, and the ratio is also inverted.
- **D:** Correct timing but inverted ratio: it scales returns up when current volatility is lower.

Q111 BT answers L2.T5.79.1

REBUILT — Original answer was right but distractors (8%, 18%) were arbitrary; replaced with options from realistic mistakes and stated the Cholesky convention.

Two historical returns form a 2x1 vector with r1 = 10.0% and r2 = 12.0%. The historical correlation between the two series was 0.0, but the current correlation has risen to 0.40. Using Dowd's correlation-weighted historical simulation (Cholesky decomposition of the correlation matrices, with unit-scale returns as in Dowd's example), what are the correlation-adjusted returns?

A. r1 = 10.0% and r2 = 11.0%

B. r1 = 10.0% and r2 = 12.0%

C. r1 = 10.0% and r2 = 15.0%

D. r1 = 10.0% and r2 = 16.0%

Answer: C

Step by step

1. Historical correlation 0 means the historical Cholesky factor is the identity, so adjusted returns = current factor × historical returns.
2. Current factor for ρ = 0.40: $L = \begin{pmatrix} 1 & 0 \\ 0.40 & \sqrt{1-0.40^2} \end{pmatrix}$, with $\sqrt{0.84} = 0.9165$.
3. $r_1^{adj} = 1 \times 10\% = 10.0\%$.
4. $r_2^{adj} = 0.40 \times 10\% + 0.9165 \times 12\% = 4.00\% + 11.00\% = 15.0\%$ (exactly 14.998%).

Why C is correct: Correlation weighting re-mixes the historical returns so the pair carries the current correlation; with higher correlation, r2 absorbs part of r1, giving 15.0%.

Why the others are wrong

- **A:** Used only $\sqrt{1-\rho^2}\, r_2 = 11.0\%$ and omitted the $\rho r_1 = 4\%$ term.
- **B:** Left returns unchanged, ignoring that the correlation changed from 0 to 0.40.
- **D:** Added $\rho r_1 = 4\%$ to the full $r_2 = 12\%$ without scaling by $\sqrt{1-\rho^2}$.

Q112 BT answers L2.T5.79.2

In regard to Dowd's correlation-weighted historical simulation, EACH of the following is TRUE EXCEPT:

A. The covariance/correlation matrix "add-on" to the non-parametric historical simulation makes this a semi-parametric approach

B. It can be combined with volatility-weighted historical simulation

C. The weighting takes into account information about both volatilities and correlations
D. Because actual historical returns are used, the "curse of dimensionality" created by the need for many pairwise correlations is avoided

Answer: D
Step by step
1. A: a parametric covariance/correlation update is added to non-parametric HS, so it is semi-parametric: true.
2. B: the updated variances can be combined with volatility weighting: true.
3. C: the updated matrix holds variances (diagonal) and covariances/correlations (off-diagonal), so both are used: true.
4. D: the approach needs historical and current correlation matrices for all asset pairs, so dimensionality is not avoided: false.

Why D is correct: Correlation weighting requires estimating the full correlation matrix, so it still faces the curse of dimensionality; only D is false.
Why the others are wrong
- **A:** True: the parametric correlation adjustment on top of HS makes it semi-parametric.
- **B:** True: it extends volatility-weighted HS and can be used with it.
- **C:** True: the updated matrix carries both variances and correlations, making it richer than volatility weighting.

Q113 BT answers L2.T5.80.1 · BT notes P2.T5.80.1

Filtered historical simulation includes EACH of the following steps EXCEPT:
A. Specify a conditional volatility model, e.g., GARCH(1,1)
B. Convert actual historical returns into volatility-standardized returns
C. Bootstrap (simulate) from the series of standardized returns
D. Fit a parametric VaR model to the standardized returns

Answer: D
Step by step
1. Step 1: fit a conditional volatility model (GARCH) - option A.
2. Step 2: standardize each historical return by its conditional volatility - option B.
3. Step 3: bootstrap standardized returns and rescale by the current volatility forecast - option C.
4. Step 4: read VaR/ES from the simulated distribution; no parametric VaR model is fitted, so D is not a step.

Why D is correct: FHS reads VaR off a bootstrapped empirical distribution; it never fits a parametric VaR distribution to the standardized returns.
Why the others are wrong
- **A:** This is the first step of FHS.
- **B:** Standardizing returns by conditional volatility is step two.
- **C:** Bootstrapping is the simulation step.

Q114 BT answers L2.T5.80.2 · BT notes P2.T5.80.2

Which of the following most nearly describes the filtered historical simulation approach?
A. Pure parametric
B. Pure simulation-based
C. Empirical
D. Semi-parametric

Answer: D
Step by step
1. FHS uses a parametric conditional volatility model (e.g., GARCH) to filter returns.
2. The standardized residuals are then resampled non-parametrically by bootstrap.
3. A parametric filter plus non-parametric resampling is semi-parametric.

Why D is correct: Combining a parametric volatility model with non-parametric bootstrapping makes FHS semi-parametric.
Why the others are wrong
- **A:** Ignores the bootstrap of empirical standardized returns; no parametric return distribution is assumed.

- **B:** Ignores the GARCH filtering step, which is parametric.
- **C:** Ignores that returns are first transformed by a parametric volatility model, so it is not purely empirical.

Q115 BT answers L2.T5.80.3 · BT notes P2.T5.80.3

Dowd cites EACH of the following as ADVANTAGES ("attractions") of the filtered historical simulation approach EXCEPT:

A. It is fast

B. It incorporates correlations without needing a variance-covariance matrix or a conditional distribution of asset returns

C. Unlike volatility-weighted HS, which can produce VaR/ES estimates above the maximum historical loss, it will not produce VaR/ES in excess of the historical maximum

D. It combines the attractions of non-parametric HS with the ability to account for changing volatility conditions

Answer: C

Step by step

1. A, B and D are listed attractions (speed, no covariance matrix needed, HS plus time-varying volatility).
2. For C: both volatility-weighted HS and FHS can produce estimates beyond the largest historical loss, and this is viewed as an advantage (can predict losses outside the historical range).
3. C claims FHS cannot do this, so it is false.

Why C is correct: FHS can predict losses outside the historical range; the statement in C denies that advantage.

Why the others are wrong

- **A:** Genuine attraction: computationally reasonable even for large portfolios.
- **B:** Genuine attraction: no covariance matrix required.
- **D:** Genuine attraction: this is the essence of FHS.

Q116 BT answers P2.T5.22.4.1 · BT notes P2.T5.22.4.1

A portfolio's 250 daily losses over the last year are sorted; the worst five (USD millions) are 9.60, 7.40, 5.50, 4.00 and 3.00. The basic historical simulation 99.0% one-day VaR is 5.50 million. The three worst losses were recent: 9.60 occurred 7 days ago, 7.40 occurred 5 days ago and 5.50 occurred 3 days ago. Using age-weighted (hybrid) HS with decay λ = 0.850, weight on an observation i days old $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$, and no interpolation, what is the 99.0% VaR?

A. 4.00 million

B. 5.50 million

C. 7.40 million

D. 9.60 million

Answer: D

Step by step

1. With n = 250, $1-\lambda^{250} \approx 1$.
2. Worst loss (7 days old): $w(7) = 0.85^6 \times 0.15 = 5.66\%$.
3. The tail probability for 99% VaR is 1%, and the worst loss alone has weight 5.66% > 1%.
4. Hence VaR is the worst loss, 9.60 million (for reference: 7.40 adds 7.83% to reach 13.49%; 5.50 adds 10.84% to reach 24.32%).

Why D is correct: Recent observations carry heavy weights, so the single worst loss already exceeds the 1% tail mass and defines the VaR.

Why the others are wrong

- **A:** Went further down the ranking than the weights justify, ignoring that the worst loss alone covers the 1% tail.
- **B:** Basic (equal-weight) HS answer: 1% of 250 = 2.5 observations, about the 3rd worst; the decay weights were ignored.
- **C:** Skipped the worst loss as if its weight were below 1%; in fact w(7) = 5.66% already exceeds the tail.

Q117 BT answers P2.T5.22.4.2 · BT notes P2.T5.22.4.2

Peter manages a large portfolio and wants to generate a one-day value at risk (VaR), but he has the following preferences with respect to the approach:
- He would like the approach to incorporate the cross-sectional correlation structure on a given day
- He would like to be able to incorporate autocorrelations in a long-term forecast that will extend over at least 20 trading days
- If the current regime suggests, he would like to be able to produce VaR estimates that exceed the greatest loss in the historical sample

Given these preferences, which of the following approaches is BEST?

A. Basic HS
B. Bootstrap HS
C. Filtered HS
D. Volatility-weighted HS

Answer: C

Step by step

1. Requirement 3: VaR above the worst historical loss requires rescaling returns by current-versus-historical volatility (or simulating from a fitted model). Basic HS and bootstrap HS only reuse observed losses, so they fail.
2. Requirement 1: cross-sectional correlation on a given day is kept by resampling/rescaling all assets' returns for the same date together; this is satisfied by basic, bootstrap, volatility-weighted and filtered HS.
3. Requirement 2: multi-day forecasts with autocorrelation/volatility dynamics need a time-series model (GARCH) to simulate the volatility path forward over 20+ days. Only filtered HS combines a GARCH filter with bootstrapped standardized residuals.
4. Only filtered HS meets all three, so the answer is C.

Why C is correct: Filtered HS fits a GARCH-type model, bootstraps the standardized same-date residuals (keeping cross-sectional correlation), and re-rescales them with simulated future volatility, so it handles multi-day horizons and can exceed the historical maximum loss.

Why the others are wrong

- **A:** Basic HS has no volatility adjustment, so it cannot produce a VaR beyond the worst observed loss and has no model for multi-day dynamics; it only satisfies the correlation requirement.
- **B:** Bootstrap HS resamples observed returns with replacement, so it cannot exceed the worst historical loss and treats draws as independent, ignoring autocorrelation/volatility clustering.
- **D:** Volatility-weighted HS can exceed the historical maximum and keeps same-day correlation, but it only rescales by current volatility and has no volatility model to project dynamics over a 20+ day horizon; choosing it ignores the second requirement.

Q118 GARP PE2 Q15 · Mock 4 Q16

An analyst at a commercial bank is evaluating how the bank applies historical simulation (HS) to estimate VaR and ES. The analyst focuses on the approaches used for weighting past return observations, including the age-weighted, volatility-weighted, correlation-weighted, and filtered HS approaches. Which of the following statements is correct regarding the given weighting approach?

A. The age-weighted approach typically specifies that observations that occurred after the cutoff date are given an equal weight while observations that occurred before the cutoff date are given a weight of zero.
B. The volatility-weighted approach adjusts historical returns in the sample by increasing them if the historical volatility forecast was higher than the current volatility forecast, and decreasing them if the historical volatility forecast was lower than the current volatility forecast.
C. The correlation-weighted approach uses matrix multiplication to adjust historical portfolio returns so that these returns reflect current volatilities and current correlations.
D. The filtered HS approach allows the historical returns data to be trimmed or customized based on day of the week, magnitude of return, or any other characteristic of the data.

Answer: C

Step by step

1. Age-weighting assigns declining weights $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$ to older observations, so a hard cutoff with equal weights inside and zero outside is plain (equal-weighted) HS, not age-weighting. A is out.

2. Volatility-weighting scales each historical return by σ_T/σ_t (current over then-current volatility): it is scaled UP when current volatility exceeds the historical forecast and DOWN when it is lower. B states the reverse.
3. Correlation-weighting 'multiplies' the historical returns by a matrix built from the revised (current) variance-covariance matrix, so the adjusted returns reflect both current volatilities (diagonal) and current correlations (off-diagonal). C is correct.
4. Filtered HS combines HS with a conditional volatility model (GARCH), standardizes returns by forecast volatility and bootstraps them; it does not trim the data by day of week or return size. D is out.

Why C is correct: Correlation weighting generalizes volatility weighting by using matrix multiplication to update both variances and covariances/correlations of historical returns to current conditions.

Why the others are wrong

- **A:** Describes the traditional equal-weighted window with a cutoff, not age-weighting, which uses geometrically declining weights.
- **B:** Reverses the direction of the volatility adjustment; historical returns are scaled up when current volatility is higher than the historical forecast, not when historical volatility was higher.
- **D:** Confuses filtered HS with data screening; FHS filters returns through a GARCH-type volatility model and bootstraps standardized residuals, it does not trim observations.

Q119 Schweser Module Quiz 2.1 Q2

All of the following approaches improve the traditional historical simulation approach for estimating VaR except the:

A. volatility-weighted historical simulation.
B. age-weighted historical simulation.
C. market-weighted historical simulation.
D. correlation-weighted historical simulation.

Answer: C

Step by step

1. The Schweser reading lists four improvements to equal-weighted HS: age-weighted, volatility-weighted, correlation-weighted and filtered HS.
2. Volatility-weighted (A), age-weighted (B) and correlation-weighted (D) are all on that list.
3. 'Market-weighted' historical simulation is not one of the approaches, so C is the exception.

Why C is correct: There is no market-weighted HS approach among the recognized refinements (age, volatility, correlation, filtered).

Why the others are wrong

- **A:** A genuine improvement: rescales past returns to current volatility; picking it misreads the 'except' wording.
- **B:** A genuine improvement: gives recent observations more weight and reduces ghost effects; picking it misreads the 'except' wording.
- **D:** A genuine improvement: updates historical returns for current volatilities and correlations; picking it misreads the 'except' wording.

Q120 Schweser Module Quiz 2.1 Q3

Which of the following statements about age-weighting is most accurate?

A. The age-weighting procedure incorporates estimates from GARCH models.
B. If the decay factor in the model is close to 1, there is persistence within the dataset.
C. When using this approach, the weight assigned on day i is equal to $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^i)$.
D. The number of observations should at least exceed 250.

Answer: B

Step by step

1. Weights decay geometrically: $w(i) = \lambda^{i-1}(1-\lambda)/(1-\lambda^n)$, with n the number of observations in the window.
2. A decay factor λ near 1 means slow decay: old observations keep substantial weight (persistence); $\lambda = 1$ collapses to equal-weighted HS. B is correct.
3. Check of C: the denominator must use n, not i. E.g. $\lambda = 0.95, n = 10$: with $1-\lambda^n$ the weights sum to 1; with $1-\lambda^i$ they do not. C is wrong.
4. GARCH estimates enter filtered HS (and volatility weighting), not age-weighting; and no minimum of 250 observations is prescribed.

Why B is correct: With λ close to 1 the weights decline slowly, so past observations remain influential, which is persistence.

Why the others are wrong

- **A:** Confuses age-weighting with volatility-weighted or filtered HS, which use GARCH/EWMA volatility forecasts.
- **C:** Puts i instead of n in the denominator exponent; the normalizing constant must be $1-\lambda^n$ so the weights sum to one.
- **D:** Invents a minimum sample size; the reading prescribes none (a large sample may even contain unrepresentative old data).

Q121

Schweser Module Quiz 2.1 Q4

Which of the following statements about volatility-weighting is true?

A. Historic returns are adjusted, and the VaR calculation is more complicated.
B. Historic returns are adjusted, and the VaR calculation procedure is the same.
C. Current period returns are adjusted, and the VaR calculation is more complicated.
D. Current period returns are adjusted, and the VaR calculation is the same.

Answer: B

Step by step

1. Volatility weighting replaces each historical return by $r_t^* = r_t\,\sigma_T/\sigma_t$, where σ_t is the volatility forecast at the time of the return and σ_T is the current forecast.
2. So it is the historical returns that are adjusted, not current-period returns.
3. VaR/ES are then computed in the usual HS way on the adjusted returns (sort and take the quantile); only the input data change.

Why B is correct: Volatility weighting rescales historical returns to current volatility and feeds them into the unchanged HS procedure.

Why the others are wrong

- **A:** Right that historical returns are adjusted, but the VaR step is not more complicated; the standard HS quantile is applied to the adjusted data.
- **C:** Wrong on both counts: current returns are not the ones adjusted, and the VaR calculation is not more complex.
- **D:** Correct that the procedure is unchanged, but it is historical returns, not current-period returns, that are rescaled.

LO 2.d

Identify advantages and disadvantages of non-parametric estimation methods.

Q122

AnalystPrep Q.6465

REBUILT — Original options were vague (bias, narrow intervals and amplified noise were all arguable); rebuilt around the clear, reading-supported limitation that resamples cannot exceed the worst observed loss.

A risk manager uses bootstrap historical simulation on a small loss dataset in which the worst observed daily loss is 4.8%. Which of the following is the key limitation of the approach in this setting?

A. Resamples cannot contain a loss larger than 4.8%, so the method cannot reflect plausible losses worse than any observed.
B. Resamples require the returns to be normally distributed.
C. The bootstrap estimate is less precise than the raw historical simulation estimate.
D. Each resample is drawn without replacement, which shrinks the sample size.

Answer: A

Step by step

1. Bootstrapping draws only from observations in the dataset (with replacement).
2. Therefore no resample can contain a loss beyond the largest observed (4.8%), and events absent from the sample cannot appear.
3. This is a data limitation of historical methods; the bootstrap does improve precision, is non-parametric, and samples with replacement.

Why A is correct: Because resamples only recombine observed data, the bootstrap cannot generate unobserved extreme losses.

Why the others are wrong

- **B:** Bootstrap HS is non-parametric and makes no normality assumption.
- **C:** Empirically the bootstrap gives more precise risk-measure estimates than raw HS, not less.
- **D:** Resampling is with replacement, and each resample has the same size as the original.

Q123 BT answers P2.T5.710.2 · BT notes P2.T5.710.2

Which of the following is the chief drawback or limitation of the standard (i.e., unmodified) bootstrap procedure?

A. It employs a random number generator, which is too arbitrary.
B. It is difficult to compute standard errors of estimators.
C. It presupposes that observations are independent over time.
D. It cannot be implemented parametrically; it requires a non-parametric dataset.

Answer: C

Step by step

1. Standard bootstrap resamples observations at random with replacement, treating each observation as an independent draw from the same distribution (i.i.d.).
2. If returns show serial dependence or time-varying volatility (e.g., volatility clustering), resampling destroys that structure and ignores it.
3. Hence the key limitation is the independence assumption (it is violated by data with seasonality or volatility clustering).

Why C is correct: The standard bootstrap assumes i.i.d. observations, which is violated by serial dependence or volatility clustering in financial returns.

Why the others are wrong

- **A:** Randomness is the mechanism of resampling, not a flaw; repeating many draws averages it out.
- **B:** Reverse of the truth: the bootstrap makes standard errors and confidence intervals easy to obtain without closed-form formulas.
- **D:** False: a bootstrap can also be implemented parametrically (resampling from a fitted distribution).

Q124 BT answers L2.T5.81.1 · BT notes P2.T5.81.1

REBUILT — Option D (easy confidence intervals) is not a Schweser-listed advantage and is debatable; replaced it with the fat-tail/skewness advantage so only C is the non-advantage.

In regard to non-parametric estimation methods, EACH of the following is an ADVANTAGE EXCEPT:

A. Intuitive and easy to implement on a spreadsheet
B. Avoids the dimension problems and complex variance-covariance matrices of parametric methods
C. Good at allowing for plausible events that did not actually occur in the sample period
D. Not hindered by skewness or fat tails in the data

Answer: C

Step by step

1. Recall the advantages: intuitive and computationally simple (even on a spreadsheet); not hindered by skewness, fat tails and other parametric violations; avoids covariance matrices and dimension problems; data readily available; can combine weighting schemes.
2. A, B and D are on this list.
3. Non-parametric estimates come only from the historical sample, so they cannot accommodate plausible large-impact events that did not occur in the window. This is a listed disadvantage, so C is the exception.

Why C is correct: Because the estimate is driven entirely by observed data, events absent from the sample window have no influence; this is a disadvantage, not an advantage.

Why the others are wrong

- **A:** A true advantage; simple to build in a spreadsheet.
- **B:** A true advantage; no variance-covariance matrix is needed, so no curse of dimensionality.
- **D:** A true advantage; the empirical distribution carries skewness and fat tails automatically.

Q125

BT answers L2.T5.81.2 · BT notes P2.T5.81.2

REBUILT — Option B (ghosting/shadow effects) is outside the Schweser reading; replaced it with the Schweser disadvantage that new instruments or markets lack sufficient data.

In regard to non-parametric estimation methods, EACH of the following is a DISADVANTAGE EXCEPT:

A. Difficult to communicate to senior management
B. Cannot easily be applied to new instruments or markets that lack sufficient history
C. More constrained than parametric methods by the largest loss in the historical dataset
D. Difficulty detecting abrupt shifts (e.g., a permanent change in exchange rate risk) during the sample period

Answer: A

Step by step

1. Disadvantages: dependence on historical data; too-high estimates after volatile periods and too-low after quiet ones; difficulty detecting structural shifts; cannot handle events outside the sample; difficulty estimating losses beyond the maximum loss; need sufficient data.
2. B, C and D match these items.
3. Non-parametric methods are intuitive and simple, so communication is easy. A is an advantage, hence the exception.

Why A is correct: Intuitiveness is a core advantage of non-parametric methods, so 'difficult to communicate' is not a disadvantage.

Why the others are wrong

- **B:** A genuine disadvantage: enough data is needed.
- **C:** A genuine disadvantage: historical simulation cannot produce losses beyond the sample maximum.
- **D:** A genuine disadvantage: structural shifts are hard to detect in the sample.

Q126

BT answers P2.T5.22.4.3 · BT notes P2.T5.22.4.3

REBUILT — Original options were partly nonsensical (OHS with Richardson extrapolation, pseudo-random bootstrap) and the answer was hedged; rebuilt around the data-sufficiency / beyond-sample-maximum disadvantage.

A risk manager has only 63 daily P/L observations for a new portfolio, all from a low-volatility regime. The board wants VaR at 99.5%, 99.9% and 99.99% confidence. Other portfolios are reported with non-parametric methods because they are intuitive and simple. Which approach is most appropriate here?

A. Basic historical simulation, since it is intuitive and consistent with the other portfolios
B. Bootstrap historical simulation with many resamples to generate additional tail observations
C. Age-weighted historical simulation to emphasize the most recent observations
D. A parametric (distributional) approach, because the sample is too short and too calm to support non-parametric estimates at such confidence levels

Answer: D

Step by step

1. Each observation carries probability $1/63 = 1.59\%$.
2. Tail probabilities required: 0.5%, 0.1%, 0.01%, all far below 1.59%. Expected observations beyond them: $63 \times 0.005 = 0.32$, $63 \times 0.001 = 0.06$, $63 \times 0.0001 = 0.006$.
3. So these VaRs lie beyond the worst observed loss; non-parametric methods (including resampling and reweighting) cannot produce losses larger than the sample maximum, and a quiet regime understates risk.
4. A distributional assumption (possibly with tail modeling) is needed to extrapolate, so D.

Why D is correct: Non-parametric estimates need sufficient data and cannot extrapolate beyond the largest sample loss; with 63 calm observations a parametric model is required.

Why the others are wrong

- **A:** Plain HS cannot resolve probabilities below $1/63 = 1.59\%$ and reflects only the quiet regime.
- **B:** Bootstrapping resamples existing observations and never creates losses beyond the sample maximum.
- **C:** Age-weighting only reweights the same 63 observations; it does not extend the tail or fix the short, calm sample.

Q127 BT answers P2.T8.3.2 · BT notes P2.T9.3.2

A bank is proposing to add Position (A) to Portfolio (P). (A) is large relative to (P), and both contain heavy-tailed (non-normal) returns that defy distributional assumptions. The bank's risk manager needs a method that is computationally simple and easy to communicate. Which is the best method?

A. Parametric VaR with marginal VaR
B. Historical simulation
C. Monte Carlo simulation
D. Delta-normal VaR

Answer: B

Step by step

1. The position is large, so local (linear) valuation is inadequate: this rules out delta-normal and parametric approaches.
2. Returns are heavy-tailed, so a normal assumption is inappropriate; a non-parametric method avoids distributional assumptions.
3. The choice is between historical simulation and Monte Carlo. HS uses actual data with full revaluation and is intuitive, whereas Monte Carlo needs a specified process and heavy computation. HS wins.

Why B is correct: Historical simulation is non-parametric, handles fat tails and large positions via full revaluation, and is simple to explain.

Why the others are wrong

- **A:** Parametric methods assume normality and linear approximation, both inappropriate for a large position with heavy tails.
- **C:** Monte Carlo can handle non-normality but requires a distributional model and is computationally heavy and harder to explain.
- **D:** Delta-normal assumes normal returns and linear exposures, both violated here.

Q128 GARP PE1 Q53

A risk analyst at an investment bank is evaluating the historical simulation approach to estimating ES. Which of the following statements accurately describes an advantage of the historical simulation approach to estimating ES?

A. Since the data reflect actual historical events, an ES estimate derived from this data is more accurate than an estimate derived from using parametric methods.
B. The precision of ES estimates and their responsiveness to new market observations can typically be enhanced by increasing the length of the sample period.
C. Major shifts in markets, their structure, or the factors that influence them can be ignored since any effects of these shifts will be included in historical datasets.
D. Features such as fat tails and skewness are easily accommodated without making assumptions about the distribution.

Answer: D

Step by step

1. Historical simulation uses the empirical distribution, so no distributional form is imposed.
2. Hence fat tails and skewness present in the data pass directly into the ES estimate: D.
3. A, B and C each overstate: accuracy depends on the sample; a longer window trades responsiveness for precision; structural shifts enter the window only slowly.

Why D is correct: No parametric assumption is made, so empirical skewness and fat tails are captured automatically.

Why the others are wrong

- **A:** Real data does not guarantee accuracy; the estimate depends on whether the window was calm or turbulent.
- **B:** A longer window raises precision but lowers responsiveness; they trade off.
- **C:** Structural shifts cannot be ignored; they are absorbed only slowly and are hard to detect.

Q129 GARP Mock 2 Q69

A hedge fund manager has asked her risk analyst to incorporate nonparametric methods into their statistical modeling program. The analyst, used to parametric methods, is concerned about using nonparametric methods. Which of the following statements would accurately reflect the analyst's concerns?

A. Historical data is a relatively small input to the calculations.

B. Model outputs tend to exhibit positive skewness and fat tails.
C. These methods may be less reliable for newer markets or instruments.
D. Volatile data periods lead to value at risk (VaR) and expected shortfall (ES) estimates that are too low.

Answer: C
Step by step
1. Non-parametric estimation relies critically on historical data.
2. New markets or instruments lack sufficient history, so estimates are unreliable: C.
3. A reverses the dependence; B confuses outputs with imposed shape; D reverses the direction of the volatile-period bias.

Why C is correct: Reliance on a sufficient historical sample makes the methods weak where history is short.
Why the others are wrong
- **A:** History is the main input, not a small one.
- **B:** The methods impose no shape; any skew or fat tails come from the data and are not a concern.
- **D:** Volatile windows make VaR/ES too high; quiet windows make them too low.

Q130 Schweser Module Quiz 2.1 Q5

All of the following items are generally considered advantages of nonparametric estimation methods except:
A. ability to accommodate skewed data.
B. availability of data.
C. use of historical data.
D. little or no reliance on covariance matrices.

Answer: C
Step by step
1. Recall the advantages in the reading: intuitive and simple, not hindered by skewness or fat tails, no complex variance-covariance matrices, and readily available data that needs no adjustment.
2. Recall the disadvantages: results depend critically on historical data (quiet periods understate VaR/ES, volatile periods overstate it), structural shifts are hard to detect, and events not in the sample cannot be captured.
3. Option C, reliance on historical data, is therefore a disadvantage, so it is the exception.

Why C is correct: Nonparametric methods let the historical sample speak for itself, so their results are only as good as that sample; dependence on historical data is their central weakness, not a strength.
Why the others are wrong
- **A:** This is an advantage: no distributional assumption is imposed, so skewness and fat tails are captured. Choosing it confuses nonparametric with parametric (normal) methods.
- **B:** This is an advantage: historical data is readily available and needs no adjustments. Picking it overlooks that availability is a practical strength.
- **D:** This is an advantage: nonparametric methods avoid estimating variance-covariance matrices and the associated dimension problems. Picking it confuses them with delta-normal approaches.

Q131 BT answers P2.T8.3.2

A bank is proposing to add Position (A) to Portfolio (P). (A) is large relative to (P), and both contain heavy-tailed (i.e., non-normal risk factors) returns that defy distributional assumptions. The bank's Risk Manager needs a method that is computationally simple and easy to communicate. Which is the best method given the situation and the risk manager's preference?
A. Parametric VaR with marginal VaR
B. Historical Simulation
C. Monte Carlo Simulation
D. Delta-normal VaR

Answer: B
Step by step
1. Heavy-tailed, non-normal returns rule out methods that assume normality (delta-normal, parametric VaR).
2. A large position means a marginal (local, linear) approximation of the portfolio change is unreliable, so a full-revaluation approach is needed.

3. That leaves historical simulation and Monte Carlo as full-revaluation, distribution-free-capable methods.
4. Monte Carlo requires specifying a joint distribution and many simulated paths, so it is not computationally simple or easy to communicate.
5. Historical simulation uses actual past returns (captures fat tails and skew), is computationally simple (even on a spreadsheet) and intuitive to explain: choose B.

Why B is correct: Nonparametric historical simulation is not hindered by skewness or fat tails, avoids covariance matrices, is simple and intuitive, and fully revalues the portfolio with the new position.

Why the others are wrong

- **A:** Parametric VaR assumes a distribution (normal) and marginal VaR is a local linear approximation; both fail for heavy tails and a large position.
- **C:** Handles non-normality and large positions but needs a specified joint distribution, many simulations and is harder to explain, so it fails the 'simple and easy to communicate' requirement.
- **D:** Delta-normal assumes normal returns and linear exposures; it is simple but inappropriate for heavy tails and a large position.

Reading 3. Parametric Approaches (II): Extreme Value

Kevin Dowd, Measuring Market Risk, 2nd Edition (West Sussex, UK: John Wiley & Sons, 2005). Chapter 7.

LO	Learning objective	Qs
3.a	explain the importance and challenges of extreme values in risk management.	5
3.b	describe extreme value theory (EVT) and its use in risk management.	17
3.c	describe the peaks-over-threshold (POT) approach.	8
3.d	compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks.	12
3.e	discuss the application of the generalized Pareto (GP) distribution in the POT approach.	13
3.f	explain the multivariate EVT for risk management.	2

LO 3.a

Explain the importance and challenges of extreme values in risk management.

Q132 AnalystPrep Q.2176

Modeling extreme events in finance and risk management is difficult. What is the primary reason why modeling extreme events is typically problematic?

A. A lack of models that can estimate the effects of certain extreme but possible events.
B. A lack of qualified personnel to oversee the modeling process.
C. A lack of credible, reliable input data.
D. The fact that extreme event modeling requires a considerable investment of time and expertise.

Answer: C

Step by step

1. Extreme events are rare by definition, so the sample contains very few tail observations, and some possible extremes have never occurred.
2. Without data, the analyst must assume a distribution that is fitted mainly to the central observations.
3. The assumed distribution will not match the true tail, so tail estimates are unreliable: the core challenge is data scarcity.

Why C is correct: The fundamental challenge is too few (or no) tail observations to build a credible, reliable model.

Why the others are wrong

- **A:** Models for extremes do exist (e.g., EVT); the problem is the data to calibrate them, not the absence of models.
- **B:** Staffing is not the fundamental statistical obstacle discussed in the reading.
- **D:** Time and expertise are practical costs, not the primary reason modeling is unreliable.

Q133 AnalystPrep Q.3994

An extreme value is one with a low probability of occurrence but potentially catastrophic effects. The main challenge posed by extreme values is that:

A. They do not conform to any of the established loss distributions.
B. They can only be fully characterized by multiple loss distributions.
C. They are so large that the resulting loss estimates are infinitely large.
D. There are only a few observations from which a credible, reliable analytical model can be built.

Answer: D

Step by step

1. Extreme values occur rarely, so few observations fall in the tail.
2. With so little data, a reliable model of the tail is hard to build, and some extremes have never been observed.
3. The other options assert properties of extremes that the reading does not state.

Why D is correct: The challenge is data scarcity in the tail, which makes any fitted model subject to considerable error.

Why the others are wrong

- **A:** Extremes can be described by distributions (e.g., generalized Pareto); the issue is sparse data, not nonconformity.

- **B:** Nothing implies multiple distributions are required; this is invented.
- **C:** Loss estimates from a fitted tail model are finite; infinite estimates are not the challenge.

Q134 BT answers L2.T5.86.1 · BT notes P2.T5.805.3

In regard to extreme losses and extreme value theory (EVT), Dowd says EACH of the following is TRUE EXCEPT:

A. The central limit theorem (CLT) does not apply.
B. EVT estimates are subject to considerable model risk.
C. EVT cannot, by definition, provide loss estimates beyond (in excess of) the worst actual observed loss in the historical sample.
D. EVT is inherently difficult because our sample contains few relevant loss observations.

Answer: C

Step by step

1. A: CLT concerns averages; extremes are governed by different limit laws, so CLT does not apply (true).
2. B: Fitting a tail from few data points gives high model and parameter risk (true).
3. D: Tail data are scarce (true).
4. C: EVT is designed precisely to extrapolate beyond the worst observed loss, so the statement is false and is the EXCEPT answer.

Why C is correct: EVT fits a parametric tail model and extrapolates beyond the sample maximum; it is not capped at the worst observed loss.

Why the others are wrong

- **A:** True statement, so not the exception: extremes are not described by the CLT's normal limit.
- **B:** True statement: tail estimates are highly sensitive to the model and parameters.
- **D:** True statement: few relevant observations is the core difficulty.

Q135 BT answers L2.T5.86.3 · BT notes P2.T5.805.3

REBUILT — Option A was OCR/printed as 'confidence intervals for our estimated stress tests'; corrected to 'estimated risk measures' and keyed D (merged two copies, one with no key).

In regard to EVT estimates, which of the following "solutions" does Dowd recommend for coping with the limitations of data paucity in the tail?

A. Estimate confidence intervals for our estimated risk measures.
B. Subject our extreme risk measures to stress tests.
C. Acknowledge that a lack of data inevitably limits the reliability of our estimates (the humility principle).
D. All of the above.

Answer: D

Step by step

1. Scarce tail data means EVT estimates are highly uncertain.
2. Dowd's responses: (i) quantify the uncertainty with confidence intervals around the estimates; (ii) stress test the extreme risk measures; (iii) accept humbly that data limits reliability.
3. All three are recommended, so D.

Why D is correct: Dowd offers all three coping mechanisms: confidence intervals, stress testing, and humility about what the data can support.

Why the others are wrong

- **A:** Valid but incomplete; stress testing and humility are also recommended.
- **B:** Valid but incomplete; confidence intervals and humility are also recommended.
- **C:** Valid but incomplete; the other two measures are also recommended.

Q136 AnalystPrep Q.2175

Extreme events have a very low probability of occurrence but are nonetheless taken very seriously in finance. Which of the following best explains why?

A. Extreme events tend to recur at rather regular time intervals.
B. Extreme events rarely have warning signs and thus markets cannot prepare for them in any way.
C. Extreme events are normally very costly and can create a ripple effect on the global market.

D. No models have been developed to accurately predict and estimate the effects of extreme events in qualitative terms.

Answer: C

Step by step

1. Importance is about impact: probability is small but the loss given occurrence is huge (market crashes, failures of major institutions, crises).
2. Such losses can cascade through the system, so identifying them is crucial to risk management.
3. Only C states this.

Why C is correct: Extreme events are rare but very costly and can spread through the market, so they must be identified and managed.

Why the others are wrong

- **A:** Extremes are rare and irregular; regular recurrence is not claimed.
- **B:** Overstated: markets can prepare using stress tests and EVT, and the reading does not say preparation is impossible.
- **D:** Models such as EVT exist; the issue is data and reliability, not the absence of all models.

LO 3.b

Describe extreme value theory (EVT) and its use in risk management.

Q137

AnalystPrep Q.4000

REBUILT — Original distractors B and D were implausible numbers; replaced with results of realistic formula mistakes.

For the standardized Gumbel distribution ($\mu = 0, \sigma = 1$), the quantile function is $x_p = -\ln[-\ln(p)]$. Which of the following gives the 5%, 95% and 99% quantiles, respectively?

A. -1.0972; 2.9702; 4.6001

B. 1.0972; -2.9702; -4.6001

C. -2.9702; 1.0972; 1.5272

D. 0.0513; 2.9957; 4.6052

Answer: A

Step by step

1. Quantile: $x_p = -\ln[-\ln(p)]$.
2. p = 0.05: $-\ln(0.05) = 2.9957$; $\ln(2.9957) = 1.0972$; so $x = -1.0972$.
3. p = 0.95: $-\ln(0.95) = 0.05129$; $\ln(0.05129) = -2.9702$; so $x = 2.9702$.
4. p = 0.99: $-\ln(0.99) = 0.010050$; $\ln(0.010050) = -4.6001$; so $x = 4.6001$.

Why A is correct: The Gumbel (GEV with ξ = 0) CDF is $\exp[-\exp(-x)]$; inverting it gives $x = -\ln[-\ln p]$, which increases with p.

Why the others are wrong

- **B:** Drops the leading minus sign (computes $\ln[-\ln p]$), so every quantile has the wrong sign.
- **C:** Uses the Gumbel-for-minima form $\ln[-\ln(1-p)]$, which reverses the tails (the 5% value becomes -2.9702).
- **D:** Uses the exponential quantile $-\ln(1-p)$ instead of the double-log Gumbel quantile.

Q138

AnalystPrep Q.4004

REBUILT — Replaced arbitrary distractors with results of realistic formula mistakes.

A risk analyst computes VaR for the US stock market under the GEV distribution using the Frechet form $\text{VaR}_\alpha = \mu - \frac{\sigma}{\xi}\left[1 - (-n \ln \alpha)^{-\xi}\right]$ with location μ = 4.0%, scale σ = 0.80, tail index ξ = 0.5 and sample size n = 100. Which is nearest to the implied 99.90% VaR?

A. 0.5416%

B. 7.4584%

C. 2.9061%

D. 5.7292%

Answer: B

Step by step

1. ξ > 0, so use the Frechet form; $\sigma/\xi = 0.8/0.5 = 1.6$.
2. $-100\ln(0.999) = 0.10005$.
3. $0.10005^{-0.5} = 3.1615$.
4. VaR = 4 - 1.6(1 - 3.1615) = 4 + 3.4584 = 7.4584%.

Why B is correct: The n-scaled tail-probability term raised to -ξ is large, so VaR sits well above μ.

Why the others are wrong

- **A:** Flips the sign: 4 + 1.6(1 - 3.1615) = 0.5416%.
- **C:** Uses exponent +ξ ($0.10005^{0.5} = 0.3163$): 4 - 1.6(0.6837) = 2.9061%.
- **D:** Forgets to divide σ by ξ (uses 0.8): 4 - 0.8(1 - 3.1615) = 5.7292%.

Q139

BT Mock Q61

REBUILT — Options A and D were giveaways (A below the 99.5% VaR; D unrelated); replaced with results of realistic mistakes.

Following Dowd, if ξ > 0 the tail of the loss distribution follows a power law, and assuming k is approximately constant the tail probability simplifies to $F(x) \approx kx^{-\alpha}$. Patricia sets the exponent α = 4.0 and observes that the 99.50% VaR is USD 6.0 million. If the loss tail follows the power law, which is nearest to the implied 99.90% VaR?

A. USD 4.0 million
B. USD 9.0 million
C. USD 13.4 million
D. USD 30.0 million

Answer: B

Step by step

1. Tail probabilities: 0.50% at the 99.5% VaR and 0.10% at the 99.9% VaR.
2. From $p = kx^{-\alpha}$: $x_2/x_1 = (p_1/p_2)^{1/\alpha}$.
3. $p_1/p_2 = 0.005/0.001 = 5$, so $x_2/x_1 = 5^{1/4} = 1.4953$.
4. $\text{VaR}_{99.9\%} = 6.0 \times 1.4953 = 8.97 \approx$ USD 9.0 million.

Why B is correct: Under a power-law tail, VaR scales with the fourth root of the ratio of tail probabilities, giving a modest increase from 6.0 to about 9.0.

Why the others are wrong

- **A:** Inverts the ratio ($6 \times 0.2^{1/4} = 4.0$): VaR would fall as confidence rises, which is impossible.
- **C:** Uses α = 2 ($6 \times 5^{1/2} = 13.4$) instead of α = 4.
- **D:** Multiplies by the probability ratio 5 and ignores the 1/α root ($6 \times 5 = 30$).

Q140

AP Mock 2 Q30

Salim Abdalla, FRM, recommends the use of extreme value theory. While discussing EVT, the following statements are made:

I. To apply EVT, the underlying loss distribution can be any of the commonly used distributions, e.g. normal or lognormal, and will usually be unknown.

II. If the tail parameter of the generalized extreme value (GEV) distribution goes to zero, then the GEV essentially becomes a Gumbel distribution.

III. If the tail parameter of the GEV distribution is less than zero, the GEV becomes the normal distribution.

IV. As the threshold value is increased, the distribution of exceedances converges to a generalized Gumbel distribution.

Which of the statements above are incorrect?

A. I and III.
B. II and IV.
C. III and IV.
D. I and II.

Answer: C

Step by step

1. I: EVT works whatever the parent distribution (normal, lognormal, ...), which is usually unknown. Correct.
2. II: ξ = 0 gives the Gumbel distribution. Correct.
3. III: ξ < 0 gives the Weibull (tails lighter than normal), not the normal. Incorrect.

4. IV: By the Gnedenko-Pickands-Balkema-de Haan theorem, exceedances over a rising threshold converge to the generalized Pareto distribution, not a 'generalized Gumbel'. Incorrect.
5. Incorrect statements: III and IV.

Why C is correct: $\xi < 0$ corresponds to the Weibull, and threshold exceedances converge to the generalized Pareto, so III and IV are the false statements.

Why the others are wrong

- **A:** Treats I as false; EVT does not require a known parent distribution.
- **B:** Treats II as false; $\xi \to 0$ does give the Gumbel, and it misses that III is also false.
- **D:** Marks the two correct statements (I and II) as incorrect.

Q141 AP Mock 3 Q52

REBUILT — Distractors replaced with results of realistic formula mistakes (source options were arbitrary and B showed VaR falling with confidence); stem now states the formula.

For a US stock market index (returns in percent), the location parameter of the limiting distribution is μ = 2%, the scale parameter is σ = 0.6, and the tail index is ξ = 0.4. Using the Fréchet VaR formula $\text{VaR}_\alpha = \mu - \frac{\sigma}{\xi}\left[1 - (-n \ln \alpha)^{-\xi}\right]$ with n = 100, what are the estimated 95% VaR and 99.5% VaR, respectively?

A. 1.71%; 2.19%
B. 2.72%; 1.52%
C. 1.28%; 2.48%
D. 3.38%; 1.64%

Answer: C

Step by step

1. $\sigma/\xi = 0.6/0.4 = 1.5$.
2. 95%: $-100\ln(0.95) = 5.1293$; $5.1293^{-0.4} = 0.5200$; VaR = 2 - 1.5(1 - 0.5200) = 1.28%.
3. 99.5%: $-100\ln(0.995) = 0.50125$; $0.50125^{-0.4} = 1.3182$; VaR = 2 - 1.5(1 - 1.3182) = 2.477%.
4. VaR rises with the confidence level, as it must.

Why C is correct: Plugging α into the Fréchet quantile formula with n = 100 gives 1.28% and 2.48%, increasing with confidence.

Why the others are wrong

- **A:** Forgets to divide σ by ξ (uses σ = 0.6 as the multiplier): 1.71% and 2.19%.
- **B:** Flips the sign of the bracket term (adds instead of subtracts): 2.72% and 1.52%; VaR falls with confidence, a red flag.
- **D:** Uses exponent +ξ instead of -ξ: 3.38% and 1.64%; again VaR falls as confidence rises.

Q142 AnalystPrep Q.2177

In risk management, accurately capturing the impact of extreme events is crucial for assessing potential risks. Extreme events can best be modeled through the application of:

A. The central limit theorem
B. The standard normal distribution
C. Extreme-value theorems
D. The exponential distribution

Answer: C

Step by step

1. EVT is built to describe the distribution of maxima/minima (tail behavior) of a sample, e.g. via the Fisher-Tippett theorem.
2. The CLT, normal and exponential distributions describe central or thin-tailed behavior, not extremes.

Why C is correct: Extreme-value theorems (Fisher-Tippett/GEV, POT) specifically model the tail and extreme outcomes rather than central tendency.

Why the others are wrong

- **A:** The CLT governs the distribution of sample means (central tendency), not extremes.
- **B:** The normal has thin tails and underestimates extreme-event probabilities.
- **D:** The exponential models waiting times and is not a limiting law for sample extremes in general.

Q143 AnalystPrep Q.2179

The cumulative distribution function of the generalized extreme value (GEV) distribution is:

$H(x) = \exp\left[-\left(1+\xi\frac{x-\mu}{\sigma}\right)^{-1/\xi}\right]$ if $\xi \neq 0$, and $H(x) = \exp\left[-\exp\left(-\frac{x-\mu}{\sigma}\right)\right]$ if $\xi = 0$,

where $1+\xi(x-\mu)/\sigma > 0$. If $\xi > 0$, the GEV becomes the:

A. Frechet distribution
B. Pareto distribution
C. Gumbel distribution
D. Weibull distribution

Answer: A

Step by step

1. The sign of the tail index ξ determines the GEV type.
2. $\xi > 0$: Frechet (heavy tails, power-law decay, like the t and Pareto).
3. $\xi = 0$: Gumbel; $\xi < 0$: Weibull.

Why A is correct: A positive tail index gives the heavy-tailed Frechet distribution.

Why the others are wrong

- **B:** Pareto is a heavy-tailed parent distribution (in the Frechet domain of attraction), not a GEV case.
- **C:** Gumbel corresponds to $\xi = 0$.
- **D:** Weibull corresponds to $\xi < 0$.

Q144 AnalystPrep Q.2180

If $\xi < 0$, the GEV becomes the Weibull distribution, but this distribution is rarely used to model financial returns mainly because:

A. Its cumulative distribution has heavier than normal tails and very few empirical financial returns are heavy-tailed.
B. Its cumulative distribution has lighter than normal tails and very few empirical financial returns are light-tailed.
C. It is asymmetric.
D. It is symmetric.

Answer: B

Step by step

1. For $\xi < 0$ the GEV (Weibull) corresponds to parent distributions with tails lighter than the normal.
2. Empirical financial returns are typically heavy-tailed, so a light-tailed limiting law rarely fits.

Why B is correct: The Weibull case describes tails lighter than normal, whereas financial returns are almost always heavy-tailed.

Why the others are wrong

- **A:** Wrong tail direction: the Weibull has lighter, not heavier, tails (heavier tails are the Frechet case), and financial returns are mostly heavy-tailed.
- **C:** Asymmetry is not the reason; financial returns are often skewed too.
- **D:** The GEV family is not symmetric, and symmetry is not the stated reason.

Q145 AnalystPrep Q.2182

REBUILT — Original distractors were arbitrary (e.g. -0.9 as a giveaway); replaced with results of realistic mistakes. Stem now states the quantile formula.

For the standardized Frechet distribution ($\mu = 0, \sigma = 1$) with $\xi = 0.3$, the p-quantile is $x_p = -\frac{1}{\xi}\left[1-(-\ln p)^{-\xi}\right]$.

The 5% quantile is closest to:

A. 1.2993
B. -0.2805
C. 4.7924
D. -0.9349

Answer: D

Step by step

1. $-\ln(0.05) = 2.9957$.

2. $2.9957^{-0.3} = 0.7195$.
3. $x_{0.05} = -(1/0.3)(1 - 0.7195) = -3.3333 \times 0.2805 = -0.9349$.

Why D is correct: Substituting p = 0.05 and ξ = 0.3 into the Frechet quantile gives -0.9349.

Why the others are wrong

- **A:** Uses exponent +ξ ($2.9957^{0.3} = 1.3898$), giving $-3.3333(1 - 1.3898) = +1.2993$.
- **B:** Omits the 1/ξ multiplier: -(1 - 0.7195) = -0.2805.
- **C:** Uses p = 0.95 instead of 0.05 ($-\ln 0.95 = 0.05129$), giving +4.7924.

Q146

AnalystPrep Q.3993

REBUILT — Tail index was printed as 0.20% (units error); restated as unitless 0.20 and replaced arbitrary distractors with formula-mistake results.

A risk analyst computes VaR for the US stock market under the GEV distribution using the Frechet form $\text{VaR}_\alpha = \mu - \frac{\sigma}{\xi}\left[1 - (-n \ln \alpha)^{-\xi}\right]$ with location μ = 3.0%, scale σ = 0.80, tail index ξ = 0.20 and sample size n = 100. Which is nearest to the implied 99.90% VaR?

A. 0.6611%
B. 1.5241%
C. 3.4678%
D. 5.3389%

Answer: D

Step by step

1. $\sigma/\xi = 0.8/0.2 = 4$.
2. $-n\ln(0.999) = 100 \times 0.0010005 = 0.10005$.
3. $0.10005^{-0.2} = 1.5847$.
4. VaR = 3 - 4(1 - 1.5847) = 3 + 2.3389 = 5.3389%.

Why D is correct: Because the bracket is negative (the power term exceeds 1), VaR lies above the location parameter, at 5.34%.

Why the others are wrong

- **A:** Flips the sign (adds the correction instead of subtracting): 3 + 4(1 - 1.5847) = 0.6611%.
- **B:** Uses exponent +ξ: $0.10005^{0.2} = 0.6311$, giving 3 - 4(0.3689) = 1.5241%.
- **C:** Forgets to divide σ by ξ (uses 0.8): 3 - 0.8(1 - 1.5847) = 3.4678%.

Q147

AnalystPrep Q.4009 · Schweser Module Quiz 3.1 Q1

According to the Fisher-Tippett theorem, as the sample size n gets large, the distribution of (normalized) extremes converges to a:

A. normal distribution.
B. uniform distribution.
C. generalized Pareto distribution.
D. generalized extreme value distribution.

Answer: D

Step by step

1. Fisher-Tippett concerns the limit law of the maximum (or minimum) of n i.i.d. observations, suitably normalized.
2. The only non-degenerate limit is the GEV family, which nests Fréchet ($\xi > 0$), Gumbel ($\xi = 0$) and Weibull ($\xi < 0$).
3. So the answer is the generalized extreme value distribution.

Why D is correct: Just as the CLT gives the normal for sample means, Fisher-Tippett gives the GEV as the limit law for sample extremes, regardless of the parent distribution.

Why the others are wrong

- **A:** The normal is the CLT limit for averages, not for extremes; confuses the two limit theorems.
- **B:** The uniform is not a limit law for extremes; no theorem supports it.
- **C:** The GPD is the limit for exceedances over a high threshold (POT approach), not for block maxima.

Q148 BT answers L2.T5.86.2 · BT notes P2.T5.805.3

Which of the following is TRUE about extreme value theory (EVT), according to Dowd?

A. EVT can produce an estimate for expected shortfall (ES) but NOT value at risk (VaR).
B. EVT can produce an estimate for value at risk (VaR) but NOT expected shortfall (ES).
C. EVT makes use of all information in the entire distribution.
D. The application of EVT methods involves much subjectivity, and estimates are very uncertain and prone to small-sample effects, biases and non-linearities.

Answer: D

Step by step

1. EVT fits a distribution to the tail only, so both a tail quantile (VaR) and the conditional tail mean (ES) can be derived from it: A and B are false.
2. EVT deliberately ignores the body of the distribution and uses only extreme observations: C is false.
3. Because extremes are rare, parameter estimates (e.g., the tail index, threshold or block size choice) are subjective and uncertain, with small-sample bias: D is true.

Why D is correct: EVT is a powerful but data-scarce tool; its estimates depend on subjective choices (block size, threshold) and few extreme observations, so they are highly uncertain.

Why the others are wrong

- **A:** EVT gives VaR too (a tail quantile); the error is thinking only the average tail loss is obtainable.
- **B:** EVT also gives ES once the tail distribution is fitted; the error is thinking ES needs the whole distribution.
- **C:** EVT uses only tail observations and discards the body; the claim describes historical simulation or parametric methods.

Q149 BT answers P2.T5.22.6.2 · BT notes P2.T5.22.6.2

REBUILT — Source set tail index xi = 4.0 but solved with exponent xi (inverse of the stated formula; the stated formula would give about $3,750 million); rebuilt with xi = 0.25 so the exponent 1/xi = 4 gives a sensible answer, and replaced options.

For a heavy-tailed loss distribution ($\xi > 0$), the tail probability is approximated by the power law $F(x) = P(X > x) \approx k\,x^{-1/\xi}$, where k is constant. An analyst estimates a tail index $\xi = 0.25$ and finds the 99.5% VaR is $6.0 million. Which is nearest to the implied 99.9% VaR under the power law?

A. $4.0 million
B. $9.0 million
C. $30.0 million
D. $3,750 million

Answer: B

Key correction: Source key (B, $9.0) is only consistent with an exponent of 4 on x; with the stated xi = 4.0 and F(x) = k x^(-1/xi) the result would be $3,750 million, so the item was rebuilt with xi = 0.25.

Step by step

1. Exponent: $1/\xi = 1/0.25 = 4$, so $F(x) = kx^{-4}$.
2. Same k at both quantiles: $0.005 \times 6.0^4 = 0.001 \times x^4$.
3. $x^4 = 6.0^4 \times 5 = 6480$, so $x = 6.0 \times 5^{1/4} = 6.0 \times 1.4953 = 8.97$.
4. Nearest option: $9.0 million.

Why B is correct: Scaling VaR by the tail-probability ratio raised to ξ: $\text{VaR}_{99.9\%} = 6.0 \times (0.005/0.001)^{\xi} = 6.0 \times 5^{0.25} \approx 8.97$.

Why the others are wrong

- **A:** Inverted the power: $6.0 \times 5^{-0.25} = 4.0$; a higher confidence level must give a higher VaR.
- **C:** Scaled VaR linearly with the probability ratio, $6.0 \times 5 = 30$, ignoring the power-law exponent.
- **D:** Used $5^{1/\xi} = 5^4$ instead of 5^{ξ}, i.e., applied the exponent $1/\xi$ to the scaling instead of ξ.

Q150 BT answers P2.T5.22.6.3 · BT notes P2.T5.22.6.3

REBUILT — Option D (8.3%) was not produced by its stated mistake (a +xi exponent gives 0.65%) or by any natural error; replaced D with 10.0%, the result of keeping mu = 3.00% from the worked example.

The VaR under the block maxima (GEV) approach at confidence level α is $\mu - \frac{\sigma}{\xi}\left[1 - (-n \ln \alpha)^{-\xi}\right]$. As an example, with n = 120, $\mu = 3.00\%$, $\sigma = 1.33\%$, $\xi = 0.80$, the 99.90% VaR is 10.40%. Now the block parameters are n = 160, $\mu = 2.00\%$, $\sigma = 1.50\%$ and $\xi = 0.90$. Which is nearest to the 99.90% VaR?

A. 11.6%
B. 9.0%
C. 1.4%
D. 10.0%

Answer: B
Step by step
1. $-n \ln \alpha = -160 \times \ln(0.999) = 160 \times 0.0010005 = 0.16008$.
2. $(0.16008)^{-0.9} = 5.2011$.
3. $1 - 5.2011 = -4.2011$.
4. $\sigma/\xi = 1.50/0.90 = 1.6667$, so VaR $= 2.00 - 1.6667 \times (-4.2011) = 2.00 + 7.0019 = 9.00\%$.

Why B is correct: Substituting all four new parameters (n = 160, $\mu = 2.00\%$, $\sigma = 1.50\%$, $\xi = 0.90$) into the GEV VaR formula gives 9.002%, about 9.0%.

Why the others are wrong
- **A:** Used n = 120 from the worked example instead of n = 160 (gives 11.56%).
- **C:** Used 99% instead of 99.90% as the confidence level (gives 1.42%).
- **D:** Kept the worked-example location $\mu = 3.00\%$ instead of 2.00%: $3.00 + 7.0019 = 10.00\%$.

Q151

BT answers P2.T7.804.2 · BT notes P2.T5.804.2

REBUILT — Original issue II (clustering vs i.i.d.) is debatable and option C was a giveaway with a muddled claim; dropped issue II and rewrote options so exactly one is defensible.

Sandra is fitting a distribution to the extreme loss tail of historical returns using block maxima. Her situation includes:
I. The parent distribution F(x) is actually unknown.
II. The parent (non-extreme) distribution is well described by a Student's t, i.e., non-normal.
III. End users might prefer the extreme tail to have exponential tails (Gumbel).
IV. End users want the extreme tail distribution to exhibit positive (right) skew.
Which of these issues, in theory, DISQUALIFIES the generalized extreme value (GEV) distribution?

A. Only I, because F(x) must be specified.
B. Only II, because the parent must be normal.
C. Both III and IV, because GEV cannot be a Gumbel or have skew.
D. None of these issues disqualify the GEV.

Answer: D
Step by step
1. Fisher-Tippett gives the GEV as the limit of normalized maxima without needing to know the parent F(x): I is not disqualifying.
2. The theorem does not require a normal parent; the t parent just implies a heavy tail ($\xi > 0$, Fréchet): II is fine.
3. The GEV nests the Gumbel ($\xi = 0$, exponential tail): III is fine.
4. GEV maxima distributions are right-skewed (especially Fréchet), so IV is fine. None disqualifies.

Why D is correct: The GEV is the universal limit law for maxima whatever the parent, and it nests Gumbel, Fréchet and Weibull, so none of the issues is a barrier.

Why the others are wrong
- **A:** Confuses EVT with parametric VaR; like the CLT, EVT does not require the parent distribution to be known.
- **B:** Wrongly assumes a normal parent is required; the t parent merely implies a Fréchet-type heavy tail.
- **C:** The Gumbel is the $\xi = 0$ special case of the GEV, and GEV maxima are positively skewed, so neither is disqualifying.

Q152

BT answers P2.T7.804.3 · BT notes P2.T5.804.2

REBUILT — Stem restated tail index as 0.60 (source printed '0.60%' but used 0.60) and the arbitrary distractors were replaced with results of realistic formula errors; the two source copies were identical and merged.

To obtain value at risk (VaR) from the generalized extreme-value (GEV) distribution for a heavy-tailed (Frechet) case, Dowd's derivation gives:
$\text{VaR} = \mu - \frac{\sigma}{\xi}\left[1 - (-n \ln \alpha)^{-\xi}\right]$
Assume location $\mu = 3.0\%$, scale $\sigma = 0.70\%$, tail index $\xi = 0.60$, and sample (block) size $n = 60$. Which is closest

to the implied 99.90% VaR?

A. 2.05%
B. 4.99%
C. 5.14%
D. 8.14%

Answer: D

Step by step

1. $\alpha = 0.999$, so $-n \ln \alpha = -60 \times \ln(0.999) = 60 \times 0.0010005 = 0.060030$.
2. Raise to the power $-\xi = -0.60$: $0.060030^{-0.60} = 5.4073$.
3. Bracket: $1 - 5.4073 = -4.4073$.
4. Multiplier: $\sigma/\xi = 0.70/0.60 = 1.1667$.
5. VaR $= 3.0 - 1.1667 \times (-4.4073) = 3.0 + 5.1418 = 8.14\%$.

Why D is correct: The Frechet VaR adds a positive tail term $(\sigma/\xi)[(-n \ln \alpha)^{-\xi} - 1]$ to the location parameter; at 99.90% the base is small, so the heavy-tail power term is large.

Why the others are wrong

- **A:** 2.05% flips the sign of the exponent (uses $+\xi$, giving $0.0600^{0.60} = 0.1843$), which removes the heavy-tail effect.
- **B:** 4.99% treats the tail index as 0.60% = 0.006 instead of 0.60 in both the exponent and the σ/ξ ratio.
- **C:** 5.14% is only the tail term $(\sigma/\xi)[(-n \ln \alpha)^{-\xi} - 1]$; it forgets to add the location parameter $\mu = 3.0\%$.

Q153 GARP PE2 Q66 · Mock 4 Q67

REBUILT — Merged two near-identical items sharing the correct option; replaced the out-of-syllabus 'moment-based method' distractor and the weak distractors with ones based on the Schweser methods (ML, regression, Hill).

A risk analyst at an investment bank compares methods for estimating the parameters of an extreme value (EV) distribution for operational losses, covering maximum likelihood, regression and semi-parametric approaches. Which of the following statements is correct?

A. The regression method orders the extreme observations and obtains least squares estimates of the EV parameters.
B. The maximum likelihood method estimates the tail index by averaging a chosen number of the most extreme observations.
C. The Hill estimator is insensitive to how many tail observations are included.
D. Maximum likelihood is simple to apply because it needs no optimization of a likelihood function.

Answer: A

Step by step

1. Schweser: the regression approach orders the extreme values from lowest to highest and fits the parameters by least squares; it is easier than ML.
2. ML maximizes a complex likelihood or log-likelihood function.
3. The semi-parametric Hill estimator of the tail index averages log-exceedances of the top k observations, and choosing k is its hardest step.
4. Hence only A is correct.

Why A is correct: Regression estimation ranks the extreme values and fits the EV parameters by least squares, making it easier than ML.

Why the others are wrong

- **B:** Describes the Hill (semi-parametric) estimator, not ML; ML maximizes a likelihood.
- **C:** The Hill estimator is highly sensitive to the number of tail observations chosen, which is its main challenge.
- **D:** ML involves maximizing a complex likelihood function, so it is the harder, not simpler, method.

LO 3.c

Describe the peaks-over-threshold (POT) approach.

Q154 BT answers L2.T5.87.1

If X is a random i.i.d. loss with distribution function F(x), and u is a threshold value of X, which expression defines the distribution of excess losses used in the peaks-over-threshold (POT) approach?

A. $P\{X \leq x \mid X > u\}$
B. $P\{X \leq x \mid X = u\}$
C. $P\{X - u \leq x \mid X > u\}$
D. $P\{X - u \leq x \mid X = u\}$

Answer: C

Step by step

1. POT focuses on losses that exceed the threshold, so the conditioning event is $X > u$.
2. The variable modeled is the excess over the threshold, $X - u$, not the loss itself.
3. Hence $F_u(x) = P\{X - u \leq x \mid X > u\}$.

Why C is correct: POT models the conditional distribution of the excess $X - u$ given that the threshold is exceeded.

Why the others are wrong

- **A:** Conditions correctly on X > u but models the loss X rather than the excess X - u.
- **B:** Conditions on X exactly equal to u (a zero-probability event) and models X, not the excess.
- **D:** Models the excess correctly but conditions on X = u instead of X > u.

Q155 BT answers P2.T5.22.5.3 · BT notes P2.T5.22.5.3

REBUILT — Original distractors (3.90, 8.40) did not correspond to identifiable errors; replaced with options produced by realistic formula mistakes, and corrected source typo 45/1,000.

Peter wants to estimate value at risk (VaR) under the peaks-over-threshold (POT) approach. He selects a threshold of u = 3.0% and observes N(u) = 45 exceedances out of n = 750 days, so N(u)/n = 45/750 = 6.0%. The scale parameter is β = 0.90% and the tail index is ξ = 0.250. Which is nearest to the 99.50% VaR?

A. 3.10%
B. 5.64%
C. 6.10%
D. 9.70%

Answer: C

Step by step

1. POT VaR: $\text{VaR} = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-\alpha)\right)^{-\xi} - 1\right]$.
2. $\frac{n}{N_u}(1-\alpha) = 0.005/0.06 = 0.08333$.
3. $0.08333^{-0.25} = 1.8612$, so bracket = 0.8612.
4. $\beta/\xi = 0.90/0.25 = 3.6$; $3.6 \times 0.8612 = 3.100$.
5. $\text{VaR} = 3.0 + 3.100 = 6.10\%$.

Why C is correct: VaR equals the threshold plus the GP-based excess quantile, scaled by the exceedance frequency N(u)/n.

Why the others are wrong

- **A:** Reported only the excess term 3.10% and forgot to add the threshold u = 3.0%.
- **B:** Used N(u)/n = 45/1,000 = 4.5% instead of 45/750 = 6.0%.
- **D:** Dropped the '-1' inside the bracket: 3.0 + 3.6 x 1.8612 = 9.70%.

Q156 BT answers P2.T7.805.1 · BT notes P2.T5.805.1

REBUILT — Reworded the awkward 'increase EXCEPT decrease' stem into a direct 'which decreases ES' question; numbers verified.

Robert is using the peaks-over-threshold (POT) approach to estimate value at risk (VaR) and expected shortfall (ES). His base model:

Parameter	Value
Loss threshold, u	5.00%
No. of observations, N	1,350
No. of observations exceeding threshold, N(u)	64
N(u)/N	4.74%
Scale, β	0.70
Shape (tail index), ξ	0.20

The base 99.0% ES is 7.47%. Considering each of the following changes one at a time (ceteris paribus), which change will DECREASE the 99.0% ES?

A. Increase the confidence level from 99.0% to 99.9%
B. Increase the scale parameter from β = 0.70 to β = 0.85
C. Double the tail index from ξ = 0.20 to ξ = 0.40
D. Increase N from 1,350 to 1,500 with no additional exceedances, so N(u)/N falls

Answer: D

Step by step

1. Formulas: VaR $= u + \frac{\beta}{\xi}\left[\left(\frac{N}{N_u}(1-\alpha)\right)^{-\xi} - 1\right]$, ES $= \frac{\text{VaR}+\beta-\xi u}{1-\xi}$.
2. Base: VaR = 6.278%, ES = 7.472%.
3. 99.9% confidence: VaR = 9.072%, ES = 10.97% (up).
4. β = 0.85: VaR = 6.552%, ES = 8.00% (up).
5. ξ = 0.40: VaR = 6.511%, ES = 8.69% (up).
6. N = 1,500: N(u)/N = 4.27%, VaR = 6.178%, ES = 7.35% (down).

Why D is correct: A smaller exceedance frequency N(u)/N means the 1% tail lies closer to the threshold, so VaR and ES fall.

Why the others are wrong

- **A:** A higher confidence level pushes the quantile further into the tail, raising ES to about 10.97%.
- **B:** ES is increasing in the scale parameter; a larger β widens the excess distribution (ES about 8.00%).
- **C:** A fatter tail (larger ξ) raises ES here, to about 8.69%; the 1/(1-ξ) factor grows.

Q157

BT notes P2.T5.804.1

REBUILT — Removed commentary about GARP practice exam; replaced distractors with values from identifiable formula errors; verified VaR of 7.00%.

A risk manager applies the peaks-over-threshold (POT) approach with the following parameters (all in % terms):

Parameter	Value
Loss threshold, u	6.00%
No. of observations, N	900
No. of observations exceeding threshold, N(u)	27
N(u)/N	3.00%
Scale, β	0.80
Shape (tail index), ξ	0.23

At the 99.0% confidence level the POT VaR is 7.00%. Which is nearest to the corresponding 99.0% expected shortfall (ES)?

A. 7.80%
B. 8.34%
C. 9.09%
D. 10.13%

Answer: B

Step by step

1. Check VaR: $6 + \frac{0.80}{0.23}[(0.01/0.03)^{-0.23} - 1] = 7.00\%$.
2. POT ES: ES $= \frac{\text{VaR}+\beta-\xi u}{1-\xi}$.
3. Numerator: $7.00 + 0.80 - 0.23 \times 6.00 = 6.42$.
4. Denominator: $1 - 0.23 = 0.77$.
5. ES $= 6.42/0.77 = 8.34\%$.

Why B is correct: ES = VaR/(1-ξ) + (β - ξu)/(1-ξ): the expected excess beyond VaR under a GP tail, which is above

VaR.

Why the others are wrong

- **A:** Computed VaR + β = 7.80, skipping the -ξu term and the division by (1-ξ).
- **C:** Used VaR/(1-ξ) = 7/0.77 = 9.09, omitting the (β - ξu) term.
- **D:** Divided (VaR + β)/(1-ξ) = 7.8/0.77 = 10.13, forgetting to subtract ξu.

Q158

BT answers L2.T5.89.8

REBUILT — Original 'compatible with ES' was ambiguous for GEV; rebuilt around computing VaR and ES from fitted POT parameters.

A risk manager has estimated the generalized Pareto parameters (scale β and tail index $\xi < 1$) from losses exceeding threshold u, along with the exceedance count N_u out of n observations. Which statement is correct?

A. Both VaR and expected shortfall can be computed from these estimates.

B. Only VaR can be computed; expected shortfall needs a separate non-parametric estimate.

C. Only expected shortfall can be computed; VaR needs a separate non-parametric estimate.

D. Neither can be computed because POT only describes the distribution of excesses, not the tail of the loss itself.

Answer: A

Step by step

1. The POT tail estimate is: $\text{VaR} = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-\alpha)\right)^{-\xi} - 1\right]$.
2. Expected shortfall follows in closed form as $\text{ES} = \frac{\text{VaR}}{1-\xi} + \frac{\beta - \xi u}{1-\xi}$, which is finite when $\xi < 1$.
3. Both measures therefore come directly from the fitted parameters.

Why A is correct: Because POT gives a fully specified parametric tail beyond u, both VaR and ES are available in closed form.

Why the others are wrong

- **B:** Wrong: ES has a closed-form expression from the same parameters.
- **C:** Wrong: VaR is the starting point and has the standard POT formula.
- **D:** Wrong: adding u and N_u/n converts the excess distribution into the tail of losses.

Q159

AnalystPrep Q.4011

In setting the threshold in the POT approach, which of the following statements is most accurate? Setting the threshold relatively low makes the model:

A. more applicable but decreases the number of observations in the modeling procedure.

B. less applicable and decreases the number of observations in the modeling procedure.

C. more applicable but increases the number of observations in the modeling procedure.

D. less applicable but increases the number of observations in the modeling procedure.

Answer: D

Step by step

1. The GPBdH theorem says excess losses converge to a GP distribution only as u becomes large.
2. A lower u moves away from that asymptotic regime, so the GP model is less applicable (more bias).
3. A lower u puts more observations above the threshold, giving more data for estimating β and ξ (lower variance).

Why D is correct: A low threshold trades GP validity for sample size: more exceedances but a poorer GP approximation.

Why the others are wrong

- **A:** Reverses both effects: a low threshold does not improve applicability and cannot reduce the count of exceedances.
- **B:** Correct on applicability but wrong on count; a lower threshold increases exceedances.
- **C:** Correct on count but wrong on applicability; a low threshold weakens the GP approximation.

Q160

Schweser Module Quiz 3.1 Q3

In setting the threshold in the POT approach, which of the following statements is the most accurate? Setting the threshold relatively high makes the model:

A. more applicable, but decreases the number of observations in the modeling procedure.

B. less applicable, and decreases the number of observations in the modeling procedure.
C. more applicable, but increases the number of observations in the modeling procedure.
D. less applicable, but increases the number of observations in the modeling procedure.

Answer: A
Step by step
1. GPBdH: the GP approximation improves as u gets larger, so a high threshold makes the model more applicable.
2. A higher u leaves fewer exceedances N_u, so there are fewer observations to estimate β and ξ.

Why A is correct: A high threshold improves the GP fit but reduces sample size; the threshold must balance both.
Why the others are wrong
- **B:** Wrong on applicability: a high threshold improves, not harms, the GP approximation.
- **C:** Wrong on count: raising u reduces the number of exceedances.
- **D:** Reverses both effects.

Q161 BT Mock Q76 · BT answers P2.T5.22.5.2 · BT notes P2.T5.22.5.2

Ralph applies the peaks-over-threshold approach with n = 1,000 days. At a threshold of u = 4.0% he observes N(u) = 28 exceedances (2.80%), fits $\beta = 0.80\%$ and $\xi = 0.310$, and obtains a 99.90% VaR of 8.670%. He then raises the threshold to u = 5.0%, keeping β and ξ, with N(u) = 17 (1.70%); the revised 99.90% VaR is 8.630%. In regard to choosing a threshold, which statement is TRUE?

A. The higher threshold offers the benefit of lower bias but the drawback of higher variance.
B. The higher threshold offers the benefit of lower variance but the drawback of higher bias.
C. Because the desired confidence is 99.90% and 0.10% = 1/1,000, the optimal threshold should be set equal to the dataset's second-worst return.
D. He can avoid a subjective choice of the threshold by applying an analytical maximum log-likelihood function to all three parameters simultaneously.

Answer: A
Step by step
1. Check: $4 + (0.8/0.31)\{[(0.001)(1000/28)]^{-0.31} - 1\} = 8.670\%$; with u = 5 and N_u = 17: 8.630%.
2. Raising u moves further into the tail where the GP theorem holds better: lower bias.
3. But exceedances fall from 28 to 17, so parameter estimates are noisier: higher variance.
4. Threshold choice has no analytical solution (MLE needs u fixed first), and setting u at the second-worst return leaves too few points.

Why A is correct: POT has a bias-variance trade-off: higher u reduces bias but leaves fewer exceedances and so raises variance.
Why the others are wrong
- **B:** Reversed: fewer exceedances raise variance, a higher threshold reduces bias.
- **C:** Confuses the threshold with the quantile being estimated; a threshold at the 2nd-worst return leaves almost no exceedances to fit.
- **D:** MLE estimates β and ξ given a chosen u; u itself must be chosen judgmentally (mean excess and stability plots).

LO 3.d

Compare and contrast the generalized extreme value (GEV) and POT approaches to estimating extreme risks.

Q162 Schweser Module Quiz 3.1 Q2

Compared with the generalized extreme value (GEV) approach, the peaks-over-threshold (POT) approach generally requires:

A. more estimated parameters than GEV and shares one parameter with GEV
B. fewer estimated parameters than GEV and shares one parameter with GEV
C. more estimated parameters than GEV and shares no parameters with GEV
D. fewer estimated parameters than GEV and shares no parameters with GEV

Answer: B

Step by step

1. GEV has three parameters: location, scale, and tail index ξ.
2. POT (generalized Pareto) has two: scale β and tail index ξ, plus the choice of threshold u.
3. The tail index ξ is common to both, so POT has fewer parameters and shares ξ.

Why B is correct: GEV needs an extra location parameter; both approaches share the tail index ξ.

Why the others are wrong

- **A:** POT does not need more parameters; GEV needs the additional location parameter.
- **C:** Wrong on both counts: POT has fewer parameters and they share ξ.
- **D:** Correct about fewer parameters, but the tail index ξ is shared by GEV and POT.

Q163

AnalystPrep Q.4003

REBUILT — Choice D described what POT does rather than why it is preferred (a second true statement); replaced it with a false statement so only B answers the question.

In practice, risk analysts often prefer the peaks-over-threshold (POT) approach over the generalized extreme value (GEV) block-maxima approach because the POT approach:

A. Does not require any parameter estimation
B. Makes more efficient use of the available data
C. Does not require the analyst to choose a threshold
D. Is based on the distribution of a single maximum per block

Answer: B

Step by step

1. GEV (block maxima) keeps only the largest loss in each block and discards the rest, so a block with several extreme losses contributes only one.
2. POT uses every observation exceeding the threshold, so more extreme data inform the tail estimate.
3. Hence the main advantage of POT is more efficient use of data (it also needs one fewer parameter).

Why B is correct: POT fits the generalized Pareto distribution to all exceedances over a high threshold, whereas block maxima discard potentially valuable extreme observations.

Why the others are wrong

- **A:** Wrong: POT still estimates the scale and shape parameters of the GPD.
- **C:** Wrong: POT requires choosing a threshold; this is its disadvantage (bias/variance trade-off).
- **D:** Wrong: this describes the GEV block-maxima approach, which is the source of its data loss.

Q164

AnalystPrep Q.4012 · BT answers L2.T5.89.2

The peaks-over-threshold (POT) approach generally requires:

A. fewer estimated parameters than the GEV approach and shares one parameter with the GEV.
B. fewer estimated parameters than the GEV approach and does not share any parameters with the GEV approach.
C. more estimated parameters than the GEV approach and shares one parameter with the GEV.
D. more estimated parameters than the GEV approach and does not share any parameters with the GEV approach.

Answer: A

Step by step

1. GEV parameters: location, scale, and tail (shape) index ξ = 3 parameters.
2. POT (generalized Pareto) parameters: scale β and tail index ξ = 2 parameters, so fewer than GEV.
3. The tail index ξ is the same in both, so one parameter is shared.

Why A is correct: POT drops the location parameter (the threshold is chosen, not estimated) while both approaches share the tail index ξ.

Why the others are wrong

- **B:** Wrong: ignores that ξ is common to both approaches.
- **C:** Wrong: reverses the parameter count; GEV is the one with the additional parameter.
- **D:** Wrong: reverses the count and also ignores the shared tail index.

Q165 BT answers L2.T5.89.1

REBUILT — Source key said neither approach is superior but conceded POT is defensible; stem reworded so exactly one answer follows from the Schweser text (choice depends on the data).

Which statement about choosing between the GEV (block maxima) and POT approaches to extreme risk is most accurate?

A. GEV is superior in virtually all circumstances because it avoids choosing a threshold.
B. POT is superior in virtually all circumstances because it always uses more data.
C. Neither approach is superior in all circumstances; each has advantages and the choice depends on the data and the trade-offs involved.
D. They are unrelated methods and cannot be compared.

Answer: C

Step by step

1. Both approaches come from extreme value theory and share the tail index ξ.
2. POT needs fewer parameters and wastes less data, but requires choosing a threshold, which adds uncertainty.
3. GEV avoids a threshold but loses data. The nature of the data can favor either, so neither dominates.

Why C is correct: Each approach has distinct strengths and weaknesses, and the preferred choice depends on the data.

Why the others are wrong

- **A:** Wrong: GEV avoids a threshold but wastes data and needs an extra parameter.
- **B:** Wrong: POT's data efficiency is offset by threshold-choice uncertainty, so it is not universally superior.
- **D:** Wrong: both stem from EVT and share the tail parameter ξ.

Q166 BT answers L2.T5.89.6

Which EVT approach is parametric?

A. GEV (block maxima)
B. Peaks-over-threshold (POT)
C. Both GEV and POT
D. Neither GEV nor POT

Answer: C

Step by step

1. GEV fits a three-parameter distribution (location, scale, tail index) to block maxima.
2. POT fits a two-parameter generalized Pareto distribution to exceedances.
3. Both assume a parametric form, so both are parametric.

Why C is correct: Both approaches impose a parametric distribution (GEV or generalized Pareto) on the extremes.

Why the others are wrong

- **A:** Incomplete: POT is parametric too.
- **B:** Incomplete: GEV is parametric too.
- **D:** Wrong: neither is non-parametric like historical simulation.

Q167 BT answers L2.T5.89.3

Which EVT approach is more likely to imply a loss of potentially useful data?

A. GEV (block maxima)
B. Peaks-over-threshold (POT)
C. Both GEV and POT
D. Neither GEV nor POT

Answer: A

Step by step

1. GEV block maxima keeps only the largest loss in each block.
2. Other large losses in the same block are discarded.
3. POT uses all exceedances over the threshold.

Why A is correct: Block maxima discard all but one observation per block, including other extreme ones.

Why the others are wrong

- **B:** Wrong: POT uses every exceedance; its issue is threshold choice.
- **C:** Wrong: only GEV wastes data.
- **D:** Wrong: GEV does.

Q168 BT answers L2.T5.89.4

Which EVT approach has the disadvantage of requiring the analyst to grapple with the problem of choosing a threshold?

A. GEV (block maxima)
B. Peaks-over-threshold (POT)
C. Both GEV and POT
D. Neither GEV nor POT

Answer: B

Step by step

1. POT defines excesses relative to a threshold u, which must be chosen.
2. Too high a u leaves few exceedances (high variance); too low a u breaks the GP approximation (bias).
3. GEV uses blocks, not a threshold (although block size must be chosen).

Why B is correct: Only POT depends on a threshold choice, which adds uncertainty.

Why the others are wrong

- **A:** Wrong: GEV chooses a block length, not a threshold.
- **C:** Wrong: GEV has no threshold.
- **D:** Wrong: POT has this problem.

Q169 BT answers L2.T5.89.7

Which EVT approach is capable of modeling a light or heavy tail, depending on the value of the tail (shape) index?

A. GEV (block maxima)
B. Peaks-over-threshold (POT)
C. Both GEV and POT
D. Neither GEV nor POT

Answer: C

Step by step

1. GEV: $\xi > 0$ Frechet (heavy), $\xi = 0$ Gumbel (light), $\xi < 0$ Weibull (thin, bounded tail).
2. POT: the generalized Pareto distribution shares the same ξ and likewise gives heavy or light tails depending on its sign.
3. Both therefore model either tail type.

Why C is correct: Both approaches share the tail index ξ, whose value determines the tail heaviness.

Why the others are wrong

- **A:** Incomplete: POT also uses ξ.
- **B:** Incomplete: GEV also uses ξ.
- **D:** Wrong: both use ξ.

Q170 BT answers P2.T5.22.5.1 · BT notes P2.T5.22.5.1

The board has asked Claire to develop risk estimates at very high confidence levels, e.g., 99.5% and 99.9%. Given the realities of her dataset (e.g., high time dependency), she may need to rely on extreme value theory (EVT). She recognizes the two branches of univariate EVT are block maxima (GEV) and peaks-over-threshold (POT), and she is comparing them with non-parametric approaches such as historical simulation.

With respect to the two univariate EVT approaches, each of the following statements is true of both approaches, EXCEPT which statement is false?

A. Each of the approaches automatically and easily fits distributions to a stochastic process with autocorrelated clusters
B. Each of the fitted EV parametric distributions has a parameter that can be increased to model increasingly heavier tails
C. She can fit a parametric distribution of extreme values to actual datasets without knowledge of the underlying distribution

D. Unlike basic historical simulation, the EVT VaR or ES can return value(s) that exceed the greatest loss in the actual historical dataset

Answer: A

Step by step

1. Both GEV (block maxima) and POT are derived assuming losses are i.i.d.; clustering (time dependency) violates this.
2. Time dependency is not handled automatically by either method; it requires extra work (e.g., longer blocks in GEV, declustering/extensions in POT). So A is false.
3. The tail index ξ is shared by GEV and the generalized Pareto; a larger ξ means a heavier tail, so B is true.
4. The Fisher-Tippett and GPBdH theorems apply whatever the parent distribution, so C is true.
5. Fitted EV distributions extrapolate beyond the sample, so VaR/ES can exceed the worst observed loss (historical simulation cannot), so D is true.

Why A is correct: Both approaches rest on an i.i.d. assumption, so autocorrelated clusters are not fitted automatically or easily; the other three statements describe genuine shared features of EVT.

Why the others are wrong

- **B:** True: ξ is the tail index in both GEV and GP, and a higher value gives heavier tails. Choosing it means mistaking a true statement for the false one.
- **C:** True: the EV theorems are distribution-free regarding the parent distribution. Choosing it confuses this with needing to know the parent distribution.
- **D:** True: parametric extrapolation allows estimates beyond the largest observed loss; only basic historical simulation is capped by the sample maximum.

Q171 BT answers P2.T5.22.6.1 · BT notes P2.T5.22.6.1

In applying extreme value theory (EVT) to their investment firm's financial returns, Patricia and Peter are deciding between the classic block maxima (GEV) approach and the modern peaks-over-threshold (POT) approach. Patricia, who is more familiar with block maxima, makes the following points about that approach, but one is inaccurate. Each of the following is true EXCEPT which statement is false?

A. If the parent distribution is a Levy distribution, the extremes fall in the domain of attraction of the Frechet distribution

B. For a given historical window, shorter blocks (and therefore more blocks) achieve lower variance but with more bias

C. Block maxima is superior to POT because it uses the data more efficiently, and its Gumbel instance avoids the trade-off between bias and variance

D. Time dependency in the loss data can be addressed with longer (and therefore fewer) blocks, giving less bias but at the cost of higher variance

Answer: C

Step by step

1. Levy is a heavy-tailed parent, so its extremes converge to the Frechet case ($\xi > 0$); A is true.
2. Block size trade-off: shorter blocks give more block maxima (lower variance) but the i.i.d./GEV approximation is worse (more bias); B is true.
3. Longer blocks give fewer maxima (higher variance) but better GEV approximation and weaker serial dependence between maxima (less bias); D is true.
4. Block maxima keeps only one observation per block, so it uses data less efficiently than POT, and the bias-variance trade-off exists whichever GEV case (Gumbel $\xi = 0$ included) is used. C is false.

Why C is correct: Block maxima discards all but the largest loss in each block and faces an unavoidable bias-variance trade-off in choosing block length, so claiming superior efficiency and no trade-off is false.

Why the others are wrong

- **A:** True: heavy-tailed parents such as Levy (like t or Pareto) lie in the Frechet domain; choosing it wrongly assumes only light-tailed parents give GEV limits.
- **B:** True: shorter blocks mean more data points (lower variance) but more bias; choosing it reverses the direction of the trade-off.
- **D:** True: longer blocks reduce bias but increase variance; choosing it reverses the direction of the trade-off.

Q172 BT answers P2.T7.805.3 · BT notes P2.T5.805.3

REBUILT — Original options (particularly B and C) embedded wrong rationales and the key rested on a loosely-supported reading of time dependency; rebuilt around the Schweser POT-vs-GEV considerations (fewer parameters, less data loss, threshold choice).

Sally is comparing the generalized extreme value (GEV) block maxima method with the peaks-over-threshold (POT) method to model extreme losses in her firm's loss database. Consider the following considerations:
I. She wants to avoid discarding data by retaining only one extreme loss per block.
II. She prefers a model with fewer parameters to estimate.
III. She wants to avoid having to make a subjective tuning choice (block length for GEV, threshold for POT).
IV. She does not know the parent distribution of the losses.
Which of the considerations, taken individually, favor POT over GEV?

A. I only
B. I and II only
C. I, II and III only
D. II, III and IV only

Answer: B

Step by step

1. I: GEV commonly uses only the maximum of each block, losing data; POT uses all exceedances over the threshold. Favors POT.
2. II: GEV estimates location, scale and tail index; POT (generalized Pareto) estimates only scale and tail index. Favors POT.
3. III: GEV requires choosing block length, POT requires choosing the threshold; both involve a subjective choice, so this does not favor either.
4. IV: Both the Fisher-Tippett and GPBdH theorems hold without knowing the parent distribution, so this does not distinguish them.
5. Therefore I and II only.

Why B is correct: POT needs one fewer parameter and typically wastes less data than block maxima; the tuning-choice and unknown-parent issues apply to both methods.

Why the others are wrong

- **A:** Omits II; forgets that GEV carries an extra location parameter.
- **C:** Adds III, but POT's threshold choice is just as subjective as GEV's block-length choice, so there is no advantage.
- **D:** Drops I and adds III and IV; treats the unknown parent distribution and tuning choice as POT advantages although they apply equally to both.

Q173 GARP PE2 Q57 · Mock 4 Q58

The CRO of an investment bank is preparing to lead a panel discussion at a risk management conference on estimating extreme risk measures. The CRO plans to discuss the main branches of univariate extreme value theory as well as the characteristics of the generalized extreme value (GEV) approach and the peaks-over-threshold (POT) approach. Which of the following statements would be correct for the CRO to include in the discussion?

A. The block maxima approach can avoid the data loss issues that sometimes arise under the POT approach, which constitutes an advantage of GEV compared to POT.
B. The additional parameter that is included in the POT approach, but not the GEV approach, is a location parameter that measures the central tendency of the extreme loss distribution.
C. An advantage of the POT approach is that the distribution of extreme losses over a high threshold is assumed to always have the same form regardless of the distribution of all losses.
D. A drawback of the GEV approach is that the parameter estimates describing the extreme loss distribution suffer from the fact that the choice of the threshold over which losses are considered extreme is subjective.

Answer: C

Step by step

1. By the GPBdH theorem, excesses over a sufficiently high threshold follow a generalized Pareto distribution whatever the parent distribution, so C is correct.
2. Block maxima keeps one value per block, so it loses data relative to POT; A reverses this.
3. The extra parameter belongs to GEV (location μ); POT uses only β and ξ. B reverses this.
4. Threshold choice is a POT issue; GEV's subjective choice is block length. D misattributes it.

Why C is correct: The GPBdH theorem gives a universal generalized Pareto form for exceedances over a high

threshold, independent of the parent distribution.

Why the others are wrong

- **A:** Reversed: block maxima is the approach that tends to lose data, so this is a disadvantage of GEV.
- **B:** The location parameter is the extra parameter in GEV, not POT; POT has fewer parameters.
- **D:** The threshold is chosen in POT, not GEV; GEV's tuning choice is block length.

LO 3.e

Discuss the application of the generalized Pareto (GP) distribution in the POT approach.

Q174

GARP PE1 Q57 · Mock 3 Q24

The CRO of a regional bank expresses concern that the bank's internal risk models do not adequately assess potential extreme losses. A risk analyst asks whether a model based on extreme value theory (EVT) would address the concern. Which of the following is correct when applying EVT and examining distributions of losses exceeding a threshold value?

A. As the threshold value is increased, the distribution of losses over a fixed threshold value converges to a generalized Pareto distribution.

B. If the tail parameter value of the generalized extreme-value (GEV) distribution goes to infinity, then the GEV essentially becomes a normal distribution.

C. To apply EVT, the underlying loss distribution must be either normal or lognormal.

D. The number of exceedances decreases as the threshold value decreases, which causes the reliability of the parameter estimates to increase.

Answer: A

Step by step

1. Recall the Gnedenko-Pickands-Balkema-de Haan theorem: as the threshold u becomes large, the distribution of excesses $F_u(x)$ converges to a generalized Pareto (GP) distribution.
2. This holds for a wide class of parent distributions, so statement A is correct.
3. Check the others: $\xi = 0$ (not infinity) gives the light-tailed Gumbel; EVT needs no normality assumption; lowering the threshold increases the number of exceedances.

Why A is correct: The GPBdH theorem says excess losses over a sufficiently high threshold follow a GP distribution irrespective of the parent distribution.

Why the others are wrong

- **B:** The GEV tail index equal to zero (not infinity) gives the Gumbel, whose light tails resemble normal or lognormal; infinity is meaningless here.
- **C:** EVT is valuable precisely because it models tails without assuming a normal or lognormal parent; those are only examples of the $\xi = 0$ case.
- **D:** Reverses the relationship: lowering the threshold increases exceedances (more data, lower variance of estimates) but worsens the GP approximation (more bias).

Q175

AnalystPrep Q.4010

As the threshold level, u, gets large, the Gnedenko-Pickands-Balkema-deHaan (GPBdH) theorem states that the distribution of above-threshold observations converges to:

A. a normal distribution

B. a generalized extreme value distribution

C. a generalized Pareto distribution

D. a uniform distribution

Answer: C

Step by step

1. The POT approach studies the excess distribution $F_u(x) = P\{X - u \leq x \mid X > u\}$.
2. GPBdH: for a sufficiently high threshold u, $F_u(x)$ converges to the generalized Pareto (GP) distribution with scale β and tail index ξ.
3. So the limit for threshold exceedances is the GP distribution.

Why C is correct: The GPBdH theorem is the POT counterpart of the Fisher-Tippett theorem: excesses over a high threshold follow a GP distribution whatever the (unknown) parent distribution.

Why the others are wrong

- **A:** Normal is a common parent-distribution assumption but does not describe the extreme tail; confuses the body with the tail.
- **B:** GEV describes block maxima (Fisher-Tippett), not threshold exceedances; confuses the two EVT approaches.
- **D:** A uniform distribution has bounded, flat density and has no role in extreme value theory.

Q176

AnalystPrep Q.4005

Assume the following observed parameter values for a POT/generalized Pareto model: $\beta = 0.60$; $\xi = 0.30$; $u = 1\%$; $N_u/n = 5\%$. Compute the VaR at 99% confidence and the corresponding expected shortfall.

A. VaR = 2.25%; ES = 3.5%
B. VaR = 2.2548%; ES = 0.5252%
C. VaR = 2.2413%; ES = 3.6304%
D. VaR = 1.5825%; ES = 2.2385%

Answer: C

Step by step

1. $\text{VaR} = u + \frac{\beta}{\xi}\left\{\left[\frac{n}{N_u}(1-c)\right]^{-\xi} - 1\right\}$
2. $(1-c)\,n/N_u = 0.01/0.05 = 0.20$, so $0.20^{-0.30} = 1.6207$
3. $\text{VaR} = 1 + (0.60/0.30)(1.6207 - 1) = 1 + 2(0.6207) = 2.2413\%$
4. $\text{ES} = \frac{\text{VaR}+\beta-\xi u}{1-\xi}$
5. $\text{ES} = (2.2413 + 0.60 - 0.30 \times 1)/0.70 = 2.5413/0.70 = 3.6304\%$

Why C is correct: With the GP tail, VaR is the threshold plus the scaled quantile of the excess distribution, and ES is VaR/(1-ξ) plus the adjustment (β-ξu)/(1-ξ), which exceeds VaR.

Why the others are wrong

- **A:** Rounded figures that do not come from the POT formulas; the tail-adjusted ES is 3.63%, not 3.5%.
- **B:** An ES below the VaR is impossible, since ES is the average loss given the VaR is exceeded; the ES step was applied incorrectly.
- **D:** Both numbers are too low and not reproducible from the formulas; VaR must exceed the threshold by 2(0.6207) = 1.2413 points, not 0.58.

Q177

AnalystPrep Q.2957

REBUILT — Original distractors were arbitrary numbers unrelated to any calculation; replaced with results of realistic formula errors (same inputs and correct answer).

Using empirical values for contracts on futures clearing companies, a risk manager fits a generalized Pareto tail with $\beta = 0.7$, $\xi = 0.12$, threshold $u = 3\%$ and $N_u/n = 5\%$. What are the POT VaR and expected shortfall at the 99.5% confidence level, respectively (in %)?

A. VaR = 1.856; ES = 2.496
B. VaR = 4.856; ES = 5.905
C. VaR = 3.223; ES = 4.049
D. VaR = 8.183; ES = 9.685

Answer: B

Step by step

1. $(1-c)\,n/N_u = 0.005/0.05 = 0.10$; $0.10^{-0.12} = 1.3183$
2. $\text{VaR} = 3 + (0.7/0.12)(1.3183 - 1) = 3 + 5.8333(0.3183) = 4.8565\%$
3. $\text{ES} = (4.8565 + 0.7 - 0.12 \times 3)/0.88 = 5.1965/0.88 = 5.9051\%$

Why B is correct: VaR is the threshold plus the GP excess quantile at tail probability (1-c)n/N_u, and ES grosses VaR up by 1/(1-ξ) with a (β-ξu) adjustment.

Why the others are wrong

- **A:** Omitted the threshold u from VaR (VaR = 1.856), then applied the ES formula to that.
- **C:** Used β instead of β/ξ as the multiplier (3 + 0.7 x 0.3183 = 3.223).
- **D:** Forgot to scale by n/N_u, using (1-c) = 0.005 directly inside the bracket, which overstates the tail quantile.

Q178 AnalystPrep Q.4006

A firm uses the peaks-over-threshold approach to estimate VaR for a commodity futures position. Empirical analysis gives:

Parameter	Value
Loss threshold, u	5.00%
No. of observations, N	700
No. of observations that exceed threshold, N(u)	21
N(u)/N	3.00%
Scale, β	0.70
Shape (tail index), ξ	0.25

Determine the VaR at the 99% confidence level.

A. 0.02225
B. 0.04125
C. 0.05885
D. 0.05151

Answer: C

Step by step

1. $N_u/n = 21/700 = 3\%$; $(1-c)\,n/N_u = 0.01/0.03 = 0.3333$
2. $0.3333^{-0.25} = 1.3161$
3. VaR $= 5 + (0.70/0.25)(1.3161 - 1) = 5 + 2.8(0.3161) = 5.885\% = 0.05885$

Why C is correct: VaR equals the threshold plus the scaled GP quantile of the excess, so it must exceed the 5% threshold.

Why the others are wrong

- **A:** Below the 5% threshold, so it cannot be a 99% VaR when only 3% of observations exceed 5%; the threshold and excess term were mishandled.
- **B:** Also below the threshold; since 1-c = 1% < 3%, the 99% VaR lies beyond u.
- **D:** Above the threshold but the excess is only 0.151 points, far below the correct 0.885; it does not follow from the POT formula.

Q179 AnalystPrep Q.4007

REBUILT — Options mixed decimals with implausible values (two were below the stated VaR, a giveaway); rewrote in percent with distractors from realistic formula errors.

A risk manager uses the POT approach with these parameters:

Parameter	Value
Loss threshold, u	5.00%
No. of observations, N	700
No. of observations that exceed threshold, N(u)	21
N(u)/N	3.00%
Scale, β	0.70
Shape (tail index), ξ	0.25

The 99.0% POT VaR is 5.885%. Which is nearest to the corresponding 99.0% expected shortfall?

A. 7.85%
B. 7.11%
C. 8.78%
D. 6.82%

Answer: B

Step by step

1. ES $= \frac{\text{VaR} + \beta - \xi u}{1 - \xi}$
2. ES $= (5.885 + 0.70 - 0.25 \times 5)/(1 - 0.25)$
3. $= 5.335/0.75 = 7.113\%$
4. Equivalent form: $5.885 + [0.70 + 0.25(5.885 - 5)]/0.75 = 5.885 + 1.228 = 7.113\%$

Why B is correct: ES = VaR/(1-ξ) + (β-ξu)/(1-ξ); the second term is the mean-excess adjustment above VaR.

Why the others are wrong

- **A:** Used VaR/(1-ξ) = 7.847 only, ignoring the (β-ξu)/(1-ξ) term.

- **C:** Forgot to subtract ξu: (5.885+0.70)/0.75 = 8.78.
- **D:** Added β/(1-ξ) to VaR without the shape adjustment: 5.885 + 0.933 = 6.82.

Q180

BT answers L2.T5.87.2

REBUILT — OCR parameters were damaged (xi = 0.75, Nu/n = 70% did not reproduce the key); reconstructed as xi = 0.15, Nu/n = 10%, which reproduces 7.40% and 9.97%, and replaced uncertain/arbitrary distractors with formula-error results.

Assume the following generalized Pareto parameters under the POT approach: scale $\beta = 0.9$, shape $\xi = 0.15$, threshold $u = 4.0\%$, and fraction of observations above the threshold $N_u/n = 10.0\%$. What are, respectively, the 99.5% and 99.9% VaR?

A. 11.28% and 14.91%
B. 3.40% and 5.97%
C. 7.40% and 9.97%
D. 4.51% and 4.90%

Answer: C

Step by step

1. $\text{VaR} = u + \frac{\beta}{\xi}\left\{\left[\frac{n}{N_u}(1-c)\right]^{-\xi} - 1\right\}$
2. 99.5%: $(0.005/0.10)^{-0.15} = 0.05^{-0.15} = 1.5673$; VaR $= 4 + 6(0.5673) = 7.404\%$
3. 99.9%: $(0.001/0.10)^{-0.15} = 0.01^{-0.15} = 1.9953$; VaR $= 4 + 6(0.9953) = 9.972\%$

Why C is correct: The tail probability scaled by n/N_u is the share of the exceedance distribution beyond VaR; plugging into the GP quantile and adding u gives the VaR.

Why the others are wrong

- **A:** Forgot to divide (1-c) by N_u/n, using 0.005 and 0.001 directly, which overstates VaR (11.28%, 14.91%).
- **B:** Omitted the threshold u (3.40% and 5.97% are only the excess terms).
- **D:** Used β instead of β/ξ as the multiplier (4 + 0.9 x 0.5673 = 4.51; 4 + 0.9 x 0.9953 = 4.90), understating the excess.

Q181

BT answers L2.T5.87.3

REBUILT — Original distractors C and D did not correspond to any realistic error; replaced with VaR/(1-xi) and (VaR+beta)/(1-xi) results.

Assume the following generalized Pareto (GP) parameters under the peaks-over-threshold (POT) approach: scale β = 0.9, shape (tail index) ξ = 0.15, threshold u = 4.0%, and Nu/n = 10.0%. What are, respectively, the 99.5% and 99.9% expected shortfall (ES)?

A. 7.40% (99.5%) and 9.97% (99.9%)
B. 9.06% and 12.08%
C. 8.71% and 11.73%
D. 9.77% and 12.79%

Answer: B

Step by step

1. $\text{VaR} = u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-c)\right)^{-\xi} - 1\right]$, with $n/N_u = 10$.
2. 99.5%: $4 + 6[(10 \times 0.005)^{-0.15} - 1] = 4 + 6(1.5674 - 1) = 7.404\%$.
3. 99.9%: $4 + 6[(10 \times 0.001)^{-0.15} - 1] = 9.972\%$.
4. $\text{ES} = (\text{VaR} + \beta - \xi u)/(1-\xi)$, with $\beta - \xi u = 0.9 - 0.6 = 0.3$.
5. 99.5%: $(7.404 + 0.3)/0.85 = 9.06\%$; 99.9%: $(9.972 + 0.3)/0.85 = 12.08\%$.

Why B is correct: ES for a GP tail equals VaR/(1−ξ) plus (β−ξu)/(1−ξ); ES exceeds VaR because it averages losses beyond VaR.

Why the others are wrong

- **A:** These are the VaRs (7.40%, 9.97%); the question asks for ES, which is larger.
- **C:** VaR/(1−ξ) only: 7.404/0.85 = 8.71% and 9.972/0.85 = 11.73%; omits the (β−ξu)/(1−ξ) term.
- **D:** (VaR+β)/(1−ξ): (7.404+0.9)/0.85 = 9.77% and 12.79%; forgets to subtract ξu.

Q182 BT answers L2.T5.88.1

The FRM reviews two EVT approaches: peaks-over-threshold (POT) and block maxima. POT is characterized by a generalized Pareto (GP) distribution, while block maxima is characterized by a generalized extreme-value (GEV) distribution. Which of the following describes the necessary parameters for the POT GP distribution?

A. Only scale and tail/shape index parameters
B. Scale and tail/shape index parameters, plus the threshold (u) must be specified
C. Only scale, location, and tail/shape index parameters
D. Scale, location, and tail/shape index parameters, plus the threshold (u) must be specified

Answer: B

Step by step

1. The GP distribution has two parameters: scale $\beta > 0$ and tail (shape) index ξ.
2. Unlike GEV it has no location parameter; the location role is played by the threshold.
3. The analyst must also choose the threshold u, which determines Nu, the number of exceedances.
4. So: scale and shape, plus a specified threshold.

Why B is correct: GP excess losses need β and ξ; the threshold u is a user-chosen input that replaces GEV's location parameter.

Why the others are wrong

- **A:** Omits the threshold u, which must be selected and drives Nu and the VaR formula.
- **C:** Location parameter belongs to GEV (block maxima), not GP; also omits u.
- **D:** Mixes GEV's location parameter into the GP model.

Q183 BT answers L2.T5.88.2

In the peaks-over-threshold (POT) approach, as the analyst selects a HIGHER threshold level, what is the trade-off encountered?

A. More data points but less help in selecting which distribution to employ
B. More data points but less convergence to the generalized Pareto distribution
C. Better indication of which distribution to employ but fewer data points on which to make estimates
D. Better convergence to the generalized Pareto distribution but fewer data points on which to make estimates

Answer: D

Step by step

1. GPBdH theorem: the excess distribution converges to GP as u gets large, so a higher u improves the GP approximation.
2. A higher u leaves fewer exceedances Nu, so parameter estimates are less precise.
3. Hence better convergence, fewer data points.

Why D is correct: The threshold must be high enough for GP convergence but low enough to retain enough observations for estimation.

Why the others are wrong

- **A:** A higher threshold reduces, not increases, data points; and choice of distribution is not the issue.
- **B:** Reverses both effects: higher u gives fewer data points and better convergence.
- **C:** Fewer data points is right, but the benefit is GP convergence, not a better indication of which distribution to use.

Q184 BT answers P2.T7.805.2 · BT notes P2.T5.805.2

Barbara the Risk Manager is selecting a threshold in her peaks-over-threshold (POT) extreme value theory model. She realizes that the choice of threshold, in Dowd's words, "is the weak spot of POT theory: it is inevitably arbitrary and therefore judgmental." Her data set contains 1,000 observations. At current calibration the threshold implies 50 losses, so $N(u)/N = 50/1{,}000 = 5.0\%$. Regarding adjusting the threshold, which statement is TRUE?

A. If she increases the threshold, the bias increases
B. If she increases the threshold, the variance decreases
C. She should employ maximum likelihood estimation (MLE) to determine the objectively optimal threshold
D. If she increases the threshold for the same dataset, she should expect N(u)/N to decrease but with unclear effect on VaR and ES (they may increase or decrease)

Answer: D

Step by step

1. Raising u with the same 1,000 observations leaves fewer exceedances, so N(u)/N falls below 5.0%.
2. VaR $= u + \frac{\beta}{\xi}\left[\left(\frac{n}{N_u}(1-c)\right)^{-\xi} - 1\right]$: a larger u pushes VaR up, a smaller N_u/n pushes it down, so the net effect is unclear.
3. A higher threshold improves GP convergence, so bias falls (not rises); fewer exceedances raise estimator variance (not lower it).
4. MLE estimates β and ξ given u; it does not remove the judgment in choosing u.

Why D is correct: N(u)/N mechanically falls when u rises, and u and Nu/n move VaR/ES in opposite directions.

Why the others are wrong

- **A:** Reverses the bias effect: a higher threshold reduces bias because GP convergence is better.
- **B:** Reverses the variance effect: fewer exceedances increase variance.
- **C:** MLE fits parameters given u; threshold choice stays subjective.

Q185

Schweser Module Quiz 3.1 Q4

REBUILT — Source explanation was missing and options A and D were not traceable to errors; kept the stem and replaced them with realistic miscalculations.

A researcher using the POT approach observes the following parameter values: β = 0.9, ξ = 0.15, u = 2%, and Nu/n = 4%. The 5% VaR in percentage terms is:

A. 5.404
B. 1.802
C. 2.204
D. 7.802

Answer: B

Step by step

1. n/Nu = 1/0.04 = 25; (n/Nu)(1−c) = 25 × 0.05 = 1.25.
2. $1.25^{-0.15} = 0.96708$.
3. β/ξ = 0.9/0.15 = 6.
4. VaR $= 2 + 6(0.96708 - 1) = 2 - 0.1975 = 1.802\%$.

Why B is correct: Apply the POT VaR formula with exponent −ξ on (n/Nu)(1−c); the bracket is below 1 here, so the term is slightly negative.

Why the others are wrong

- **A:** Dropped n/Nu and used (1−c) = 0.05 alone: 2 + 6(0.05^−0.15 − 1) = 5.404.
- **C:** Used exponent +ξ instead of −ξ: 2 + 6(1.25^0.15 − 1) = 2.204.
- **D:** Forgot the −1 inside the bracket: 2 + 6 × 0.96708 = 7.802.

Q186

Schweser Module Quiz 3.1 Q5

REBUILT — Source explanation missing and distractors untraceable; kept stem, rebuilt options from realistic formula errors.

Given a VaR equal to 2.56, a threshold of 1%, a shape parameter equal to 0.2, and a scale parameter equal to 0.3, what is the expected shortfall?

A. 3.325
B. 3.200
C. 3.575
D. 2.217

Answer: A

Step by step

1. ES = VaR/(1−ξ) + (β − ξu)/(1−ξ).
2. VaR/(1−ξ) = 2.56/0.8 = 3.200.
3. β − ξu = 0.3 − 0.2×1 = 0.1; 0.1/0.8 = 0.125.
4. ES = 3.200 + 0.125 = 3.325.

Why A is correct: GP-tail ES scales VaR by 1/(1−ξ) and adds an adjustment for scale and threshold.

Why the others are wrong

- **B:** Only VaR/(1−ξ); omitted the (β−ξu)/(1−ξ) term.
- **C:** Forgot to subtract ξu: (2.56+0.3)/0.8 = 3.575.

- **D:** Divided by (1+ξ) instead of (1−ξ): 2.66/1.2 = 2.217.

LO 3.f

Explain the multivariate EVT for risk management.

Q187

BT answers L2.T5.90.1

REBUILT — Original option A (specification of the distributions) was described by the source itself as 'a good answer', giving two defensible options; reworded the stem to 'central focus' and replaced distractors with clearly wrong ones.

According to Dowd, what is the central focus of multivariate extreme value theory (EVT) when it is applied to several risk factors at once?

A. Fitting a separate generalized Pareto distribution to each risk factor and ignoring joint behavior
B. Estimating the pairwise linear correlation matrix under an elliptical distribution
C. Calibrating the central limit theorem to the sample means of the risk factors
D. Modeling the tail dependence structure among the extreme values, using extreme value copulas

Answer: D

Key correction: Source key (D) is correct, but its explanation also called (A) 'a good answer'; the original option A was ambiguous and has been replaced.

Step by step

1. Multivariate EVT has the same goal as univariate EVT (estimating extremes) but applies it to more than one random variable at once.
2. Extreme values of different variables tend to occur together, so the added feature is tail dependence.
3. Per Dowd/Schweser, the limiting distribution of multivariate extremes is a member of the EV copula family, so the task is to model tail dependence with an EV copula.
4. Hence the central focus is modeling the tail dependence structure.

Why D is correct: Univariate EVT already handles each marginal tail; what multivariate EVT adds is the joint behavior of extremes, captured by tail dependence via EV copulas.

Why the others are wrong

- **A:** Treating each factor separately is univariate EVT; it ignores tail dependence, which is the whole point of the multivariate extension.
- **B:** Linear correlation and elliptical/covariance-matrix assumptions are of limited use for extremes; they do not capture tail dependence.
- **C:** The CLT describes the distribution of averages (central values), whereas EVT concerns the tails; it is the wrong tool.

Q188

BT answers L2.T5.90.2

In regard to multivariate EVT, each of the following is true EXCEPT:

A. As the number of random variables increases, multivariate EVT suffers from a "curse of dimensionality"
B. We can model multivariate extreme value dependence by assuming an EV copula
C. We can generally assume that extreme events are independent
D. As dimensionality increases, there are fewer multivariate extreme observations but more parameters to estimate

Answer: C

Step by step

1. A: adding variables makes joint extremes rarer and parameters more numerous, the curse of dimensionality. True.
2. B: the limiting distribution of multivariate extremes is an EV copula, so dependence is modeled by assuming one. True.
3. C: extreme events are typically dependent (tail dependence, e.g., a disaster hitting many markets), so assuming independence is false.
4. D: with two variables each extreme 1 in 100, a joint extreme occurs about 1 in 10,000 observations (1 in 1,000,000 with three), while parameters grow. True.
5. The false statement is C.

Why C is correct: The whole rationale for multivariate EVT is that extremes are dependent, so independence cannot generally be assumed.

Why the others are wrong

- **A:** True statement, so not the exception; dimensionality makes joint extremes rarer and models heavier.
- **B:** True statement; EV copulas are the limiting family for multivariate extremes.
- **D:** True statement; joint extreme observations become scarce (1 in 100^n) while parameters to estimate increase.

Reading 4. Backtesting VaR

Philippe Jorion, Value at Risk: The New Benchmark for Managing Financial Risk, 3rd Edition (New York, NY: McGraw-Hill, 2007). Chapter 6.

LO	Learning objective	Qs
4.a	describe backtesting and exceptions and explain the importance of backtesting VaR models.	4
4.b	explain the significant difficulties in backtesting a VaR model.	4
4.c	evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test.	17
4.d	identify and describe Type I and Type II errors in the context of a backtesting process.	17
4.e	explain the need to consider conditional coverage in the backtesting framework.	6
4.f	describe the Basel rules for backtesting.	8

LO 4.a

Describe backtesting and exceptions and explain the importance of backtesting VaR models.

Q189

AnalystPrep Q.1499

REBUILT — Source compared the number of exceptions with a significance level (a probability); reworded to compare the exception rate with 1 minus the confidence level and added a concrete case.

A risk manager backtests a 99% one-day VaR model over 500 trading days. Which outcome is most consistent with an accurate model?

A. About 5 exceptions, so that the exception rate is close to 1%
B. About 25 exceptions, so that the exception rate is close to 5%
C. Zero exceptions, since a loss should never exceed VaR
D. The number of exceptions is irrelevant if the average daily loss is below VaR

Answer: A

Step by step

1. An exception occurs when the actual loss exceeds the VaR estimate.
2. At 99% confidence the expected exception probability is 1 - 0.99 = 1%.
3. Expected exceptions = 500 x 0.01 = 5, so an exception rate near 1% (about 5 exceptions, within sampling error) is consistent with an accurate model.

Why A is correct: An accurate model produces exceptions at a rate equal to the significance level, 1 - confidence level, apart from sampling variation.

Why the others are wrong

- **B:** 25 exceptions is a 5% rate (500 x 0.05), which is the rate for a 95% VaR, and would indicate the model understates risk at 99%.
- **C:** Zero exceptions is not the goal: too few exceptions signals an overly conservative model that overstates risk and ties up capital.
- **D:** Backtesting looks at the frequency of losses beyond VaR, not at the average loss; the exception count is the key test statistic.

Q190

BT answers L2.T5.57.1

REBUILT — Original asked which VaR method is 'most conducive' to a backtest, which is loosely worded; recast as a clear statement about the independence of backtesting from the VaR method.

A bank reports a daily 99% VaR of $1 million and will run a backtest at year end (250 trading days). Which statement about the backtest is correct?

A. It is only meaningful if VaR is computed with the parametric (delta-normal) method
B. It is easier with historical simulation because the VaR is based on actual returns
C. It requires Monte Carlo VaR because only then can the expected number of exceptions be derived
D. It works the same for any VaR method: daily VaR is compared with the realized P&L and the exceptions counted against the expected 2.5

Answer: D

Step by step

1. Backtesting compares each day's VaR forecast with the subsequent realized P&L, whatever model produced the VaR.
2. Expected exceptions = 250 x (1 - 0.99) = 2.5.
3. The method only changes the VaR number, not the backtest procedure, so no method is more conducive.

Why D is correct: The backtest depends only on the stated confidence level and the realized outcomes, so it is independent of the VaR approach.

Why the others are wrong

- **A:** Parametric VaR gets no special treatment; assuming normality is a modelling choice and does not affect how exceptions are counted.
- **B:** Historical simulation VaR is no more testable; using actual past returns does not change the exception count or expected frequency.
- **C:** The expected exceptions come from the confidence level (2.5 = 250 x 1%), not from simulation.

Q191

BT answers P2.T7.506.2

REBUILT — Original false statement had two defensible answers (VaR can rank portfolios; backtests have low power); replaced with unambiguous statements on backtesting.

Which of the following statements about backtesting VaR models is FALSE?

A. Backtesting compares VaR forecasts with subsequently realized profits and losses

B. A significantly higher number of exceptions than expected suggests the model understates risk

C. Under the Basel internal models approach, banks must backtest their VaR models

D. A model with zero exceptions over a long sample is necessarily the most accurate

Answer: D

Step by step

1. A, B and C describe the definition of backtesting, the interpretation of too many exceptions, and the Basel requirement; all true.
2. D: at 99% confidence about 1% of days should be exceptions (e.g., 2.5 of 250), so zero exceptions over a long sample indicates an overly conservative model that overstates risk, not an accurate one.

Why D is correct: Accuracy means exceptions occur at the expected rate; too few exceptions is as much a calibration failure as too many.

Why the others are wrong

- **A:** True; this is the definition of backtesting.
- **B:** True; excess exceptions mean losses exceeded VaR too often, so risk is underestimated.
- **C:** True; Basel requires backtesting, with a multiplier add-on for excess exceptions.

Q192

Mock 4 Q50

An analyst at a bank is asked to evaluate the efficacy of the VaR model that the bank used over the last year. The analyst finds that the bank experienced more VaR exceptions than were forecast by the bank's VaR model and examines both the reason this occurred and its potential impact on the bank. Which of the following is correct regarding the analyst's examination of the situation?

A. The confidence level of the backtest performed on the VaR model was set too low.

B. The model is most likely overestimating the market risk faced by the bank.

C. The risk-taking units of the bank will likely be allocated less capital than they should be.

D. The bank's regulator will impose a financial penalty on the bank.

Answer: C

Step by step

1. More exceptions than forecast means actual losses exceeded VaR too often.
2. This indicates the model underestimates risk.
3. Capital allocation is driven by VaR, so units will be allocated less capital than their true risk warrants.

Why C is correct: Too many exceptions means VaR is understated, and capital based on that VaR is too low.

Why the others are wrong

- **A:** The backtest confidence level affects when the model is rejected, not how many exceptions occurred.

- **B:** Overestimation would produce fewer exceptions than forecast, not more.
- **D:** Basel's response is a higher multiplier on market risk capital depending on the zone, not a fine.

LO 4.b

Explain the significant difficulties in backtesting a VaR model.

Q193 AP Mock 1 Q12

Backtesting VaR models often involves limited data, particularly when considering extreme events. This can lead to issues with the statistical power of standard backtesting tests. What technique is recommended to address this small-sample problem and ensure a correctly sized test?

A. Increasing the VaR confidence level (e.g., from 99% to 99.9%).
B. Using asymptotic (chi-squared) distributions for p-value calculation.
C. Employing Monte Carlo simulated p-values instead of relying on asymptotic distributions.
D. Aggregating data across different time periods to increase the sample size.

Answer: C

Step by step

1. The backtest sample is limited (e.g., about 250 days) and exceptions are rare, so the sample may not be representative.
2. Kupiec/Christoffersen statistics are compared with asymptotic chi-squared critical values, which are inaccurate with few exceptions, so the test is mis-sized.
3. Simulating the test statistic under the null hypothesis at the actual sample size gives exact (correctly sized) p-values: Monte Carlo p-values.

Why C is correct: Simulating the statistic's null distribution at the real sample size replaces the unreliable large-sample approximation.

Why the others are wrong

- **A:** A higher confidence level lowers expected exceptions (e.g., 250 x 0.001 = 0.25), worsening the small-sample problem.
- **B:** The asymptotic chi-squared approximation is the source of the poor sizing, not the cure.
- **D:** Pooling different periods mixes non-identical conditions and breaks the i.i.d. exception assumption, biasing the test.

Q194 AnalystPrep Q.6477

Prior to 2013, some banks included fees, commissions, and intraday trading revenue in the profit and loss (P&L) used for VaR backtesting. What problem did this practice introduce?

A. It could have masked actual portfolio losses, making the VaR model appear more accurate than it was.
B. It could have led to an underestimation of VaR, resulting in insufficient capital reserves.
C. It made it impossible to compare VaR backtesting results across different banks.
D. It increased the computational complexity of VaR calculations significantly.

Answer: A

Step by step

1. VaR is a forecast for a static end-of-day portfolio; it does not model fees, commissions or intraday trading.
2. Those items are typically positive revenue that offsets trading losses in the actual P&L.
3. Fewer measured exceptions then occur, so the model looks more accurate than it is.

Why A is correct: Non-risk revenue contaminates P&L, hiding losses and understating exceptions.

Why the others are wrong

- **B:** The contamination affects the backtest P&L, not the VaR estimate itself; VaR is not computed from these items.
- **C:** Comparison is distorted, not impossible; the issue is bias in exception counts.
- **D:** VaR calculation is unchanged; only the P&L series used for comparison is affected.

Q195

BT answers L2.T5.58.2

REBUILT — Original keyed hypothetical return, but Jorion/Schweser also accept cleaned returns, giving two defensible answers; reworded to ask which series' failure points to a flaw in the VaR methodology and replaced the cleaned option.

A risk manager compares VaR forecasts with several return series. VaR assumes a static portfolio, so which return series, if it fails the backtest, indicates that the VaR modeling methodology itself must be adjusted?

A. Actual returns
B. Hypothetical returns
C. Long-horizon returns
D. Returns including funding costs and fee income only

Answer: B

Step by step

1. VaR is built on a portfolio held constant over the holding period.
2. The comparable series is the hypothetical return: the return if positions had stayed fixed (frozen) over the period.
3. Actual returns also include intraday trading, fees, commissions, spreads and interest, so an actual-return failure may not reflect the model.
4. Hence a failure on hypothetical returns signals a methodology problem.

Why B is correct: Hypothetical returns isolate the model's own assumption (a constant portfolio) from non-modeled P&L effects.

Why the others are wrong

- **A:** Actual returns contain intraday trading and fee effects, so failing them need not mean the model is wrong.
- **C:** Long-horizon returns accumulate more non-modeled effects; backtesting is best done at a short daily horizon.
- **D:** Funding costs and fee income are exactly the non-mark-to-market items stripped out, not a valid backtest return series.

Q196

GARP PE1 Q31 · Mock 3 Q49

An analyst at bank LKS has been asked to validate the bank's VaR model through backtesting. The analyst uses two sets of returns data to compare predicted and actual losses. Which of the following correctly describes the two most appropriate sets of returns data to use?

A. The cleaned returns, which are the actual returns minus any profit and loss from intraday trades, and the actual returns, which correspond to the total returns on the bank's trading portfolio
B. The actual returns, which correspond to the total return on the bank's trading portfolio, and the hypothetical returns, which represent the returns obtained from freezing the starting positions in the bank's trading portfolio
C. The hypothetical returns, which represent the returns obtained from freezing the starting positions in the bank's trading portfolio, and the cleaned returns, which are the actual returns minus any profit and loss from intraday trades
D. The trading returns, which are the actual returns minus any fees and commissions, and the hypothetical returns, which represent the actual returns obtained from freezing the starting positions in the bank's trading portfolio

Answer: B

Key correction: Mock 3 Q49 (a variant with garbled definitions) keyed C; correct is the pairing of actual and hypothetical returns, as in GARP PE1 Q31 (B).

Step by step

1. Schweser: both actual and hypothetical returns should be backtested.
2. Actual returns = total realized portfolio return; hypothetical = return on frozen starting positions.
3. Option B pairs both with correct definitions.

Why B is correct: Backtesting both series separates model error (hypothetical fails) from intraday trading and fee effects (only actual fails).

Why the others are wrong

- **A:** Cleaned returns remove fees, commissions and non-marked items (not intraday trading P&L), and it omits hypothetical returns.
- **C:** Omits actual returns and misdefines cleaned returns as actual minus intraday P&L.
- **D:** 'Trading returns' is not a standard series and hypothetical returns are not 'actual returns'.

LO 4.c

Evaluate the accuracy of a VaR model based on exceptions or failure rates by using a model verification test.

Q197 BT Mock Q56 · BT answers P2.T5.22.8.2 · BT notes P2.T5.22.8.2

Betty's firm allocates an economic capital cushion based on a one-day 99.5% VaR model. She will conduct a basic exception (frequency) backtest in which the number of tail losses follows a binomial distribution. Her backtest window is T = 500 days. Which statement is TRUE?

A. To increase the power of her backtest, she can decrease the VaR confidence level and apply a higher (k) capital multiplier.

B. At this high VaR confidence level and relatively short backtest window, the normal approximation is a better fit to test the model's bias than the binomial distribution.

C. She can achieve more powerful tests of the same hypothesis, without any additional information, by using several available extensions.

D. Although VaR is always a one-tailed quantile, the backtest is always a two-tailed backtest with a lower and an upper cutoff.

Answer: A

Step by step

1. Expected exceptions = $500 \times 0.005 = 2.5$, far too few for a powerful test.
2. At 95% the expected count is $500 \times 0.05 = 25$, giving much more power.
3. To keep capital unchanged, the lower-confidence VaR is scaled up by a larger multiplier k, so A is correct.

Why A is correct: Rare exceptions at 99.5% give little power; testing at a lower confidence level and scaling by k is the standard remedy.

Why the others are wrong

- **B:** The normal approximation needs roughly $np \geq 10$; here $np = 2.5$, so the skewed binomial is the better fit.
- **C:** More powerful extensions (e.g., independence or duration tests) need extra information such as exception timing.
- **D:** The backtest can be one- or two-tailed; 'always two-tailed' is false, and the Basel zone approach is effectively one-sided.

Q198 AP Mock 1 Q21 · AnalystPrep Q.2637

REBUILT — Two sources asked the same thing in opposite directions (one asked for the rejection count, the other 'least acceptable' count, which is mis-worded); merged into one unambiguous 'maximum accepted exceptions' question with cleaner options.

A risk analyst is backtesting a bank's daily 98% VaR model over one year of 252 trading days. The backtest uses the normal approximation to the binomial and a two-tailed test at the 95% confidence level (critical value 1.96). What is the maximum number of exceptions (daily losses exceeding VaR) for which the analyst would still conclude that the model is correctly calibrated?

A. 5

B. 8

C. 9

D. 10

Answer: C

Key correction: AP Mock 1 Q21 key (C = 10) answered a different framing (the smallest count that is rejected); AnalystPrep Q.2637 asked for the 'least acceptable' count but keyed 9, which is the maximum acceptable count. Under the maximum-accepted wording, the correct answer is 9.

Step by step

1. $p = 1 - 0.98 = 0.02, T = 252$; expected exceptions $pT = 5.04$.
2. Standard deviation $\sqrt{p(1-p)T} = \sqrt{252(0.02)(0.98)} = 2.222$.
3. Acceptance region upper bound: $5.04 + 1.96(2.222) = 9.396$.
4. Check: $x = 9 \Rightarrow z = (9 - 5.04)/2.222 = 1.78 < 1.96$ (accept); $x = 10 \Rightarrow z = 2.23 > 1.96$ (reject).
5. Maximum accepted number of exceptions = 9.

Why C is correct: The model is rejected when the z-score exceeds 1.96, i.e. when exceptions exceed 9.396; the largest integer count inside the acceptance region is 9.

Why the others are wrong

- **A:** 5 is just the expected number of exceptions (0.02×252); the acceptance region extends well above it.
- **B:** 8 comes from using the one-tailed 1.645 cutoff ($5.04 + 1.645 \times 2.222 = 8.69$) instead of the two-tailed 1.96.
- **D:** 10 is the smallest count that is rejected (z = 2.23), not the largest count that is accepted.

Q199

AP Mock 2 Q14

A large investment firm has a trading book whose size changes with perceived trading opportunities. The firm's chief risk analyst reports a one-day VaR at the 95% confidence level of GBP 110 million. Assuming losses are independently and identically distributed, which of the following would be convincing evidence that the analyst is estimating VaR poorly?

A. Over the past 250 days, there are no exceptions.
B. Over the past 250 days, the mean loss is GBP 132 million.
C. Over the past 250 days, the largest loss is GBP 1.1 billion.
D. Over the past 250 days, there are seven exceptions.

Answer: A

Step by step

1. $p = 0.05, T = 250$: expected exceptions $= 12.5$, standard deviation $= \sqrt{250(0.05)(0.95)} = 3.446$.
2. Zero exceptions: $z = (0 - 12.5)/3.446 = -3.63$, far outside ± 1.96 (probability $0.95^{250} \approx 2.7 \times 10^{-6}$).
3. Seven exceptions: $z = (7 - 12.5)/3.446 = -1.60$, inside ± 1.96; not evidence of a poor model.
4. Mean or maximum loss size is not what an exception-based backtest evaluates.

Why A is correct: Exception-count backtests compare the observed number of VaR breaches with the expected 12.5; zero breaches is statistically far too few, indicating VaR is grossly overstated.

Why the others are wrong

- **B:** A mean loss above VaR is not an exception count and, given a changing book, says nothing about the frequency of VaR breaches.
- **C:** Exception tests count breaches, not their size; one large loss can occur in a correctly calibrated model (the tail beyond VaR is unbounded).
- **D:** Seven exceptions give $z = -1.60$, inside the ± 1.96 acceptance region, so the model is not rejected.

Q200

AP Mock 2 Q51

REBUILT — Added the normal-approximation instruction (exact binomial would give a different cutoff at such a small expected count) and replaced the giveaway option 0 with plausible distractors.

A risk manager backtests a bank's one-day 99.5% VaR model over one year of 252 trading days, using the normal approximation to the binomial and a two-tailed test at the 99% confidence level (critical value 2.576). What is the maximum number of exceptions for which the model would be judged well calibrated?

A. 2
B. 3
C. 4
D. 5

Answer: C

Step by step

1. $p = 0.005, T = 252$: expected exceptions $pT = 1.26$.
2. Standard deviation $\sqrt{252(0.005)(0.995)} = 1.120$.
3. Upper bound $1.26 + 2.576(1.120) = 4.14$.
4. $x = 4 \Rightarrow z = 2.45 < 2.576$ (accept); $x = 5 \Rightarrow z = 3.34 > 2.576$ (reject).
5. Maximum acceptable = 4.

Why C is correct: The z-score of 4 exceptions (2.45) is within the 99% two-tailed critical value, while 5 exceptions (3.34) is not.

Why the others are wrong

- **A:** 2 is the expected count 1.26 rounded up; the acceptance region extends to 4.14.
- **B:** 3 results from using a smaller cutoff (e.g. 1.645 or 2.326: $1.26 + 2.326 \times 1.120 = 3.87$) instead of 2.576.
- **D:** 5 is the first rejected count (z = 3.34), or comes from rounding 4.14 up instead of down.

Q201

AnalystPrep Q.6443

REBUILT — Original distractor 'likelihood ratio test' is defensible (Kupiec's test is itself an LR test); replaced with clearly wrong alternatives so exactly one answer is correct.

The risk management team at Horizon Bank is validating its VaR model. As a first step they want to check only whether the observed frequency of exceptions (losses exceeding VaR) matches the frequency implied by the model's confidence level, ignoring the timing of exceptions. Which test is most appropriate?

A. Unconditional coverage test (Kupiec)
B. Independence test for clustering of exceptions
C. Stress test
D. Comparison of average loss with expected shortfall

Answer: A

Step by step

1. The question concerns only the number (frequency) of exceptions versus $p = 1 - c$.
2. That is exactly the unconditional coverage (proportion of failures) test, based on a log-likelihood ratio compared with a chi-squared(1) critical value.
3. Timing/clustering belongs to the independence component of conditional coverage; stress tests and ES comparisons do not test exception frequency.

Why A is correct: Unconditional coverage tests whether the observed failure rate equals the expected rate, without regard to when exceptions occur.

Why the others are wrong

- **B:** The independence test examines whether exceptions cluster in time, not whether their overall frequency is right.
- **C:** A stress test assesses losses under extreme scenarios; it does not compare realised exceptions with the VaR confidence level.
- **D:** Comparing average loss to expected shortfall evaluates tail severity, not the frequency of VaR exceptions.

Q202

BT answers L2.T5.58.1

In regard to Jorion's value at risk (VaR) backtest based on exceptions (failure rates), EACH of the following is true EXCEPT:

A. It is nonparametric.
B. If the bank's VaR model is accurate, the sample failure rate is an unbiased estimator of (1 - confidence).
C. The significance test of the backtest requires the assumption of a normal distribution in order to avoid a numerical/simulation approach.
D. The hypothesis test of the model's accuracy can assume a binomial distribution, which is easy to use.

Answer: C

Step by step

1. Counting exceptions requires no distributional assumption about returns, so the test is nonparametric (A true).
2. Failure rate N/T converges to $p = 1 - c$ for an accurate model (B true).
3. Exceptions are Bernoulli trials, so N is binomial (D true); the exact binomial can be used directly.
4. The normal distribution is only a convenient large-sample approximation to the binomial, not a requirement (C false).

Why C is correct: The normal is just an approximation to the binomial for large T; the test does not require it, so statement C is the false one.

Why the others are wrong

- **A:** True: the exception count approach makes no assumption about the return distribution.
- **B:** True: the failure rate is an unbiased measure whose expectation equals $1 - c$ for a correct model.
- **D:** True: each day is a Bernoulli trial with exception probability p, so the count is binomial.

Q203

BT answers L2.T5.59.1

REBUILT — Source distractor statistics (e.g. 1.25) corresponded to no realistic computation; options rebuilt around genuine errors (variance vs. SD, wrong cutoff, omitted factor).

A bank backtests its 95% daily VaR and observes 19 exceptions over 250 trading days. Using the normal approximation to the binomial, what is the decision and test statistic for a 90% two-tailed test of the model?

A. Accept, with a test statistic of 0.55
B. Accept, with a test statistic of 1.89
C. Reject, with a test statistic of 1.89
D. Reject, with a test statistic of 1.84

Answer: C
Step by step
1. $p = 0.05, T = 250$: expected exceptions $= 12.5$.
2. Standard deviation $\sqrt{250(0.05)(0.95)} = 3.446$.
3. $z = (19 - 12.5)/3.446 = 1.886 \approx 1.89$.
4. 90% two-tailed critical value is 1.645; $1.89 > 1.645$, so reject.

Why C is correct: The z-statistic of 1.89 exceeds the 90% two-tailed critical value of 1.645, so the model is rejected.
Why the others are wrong
- **A:** 0.55 arises from dividing by the variance (11.875) instead of the standard deviation; the decision is also wrong.
- **B:** The statistic is right but the decision uses the 95% cutoff (1.96) instead of 1.645 for a 90% test.
- **D:** 1.84 results from omitting the $(1 - p)$ factor: $6.5/\sqrt{250 \times 0.05} = 1.84$; the correct statistic is 1.89.

Q204

BT answers L2.T5.59.2

REBUILT — Stem depended on the preceding question for the VaR level; made self-contained and replaced options with ones tied to realistic errors.

A bank backtests its 95% daily VaR model over two years (T = 500 trading days). It applies the normal approximation to the binomial with a two-tailed test at the 95% confidence level (critical value 1.96). What is the largest number of exceptions for which the bank still accepts the model?

A. 33
B. 34
C. 35
D. 37

Answer: B
Step by step
1. $p = 0.05, T = 500$: expected exceptions $= 25$; standard deviation $\sqrt{500(0.05)(0.95)} = 4.873$.
2. Upper bound $25 + 1.96(4.873) = 34.55$.
3. $x = 34 \Rightarrow z = 1.847 < 1.96$ (accept); $x = 35 \Rightarrow z = 2.05 > 1.96$ (reject).
4. Maximum accepted = 34.

Why B is correct: The cutoff 34.55 must be rounded down: 34 exceptions give z = 1.85 (accept) and 35 give z = 2.05 (reject).
Why the others are wrong
- **A:** 33 comes from the one-tailed 1.645 cutoff ($25 + 1.645 \times 4.873 = 33.0$).
- **C:** 35 comes from rounding 34.55 up; 35 exceptions are already rejected (z = 2.05).
- **D:** 37 comes from using the 99% cutoff 2.576 ($25 + 2.576 \times 4.873 = 37.6$).

Q205

BT answers P2.T5.22.7.3 · BT notes P2.T5.22.7.3

Mary backtests her firm's one-day 99.0% VaR model using four years of data (n = 1,000 daily observations) and the binomial distribution with p = 0.010. She finds Pr(X <= 4) = 2.87% and Pr(X >= 17) = 2.64%, so the two-sided 95.0% acceptance region is 4 < N < 17 (consistent with Jorion's table). She observes 16 exceptions and accepts the model. She then realizes the firm is concerned only about a VaR that is too aggressive (too many exceptions), so she instead runs a one-sided 95.0% test of the 99.0% VaR model. What is her decision?

A. Reject the model, because 16 exceptions fall in the one-sided rejection region
B. Accept the model, because 16 exceptions fall in the acceptance region
C. The test is invalid because the 95.0% test confidence level differs from the 99.0% VaR confidence level
D. The test is invalid because the alternative should be Ha: p < 0.010 rather than Ha: p > 0.010

Answer: A
Step by step
1. A model that is too aggressive (VaR too low) produces too many exceptions, so use Ha: p > 0.010 and put the

entire 5.0% in the upper tail.
2. Binomial (n = 1,000, p = 0.010): Pr(X >= 17) = 2.64%, Pr(X >= 16) = 4.79% < 5.0%, but Pr(X >= 15) = 8.24% > 5.0%.
3. So the one-sided rejection region is X >= 16; Mary's 16 exceptions fall in it.
4. Decision: reject the model.

Why A is correct: Moving from a two-sided to a one-sided test shifts the whole 5% rejection probability into the upper tail, lowering the critical count from 17 to 16.

Why the others are wrong

- **B:** Uses the two-sided region (4 < N < 17), where 16 is accepted; the one-sided region starts at 16.
- **C:** The test confidence level for the decision rule is separate from the VaR confidence level, so a 95% test of a 99% VaR is perfectly valid.
- **D:** Too many exceptions is the concern, so the alternative is p > 0.010; the lower tail is irrelevant.

Q206

BT answers P2.T5.712.1 · BT notes P2.T5.712.1

A risk manager is backtesting a one-day 97.0% VaR model over a two-year horizon of 500 trading days. The backtest is two-tailed with 95.0% confidence, and the binomial distribution is approximated by a normal distribution. What is the maximum number of exceptions that can be observed while still concluding that the model is correctly calibrated?

A. 15 exceptions
B. 19 exceptions
C. 22 exceptions
D. 25 exceptions

Answer: C

Step by step

1. Exception probability: $p = 1 - 0.97 = 0.03$; sample $T = 500$.
2. Expected exceptions: $Tp = 500 \times 0.03 = 15$.
3. Standard deviation: $\sigma = \sqrt{Tp(1-p)} = \sqrt{500 \times 0.03 \times 0.97} = \sqrt{14.55} = 3.814$.
4. Two-tailed 95% critical value $z = 1.96$: upper cutoff $= 15 + 1.96 \times 3.814 = 22.48$.
5. Whole exceptions only: 22 is inside the acceptance region, 23 is outside, so the maximum is 22.

Why C is correct: Under the normal approximation the acceptance region is the expected count plus or minus 1.96 standard deviations; the largest whole number below 22.48 is 22.

Why the others are wrong

- **A:** 15 is only the expected number of exceptions, not the upper limit of the acceptance region.
- **B:** 19 results from adding just one standard deviation (15 + 3.8 = 18.8) instead of 1.96.
- **D:** 25 comes from using the 99% two-tailed value 2.576 (15 + 2.576 × 3.814 = 24.8) rather than 1.96.

Q207

BT answers P2.T5.712.2 · BT notes P2.T5.712.2

REBUILT — Original had ambiguous 'two yes/two no' wording and a weak distractor; reworded the stem and replaced the 'four or more' option with the 5%-significance cutoff error.

A risk manager backtests a 97.0% one-day VaR model over 250 trading days and observes exactly one exception. The backtest is one-tailed with 99.0% confidence (1% significance), so the model is rejected if the probability of observing this few exceptions or fewer is below 1%. Which conclusion is best?

A. Reject the model: P(0 or 1 exception) is about 0.43%, so at least two exceptions are needed to be in the acceptance region
B. Reject the model: the acceptance region requires at least three exceptions
C. Do not reject the model: the normal-based lower cutoff is 0.55 exceptions, so one exception lies inside the acceptance region
D. Do not reject the model: the asymmetry of the binomial means too few exceptions can never lead to rejection for this p and sample size

Answer: A

Step by step

1. Expected exceptions $= 250 \times 0.03 = 7.5$; the concern here is too few exceptions (left tail).
2. Exact binomial: $P(X \leq 1) = 0.43\% < 1\%$, so one exception is in the rejection region.

3. $P(X \leq 2) = 1.90\% > 1\%$, so the acceptance region starts at two exceptions.
4. Normal check: $\sigma = \sqrt{250 \times 0.03 \times 0.97} = 2.697$; one-tailed $z = 2.326$ gives lower cutoff $7.5 - 2.326 \times 2.697 = 1.23$. One exception is below 1.23, so reject again.
5. Conclusion: reject; at least two exceptions are needed.

Why A is correct: With 7.5 expected exceptions, seeing only one has a cumulative probability of 0.43%, below the 1% threshold, so the VaR model is likely too conservative and is rejected.

Why the others are wrong

- **B:** Three is the cutoff only at 5% significance (P(X ≤ 2) = 1.90% < 5% but P(X ≤ 3) = 5.65% > 5%); the 99% test uses 1%.
- **C:** 0.55 is the two-tailed 99% cutoff (7.5 − 2.576 × 2.697); this test is one-tailed (z = 2.326, cutoff 1.23), and one exception is below 1.23, so it is rejected.
- **D:** Asymmetry of the binomial does not preclude left-tail rejection here; P(X ≤ 1) = 0.43% is clearly in the rejection region.

Q208

BT answers P2.T5.712.3 · BT notes P2.T5.712.3

REBUILT — Original 'could accept EXCEPT which is false' wording was confusing; restated as a plain 'which statement is FALSE' question with unchanged content.

A risk manager backtests a 95.0% VaR model over only 60 days and observes zero exceptions. Given a correct model, the probability of zero exceptions is $0.95^{60} = 4.61\%$, so under a one-tailed test with 95.0% confidence (5% significance) the model is rejected (too few exceptions). Which of the following statements about alternative scenarios, all else equal, is FALSE?

A. If zero exceptions occurred for a 97.0% VaR model, the model would not be rejected

B. If the backtest confidence were lowered to 90.0%, the model would not be rejected

C. If zero exceptions occurred over a smaller sample of 50 days, the model would not be rejected

D. If a two-tailed 95.0% test with the normal approximation were used, the model would not be rejected (although the approximation is dubious for this sample)

Answer: B

Step by step

1. Baseline: $P(0) = 0.95^{60} = 4.61\% < 5\%$, so reject.
2. A: $0.97^{60} = 16.08\% > 5\%$, not rejected, so A is true.
3. B: lowering confidence to 90% raises the significance level to 10%; $4.61\% < 10\%$, so the model is still rejected (the rejection region grows). B is false.
4. C: $0.95^{50} = 7.69\% > 5\%$, not rejected, so C is true.
5. D: $\mu = 3$, $\sigma = \sqrt{60 \times 0.05 \times 0.95} = 1.688$; lower cutoff $3 - 1.96 \times 1.688 = -0.31 < 0$, so zero exceptions is acceptable; D is true.

Why B is correct: Lowering the confidence level widens the rejection region (a larger significance level), so a result that already fails at 95% must also fail at 90%.

Why the others are wrong

- **A:** True statement: a lower exception probability makes zero exceptions much more likely (16.08%), so it is accepted.
- **C:** True statement: a smaller sample makes zero exceptions more likely (7.69% > 5%), so it is accepted.
- **D:** True statement: the two-tailed normal lower cutoff is negative (−0.31), so zero is inside the acceptance region.

Q209

Mock 1 Q75

REBUILT — Stem did not specify the normal approximation (exact binomial would allow 10 exceptions); added it and replaced weak options with ones reflecting realistic errors.

A risk manager is backtesting a one-day 98% VaR model over one year of 250 trading days at a 95% confidence level (two-tailed). Using the normal approximation to the binomial, what is the maximum number of exceptions that is consistent with the model being correctly calibrated?

A. 5 exceptions

B. 9 exceptions

C. 10 exceptions

D. 15 exceptions

Answer: B

Step by step

1. $p = 0.02, T = 250$: expected exceptions $= 5$.
2. $\sigma = \sqrt{250 \times 0.02 \times 0.98} = \sqrt{4.9} = 2.214$.
3. Upper cutoff $= 5 + 1.96 \times 2.214 = 9.34$.
4. Round down to whole exceptions: 9 is accepted; 10 gives $z = (10 - 5)/2.214 = 2.26 > 1.96$, rejected.

Why B is correct: The acceptance region extends to 9.34 exceptions, so 9 is the largest whole number that does not reject the model.

Why the others are wrong

- **A:** 5 is the expected number of exceptions, not the upper limit of the acceptance region.
- **C:** 10 results from rounding 9.34 up; but 10 exceptions gives z = 2.26, outside the 1.96 limit.
- **D:** 15 results from using the variance (4.9) instead of σ: 5 + 1.96 × 4.9 = 14.6.

Q210

Mock 2 Q31

REBUILT — Options 19/36 were not tied to any realistic error; replaced with options reflecting rounding, wrong z and expected-value errors, and added the normal-approximation instruction.

A risk analyst backtests a bank's one-day 99% VaR model over eight years (250 trading days per year) at a 95% two-tailed confidence level. Assume i.i.d. daily returns and use the normal approximation to the binomial. What is the highest number of exceptions that is acceptable to conclude the model is correctly calibrated?

A. 20 exceptions
B. 28 exceptions
C. 29 exceptions
D. 31 exceptions

Answer: B

Step by step

1. $T = 8 \times 250 = 2000, p = 0.01$: expected exceptions $= 20$.
2. $\sigma = \sqrt{2000 \times 0.01 \times 0.99} = \sqrt{19.8} = 4.450$.
3. Upper cutoff $= 20 + 1.96 \times 4.450 = 28.72$.
4. Round down: 28 is accepted; 29 exceeds 28.72 and is rejected.

Why B is correct: The acceptance region ends at 28.72 exceptions, so the maximum whole number consistent with correct calibration is 28.

Why the others are wrong

- **A:** 20 is only the expected number of exceptions.
- **C:** 29 comes from rounding 28.72 to the nearest integer; 29 lies outside the acceptance region.
- **D:** 31 uses the 99% critical value 2.576 (20 + 2.576 × 4.450 = 31.5) instead of 1.96.

Q211

Schweser Module Quiz 4.1 Q1

REBUILT — Replaced distractors (3.7 etc.) that corresponded to no realistic mistake.

A VaR model is built at a 97.5% confidence level and backtested over a 252-day period. How many exceptions are expected?

A. 2.5
B. 6.3
C. 12.6
D. 245.7

Answer: B

Step by step

1. Exception probability $p = 1 - 0.975 = 0.025$.
2. Expected exceptions $= pT = 0.025 \times 252 = 6.3$.

Why B is correct: The expected number of exceptions is the tail probability times the number of observations.

Why the others are wrong

- **A:** 2.5 is the exception percentage (2.5%) not scaled by the 252 observations.
- **C:** 12.6 uses 5% (a 95% confidence level) instead of 2.5%: 0.05 × 252.
- **D:** 245.7 multiplies the confidence level (0.975) rather than the tail probability (0.025) by 252, giving the expected number of non-exceptions.

Q212 Schweser Module Quiz 4.1 Q3

Which of the following statements regarding verification of a VaR model by examining its failure rates is false?

A. The frequency of exceptions can be determined with the confidence level used for the model.

B. According to Kupiec (1995), we should reject the hypothesis that the model is correct if the log-likelihood ratio (LR) > 3.84.

C. Backtesting VaR models with a higher probability of exceptions is difficult because the number of exceptions is not high enough to provide meaningful information.

D. The range for the number of exceptions must strike a balance between the chances of rejecting an accurate model (a Type I error) and the chances of failing to reject an inaccurate model (a Type II error).

Answer: C

Step by step

1. A: the expected exception frequency is 1 minus the confidence level, so A is true.
2. B: the Kupiec LR is compared with the chi-square(1) critical value 3.84 at 95% confidence, so B is true.
3. C: it is models with a LOWER probability of exceptions (higher VaR confidence, e.g. 99%+) that are hard to backtest, because few exceptions occur and they carry little information. A higher exception probability yields more exceptions, so C is false.
4. D: choosing the acceptance range trades off Type I against Type II errors, so D is true.

Why C is correct: Rare exceptions (high confidence levels) give too little data for a meaningful backtest; statement C reverses this.

Why the others are wrong

- **A:** True: the expected failure rate equals one minus the VaR confidence level.
- **B:** True: 3.84 is the 95% chi-square critical value (1 degree of freedom) used in Kupiec's test.
- **D:** True: widening the range reduces Type I errors but raises Type II errors, and vice versa.

Q213 Schweser Module Quiz 4.1 Q4

REBUILT — Original distractors (0.62, 1.64, 2.86) did not correspond to realistic errors; replaced with options produced by specific common mistakes.

A risk manager is backtesting a VaR model at the 95% confidence level using 252 daily returns and observes 17 exceptions. Using the model verification z-score $z = \frac{x - pT}{\sqrt{p(1-p)T}}$, where $p = 1 - c$, what is the z-score for this sample?

A. 0.37

B. 1.24

C. 1.27

D. 1.96

Answer: C

Step by step

1. Exception probability: $p = 1 - 0.95 = 0.05$; $T = 252$; $x = 17$.
2. Expected exceptions: $pT = 0.05 \times 252 = 12.6$, so $x - pT = 4.4$.
3. Variance of the count: $p(1-p)T = 0.05 \times 0.95 \times 252 = 11.97$; standard deviation $= \sqrt{11.97} = 3.460$.
4. $z = 4.4/3.460 = 1.27$, below the 1.96 critical value, so the model is not rejected at the 95% test level.

Why C is correct: The z-score standardizes the excess exceptions over the binomial expectation by the binomial standard deviation $\sqrt{p(1-p)T}$.

Why the others are wrong

- **A:** Divided by the variance (11.97) rather than the standard deviation: 4.4/11.97 = 0.37.
- **B:** Dropped the (1 - p) factor from the standard deviation: $4.4/\sqrt{0.05 \times 252}$ = 4.4/3.550 = 1.24.
- **D:** This is the 95% two-tailed critical value, not the z-score; it is the threshold the computed z is compared against.

LO 4.d

Identify and describe Type I and Type II errors in the context of a backtesting process.

Q214 Mock 3 Q27

A new risk analyst assists in backtesting the bank's VaR model. The bank uses a 1-day VaR at 95% confidence but is considering a change to 99% confidence, as in the Basel framework. Which statement about this change is correct?

A. The decision to accept or reject a VaR model from a two-tailed backtest at the 95% test confidence level is less reliable for a 99% VaR model than for a 95% VaR model.

B. The 95% VaR model is less likely to be rejected by a backtest than the 99% VaR model.

C. When using a two-tailed 90% confidence test in a backtest, there is a smaller probability of incorrectly rejecting a 95% VaR model than a 99% VaR model.

D. Using a 99% VaR model will lower the probability of committing both Type I error and Type II error.

Answer: A

Step by step

1. Over 250 days a 95% VaR expects $250 \times 0.05 = 12.5$ exceptions, standard deviation $\sqrt{250 \times 0.05 \times 0.95} = 3.45$ (28% of the mean).
2. A 99% VaR expects $250 \times 0.01 = 2.5$ exceptions, standard deviation $\sqrt{250 \times 0.01 \times 0.99} = 1.57$ (63% of the mean).
3. Relative noise is much larger at 99%, so accept/reject decisions are less reliable: A.

Why A is correct: With few expected exceptions the count is very noisy, so the backtest has less power and is less reliable.

Why the others are wrong

- **B:** Rejection probability depends on model accuracy and sample, not on the stated confidence level of a correct model.
- **C:** Type I error equals the chosen test significance level (here 10%) regardless of the VaR level.
- **D:** Changing the VaR level cannot reduce both errors; at fixed sample size lowering one raises the other, and only more data helps.

Q215 AP Mock 1 Q10

A model gives a VaR value of USD 5 million for a portfolio at a 99% confidence level. A one-year backtest, with statistical significance judged at the 95% level, reveals that losses exceeded USD 5 million on 8 occasions. The model is accepted as accurate. Assuming 252 days in a year, which of these statements is most likely true?

A. A Type I error has occurred.

B. A Type II error has occurred.

C. Both Type I and Type II errors have occurred.

D. The model has been accepted correctly.

Answer: B

Step by step

1. Expected exceptions at 99% VaR: $252 \times 0.01 = 2.52$.
2. Standard deviation: $\sqrt{252 \times 0.01 \times 0.99} = 1.58$.
3. $z = (8 - 2.52)/1.58 = 3.47$, far above the 1.96 critical value at the 95% test level, so the model should be rejected.
4. The model was accepted although the evidence says it is inaccurate: failing to reject a bad model is a Type II error. Only one decision was made, so only one type of error can be involved.

Why B is correct: Type II error = accepting (failing to reject) an inaccurate model; Type I error = rejecting an accurate one. The data clearly contradict the model, yet it was accepted.

Why the others are wrong

- **A:** Type I error requires rejecting an accurate model; here the model was accepted, not rejected.
- **C:** One accept/reject decision can produce only one kind of error; both cannot occur simultaneously.
- **D:** With $z = 3.47 > 1.96$ the exception count is inconsistent with a 99% VaR model, so acceptance is not correct. A candidate who uses the 5% failure rate of the 95% test level instead of 1% would wrongly conclude the count is normal.

Q216

AP Mock 3 Q11

REBUILT — Stem called the VaR 'annual' although exceptions are counted daily, and 'confidence interval' was loosely used for the test level; reworded to a daily VaR and a 95% test level.

A model gives a daily VaR value of USD 9.5 million for a portfolio at a 99% confidence level. A one-year (224-trading-day) backtest, with statistical significance judged at the 95% level, reveals that losses exceeded USD 9.5 million on 12 occasions. The model is accepted as accurate. Which of these statements is most likely true?

A. A Type I error has occurred.
B. A Type II error has occurred.
C. Both Type I and Type II errors have occurred.
D. The model has been accepted correctly.

Answer: B

Step by step

1. Expected exceptions at 99% VaR: $224 \times 0.01 = 2.24$.
2. Standard deviation: $\sqrt{224 \times 0.01 \times 0.99} = 1.489$.
3. $z = (12 - 2.24)/1.489 = 6.55$, far above the 1.96 critical value at the 95% test level, so the model should be rejected.
4. The model was accepted although the evidence says it is inaccurate: failing to reject a bad model is a Type II error. Only one decision was made, so only one type of error can be involved.

Why B is correct: Type II error = accepting (failing to reject) an inaccurate model; Type I error = rejecting an accurate one. The data clearly contradict the model, yet it was accepted.

Why the others are wrong

- **A:** Type I error requires rejecting an accurate model; here the model was accepted, not rejected.
- **C:** One accept/reject decision can produce only one kind of error; both cannot occur simultaneously.
- **D:** With $z = 6.55 > 1.96$ the exception count is inconsistent with a 99% VaR model, so acceptance is not correct. A candidate who uses the 5% failure rate of the 95% test level instead of 1% would wrongly conclude the count is normal.

Q217

AP Mock 2 Q45 · AP Mock 3 Q53

An intern has been tasked with backtesting a firm's VaR model. The firm has always computed 1-day VaR at the 99% confidence level. In a departure from Basel Committee recommendations, the intern proposes shifting to the 95% level. What will be the implication of making this switch?

A. Decisions on whether to accept or reject VaR models based on backtesting will be more reliable.
B. The power of statistical tests will decrease.
C. The probability of making both Type I and Type II errors will increase.
D. The 95% VaR model is less likely to be rejected using backtesting than the 99% VaR model.

Answer: A

Key correction: AP Mock 3 Q53 is the same question with reworded options (its key B has the same content as A here).

Step by step

1. Over 250 days, a 99% VaR gives about $250 \times 0.01 = 2.5$ expected exceptions; a 95% VaR gives $250 \times 0.05 = 12.5$.
2. With more expected exceptions, the sample distinguishes a correct from an incorrect model better, so the power of the test rises and the Type II error probability falls.
3. The Type I error probability is fixed by the chosen test significance level, so it does not rise.
4. Therefore accept/reject decisions become more reliable.

Why A is correct: A higher exception frequency gives the backtest more information, raising power and reducing the chance of accepting a bad model, while the Type I rate is set by the test level.

Why the others are wrong

- **B:** Reverses the effect: more expected exceptions increase, not decrease, test power.
- **C:** Neither error probability rises; Type II falls and Type I is fixed by the significance level.
- **D:** A correct model is rejected with the same probability (the test significance level) at either VaR level; an incorrect 95% model is more, not less, likely to be rejected because power is higher.

Q218

AnalystPrep Q.6473

REBUILT — Original asked a vague 'which error is more costly' with an arguable key; rebuilt as a precise statement of the consequence of each error for a regulator.

A regulator backtests a bank's internal VaR model using a null hypothesis that the model is accurate. Which statement correctly describes the consequences of the two possible errors?

A. A Type I error means an inaccurate model is accepted, leaving risk and capital understated; a Type II error means an accurate model is rejected, causing unnecessary penalties.

B. A Type I error means an accurate model is rejected, causing unnecessary penalties or recalibration; a Type II error means an inaccurate model is accepted, leaving risk and capital understated.

C. Both errors lead to understated risk, so the regulator should minimise both by increasing the confidence level of the VaR.

D. Type I and Type II errors always have the same probability, so their consequences cancel out in the backtest.

Answer: B

Step by step

1. Null hypothesis: the model is accurate.
2. Type I error = rejecting a true null, i.e. rejecting an accurate model; the cost is unnecessary penalties or recalibration.
3. Type II error = failing to reject a false null, i.e. accepting an inaccurate model; risk is understated and capital may be inadequate, typically the more serious concern for regulators.
4. Only B matches these definitions.

Why B is correct: Type I rejects a good model; Type II accepts a bad one, which hides understated risk.

Why the others are wrong

- **A:** Swaps the definitions of Type I and Type II.
- **C:** Rejecting an accurate model does not understate risk, and raising the VaR confidence level reduces test power rather than minimising both errors.
- **D:** The two error probabilities are generally not equal; reducing one tends to raise the other, and the consequences do not cancel.

Q219

BT answers L2.T5.57.2 · BT notes P2.T5.57.2

REBUILT — Merged duplicate stems (one without an answer); fixed OCR-doubtful figures by stating a one-day 99% VaR of USD 1 million and 1,000 daily observations.

A bank's internal one-day 99% VaR is USD 1 million. A regulator backtests the model with the null hypothesis that the VaR model is accurate. Out of 1,000 daily observations, 25 exceptions are observed (the actual loss exceeded VaR 25 times). Which statement is most appropriate?

A. We will probably call the VaR model accurate, but we risk a Type I error.

B. We will probably call the VaR model accurate, but we risk a Type II error.

C. We will probably call the VaR model inaccurate, but we risk a Type I error.

D. We will probably call the VaR model inaccurate, but we risk a Type II error.

Answer: C

Step by step

1. Expected exceptions if accurate: $1000 \times 0.01 = 10$.
2. Probability of 25 or more exceptions if the model is accurate: $P(X \geq 25) = 1 - \text{BINOM.DIST}(24, 1000, 0.01, \text{TRUE}) \approx 0.004\%$ (computed in Python: 4.2e-5).
3. This is extremely unlikely, so we reject the null and call the model inaccurate.
4. Rejecting a null that might actually be true is a Type I error (its probability here is tiny, but it is the only error that can arise from a rejection). A Type II error can only arise when the null is not rejected.

Why C is correct: 25 versus 10 expected exceptions leads to rejection; the error risked when rejecting the null is Type I.

Why the others are wrong

- **A:** Accepting is the wrong decision (25 vs 10 expected exceptions), and Type I is not the error associated with acceptance.
- **B:** Wrong decision: the data call for rejection, though the Type II label is the one tied to acceptance.
- **D:** Correct decision but wrong error: a Type II error arises only when the null is not rejected.

Q220 BT answers L2.T5.57.3

Backtesting underpins the Basel Committee's decision to allow internal VaR models, because without it banks might have an incentive to understate their risk. Under a 99% VaR, Basel's backtest green zone is 0 to 4 exceptions (out of 250), the yellow zone 5 to 9, and the red zone 10 or more. Each of the following is true EXCEPT:

A. By shrinking the green zone (e.g., 0 to 2 exceptions), Basel can maximize the probability of catching banks that willfully understate their risk.

B. By shrinking the red zone (e.g., 13 or more), Basel can better avoid unduly penalizing banks whose VaR is exceeded simply due to bad luck.

C. By expanding the red zone (e.g., 7 or more), Basel can better avoid unduly penalizing banks whose VaR is exceeded simply due to bad luck.

D. By expanding the red zone (e.g., 7 or more), Basel can maximize the probability of catching banks that willfully understate their risk.

Answer: C

Step by step

1. If the model is accurate, $P(N \geq 5) = 10.8\%$, $P(N \geq 7) = 1.37\%$, $P(N \geq 10) = 0.025\%$, $P(N \geq 13) \approx 0.0002\%$ (binomial, $n = 250$, $p = 1\%$; computed in Python).
2. A larger red zone (7+) means more correct models fall into it: Type I error rises (more good banks penalized) while Type II error falls (fewer cheating banks escape).
3. So expanding the red zone supports catching cheaters (D true) but does the opposite of protecting unlucky banks, making C false.
4. Shrinking the red zone (B) and shrinking the green zone (A) are consistent with their stated aims.

Why C is correct: Widening the rejection region trades a lower Type II error for a higher Type I error, so it penalizes unlucky good banks more, not less.

Why the others are wrong

- **A:** True: a smaller green zone lowers Type II error, catching more understating banks.
- **B:** True: a smaller red zone lowers Type I error, sparing unlucky banks.
- **D:** True: a bigger red zone lowers Type II error.

Q221 BT answers L2.T5.57.3 (sub 3, ref L2.T5.58.3)

A bank wants to increase the reliability of the backtest by reducing the probability of accepting a bad VaR model. How can it do this?

A. Increase the VaR model's confidence level, e.g., 95% to 99%.

B. Decrease the VaR model's confidence level, e.g., 99% to 95%.

C. Decrease the sample size in order to observe more recent data.

D. Increase the two-tailed non-rejection confidence level (i.e., the significance test of the observed number of exceptions versus the null hypothesis).

Answer: B

Step by step

1. The probability of accepting a bad model is the Type II error; reducing it means raising test power.
2. Power increases when more exceptions are expected: lowering VaR confidence from 99% to 95% raises expected exceptions in 250 days from 2.5 to 12.5.
3. Reducing sample size or raising VaR confidence reduces the information in the sample; widening the non-rejection region makes acceptance more likely.

Why B is correct: A lower VaR confidence level produces more exceptions, giving the test more power to detect an inaccurate model.

Why the others are wrong

- **A:** Fewer exceptions are expected at 99%, so the test has less power and Type II error rises.
- **C:** A smaller sample lowers power; reliability improves with a larger sample.
- **D:** A higher non-rejection confidence widens the acceptance region, reducing Type I error but increasing Type II error.

Q222 BT answers L2.T5.60.1

Basel's traffic-light market risk backtest (green/yellow/red zones) tests the null hypothesis that the bank's 99% VaR model is accurate, with the yellow zone beginning at 5 exceptions in 250 days. Which of the following is MOST difficult to quantify?

A. Probability of Type I error
B. Probability of Type II error
C. Probability of accepting an accurate VaR model
D. Significance level of backtest

Answer: B

Step by step

1. Under the null (true p = 1%), $P(N \geq 5) = 10.8\%$ = Type I error = significance level (computed in Python: 0.1078).
2. Probability of accepting an accurate model $= 1 - 0.108 = 89.2\%$.
3. Type II error needs a specific alternative (true coverage might be 98%, 97%, 92%...). For example at a true 97% coverage, $P(N \leq 4) = 12.8\%$, but it changes with every alternative, so no single number exists.

Why B is correct: Type I error is computed under a single, known null; Type II error depends on the unknown true failure rate.

Why the others are wrong

- **A:** Directly computable: 10.8% from the binomial under the null.
- **C:** Computable as one minus the Type I error, 89.2%.
- **D:** The significance level equals the Type I error probability (10.8%), so it is quantifiable.

Q223 BT answers L2.T5.60.2

REBUILT — Original distractors 8.2% and 12.6% were arbitrary (OCR-damaged); replaced with off-by-one binomial tail probabilities.

Under the Basel II internal models approach with 250 days of data, the green zone is currently four or fewer exceptions, with five or more exceptions (yellow or red zone) signifying rejection of the VaR model. Assuming the model is truly a correct 99% VaR model, the probability of a Type I error is currently 10.8%. If the Committee widened the green zone to five or fewer exceptions (so that only six or more exceptions reject the model), what would the probability of a Type I error become?

A. 10.8%
B. 4.1%
C. 24.2%
D. 1.4%

Answer: B

Step by step

1. A Type I error is rejecting a correct model, so evaluate the rejection region under the null: $X \sim \text{Bin}(n = 250, p = 0.01)$.
2. New rejection region: six or more exceptions, so $P(X \geq 6) = 1 - P(X \leq 5)$.
3. $P(X \leq 5) = 0.9588$, so $P(X \geq 6) = 1 - 0.9588 = 0.0412 \approx 4.1\%$.
4. The Type I error falls from 10.8% to 4.1% (at the cost of a higher Type II error).

Why B is correct: Widening the green zone shrinks the rejection region, so a correct model is wrongly rejected less often; the probability is the binomial tail $P(X \geq 6)$.

Why the others are wrong

- **A:** This is the current Type I error, $P(X \geq 5)$; it ignores the change in the cutoff.
- **C:** This is $P(X \geq 4) = 24.2\%$, which moves the cutoff the wrong way (narrowing the green zone to three).
- **D:** This is $P(X \geq 7) = 1.4\%$, an off-by-one error that treats six exceptions as acceptable.

Q224 BT answers L2.T5.60.3 · BT notes P2.T5.60.3

Assume a bad 99% VaR model has a true level of coverage of 97%; that is, losses exceed VaR 3% of the time rather than 1%. Assume a yellow or red zone outcome signifies rejection of the model. Currently the green zone is four or fewer exceptions in 250 days, giving a Type II error probability of 12.8%. If the Committee narrowed the green zone to three or fewer exceptions, what would the probability of a Type II error become?

A. 1.9%

B. 5.7%
C. 10.8%
D. 12.8%

Answer: B
Step by step
1. A Type II error is accepting a bad model, i.e. landing in the green zone when the true p = 3%: $X \sim \text{Bin}(250, 0.03)$.
2. New green zone: $X \leq 3$.
3. $P(X \leq 3) = 0.0565 \approx 5.7\%$.
4. Check: $P(X \leq 4) = 12.8\%$, matching the current figure in the stem.

Why B is correct: Type II error is the probability the bad model produces few enough exceptions to be accepted; shrinking the green zone cuts that probability to $P(X \leq 3 \mid p = 3\%) = 5.7\%$.

Why the others are wrong
- **A:** This is $P(X \leq 2) = 1.9\%$, an off-by-one error excluding three exceptions.
- **C:** This is the Type I error at 99% confidence, $P(X \geq 5 \mid p = 1\%)$, not a Type II error.
- **D:** This is the current Type II error (green zone up to four); it ignores the narrowing of the green zone.

Q225

BT answers L2.T5.60.6 · BT notes P2.T5.60.6

If a regulator wants to minimize the probability of accepting a bad internal market risk value at risk (VaR) model, EACH of the following, in theory, will help that goal EXCEPT for:

A. Shrink (expand) the green (red) zone
B. Increase sample size
C. Decrease VaR confidence level
D. Increase VaR confidence level

Answer: D
Step by step
1. Accepting a bad model is a Type II error; the goal is to reduce it (raise test power).
2. Smaller green zone: fewer outcomes lead to acceptance, so Type II error falls.
3. Larger sample: more observations make the exception count more informative, so power rises.
4. Lower VaR confidence level (e.g. 95%): more expected exceptions, so a bad model is easier to distinguish from a good one.
5. Higher confidence level (e.g. 99.9%): even fewer expected exceptions, so a bad model is harder to detect and Type II error rises. This is the EXCEPT answer.

Why D is correct: A higher VaR confidence level produces fewer tail exceptions, which weakens the test and raises the chance of accepting a bad model.

Why the others are wrong
- **A:** This helps: narrowing the acceptance region lowers Type II error (at the cost of more Type I error).
- **B:** This helps: a larger sample raises the power of the backtest.
- **C:** This helps: a lower confidence level yields more exceptions and so more statistical power.

Q226

BT answers P2.T5.22.7.1 · BT notes P2.T5.22.7.1

Mary conducts a backtest of her firm's one-day 99.0% VaR model using 500 daily observations (two years). She applies the binomial distribution with n = 500 and p = 0.010, shown below. She selects a cutoff of nine exceptions: she rejects the model as inaccurate if she observes nine or more exceptions (X >= 9). If the model is accurate, what is the probability that she commits a Type I error?

No. of exceptions (out of 500)	pdf	CDF
0	0.66%	0.66%
1	3.32%	3.98%
2	8.36%	12.34%
3	14.02%	26.36%
4	17.60%	43.96%
5	17.64%	61.60%
6	14.70%	76.29%
7	10.48%	86.77%
8	6.52%	93.29%
9	3.60%	96.89%
10	1.79%	98.68%

A. 1.79%
B. 3.11%
C. 3.60%
D. 6.71%

Answer: D

Step by step

1. Type I error = rejecting an accurate model, so use the null distribution (p = 1%).
2. Rejection region: $X \geq 9$.
3. $P(X \geq 9) = 1 - P(X \leq 8) = 1 - 0.9329 = 0.0671$.
4. Type I error = 6.71%.

Why D is correct: Under the null, the probability of falling in the rejection region is the upper tail $1 - \text{CDF}(8)$.

Why the others are wrong

- **A:** This is the pdf at 10 exceptions, an arbitrary single-point probability.
- **B:** This is $1 - \text{CDF}(9) = P(X \geq 10)$, an off-by-one error that uses the CDF at 9 instead of 8.
- **C:** This is the pdf at exactly nine exceptions; the rejection region includes all outcomes of nine or more.

Q227

BT answers P2.T5.22.7.2 · BT notes P2.T5.22.7.2

Peter backtests his firm's one-day 99.0% VaR model over 500 daily observations. He rejects the model if he observes nine or more exceptions (X >= 9) and otherwise accepts it. Below are the binomial pdf and CDF (n = 500) for a model whose true coverage is inaccurate, at 98.0% (p = 0.02), 97.0% and 96.0%.

No. of exceptions	98.0% pdf	98.0% CDF	97.0% pdf	97.0% CDF	96.0% pdf	96.0% CDF
0	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
1	0.04%	0.05%	0.00%	0.00%	0.00%	0.00%
2	0.21%	0.26%	0.00%	0.00%	0.00%	0.00%
3	0.72%	0.98%	0.01%	0.02%	0.00%	0.00%
4	1.83%	2.81%	0.06%	0.08%	0.00%	0.00%
5	3.71%	6.52%	0.18%	0.25%	0.00%	0.01%
6	6.24%	12.76%	0.45%	0.70%	0.02%	0.02%
7	8.99%	21.75%	0.98%	1.68%	0.04%	0.07%
8	11.30%	33.05%	1.86%	3.54%	0.11%	0.18%
9	12.61%	45.67%	3.15%	6.69%	0.26%	0.44%
10	12.64%	58.30%	4.79%	11.48%	0.53%	0.97%

If the firm's true coverage level is 98.0%, what is the implied probability of a Type II error?

A. 2.81%
B. 11.30%
C. 33.05%
D. 66.95%

Answer: C

Step by step

1. Type II error = accepting an inaccurate model; Peter accepts when $X \leq 8$.
2. Use the 98.0% (p = 0.02) column: $P(X \leq 8) = \text{CDF}(8) = 33.05\%$.
3. Check: $1 - 33.05\% = 66.95\%$ is the power (probability of correctly rejecting).

Why C is correct: The Type II error probability is the chance the inaccurate model still produces eight or fewer

exceptions, read from the CDF at 8 for the 98% column.

Why the others are wrong

- **A:** This is the 98% CDF at four exceptions, the wrong cutoff (confused with the Basel green-zone cutoff).
- **B:** This is the pdf at exactly eight exceptions rather than the cumulative probability of eight or fewer.
- **D:** This is 1 - 33.05%, the power of the test (probability of rejecting the bad model), not the Type II error.

Q228

BT answers P2.T5.22.8.1 · BT notes P2.T5.22.8.1

Bertha backtests a one-day 97.5% VaR of $1.40 million over the prior 250 days using a 95.0% confident two-tailed test. Expected exceptions are 2.5% × 250 = 6.25. She sets the acceptance region at 2 < X < 12, so the lower rejection region is X ≤ 2 and the upper rejection region is X ≥ 12, where X is the number of exceptions. Which of the following statements is TRUE?

A. The probability of a Type II error is about 7.44%.

B. Bertha can increase the power of her test by shrinking the acceptance region to 3 < X < 11.

C. Bertha's test will be more powerful if she shifts, ceteris paribus, from backtesting the 97.5% VaR to backtesting the 99.0% VaR.

D. If Bertha doubles the sample size to N = 500 days, the acceptance region will remain 2 < X < 12, but the one-day 97.5% VaR would approximately double to about $2.80 million.

Answer: B

Step by step

1. Under the null (p = 2.5%, N = 250) the acceptance region is X = 3,...,11. Type I error = P(X ≤ 2) + P(X ≥ 12) = 1 − 0.9256 = 7.44% (binomial).
2. Type II error needs a specific alternative (a model that is actually bad); with true p = 4%, P(accept 3 ≤ X ≤ 11) = 69.7%. So 7.44% is the Type I error, not Type II.
3. Shrink the acceptance region to 4 ≤ X ≤ 10: with true p = 4% the probability of wrongly accepting falls to 57.4%, so power (1 − Type II) rises from 30.3% to 42.6%; the cost is a higher Type I error (17.9% vs 7.44%).
4. Moving to a 99% VaR lowers expected exceptions to 2.5, so a bad model is harder to distinguish from a good one: power falls.
5. At N = 500 expected exceptions are 12.5, so the acceptance region shifts to about 6 to 20; the dollar VaR depends on the P/L distribution, not on N.

Why B is correct: Power = 1 − P(Type II). A narrower acceptance region rejects more often, so a bad model is more likely to be caught, at the price of a higher Type I error.

Why the others are wrong

- **A:** 7.44% is the Type I error (rejecting a correct model, computed under the null p = 2.5%); confusing Type I with Type II.
- **C:** A higher VaR confidence level means fewer expected exceptions, so the test has less ability to detect a mis-specified model; power falls, not rises.
- **D:** Doubling N to 500 would scale expected exceptions to 12.5 and move the acceptance region (it does not stay at 2 < X < 12); the VaR level is a property of the P/L distribution and does not double with sample size.

Q229

BT answers P2.T5.713.2 · BT notes P2.T5.713.2

Barbara simulates the backtest of a proposed VaR model. The Risk Committee asked her to reduce the probability of a Type II error. The previous version assumed a 97.5% VaR model, a two-tailed backtest confidence level of 95.0%, and a sample of two years of daily P/L (500 trading days). Which of the following adjustments is the MOST EFFECTIVE way to reduce the Type II error rate?

A. Decrease the sample size from 500 days to 250 days

B. Increase the cutoff (the number of acceptable exceptions)

C. Increase the VaR confidence level from 97.5% to 99.0%

D. Decrease the two-tailed backtest confidence level from 95.0% to 90.0%

Answer: D

Step by step

1. Type II error = failing to reject a bad model. It falls only if the acceptance region shrinks or the test gains power.
2. Illustration (N = 500, p0 = 2.5%, true p = 3.5%): at 95% two-tailed the rejection region is X ≤ 5 or X ≥ 21, Type I = 3.0%, Type II = 77.2%.
3. At 90% two-tailed the rejection region becomes X ≤ 6 or X ≥ 19, Type I = 8.3%, Type II = 60.9%: Type II falls,

Type I rises.
4. Cutting N to 250 gives (95%) Type II = 82.9%, worse. Raising the cutoff to X ≥ 24 raises Type II to 92.3% (acceptance 6-23). Testing a 99% VaR (N = 500, p0 = 1%, true 1.4%) gives Type II about 90%, worse.

Why D is correct: Lowering the backtest confidence level narrows the acceptance region, so bad models are rejected more often; the trade-off is a higher Type I error.

Why the others are wrong

- **A:** Less data lowers power; at N = 250 the Type II error rises (about 83% vs 77% in the illustration).
- **B:** Raising the cutoff widens the acceptance region, which accepts more bad models and increases Type II error.
- **C:** A 99% VaR produces fewer expected exceptions, making deviations harder to detect, so Type II error increases.

Q230

GARP PE2 Q70 · Mock 4 Q71

A model validation team at a bank is backtesting the bank's VaR model. One team member is concerned that the validation could result in a Type I or a Type II error. Which of the following is correct regarding Type I and Type II errors?

A. The probability of committing a Type II error decreases when the sample size increases and the level of significance is held constant.

B. A backtest is considered statistically powerful if it minimizes the probability of committing a Type II error regardless of the probability of committing a Type I error.

C. A Type I error is committed when an incorrectly specified model is accepted.

D. A Type II error is committed when a correctly specified model is rejected.

Answer: A

Step by step

1. Definitions: Type I = rejecting a correct model; Type II = failing to reject (accepting) an incorrect model.
2. With the significance level (Type I rate) fixed, a larger sample makes the distribution of the failure rate tighter, so the test separates good from bad models better; power rises and Type II falls.
3. Power is judged jointly with Type I error; Type II alone can be driven to zero by rejecting every model.

Why A is correct: More data at a fixed significance level lowers the Type II error probability.

Why the others are wrong

- **B:** Ignoring Type I error is meaningless: a test that rejects every model has zero Type II error but is useless; power must be assessed at a given Type I level.
- **C:** This is the definition of Type II error; the two are reversed.
- **D:** This is the definition of Type I error; the two are reversed.

LO 4.e

Explain the need to consider conditional coverage in the backtesting framework.

Q231

AP Mock 2 Q25

REBUILT — Options C and D named specific causes that could also be defended as explanations of clustering; rewrote them as clearly weaker or irrelevant so only the independence-failure diagnosis is correct.

A bank backtests its 99% one-day VaR model and observes that several exceptions occurred within a short span of consecutive days, although the total number of exceptions over the sample is close to the expected number. What does this clustering of exceptions MOST likely indicate?

A. The confidence level is set too low, so the model produces too many exceptions overall.

B. The exceptions are not independent over time, so the model does not adequately capture time-varying risk such as volatility clustering.

C. The total number of exceptions is too large, so the model fails the unconditional coverage test.

D. The sample is too small, so a Type II error is certain to occur.

Answer: B

Step by step

1. The total count of exceptions is close to the expected count, so unconditional coverage is satisfied.

2. Clustering concerns the timing of exceptions: they should be spread evenly through time (independent).
3. Bunching signals that the model does not respond to changing conditions (e.g., volatility clustering, changed correlations or positions), which is a failure of the independence component of conditional coverage.

Why B is correct: Conditional coverage requires the right number of exceptions and that they be serially independent; bunching violates independence even when the count is right.

Why the others are wrong

- **A:** A confidence level that is too low would change the number of exceptions, not their timing; the stem says the count is about right.
- **C:** Misreads the stem: the count is stated to be near expectation, so unconditional coverage is not the issue; clustering is a timing issue.
- **D:** Sample size affects test power but does not make an error 'certain'; it does not explain why the exceptions bunch.

Q232 AnalystPrep Q.6442

The risk management team at Horizon Bank validates its VaR model with backtesting. It starts with an unconditional coverage test to check whether the observed frequency of exceptions matches the model's predicted exception rate. The team is also concerned that exceptions may cluster in certain periods, indicating that the model does not account for time-varying risk. Which of the following would address this concern?

A. Perform a conditional coverage test.
B. Compare the VaR model against historical market data.
C. Conduct a stress test to evaluate the model under extreme scenarios.
D. Use a rolling window analysis to assess model performance over time.

Answer: A

Step by step

1. Unconditional coverage only compares the number of exceptions with the expected number; timing is ignored.
2. Clustering is a violation of independence of exceptions.
3. The conditional coverage test adds an independence test: $LR_{cc} = LR_{uc} + LR_{ind}$, with LR_{cc} compared with the chi-squared (2 d.f.) cutoff of 5.99 and LR_{ind} with 3.84.

Why A is correct: Only the conditional coverage framework explicitly tests whether exceptions are serially independent, which is what clustering violates.

Why the others are wrong

- **B:** Comparing with historical data is what the backtest already does; it adds no test of independence.
- **C:** Stress testing examines extreme hypothetical scenarios and does not test the timing pattern of observed exceptions.
- **D:** Rolling windows show how performance changes, but without a statistical independence test they do not establish whether exceptions cluster.

Q233 BT answers L2.T5.59.3

A conditional coverage model would be appropriate for EACH of the following EXCEPT in the case of:

A. Serial independence of exceptions
B. Serial correlation of exceptions
C. Time variation of exceptions
D. Recent volatility clustering of exceptions

Answer: A

Step by step

1. Conditional coverage extends the unconditional test to allow for time variation and dependence of exceptions.
2. Serial correlation, time variation and volatility clustering are all forms of dependence/time variation, so a conditional model is needed.
3. Serial independence means exceptions are already unrelated over time, so the unconditional framework suffices.

Why A is correct: A conditional framework is needed only when exceptions are dependent or time-varying; independence removes that need.

Why the others are wrong

- **B:** Serial correlation means exceptions depend on past exceptions, which is exactly what conditional coverage

addresses.
- **C:** Time variation is the very feature conditioning is meant to capture.
- **D:** Volatility clustering produces bunched exceptions, a classic reason to use conditional coverage.

Q234

BT answers P2.T5.713.3 · BT notes P2.T5.713.3

An analyst evaluates the latest backtest results of his firm's VaR model. He concludes that the firm should shift to a VaR model that uses filtered historical simulation with GARCH volatility, and his primary reason is that his statistical test shows that a conditional coverage framework is necessary. Which of the following is the MOST likely proximate justification for a shift to a conditional coverage model?

A. Extreme losses are bunching (i.e., not independent) in the sample.

B. His test provides evidence of a long-run (unconditional) volatility in the series.

C. The historical sample is not large enough to give sufficiently low Type I and Type II error rates.

D. His Kupiec test implies a confidence region that extends to zero exceptions, so he cannot evaluate outcomes with few exceptions.

Answer: A

Step by step

1. The basic (unconditional) backtest treats daily exceptions as independent Bernoulli trials.
2. Evidence that exceptions bunch contradicts independence, i.e., there is time variation in risk.
3. A conditional model (e.g., GARCH-filtered HS) that updates volatility through time is the remedy.

Why A is correct: Bunching of exceptions violates the independence assumption and signals time-varying risk, which is the reason to move to conditional coverage.

Why the others are wrong

- **B:** A long-run volatility is a constant, unconditional feature and does not call for a conditional model.
- **C:** Small sample size is a power problem (Type I/II errors) and is unrelated to the timing of exceptions.
- **D:** The Kupiec confidence region is a feature of unconditional coverage testing at low exception counts; it does not motivate conditioning.

Q235

BT notes P2.T5.22.8.3 · BT answers P2.T5.22.8.3

An analyst backtests a VaR model using Christoffersen's approach, with joint statistic $LR_{cc} = LR_{uc} + LR_{ind}$. The results are: $LR_{uc} = 2.58$, below the $\chi^2(1)$ cutoff of 3.841, so the null is not rejected; and $LR_{ind} = 15.36$, above the cutoff, so the null is rejected. The analyst hopes that the model is unbiased and that exceptions do not cluster. What do the results say?

A. Good news: the VaR model is unbiased and there is no clustering.

B. Mixed news: the VaR model is unbiased, but there is clustering.

C. Mixed news: there is no clustering, but the VaR model is biased.

D. Bad news: the VaR model is biased and there is clustering.

Answer: B

Step by step

1. LR_{uc} tests whether the exception frequency equals the expected frequency: 2.58 < 3.841, so we fail to reject correct coverage (model unbiased).
2. LR_{ind} tests independence of exceptions: 15.36 > 3.841, so we reject independence (exceptions cluster).
3. Check: $LR_{cc} = 2.58 + 15.36 = 17.94 > 5.99$ ($\chi^2(2)$), so the model fails conditional coverage overall, driven by the independence term.
4. Result: unbiased coverage but clustered exceptions, i.e., mixed news.

Why B is correct: Failing to reject LR_{uc} means the frequency is acceptable, while rejecting LR_{ind} means exceptions are serially dependent.

Why the others are wrong

- **A:** Ignores the rejection of the independence null; it only reads the LR_{uc} result.
- **C:** Reverses the two tests: a rejected LR_{ind} means clustering, and a non-rejected LR_{uc} means no bias.
- **D:** Treats the non-rejection of LR_{uc} as a rejection, i.e., misreads 2.58 against the 3.841 cutoff.

Q236 Schweser Module Quiz 4.1 Q2

Unconditional testing does not reflect the:

A. size of the portfolio.
B. number of exceptions.
C. confidence level chosen.
D. timing of the exceptions.

Answer: D

Step by step

1. Unconditional coverage compares the observed number of exceptions with the number expected at the chosen confidence level.
2. It treats each day alike and does not consider when the exceptions occurred.
3. Timing (bunching) is captured only by adding the independence test, giving conditional coverage.

Why D is correct: Unconditional testing counts exceptions only; it is blind to whether they cluster in time.

Why the others are wrong

- **A:** Portfolio size is embedded in the VaR and P&L figures that define an exception, so it is reflected.
- **B:** The number of exceptions is exactly what unconditional testing examines.
- **C:** The confidence level sets the expected exception rate used as the null in the test.

LO 4.f

Describe the Basel rules for backtesting.

Q237 BT answers L2.T5.60.4 · BT notes P2.T5.60.4

REBUILT — Source option B was internally inconsistent ('0 to +1.0' vs '3.0 to 3.85') and one source had no key; rewrote option B to the Basel yellow-zone plus-factors of +0.40 to +0.85.

If a bank's backtest produces exceptions in the yellow zone (five to nine exceptions out of 250 days at 99% VaR), how is the VaR multiplier treated?

A. The multiplier remains at 3.0.
B. The multiplier might be increased by between +0.40 and +0.85 (to between 3.40 and 3.85), at the supervisor's discretion depending on the cause of the exceptions.
C. The multiplier must be increased to 4.0 or greater.
D. An additional stressed VaR must be added to the standard VaR.

Answer: B

Step by step

1. Yellow zone = 5 to 9 exceptions; the plus-factor scale runs from +0.40 (5 exceptions) to +0.85 (9 exceptions).
2. The base multiplier is 3, so the possible range is $3 + 0.40 = 3.40$ to $3 + 0.85 = 3.85$.
3. Application depends on the cause: model integrity/accuracy problems should attract the penalty, intraday trading should be considered, bad luck gives no guidance.
4. Only the red zone (10 or more) makes the increase to 4.0 automatic.

Why B is correct: In the yellow zone the increase is graduated (+0.40 to +0.85) and applied with supervisory discretion based on why the exceptions occurred.

Why the others are wrong

- **A:** Ignores the penalty: 3.0 is the green-zone multiplier, and in the yellow zone a plus-factor is presumed unless the bank shows it is not warranted.
- **C:** Describes the red zone (10 or more exceptions), where the multiplier rises to 4.0 automatically.
- **D:** Stressed VaR is a separate capital add-on and is not the backtesting penalty mechanism.

Q238 BT answers L2.T5.60.5 · BT notes P2.T5.60.5

In regard to backtest outcomes in the yellow zone, which of the following is MOST accurate?

A. The increase to the multiplier is specified (e.g., +0.65 at 7 exceptions), and the supervisor must apply it in all cases.
B. The increase to the multiplier is specified, but the supervisor has discretion in whether to apply it.

C. The increase to the multiplier is subjective, but the supervisor must apply it in all cases.
D. The increase to the multiplier is subjective and the supervisor has discretion in whether to apply it.

Answer: B
Step by step
1. Basel specifies a scale of plus-factors: +0.40 (5), +0.50 (6), +0.65 (7), +0.75 (8), +0.85 (9 exceptions), so the size of the increase is given, not subjective.
2. Applying it in the yellow zone is subject to supervisors' judgement of the cause (model integrity/accuracy: should apply; intraday trading: consider; bad luck: no guidance).
3. Only in the red zone (10 or more, +1.0) is the increase automatic.
4. So: increase is specified, application is discretionary.

Why B is correct: The yellow-zone plus-factors are fixed by the Basel table, but whether they are imposed depends on supervisory discretion about the cause of the exceptions.
Why the others are wrong
- **A:** Right on size but wrong on automatic application; automatic penalty applies only in the red zone.
- **C:** Reverses the two features: the size is specified, and application is not mandatory in all cases.
- **D:** Right on discretion but wrong that the size is subjective; the plus-factors are tabulated.

Q239

BT notes P2.T5.57.3

REBUILT — Original question was tangled, quote-heavy and unkeyed; rewrote as a clear EXCEPT item on the Type I/Type II trade-off in setting Basel zone thresholds, with a supporting binomial calculation.

Under Basel's backtest, a 99% one-day VaR is tested over 250 days (green 0-4, yellow 5-9, red 10 or more exceptions). Regulators want to catch banks that understate risk without unduly penalizing banks with a correct model that are merely unlucky. EACH of the following is true EXCEPT:

A. Shrinking the green zone (e.g., to 0-2 exceptions) increases the probability of catching banks that understate their risk.
B. Raising the red-zone threshold (e.g., to 13 or more exceptions) makes it less likely that a bank with a correct model is penalized through bad luck.
C. Lowering the red-zone threshold (e.g., to 7 or more exceptions) makes it less likely that a bank with a correct model is penalized through bad luck.
D. Lowering the red-zone threshold (e.g., to 7 or more exceptions) increases the probability of catching banks that understate their risk.

Answer: C
Step by step
1. For a correct model, exceptions $\sim$ Binomial(250, 0.01). $P(X \geq 10) \approx 0.025\%$, $P(X \geq 7) \approx 1.4\%$, $P(X \geq 13) \approx 0.0002\%$.
2. Lowering the threshold to 7 raises the chance a correct model is wrongly flagged (Type I error) from about 0.025% to about 1.4%, so C is false.
3. The same lowering makes it easier to flag a truly deficient model (fewer Type II errors), so D is true.
4. Raising the threshold to 13 cuts false penalties (B true); shrinking the green zone to 0-2 flags more deficient banks (A true).

Why C is correct: Tighter thresholds (smaller green zone, lower red threshold) trade more Type I errors for fewer Type II errors; looser thresholds do the reverse.
Why the others are wrong
- **A:** A true statement: a tighter green zone catches more faulty models, at the cost of more false penalties.
- **B:** A true statement: a higher red threshold lowers the Type I (bad luck) penalty probability.
- **D:** A true statement: a lower red threshold catches more understating banks; the error is to think this also protects unlucky banks.

Q240

GARP PE1 Q58 · Mock 3 Q23

A regulatory analyst at an investment bank is reviewing the Basel rules for backtesting VaR models. Under the Basel framework, a penalty can be given to banks that have more than four exceptions to their 1-day 99% VaR over the last 250 trading days. Which of the following scenarios is most likely to result in a penalty?

A. A large move in interest rates occurs in conjunction with a small move in correlations.
B. The bank's model calculates interest rate risk based on the median duration of the bonds in the portfolio.

C. A sudden market crisis in an emerging market leads to losses in the bank's equity positions in that country.
D. A sudden devastating earthquake causes major losses in the bank's key area of operation.

Answer: B
Key correction: Mock 3 Q23 (a variant with different options) keyed C (emerging-market crisis); under the Basel categories that is bad luck, so the model-deficiency scenario in B is the one that attracts a penalty.
Step by step
1. Basel classes causes of exceptions: basic integrity of model lacking and model accuracy needs improvement (penalty should apply), intraday trading (consider), bad luck (no penalty guidance).
2. A large rate move with a small correlation move, an emerging-market crisis, and an earthquake are unforeseeable market/event shocks: bad luck.
3. Using the median duration to measure interest rate risk is a model-accuracy flaw that the bank could fix.
4. Hence B is the scenario most likely to be penalized.

Why B is correct: Penalties target avoidable model deficiencies (integrity or accuracy), not exceptions caused by unforeseeable market or natural events.
Why the others are wrong
- **A:** Unusual market conditions (rate and correlation moves) are classified as bad luck.
- **C:** A market crisis in one country is bad luck; the clustering of exceptions does not turn it into a model flaw under Basel's categories.
- **D:** Natural disasters are bad luck, and exceptions from them may be excluded.

Q241

Mock 1 Q48

REBUILT — Source distractors were unexplained; stated the z-values used and rebuilt distractors from realistic errors (no multiplier, wrong multiplier, no sqrt-10 scaling).

A risk manager computes the market risk capital charge for a trading portfolio under the internal models approach, $\max(\text{VaR}_{t-1},\ k \times \text{VaR}_{avg,60})$. The bank's own VaR is at a 95% confidence level and a one-day horizon:
VaR of previous trading day: USD 40,000
Average VaR of last 60 trading days: USD 22,000
Backtesting zone: red
Assume normally distributed returns, scale to the regulatory 99% confidence level and 10-day horizon using z-values of 1.65 (95%) and 2.33 (99%) and the square-root-of-time rule. What is the market risk charge?

A. USD 178,621
B. USD 124,267
C. USD 294,724
D. USD 392,966

Answer: D
Key correction: Source key D was correct; its working gave 392,944 because of rounding.
Step by step
1. Scaling factor $= (2.33/1.65)\sqrt{10} = 1.4121 \times 3.1623 = 4.4655$.
2. $\text{VaR}_{t-1} = 40{,}000 \times 4.4655 = 178{,}621$; $\text{VaR}_{avg} = 22{,}000 \times 4.4655 = 98{,}241$.
3. Red zone: $k = 4$, so $k \times \text{VaR}_{avg} = 392{,}966$.
4. Charge $= \max(178{,}621;\ 392{,}966) = 392{,}966$.

Why D is correct: The charge is the larger of yesterday's VaR and k times the 60-day average VaR, with k = 4 in the red zone, after rescaling to 99%/10-day.
Why the others are wrong
- **A:** Reports only yesterday's scaled VaR (178,621) and ignores the multiplier term.
- **B:** Forgets the square-root-of-10 scaling: $4 \times 22{,}000 \times 1.41212 = 124{,}267$.
- **C:** Uses the green-zone multiplier of 3: $3 \times 98{,}241 = 294{,}724$.

Q242

Mock 1 Q78

REBUILT — Source distractors were unexplained; kept the data, specified the regulatory 99% row and rebuilt options from realistic errors (k=3, wrong confidence level).

A bank uses VaR and stressed VaR (SVaR) for its market risk capital requirement: $\max(\text{VaR}_{t-1},\ k\,\overline{\text{VaR}}_{60}) + \max(\text{SVaR}_{t-1},\ k\,\overline{\text{SVaR}}_{60})$. The regulatory 10-day figures are taken at the 99% confidence level. Data:

Confidence level	Latest 10-day VaR	Latest 10-day SVaR	Avg 10-day VaR, last 60 days	Avg 10-day SVaR, last 60 days
95%	538	844	243	610
99%	756	1,039	544	865
99.9%	962	1,527	726	1,229

The VaR backtest shows 10 exceptions over the past 250 days. What is the overall market risk capital charge?

A. 4,227
B. 5,636
C. 7,820
D. 3,412

Answer: B

Key correction: Source key C (5,636) was correct; it is option B in this rebuilt version.

Step by step

1. 10 exceptions is the red zone, so $k = 4$ (3 + 1.0).
2. VaR term: max(756, $4 \times 544 = 2{,}176$) = 2,176.
3. SVaR term: max(1,039, $4 \times 865 = 3{,}460$) = 3,460.
4. Total = 2,176 + 3,460 = 5,636.

Why B is correct: Red-zone backtesting sets k = 4, which drives both the VaR and SVaR terms above the latest-day values.

Why the others are wrong

- **A:** Uses the green-zone multiplier 3: 1,632 + 2,595 = 4,227.
- **C:** Reads the 99.9% row: 2,904 + 4,916 = 7,820.
- **D:** Reads the 95% row: 972 + 2,440 = 3,412.

Q243

Schweser Module Quiz 4.2 Q1

The Basel Committee has established four categories of causes for exceptions. Which of the following does not apply to one of those categories?

A. The sample is small.
B. Intraday trading activity.
C. Model accuracy needs improvement.
D. The basic integrity of the model is lacking.

Answer: A

Step by step

1. Basel's four categories: basic integrity of the model lacking, model accuracy needs improvement, intraday trading activity, and bad luck.
2. B, C and D are three of these; sample size is not a category.
3. So A is the one that does not apply.

Why A is correct: Basel classifies exception causes as model integrity, model accuracy, intraday trading, and bad luck; sample size is a statistical issue, not a Basel category.

Why the others are wrong

- **B:** Intraday trading activity is a Basel category (penalty should be considered).
- **C:** Model accuracy needs improvement is a Basel category (penalty should apply).
- **D:** Basic model integrity lacking is a Basel category (penalty should apply).

Q244

Schweser Module Quiz 62.1 Q2

Saugatuck National Bank uses the internal model-based approach to set market risk capital as prescribed by the 1996 Amendment to the 1988 Basel Accord. The bank has backtested its 99% one-day VaRs against the actual losses over the last 250 trading days. Based on the results of the backtesting, the bank recorded 11 exceptions. Based on these results, the multiplicative factor (m) in the model should be set:

A. less than 3.
B. equal to 3.
C. between 3.1 and 3.9.
D. equal to 4.

Answer: D

Step by step

1. Backtest window is 250 days at 99%, so the expected number of exceptions is $250 \times 0.01 = 2.5$.
2. Basel zones: green = 0-4 exceptions (multiplier 3.00); yellow = 5-9 exceptions (multiplier 3.40 to 3.85); red = 10 or more exceptions.
3. The bank has 11 exceptions, which is 10 or more, so it is in the red zone.
4. The red zone carries an automatic plus factor of 1, so $m = 3 + 1 = 4$.

Why D is correct: Ten or more exceptions in 250 days is the Basel red zone, which automatically raises the multiplier to its maximum of 4.

Why the others are wrong

- **A:** The multiplier is never below 3; it starts at 3 and only increases with exceptions.
- **B:** m = 3 applies only to the green zone (0-4 exceptions); 11 exceptions is far beyond that.
- **C:** The 3.1-3.9 range is only an approximation of the yellow zone (5-9 exceptions, 3.40-3.85); 11 exceptions exceeds that zone.

Reading 5. VaR Mapping

Philippe Jorion, Value at Risk: The New Benchmark for Managing Financial Risk, 3rd Edition (New York, NY: McGraw-Hill, 2007). Chapter 11.

LO	Learning objective	Qs
5.a	explain the principles underlying VaR mapping and describe the mapping process.	17
5.b	explain how the mapping process captures general and specific risks, and calculate these risks in a portfolio given a set of primitive risk factors.	6
5.c	differentiate among the three methods for mapping portfolios of fixed-income securities.	16
5.d	summarize how to map a fixed-income portfolio into positions of standard instruments.	2
5.e	describe how mapping of risk factors can support stress testing.	2
5.f	explain how VaR can be calculated and used relative to a performance benchmark.	7
5.g	describe the method of mapping forwards, forward rate agreements, interest rate swaps, and options.	23

LO 5.a

Explain the principles underlying VaR mapping and describe the mapping process.

Q245

Mock 2 Q43

REBUILT — Original distractors (1.5M, 3.0M) did not correspond to realistic mistakes; replaced with plausible counting errors and tightened the stem.

An equity portfolio holds 3,000 stocks. Each stock has a market component and a firm-specific component, and each stock is treated as its own risk factor. How many distinct covariance terms (between different risk factors) would be needed to evaluate the correlations among all of these risk factors without using mapping?

A. 3,000
B. About 4.5 million
C. About 8.997 million
D. 9.0 million

Answer: B

Step by step

1. With $N = 3{,}000$ risk factors, each distinct pair needs one covariance term.
2. Number of pairs = $N(N-1)/2 = 3{,}000 \times 2{,}999/2 = 4{,}498{,}500 \approx 4.5$ million.
3. Mapping the market component of each stock onto a few general factors (e.g. a market index) removes the need for this huge matrix.

Why B is correct: The covariance matrix is symmetric, so only $N(N-1)/2$ off-diagonal pairs are distinct; this explosion in parameters is the motivation for VaR mapping.

Why the others are wrong

- **A:** Counts only one term per stock (the variances) and ignores covariances between different factors.
- **C:** Uses $N(N-1) = 8{,}997{,}000$, counting each pair twice and ignoring symmetry.
- **D:** Uses $N^2 = 9{,}000{,}000$, counting the full matrix including the diagonal variances and both triangles.

Q246

AnalystPrep Q.1511

REBUILT — Original had a giveaway answer and distractors that were obviously wrong; rewrote the stem and options to test the core principle with plausible alternatives.

Which statement best describes the principle underlying VaR mapping?

A. Only the most extreme (abnormal) risk factors are retained, because normal factors do not contribute to tail losses.
B. Each position is mapped to every risk factor that could affect its price, so that no approximation error is introduced.
C. Every portfolio is mapped to exactly five risk factors regardless of its composition.
D. The current values of many positions are replaced by exposures to a smaller set of selected primitive risk factors, because modeling each position separately is impractical.

Answer: D

Step by step

1. VaR mapping first measures current positions, then replaces them with exposures to common risk factors.
2. Purpose: aggregate risk when valuing and modeling each position separately would require too many computations.
3. The number of factors is a choice trading accuracy against complexity, not a fixed number or all conceivable factors.

Why D is correct: Mapping aggregates many positions into exposures on a limited set of primitive risk factors, simplifying risk measurement.

Why the others are wrong

- **A:** Mapping covers the factors that drive the portfolio's normal risk; it does not filter to only abnormal factors.
- **B:** Using every factor defeats the simplification purpose; mapping deliberately accepts some approximation.
- **C:** The number of factors is chosen to fit the portfolio (trade-off of accuracy vs speed); five is arbitrary.

Q247

AnalystPrep Q.1512

REBUILT — Original stem was garbled and option C/D were not true alternatives; rewrote as a clear situation-recognition question.

VaR mapping is especially useful when historical prices of the instrument itself are not a reliable guide to its risk. Which position best illustrates such a case?

A. A large holding of an index fund whose price history is long and whose characteristics are stable.

B. A cash deposit in the bank's domestic currency.

C. A fixed-rate bond whose maturity and interest rate sensitivity shrink as it ages, or an option whose sensitivity to the underlying changes with the market.

D. A single stock position exposed to one risk factor and with unchanged characteristics.

Answer: C

Step by step

1. Historical price series describe an instrument only if its risk characteristics are stable.
2. A bond's duration falls as it ages, and an option's delta/gamma shift with the market, so past prices of the instrument misstate current risk.
3. Mapping handles this by expressing the position as exposures to risk factors (e.g. current spot yields, the underlying) matching its current profile.

Why C is correct: Mapping handles instruments whose characteristics change over time (bonds, options) by mapping them to risk factors that reflect their current profile.

Why the others are wrong

- **A:** A stable instrument with a long history can use historical prices directly; mapping is not needed for that reason.
- **B:** Domestic cash carries no market risk and is not a risk factor; it does not raise a historical-data problem.
- **D:** Unchanged characteristics and a single factor means historical data are adequate.

Q248

AnalystPrep Q.1513

REBUILT — Original options A and B were both arguably defensible; reworded so only the cash-not-a-risk-factor answer is right.

A position with market value V_m is mapped to risk exposures X_1, X_2, X_3 on three risk factors. The exposures sum to less than V_m, and no relevant risk factor has been omitted. What does this imply about the remainder of the value?

A. The remainder is an omitted risk factor that must be added before VaR can be computed.

B. The remainder is held in cash, which is not a risk factor and so contributes no market risk.

C. The remainder is specific risk that is unmapped and must be estimated as a separate risk factor.

D. The remainder is allocated to a second, separate set of risk factors.

Answer: B

Step by step

1. Mapping must preserve the position's market value.
2. If exposures to the risk factors sum to less than V_m and no factor was omitted, the difference must be value that is not exposed to a market risk factor.
3. That remainder is treated as cash, which is not a risk factor.

Why B is correct: Value not allocated to any risk factor, with no factors omitted, is cash, which carries no market risk factor.

Why the others are wrong
- **A:** Contradicts the stem, which says no factors are omitted.
- **C:** Specific risk is a residual of the allocated exposures, not the unallocated value.
- **D:** There is no separate set of factors; the stem says the relevant factors are already included.

Q249

AnalystPrep Q.1514

REBUILT — Original options B and C were nearly equivalent; rewrote options so only the accuracy-versus-complexity trade-off is correct.

When selecting the primitive risk factors for a VaR mapping, the choice should mainly reflect:

A. A trade-off between a better approximation of risk (more factors) and faster, simpler modeling (fewer factors).

B. The use of as many factors as possible, so that specific risk is eliminated entirely.

C. The use of a single market factor, because specific risk does not depend on which factors are chosen.

D. Only the availability of long historical data for each factor, regardless of whether it explains the portfolio's exposures.

Answer: A

Step by step
1. More risk factors give a better approximation of portfolio risk but make modeling more time consuming.
2. Fewer factors are faster but leave more residual (specific) risk.
3. The choice is therefore a trade-off, and the amount of specific risk depends on the factors chosen.

Why A is correct: Factor choice balances approximation quality against modeling time and complexity.

Why the others are wrong
- **B:** More factors reduce but never eliminate specific risk, and they raise modeling cost, defeating mapping's purpose.
- **C:** Specific risk depends on the factors chosen: adding factors (credit, currency) reduces it.
- **D:** Data availability matters, but factors must also capture the portfolio's exposures.

Q250

AnalystPrep Q.1515

Once risk factors are selected, positions are mapped onto them using the most suitable mapping approach. Whichever approach is chosen, what should the mapping preserve?

A. Only the market value of the position.

B. The par value and the market risk of the position.

C. The market value and the interest rate risk of the position.

D. The market value and the market risk of the position.

Answer: D

Step by step
1. A mapping replaces the position with exposures to risk factors.
2. For the replacement to be valid it must reproduce the position's current market value.
3. It must also reproduce the position's market risk, i.e. how the value responds to the risk factors.

Why D is correct: A valid mapping preserves both the market value and the market risk of the position.

Why the others are wrong
- **A:** Matching value alone ignores risk, which is the purpose of mapping.
- **B:** Par value is not the economic value; the mapping uses market value.
- **C:** Interest rate risk is only one type of market risk (and irrelevant for equity or FX positions).

Q251

BT Mock Q51 · BT notes P2.T5.22.9.3

REBUILT — Merged duplicates; option A relied on a debatable '3 maturities + 6 ratings = nine factors' count, so it was reworded to a defensible statement; the false statement is unchanged.

A bank is mapping several portfolios to risk factors: a fund investing in IPOs; a fund investing in emergent assets with little historical data; a small equity portfolio with fewer than 30 positions; a large equity portfolio with 1,000 positions; and a corporate bond portfolio with six ratings and maturities of 3, 5 and 7 years. Each of the following statements is true EXCEPT:

A. The large equity portfolio is a good candidate for mapping because a small set of general and specific factors needs far fewer estimates than a full covariance matrix of 1,000 positions.

B. The corporate bond portfolio can be mapped to a small set of general risk factors, such as maturity vertices and rating spreads, rather than one factor per bond.
C. Equity mapping contains model risk, including factor-sensitivity (beta) estimation error, subjective choice of risk factors, and omission of specific risk.
D. The IPO and emergent-asset funds cannot be mapped because they lack historical data, so sensitivities cannot be estimated.

Answer: D
Step by step
1. Mapping is useful precisely when there is insufficient historical data: such positions are expressed as exposures to similar risk factors that do have histories. So D is false.
2. A: a 1,000-position covariance matrix needs $1000 \times 1001/2 = 500{,}500$ terms (with variances), versus far fewer factor estimates. True.
3. B: bonds are mapped to a small set of general factors rather than per bond. True.
4. C: betas are estimated with error, factor choice is subjective and specific risk may be omitted. True.

Why D is correct: Lack of history is a reason to use mapping, not a barrier to it, so the claim that IPO and emergent-asset funds cannot be mapped is false.
Why the others are wrong
- **A:** True statement; the factor approach cuts the parameter count dramatically.
- **B:** True statement; fixed income maps to a small set of general factors.
- **C:** True statement; these are genuine model-risk sources.

Q252

BT answers L2.T5.61.1 · BT notes P2.T5.61.1

Each of the following is a valid reason to map portfolio positions to risk factors (VaR mapping) EXCEPT:
A. In a large portfolio there are too many positions to model individually.
B. Risk factors are common to various positions and/or instruments.
C. Risk factors allow for more accurate pricing (valuation) of the portfolio.
D. It is typically easier to stress test the risk factors.

Answer: C
Step by step
1. Mapping aggregates many positions into a small set of common risk factors (A and B are principles of mapping).
2. Fewer, common factors are easier to shock in stress tests (D).
3. Mapping simplifies risk measurement and tolerates approximation; it does not improve pricing accuracy, and if anything it is less accurate than full valuation (C is not a reason).

Why C is correct: Mapping is a risk-measurement simplification, not a valuation tool; it does not make pricing more accurate.
Why the others are wrong
- **A:** A valid reason: too many positions make position-by-position computation impractical.
- **B:** A valid reason: common factors let positions be aggregated.
- **D:** A valid reason: stress scenarios are simpler on a small set of factors.

Q253

BT answers L2.T5.61.2 · BT notes P2.T5.61.2

REBUILT — Original had two defensible false statements (option C 'only solution' and option D on options) and relied on Jorion material outside the Schweser text; rebuilt around the Schweser VaR mapping principles with one clearly false statement.

Each of the following is consistent with the principles underlying VaR mapping EXCEPT:
A. Mapping aggregates risk exposures when it is impractical to treat each position separately.
B. Mapping helps when there is insufficient price history for an investment, such as a recent IPO.
C. Mapping helps with positions whose characteristics change over time, such as bonds that move toward maturity.
D. Aggregating many positions into one risk factor is as acceptable for pricing a portfolio as it is for measuring its risk.

Answer: D
Step by step

1. Recall the principles: mapping aggregates exposures, simplifies them into primitive risk factors, handles changing characteristics and missing history.
2. A, B and C each restate one of these principles.
3. D claims aggregation is acceptable for pricing; the reading states that prices cannot be aggregated, only risk exposures can, so D is the exception.

Why D is correct: VaR mapping aggregates positions into common risk factors for risk measurement only; pricing needs position-level precision and cannot be aggregated.

Why the others are wrong

- **A:** True: it is the first stated principle (too many computations to treat each position separately). Not the exception.
- **B:** True: with no usable price history (IPO), mapping to risk factors replaces position-level history.
- **C:** True: as a bond matures, its exposure is re-mapped to the relevant spot yields.

Q254

BT answers L2.T5.61.3 · BT notes P2.T5.61.3

REBUILT — Option B ('price of commodity forward') was ambiguous because a forward price can also be treated as a risk factor; reworded options so exactly one is a position value rather than a primitive factor.

Which of the following is the LEAST likely to be a primitive risk factor in a VaR mapping?

A. The market value of a commodity forward position
B. The spot price of the commodity
C. The spot foreign exchange rate
D. A zero-coupon yield

Answer: A

Step by step

1. Primitive risk factors are market variables (spot prices, FX rates, zero yields, equity prices) onto which position values are mapped.
2. B, C and D are market variables.
3. A is the value of a position, which is what gets mapped onto primitive factors (spot price and interest rate); it is not itself a primitive factor.

Why A is correct: A position's market value is the thing allocated to risk factors; primitive risk factors are underlying market variables.

Why the others are wrong

- **B:** A spot commodity price is a standard primitive factor; picking it confuses the underlying with the derivative.
- **C:** Spot FX rates are classic primitive factors.
- **D:** Zero-coupon yields are the primitive factors for fixed-income mapping.

Q255

BT answers L2.T5.61.4 · BT notes P2.T5.61.4

REBUILT — Original option B ('distribution for each position') was subtly confusable with constructing risk factor distributions, which is a genuine step; replaced with an option describing position-level modelling that mapping is designed to avoid.

Each of the following is a step in the VaR mapping process EXCEPT:

A. Measure the current market value of each position.
B. Allocate the market value of each position to one or more common risk factors.
C. Sum the allocated exposures across positions for each risk factor.
D. Build a separate model from the full price history of every individual position.

Answer: D

Step by step

1. Mapping steps: value the positions, allocate each to common risk factors, then sum exposures by factor (yielding a vector of factor exposures) and apply risk-factor distributions.
2. A, B, C match these steps.
3. D is the position-by-position approach that mapping replaces, so it is not a step.

Why D is correct: Mapping works at the risk-factor level; modelling each position's own history defeats the purpose of mapping.

Why the others are wrong

- **A:** Valuing current positions is the first step.

- **B:** Allocation of each position to factors is the core step.
- **C:** Summing by factor produces the exposure vector (Figure 5.2).

Q256 BT answers P2.T5.714.1 · BT notes P2.T5.714.1

REBUILT — Item IV (stale prices) is outside the Schweser reading's rationale for mapping, so it was replaced with a development that clearly does not justify mapping; answer options adjusted.

A firm currently computes VaR without mapping. Which of these developments would support adopting VaR mapping?

I. The equity book increasingly holds recent IPOs with little price history.
II. The portfolio holds bonds and other exposures whose risk characteristics change over time.
III. The portfolio has grown to thousands of positions, making position-by-position risk computation impractical.
IV. The portfolio holds three large, liquid equity positions, each with more than 20 years of clean daily price history.

A. I only
B. I and III only
C. I, II and III only
D. I, II, III and IV

Answer: C

Step by step

1. I: insufficient history favours mapping.
2. II: changing characteristics (e.g. maturing bonds) favour mapping to current spot yields.
3. III: too many positions to treat individually favours aggregation into risk factors.
4. IV: few positions with ample history can be modelled directly, so no case for mapping. Supported: I, II, III.

Why C is correct: Mapping is justified by missing history, time-varying characteristics and portfolio size; IV has none of these.

Why the others are wrong

- **A:** Ignores II and III, which are both stated rationales.
- **B:** Omits II; bonds approaching maturity are a textbook case for mapping.
- **D:** Includes IV, a small portfolio with ample history that needs no mapping.

Q257 BT answers P2.T5.714.3 · BT notes P2.T5.714.3

REBUILT — Original key (B, remainder to cash) and distractors relied on Jorion material not in the Schweser reading and A was debatable; rebuilt on the Schweser principles with one unambiguous true statement.

Which of the following statements about VaR mapping is TRUE?

A. Pricing a portfolio may aggregate positions into a single risk factor, just as risk measurement does.
B. Aggregating positions into risk factor exposures is acceptable for risk measurement even though prices themselves cannot be aggregated.
C. Mapping is useful only when long price histories are available for each position.
D. After mapping, the number of variables whose covariances must be estimated equals the number of positions.

Answer: B

Step by step

1. Mapping replaces position values with exposures to a smaller set of common risk factors.
2. The reading states that VaR measurement differs from pricing: prices cannot be aggregated, but aggregation to risk factors is fine for risk.
3. That matches B; A reverses it, C contradicts the IPO/no-history use, D contradicts the reduction in variables.

Why B is correct: Risk measurement tolerates aggregation to risk factors; pricing does not.

Why the others are wrong

- **A:** Reverses the principle: prices cannot be aggregated.
- **C:** Mapping is specifically useful when history is lacking (e.g. IPOs).
- **D:** Mapping reduces the variables to the number of risk factors, not positions.

Q258 BT answers P2.T7.507.1

According to Malz, which of the following statements about the mapping of risk factors to positions is incorrect?

A. Mapping is the process of assigning risk factors to positions; in order to compute risk measures, we have to assign risk factors to securities.

B. When the underlying position and its hedge can be mapped to the same risk factor (or set of risk factors), the appropriate measure of value at risk (VaR) is zero.

C. Mapping decisions are often pragmatic; for example, fixed-income cash-flow mappings are more accurate than duration mappings but are more complex and require more risk factors.

D. As a practical matter, it may be difficult to find data that address certain risk factors, a problem that may merely mirror the real-world challenge of hedging or expressing some trade ideas.

Answer: B

Step by step

1. Mapping assigns each position's exposure to a (smaller) set of risk factors so that risk measures can be computed: A is true.
2. If a position and its hedge map to the same factor(s), the modelled exposures offset exactly, so the model reports zero risk.
3. In reality the two instruments differ in features the mapping ignores (basis, credit, liquidity, idiosyncratic risk), so real risk remains and VaR is not zero: B is the false statement.
4. C (cash-flow mapping is more precise but needs more factors than duration mapping) and D (data limits for some factors) are true statements of practical trade-offs.

Why B is correct: Mapping compresses instruments into common factors, so a hedge mapped to the same factor looks like a perfect offset; residual basis/specific risk is ignored, meaning VaR of zero is an artefact of the mapping, not the true risk.

Why the others are wrong

- **A:** True: this is the definition of mapping, so it is not the EXCEPT answer.
- **C:** True: cash-flow mapping uses many more vertices (more factors, more complexity) and is more accurate than a single-factor duration mapping.
- **D:** True: data availability limits the factors that can be used, mirroring the real-world difficulty of hedging some trade ideas.

Q259 BT notes P2.T5.61.5 · BT answers L2.T5.61.5

REBUILT — Original 'Both/Neither' stem was vague and gave away the answer; rewritten with concrete wording and plausible single-method distractors.

In VaR mapping, a position's exposure to a primitive risk factor (for example, a stock's exposure to a market index or a derivative's exposure to its underlying) can be determined in which of the following ways?

A. Only analytically, as the partial derivative (sensitivity) of the position value with respect to the risk factor.

B. Only statistically, by regressing the position's returns on the risk factor's returns.

C. Either analytically, as a partial derivative of the position value with respect to the risk factor, or empirically, as a regression slope estimated from historical returns.

D. By setting the exposure equal to the position's market value, regardless of how the value responds to the factor.

Answer: C

Step by step

1. Mapping replaces a position with exposures to risk factors, so we need the sensitivity of position value to each factor.
2. When a pricing formula is available (e.g., a forward or a bond), the exposure is the partial derivative (delta/duration) of value with respect to the factor.
3. When no formula is practical (e.g., a stock vs. the market index), the exposure is the regression slope β of the position's return on the factor's return (Jorion).
4. Hence both methods are valid: answer C. Setting exposure to market value ignores sensitivity.

Why C is correct: Exposure is a sensitivity, which can be obtained exactly from a valuation formula (partial derivative) or estimated from data by regression, so both approaches are used.

Why the others are wrong

- **A:** Too restrictive: analytical derivatives are not available for many positions such as stocks, where β is estimated by regression.
- **B:** Too restrictive: regression is not the only way; for instruments with known pricing formulas the exposure is computed exactly as a partial derivative.

- **D:** Confuses market value with exposure; a position's exposure to a factor equals market value only when sensitivity is exactly one (e.g., $\beta = 1$).

Q260

BT notes P2.T5.22.9.1

REBUILT — Original distractors (hierarchical-based logistic clusters, NLP) were nonsensical or irrelevant; stem made specific to reducing correlated yield-curve factors and distractors made plausible ML techniques.

A risk manager is mapping a large fixed-income portfolio to risk factors. The yield curve has 20 highly correlated vertices, and she wants to reduce them to a few uncorrelated primitive risk factors (such as level, slope and curvature) that explain most of the variation in rate changes. Which technique is most appropriate?

A. Natural language processing of central bank statements
B. Random forest classification of bonds into investment-grade and high-yield
C. Principal component analysis (PCA) of yield changes
D. Logistic regression of default events on bond ratings

Answer: C

Step by step

1. Goal: dimension reduction, replacing many correlated factors by a few primitive ones, which is the essence of mapping.
2. PCA decomposes the covariance matrix of yield changes into orthogonal components; for yield curves the first three (level, slope, curvature) typically explain the vast majority of variance.
3. NLP deals with text, random forests and logistic regression are supervised classifiers: none constructs a small set of uncorrelated factors from correlated ones.
4. Answer: C.

Why C is correct: PCA extracts a few uncorrelated components from correlated risk-factor changes, making it the natural tool for reducing the number of primitive factors in mapping.

Why the others are wrong

- **A:** NLP processes text; it does not reduce the dimension of numerical yield-curve factors.
- **B:** A supervised classification tool for labelling bonds; it does not produce uncorrelated risk factors.
- **D:** Models a binary outcome (default); it is not a dimension-reduction method for yield curve movements.

Q261

AnalystPrep Q.2501

REBUILT — Original was a two-statement true/true item with a giveaway 'None of the above' option; rebuilt as a scenario that tests the mapping process and rationale with plausible distractors.

A risk manager must compute the VaR of a large portfolio that contains a recently listed IPO stock with only three months of price history, plus hundreds of bonds with differing maturities and coupons. Which approach is most consistent with the principles of VaR mapping?

A. Estimate the IPO's volatility from its three months of returns and treat each bond as a separate risk factor, so that no information is lost.
B. Measure the current value of each position, replace each position with exposures to a small set of common risk factors (e.g., a proxy equity index for the IPO and standard-maturity interest rate vertices for the bonds), and compute VaR from those exposures and the risk factors' covariance matrix.
C. Exclude the IPO from the VaR calculation until it has a full history, and map only the bonds because mapping is unsuitable for positions with limited data.
D. Map the portfolio to a larger set of risk factors than the number of positions, so that each position's idiosyncratic risk is captured by its own factor.

Answer: B

Step by step

1. Mapping starts by measuring all current positions in the portfolio.
2. Each position is then expressed as exposures to a limited set of common (primitive) risk factors, such as equity indices or interest rates at standard maturities.
3. For the IPO, a short history makes its own return series unreliable, so a proxy factor (e.g., an index of comparable firms) replaces it. For the bonds, whose characteristics change as maturity shortens, exposures to fixed vertices on the yield curve replace the individual securities.
4. VaR is then computed from the factor exposures and the volatilities and correlations of the factors, which greatly reduces the number of variables to model.

5. Option B describes exactly this process.

Why B is correct: VaR mapping replaces many positions by exposures to a few common risk factors, which simplifies the problem and handles positions with insufficient or irrelevant history (IPOs, aging bonds) through proxies.

Why the others are wrong

- **A:** Defeats the purpose of mapping: three months of data give unreliable risk estimates, and modelling each bond separately keeps the dimension of the problem huge and ignores that bond characteristics change over time.
- **C:** Mapping is precisely the tool for positions lacking sufficient history; excluding the IPO would understate portfolio risk.
- **D:** Mapping reduces, not increases, the number of variables; using more factors than positions would add complexity and is the opposite of the principle.

LO 5.b

Explain how the mapping process captures general and specific risks, and calculate these risks in a portfolio given a set of primitive risk factors.

Q262

AnalystPrep Q.2644 · BT answers L2.T5.62.1 (sub-question L2.T5.62.2)

REBUILT — Merged two duplicate items (AnalystPrep Q.2644 and BT L2.T5.62.2); reworded the stem to state total risk is fixed and to name the added factors.

A risk manager maps a bond portfolio first onto a single general risk factor (duration only) and later adds further general risk factors such as credit spreads and currencies. Total portfolio risk is unchanged. Compared with the single-factor mapping, the specific (residual) risk in the multi-factor mapping will most likely be:

A. greater, because more factors add more estimation error.
B. smaller, because the additional general factors explain part of what was previously unexplained.
C. unchanged, because specific risk is independent of the choice of general factors.
D. impossible to determine without knowing the betas of the portfolio on each added factor.

Answer: B

Step by step

1. Total risk = general (factor-explained) risk + specific (residual) risk.
2. Specific risk is whatever the chosen general factors fail to explain, so its size depends on which general factors are used.
3. Adding factors (credit spread, currency) moves variance from the residual into the explained component.
4. With total risk fixed, more general-factor detail therefore means less specific risk.

Why B is correct: Specific risk is defined as the residual left after the general risk factors; the more precisely general risk is described, the smaller that residual is for a given total risk.

Why the others are wrong

- **A:** Confuses model complexity with risk allocation; extra factors reduce, not increase, the unexplained residual.
- **C:** Treats specific risk as a fixed property of the assets; in Jorion's framework it is a function of the general factors chosen.
- **D:** The direction of the effect does not depend on the particular betas; adding explanatory factors cannot raise the residual for a given total risk.

Q263

AnalystPrep Q.1530

REBUILT — Original was a vague, formula-garbled conceptual item duplicating the concept in item 1; rebuilt as a numerical decomposition of portfolio variance into general and specific components.

A portfolio holds four stocks with equal weights of 25%. Each stock is mapped to the market factor, and the portfolio beta is 0.90. The market return volatility is 2.0% per day, and each stock's idiosyncratic return has a volatility of 3.0% per day, uncorrelated with the market and with each other. Using $V(R_p) = \beta_p^2 V(R_m) + \sum w_i^2 \sigma_{\varepsilon i}^2$, the daily volatility of the portfolio return is closest to:

A. 1.80%
B. 2.34%
C. 2.42%
D. 3.50%

Answer: B

Step by step

1. General market variance: $0.90^2 \times 0.02^2 = 0.81 \times 0.0004 = 0.000324$.
2. Specific variance: $4 \times 0.25^2 \times 0.03^2 = 4 \times 0.0625 \times 0.0009 = 0.000225$.
3. Total variance: $0.000324 + 0.000225 = 0.000549$.
4. Volatility: $\sqrt{0.000549} = 2.343\%$.

Why B is correct: Portfolio variance is the sum of squared-beta times market variance plus the sum of squared-weight times residual variance; the square root gives volatility.

Why the others are wrong

- **A:** Only the general market risk: $\sqrt{0.000324} = 1.80\%$; omits the specific risk component.
- **C:** Used β_p instead of β_p^2 in the market term: $\sqrt{0.9 \times 0.0004 + 0.000225} = 2.42\%$.
- **D:** Used w_i instead of w_i^2 in the specific term: $\sqrt{0.000324 + 0.0009} = 3.50\%$.

Q264

BT answers L2.T5.62.1

REBUILT — Kept the question; replaced the weak distractor 'Eight' with the plausible product error (twelve) and stated the counting convention (risk-free yield factors plus credit-spread factors).

A portfolio of corporate bonds contains bonds with four different credit ratings and maturities of 1, 5, and 10 years. Using Jorion's approach (risk-free yield factors by maturity plus credit-spread factors by rating), how many general risk factors is the portfolio exposed to?

A. Three
B. Four
C. Seven
D. Twelve

Answer: C

Step by step

1. Risk-free (government) yield factors: one per maturity vertex = 3.
2. Credit-spread factors: one per rating = 4.
3. General risk factors = 3 + 4 = 7; idiosyncratic issuer risk is specific risk and is not counted.

Why C is correct: Yield-curve factors and rating spread factors add; they are not multiplied, since a bond's spread depends on rating and its yield on maturity.

Why the others are wrong

- **A:** Counts only the maturity (yield) factors and ignores credit spreads.
- **B:** Counts only the rating (spread) factors and ignores the yield curve.
- **D:** Multiplied maturities by ratings (3 × 4) instead of adding the two sets of factors.

Q265

BT notes P2.T5.714.2 · BT answers P2.T5.714.2

A portfolio of corporate bonds has maturities of 1, 5, 10, and 15 years. The bonds carry the following S&P ratings (Moody's equivalents in parentheses): AA (Aa); A (A); BBB (Baa); BB (Ba); and CCC (Caa). How many general risk factors is the portfolio exposed to? (Based on Jorion Question 11.4.)

A. Five: corporate bond yields at short and long maturities; plus rating migrations at three different general credit ratings
B. Six: corporate bond yields at short, medium, and long maturities; plus rating migrations at three different general credit ratings
C. Nine: corporate bond yields at four maturities; plus rating migrations at five different credit ratings
D. Nine: government bond yields at four maturities; plus credit spreads for five different credit ratings

Answer: D

Step by step

1. Risk-free yield factors: one for each of the 4 maturities = 4.
2. Credit-spread factors: one for each of the 5 ratings = 5.
3. Total general factors = 4 + 5 = 9.
4. The general factors are government yields and credit spreads; option C also gives 9 but wrongly treats corporate yields and rating migrations as the factors, double counting credit risk.

Why D is correct: Corporate bond risk decomposes into the risk-free curve (by maturity) plus a spread for each

rating, giving 4 + 5 = 9 general factors.

Why the others are wrong

- **A:** Only 5 factors: lumps maturities into short/long and credit ratings into three groups, ignoring the four maturities and five ratings given.
- **B:** Only 6 factors: groups maturities into three and ratings into three, not matching the portfolio's 4 maturities and 5 ratings.
- **C:** Right count but wrong factors: corporate yields already embed credit spreads, and rating migration is a credit-event risk, not the Jorion spread factor; the risk-free yields should be used with separate spreads.

Q266

BT notes P2.T5.22.9.2

REBUILT — Original distractors ($3.80m, $6.35m, $9.05m) were arbitrary giveaways; replaced with options from realistic errors.

An asset manager's $80.0 million equity portfolio has four equally weighted positions of $20.0 million each. The market's daily volatility is 1.0%. The betas of the positions with respect to the market factor are 0.50, 0.90, 1.10, and 1.30. Assume zero-mean normal returns and a well-diversified portfolio, so that specific risk is zero. The portfolio's one-day 95% VaR is closest to:

A. $1.32 million
B. $1.25 million
C. $1.77 million
D. $3.95 million

Answer: B

Step by step

1. Portfolio beta exposure: $20 \times (0.50 + 0.90 + 1.10 + 1.30) = 20 \times 3.80 = \$76.0$ million.
2. Market volatility: 1.0% per day; 95% z-value = 1.645.
3. VaR = $1.645 \times 0.01 \times 76.0 = \$1.250$ million.

Why B is correct: With specific risk zero, all risk is market risk, so VaR is z times market volatility times the beta-adjusted exposure of $76 million.

Why the others are wrong

- **A:** Ignored betas (effectively beta = 1): $1.645 \times 0.01 \times 80 = 1.32$.
- **C:** Used the 99% z-value 2.326 instead of 1.645: $2.326 \times 0.01 \times 76 = 1.77$.
- **D:** Scaled the one-day VaR by $\sqrt{10}$ (a ten-day horizon): $1.25 \times 3.162 = 3.95$.

Q267

Schweser Module Quiz 5.1 Q1

Which of the following could be considered a general risk factor?
I. Exchange rates.
II. Zero-coupon bonds.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: A

Step by step

1. General (primitive) risk factors are underlying market variables such as exchange rates, interest rates, equity indices.
2. Exchange rates qualify.
3. Zero-coupon bonds are instruments used as building blocks to map bond cash flows; the risk factor is the zero-coupon interest rate, not the bond itself.

Why A is correct: A risk factor is a market variable that drives value; a zero-coupon bond is a position/instrument used in mapping, whereas its yield is the factor.

Why the others are wrong

- **B:** Mistakes a mapping instrument for the factor and rejects exchange rates, which are a standard general factor.
- **C:** Also counts the zero-coupon bond itself as a factor; the factor is its interest rate.
- **D:** Rejects exchange rates, which are clearly general risk factors.

LO 5.c

Differentiate among the three methods for mapping portfolios of fixed-income securities.

Q268

AP Mock 2 Q69

REBUILT — Original distractors (1.2235, 1.7765) were arbitrary numbers; replaced with results of realistic mistakes (undiversified sum, zero-correlation sum, forgetting to scale to USD 200m).

The table gives 95% VaR percentages for zero-coupon bonds of various maturities:

Maturity (years)	1	2	3	4	5
VaR (%)	0.4777	0.9961	1.4264	1.9618	2.4120

A portfolio consists of a USD 100 million bond maturing in one year and a USD 100 million bond maturing in three years. Using the principal mapping method, what is the VaR of this portfolio?

A. USD 1.9041 million
B. USD 1.5043 million
C. USD 1.9922 million
D. USD 0.9961 million

Answer: C

Step by step

1. Principal mapping maps the whole principal to a single zero at the average maturity: $(1 + 3)/2 = 2$ years.
2. VaR of the 2-year vertex is 0.9961% of value.
3. Portfolio value = USD 200 million.
4. VaR $= 200 \times 0.9961\% =$ USD 1.9922 million.

Why C is correct: Principal mapping treats the entire USD 200m as one zero-coupon position at the average principal maturity, so VaR is simply the 2-year vertex VaR% times total value.

Why the others are wrong

- **A:** Adds the separate 1-year and 3-year VaRs (0.4777 + 1.4264 = 1.9041), i.e. assumes perfect correlation of separately mapped positions; that is not principal mapping.
- **B:** Combines the 1-year and 3-year VaRs assuming zero correlation, $\sqrt{0.4777^2 + 1.4264^2} = 1.5043$; again not principal mapping.
- **D:** Applies the 2-year VaR% to only USD 100m, forgetting that the portfolio is USD 200m.

Q269

AnalystPrep Q.1516

In a two-bond portfolio example, the VaR computed with duration mapping is 2.80 while the VaR computed with cash flow mapping is 2.67. The difference between these values is due to the fact that:

A. risk measures are not perfectly linear with maturity and correlations between vertices are below unity.
B. risk measures are perfectly linear with maturity and correlations are below unity.
C. risk measures are perfectly linear with maturity and correlations are above unity.
D. risk measures are not perfectly linear with maturity and correlations are equal to unity.

Answer: A

Step by step

1. Duration mapping represents the portfolio by one zero at the portfolio duration, so it ignores the actual cash flow timing and any diversification across vertices.
2. Cash flow mapping spreads value over the actual vertices and uses the correlation matrix.
3. Cash flow VaR is lower (2.67 < 2.80) because (i) VaR is not linear in maturity, so the duration-based interpolation differs from the true vertex risks, and (ii) vertex correlations are below 1, giving diversification.
4. Correlations cannot exceed 1, and correlations of 1 would give no diversification benefit.

Why A is correct: Lower cash flow VaR reflects diversification from imperfect correlation across vertices plus non-linearity of risk in maturity, which duration mapping's single-vertex interpolation ignores.

Why the others are wrong

- **B:** If risk were perfectly linear in maturity, interpolation at the duration would be exact; the stated cause (and 'perfectly linear') is wrong for this example.
- **C:** Correlations cannot exceed unity, and linearity is not perfect.
- **D:** Correlation of exactly 1 would remove diversification, so cash flow mapping would not give a lower VaR.

Q270 AnalystPrep Q.1520

Which method of mapping fixed-income portfolios decomposes the risk of a bond into the risk of each of the bond's cash flows?

A. Principal mapping
B. Duration mapping
C. Cash flow mapping
D. Present value mapping

Answer: C

Step by step

1. Principal mapping uses only principal repayments at an average maturity.
2. Duration mapping replaces the bond with one zero at the bond's duration.
3. Cash flow mapping takes the present value of each cash flow and assigns it to the zero-coupon risk factor of the same maturity, so each cash flow's risk is captured.

Why C is correct: Only cash flow mapping treats every coupon and principal payment as its own risk exposure at its own vertex.

Why the others are wrong

- **A:** Principal mapping ignores coupons and uses only principal timing.
- **B:** Duration mapping collapses all cash flows into one position at the duration.
- **D:** 'Present value mapping' is not one of the three standard methods (principal, duration, cash flow).

Q271 BT answers L2.T5.63.1 (q0433)

REBUILT — Original distractors were not tied to specific errors; replaced with results of realistic mistakes.

Assume a flat yield curve with spot rate of 4.0% at all maturities and normally distributed yield volatility of 1.0%. We are mapping a two-bond portfolio. Both bonds have a $100 million face value and pay an ANNUAL 4% coupon. One bond has a one-year maturity; the other has a five-year maturity. See below.

Portfolio Assumptions:	Bond #1	Bond #2
Face	$100	$100
Maturity	5.0	1.0
Coupon	4.0%	4.0%
Yield	4.0%	4.0%

Flat yield curve at 4.0% and risk (returns VaRs) based on 1.0% yield volatility and 95.0% confidence:

Term (years)	Yield (%)	Yield VaR (%)	Mac Duration (years)	Modified Duration (years)	Returns VaR (%)
1	4.0%	1.645%	1.0	0.962	1.58%
2	4.0%	1.645%	2.0	1.923	3.16%
3	4.0%	1.645%	3.0	2.885	4.74%
4	4.0%	1.645%	4.0	3.846	6.33%
5	4.0%	1.645%	5.0	4.808	7.91%

If we use principal mapping, what is the portfolio's 95% value at risk (VaR)?

A. $4.7 million
B. $7.9 million
C. $9.9 million
D. $9.5 million

Answer: D

Step by step

1. Both bonds are priced at par (coupon = yield), so portfolio value = $200 million.
2. Average maturity = $(1 + 5)/2 = 3$ years; the 3-year row gives modified duration 2.885 (3/1.04).
3. Returns VaR = $1.645 \times 1.0\% \times 2.885 = 4.745\%$.
4. VaR = $4.745\% \times 200 = \$9.49$ million, about $9.5 million.

Why D is correct: Principal mapping puts the full $200m at the average maturity of 3 years, whose return VaR is 4.745%.

Why the others are wrong

- **A:** Applies 4.745% to only $100m, forgetting that both bonds ($200m) are mapped.
- **B:** Uses the 5-year row's 7.91% on $100m, ignoring the 1-year bond and the averaging.
- **C:** Uses Macaulay duration 3.0 instead of modified 2.885: $3 \times 1.645\% \times 200 = 9.87$.

Q272

BT answers L2.T5.63.1 (q0434)

Assume a flat yield curve with spot rate of 4.0% at all maturities and normally distributed yield volatility of 1.0%. We are mapping a two-bond portfolio. Both bonds have a $100 million face value and pay an ANNUAL 4% coupon. One bond has a one-year maturity; the other has a five-year maturity. See below.

Portfolio Assumptions:	Bond #1	Bond #2
Face	$100	$100
Maturity	5.0	1.0
Coupon	4.0%	4.0%
Yield	4.0%	4.0%

Flat yield curve at 4.0% and risk (returns VaRs) based on 1.0% yield volatility and 95.0% confidence:

Term (years)	Yield (%)	Yield VaR (%)	Mac Duration (years)	Modified Duration (years)	Returns VaR (%)
1	4.0%	1.645%	1.0	0.962	1.58%
2	4.0%	1.645%	2.0	1.923	3.16%
3	4.0%	1.645%	3.0	2.885	4.74%
4	4.0%	1.645%	4.0	3.846	6.33%
5	4.0%	1.645%	5.0	4.808	7.91%

Under these assumptions (flat yield curve, constant 1.0% yield volatility), why can we expect cash flow mapping to produce a lower diversified VaR than either duration or principal mapping?

A. The risk measures are non-linear

B. Due to imperfect correlations between pairwise risk factors

C. Fewer total cash flows will be mapped

D. We cannot expect a lower diversified VaR

Answer: B

Step by step

1. With a flat curve and constant yield volatility, vertex risk = $1.645 \times 1.0\% \times M/1.04$, which is linear in maturity, so non-linearity cannot explain a gap.
2. Principal and duration mapping use a single risk factor, which implicitly treats all cash flows as perfectly correlated.
3. Cash flow mapping combines vertex VaRs with a correlation matrix below 1, so the diversified VaR is lower.

Why B is correct: The reduction comes from diversification: vertex correlations below 1 reduce the aggregate VaR relative to a single-factor mapping.

Why the others are wrong

- **A:** Here risk is linear in maturity by construction, so non-linearity is not the source (it matters in other examples).
- **C:** The same total present value ($200m) is mapped; cash flows are not fewer, they are spread over more vertices.
- **D:** A lower diversified VaR is expected because correlations are less than 1.

Q273

BT answers L2.T5.63.2

REBUILT — Source explanation used an inconsistent duration (2.875 vs correct 2.815) and a flawed alternative; recomputed and replaced options with results of realistic mistakes.

Assume a flat yield curve with spot rate of 4.0% at all maturities and normally distributed yield volatility of 1.0%. We are mapping a two-bond portfolio. Both bonds have a $100 million face value and pay an ANNUAL 4% coupon. One bond has a one-year maturity; the other has a five-year maturity. See below.

Portfolio Assumptions:	Bond #1	Bond #2
Face	$100	$100
Maturity	5.0	1.0
Coupon	4.0%	4.0%
Yield	4.0%	4.0%

Flat yield curve at 4.0% and risk (returns VaRs) based on 1.0% yield volatility and 95.0% confidence:

Term (years)	Yield (%)	Yield VaR (%)	Mac Duration (years)	Modified Duration (years)	Returns VaR (%)
1	4.0%	1.645%	1.0	0.962	1.58%
2	4.0%	1.645%	2.0	1.923	3.16%
3	4.0%	1.645%	3.0	2.885	4.74%
4	4.0%	1.645%	4.0	3.846	6.33%
5	4.0%	1.645%	5.0	4.808	7.91%

If we use duration mapping with linear interpolation, what is the portfolio's 95% VaR?

A. $8.56 million
B. $9.26 million
C. $8.90 million
D. $9.49 million

Answer: C

Key correction: Source explanation quoted a portfolio duration of 2.875; the correct Macaulay duration is 2.815 (it still leads to about $8.90m).

Step by step

1. Both bonds are at par; the 1-year bond has Macaulay duration 1.0. The 5-year 4% annual bond has Macaulay duration $\frac{1.04}{0.04}\left(1-1.04^{-5}\right)=4.630$.
2. Portfolio duration (equal $100m values) = $(4.630+1.0)/2=2.815$ years.
3. Table rows: 2 years gives $1.923\times 1.645\%=3.163\%$; 3 years gives $2.885\times 1.645\%=4.746\%$.
4. Interpolate at 2.815: $3.163\%+0.815\times(4.746\%-3.163\%)=4.452\%$.
5. VaR $=4.452\%\times 200=\$8.90$ million.

Why C is correct: Duration mapping replaces the portfolio with one zero at its Macaulay duration (2.815 years) and reads its VaR% by interpolating the vertex VaRs.

Why the others are wrong

- **A:** Interpolates using the modified duration (2.815/1.04 = 2.707) as if it were the vertex maturity, understating the risk.
- **B:** Multiplies Macaulay duration 2.815 directly by 1.645% (without dividing by 1.04) instead of interpolating the table.
- **D:** This is the principal-mapping answer (3-year vertex, 4.745% x 200), not the duration-mapped one.

Q274

BT answers L2.T5.63.3

Assume a flat yield curve with spot rate of 4.0% at all maturities and normally distributed yield volatility of 1.0%. We are mapping a two-bond portfolio. Both bonds have a $100 million face value and pay an ANNUAL 4% coupon. One bond has a one-year maturity; the other has a five-year maturity. See below.

Portfolio Assumptions:	Bond #1	Bond #2
Face	$100	$100
Maturity	5.0	1.0
Coupon	4.0%	4.0%
Yield	4.0%	4.0%

Flat yield curve at 4.0% and risk (returns VaRs) based on 1.0% yield volatility and 95.0% confidence:

Term (years)	Yield (%)	Yield VaR (%)	Mac Duration (years)	Modified Duration (years)	Returns VaR (%)
1	4.0%	1.645%	1.0	0.962	1.58%
2	4.0%	1.645%	2.0	1.923	3.16%
3	4.0%	1.645%	3.0	2.885	4.74%
4	4.0%	1.645%	4.0	3.846	6.33%
5	4.0%	1.645%	5.0	4.808	7.91%

If we use cash flow mapping, the portfolio is mapped to how many vertices?

A. Five
B. Six
C. Ten
D. Eleven

Answer: A

Step by step

1. The 5-year annual-coupon bond has cash flows at years 1, 2, 3, 4 and 5.
2. The 1-year bond has a single cash flow at year 1, which coincides with an existing date.
3. Distinct dates = years 1 through 5 = 5 vertices.

Why A is correct: Cash flows are aggregated by date, so overlapping dates share a vertex; only years 1-5 appear.

Why the others are wrong

- **B:** Counts the total number of cash flows (5 + 1 = 6) without merging the two year-1 cash flows.
- **C:** Treats each bond's cash flows as separate vertices (5 + 5), but the one-year bond has only one cash flow and

it shares the year-1 vertex.
- **D:** Overcounts by adding a time-0 vertex; vertices are only the distinct cash flow dates (years 1 to 5).

Q275

BT answers P2.T5.715.1 · BT notes P2.T5.715.1

REBUILT — Merged duplicate; kept data and correct answer but replaced arbitrary distractors with values produced by realistic mapping mistakes.

Consider a $200.00 million portfolio that is equally allocated between two bonds: the first bond pays a 5.0% annual coupon and matures in five years; the second bond pays a 3.0% annual coupon and matures in three years. Both bonds are currently priced at par, so their market value is $100.00 million each. The spot rate term structure is depicted below: 1.20% at one year, 2.00% at two years, 3.04% at three years, 4.50% at four years, and 5.20% at five years. Finally, the returns VaR (aka, risk) is based on a yield volatility of 100 basis points at all maturities and a confidence level of 99.0%. In this way, for example, the risk at one year is given by 1.0%*2.33*(1/1.0120) = 2.30% and the risk at two years is given by 1.0%*2.33*(2/1.020) = 4.57%.

Portfolio with Two Bonds: 5-Year 5.0% Par Bond (#1) and 3-Year 3.0% Par Bond (#2)

Term	Cash Flow Bond #1	Cash Flow Bond #2	Spot Rate	Risk (aka, Returns VaR)
1.0	$5.00	$3.00	1.20%	2.30%
2.0	$5.00	$3.00	2.00%	4.57%
3.0	$5.00	$103.00	3.04%	6.78%
4.0	$5.00	| 4.50%	8.92%	
5.0	$105.00	| 5.20%	11.07%	

Please note that compound frequency is annual, following the annual coupon payment. Under the principal mapping approach, which is NEAREST to the portfolio value at risk (VaR)?

A. $8.92 million
B. $13.57 million
C. $17.84 million
D. $22.15 million

Answer: C

Step by step

1. Principal mapping places the whole portfolio at its principal-weighted average maturity.
2. Average maturity = 0.5(5) + 0.5(3) = 4.0 years.
3. Risk at 4 years = 1.0% × 2.33 × 4/1.045 = 8.92% (given in the table).
4. Portfolio VaR = $200 million × 8.92% = $17.84 million.

Why C is correct: Principal mapping treats the entire $200 million as a single zero-coupon position at the average maturity of the principals (4 years) and applies that vertex's VaR percentage.

Why the others are wrong

- **A:** Applied the 4-year risk of 8.92% to only $100 million (one bond) instead of the whole $200 million portfolio.
- **B:** Mapped to the 3-year vertex (6.78%), i.e., the maturity of the shorter bond, instead of the 4-year average maturity.
- **D:** Mapped to the 5-year vertex (11.07%), i.e., the longest maturity, instead of the average maturity.

Q276

BT answers P2.T5.715.2 · BT notes P2.T5.715.2

REBUILT — Merged duplicate; kept data and correct answer, stated the modified-duration convention, and replaced arbitrary distractors with mistake-driven values.

A portfolio with a current value of $171.30 million is allocated between two bonds:
- The first bond is a zero-coupon bond with five years to maturity, and it has a value of $71.30 million. Its annual yield to maturity is 7.0%, as 1/1.07^5 equals about 0.7130.
- The second bond matures in one year and pays a 4.0% annual coupon. It has a yield of 4.0%, such that its market value is $100.00 million.

The spot rate term structure is depicted below: 4.00% at one year, 5.00% at two years, 5.50% at three years, 6.00% at four years, and 7.00% at five years. Finally, the returns VaR (aka, risk) is based on a yield volatility of 100 basis points at all maturities and a confidence level of 95.0%. In this way, for example, the risk at one year is given by 1.0%*1.645*(1/1.040) = 1.58%, and the risk at two years is given by 1.0%*1.645*(2/1.050) = 3.133%.

Portfolio with Two Bonds: 5-Year Zero-Coupon (#1) and 1-Year 4.0% Par Bond (#2)

Term	Cash Flow Bond #1	Cash Flow Bond #2	Spot Rate	Risk (aka, Returns VaR)
1.0	$0.00	$104.00	4.00%	1.58%
2.0	$0.00	| 5.00%	3.13%	
3.0	$0.00	| 5.50%	4.68%	
4.0	$0.00	| 6.00%	6.21%	
5.0	$100.00	| 7.00%	7.69%	

Please note that compound frequency is annual, following the annual coupon payment. Using modified duration (zero-coupon bond duration divided by 1 + yield) and linear interpolation between vertices, under the duration mapping approach, which is NEAREST to the portfolio value at risk (VaR)?

A. $6.71 million
B. $7.13 million
C. $8.01 million
D. $13.17 million

Answer: A

Step by step

1. Value weights: $w_1 = 71.30/171.30 = 0.4162$ (5-year zero), $w_2 = 100/171.30 = 0.5838$ (1-year bond).
2. Modified durations: 5-year zero $5/1.07 = 4.673$; 1-year bond $1/1.04 = 0.9615$.
3. Portfolio modified duration = $0.4162(4.673) + 0.5838(0.9615) = 2.506$ years.
4. Interpolate risk between 2 years (3.133%) and 3 years (4.678%): $3.133\% + 0.506(4.678\% - 3.133\%) = 3.915\%$.
5. VaR = $\$171.30 \times 3.915\% = \$6.71$ million.

Why A is correct: Duration mapping replaces the portfolio with one zero-coupon bond whose maturity equals the portfolio's (modified) duration, 2.506 years, and uses the interpolated risk for that maturity.

Why the others are wrong

- **B:** Used the value-weighted Macaulay maturity/duration of 2.665 years (no division by 1 + yield), giving risk of 4.16% and $7.13 million; this is also the principal-mapping result.
- **C:** Rounded the mapping point up to 3 years (risk 4.678%) rather than interpolating at 2.506 years.
- **D:** Mapped the whole portfolio to the 5-year vertex (7.687%), i.e., the longest maturity, ignoring the 1-year bond.

Q277

BT answers P2.T5.715.3 · BT notes P2.T5.715.3

REBUILT — Merged duplicate; options C and D differed in rounding of exhibit values as well as the label, so C and D now use identical numbers and differ only in the diversified/undiversified label; the unverifiable correlation-matrix figure was removed.

Consider a $200.00 million portfolio that is equally allocated between two bonds:
- The first $100.00 million bond pays a 6.0% annual coupon and matures in five years. It is priced at par (its annual yield is, therefore, 6.0%)
- The second $100.00 million bond pays a 4.0% annual coupon and matures in one year. It is priced at par (its annual yield is, therefore, 4.0%)

Portfolio with Two Bonds: 5-Year 6% Par (#1) and 1-Year 4.0% Par Bond (#2)

Term	FV Cash Flow Bond #1	FV Cash Flow Bond #2	PV of CF (x)	Spot Rate	Risk (aka, Returns VaR)
1.0	$6.00	$104.00	$105.77	4.000%	1.550%
2.0	$6.00	$0.00	$5.48	4.618%	3.110%
3.0	$6.00	$0.00	$5.15	5.192%	4.660%
4.0	$6.00	$0.00	$4.80	5.716%	6.210%
5.0	$106.00	$0.00	$78.79	6.112%	7.760%
Total	|	$200.00			

The exhibit above displays the spot rate term structure and assumptions for risk (aka, returns VaR) at each maturity (these assumptions mostly follow Jorion's own example in Chapter 11 EXCEPT the underlying set of yield VaR assumptions have been increased).

Please note that compound frequency is annual, following the annual coupon payment. If we utilize the cash flow mapping approach, which of the following is TRUE about the resultant portfolio value at risk (VaR)? Assume the correlations among the risk factors are less than 1.

A. The portfolio is mapped to a single risk factor, the 5-year vertex, so VaR is approximately $\$200.00 \times 7.760\% = \$15.52$ million
B. The undiscounted (future value) cash flows are mapped to the five vertices, so VaR is approximately $110(1.550\%) + 6(3.110\%) + 6(4.660\%) + 6(6.210\%) + 106(7.760\%) = \$10.77$ million
C. The present values are mapped to the five vertices and $105.77(1.550\%) + 5.48(3.110\%) + 5.15(4.660\%) +$

$4.80(6.210\%) + 78.79(7.760\%) = \$8.46$ million is the diversified portfolio VaR

D. The present values are mapped to the five vertices and $105.77(1.550\%) + 5.48(3.110\%) + 5.15(4.660\%) + 4.80(6.210\%) + 78.79(7.760\%) = \$8.46$ million is the undiversified portfolio VaR

Answer: D

Step by step

1. Cash-flow mapping maps the present value of each cash flow to the zero-coupon vertex of the same maturity.
2. Individual VaRs: $105.77(1.55\%) = 1.639$; $5.48(3.11\%) = 0.170$; $5.15(4.66\%) = 0.240$; $4.80(6.21\%) = 0.298$; $78.79(7.76\%) = 6.114$.
3. Sum = \$8.462 million. Adding individual VaRs assumes perfect correlation (all correlations = 1), so it is the undiversified VaR.
4. The diversified VaR uses the correlation matrix, $\sqrt{x'Rx}$, and is lower when correlations are below 1.

Why D is correct: Summing the vertex VaRs of the PV-mapped cash flows assumes perfect correlation, which gives the undiversified VaR; only the correlation-adjusted figure is the diversified VaR.

Why the others are wrong

- **A:** Mapped everything to one factor at the longest vertex; that is neither principal, duration nor cash-flow mapping and overstates VaR.
- **B:** Applied the VaR percentages to undiscounted future cash flows (\$110, \$6, ..., \$106) instead of their present values, overstating VaR (\$10.77 million).
- **C:** The \$8.46 million figure is right, but a straight sum of vertex VaRs assumes correlations of 1 and is the undiversified VaR, not the diversified one.

Q278

BT notes P2.T5.22.10.1

REBUILT — Source had no answer and the original statements B, C and D were all defensibly true; rewrote option C as a clearly false claim about duration mapping.

Ralph manages a fixed-income portfolio of high-quality corporate bonds (mostly coupon-bearing, some zero-coupon). His colleague Mary is designing a VaR mapping process using three approaches: principal mapping, duration mapping, and cash flow mapping. In regard to these three approaches, each of the following statements is true EXCEPT which is false?

A. In all three approaches the primitive instruments are zero-coupon bonds.

B. Principal mapping is the simplest approach, but it tends to overstate the risk of a coupon bond because it ignores the earlier coupon payments.

C. Duration mapping gives the exact change in portfolio value for any change in the term structure, including non-parallel shifts.

D. The diversified VaR from cash flow mapping rises as the pairwise correlations among the risk factors rise, and equals the undiversified VaR when all correlations are 1.

Answer: C

Step by step

1. A: each method maps positions onto zero-coupon risk factors - true.
2. B: principal mapping treats all value as received at the average principal maturity, which is longer than the duration, so it overstates VaR - true.
3. C: duration mapping is a single-factor linear approximation that relies on parallel shifts; it is not exact for non-parallel moves - FALSE.
4. D: VaR $= \sqrt{x'Rx}$ increases with correlations and reaches the sum of vertex VaRs at correlation 1 - true.

Why C is correct: Duration mapping uses one risk factor and so captures only parallel shifts approximately; cash flow mapping is needed to capture non-parallel moves.

Why the others are wrong

- **A:** True statement: all three methods use zero-coupon bonds as the primitive instruments.
- **B:** True statement: ignoring coupons places the value at a longer maturity than the duration, overstating risk.
- **D:** True statement: higher correlations reduce diversification, and with correlations of 1 diversified VaR equals undiversified VaR.

Q279

GARP PE2 Q79

An analyst at a fixed-income investment company is evaluating different ways the company uses to estimate the VaR of its corporate bond portfolios. The portfolios consist of a large number of bonds with a wide range of maturities. The analyst examines the possibility of using a mapping approach to simplify the estimation process. Which of

the following statements would the analyst be correct to make regarding the approaches to mapping fixed-income portfolios?

A. The VaR estimated using the principal mapping approach understates the true risk of a portfolio since it ignores coupon payments and any risk associated with them.

B. The VaR estimated using the duration mapping approach replaces the portfolio with a zero-coupon bond whose maturity equals the duration of the portfolio.

C. The VaR estimated using the principal mapping approach differs from the undiversified VaR estimated using the duration mapping approach due to an adjustment made for correlations.

D. The VaR estimated using the cash-flow mapping approach is less accurate than the VaR estimated using the duration mapping approach since it does not account for the timing of cash flows.

Answer: B

Step by step

1. Duration mapping: replace the portfolio with a single zero-coupon bond whose maturity equals the portfolio duration, then use that vertex's risk - statement B.
2. Principal mapping uses the average principal maturity, which exceeds the duration for coupon bonds, so it overstates (not understates) risk - A false.
3. Principal and duration mapping are both single-factor methods; neither involves a correlation adjustment - C false.
4. Cash flow mapping preserves the timing of every cash flow and is the most precise method - D false.

Why B is correct: Duration mapping is defined as mapping the portfolio to one zero-coupon bond with maturity equal to the portfolio duration.

Why the others are wrong

- **A:** Reverses the direction of the bias: ignoring coupons places value at a longer maturity, so principal mapping overstates risk.
- **C:** Correlation adjustments arise only in multi-vertex cash flow mapping (diversified vs undiversified VaR); the single-factor methods differ because they use different mapping maturities.
- **D:** Cash flow mapping maps each payment at its own maturity, so it accounts for timing best and is the most accurate.

Q280 Mock 4 Q80

A risk analyst is choosing among the three approaches to mapping a portfolio of coupon-bearing bonds onto a set of zero-coupon risk factors: principal mapping, duration mapping and cash-flow mapping. Which of the following correctly describes the relationship between these approaches?

A. Cash-flow mapping produces the most accurate VaR estimate because it assigns every individual payment to the risk factor matching its own maturity.

B. Principal mapping produces the most accurate VaR estimate because it avoids the estimation error introduced by decomposing the portfolio into many cash flows.

C. Duration mapping produces a more accurate VaR estimate than cash-flow mapping because it captures the convexity of the portfolio.

D. All three approaches produce identical VaR estimates for a portfolio of option-free bonds, since the underlying cash flows are the same.

Answer: A

Step by step

1. Principal mapping keeps only the principal and its maturity: crudest.
2. Duration mapping uses one zero at the portfolio duration: good for parallel shifts only.
3. Cash-flow mapping maps each payment to its own vertex, capturing the full term structure: most accurate (but most complex).
4. So A is correct; the others misstate accuracy or equivalence.

Why A is correct: Cash-flow mapping retains the amount and timing of every payment and uses all relevant vertices, so it is the most precise of the three approaches.

Why the others are wrong

- **B:** Decomposing contractual cash flows adds no estimation error; principal mapping is the least precise because it ignores coupons.
- **C:** Duration mapping is a linear, single-factor approximation and does not capture convexity; it is less accurate than cash-flow mapping.

- **D:** The methods give different VaRs because they place the value on different risk factors (average maturity, duration, or each vertex).

Q281

Schweser Module Quiz 5.2 Q1

Which of the following methods is not one of the three approaches for mapping a portfolio of fixed-income securities onto risk factors?

A. Principal mapping.
B. Duration mapping.
C. Cash flow mapping.
D. Present value mapping.

Answer: D

Step by step

1. The three approaches are principal, duration and cash flow mapping.
2. Present value mapping is not one of them.

Why D is correct: The reading names exactly three fixed-income mapping methods: principal, duration and cash flow mapping.

Why the others are wrong

- **A:** Principal mapping is one of the three approaches.
- **B:** Duration mapping is one of the three approaches.
- **C:** Cash flow mapping is one of the three approaches.

Q282

Schweser Module Quiz 5.2 Q3

REBUILT — The 95% VaR-percentage table was missing from the source; reconstructed it (keeping the Schweser 2-year and 3-year values) and replaced unexplained distractors with error-based ones.

The 95% confidence VaR percentages for zero-coupon bonds are:

Maturity (years)	1	2	3	4	5
VaR %	0.4696	0.9868	1.4841	1.9728	2.4660

A bond portfolio consists of a $100 million bond maturing in two years and a $100 million bond maturing in four years. What is the VaR of this portfolio using the principal mapping method?

A. $1.484 million
B. $1.974 million
C. $3.946 million
D. $2.968 million

Answer: D

Step by step

1. Principal mapping uses only the principal repayments, so the whole portfolio is placed at its weighted-average maturity.
2. Weights are 50% each: average life = $0.5(2) + 0.5(4) = 3$ years.
3. Total market value = $100 + 100 = 200$ million; the 3-year zero VaR percentage is 1.4841%.
4. VaR $= 200 \times 0.014841 = 2.968$ million.

Why D is correct: Principal mapping collapses the portfolio into one zero-coupon position at the average maturity of the principal payments, so the VaR is the full portfolio value times that zero's VaR percentage.

Why the others are wrong

- **A:** Applied the 3-year VaR percentage to only $100 million instead of the full $200 million portfolio.
- **B:** Used the 2-year VaR percentage (0.9868%) on $200 million, i.e. took the shorter bond's maturity rather than the average life.
- **C:** Used the 4-year VaR percentage (1.9728%) on $200 million, i.e. took the longer bond's maturity rather than the average life.

Q283 BT answers P2.T5.717.1 · BT notes P2.T5.717.1

REBUILT — Merged two identical items; kept the stem and answer but replaced giveaway/unexplained distractors with error-based ones and corrected the rounded dollar duration.

A portfolio manager evaluates the risk of the following two-bond portfolio:

Bond	Price	Modified Duration	Number Held
7-year bond	$103.0	5.80	300
20-year bond	$96.0	16.50	500

Assume specific risk is negligible and the volatility of changes in market yields is 50 basis points. Using duration mapping onto a single yield factor, which is NEAREST to the volatility of the portfolio value?

A. $896
B. $4,060
C. $4,856
D. $9,712

Answer: C

Step by step

1. Duration mapping: portfolio dollar duration is the sum of position values times modified duration.
2. 7-year bond: $103 \times 300 \times 5.80 = 179{,}220$.
3. 20-year bond: $96 \times 500 \times 16.50 = 792{,}000$.
4. Total dollar duration = 971,220 per unit change in yield (100%).
5. Volatility = $971{,}220 \times 0.0050 = 4{,}856.1$, about $4,856.

Why C is correct: Both bonds are driven by the same yield factor, so their dollar durations add (perfect correlation) and are scaled by the 50 bp yield volatility.

Why the others are wrong

- **A:** Included only the 7-year bond ($179{,}220 \times 0.005 = 896$) and ignored the 20-year bond.
- **B:** Treated the two bonds' risks as uncorrelated, combining as $\sqrt{179{,}220^2 + 792{,}000^2} \times 0.005$; a single yield factor implies perfect correlation, so the risks add.
- **D:** Used a 100 bp yield volatility instead of 50 bp ($971{,}220 \times 0.01 = 9{,}712$).

LO 5.d

Summarize how to map a fixed-income portfolio into positions of standard instruments.

Q284 BT Mock 1 Q18

REBUILT — Source option D was off by 1 USD from the exact result and the compounding convention was unstated; stem now states semiannual compounding and options were rebuilt around realistic errors.

A portfolio manager holds a short position in a one-year bond with USD 150 million face value and a 6% annual coupon paid semiannually. Zero-coupon rates (annualized, semiannual compounding) are 3% for 6 months and 4.1% for 12 months. After decomposing the bond into positions in the two standard instruments (6-month and 12-month zero-coupon bonds), what is the total present value of the standard instruments?

A. −USD 161,543,112
B. −USD 152,848,483
C. −USD 152,724,027
D. −USD 152,788,592

Answer: D

Key correction: Source key (D, −USD 152,788,591) is right in substance but its figure is off by USD 1 from the exact value of −USD 152,788,592.

Step by step

1. Semiannual coupon = 150,000,000 × 6%/2 = USD 4,500,000 at 6 months; at 12 months the cash flow is 4,500,000 + 150,000,000 = USD 154,500,000.
2. Discount the 6-month flow at 3%/2 = 1.5%: 4,500,000/1.015 = 4,433,498.
3. Discount the 12-month flow with 4.1%/2 = 2.05% per period over 2 periods: $154{,}500{,}000/1.0205^2 = 148{,}355{,}095$.
4. Total PV of the long bond = 152,788,592; the position is short, so the standard instruments have PV of −USD

152,788,592.

Why D is correct: Decomposing the bond into zero-coupon cash flows and discounting each at its own spot rate (semiannually compounded) gives the bond value; a short position flips the sign.

Why the others are wrong

- **A:** Uses a full 6% (USD 9m) coupon at 6 months instead of halving it ($9m/1.015 + 159m/1.0205^2$), overstating the value.
- **B:** Discounts the 12-month flow with annual compounding (154.5m/1.041) instead of semiannual compounding.
- **C:** Discounts the 6-month flow at 3% annual (÷1.03) instead of 1.5% per half-year.

Q285

Schweser Module Quiz 5.2 Q2

If the zero-coupon bonds (standard instruments) into which a fixed-income portfolio is mapped are perfectly correlated, portfolio VaR will equal:

A. marginal VaR.
B. component VaR.
C. undiversified VaR.
D. diversified VaR.

Answer: C

Step by step

1. Portfolio VaR combines the VaRs of the mapped positions using their correlations: $\text{VaR}_P^2 = \mathbf{V}'\mathbf{R}\mathbf{V}$.
2. With all correlations equal to 1, the formula collapses to the simple sum of the individual VaRs: $\text{VaR}_P = \sum_i \text{VaR}_i$.
3. That sum is by definition the undiversified VaR.

Why C is correct: With correlations of 1 there is no diversification benefit, so portfolio VaR is the sum of the individual VaRs, i.e. undiversified VaR.

Why the others are wrong

- **A:** Marginal VaR is the change in portfolio VaR from a small change in one position; it is not the total.
- **B:** Component VaR is one position's contribution to portfolio VaR; components sum to diversified VaR, not undiversified.
- **D:** Diversified VaR is the portfolio VaR computed with the actual correlations (below 1), which is lower than the sum of the individual VaRs; the term for the simple sum that ignores diversification, which is what perfect correlation gives, is undiversified VaR.

LO 5.e

Describe how mapping of risk factors can support stress testing.

Q286

BT answers L2.T5.64.1

REBUILT — Original had OCR-damaged options (one reconstructed with uncertainty) and ambiguous units; rebuilt with explicit $ million units and distractors that each reflect a realistic error.

Spot (zero) rates are: 3.0% (1 year), 3.2% (2 years), 3.4% (3 years), 3.6% (4 years) and 3.8% (5 years), with annual compounding. A portfolio holds two bonds, each of $100 million face value: a 1-year bond with a 3% annual coupon and a 5-year bond with a 5% annual coupon. The portfolio is mapped by cash-flow mapping to the five zero-coupon vertices. Yield volatility is 1.0% (normally distributed) at every maturity. To stress test, the analyst assumes the zero-coupon rates are perfectly correlated and stresses each zero by its 99% VaR (z = 2.326), where a zero's VaR % = z × yield volatility × T/(1 + y). What is the resulting 99% undiversified VaR of the portfolio (in $ million)?

A. $9.22 million
B. $10.06 million
C. $13.04 million
D. $23.03 million

Answer: C

Step by step

1. Cash flows by vertex: year 1 = 103 + 5 = 108; years 2-4 = 5; year 5 = 105.

2. PV of each flow at its spot rate: 104.85, 4.69, 4.52, 4.34, 87.14; total PV = 205.55.
3. Zero VaR % = 2.326 × 1% × T/(1+y): 2.26%, 4.51%, 6.75%, 8.98%, 11.20%.
4. VaR in dollars per vertex = PV × VaR %: 2.37, 0.21, 0.31, 0.39, 9.76.
5. Perfect correlation means the VaRs add: 2.37 + 0.21 + 0.31 + 0.39 + 9.76 = 13.04. Equivalently, revalue after shocking each zero down by its VaR: 205.55 - 192.51 = 13.04.

Why C is correct: With perfectly correlated zeros, portfolio VaR equals the sum of the individual vertex VaRs, which equals the loss from revaluing every zero after reducing its value by its own VaR %.

Why the others are wrong

- **A:** Uses the 95% deviate (1.645) instead of 2.326, which scales the answer down to 9.22.
- **B:** Combines the vertex VaRs as if uncorrelated (square root of the sum of squares = 10.06), contradicting the perfect-correlation assumption.
- **D:** Applies the 5-year zero's 11.20% VaR to the entire 205.55 portfolio value instead of using each vertex's own VaR %.

Q287

BT answers P2.T5.716.1 · BT notes P2.T5.716.1

The table below shows a stress test (Jorion, Table 11-3) of a $200.0 million two-bond portfolio: one bond is a $100.0 million 5-year 6.0% annual coupon issue, and the second is a $100.0 million 1-year 4.0% annual coupon issue.

Term	Cash Flows	Old Spot Rate	Old Disc. Factor	PV of Flows	Risk (%)	New Zero Value	New PV of Flows
1	$110.00	4.00%	0.9615	$105.77	0.47%	0.9570	$105.27
2	$6.00	4.62%	0.9137	$5.48	0.99%	0.9046	$5.43
3	$6.00	5.19%	0.8591	$5.15	1.48%	0.8464	$5.08
4	$6.00	5.72%	0.8006	$4.80	1.97%	0.7849	$4.71
5	$106.00	6.11%	0.7433	$78.79	2.43%	0.7253	$76.88
Total	|	| $200.00	|	$197.37			

Each of the following statements about this stress test is true EXCEPT which is false?

A. The stressed portfolio loss is about $2.63 million.

B. Because this stress test does not use mapping to risk factors, the VaRs of the zero-coupon vertices are not used; it is essentially a simulation that fully re-prices the component bonds.

C. The stressed portfolio loss equals the undiversified portfolio VaR because the stress implicitly assumes all zeros are perfectly correlated.

D. The stressed loss will increase if the confidence level is increased and/or the underlying yield volatility assumptions are increased.

Answer: B

Step by step

1. Loss = original value - stressed value = 200.00 - 197.37 = 2.63 million, so A is true.
2. The cash flows are mapped to five zero-coupon risk factors (e.g. the year-1 $110 = $104 + $6 is mapped to the 1-year zero), and each zero is shocked down by its own VaR % (the 'Risk (%)' column). So the stress test uses mapping and the zero VaRs; B is false.
3. Shocking every zero by its VaR at once is the perfect-correlation case, so the loss equals the undiversified VaR; C is true.
4. Risk % = z × σ × T/(1+y), which rises with z (confidence) and σ, so the stressed loss rises; D is true.

Why B is correct: B contradicts the method: the stress is built directly from the mapped zero-coupon risk factors and their VaR percentages, not from an unmapped full repricing.

Why the others are wrong

- **A:** True: 200.00 - 197.37 = 2.63; not the exception.
- **C:** True: simultaneous VaR shocks to all zeros equal the sum of vertex VaRs (perfect correlation); not the exception.
- **D:** True: each zero's Risk % is proportional to the deviate and yield volatility, so a higher confidence level or volatility enlarges the loss; not the exception.

LO 5.f

Explain how VaR can be calculated and used relative to a performance benchmark.

Q288 BT notes P2.T5.22.10.2

REBUILT — Original option B (stress testing cannot capture imperfect correlations) is defensible per Jorion/Schweser, giving two defensible answers; reworded B to be clearly false and added the missing squared term in D.

VaR mapping lends itself to stress testing and to performance benchmarking against an index. Consider both extensions:

I. Stress testing a fixed-income portfolio by shocking the zero-coupon vertices (risk factors) of the spot rate term structure by their VaRs

II. Performance benchmarking a fixed-income portfolio against a well-known bond index

Which of the following statements is TRUE?

A. The stress test is equivalent to the undiversified VaR at a sufficiently high confidence level, whatever the correlations between vertices.

B. Shocking each zero by its own VaR remains a valid way to obtain the portfolio VaR when correlations between maturities are less than one.

C. In performance benchmarking, minimizing the absolute VaR necessarily minimizes the relative VaR (tracking error VaR).

D. The variance improvement from benchmarking can be measured as $1 - (\text{TE-VaR/absolute VaR})^2$, analogous to R^2 in a regression.

Answer: D

Step by step

1. Stress by VaR shocks gives the undiversified VaR only when all zeros are perfectly correlated; the confidence level does not create that equivalence, so A is false.
2. The method ceases to be viable if correlations are anything other than 1, so B is false.
3. Absolute risk and tracking error are different: a low-VaR portfolio can still deviate strongly from the index (nonparallel yield shifts), so C is false.
4. Variance improvement = 1 - (TE-VaR/VaR of the portfolio)^2, the fraction of variance removed by tracking the benchmark, like R-squared; D is true.

Why D is correct: The tracking-error VaR is the residual risk relative to the benchmark, so one minus its squared ratio to absolute VaR is the share of variance explained, directly analogous to R-squared.

Why the others are wrong

- **A:** Equivalence with undiversified VaR comes from the perfect-correlation assumption, not from raising the confidence level.
- **B:** Stressing each zero by its own VaR implicitly assumes correlation of 1; with imperfect correlation it is not valid.
- **C:** Confuses absolute risk with relative risk; tracking error depends on mismatch to the index, so minimum absolute VaR need not give minimum TE-VaR.

Q289 AnalystPrep Q.1517

REBUILT — Original omitted the confidence level and horizon and used a muddled min/max and normal/abnormal framing; rewritten as a clean interpretation of tracking error VaR at a stated confidence level.

A portfolio manager benchmarks portfolio A against an index. The tracking error VaR, $\alpha\sqrt{(x - x_0)'\Sigma(x - x_0)}$, is computed at the 95% confidence level over a one-month horizon and equals \$0.63 million. Which interpretation is most appropriate?

A. Over one month, there is a 5% chance that portfolio A will underperform the index by more than \$0.63 million, under normal market conditions.

B. Over one month, portfolio A will never deviate from the index by more than \$0.63 million, in either direction.

C. Portfolio A has a one-month 95% absolute VaR of \$0.63 million.

D. Over one month, the minimum deviation between portfolio A and the index is \$0.63 million with 95% confidence.

Answer: A

Step by step

1. Tracking error VaR is the VaR of the position difference $x - x_0$ between the portfolio and the benchmark, using the covariance matrix Σ and the confidence multiplier α (1.645 at 95%).
2. So it is a quantile of the relative P&L: with 95% confidence, the shortfall versus the index over the horizon will not exceed \$0.63 million.
3. Equivalently, there is a 5% chance the shortfall exceeds \$0.63 million, under the usual normal-market (delta-

normal) assumptions.

Why A is correct: Tracking error VaR is a VaR applied to the portfolio-minus-benchmark position, so it is a probabilistic worst-case relative loss at the chosen confidence level under normal conditions.

Why the others are wrong

- **B:** VaR is not a guaranteed maximum; by construction it is exceeded with probability 1 minus the confidence level, and it measures the adverse tail only.
- **C:** Confuses relative VaR with absolute VaR; absolute VaR uses the portfolio positions x, tracking error VaR uses x minus x0.
- **D:** VaR is a maximum (worst-case quantile) loss measure, not a minimum deviation; the minimum deviation could be zero.

Q290

AnalystPrep Q.2826

REBUILT — Cleaned the stem (the 'cash flow decomposition with duration of 5 years' wording was irrelevant) and rebuilt distractors so each corresponds to a specific calculation error.

A portfolio of zero-coupon bonds is benchmarked against a bond index. At the 95% confidence level the index has an absolute VaR of $2.080 million, and the tracking error VaR of the portfolio relative to the index is $1.09 million. What is the variance improvement of the portfolio relative to the original index?

A. 27.5%
B. 47.6%
C. 72.5%
D. 52.4%

Answer: C

Step by step

1. Variance improvement $= 1 - (\text{TE VaR}/\text{VaR}_{index})^2$, analogous to R-squared.
2. Ratio: $1.09/2.080 = 0.5240$.
3. Square: $0.5240^2 = 0.2746$.
4. Improvement: $1 - 0.2746 = 0.7254 \approx 72.5\%$.

Why C is correct: Because VaR scales with standard deviation, the ratio of variances is the squared VaR ratio; the share of index variance removed is one minus that.

Why the others are wrong

- **A:** Stopped at the squared ratio (27.5%), which is the unexplained variance remaining, not the improvement.
- **B:** Forgot to square the ratio: $1 - 0.524 = 47.6\%$.
- **D:** Reported the unsquared ratio $1.09/2.08 = 52.4\%$ itself.

Q291

BT answers L2.T5.65.2

Which of the following is a true statement if we are benchmarking a fixed-income portfolio relative to a fixed-income index?

A. Minimizing absolute VaR will also tend to minimize tracking error VaR
B. Zero tracking error implies no underperformance vis-a-vis benchmark
C. Duration mapping is insufficient to minimize tracking error
D. Portfolios with the same duration should have similar/same tracking error VaRs

Answer: C

Step by step

1. Tracking error VaR is minimized when the portfolio's cash flows by maturity closely match those of the index, not merely its duration.
2. Matching duration only controls parallel yield shifts; non-parallel term-structure moves still generate tracking error, so a finer decomposition by maturity is needed.
3. Test each option: A fails (barbells have the lowest absolute VaR but the highest tracking error); B fails (a portfolio that constantly lags by a fixed amount has zero tracking error); D fails (equal-duration portfolios can have very different tracking error); C holds.

Why C is correct: Tracking error comes from mismatched exposures along the term structure, so duration (one number) cannot eliminate it; cash-flow-level matching is required.

Why the others are wrong

- **A:** Absolute VaR and tracking error VaR measure different things; the barbell has the lowest absolute VaR yet

the highest tracking error.
- **B:** Tracking error measures dispersion of the difference, not its sign; a constant underperformance has zero tracking error.
- **D:** Same duration only matches parallel shifts; portfolios with identical duration (e.g., barbell versus bullet) differ greatly in tracking error VaR.

Q292

BT answers P2.T5.716.3 · BT notes P2.T5.716.3

REBUILT — Merged the two duplicate sources; corrected option C's arithmetic (1-(0.43/1.99)^2 = 95.3%, not 95.4%) so the true statements are unambiguously true, and dropped the unneeded Risk (%) column that had an OCR-ambiguous value.

The table recreates Jorion's example of benchmarking a $100.0 million bond portfolio against a bond index (monthly tracking error VaR, 95% level; positions in $ millions).

Vertex	Position Index	Portfolio #1	#2	#3	#4	#5
<1m	1.05	0.0	0.0	0.0	0.0	84.8
3m	1.35	0.0	0.0	0.0	0.0	0.0
6m	2.49	0.0	0.0	0.0	0.0	0.0
1	13.96	0.0	0.0	0.0	59.8	0.0
2	24.83	0.0	0.0	62.6	0.0	0.0
3	15.40	0.0	59.5	0.0	0.0	0.0
4	11.57	38.0	0.0	0.0	0.0	0.0
5	7.62	62.0	0.0	0.0	0.0	0.0
7	6.43	0.0	40.5	0.0	0.0	0.0
9	4.51	0.0	0.0	37.4	0.0	0.0
10	3.34	0.0	0.0	0.0	40.2	0.0
15	3.00	0.0	0.0	0.0	0.0	0.0
20	3.15	0.0	0.0	0.0	0.0	0.0
30	1.31	0.0	0.0	0.0	0.0	15.2
Total	100.00	100.00	100.00	100.00	100.00	100.00
Duration	4.62	4.62	4.62	4.62	4.62	4.62
Absolute VaR	1.99	2.25	2.16	2.04	1.94	1.71
Tracking Error VaR	0.00	0.43	0.29	0.16	0.20	0.81

All of the following statements about this exercise are true EXCEPT:

A. If the goal is to minimize risk, Portfolios #1 and #2 are superior to the others.

B. The barbell portfolio (#5) has the lowest absolute VaR, consistent with imperfect correlation between distant maturities.

C. The variance improvement of Portfolio #1 relative to the index is about $1 - (0.43/1.99)^2 = 95.3\%$.

D. If the goal is to minimize tracking error, the best portfolio is #3.

Answer: A

Step by step

1. A: #1 and #2 have the highest absolute VaR (2.25, 2.16) and higher tracking error (0.43, 0.29) than #3 (2.04, 0.16) and #4 (1.94, 0.20), so they are dominated, not superior. A is false.
2. B: #5 has the lowest absolute VaR (1.71) thanks to its 30-year/short-end mix; true.
3. C: $0.43/1.99 = 0.2161$; squared $= 0.0467$; $1 - 0.0467 = 95.3\%$; true.
4. D: #3 has the lowest tracking error VaR (0.16); true.

Why A is correct: Portfolios #1 and #2 are worse than #3 and #4 on both absolute and tracking-error VaR, so the claim that they are superior is the only false statement.

Why the others are wrong

- **B:** True: #5 has absolute VaR of 1.71, the lowest, but also the highest tracking error (0.81), showing the trade-off.
- **C:** True: the variance improvement formula gives 95.3%; one might wrongly reject it by forgetting to square the ratio.
- **D:** True: 0.16 is the smallest tracking error VaR; one might wrongly pick #5 by confusing lowest absolute VaR with lowest tracking error.

Q293

Schweser Module Quiz 5.3 Q1

REBUILT — Benchmark table was missing; reconstructed a $100 million benchmark (largest cash flows at 3 and 5 years) and stated that each pair is weighted to match benchmark duration (4.25); pairs chosen so each can bracket that duration.

A $100 million benchmark bond portfolio has the following market value cash flows by maturity (zero-coupon

equivalents).

Maturity (years)	1	2	3	4	5	7	10
Market value ($ millions)	5	10	30	10	30	10	5

Each candidate portfolio consists of two zero-coupon bonds weighted so that its duration equals the benchmark's duration of 4.25 years. Which combination would most likely have the smallest tracking error VaR?

A. 1 year and 10 year
B. 2 year and 7 year
C. 3 year and 5 year
D. 4 year and 7 year

Answer: C

Step by step

1. Benchmark duration: $0.05 + 0.20 + 0.90 + 0.40 + 1.50 + 0.70 + 0.50 = 4.25$.
2. Every pair brackets 4.25 so duration can be matched; the remaining tracking error comes from non-parallel shifts and cash-flow mismatch.
3. Tracking error VaR is smallest when the zero-coupon positions coincide with the benchmark's largest cash flows: $30 million at 3 and 5 years (60% of the portfolio).
4. 3 and 5 years hold exactly those cash flows; the other pairs miss at least one of them, and 1/10-year is a barbell far from the benchmark's cash flows, so it leaves the most non-parallel exposure.

Why C is correct: With duration matched, tracking error VaR depends on how closely the cash flows line up with the benchmark's; the 3- and 5-year buckets carry most of the value.

Why the others are wrong

- **A:** A 1/10-year barbell has low absolute VaR appeal but is furthest from the benchmark's concentrated 3/5-year cash flows, giving the largest tracking error.
- **B:** Captures only the 2-year ($10m) and 7-year ($10m) buckets, missing the two largest cash flows.
- **D:** Captures only the 4-year ($10m) and 7-year ($10m) buckets; close in maturity but still misses the $30m 3- and 5-year flows.

Q294 BT answers L2.T5.65.1

A fund with value of $10 million is benchmarked against an index. The fund's volatility is 10.0% and normally distributed. The benchmark volatility is 5.0% and normally distributed. The fund's active risk (tracking error) is 8.0% and normally distributed. What is the 95% tracking error value at risk?

A. $820,000
B. $1.32 million
C. $1.65 million
D. $1.96 million

Answer: B

Step by step

1. Tracking error VaR uses the volatility of the active (fund minus benchmark) return, i.e., 8.0%.
2. At 95% (one-tailed) the multiplier is 1.645.
3. TE VaR $= 1.645 \times 8.0\% \times \$10$ million $= \$1.316$ million $\approx \$1.32$ million.

Why B is correct: Tracking error VaR is the VaR of the active return distribution, so it scales active risk, not fund or benchmark volatility.

Why the others are wrong

- **A:** Used the benchmark volatility: $1.645 \times 5\% \times 10 = \$0.82$ million.
- **C:** Used the fund's own volatility: $1.645 \times 10\% \times 10 = \$1.645$ million (absolute VaR).
- **D:** Used the fund volatility with the 97.5% multiplier: $1.96 \times 10\% \times 10 = \$1.96$ million.

LO 5.g

Describe the method of mapping forwards, forward rate agreements, interest rate swaps, and options.

Q295

GARP PE1 Q30 · Mock 3 Q50

REBUILT — Option A mapped a forward to spot alone, which ignores the two interest rate legs; reworded to the full Jorion decomposition so exactly one option is correct.

A senior risk analyst at a large investment bank proposes simplifying the bank's portfolio VaR estimation by mapping its large number of trading positions to a smaller number of elementary risk factors. Which of the following is the most appropriate way of mapping the given position?

A. Mapping a long USD/EUR forward contract to a long EUR spot position, a long one-year EUR bill and a short one-year USD bill.

B. Mapping each position in a corporate bond portfolio to the government bond with the closest maturity.

C. Mapping zero-coupon government bonds to government bonds paying regular coupons.

D. Mapping a position in a stock market index to a position in one stock within that index.

Answer: A

Step by step

1. Mapping should decompose each instrument into elementary risk factors that capture its main sources of risk.
2. A forward to buy EUR with USD in one year is equivalent to a long EUR spot position, a long one-year EUR bill, and a short one-year USD bill (borrowing USD): three elementary factors (FX, EUR rate, USD rate). A is appropriate.
3. B drops credit spread risk, the main risk of corporate bonds, so VaR is understated.
4. C reverses the direction: coupon bonds are mapped to zero-coupon bonds (as a portfolio of zeros), not the reverse.
5. D replaces a diversified index with one stock, adding idiosyncratic risk; individual stocks are mapped to the index, not vice versa.

Why A is correct: A forward decomposes exactly into spot FX and two zero-coupon rate positions, which are elementary factors; the other options reverse the mapping direction or discard important risk.

Why the others are wrong

- **B:** Ignores the credit spread and rating-related risk of corporate bonds, so risk is understated.
- **C:** Backwards: a coupon bond is a portfolio of zero-coupon bonds, so coupon bonds map onto zeros, not zeros onto coupon bonds.
- **D:** Backwards and inaccurate: stocks map onto the index (via β); mapping the index onto one constituent introduces specific risk and loses diversification.

Q296

AnalystPrep Q.1526

REBUILT — Original options were garbled ('floating-paying bond'); rewrote the stem and options so exactly one pairing is correct.

When mapping a plain-vanilla interest rate swap to risk factors, the swap is decomposed into a fixed leg and a floating leg. Which pairing of equivalent instruments is correct?

A. Fixed leg: floating-rate note; floating leg: coupon-paying bond

B. Fixed leg: coupon-paying bond; floating leg: floating-rate note

C. Fixed leg: zero-coupon bond; floating leg: coupon-paying bond

D. Fixed leg: floating-rate note; floating leg: zero-coupon bond

Answer: B

Step by step

1. The fixed leg pays a fixed coupon at each payment date plus notional at maturity, so it is priced like a coupon-paying bond.
2. The floating leg pays a rate that resets to market, so it is priced like a floating-rate note (worth par at each reset date).
3. Only pairing B matches fixed leg = coupon bond and floating leg = FRN.

Why B is correct: The fixed leg has predetermined coupons (a fixed-coupon bond), while the floating leg's coupons reset to market so it behaves like a floating-rate note.

Why the others are wrong

- **A:** Reverses the legs: the FRN is the floating leg, not the fixed leg.
- **C:** A zero-coupon bond has no interim payments, so it cannot represent the fixed leg's periodic coupons; the floating leg is an FRN, not a coupon bond.
- **D:** Reverses the legs and wrongly treats the floating leg as a zero-coupon bond (only its next reset payment behaves like a short zero, not the whole leg).

Q297 BT answers L2.T5.66.4

REBUILT — Clarified the treatment (floating leg = FRN worth par, mapped to cash) so the count of risk-factor vertices is unambiguous.

A plain-vanilla at-the-market interest rate swap settles semi-annually and has five years to maturity. Using the cash-flow mapping approach (fixed leg as a coupon bond, floating leg as a floating-rate note), immediately before the initial reset the swap maps to how many risk factors (vertices), not including cash?

A. Nine
B. Ten
C. Nineteen
D. Twenty

Answer: B

Step by step

1. Fixed leg: coupon bond with semi-annual payments over 5 years, so payments at 0.5, 1.0, ..., 5.0 = 5 × 2 = 10 vertices.
2. Floating leg (FRN): immediately before the reset it is worth par (equal to notional), has no interest-rate risk and maps to cash.
3. Excluding cash, the risk factors are the 10 fixed-leg vertices.

Why B is correct: Right before a reset the FRN is worth par and carries no market risk (cash), leaving only the fixed leg's ten semi-annual cash flow vertices.

Why the others are wrong

- **A:** Drops one vertex (e.g., omits the final coupon-plus-principal date or counts years × 2 − 1).
- **C:** Counts ten fixed vertices plus nine floating-leg vertices, treating the floater as if it had a vertex at every payment date.
- **D:** Counts ten vertices for each leg, ignoring that the floater is par/cash at reset.

Q298 BT answers L2.T5.67.2

If an option portfolio is mapped using a Taylor series approximation of the Black-Scholes-Merton price, to which risk factors can the portfolio be mapped?

A. Spot price and volatility
B. Spot price, volatility, time to expiration, and dividend yield
C. Spot price, volatility, interest rate, and dividend yield
D. Spot price, volatility, interest rate, time to expiration, and dividend yield

Answer: D

Step by step

1. BSM is a closed-form function of S, σ, r, T and the dividend (or foreign-rate) yield.
2. A Taylor expansion takes partial derivatives with respect to each input: delta/gamma (spot), vega (volatility), rho (interest rate), theta (time) and the dividend-yield sensitivity.
3. Hence all five inputs are available as risk factors.

Why D is correct: Because the option value is differentiable in every BSM input, the Taylor series yields sensitivities to spot, volatility, rate, time and dividend yield.

Why the others are wrong

- **A:** Limits to delta/gamma and vega only; omits rho, theta and dividend sensitivity.
- **B:** Omits the interest rate (rho).
- **C:** Omits time to expiration (theta).

Q299 BT answers L2.T5.67.4

REBUILT — Stem did not state that the price decline is accompanied by a rise in volatility, making the answer assumption-dependent; added it explicitly.

A portfolio contains only SHORT PUT options and is mapped to delta and vega. A sharp decline in the underlying price occurs, accompanied by a rise in implied volatility. What is the impact on the mapped exposures?

A. Gain in value due to both risk factors
B. Gain in delta exposure mitigated by loss in vega exposure
C. Loss in delta exposure mitigated by gain in vega exposure
D. Loss in value due to both risk factors

Answer: D

Step by step

1. A put has negative delta, so a short put has positive delta: its value falls when the underlying falls (delta loss).
2. A put has positive vega, so a short put has negative vega: a volatility rise reduces its value (vega loss).
3. Both effects are losses, so the short put loses on both risk factors.

Why D is correct: A short put is long the underlying (positive delta) and short volatility (negative vega), so a price drop with a volatility rise hurts on both.

Why the others are wrong

- **A:** Treats the position as a long put (a long put gains from a price drop and a vol rise).
- **B:** Gets delta wrong (assumes the short put gains when price falls) while correctly signing vega.
- **C:** Gets delta right but treats the short put as long vega; a volatility rise hurts a short option.

Q300

BT answers L2.T5.67.3

REBUILT — Stem did not state that the price decline is accompanied by a rise in volatility; added it explicitly.

A portfolio contains only LONG CALL options and is mapped to delta and vega. A sharp decline in the underlying price occurs, accompanied by a rise in implied volatility. What is the impact on the mapped exposures?

A. Gain in value due to both risk factors
B. Gain in delta exposure mitigated by loss in vega exposure
C. Loss in delta exposure mitigated by gain in vega exposure
D. Loss in value due to both risk factors

Answer: C

Step by step

1. A long call has positive delta, so a price decline reduces its value (delta loss).
2. A long call has positive vega, so the volatility rise increases its value (vega gain).
3. The delta loss is partly offset by the vega gain.

Why C is correct: A long call loses from the lower spot (positive delta) but gains from higher volatility (positive vega), so the delta loss is mitigated by the vega gain.

Why the others are wrong

- **A:** Ignores the sign of delta: a long call does not gain when the underlying falls.
- **B:** Reverses both signs, as if the position were a long put or short volatility.
- **D:** Ignores that a long option benefits from a volatility increase.

Q301

BT answers P2.T5.716.2 · BT notes P2.T5.716.2

REBUILT — Removed the weak 'zero' distractor, stated the floater duration explicitly and replaced options with ones that map to realistic errors.

A portfolio manager enters a 7-year pay-fixed swap with notional of $100 million. The modified duration of the fixed leg is 5.90 years, and the floating leg has just been reset, with the next payment in one year (floater duration of about 1.0 year). Assume a flat term structure and an annual volatility of yield changes of 100 basis points. Using a normal approximation (z = 2.326 for 99%) and square-root-of-time scaling, what is the 99% one-month VaR of the swap?

A. $2.33 million
B. $3.29 million
C. $3.96 million
D. $11.40 million

Answer: B

Step by step

1. The pay-fixed swap = short fixed-rate bond + long floating-rate note, so net duration = $5.90 - 1.00 = 4.90$ years.
2. Monthly yield volatility = $1.00\% \times \sqrt{1/12} = 0.2887\%$.
3. VaR = $2.326 \times 4.90 \times 0.2887\% \times 100$ million = $3.29 million.

Why B is correct: The swap's rate risk is the net duration of the fixed leg less the floater (4.90), applied to monthly yield volatility at the 99% quantile.

Why the others are wrong

- **A:** Uses the 95% quantile (1.645) instead of 99%: $1.645 \times 4.90 \times 0.2887\% \times 100$ million = $2.33 million.
- **C:** Ignores the floating leg's duration and uses 5.90 years (gives $3.96m).
- **D:** Fails to scale the annual yield volatility to one month: $2.326 \times 4.90 \times 1\% \times 100$ million = $11.40 million.

Q302 BT answers P2.T5.717.2 · BT notes P2.T5.717.2

In regard to value at risk (VaR) mapping, each of the following statements is true EXCEPT which is false?

A. A forward currency position has three major risk factors: the spot price and two interest rates

B. A consumption commodity forward contract is more likely to be directly mapped to the commodity's volatile forward price than an investment commodity forward

C. A forward rate agreement can be mapped to two risk factors: a long position in a zero-coupon bond (or bill) plus a short position in a zero-coupon bond (or bill)

D. Because an interest rate swap is a bilateral contract, it cannot be effectively mapped to primitive risk factors, and simulation must be used to compute a diversified VaR

Answer: D

Step by step

1. A: a currency forward maps to the spot FX rate, the domestic bill and the foreign bill - true.
2. B: consumption commodities lack a reliable cost-of-carry arbitrage (convenience yield), so the forward price itself is a risk factor - true.
3. C: an FRA is a long zero and a short zero at the two maturities - true.
4. D: a swap is decomposed into a coupon bond (fixed leg) and an FRN (floating leg) mapped to cash-flow vertices, and diversified VaR comes from the delta-normal covariance method - false.

Why D is correct: A swap can be mapped to its cash flow vertices (fixed coupon bond plus floating-rate note), so delta-normal diversified VaR applies and simulation is not required.

Why the others are wrong

- **A:** True statement: currency forwards map to spot FX and two zero-coupon rates.
- **B:** True statement: consumption commodities have unstable convenience yields, so the forward price is mapped directly.
- **C:** True statement: an FRA is long and short zero-coupon positions at the two dates.

Q303 BT answers P2.T5.717.3 · BT notes P2.T5.717.3

In January, Stacy, a risk analyst, mapped a large long position in call options with the following profile:
- Ratio of spot price to strike price, S/K: approximately 1.0
- Implied volatility: about 23.0% per annum
- Time to expiration: six months (matures in July)
- Risk-free rate: 1.50%

Her VaR mapping was based on a Taylor series using first-order (delta) terms only, and it worked well for several months. In late June the position is only five days from maturity. Which of the following changes to her mapping is MOST likely?

A. She adds the risk-free interest rate (rho) as a risk factor because expiration is now virtually certain.

B. She adds gamma as a risk factor because the first-order linear approximation has become insufficient.

C. She adds vega as a risk factor, which was omitted at long maturities but is more relevant at short maturities.

D. She adds delta as a risk factor, which had a low profile at longer terms but becomes more relevant for at-the-money options as maturity approaches.

Answer: B

Step by step

1. The existing mapping is delta-only, i.e. linear in the underlying: $dV \approx \Delta\, dS$.
2. For an at-the-money option, gamma (the curvature of value in S) rises sharply as expiration approaches, because delta flips between about 0 and 1 over a very narrow range of S.
3. So even over a one-day horizon a linear approximation misstates the option's P&L; the second-order term $\frac{1}{2}\Gamma(dS)^2$ must be added (delta-gamma mapping).
4. Rho and vega become smaller for a 5-day option, and delta was already in the mapping.

Why B is correct: Near expiry an ATM option is highly nonlinear (large gamma), so a delta-only mapping is no longer adequate and gamma must be added.

Why the others are wrong

- **A:** Rho matters little for a 5-day option; shorter maturity makes interest-rate sensitivity smaller, not larger.
- **C:** Vega of an ATM option is an increasing function of maturity, so it shrinks as expiry nears; it is also not what breaks a linear mapping.
- **D:** Delta was already the first-order factor in the original mapping, so it cannot be 'added'; the new problem is

curvature, not delta.

Q304 BT notes P2.T5.22.10.3

A firm maps a variety of reference instruments into risk factors for VaR purposes: a forward foreign exchange (FX) contract, a consumption commodity forward, a vanilla interest rate swap, and a stock option portfolio. Which of the following statements about mapping these instruments is TRUE?

A. The forward FX contract is one of the most difficult instruments to map.

B. The consumption commodity forward contract is one of the easiest instruments to map.

C. The vanilla interest rate swap should be mapped to the dot product of its diffusion vector and its amortization vector.

D. The stock option portfolio can be mapped in many ways, from simple (delta-normal) to complex (multiple Greek risk factors).

Answer: D

Step by step

1. An option's value is nonlinear in the underlying, so it can be mapped via delta only (delta-normal) or with gamma, vega and other Greeks for greater accuracy (delta-gamma, full revaluation).
2. A forward FX contract decomposes into three simple building blocks (spot FX, foreign bill, domestic bill), which is easy, not hard.
3. A consumption commodity forward is harder because storage costs and convenience yield are unobservable and variable.
4. A swap is mapped as a fixed-rate bond (short) plus a floating-rate note, not by a 'dot product of diffusion and amortization vectors'.

Why D is correct: Options are nonlinear, so the mapping can range from a first-order delta approximation to multi-Greek approximations.

Why the others are wrong

- **A:** Reverses the facts: an FX forward maps cleanly into spot and two bill positions, making it among the easiest.
- **B:** Ignores storage cost and convenience yield, which make commodity forwards harder than financial forwards to map.
- **C:** Invented procedure; a plain swap is mapped as a fixed-coupon bond versus a floating-rate note and then cash-flow mapped to vertices.

Q305 GARP PE2 Q49

A risk manager at Bank QRP is examining the process of estimating the VaR of derivative contracts after mapping their risks to a set of factors. The bank holds forward contracts in foreign currencies, commodities, and interest rates. Which of the following statements is correct about mapping forward contracts?

A. Valuing a commodity forward through mapping is simpler and more precise than valuing a financial forward, since commodity forwards generally do not make periodic cash payments.

B. A key factor used in valuing a forward rate agreement is the convenience yield arising from the buyer's ability to lock in long-term borrowing rates.

C. When mapping the risks of a foreign currency forward contract to risk factors, volatility of the underlying spot contract is the primary source of risk.

D. The VaR of an interest rate forward contract increases as it approaches maturity, reflecting lower confidence in estimates of the future levels of the risk factors.

Answer: C

Step by step

1. A currency forward maps to three positions: long spot foreign currency exposure, long foreign bill, short domestic bill.
2. Spot FX volatility is far larger than the price volatility of the short-term bills, so the spot leg dominates the VaR.
3. Commodity forwards depend on unobservable convenience yield and storage costs, so they are harder, not easier, to map (A false).
4. Convenience yield is a commodity concept, not an FRA input (B false).
5. As a forward approaches maturity, the bills' durations shrink and their price volatility falls, so VaR of an interest rate forward decreases (D false).

Why C is correct: The forward's value tracks the spot rate adjusted for carry, so spot volatility is its main risk

driver.

Why the others are wrong

- **A:** Convenience yield and storage costs are unobservable, making commodity forwards harder and less precise to map.
- **B:** Confuses FRAs with commodity forwards; convenience yield is the benefit of holding a physical commodity.
- **D:** Reverses the effect: bill price volatility falls as maturity nears, so interest rate forward VaR decreases.

Q306

BT answers L2.T5.67.1

REBUILT — Source key (57.77 cash) was arithmetically wrong, leaving no correct option; recomputed the BSM mapping and rebuilt options around realistic errors.

Jorion shows that a European call option can be mapped using the Black-Scholes-Merton model as a long position in delta units of the stock plus a short (borrowed) cash position. A call on a non-dividend-paying stock has S = K = 100, volatility 20%, risk-free rate 4.0% (continuously compounded), and T = 0.5 years. What is the BSM mapping of the option?

A. Long $58.40 of stock and short $51.77 of cash
B. Long $58.40 of stock and short $52.82 of cash
C. Long $58.40 of stock and short $6.63 of cash
D. Long $50.00 of stock and short $43.37 of cash

Answer: A

Key correction: Source key (D: long $58.40 stock + short $57.77 cash) is wrong: the cash leg is S·N(d1) − c = 58.40 − 6.63 = 51.77, not 57.77.

Step by step

1. d1 = [ln(100/100) + (0.04 + 0.2²/2)(0.5)] / (0.2√0.5) = 0.06/0.28284 = 0.2121; d2 = d1 − 0.28284 = 0.0707.
2. N(d1) = 0.5840, N(d2) = 0.5282.
3. Call c = 100(0.5840) − 100e^(−0.04×0.5)(0.5282) = 58.40 − 51.77 = $6.63.
4. Stock leg = S·N(d1) = $58.40; cash leg = K e^(−rT) N(d2) = $51.77 (equivalently 58.40 − 6.63).
5. Mapping: long $58.40 stock, short $51.77 cash.

Why A is correct: The call value equals S·N(d1) − Ke^(−rT)N(d2): a delta-weighted stock position financed by borrowing Ke^(−rT)N(d2).

Why the others are wrong

- **B:** Used K·N(d2) = 52.82 without discounting the strike at the risk-free rate.
- **C:** Treated the option premium (6.63) as the cash borrowing, forgetting that the stock position is financed by borrowing 58.40 − 6.63.
- **D:** Used an ATM delta of 0.50 instead of N(d1) = 0.584, then subtracted the premium (50.00 − 6.63 = 43.37).

Q307

GARP PE1 Q59 · Mock 3 Q22

REBUILT — Merged two identical sources; source distractor USD 12,627 had no traceable mistake, so replaced it with the 95% (1.645) error. Recomputed 99% VaR = USD 13,715.

A fund manager owns a portfolio of options on TUV, a non-dividend-paying stock: 5,000 deep in-the-money call options and 20,000 deep out-of-the-money call options on TUV. The portfolio also contains 10,000 forward contracts on TUV. TUV currently trades at USD 52. Assume 252 trading days per year, TUV volatility of 12% per year, and that each option and forward contract is on one share. Which of the following amounts is closest to the 1-day 99% VaR of the portfolio?

A. USD 9,699
B. USD 11,557
C. USD 13,715
D. USD 32,000

Answer: C

Step by step

1. Deltas: deep ITM calls ≈ 1, deep OTM calls ≈ 0, forwards = 1.
2. Delta-equivalent shares = 5,000(1) + 20,000(0) + 10,000(1) = 15,000, i.e. exposure = 15,000 × 52 = USD 780,000.
3. Daily volatility = 0.12/√252 = 0.007559.
4. 99% VaR = 2.326 × 0.007559 × 780,000 ≈ USD 13,715, matching option C.

Why C is correct: Deep ITM options behave like the stock and deep OTM options contribute nothing, so the portfolio is a linear exposure to 15,000 shares.

Why the others are wrong

- **A:** Used the 95% multiplier 1.645 instead of 2.326.
- **B:** Used the 1.96 multiplier (97.5% one-tailed) instead of 2.326.
- **D:** Counted the OTM calls at delta 1 (35,000 shares) instead of 0, giving about USD 32,000.

Q308

AnalystPrep Q.1525

REBUILT — Source stem (5x10 FRA with 5- and 10-month bills) was inconsistent with its 6x12 options and described a sold FRA while the options described a long FRA; restated with consistent 6x12 data.

Selling a 6×12 forward rate agreement (FRA) on USD 50 million is equivalent to borrowing USD 50 million for 6 months and investing the proceeds for 12 months. Based on this replication, which of the following correctly describes a LONG 6×12 FRA in terms of bill positions?

A. Long 6-month bill and short 12-month bill
B. Short 6-month bill and long 12-month bill
C. Long 6-month bill and long 12-month bill
D. Short 6-month bill and short 12-month bill

Answer: A

Step by step

1. Sold FRA = borrow for 6 months (short the 6-month bill) + invest for 12 months (long the 12-month bill).
2. A long FRA is the opposite side of the same contract, so every position is reversed.
3. Long 6×12 FRA = long 6-month bill + short 12-month bill.

Why A is correct: The long FRA is the mirror image of the sold FRA, so it is long the shorter bill and short the longer bill.

Why the others are wrong

- **B:** This is the sold FRA (short 6-month, long 12-month); the sides were not reversed.
- **C:** Two long bills is pure investing with no borrowing leg, so it cannot replicate a locked-in forward rate.
- **D:** Two short bills is pure borrowing with no investing leg; it does not replicate an FRA.

Q309

AnalystPrep Q.1519

REBUILT — Source question was vague about the 'two economically equivalent alternatives' and had overlapping options; rebuilt as a concrete FX-forward mapping question.

A bank enters a forward contract to buy EUR 100 million in one year for USD 126.5 million. For VaR mapping, the forward is decomposed into building blocks. Which of the following decompositions is correct?

A. Long a one-year EUR bill (with exposure to EUR spot) and short a one-year USD bill with face value USD 126.5 million
B. Short a one-year EUR bill and long a one-year USD bill with face value USD 126.5 million
C. Long EUR spot only; the interest rate legs are ignored because they are riskless
D. Long a one-year EUR bill and long a one-year USD bill with face value USD 126.5 million

Answer: A

Step by step

1. Buying euros forward is equivalent to buying euros today (spot) and holding them for a year at the euro rate: a long one-year EUR bill converted at spot.
2. The USD 126.5m payable at delivery is a fixed liability, equivalent to a short one-year USD bill.
3. So the forward maps to three risk factors: EUR spot, EUR 1-year rate, USD 1-year rate; VaR then uses their volatilities and correlations.

Why A is correct: A long forward replicates buying the foreign bill financed by borrowing the domestic bill.

Why the others are wrong

- **B:** Signs reversed; this replicates a forward to sell euros.
- **C:** Ignores the rate factors; the bill legs have small but non-zero price risk and are part of the mapping.
- **D:** Treats the USD payment as an asset instead of a liability; the forward buyer is short the USD bill.

Q310 AnalystPrep Q.1522

A 5-year currency swap in which a firm pays dollars and receives euros can be mapped by decomposing it into building blocks. Which statement is correct?

A. It is equivalent to a series of 5 forward contracts to exchange a set amount of dollars for euros on each of the 5 payment dates.
B. It is equivalent to a series of any number of forward contracts to exchange a set amount of dollars per year.
C. It is equivalent to a single forward contract maturing in 5 years.
D. It is not equivalent to a portfolio of forward contracts, because a swap has a non-zero value at initiation.

Answer: A

Step by step

1. A swap is a portfolio of forwards with different maturities, one for each exchange date.
2. A 5-year annual swap has 5 exchange dates, so it maps to 5 forwards (maturities 1 to 5 years), each a combination of spot FX and domestic and foreign zero-coupon bond positions.
3. Each forward is a linear function of the same risk factors, so the delta-normal method applies to the whole portfolio.

Why A is correct: The swap's cash flows occur on 5 dates, and each date's exchange is a forward contract, so the number of forwards equals the number of payment dates.

Why the others are wrong

- **B:** The number of forwards is fixed by the number of payment dates, not arbitrary.
- **C:** A single forward captures only one exchange date and ignores the other four.
- **D:** Wrong premise: a swap is normally priced to have zero value at initiation, and equivalence to forwards does not depend on that.

Q311 AnalystPrep Q.1524

REBUILT — Original stem and options were garbled ('receive payments at a lower rate'); replaced with a numeric FRA settlement question that tests who gains and the discounted payoff.

A company buys (is long) a 6-month FRA on a notional of $10,000,000 at a contract rate of 4.00% (the buyer locks in a borrowing rate and pays fixed). At settlement, at the start of the 6-month period, the 6-month reference rate is 4.60% (simple interest, semiannual accrual). The settlement payment is made at the start of the period, i.e. discounted at the reference rate. What is the settlement amount for the buyer?

A. Receives $30,000
B. Receives $29,326
C. Receives $29,412
D. Pays $29,326

Answer: B

Key correction: AnalystPrep key C (buyer benefits when spot is above the forward rate) is consistent with this rebuild; the original wording was unclear.

Step by step

1. The reference rate (4.60%) exceeds the contract rate (4.00%), so the borrowing-rate buyer gains and the seller loses.
2. Undiscounted payoff at the end of the period: $10{,}000{,}000 \times (0.046 - 0.040) \times 0.5 = 30{,}000$.
3. The payment is made at the start of the period, so discount at the reference rate: $30{,}000/(1 + 0.046 \times 0.5) = 30{,}000/1.023 = 29{,}326$.

Why B is correct: Higher market rates than the locked rate benefit the FRA buyer; the settlement is the rate difference over the period, discounted at the market (reference) rate.

Why the others are wrong

- **A:** Forgot to discount the payoff to the settlement date.
- **C:** Discounted at the contract rate 4.00% (divisor 1.02) instead of the reference rate 4.60%.
- **D:** Sign error: the buyer gains when the reference rate exceeds the contract rate.

Q312 AnalystPrep Q.2825

REBUILT — Replaced an unexplained distractor (1.3676) with the inverted-ratio error (1.3846) so every wrong option reflects a realistic mistake; answer unchanged.

The EUR spot rate is $1.3988 per EUR. The 1-year EUR interest rate is 2.28% and the 1-year USD interest rate is 3.33% (annual compounding). In mapping a one-year EUR forward, what is the one-year forward rate (USD per EUR)?

A. 1.4132
B. 1.0103
C. 1.3846
D. 0.7076

Answer: A

Step by step

1. Interest rate parity: $F = S\,(1 + r_{USD})/(1 + r_{EUR})$.
2. $F = 1.3988 \times 1.0333/1.0228 = 1.4132$.
3. Since the USD rate is higher, the EUR trades at a forward premium, so F > S.

Why A is correct: Covered interest parity makes the forward equal spot times the ratio of domestic (USD) to foreign (EUR) gross interest factors.

Why the others are wrong

- **B:** Computed only the interest factor ratio 1.0333/1.0228 and forgot to multiply by spot.
- **C:** Inverted the ratio (used 1.0228/1.0333), giving a forward discount.
- **D:** Took the reciprocal of the forward (EUR per USD) instead of USD per EUR.

Q313 AnalystPrep Q.2828

REBUILT — Original distractors (0.78, 1.32, 1.5) were giveaways; replaced with options from realistic errors. Answer unchanged.

The spot price of an asset is 1.22. The domestic risk-free rate is 3.70% and the asset's income yield is 1.92% (both continuously compounded). What is the 1-year forward price?

A. 1.27
B. 1.20
C. 1.24
D. 1.29

Answer: C

Step by step

1. $F = Se^{(r-y)\tau}$.
2. $(r - y)\tau = 0.0370 - 0.0192 = 0.0178$.
3. $F = 1.22\,e^{0.0178} = 1.22 \times 1.01796 = 1.2419$, about 1.24.

Why C is correct: The forward compounds spot at the risk-free rate net of the income yield received by holding the asset.

Why the others are wrong

- **A:** Ignored the income yield: $1.22e^{0.037} = 1.266$.
- **B:** Reversed the sign of the net rate: $1.22e^{-0.0178} = 1.198$.
- **D:** Added the yield instead of subtracting: $1.22e^{0.0562} = 1.291$.

Q314 BT answers L2.T5.66.1

For value at risk (VaR) mapping, which are the most likely risk factors for a foreign currency forward contract?

A. Convenience yield, domestic interest rate, and foreign interest rate
B. Spot exchange rate, foreign interest rate, and storage cost
C. Spot exchange rate, domestic interest rate, and foreign interest rate
D. Forward exchange rate, spot exchange rate, and domestic interest rate

Answer: C

Step by step

1. By interest rate parity, $F = Se^{(r_d - r_f)T}$.
2. The forward maps to a long spot FX position, a long foreign bill and a short domestic bill.

3. The risk factors are the spot rate, the domestic rate and the foreign rate.

Why C is correct: A currency forward is priced off spot FX and the two interest rates, so these three drive its value.

Why the others are wrong

- **A:** Convenience yield is a commodity factor, not relevant to currency forwards.
- **B:** Storage cost is a commodity feature; it is irrelevant for currency.
- **D:** The forward rate is itself a function of the other factors, so it is not an independent risk factor, and the foreign rate is omitted.

Q315

BT answers L2.T5.66.2

REBUILT — Original options A and D were both defensible (time-varying spot-forward relationship is a consequence of convenience yield); rewrote so exactly one is correct.

Mapping a commodity forward is generally harder than mapping a currency forward. What is the main reason?

A. Commodity forward prices embed a convenience yield, which is highly variable and not tied to another observable financial variable.

B. Commodity forwards cannot be expressed as a combination of spot and bond positions.

C. Commodity forwards have no sensitivity to domestic interest rates.

D. Commodity spot prices cannot be observed, whereas FX spot rates can.

Answer: A

Key correction: Source key (A) retained; original option D was also arguably defensible, hence the rebuild.

Step by step

1. Currency forward: $F = Se^{(r_d - r_f)T}$; the extra factor r_f is an observable interest rate with available volatility and correlation data.
2. Commodity forward: $F = Se^{(r+u-c)T}$, where u is storage cost and c is convenience yield.
3. Convenience yield is not a traded rate, is unobservable and varies strongly over time, so it adds a risk factor that is hard to measure and map.

Why A is correct: Convenience yield is an additional, unobservable and highly variable risk factor with no financial-market counterpart.

Why the others are wrong

- **B:** Commodity forwards can still be decomposed into spot and zero-coupon positions; the difficulty lies in the extra factor.
- **C:** Commodity forwards are still sensitive to the interest rate through cost of carry.
- **D:** Commodity spot prices are observable; this is not the difficulty.

Q316

AnalystPrep Q.1522

A risk manager maps a 5-year currency swap in which the bank pays dollars and receives euros once a year. Which statement correctly describes how the swap is treated for VaR mapping?

A. It is equivalent to a portfolio of 5 forward contracts, one for each annual exchange date, each mapped to its own risk factors.

B. It is equivalent to a portfolio of an arbitrary number of forward contracts, since the number of forwards does not matter.

C. It is equivalent to a single forward contract maturing in 5 years, mapped to the 5-year risk factors only.

D. It cannot be represented by forward contracts because a swap has multiple payment dates.

Answer: A

Step by step

1. A currency swap exchanges a fixed amount of one currency for a fixed amount of another on each payment date.
2. Each annual exchange is exactly a forward contract on that date (long euro bill, short dollar bill, long euro spot).
3. 5 annual exchanges therefore means 5 forwards, with maturities of 1, 2, 3, 4 and 5 years.
4. Each forward is mapped to its own bills and spot position, and the delta-normal VaR is computed on the combined position.

Why A is correct: A swap is a portfolio of forwards with one forward per exchange date, so mapping a swap means mapping each of those forwards onto its general risk factors.

Why the others are wrong

- **B:** The number of forwards is fixed by the number of payment dates (5), not arbitrary; ignoring this misses the cash flow structure.

- **C:** Collapsing all exchanges into one 5-year forward ignores the intermediate payments, which carry exposure to shorter-maturity rates.
- **D:** Multiple payment dates are the reason a swap is decomposed into several forwards, not a barrier to doing so.

Q317 AnalystPrep Q.1524

REBUILT — Original options were garbled and non-numerical (buyer 'receives payments at a lower rate'); rebuilt as a self-contained FRA settlement question with numbers and a single correct cash flow.

A bank buys a 6 x 12 forward rate agreement (FRA) on a notional of $50 million at a contract rate of 4.8% (the FRA buyer locks in a borrowing rate). Six months later the prevailing 6-month rate is 5.6%. Assuming a 0.5-year accrual period and settlement at the start of the FRA period, what is the settlement?

A. The buyer receives about $194,553 from the seller.
B. The buyer pays about $194,553 to the seller.
C. The buyer receives $200,000 from the seller.
D. The buyer receives about $195,313 from the seller.

Answer: A

Step by step

1. The buyer locked in borrowing at 4.8%; the market rate is now 5.6% > 4.8%, so the FRA is in the buyer's favor and the buyer receives cash.
2. Undiscounted payoff at the end of the period: $50{,}000{,}000 \times (0.056 - 0.048) \times 0.5 = 200{,}000$.
3. FRA settlement is made at the start of the period, so discount at the current market rate: $200{,}000/(1 + 0.056 \times 0.5) = 200{,}000/1.028$.
4. Settlement $\approx$ 194,553, paid by the seller to the buyer.

Why A is correct: When the market rate exceeds the FRA rate, the buyer (borrower-hedger) gains and the seller loses; the interest differential is discounted at the market rate for the settlement date.

Why the others are wrong

- **B:** Reverses the direction: the buyer gains when rates rise above the locked-in rate, so the buyer does not pay.
- **C:** Omits discounting; the 200,000 is the interest differential at the end of the period, not its present value at settlement.
- **D:** Discounts at the contract rate 4.8% (200,000/1.024) instead of the prevailing market rate of 5.6%.

Reading 6. Validating Bank Holding Companies' Value-at-Risk Models for Market Risk

David Lynch (Editor), Iftekhar Hasan (Editor), Akhtar Siddique (Editor), Validation of Risk Management Models for Financial Institutions (Cambridge University Press, 2023). Chapter 2.

LO	Learning objective	Qs
6.a	describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process.	7
6.b	explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis.	5
6.c	describe the challenges a financial institution could face when calculating confidence intervals for VaR.	8
6.d	discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them.	8

LO 6.a

Describe some important considerations for a bank in assessing the conceptual soundness of a VaR model during the validation process.

Q318

AnalystPrep Q.6207

REBUILT — Original keyed 'shorten window to 5 years', which is defensible but arbitrary (other fixes also arguable); rebuilt so the Schweser point (equal-weighted historical VaR ignores dynamic volatility; GARCH-type updating fixes it) gives a single answer.

ABC Financial uses historical simulation VaR with an equally weighted 10-year window of portfolio returns. In a recent period of unusually high volatility, daily losses exceeded the 99% VaR on several occasions. A validation team is assessing the conceptual soundness of the model. Which change would MOST directly address the cause of the exceedances?

A. Raising the confidence level while keeping the same equally weighted history

B. Adding a stress-testing overlay and using the stress results in place of the VaR figure

C. Replacing the equally weighted history with a volatility-updating approach (e.g., GARCH-based VaR) that reflects recent changes in volatility

D. Extending the historical window from 10 to 20 years to obtain more tail observations

Answer: C

Step by step

1. Exceedances clustered in a high-volatility period: the model is slow to react to changing volatility.
2. An equally weighted 10-year history gives recent turbulent observations only a small weight, so VaR lags the current risk level.
3. Schweser notes that historical VaR does not account for dynamic volatility and that GARCH VaR models were developed to allow for recent volatility changes.
4. So a volatility-updating method addresses the cause; the other options leave the lag in place.

Why C is correct: Historical VaR with a long, equally weighted window ignores time-varying volatility; GARCH-type models update the risk estimate as volatility changes, so VaR rises in turbulent periods.

Why the others are wrong

- **A:** A higher confidence level scales VaR up but does not make it respond to current volatility; it still lags (and 99% is already the level used).
- **B:** Stress testing complements VaR but does not repair the model's failure to track volatility; replacing VaR with it is not a fix of the VaR model.
- **D:** A longer window dilutes recent volatility even more, making the lag worse.

Q319

AnalystPrep Q.6438

Zenith Securities is validating a new simulation-based VaR model for a trading portfolio of equities, fixed income, and derivatives. The validation team is assessing conceptual soundness, in particular how the model reflects changes in risk caused by trading activity. A junior analyst proposes to rely primarily on backtesting against historical profit and loss (P&L). Why is relying solely on backtesting insufficient to assess conceptual soundness?

A. Backtesting primarily evaluates the model's ability to predict future losses rather than its ability to simulate historical risks.

B. Backtesting assesses historical performance but does not evaluate how the model links position changes to risk.
C. Backtesting primarily focuses on tail risks, which may not represent the model's overall behavior.
D. Backtesting is primarily a regulatory requirement and not a useful tool for internal model validation.

Answer: B
Step by step
1. Conceptual soundness requires that VaR respond appropriately when positions change; a model that does not show a change in risk when positions change fails a crucial conceptual test.
2. Backtesting only compares VaR with realized P&L, which can be passed by chance and says nothing about the position-to-risk linkage.
3. So additional tests (pseudo-history, sensitivity analysis) are needed; B states this.

Why B is correct: Backtesting is an outcome test; conceptual soundness concerns the model's structure, including whether VaR moves correctly with positions.
Why the others are wrong
- **A:** Backtesting compares ex-ante VaR with subsequent realized P&L; this mischaracterizes it and does not give the real reason it is insufficient.
- **C:** Backtesting of exceedances does look at the tail, but that is not the validation gap; the gap is the missing position-risk link.
- **D:** Backtesting is a useful internal validation tool; calling it merely regulatory is false.

Q320 AnalystPrep Q.6463

A risk manager explains the concept of a "pseudo history" in VaR model validation to the board. Which statement best describes the purpose of creating a pseudo history of portfolio value changes?
A. To backtest the model against historical market crashes
B. To incorporate positional information so the model reflects how risk changes with trading activity
C. To test the model's ability to predict future portfolio losses under extreme market conditions
D. To generate synthetic data for improving the model's predictive accuracy

Answer: B
Step by step
1. A pseudo history applies historical market changes to the CURRENT portfolio positions.
2. Example: holding EUR 10m today versus EUR 5m historically doubles the impact of past FX moves in the pseudo history.
3. So VaR reflects current composition and changes made by trading, as in B.

Why B is correct: Pseudohistorical returns are recalculated using current portfolio composition, so VaR responds to position changes.
Why the others are wrong
- **A:** A pseudo history is not a crash backtest; it re-expresses past market moves on today's portfolio.
- **C:** It does not test forward predictive ability; that is a backtesting or stress-testing task.
- **D:** The data are not synthetic; they are actual historical market moves applied to current positions.

Q321 AnalystPrep Q.6467

A portfolio manager is told by the risk department to reduce the portfolio's VaR and responds by cutting holdings in all assets by 25%. Which approach would MOST effectively demonstrate the reduction in risk?
A. Recalculating VaR from the portfolio's historical P&L before the position reduction
B. Recalculating VaR from the portfolio's historical P&L after the position reduction
C. Recalculating VaR from a pseudo history of P&L that applies the reduced positions to all historical scenarios
D. Comparing the current VaR with the average VaR over the previous year

Answer: C
Step by step
1. VaR from actual historical P&L reflects the old positions, so cutting positions leaves it unchanged (A).
2. Applying the 25% smaller positions to every historical scenario scales each pseudo P&L by 0.75, so VaR falls by 25% for a proportional cut.

3. Only the pseudo-history approach shows the change immediately and fully.

Why C is correct: A conceptually sound VaR must respond to position changes; a pseudo history re-values today's positions over the whole historical window.

Why the others are wrong

- **A:** Pre-reduction historical P&L embeds the old positions, so VaR does not change and shows no reduction.
- **B:** Post-reduction actual P&L has only a short sample and the full window still reflects old positions; the new positions are not applied to the full history.
- **D:** Comparing with last year's average VaR confounds the position change with market-volatility changes and does not isolate the effect.

Q322

BT answers P2.T5.25.5.1

During a regulatory review, a CRO must justify how the bank's VaR model adapts to shifting portfolio risks in a dynamic trading environment. Regulators are concerned that some banks rely on historical P&L-based VaR, which may not capture risk evolution as positions change. One panel member suggests a standardized volatility model such as GARCH, while others debate whether simplifying the model sacrifices predictive accuracy. Which approach best ensures that the VaR model accurately reflects changing portfolio risk?

A. Prioritizing historical P&L stability over dynamic adjustments to avoid excessive model complexity

B. Applying GARCH-based volatility modeling across all asset classes because of its superior conditional risk measurement

C. Using a pseudo history to incorporate dynamic portfolio shifts, improving scenario-based VaR estimates

D. Relying on parametric VaR for better interpretability, as complex risk estimates can reduce decision-making clarity

Answer: C

Step by step

1. The issue is VaR failing to change when positions change.
2. A pseudo history applies historical market moves to the current portfolio, so position changes flow into VaR.
3. GARCH updates volatility but, if applied to historical P&L, still ignores the current composition; parametric VaR does not solve it either.

Why C is correct: Only recalculating historical returns on current positions makes VaR respond to trading activity.

Why the others are wrong

- **A:** Historical P&L reflects past positions; stability here means VaR ignores risk changes, the very weakness cited.
- **B:** GARCH addresses time-varying volatility, not changes in portfolio composition, and is not a universal solution.
- **D:** Interpretability does not make a parametric model respond to position changes; it is not the criterion at issue.

Q323

BT answers P2.T5.25.5.2

REBUILT — Original option B (illiquidity) is outside the reading and the options were loosely true; rebuilt the 'NOT a conceptual weakness' item around Schweser's points (normality, data window, historical returns ignoring portfolio composition).

A mid-sized investment bank trades structured credit and commodity derivatives. Its VaR model is parametric (variance-covariance), assumes normally distributed returns, and uses a one-year window of historical portfolio returns without adjusting for changes in portfolio composition. Backtesting shows exceedances in market shocks. Management is considering Monte Carlo or historical simulation but worries about computing cost. Which of the following is NOT a conceptual weakness of the current model?

A. Assuming normally distributed returns understates fat-tailed risk in structured credit and commodities.

B. A one-year window may omit extreme market conditions and past crises.

C. Monte Carlo and historical simulation require more computing power than a variance-covariance model.

D. Using unadjusted historical portfolio returns ignores changes in portfolio composition over the window.

Answer: C

Step by step

1. Conceptual weaknesses concern the model's methodology, input data and assumptions (the three validation areas in the reading).
2. A is an assumption weakness (non-normal returns); B is a data weakness (short window); D is an input weakness (no pseudo-history for changed positions).
3. C concerns the operational cost of alternatives, not a flaw in the current model's logic, so C is the one that is NOT a weakness.

Why C is correct: Computational burden of replacement models is an implementation consideration, not a defect in the current model's assumptions, data, or methodology.

Why the others are wrong

- **A:** This is a genuine assumption weakness (normality understates tail risk), so it is a weakness; the question asks for the one that is not.
- **B:** This is a data weakness (a short window can miss crises), so it is a weakness.
- **D:** This is an input weakness (without pseudohistorical returns VaR ignores composition changes), so it is a weakness.

Q324

Schweser Module Quiz 6.1 Q1

Which of the following items is crucial when evaluating the conceptual soundness of a VaR model?

A. The portfolio return distribution should be normal.
B. Inputs need to be adjusted to determine the change in VaR.
C. The model should use actual historical returns to calculate VaR.
D. The model should be designed to meet specific risk management objectives.

Answer: D

Step by step

1. Validation covers methodology, input data and assumptions, and the model must fit the bank's risk management objectives (e.g., regulatory capital versus internal risk assessment).
2. A, B and C are not requirements or describe different techniques; D matches the reading.

Why D is correct: A conceptually sound model is fit for its intended use.

Why the others are wrong

- **A:** Returns are generally non-normal; normality is not a requirement and assuming it is a weakness.
- **B:** Adjusting inputs to see how VaR changes is sensitivity analysis (LO 6.b), not the crucial soundness criterion.
- **C:** Actual historical returns ignore changing portfolio composition; pseudohistorical returns are preferred when positions change.

LO 6.b

Explain how to conduct sensitivity analysis for a VaR model, and describe the potential benefits and challenges of performing such an analysis.

Q325

AnalystPrep Q.6209

REBUILT — Original was a generic stress-testing question not tied to sensitivity analysis; rebuilt to test how to conduct a VaR sensitivity analysis on the correlation assumption.

A global bank with large foreign exchange and interest rate derivative positions is concerned that its VaR model is sensitive to the assumed correlations between market factors, which may shift during geopolitical shocks. The risk team wants to conduct a sensitivity analysis of the VaR model to this assumption. Which approach is most consistent with how a VaR sensitivity analysis is conducted?

A. Replace the VaR model with a historical simulation approach and compare the new VaR to the old VaR.
B. Lower the VaR confidence level and observe how the reported VaR changes.
C. Identify correlations as a key input, change them one at a time (holding other inputs fixed), recalculate VaR, and compare each new VaR with the original.
D. Compare the number of daily VaR exceptions with the number expected at the chosen confidence level.

Answer: C

Step by step

1. Step 1: identify the key inputs and assumptions (here, the factor correlations).
2. Step 2: adjust one input at a time, keeping all others unchanged, and recalculate VaR.
3. Step 3: compare each new VaR with the original to see how sensitive VaR is to that input and to judge model robustness.
4. Only C follows this identify-adjust-analyze procedure.

Why C is correct: Sensitivity analysis isolates one input, re-runs the model and measures the change in VaR, re-

vealing how dependent the result is on that assumption.

Why the others are wrong

- **A:** Changing the methodology changes many assumptions at once, so the difference cannot be attributed to correlations; this is model comparison/benchmarking, not sensitivity analysis.
- **B:** The confidence level is a parameter choice, not the correlation assumption; it does not test the input of concern.
- **D:** Counting exceptions is backtesting, which assesses accuracy ex post and does not vary inputs.

Q326

AnalystPrep Q.6464

A financial analyst is performing sensitivity analysis on a VaR model. What is the primary purpose of such an exercise?

A. To determine the model's backtesting performance.
B. To evaluate the accuracy of the VaR model in predicting portfolio losses.
C. To identify the key drivers of VaR and assess model robustness.
D. To assess whether the model's VaR estimates meet regulatory requirements under varying assumptions.

Answer: C

Step by step

1. Sensitivity analysis changes key inputs (positions, volatilities, correlations) one at a time and recalculates VaR.
2. The size of the change in VaR shows which inputs drive the result and whether the model is robust to its assumptions.
3. This matches C; accuracy and exception counting are backtesting functions.

Why C is correct: Varying inputs and observing the change in VaR reveals the main risk drivers and tests the model's robustness and conceptual soundness.

Why the others are wrong

- **A:** Backtesting performance is judged by comparing VaR with realized P&L, not by varying inputs.
- **B:** Predictive accuracy is also a backtesting question; sensitivity analysis does not compare to actual losses.
- **D:** Regulatory adequacy is judged through backtesting and capital rules; compliance is a secondary benefit, not the primary purpose.

Q327

BT answers P2.T5.25.5.3

A global systemically important bank (G-SIB) is conducting a sensitivity analysis on its VaR model as part of its validation process. The bank uses a hybrid approach that combines parametric methods with Monte Carlo simulations. During testing, analysts observe that small variations in input volatility assumptions lead to large fluctuations in VaR estimates, particularly for certain illiquid fixed-income securities and complex derivatives.

At the same time, the risk oversight committee is concerned that stress scenarios designed to evaluate the model's stability may not fully capture non-linear risks inherent in the bank's structured product holdings. Some senior risk managers argue that the model's heavy reliance on recent historical correlations may produce overly stable VaR estimates that do not reflect true tail risk exposure during market stress.

Given these findings, which of the following actions would best address the concerns raised while maintaining a robust and conceptually sound sensitivity analysis framework?

A. Adjust the VaR model to use a constant historical correlation matrix, ensuring that sensitivity tests show stable risk estimates over time and avoiding excessive model recalibrations.
B. Introduce factor perturbation sensitivity analysis by systematically shifting key risk factors such as implied volatility, correlations, and liquidity spreads to assess VaR stability across extreme but plausible scenarios.
C. Reduce the Monte Carlo simulation iterations to limit computational burden while maintaining historical return distributions, as excessive simulations could distort sensitivity analysis findings.
D. Rely on ex-post backtesting results to determine whether sensitivity analysis is necessary, as the primary focus should be on validating real-world P&L alignment rather than artificially constructed stress scenarios.

Answer: B

Step by step

1. Concerns: VaR is highly sensitive to volatility inputs, stress scenarios miss non-linear risk, and historical correlations may understate tail risk.
2. Sensitivity analysis addresses these by shifting key inputs (implied volatility, correlations, liquidity spreads)

and recalculating VaR.
3. B does this across extreme but plausible ranges, exposing instability and hidden non-linear risk; A, C and D each remove or weaken the testing.

Why B is correct: Systematically perturbing the key risk factors and recalculating VaR shows which assumptions drive the result and how stable VaR is under stress.

Why the others are wrong

- **A:** Fixing the correlation matrix hides the very sensitivity being tested and makes VaR look stable artificially.
- **C:** Fewer simulations increase sampling error, particularly for non-linear products, and do not test input sensitivity.
- **D:** Backtesting is ex post and does not replace sensitivity analysis, which exposes assumption weaknesses before losses occur.

Q328

Schweser Module Quiz 6.1 Q2

Which of the following actions is least likely a benefit of sensitivity analysis?

A. Model validation.
B. Regulatory compliance.
C. Improved decision-making.
D. Reducing regulatory capital.

Answer: D

Step by step

1. The benefits of VaR sensitivity analysis are risk assessment, model validation, regulatory compliance and improved decision-making.
2. Regulatory capital depends on the VaR level, driven by portfolio exposures, not on performing the analysis.
3. So D is the least likely benefit.

Why D is correct: Sensitivity analysis informs and validates the model but does not itself change the portfolio's risk, so it does not directly reduce capital.

Why the others are wrong

- **A:** A genuine benefit: it tests robustness of the model to its assumptions.
- **B:** A genuine benefit: it demonstrates understanding of model limitations to regulators.
- **C:** A genuine benefit: knowing how inputs affect VaR supports position and mitigation decisions.

Q329

AnalystPrep Q.6209

REBUILT — Original asked for the best way to 'improve' the VaR model and keyed stress testing, but historical simulation over past crisis data was also defensible and neither choice tested LO 6.b; rebuilt so the question asks which approach is a sensitivity analysis of the correlation assumption (identify input, shift it one at a time, recalculate and compare VaR), keeping C as the key.

A global investment bank holds large foreign exchange and interest rate derivative positions. As part of validating the conceptual soundness of its VaR model, the risk team wants to find out how much the VaR estimate depends on the model's assumed correlation between FX and interest rate risk factors, which could shift sharply during geopolitical stress. Which of the following approaches is most consistent with conducting a sensitivity analysis of the VaR model for this purpose?

A. Replace the current model with a historical simulation that uses only data from past geopolitical crises, and compare the number of backtesting exceptions of the two models.
B. Lower the VaR confidence level from 99% to 95% so that the VaR estimate is less affected by uncertainty in the correlation assumption.
C. Identify the FX/interest rate correlation as a key input, change it by a set amount while holding all other inputs fixed, recalculate VaR, and compare the revised VaR with the original estimate.
D. Lengthen the historical window used to estimate correlations, and adopt whichever window length produces the lowest VaR.

Answer: C

Step by step

1. Step 1, identify key inputs: the assumed correlation ρ between the FX and interest rate factors is a key assumption of the VaR model.
2. Step 2, adjust one input at a time and recalculate VaR. Illustration with stand-alone VaRs of 10 (FX) and 8 (rates): at $\rho = 0.2$, VaR $= \sqrt{10^2 + 8^2 + 2(0.2)(10)(8)} = \sqrt{196} = 14.0$.

3. Shift the correlation to $\rho = 0.6$ with everything else unchanged: VaR $= \sqrt{100 + 64 + 96} = \sqrt{260} = 16.12$.
4. Step 3, analyze results: compare 16.12 with the original 14.0, an increase of about 15.2%. This shows how sensitive VaR is to the correlation assumption and whether that assumption needs closer validation or a model update.
5. Only option C follows this identify, adjust and recalculate, compare sequence. A, B and D change the model, the metric or the estimation window instead of isolating the effect of one input.

Why C is correct: A VaR sensitivity analysis isolates one key input or assumption, perturbs it while holding the others fixed, recalculates VaR and compares the result with the original estimate. This shows how much the output depends on that assumption, so it tests the model's robustness and conceptual soundness directly.

Why the others are wrong

- **A:** Swapping in a different model and counting exceptions is a benchmarking or backtesting exercise, not a sensitivity analysis. It also depends on having enough relevant crisis data, and it changes many inputs at once, so the correlation effect cannot be isolated.
- **B:** Lowering the confidence level just shrinks the reported VaR and tells you nothing about how VaR responds to the correlation assumption. It treats a lower risk number as if it dealt with model uncertainty.
- **D:** A longer window dilutes recent changes in correlation, and choosing the window that gives the lowest VaR is result-shopping, not validation. No single input is isolated, so you cannot measure how sensitive VaR is to it.

LO 6.c

Describe the challenges a financial institution could face when calculating confidence intervals for VaR.

Q330

AP Mock 3 Q1

REBUILT — Distractors A, C and D were arguable (resampling bias, extreme events, normality); reworded so the computational burden of re-estimating GARCH per resample is the only defensible challenge.

A bank builds a confidence interval for its GARCH VaR by bootstrapping: it resamples the standardized residuals with replacement, rebuilds a return series, re-estimates the GARCH parameters and recomputes VaR, repeating this thousands of times. Which of the following is the main practical challenge of this approach?

A. Resampling with replacement systematically biases the VaR estimates toward zero.

B. The standardized residuals must be assumed to be normally distributed before they can be resampled.

C. Re-estimating the GARCH model by numerical likelihood maximization for every bootstrap sample is computationally intensive.

D. Resampled histories cannot contain the extreme observations that appear in the original data.

Answer: C

Step by step

1. The bootstrap draws a pseudo-history with replacement and computes a VaR for each draw; the collection of VaRs gives the distribution used for the interval.
2. For GARCH VaR each draw requires refitting the parameters (numerical maximum likelihood) and then recomputing the volatility path and VaR.
3. With 1,000 to 10,000 resamples this means thousands of nonlinear optimizations, so run-time is the practical obstacle.

Why C is correct: Each bootstrap sample needs a full GARCH re-estimation, so the number of optimizations equals the number of resamples.

Why the others are wrong

- **A:** The bootstrap is not systematically biased; this confuses sampling variability with bias.
- **B:** Bootstrapping is nonparametric and uses the empirical residual distribution, so no normality is needed; this confuses it with the parametric normal-density approach.
- **D:** Extreme observations in the data are part of the sample and can be redrawn; the real limitation is only that losses beyond those observed cannot be generated.

Q331 BT answers P2.T5.25.6.1 · BT notes P2.T5.25.6.1

REBUILT — Double-negative stem and a wrong source key; restated as 'which statement is NOT a valid reason' with the single false statement as the answer.

A bank's historical-simulation 99% VaR confidence intervals were found to be too narrow in a regulatory stress test. A group in the risk team proposes simply extending the historical lookback period. Which of the following is NOT a valid reason why extending the lookback period is not a foolproof solution?

A. Longer historical periods may dilute the impact of recent extreme events, leading to an underestimation of current risk.

B. Increasing the lookback period will always improve model accuracy by incorporating a wider range of tail risk scenarios.

C. Market dynamics evolve over time, and older data may not reflect current volatility regimes, leading to misleading risk estimates.

D. Historical simulation assumes that past returns are representative of future risks, which may not hold in rapidly changing environments.

Answer: B

Key correction: Source key (BT answers and notes) said A; correct is B because A, C and D are all genuine limitations of a longer lookback, while B ('always improves') is the only false statement and therefore the only one that is not a valid reason.

Step by step

1. The stem asks for the statement that is NOT a valid reason, so find the false statement.
2. A: older data can swamp recent stress observations, so current risk is understated. Valid reason.
3. C: old regimes may be unrepresentative of current volatility. Valid reason.
4. D: historical simulation relies on the past being a guide to the future. Valid reason.
5. B: more data does tighten the interval statistically, but 'always improves accuracy' is false because stale or non-representative data can make estimates worse. So B is the answer.

Why B is correct: A longer window helps statistically, but it is not guaranteed to improve accuracy; the absolute word 'always' makes B the one false statement.

Why the others are wrong

- **A:** A valid reason (dilution of recent extremes), so it cannot be the 'NOT' answer; picking it is the source-key error.
- **C:** A valid reason (regime change makes old data stale).
- **D:** A valid reason (the stationarity assumption of historical simulation).

Q332 AnalystPrep Q.6436

Which of the following is most likely a key challenge in calculating confidence intervals for VaR using order statistics?

A. The difficulty in incorporating time-varying volatility into the estimation.

B. The need to assume that the data points are not independent.

C. The requirement of complex distributional assumptions.

D. The discrete nature of the binomial distribution.

Answer: D

Step by step

1. Order-statistic intervals rank the pseudo-returns and use the binomial distribution to choose the ranks that bracket the VaR quantile.
2. Because the binomial is discrete, only certain ranks and coverage probabilities are available, so the interval is coarse (it may be wider than desired or miss the exact confidence level).
3. The method is nonparametric (no distributional assumptions) and assumes independent observations, which rules out B and C; time-varying volatility is a feature of the VaR model, not specific to order statistics.

Why D is correct: The discreteness of the binomial limits the available ranks, so exact confidence levels and tight intervals are hard to obtain.

Why the others are wrong

- **A:** Time-varying volatility is a limitation of the underlying historical simulation, not a challenge specific to the order-statistic interval.
- **B:** The method assumes independence, not non-independence.
- **C:** Order statistics is nonparametric, so it needs no complex distributional assumption; that applies to parametric intervals.

Q333 AnalystPrep Q.6440

REBUILT — Options C and D were arguable (overfitting, residual distribution); reworded so the conceptual challenge of treating the estimated GARCH model as the true model is clearly the only answer, and the stem now asks for the conceptual rather than computational challenge.

A validation team constructs a confidence interval for GARCH VaR with a residual bootstrap. Setting aside run-time, which of the following is the most significant conceptual limitation of the resulting interval?

A. The GARCH structure cannot capture time-varying volatility within the bootstrap samples.
B. Resampling treats the estimated GARCH model as the true data-generating process, so model misspecification is not reflected in the interval.
C. The bootstrap requires the standardized residuals to be normal in order to preserve the model's distribution.
D. Resampling with replacement makes the VaR estimates from different samples perfectly correlated.

Answer: B

Step by step

1. The bootstrap resamples standardized residuals from the fitted model and rebuilds histories using the estimated parameters.
2. Every pseudo-history is therefore generated assuming the fitted GARCH is correct, so the interval captures sampling error but not model (specification) risk.
3. If the model is misspecified, the interval can be too narrow and unreliable even though it looks precise.

Why B is correct: The interval is conditional on the fitted GARCH being the true model, so model risk is ignored.

Why the others are wrong

- **A:** GARCH is specifically designed to model time-varying volatility.
- **C:** The bootstrap uses the empirical residual distribution and needs no normality assumption.
- **D:** Resamples are drawn independently, which is what gives a distribution of VaR estimates; they are not perfectly correlated.

Q334 AnalystPrep Q.6468

REBUILT — Source's correct option rested on filtered historical simulation, which the reading does not cover; rebuilt around the reading's finding that GARCH VaR gives tighter intervals than historical simulation.

A risk analyst compares confidence intervals for one-day 99% VaR of a bank's P&L estimated by historical simulation, GARCH, order statistics and bootstrapping. Which of the following statements about the width of the confidence intervals is correct?

A. Confidence intervals for VaR are symmetric around the point estimate.
B. GARCH VaR tends to produce tighter confidence intervals than historical simulation VaR.
C. Using more data generally leads to wider confidence intervals.
D. Order statistics consistently produces tighter confidence intervals than bootstrapping.

Answer: B

Step by step

1. Recall the empirical findings: VaR intervals are not symmetric; more data gives tighter intervals; GARCH VaR gives tighter intervals than historical simulation; neither order statistics nor bootstrap is consistently tighter.
2. Statement B matches the GARCH finding.
3. A contradicts asymmetry; C reverses the effect of sample size; D claims a consistent ranking that the evidence does not show.

Why B is correct: GARCH reflects current volatility, so it produces more precise VaR estimates and narrower intervals than plain historical simulation.

Why the others are wrong

- **A:** Intervals are not symmetric.
- **C:** More data gives more precise estimates, so intervals are tighter, not wider.
- **D:** Neither nonparametric method is consistently tighter than the other.

Q335 AnalystPrep Q.6469

REBUILT — Original options C and D were both defensible; rewrote so exactly one statement is correct and the others contradict the reading.

A risk manager uses historical simulation to estimate one-day 99% VaR for a portfolio from (i) the most recent calm one-year period, (ii) a one-year stress period and (iii) a much longer history. Which statement about confidence interval width is correct?

A. The interval is widest for the longer history, because it contains more regimes.
B. The interval is narrowest for the stress period, because large losses make the quantile easy to locate.
C. The interval is the same for all three periods, because historical simulation makes no distributional assumption.
D. The interval for the stress period is wider than for the calm one-year period, while the longer history gives a tighter interval than a one-year sample.

Answer: D
Step by step
1. During stress periods historical simulation VaR is less precise, so the stress-period interval is wider than the calm one-year interval.
2. More data gives tighter intervals, so the longer history beats either one-year sample on precision.
3. D states both facts; A and B reverse them and C ignores that sample size and volatility affect sampling error even for nonparametric methods.

Why D is correct: Stress raises estimation uncertainty and a larger sample reduces it, so D combines both findings.
Why the others are wrong
- **A:** Reverses the sample-size effect; more data tightens the interval.
- **B:** Reverses the stress effect; stress periods give wider intervals.
- **C:** Nonparametric does not mean insensitive; the dispersion of tail observations and sample size still drive interval width.

Q336 AnalystPrep Q.6471

A risk analyst is comparing confidence intervals for historical simulation VaR constructed with order statistics and with bootstrapping. According to empirical studies, which statement best describes their relative performance?
A. Order statistics consistently produces significantly tighter confidence intervals than bootstrapping.
B. Bootstrapping consistently produces significantly tighter confidence intervals than order statistics.
C. Neither method consistently produces tighter confidence intervals than the other.
D. Bootstrapping is applicable only to parametric VaR methods, not to historical simulation.

Answer: C
Step by step
1. The reading states that for the two nonparametric methods neither produces a tighter confidence interval.
2. So C is correct; A and B assert a consistent ranking.
3. D is false: bootstrapping resamples the pseudo-history with replacement, which is precisely how it is applied to historical simulation.

Why C is correct: Empirical evidence shows no consistent winner between order statistics and bootstrap.
Why the others are wrong
- **A:** Claims order statistics is consistently tighter; no such evidence.
- **B:** Claims bootstrap is consistently tighter; no such evidence.
- **D:** Bootstrap is nonparametric and is applied directly to the historical (pseudo) return history.

Q337 Schweser Module Quiz 6.1 Q3

Which of the following findings is incorrect regarding the empirical analysis of VaR confidence intervals?
A. Confidence intervals are not symmetric.
B. Larger datasets lead to tighter confidence intervals.
C. Order statistics produces tighter confidence intervals for VaR compared to bootstrap techniques.
D. GARCH VaR tends to produce much tighter confidence intervals compared to historical simulation VaR.

Answer: C
Step by step
1. The question asks for the INCORRECT finding, so test each statement against the empirical findings on VaR confidence intervals.
2. A: VaR confidence intervals are asymmetric (the tail is sparse on the loss side) - true.
3. B: more data gives more precise quantile estimates, hence tighter intervals - true.
4. C: the two nonparametric methods (order statistics and bootstrap) are found to give no tighter interval than one another - the statement is false.

5. D: GARCH VaR responds to current volatility and gives tighter intervals than historical simulation VaR (which is especially wide in stress periods) - true.

Why C is correct: Neither order statistics nor bootstrap produces a tighter confidence interval than the other, so the claim that order statistics is tighter is the incorrect finding.

Why the others are wrong

- **A:** This is a correct finding; picking it means wrongly assuming intervals are symmetric like a normal-based interval.
- **B:** This is a correct finding; a larger sample reduces sampling error in the quantile estimate.
- **D:** This is a correct finding; picking it reflects doubting that conditional-volatility (GARCH) estimation is more precise than unconditional historical simulation.

LO 6.d

Discuss the challenges in benchmarking VaR models and various approaches proposed to overcome them.

Q338

BT answers P2.T5.25.6.3 · BT notes P2.T5.25.6.3

REBUILT — Original asked for the best hybrid VaR/ES framework, which is not supported by the reading and is vague; rebuilt to test the reading's point that plotting two models reveals only conservatism and formal evaluation needs statistical tests.

A bank is moving from its current VaR model to a new one and wants to benchmark the two. The risk team's proposal is to plot the daily VaR of both models over the same period and compare the lines. Which statement best describes the information gained from this approach and what a more formal evaluation would require?

A. The plot shows which model is more accurate; no further testing is needed.
B. The plot mainly reveals which model is more conservative; a formal evaluation requires statistical tests.
C. The plot proves that the models' errors are independent and identically distributed.
D. The plot is the only valid approach because banks routinely maintain several competing models.

Answer: B

Step by step

1. Schweser: the common transition-period approach is to plot the two models' VaR.
2. Such a plot reveals very little beyond which approach is more conservative (higher VaR).
3. Accuracy cannot be read off a chart; formal benchmarking evaluation involves statistical tests (e.g., backtesting or loss-function comparison).

Why B is correct: A visual comparison shows only relative conservatism, not accuracy, so statistical tests are needed for a formal comparison.

Why the others are wrong

- **A:** Confuses a higher or lower line with accuracy; the plot says nothing about how often or by how much VaR is breached.
- **C:** Because trading portfolios change frequently, errors are generally not i.i.d.; a plot proves nothing about this.
- **D:** Banks seldom have even one competing model because building it is costly; benchmarking is not routine, and a plot is not the only valid approach.

Q339

AnalystPrep Q.6439

A risk manager at Gamma Investments is comparing two different VaR models: a historical simulation model and a Monte Carlo simulation model. They have plotted the VaR outputs of both models over a period of time. However, they are unsure how to formally assess which model is more accurate. Which of the following approaches would be MOST appropriate for Gamma Investments to rigorously compare the two VaR models?

A. Observing the visual differences between the plotted VaR outputs.
B. Using a loss function-based comparison.
C. Comparing the computational speed of the two models.
D. Evaluating which model produces a higher VaR estimate.

Answer: B

Step by step

1. Plotting only reveals which model is more conservative, so it is not a formal assessment.

2. Formal benchmarking uses statistical tests; a loss function-based comparison scores each model's VaR against realized P&L day by day.
3. Speed and level of VaR do not measure accuracy.

Why B is correct: A loss function evaluates each model's VaR against realized outcomes, giving a rigorous statistical basis for comparing accuracy.

Why the others are wrong

- **A:** A visual comparison is informal and shows only relative conservatism.
- **C:** Computational speed is an efficiency attribute, not accuracy.
- **D:** A higher VaR only means more conservative, not more accurate; it may overstate risk and tie up capital.

Q340

AnalystPrep Q.6444

A bank is comparing its positional VaR model to a P&L-based VaR model using a loss function-based approach. They are using the regulatory loss function. What is the primary focus of the regulatory loss function in this comparison?

A. Ensuring that the VaR model aligns with regulatory guidelines for acceptable confidence levels.
B. Penalizing underestimation of risk (exceptions) more heavily than overestimation.
C. Ensuring that the VaR estimates are consistent across different time horizons.
D. Balancing the trade-off between false positives (overestimations) and false negatives (underestimations) in risk predictions.

Answer: B

Step by step

1. The regulatory loss function is built around VaR exceptions, which drive regulatory capital multipliers.
2. It therefore charges a loss when realized loss exceeds VaR and nothing (or little) when VaR is merely conservative.
3. So underestimation is penalized more heavily than overestimation.

Why B is correct: Because exceptions raise capital requirements, the regulatory loss function is asymmetric and penalizes underestimation of risk more.

Why the others are wrong

- **A:** It does not test confidence-level compliance; it scores performance given the chosen level.
- **C:** Time-horizon consistency is not what the function measures.
- **D:** A symmetric trade-off between over- and underestimation describes a different (symmetric) loss function, not the regulatory one.

Q341

AnalystPrep Q.6461

REBUILT — Original option A was arguably defensible and the key hinged on a loose reading; rewrote options so only the sign-count description is correct.

A financial institution is comparing two VaR models, A and B, using the sign test on daily loss differentials (loss of A minus loss of B). Which statement correctly describes what the sign test examines?

A. Whether each model's exception rate equals one minus its VaR confidence level.
B. Whether the two models' exceptions are clustered in time.
C. Whether the number of positive loss differentials is consistent with a 50% probability, i.e., neither model has systematically lower loss.
D. Whether the average size of the loss differential is different from zero, assuming normally distributed differentials.

Answer: C

Step by step

1. Compute the daily loss differential between the two models.
2. Under the null of equal performance, a positive and a negative differential are each equally likely ($p = 0.5$).
3. The sign test counts the positive differentials and compares the count with a binomial distribution (or its normal approximation); it uses signs only, not magnitudes.

Why C is correct: The sign test is a nonparametric test of whether the count of positive loss differentials is consistent with $p = 0.5$ under equal model performance.

Why the others are wrong

- **A:** That describes a Kupiec-type unconditional coverage backtest of a single model, not a comparison of two models.
- **B:** Clustering of exceptions is tested by independence tests, not the sign test.

- **D:** That describes a t-test on the mean differential; the sign test uses only signs and makes no normality assumption.

Q342

AnalystPrep Q.6462

REBUILT — Original (60 of 100) was a conceptual next-step question with a borderline outcome (exact binomial and normal approximation disagree) and a partly defensible option D; rebuilt as a numeric test with a clear-cut result.

A risk manager uses the sign test to compare two VaR models over n = 100 trading days. Model X's loss exceeds Model Y's loss on 65 of the days. Under the null hypothesis of equal performance, positive and negative loss differentials are equally likely. Using the normal approximation to the binomial and a two-tailed 5% test (critical value 1.96), what is the test statistic and conclusion?

A. z = 1.50; fail to reject the null
B. z = 3.00; reject the null
C. z = 0.30; fail to reject the null
D. z = 15.0; reject the null

Answer: B

Step by step

1. Under H0, the count of positives ~ Binomial(100, 0.5): mean = 50, variance = 100 × 0.5 × 0.5 = 25, st. dev. = 5.
2. z = (65 − 50) / 5 = 3.00.
3. 3.00 > 1.96, so reject the null: the models do not perform equally.

Why B is correct: With mean 50 and standard deviation 5, 65 positives is 3 standard deviations above expectation, beyond the 1.96 critical value.

Why the others are wrong

- **A:** Divided by 10 (the square root of n) instead of the binomial standard deviation of 5, which understates the statistic.
- **C:** Divided the 0.15 excess proportion by 0.5 and never scaled by the standard error, giving 0.30.
- **D:** Used the raw excess of 15 positives over the mean as the statistic, without dividing by the standard deviation of 5.

Q343

AnalystPrep Q.6437

Which of the following is a key challenge in benchmarking VaR models?

A. The difficulty of accessing granular historical data for accurate backtesting.
B. The assumption of independent and identically distributed (i.i.d.) errors.
C. The inability of VaR models to capture tail risk accurately.
D. The consistency of model performance across various market conditions.

Answer: B

Step by step

1. Statistical comparison tests of two VaR models typically assume i.i.d. errors.
2. Trading portfolios change frequently, so errors are not independent and identically distributed, especially for regression-based results.
3. This violated assumption is the challenge specific to formal benchmarking.

Why B is correct: Frequent portfolio changes violate the i.i.d. error assumption underlying statistical tests that compare VaR models.

Why the others are wrong

- **A:** Data access is a backtesting practicality, not the benchmarking challenge cited in the reading.
- **C:** Tail-risk limitations are a general criticism of VaR, not specific to benchmarking.
- **D:** Robustness across conditions is a validation concern, not the stated benchmarking challenge.

Q344

AnalystPrep Q.6441

REBUILT — Original options overlapped (C was arguably defensible) and omitted the reading's rationale; reworded stem and options so exactly one is correct.

Banks often do not have a second VaR model to benchmark against. An institution therefore benchmarks its positional VaR model against a GARCH(1,1) VaR fitted to the bank's daily trading P&L. Which is the primary reason this benchmark is useful?

A. It uses detailed position-level risk factors and so measures positional risk more precisely than the internal

model.

B. It is simple to build from readily available P&L data yet is hard for the more conservative positional VaR to beat on accuracy.

C. It requires no distributional or model assumptions at all.

D. It is more conservative than positional VaR, so it lowers regulatory capital.

Answer: B

Step by step

1. Building a full competing model is costly, so a simple alternative is needed.
2. P&L GARCH VaR needs only the historical P&L series, which the bank already has.
3. Because positional VaR is tied to exceedances and regulatory capital, it is more conservative and rarely beats P&L GARCH VaR on accuracy, making GARCH a challenging yet easy benchmark.

Why B is correct: The P&L GARCH VaR is cheap to construct from existing data and provides a demanding accuracy benchmark.

Why the others are wrong

- **A:** A P&L-based model ignores positions; it is not a positional model.
- **C:** GARCH still assumes a variance process and innovation distribution.
- **D:** It is positional VaR that is the more conservative model, and benchmarking does not set capital.

Q345 Schweser Module Quiz 6.1 Q4

Which of the following statements regarding benchmarking VaR models is most accurate?

A. Benchmarking VaR models is not used for validating their performance.

B. Banks routinely benchmark their VaR model against several competing models.

C. In the statistical backtesting of VaR models, the errors are independently, but not identically, distributed.

D. Benchmarking is usually conducted for only a short time period during a bank's transition to a new model.

Answer: D

Step by step

1. Recall: benchmarking compares a bank's VaR model with another model, usually briefly while moving to a new model.
2. Check each statement against the reading: A, B and C each contradict it.

Why D is correct: Benchmarking is typically a short-term exercise during a model transition, because building a competing model is resource intensive.

Why the others are wrong

- **A:** Benchmarking is crucial for validating performance.
- **B:** Banks seldom have a formal benchmarking mechanism outside transitions because building even one model is costly.
- **C:** Because portfolios change frequently, errors are neither independent nor identically distributed.

Reading 7. Beyond Exceedance-Based Backtesting of Value-at-Risk Models: Methods for Backtesting the Entire Forecasting Distribution Using Probability Integral Transform

Diana Iercosan, Alysa Shcherbakova, David McArthur and Rebecca Alper, Validation of Risk Management Models for Financial Institutions (Cambridge University Press, 2023). Chapter 4.

LO	Learning objective	Qs
7.a	identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest.	11
7.b	explain how to derive probability integral transforms (PITs) in the context of validating a VaR model.	5
7.c	describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model.	8
7.d	describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test.	9

LO 7.a

Identify the properties of an exceedance-based backtest that indicate a VaR model is accurate, and describe how these properties are reflected in a PIT-based backtest.

Q346

AnalystPrep Q.6475

REBUILT — Original relied on DQ/LDQ tests outside the reading and option A was arguably defensible; reframed around the exceedance-count limitation that PIT-based backtests address, with realistic distractors.

Exceedance-based backtests of a VaR model often have low statistical power when the sample is small and the VaR confidence level is high, because exceedances are rare. Which of the following is the MOST effective way to address this power problem?

A. Use overlapping windows to create more backtest observations from the same data.

B. Tighten the significance level of the test from 5% to 1%.

C. Rely on the unconditional coverage test alone, since it uses the total number of exceedances.

D. Use a PIT-based backtest, which draws on the entire forecast distribution rather than only the exceedance indicator.

Answer: D

Step by step

1. An exceedance backtest reduces each day's P&L to a 0/1 indicator (exceedance or not), discarding information on how far inside or outside the VaR the P&L fell.
2. With a 99% VaR and about 250 observations, only about 2.5 exceedances are expected, so the test has little ability to reject a wrong model.
3. A PIT-based backtest converts every P&L observation into a PIT value using the full forecast distribution, so every observation carries information, which raises power.
4. Therefore D is the remedy; A, B and C do not add information (A adds dependence, B lowers power, C uses only the count).

Why D is correct: Power rises when each observation contributes more information; PIT-based tests use the whole forecast distribution instead of a single exceedance indicator.

Why the others are wrong

- **A:** Overlapping windows create serially dependent observations and add no new independent information, violating the independence assumption.
- **B:** A stricter significance level reduces the Type I error rate but increases the Type II error rate, i.e., lowers power.
- **C:** Unconditional coverage uses only the exceedance count, which is exactly the source of the low power.

Q347

AnalystPrep Q.6480 · AnalystPrep Q.6481

REBUILT — Merged two items using the same results table; clarified that 'failures' means the number of the 20 banks rejected by each test at 90% confidence, and rewrote the ambiguous inference so exactly one answer is defensible.

A regulator reviews backtests of the one-day 99% VaR models of 20 banks over a benign market period. The average exceedance rate across banks is 0.4%. The table shows the number of the 20 banks whose model is rejected by each

test at the 90% confidence level.

Test	Banks rejected (of 20)
Unconditional Coverage	4
Conditional Coverage	3
Dynamic Quantile	2
Logistic Dynamic Quantile	3
VaR Quantile Regression	19

Which assessment is MOST consistent with these results?

A. The models are accurately calibrated, because few banks fail the exceedance-based tests.

B. The models are on average conservative in the tail, yet the VQR rejections show deficiencies that exceedance counts do not capture.

C. The VQR results show a systemic failure, so all models should immediately be recalibrated to be more conservative.

D. Benign market conditions make all the backtest results meaningless.

Answer: B

Key correction: Source key for Q.6480 (C) said VQR is overly sensitive to minor deviations; this is not supportable since a test using more information is more powerful, so the VQR rejections signal real weaknesses. Q.6481 key (B) retained.

Step by step

1. A 99% VaR should be exceeded about 1% of the time; the observed average of 0.4% is below this, so VaR is on average too high (conservative).
2. Few banks fail UC (4), CC (3), DQ (2) and LDQ (3), so the exceedance-based tests do not flag a calibration problem; rejections that occur reflect overly few exceedances.
3. 19 of 20 banks fail the VQR test, which uses the distribution of outcomes rather than just exceedance counts, so the models are deficient in ways counts miss.
4. Hence the models are conservative at the tail but not necessarily well specified: B.

Why B is correct: Exceedance rates below 1% imply conservatism, while mass rejection by a test using more information shows misspecification that simple exceedance counts cannot detect.

Why the others are wrong

- **A:** Treats the low exceedance rate and few exceedance-test failures as proof of accuracy and ignores the 19 VQR rejections; also a rate of 0.4% is conservative, not accurate.
- **C:** The models are already too conservative, so making them more conservative is the wrong remedy; 'immediate recalibration of all' overreacts.
- **D:** Benign conditions make exceedances rarer but the tests remain informative (the 0.4% vs 1% gap and VQR results are still interpretable).

Q348

AnalystPrep Q.6483

REBUILT — Clarified that 'failures' means number of the 20 banks rejected at 90% confidence, and reworded options so the key is unambiguous.

A regulator reviews backtests of the one-day 99% VaR models of 20 banks during a benign market period. The average exceedance rate is 1.8%. The table shows the number of the 20 banks whose model is rejected by each test at the 90% confidence level.

Test	Banks rejected (of 20)
Unconditional Coverage	15
Conditional Coverage	14
Dynamic Quantile	16
Logistic Dynamic Quantile	13
VaR Quantile Regression	2

Which assessment is MOST consistent with these results?

A. The models underestimate risk: exceedances occur well above the 1% expected rate and most banks fail the exceedance-based tests.

B. The models are aggressive but adequate, since few banks fail VQR, so no action is needed.

C. Benign markets explain the high exceedance rate, so the models are well calibrated.

D. The high exceedance rate contradicts benign conditions, so the backtesting methodology itself is flawed.

Answer: A

Step by step

1. Expected exceedance rate for a 99% VaR is 1%; observed average is 1.8%, i.e., about 1.8 times too many exceedances, indicating VaR is too low.
2. UC rejects 15 of 20 banks and CC 14 of 20, confirming the exceedance count is statistically inconsistent with 99% for most banks.
3. The low VQR count (2) does not offset this, since the exceedance-based property of correct coverage clearly fails.
4. In benign conditions a well-calibrated model should have fewer, not more, exceedances, so the issue lies with the models: A.

Why A is correct: A rate above 1% that most banks fail UC/CC on shows systematic underestimation of risk; benign markets would reduce exceedances for a correct model.

Why the others are wrong

- **B:** Treats a high exceedance rate as acceptable because one test passes; it ignores the failed unconditional coverage property.
- **C:** Reverses the logic: low volatility should make exceedances rarer for a calibrated model, not more frequent.
- **D:** Blames the methodology; the exceedance backtest is standard and the results point to the models.

Q349

AP Mock 3 Q37 · AnalystPrep Q.6485

A risk management team is conducting an exceedance-based backtest on its VaR model and wants to reflect the model's key properties through a PIT-based backtest. Which attributes of the probability integral transforms are essential for validating that the exceedance-based properties are maintained?

A. Skewness and heavy tails.
B. Uniform distribution and independence.
C. Periodicity and cyclicality.
D. Concentration and clustering.

Answer: B

Step by step

1. If the forecast distribution is correct, PIT values $u_t = F_t(r_t)$ are i.i.d. Uniform(0,1).
2. Uniformity corresponds to unconditional coverage: the share of PITs below p equals p, so exceedances occur at the right frequency.
3. Independence corresponds to the independence property (no clustering of exceedances).
4. Together they mirror conditional coverage: B.

Why B is correct: A correct VaR model yields PITs that are uniform (unconditional coverage) and independent (no clustering), mirroring the exceedance-based properties.

Why the others are wrong

- **A:** Skewness and heavy tails describe the return distribution and are not PIT properties of an accurate model.
- **C:** Periodicity and cyclicality are forms of serial dependence, i.e., symptoms of a bad model.
- **D:** Concentration (non-uniformity) and clustering (dependence) are the failure symptoms, the opposite of what is required.

Q350

AnalystPrep Q.6476

A study of 20 bank holding companies finds an average exceedance rate of 0.4% for their one-day 99% VaR models. What does this suggest about the models?

A. On average they underestimate risk.
B. On average they accurately capture risk.
C. On average they overestimate risk (are conservative).
D. Further information is needed to reach any conclusion.

Answer: C

Step by step

1. A 99% VaR should be exceeded about $1 - 0.99 = 1\%$ of the time.
2. Observed 0.4% is below 1%: fewer exceedances than expected.
3. Fewer exceedances mean VaR thresholds are set too far out, so risk is overestimated: C.

Why C is correct: An exceedance rate below the nominal tail probability implies VaR is too high, i.e., conservative.

Why the others are wrong

- **A:** Underestimation produces more than 1% exceedances; the direction is reversed.
- **B:** Accuracy would give about 1%; 0.4% is well below.
- **D:** The exceedance rate alone already indicates the direction of the bias, though not its statistical significance.

Q351

AnalystPrep Q.6478

A regulator observes that 20 banks have an average exceedance rate of 0.4% for their 99% VaR models over a period described as benign for markets. Which is the MOST likely interpretation?

A. The models are accurately calibrated and effectively capture market risk.
B. The low rate may indicate excessive conservatism, potentially leading to inefficient capital allocation.
C. Benign conditions make it impossible to draw any conclusion from the exceedance rate.
D. The low rate is due to fees and commissions being included in reported P&L.

Answer: B

Step by step

1. Expected exceedance rate is 1%; 0.4% is well below.
2. Benign markets make tail events rarer, but a 0.4% rate is still informative: VaR is set higher than needed.
3. Overly high VaR means excess capital held against market risk, i.e., conservatism and inefficiency: B.

Why B is correct: Exceedances well below the nominal rate indicate overly conservative VaR, which ties up excess capital.

Why the others are wrong

- **A:** Confuses a low exceedance count with accuracy; accuracy means about 1%.
- **C:** Overstates the effect of benign conditions; the exceedance rate still conveys information.
- **D:** Unsupported speculation about P&L construction, not implied by the data.

Q352

AnalystPrep Q.6479

REBUILT — Original distractors were weak (several clearly irrelevant); rewrote options to make a single plausible criticism of exceedance-only testing.

Historically, regulators have focused on backtests based on the indicator variable for a VaR exceedance. What is the main limitation of relying only on such exceedance-based tests?

A. They use only whether the loss exceeded the VaR threshold, so a model can pass while misrepresenting the rest of the forecast distribution.
B. They can only detect models that are too conservative, not models that are too aggressive.
C. They require the P&L to be normally distributed.
D. They test independence of exceedances but not their total number.

Answer: A

Step by step

1. An exceedance test reduces each day to a 0/1 indicator for a single preselected tail cutoff.
2. It tests unconditional coverage (count), independence (clustering) and conditional coverage (both), but only at that cutoff.
3. A model can therefore pass these tests while the rest of its forecast distribution is badly wrong; PIT-based tests address this using the whole distribution: A.

Why A is correct: Exceedance indicators carry information only about one tail point, so passing them does not validate the full forecast distribution.

Why the others are wrong

- **B:** Too many exceedances (aggressive models) are detected by the unconditional coverage test as readily as too few.
- **C:** Exceedance tests make no normality assumption; they rely on binomial/Markov-chain likelihoods.
- **D:** They test both: unconditional coverage tests the count and the independence test tests clustering.

Q353

AnalystPrep Q.6486

An investment firm is assessing the predictive accuracy of its VaR model using exceedance-based backtesting. It further explores the application of PITs to validate model assumptions thoroughly. What does a proper implementation of PIT-based backtesting reveal about the VaR model?

A. It confirms the presence of higher moments affecting distribution shape.

B. It indicates uniformity and absence of temporal patterns.
C. It highlights cyclic trends and overfitting potential.
D. It shows deviations at the distribution extremes.

Answer: B

Step by step

1. If the forecast distribution is correct, each realized P&L is a draw from that forecast, so its PIT (the forecast CDF evaluated at the realization) is uniform on [0,1].
2. Correct unconditional coverage appears as a uniform PIT distribution (the tail share of PITs matches the VaR tail probability); correct independence appears as PITs with no serial pattern (no clustering of tail PITs).
3. So a well-specified model shows uniform PITs with no temporal pattern.

Why B is correct: An accurate model produces iid uniform PITs: uniformity reflects correct coverage across the distribution and absence of temporal patterns reflects independence of exceedances.

Why the others are wrong

- **A:** Higher moments (skew, kurtosis) distorting the shape would make the PIT histogram non-uniform, which signals misspecification, not accuracy.
- **C:** Cyclical patterns mean PITs are serially dependent, which is the PIT analogue of clustered exceedances and indicates a deficient model.
- **D:** Excess mass at the extremes (U-shaped or humped PIT histogram) indicates the forecast tails are too thin or too fat, i.e., a poor model.

Q354

BT answers P2.T5.25.3.1 · BT notes P2.T5.25.3.1

REBUILT — Original treated PIT kurtosis of 3 as 'uniform' (a uniform has kurtosis 1.8) and failed Y on 1.20% vs 1.00% with no sample size; rebuilt with a sample size, explicit Kupiec and independence statistics, and a qualitative PIT description.

A risk manager backtests the 99% one-day VaR of two portfolios over 750 trading days. Both portfolios have 8 exceedances (expected: 7.5). Using the Schweser/Kupiec framework (critical values: 3.84 for LRuc and LRind, 5.99 for LRcc at 95% confidence):

Portfolio	Exceedances	Pattern of exceedances	Pattern of tail PITs
A	8	All isolated; never on consecutive days	Evenly scattered, no serial pattern
B	8	Four pairs of back-to-back days	Tail PITs appear in consecutive-day pairs

Counts of day-to-day transitions: A has n00 = 733, n01 = 8, n10 = 8, n11 = 0; B has n00 = 737, n01 = 4, n10 = 4, n11 = 4. Which portfolio's backtest indicates an accurate VaR model?

A. Both A and B, since both have the same number of exceedances.
B. Only B.
C. Only A.
D. Neither A nor B.

Answer: C

Step by step

1. Unconditional coverage (both): $LR_{uc} = -2\ln\left[\frac{0.99^{742}\times 0.01^{8}}{(742/750)^{742}\times(8/750)^{8}}\right] = 0.033 < 3.84$, so both pass.
2. Independence A: $\pi_{01} = 8/741 = 0.0108$, $\pi_{11} = 0/8 = 0$, $\pi = 8/749$; $LR_{ind} = 0.173 < 3.84$, so A shows no clustering.
3. Independence B: $\pi_{01} = 4/741 = 0.0054$, $\pi_{11} = 4/8 = 0.50$ versus $\pi = 8/749 = 0.0107$; $LR_{ind} = 27.70 > 3.84$, so B rejects independence.
4. Conditional coverage: A $LR_{cc} = 0.033 + 0.173 = 0.21 < 5.99$ (pass); B $LR_{cc} = 0.033 + 27.70 = 27.73 > 5.99$ (reject).
5. PIT view: A's PITs are uniform and serially unpatterned; B's consecutive-day tail PITs reveal dependence. Only A is accurate.

Why C is correct: Accuracy requires both correct coverage and independence of exceedances; both portfolios have correct coverage but only A has independent exceedances.

Why the others are wrong

- **A:** Looks only at the exceedance count (unconditional coverage) and ignores independence; B's exceedances cluster ($LR_{ind} = 27.70$).
- **B:** Reverses the conclusion: B is the portfolio with clustered exceedances and dependent tail PITs.
- **D:** Would be right only if coverage failed; $LR_{uc} = 0.033$ is far below 3.84 for both, and A also passes independence.

Q355 Schweser Module Quiz 7.1 Q1

Which of the following characteristics must a VaR model satisfy to ensure it passes the unconditional coverage test?

A. Exceedances must occur in clusters.
B. Exceedances must be independent of prior exceedances.
C. Exceedances must follow a uniform distribution over [0,1].
D. The total number of exceedances must match the expected frequency within a confidence level.

Answer: D

Step by step

1. Unconditional coverage (Kupiec) looks only at the total number of exceedances x out of n observations.
2. The LR_{uc} statistic compares x/n with the model's tail probability p; the model is rejected if $LR_{uc} > 3.84$.
3. Hence passing requires the exceedance count to be statistically consistent with the expected frequency (e.g., 6 to 20 for a 95% VaR over 252 days).

Why D is correct: Unconditional coverage tests only whether the frequency of exceedances is consistent with the VaR confidence level, regardless of timing.

Why the others are wrong

- **A:** Clustering violates independence and would cause rejection of the independence/conditional coverage test; it is not required for coverage.
- **B:** Independence is a separate property (Markov chain LRind test), needed for conditional coverage, not unconditional coverage.
- **C:** Uniformity over [0,1] is a property of PITs in the PIT-based approach, not the defining condition of the exceedance-count test.

Q356 Schweser Module Quiz 7.1 Q2

What does it mean when exceedance-based backtest analysis on VaR models suggests that the model has conditional coverage at a given confidence level?

A. The model performs well only on a subset of positions.
B. The model requires specific distribution assumptions to adequately deliver results.
C. The model cannot be rejected on the basis of observed exceedances, nor can it be rejected for potential lack of independence.
D. The suggested rate of exceedance incidents cannot be rejected; however, there is serial correlation among the exceedance observations.

Answer: C

Step by step

1. Conditional coverage jointly tests $H_0 : \pi_{01} = \pi_{11} = p$, combining unconditional coverage and independence.
2. $LR_{cc} = LR_{uc} + LR_{ind}$ is chi-squared with 2 df; reject if $LR_{cc} > 5.99$.
3. Having conditional coverage means neither the exceedance rate nor the independence of exceedances is rejected.

Why C is correct: Conditional coverage holds when the model passes both the frequency (coverage) test and the independence test.

Why the others are wrong

- **A:** The test is applied to the portfolio's whole exceedance series; it says nothing about subsets of positions.
- **B:** Conditional coverage is a statistical property of the exceedance series and does not depend on a specific distributional assumption.
- **D:** Serial correlation among exceedances means independence is rejected, so conditional coverage fails; only coverage would hold.

LO 7.b

Explain how to derive probability integral transforms (PITs) in the context of validating a VaR model.

Q357

AnalystPrep Q.6488

REBUILT — Stem was vague ('initial step ... alignment with expectations') and distractors were jargon; restated as the ordered Schweser steps with plausible wrong-step distractors.

In the Schweser procedure for deriving a probability integral transform (PIT) for a single day, which step comes first?

A. Locate the realized P&L on the cumulative distribution function to obtain a value in [0,1].
B. Obtain the forecast distribution of P&L for the period (e.g. normal with mean 0 and standard deviation 500).
C. Test the set of PIT values for uniformity and independence.
D. Count the days on which the realized loss exceeded the VaR.

Answer: B

Step by step

1. Step 1: obtain the forecast distribution of P&L for the period.
2. Step 2: collect the realized P&L for that period.
3. Step 3: map the observation through the forecast CDF to get the PIT in [0,1].
4. Step 4: analyze the PITs for uniformity, independence and conditional coverage.
5. The forecast must exist before an observation can be located within it, so the forecast is first.

Why B is correct: A PIT is a position within a forecast distribution, so the full predictive distribution for the period must be established before anything else.

Why the others are wrong

- **A:** This is Step 3; it cannot be done until both the forecast and the realization are available.
- **C:** This is Step 4, the final analysis of the collected PITs.
- **D:** Exceedance counting belongs to exceedance-based backtesting, which looks only at the tail cutoff, not the PIT method.

Q358

AnalystPrep Q.6489

REBUILT — Original asked a vague purpose question with loosely worded answer; rebuilt to test why each day's observation is mapped through that day's own forecast CDF.

Forecast distributions of daily P&L change from day to day because volatility and positions change. Why does the PIT method pass each day's realized P&L through that day's own forecast CDF?

A. To remove outlier days before computing the VaR.
B. To re-estimate the model parameters from historical data.
C. To count how many days exceeded the VaR threshold.
D. To put observations drawn from different forecast distributions on one common [0,1] scale so they can be tested together.

Answer: D

Step by step

1. Raw P&L from different days cannot be pooled because each comes from a different forecast distribution.
2. Applying F_t to x_t gives the cumulative probability of the realized outcome under that day's forecast.
3. If every F_t is correct, all PITs share the same Uniform[0,1] distribution and can be tested jointly.

Why D is correct: The CDF transform standardizes each observation relative to its own forecast, producing a comparable uniform value for every day.

Why the others are wrong

- **A:** PIT does not remove observations; extreme outcomes are retained since they reveal tail misforecasts.
- **B:** PIT is a validation tool applied to forecasts, not a way to refine or recalibrate the model.
- **C:** Counting exceedances is the exceedance-based approach, which uses one tail cutoff rather than the entire distribution.

Q359

AnalystPrep Q.6490

REBUILT — Original conceptual question had vague distractors; made it quantitative using the Schweser example (normal, mean 0, SD 500, P&L -250).

A model forecasts tomorrow's P&L as normal with mean \$0 and standard deviation \$500. The realized P&L is -\$250. What is the PIT value for this observation?

A. 0.692
B. -0.500
C. 0.352

D. 0.309

Answer: D

Step by step

1. Standardize: $z = (-250 - 0)/500 = -0.5$.
2. The PIT is the forecast CDF at the outcome: $N(-0.5) = 0.3085$, about 0.309.
3. So about 30.9% of the forecast distribution lies at or below the realized loss.

Why D is correct: The PIT is the cumulative probability N(z) at the realized outcome, not the z-score itself.

Why the others are wrong

- **A:** This is N(+0.5) = 0.692, from ignoring the sign of the loss and measuring the upper tail.
- **B:** This is the z-score, not the PIT; the CDF has not been applied.
- **C:** This is the normal density value at 0.5 (0.352), confusing the PDF with the CDF.

Q360

BT answers P2.T5.25.3.2 · BT notes P2.T5.25.3.2

REBUILT — Original had a redundant 'actual average' line, an ambiguous 'a day in the lower tail' (Days 3, 5, 8 all qualify) and a nonsensical option; tightened to the day with the lowest PIT and rebuilt options from real mistakes.

A model forecasts daily results as normal with mean 0.5 and standard deviation 1. Realized results and PITs for ten days are:

Day	Realized result	Statistic (z)	PIT value
1	1.2	0.7	75.8%
2	-0.3	-0.8	21.2%
3	-0.8	-1.3	9.7%
4	2.0	?	93.3%
5	-0.7	-1.2	11.5%
6	0.0	-0.5	30.9%
7	1.9	1.4	91.9%
8	-1.8	-2.3	1.1%
9	0.4	-0.1	46.0%
10	1.1	0.6	72.6%

What is the statistic (z-score) for Day 4, and on which day did the realized result fall furthest into the lower tail of the forecast distribution?

A. 1.50, Day 4

B. 1.70, Day 8

C. 1.50, Day 8

D. -1.50, Day 8

Answer: C

Step by step

1. $z = (\text{realized} - \text{expected})/\text{SD} = (2.0 - 0.5)/1 = 1.50$ for Day 4; check $N(1.50) = 0.9332$, matching the 93.3% PIT.
2. The lower tail corresponds to the lowest PIT values; the smallest PIT is Day 8 at 1.1% (z = -2.3).
3. Answer: 1.50 and Day 8.

Why C is correct: The z-score standardizes the outcome using the forecast mean and SD, and the lowest PIT identifies the outcome the model considered least likely on the downside.

Why the others are wrong

- **A:** The z-score is right but Day 4 (PIT 93.3%) is in the upper tail, not the lower tail.
- **B:** Subtracted the sample average realized result (0.3, over the 10 days) instead of the forecast mean 0.5: (2 - 0.3)/1 = 1.70.
- **D:** Reversed the subtraction (0.5 - 2.0), giving the wrong sign; Day 4 is above the forecast mean.

Q361

Schweser Module Quiz 7.2 Q1

Which of the following statements correctly describes the purpose of probability integral transform (PIT) values for backtesting VaR models? PIT values:

A. are used to determine the total number of exceedances for a VaR model.

B. measure the conditional coverage property of exceedance-based models.

C. analyze exceedance independence by forming a Markov chain of observations.

D. convert observed data points into a uniform distribution for the entire forecasted distribution.

Answer: D

Key correction: Source key (Schweser Module Quiz 7.2 Q1) said C; correct is D because the source's own explanation states PITs convert observations to a uniform distribution over the entire forecast, which is option D, while the Markov-chain test applies to exceedances.

Step by step

1. Each realized P&L is located on its forecast CDF, giving a value in [0,1].
2. If the model is correct, these values are Uniform[0,1] across the entire distribution, not only at the VaR tail.
3. This uniformity is what the PIT backtest then tests, so D describes the purpose.

Why D is correct: PITs map each observation through its forecast CDF to a uniform variable, letting the whole forecast distribution be backtested.

Why the others are wrong

- **A:** Counting exceedances is the exceedance-based method; PITs use the full distribution.
- **B:** Conditional coverage is a property of exceedance tests; PITs link uniformity to unconditional coverage and independence is checked separately.
- **C:** A Markov-chain independence test is an exceedance-based (Christoffersen-type) technique, not the purpose of PIT values.

LO 7.c

Describe how the shape of the distribution of PITs can be used as an indicator of the quality of a VaR model.

Q362

AP Mock 1 Q6

REBUILT — Original options were vague jargon (B, D) and the stem was awkward; rewrote so each distractor is a realistic misconception about PIT distributions.

A risk manager computes a probability integral transform (PIT) for each day's realized P&L using that day's forecast distribution. If the VaR model's forecast distributions are correct, which description of the resulting PIT values is correct?

A. They are skewed toward 0 and 1, reflecting the fat tails of P&L.

B. They cluster around 0.5, reflecting the central tendency of P&L.

C. They are uniformly distributed over [0,1] and independent over time.

D. They are normally distributed with mean 0 and standard deviation 1.

Answer: C

Step by step

1. Each PIT is $u_t = F_t(x_t)$, the forecast CDF evaluated at the realized outcome.
2. If F_t is the true distribution of x_t, then u_t is Uniform[0,1] (any continuous variable passed through its own CDF is uniform).
3. Correct forecasts also imply the PITs are independent across days.
4. So correct models give a flat PIT histogram with about 10% of values in each decile.

Why C is correct: Passing an outcome through its own true CDF always yields a Uniform[0,1] variable, so a correct model produces uniform, independent PITs.

Why the others are wrong

- **A:** Mass piled near 0 and 1 signals forecasts that are too thin-tailed, i.e. a flawed model; it is not the sign of a correct one.
- **B:** Clustering in the middle means the forecast tails are too fat (outcomes rarely land in the extremes); this indicates misspecification.
- **D:** Confuses the PIT (a probability on [0,1]) with the z-score (standardized outcome), which would be standard normal.

Q363

AP Mock 3 Q54

A risk analyst is evaluating a VaR model by examining the shape of the probability integral transform (PIT) distribution. Why is the shape of the PIT distribution crucial in assessing model quality, and what characteristic best indicates a well-calibrated model?

A. A heavy-tailed distribution, indicating extreme event capture.

B. A bimodal distribution, indicating inconsistent risk segmentation.
C. A clustered distribution, displaying conservative risk estimates.
D. A uniform distribution, signifying consistent predictive accuracy.

Answer: D

Step by step

1. The PIT is the forecast CDF evaluated at the realized P&L, so it measures where each outcome fell within the forecast distribution.
2. If the forecast distribution is correct, outcomes fall at every percentile with equal frequency, so the PITs are uniform on [0,1] (e.g., 10% in each decile) and independent.
3. Any systematic shape (U, hump, slope) shows that the forecast under- or over-states risk in some region, so uniformity is the benchmark for good calibration.

Why D is correct: A correctly specified forecast distribution makes the realized outcomes equally likely to land in any percentile, so PITs are uniform; the shape of the histogram shows where and how the model is wrong.

Why the others are wrong

- **A:** Heavy tails are a property of return distributions, not the desired shape of PITs; excess mass in the PIT tails signals underestimated risk.
- **B:** Two humps mean outcomes concentrate in two regions of the forecast distribution, a departure from uniformity.
- **C:** Clustering (e.g., in the middle) shows the forecast is too wide; being conservative is still miscalibration.

Q364

AnalystPrep Q.6487

REBUILT — Original asked vaguely what a fund should 'prioritize' and had a weakly grounded key (recalibrate inputs); rebuilt as a concrete case on the independence requirement of PIT-based backtesting.

A fund backtests its VaR model with PITs. A histogram of 1,000 PITs is essentially flat, with about 100 observations in each decile, and a Kolmogorov-Smirnov test does not reject uniformity. However, the PITs show strong positive serial correlation: runs of high PITs tend to be followed by high PITs, and runs of low PITs by low PITs. Which conclusion is most appropriate?

A. The model is well calibrated, because uniformity of the PITs is the only requirement.
B. The model fails the independence requirement; forecasts are not adapting to changing conditions such as volatility clustering, so the dynamics of the model should be revised.
C. The forecast distributions are too narrow, so the model underestimates risk in the tails.
D. The forecast distributions are biased in location, so the mean P&L forecast should be shifted.

Answer: B

Step by step

1. A perfect forecast produces PITs that are both uniform on [0,1] and independent across periods.
2. A flat histogram satisfies the uniformity part (the analogue of unconditional coverage).
3. Serial correlation in PITs means outcomes in consecutive periods fall in similar regions of their forecast distributions: the analogue of clustered exceedances, which fails independence (conditional coverage).
4. This pattern typically arises when the model is slow to update for time-varying volatility or dynamics, so the model's dynamics should be revised.

Why B is correct: Good calibration needs both uniform and independent PITs; a flat histogram with serially correlated PITs satisfies only the first.

Why the others are wrong

- **A:** Ignores independence; uniformity alone is necessary but not sufficient.
- **C:** Too-narrow forecasts would show up as a U-shaped histogram (mass near 0 and 1), but the histogram here is flat.
- **D:** Location bias would show up as a sloped/skewed histogram, which is not observed.

Q365

AnalystPrep Q.6492 · AnalystPrep Q.6495

REBUILT — Two vague AnalystPrep items (clustering at unspecified intervals; excess tail mass with a loosely worded key) tested the same U-shape point; merged into one with concrete decile data and an unambiguous key.

A risk manager computes 1,000 PITs for a VaR model's daily P&L forecasts. Expected frequency is 100 per decile. The observed counts are 190 in the 0.0-0.1 decile, 180 in the 0.9-1.0 decile, and a combined 630 across the eight middle deciles (about 79 each). What does this indicate, and what is the appropriate adjustment?

A. The forecast distributions are too wide and risk is overstated; reduce the volatility estimate.
B. The forecast distributions are too narrow and risk is understated in both tails; widen the forecast distribution (e.g., higher volatility or fatter tails).
C. The forecasts are biased toward gains; shift the forecast mean P&L downward.
D. The model is well calibrated, since the middle deciles are close to the expected 100 and extra capital is not needed.

Answer: B

Step by step

1. Expected count per decile = 1,000 × 10% = 100.
2. The extreme deciles have 190 and 180, i.e., 19% and 18% of PITs instead of 10%: a U-shaped histogram.
3. PITs near 0 or 1 mean realized P&L fell far into the tails of the forecast distribution more often than the model implies, so tail outcomes are more frequent than forecast.
4. This is the signature of a forecast distribution that is too narrow, so risk is understated; the remedy is to widen it (higher volatility or fatter tails).

Why B is correct: A U-shaped PIT histogram means extreme outcomes occur more often than the model predicts, so the forecast distribution is too narrow and understates risk.

Why the others are wrong

- **A:** Reverses the diagnosis: too-wide forecasts produce a hump in the middle deciles with thin tails.
- **C:** Location bias produces an asymmetric slope, not equal excess mass at both ends (190 vs 180).
- **D:** The middle deciles are about 79 each, below 100, and the extreme deciles are far above 100, so the distribution is clearly not uniform.

Q366

AnalystPrep Q.6494

In the evaluation of a VaR model, an analyst detects clustering in the PIT distribution concentrated around the middle of the [0,1] interval, with few PITs near 0 or 1. What does this suggest about the model's risk assumptions?

A. The model is conservative: forecast distributions are too wide, overstating risk.
B. The model is aggressive: forecast distributions are too narrow, underestimating risk.
C. Middle clustering indicates a perfectly balanced risk model.
D. Middle clustering indicates randomness, so the model is precise in normal states.

Answer: A

Step by step

1. PIT near 0.5 means the realized P&L is close to the median of the forecast distribution.
2. If PITs pile up in the middle, outcomes are less dispersed than the forecast implies, and tail outcomes occur less often than predicted.
3. So the forecast distribution is too wide: VaR is overstated and capital is excessive; the model should be recalibrated to be less dispersed.

Why A is correct: Too many PITs near the center means realized outcomes are less variable than forecast, so the forecast distribution is too wide and risk is overstated.

Why the others are wrong

- **B:** This is the diagnosis for a U-shape (mass near 0 and 1), not a central hump.
- **C:** Only a uniform (flat) histogram indicates good calibration.
- **D:** A hump is a systematic departure from uniformity, not randomness, and it does not show precision.

Q367

AnalystPrep Q.6496

REBUILT — Original options were irrelevant distractors (yield curves, execution times) with a vague key; rebuilt with a concrete sloped PIT histogram and plausible alternative diagnoses.

An analyst computes the PITs of a VaR model's P&L forecasts over a long sample. The PIT histogram is skewed: frequencies rise steadily from the 0.0-0.1 decile (about 5% of PITs) to the 0.9-1.0 decile (about 15% of PITs). What does this indicate?

A. The forecast distributions are too narrow, so risk is understated in both tails.
B. The forecast distributions are too wide, so risk is overstated.
C. The forecast distributions are biased in location: realized P&L is systematically higher than forecast, so the forecast is too pessimistic and the location (mean) should be adjusted.

D. The PITs are serially dependent, so exceedances cluster in time.

Answer: C

Step by step

1. A PIT is the forecast CDF at the realized P&L, so a high PIT means the outcome was better (higher P&L) than most of the forecast distribution.
2. Mass rising toward 1 (15% vs 5% in the extreme deciles, versus 10% expected) shows outcomes systematically landing in the upper part of the forecast distribution.
3. A one-sided slope (asymmetry), not symmetric tails or a central hump, is a location bias: the forecast P&L distribution is centered too low (too pessimistic).
4. The model's central tendency or skew assumptions should be recalibrated.

Why C is correct: A monotone sloped PIT histogram means outcomes systematically fall on one side of the forecast distribution, a location bias.

Why the others are wrong

- **A:** Too-narrow forecasts give a symmetric U-shape with excess mass at both 0 and 1; here the low decile is under-populated.
- **B:** Too-wide forecasts give a central hump; this histogram is monotone, not humped.
- **D:** Serial dependence is judged from autocorrelation of PITs in time, not from the shape of the histogram.

Q368

BT answers P2.T5.25.3.3 · BT notes P2.T5.25.3.3

REBUILT — Exhibit (PIT histogram) was unreadable; replaced with explicit decile counts consistent with the described concentration near 0 and tightened the options.

The risk team at Horizon Capital reviews its VaR model after a period of unexpected volatility. It computes 200 PITs, where a PIT is the forecast CDF evaluated at the realized P&L (so losses map to low values). The decile counts are:

PIT bin	0.0-0.1	0.1-0.2	0.2-0.3	0.3-0.4	0.4-0.5	0.5-0.6	0.6-0.7	0.7-0.8	0.8-0.9	0.9-1.0
Count	62	28	22	18	17	15	12	10	9	7

Which observation best indicates that the model is poorly calibrated?

A. The PIT distribution is uniform on [0,1], with no clustering in the tails or center.

B. The PIT distribution has a concentration of values near 0 (31% of PITs in the lowest decile versus 10% expected), indicating frequent underestimation of large losses.

C. The PIT distribution shows slight deviations around the center but aligns well with a uniform distribution in the tails.

D. The PIT distribution has skewness near 0 and kurtosis near 3, indicating balanced calibration.

Answer: B

Step by step

1. Expected count per decile = 200 / 10 = 20.
2. The lowest decile has 62, i.e., 62/200 = 31% versus 10% expected, and the counts decline steadily toward the top decile (7).
3. PITs near 0 mean realized P&L fell in the far left tail of the forecast, so large losses occur much more often than the model predicts: risk (losses) is underestimated.
4. The histogram is far from uniform, so the model is poorly calibrated.

Why B is correct: A pile-up of PITs near 0 means realized losses repeatedly land in the extreme left tail of the forecast distribution, so losses are underestimated.

Why the others are wrong

- **A:** The counts are far from the uniform 20 per decile.
- **C:** Contradicts the data: the main deviation is in the lowest decile (tail), not the center.
- **D:** Skewness 0 and kurtosis 3 describe a normal distribution; a uniform has skewness 0 and kurtosis 1.8, and these moments say nothing here (the mean PIT is about 0.32, not 0.5).

Q369

Schweser Module Quiz 7.2 Q2

Ideally, when backtesting VaR models, a QQ plot of probability integral transforms (PITs) should be close to a(n):

A. horizontal line.

B. inverted parabola.

C. straight line from [0,0] to [1,1].
D. uniformly distributed scatterplot.

Answer: C
Step by step
1. The QQ plot graphs the fraction of PITs less than or equal to each value u in [0,1].
2. For uniform PITs that fraction equals u, so the plot is the 45-degree line from [0,0] to [1,1].
3. Deviations of the plot from that line indicate the forecast distribution is mis-specified.

Why C is correct: If PITs are uniform on [0,1], the cumulative fraction at each u equals u, producing the 45-degree line.

Why the others are wrong
- **A:** A horizontal line would mean all PITs fall in one region, an extreme non-uniform case.
- **B:** A curved shape signals over- or under-populated ranges, i.e., non-uniform PITs.
- **D:** Uniformity describes the PIT histogram, not the shape of the QQ plot.

LO 7.d

Describe backtesting using PITs, and compare the various goodness-of-fit tests that can be used to evaluate the distribution of PITs: the Kolmogorov-Smirnov test, the Anderson-Darling test, and the Cramér-von Mises test.

Q370

AP Mock 2 Q64

REBUILT — Original asked which test is cheapest in computation/sample size (not supported by the reading; chi-squared is not an LO test; Schweser says the KS p-value needs sophisticated programming). Replaced with a comparison of the three LO 7.d statistics.

A risk manager compares goodness-of-fit statistics for a PIT-based VaR backtest. Which statement is correct?

A. The KS statistic is the mean squared deviation between the actual and theoretical distributions, while the Cramér-von Mises statistic is the maximum vertical distance.
B. For a perfect distribution forecast, each of the KS, Anderson-Darling and Cramér-von Mises statistics is expected to converge toward zero as the number of observations grows.
C. The Anderson-Darling test treats all discrepancies between the distributions equally, which is why it is preferred over KS for risk management.
D. A statistic that is larger than zero but below its critical value means the PITs are exactly uniform.

Answer: B
Step by step
1. Each statistic measures the distance between the empirical PIT distribution and the theoretical uniform [0,1] distribution.
2. With a perfect forecast the PITs are truly uniform, so the distance shrinks toward zero as observations accumulate; lower is better.
3. Check the other statements against the definitions: KS = maximum vertical distance; CvM = mean squared deviation; A-D = tail-emphasising.

Why B is correct: All three statistics measure distance from the uniform distribution, so a perfect forecast drives each toward zero.

Why the others are wrong
- **A:** Swaps the definitions: KS is the maximum vertical distance, CvM is the mean squared deviation.
- **C:** Reverses the logic: it is KS that treats all discrepancies equally; A-D places higher emphasis on the tails.
- **D:** Being below the critical value only means uniformity is not rejected at that confidence level; it does not prove exact uniformity.

Q371

AnalystPrep Q.6484

REBUILT — Original asked about Anderson-Darling small-sample limitations, which the reading does not cover and whose options were vague; replaced with a computational KS question on the LO 7.d definition.

A bank backtests its VaR model with five PIT values, sorted ascending: 0.05, 0.20, 0.35, 0.40, 0.90. Under a perfectly calibrated model the PITs are uniform on [0,1]. Using the empirical CDF (which steps up by 1/5 at each observation), what is the Kolmogorov-Smirnov test statistic, i.e. the maximum vertical distance between the empirical and theoretical CDFs?

A. 0.10
B. 0.20
C. 0.40
D. 0.80

Answer: C

Step by step

1. Empirical CDF after each observation: 0.2, 0.4, 0.6, 0.8, 1.0; theoretical CDF equals the PIT value.
2. Gaps just after each jump: 0.20-0.05=0.15; 0.40-0.20=0.20; 0.60-0.35=0.25; 0.80-0.40=0.40; 1.00-0.90=0.10.
3. Gaps just before each jump (i-1)/5 vs PIT: 0.05, 0.00, -0.05, -0.20, 0.10 (absolute values all smaller than 0.40).
4. Maximum = 0.40 at the fourth observation, so D = 0.40.

Why C is correct: The KS statistic is the largest vertical gap between the empirical and uniform CDFs; here it occurs at PIT = 0.40 where the empirical CDF is 0.80.

Why the others are wrong

- **A:** 0.10 is the gap only at the last observation (1.00 - 0.90); it ignores the larger gaps earlier.
- **B:** 0.20 is the gap at the second observation only (0.40 - 0.20), not the maximum.
- **D:** 0.80 is the empirical CDF value at the fourth observation; the theoretical CDF (0.40) was not subtracted.

Q372

AnalystPrep Q.6497

REBUILT — Reworded the correct option to match the reading (KS = maximum vertical distance, treats all discrepancies equally) and removed the unsupported 'quick and simple' claim.

A financial institution evaluates the uniformity of PITs from its VaR model using the Kolmogorov-Smirnov (KS) test. Which of the following best describes the KS test and its main limitation for risk managers?

A. It is the most sensitive of the three tests to deviations in the tails of the distribution.
B. It measures the maximum vertical distance between the actual and theoretical distributions but treats all discrepancies equally, so it gives no extra weight to the tails.
C. It averages squared deviations over the whole distribution and therefore cannot detect a single large deviation.
D. It is designed specifically to correct for small-sample bias in PIT distributions.

Answer: B

Step by step

1. The KS statistic is the maximum vertical distance between the empirical and theoretical (uniform) CDFs of the PITs.
2. Because it only looks at the largest gap wherever it occurs, every location is treated alike; tail discrepancies get no extra weight.
3. Risk managers care most about tail behaviour, which is why the Anderson-Darling test is a common alternative.

Why B is correct: KS takes the maximum gap between CDFs and treats all discrepancies equally, the criticism that motivates the Anderson-Darling test.

Why the others are wrong

- **A:** The Anderson-Darling test, not KS, emphasises the tails.
- **C:** That describes the Cramér-von Mises statistic (mean squared deviation), not KS.
- **D:** KS has no small-sample bias correction purpose; it is a goodness-of-fit test on the whole PIT distribution.

Q373

AnalystPrep Q.6498 · AnalystPrep Q.6499

An analyst notices potential tail issues in a PIT distribution during backtesting of a VaR model. Planning a more nuanced evaluation, which test should the analyst focus on for enhanced tail sensitivity, and what does this tell about the VaR model quality?

A. Cramér-von Mises test for its central sensitivity.
B. Anderson-Darling test for its tail-focused evaluation.
C. Kolmogorov-Smirnov test due to widespread use.
D. Variance-Ratio test for volatility consistency.

Answer: B

Key correction: Source key for Q.6498 gave an explanation copied from the KS question (Q.6497); the answer B is correct and the explanation of Q.6499 is the right one.

Step by step

1. Tail concerns call for a test that weights tail observations more heavily.
2. The Anderson-Darling test places higher emphasis on tail observations, unlike KS (max gap, equal weighting) and CvM (mean squared deviation).
3. A large A-D statistic would signal that the model fails to capture tail behaviour.

Why B is correct: Anderson-Darling gives higher weight to tail deviations, so it is the best test for tail problems in the PIT distribution.

Why the others are wrong

- **A:** CvM averages squared deviations over the whole distribution; it is not tail-focused.
- **C:** KS treats all discrepancies equally; popularity is not tail sensitivity.
- **D:** A variance-ratio test concerns volatility, not goodness-of-fit of the PIT distribution.

Q374

AnalystPrep Q.6500

REBUILT — Original premise that CvM is 'preferred for large datasets' is not in the reading; rebuilt to test the actual CvM definition relative to KS.

A validator contrasts the Cramér-von Mises (CvM) test with the Kolmogorov-Smirnov (KS) test when evaluating the distribution of PITs. Which statement is correct?

A. CvM uses the maximum vertical distance between distributions, like KS, but applies a tail weighting.

B. CvM is very similar to KS but, instead of the maximum difference, it considers the mean squared deviation between the distributions.

C. CvM is the test with the greatest emphasis on tail observations among the three.

D. CvM tests the independence of successive PIT values rather than their distribution.

Answer: B

Step by step

1. Both KS and CvM compare the actual PIT distribution with the uniform [0,1] theoretical distribution.
2. KS looks at the maximum difference; CvM instead considers the mean squared deviation over the distribution.
3. Tail emphasis belongs to Anderson-Darling, and independence is handled separately (Berkowitz normal transform).

Why B is correct: CvM replaces KS's maximum gap with the mean squared deviation, so it assesses fit across the whole distribution.

Why the others are wrong

- **A:** Maximum distance is the KS definition; CvM has no such feature and no tail weighting.
- **C:** Anderson-Darling, not CvM, emphasises tails.
- **D:** Independence is tested separately (e.g. Berkowitz on normal-transformed PITs); CvM is a goodness-of-fit test.

Q375

BT answers P2.T5.25.4.1 · BT notes P2.T5.25.4.1

REBUILT — Chart is not reproducible; the histogram is now described in words and the stem trimmed.

Adam Knight, a risk manager, backtests a 99% one-day VaR model over 250 trading days. Losses exceeded VaR 8 times versus about 2-3 expected, mostly in volatile periods in 2022. A histogram of the PITs shows a concentration of values near zero (well above the uniform density), indicating a systematic underestimation of extreme losses. Which goodness-of-fit test is most appropriate to identify this tail problem?

A. Kolmogorov-Smirnov, because it measures overall deviation between the empirical and theoretical CDFs and so ensures calibration across all quantiles.

B. Anderson-Darling, because it places more emphasis on the tails, making it useful for detecting underestimation of extreme losses.

C. Cramér-von Mises, because it focuses equally on the centre and the tails of the PIT distribution.

D. Kupiec test, because it evaluates the full distribution of PITs.

Answer: B

Step by step

1. PITs bunched near zero and 8 vs 2-3 exceedances indicate underestimated extreme losses: a tail problem.
2. The test of choice is the one weighting tail observations more: Anderson-Darling.
3. KS (max gap) and CvM (mean squared deviation) treat the distribution more evenly and may under-react to tail discrepancies.
4. Kupiec only counts exceedance frequency; it is not a PIT-distribution test.

Why B is correct: Anderson-Darling's tail emphasis makes it the most sensitive to the underestimated extreme losses.

Why the others are wrong

- **A:** KS treats all discrepancies equally, so it is not tail-focused.
- **C:** CvM averages squared deviations across the distribution; it is not tail-weighted.
- **D:** Kupiec tests the exceedance frequency only and does not examine the PIT distribution.

Q376

BT answers P2.T5.25.4.2 · BT notes P2.T5.25.4.2

REBUILT — Sources disagreed on sample size (700 vs 100 days) and the histogram was missing; sample size removed and the PIT pattern described in words.

A risk analyst backtests a VaR model by testing the uniformity of PITs at the 99% confidence level. A PIT histogram shows clustering near 0 and 1 relative to the uniform baseline. Results:

Test	Critical Value (99%)	Observed Statistic
Kolmogorov-Smirnov	0.136	0.09
Anderson-Darling	4.0	4.8
Cramér-von Mises	0.115	0.075

Which conclusion is best supported?

A. The KS test shows the model is well calibrated, so no further concern exists.

B. The Anderson-Darling test rejects uniformity because its statistic exceeds the critical value, signalling tail miscalibration that KS and CvM do not detect.

C. The Cramér-von Mises test proves the model is well calibrated in the tails.

D. All three tests indicate the model is well calibrated because two of three statistics are below critical values.

Answer: B

Step by step

1. Compare each statistic to its critical value: KS 0.09 < 0.136 (pass); CvM 0.075 < 0.115 (pass); A-D 4.8 > 4.0 (fail).
2. Only Anderson-Darling rejects uniformity at 99%.
3. A-D weights the tails, and PIT clustering near 0 and 1 is tail miscalibration, which KS and CvM give less weight to, so they can pass.
4. Conclusion: tail miscalibration, flagged by A-D.

Why B is correct: The A-D statistic exceeds its critical value; being tail-sensitive, it detects the tail miscalibration that the equally weighted KS and CvM miss.

Why the others are wrong

- **A:** KS passing means no overall large gap, but it is not tail-focused and does not override the A-D rejection.
- **C:** CvM passing says nothing reliable about tails since it averages across the distribution.
- **D:** Majority voting is wrong: one rejection by the tail-sensitive test is enough to question the model.

Q377

BT answers P2.T5.25.4.3 · BT notes P2.T5.25.4.3

REBUILT — Original relied on unreadable charts and had a garbled/incomplete second copy; made self-contained by describing the deviation plots in text and rewriting distractors so each reflects a realistic misreading of the test output.

A risk team backtests three VaR models by computing PITs and testing them against the uniform [0,1] distribution. The maximum gap between the empirical and uniform CDFs is about 0.05 for Model X, 0.14 for Model Y and 0.20 for Model Z. The test results are:

PIT Results	KS Stat	KS p-value	CvM Stat	CvM p-value
Model X	0.0538	0.4478	0.0910	0.6312
Model Y	0.1398	0.0001	1.5855	0.0001
Model Z	0.1954	0.0000	4.0694	0.0000

Which of the following statements correctly interprets these results?

A. Model X has the most uniform PITs: its small statistics and high p-values mean uniformity is not rejected, whereas it is rejected for Models Y and Z.

B. Model Y passes both tests, because its large test statistics indicate a close fit to the uniform distribution.

C. Model Z is overly conservative, because its p-values of 0.0000 show its PITs are almost perfectly uniform.

D. Model X is the only model whose PITs show tail clustering, because it has the highest p-values.

Answer: A

Step by step

1. Under the null hypothesis, PITs are uniform on [0,1], i.e. the VaR model forecasts the full distribution correctly.
2. The KS statistic is the maximum distance and the CvM statistic is the mean squared deviation between empirical and uniform CDFs, so larger statistics mean a worse fit.
3. Model X: KS 0.0538 and CvM 0.0910 are small, with p-values 0.4478 and 0.6312, far above 5%, so uniformity is not rejected.
4. Models Y and Z: statistics are much larger and p-values are about 0 (0.0001 or 0.0000), so uniformity is rejected at any usual significance level; Z (KS 0.1954, CvM 4.0694) is worst.
5. Hence Model X is the best calibrated and statement A is correct.

Why A is correct: Goodness-of-fit statistics measure distance from uniformity, so small statistics with high p-values mean the null of uniform PITs is not rejected; Model X is the only model for which that holds.

Why the others are wrong

- **B:** Reverses the direction of the statistics: large KS/CvM values (and tiny p-values) mean a large deviation from uniformity and a rejection, not a good fit.
- **C:** Misreads the p-value: a p-value of 0.0000 means uniformity is strongly rejected, so Model Z is the worst fit; the p-value says nothing about conservatism being 'perfect'.
- **D:** High p-values mean the PITs are consistent with uniformity (no clustering); tail clustering would show up as large statistics and low p-values, as for Models Y and Z.

Q378 Schweser Module Quiz 7.2 Q3

Which of the following goodness-of-fit tests used to evaluate the distribution of probability integral transforms (PITs) best captures tail events?

A. Anderson-Darling test.
B. Cramér-von Mises test.
C. Kolmogorov-Smirnov test.
D. Tests of independence for PIT values.

Answer: A

Step by step

1. The KS test uses only the maximum vertical distance between the empirical and uniform CDFs, treating all discrepancies equally.
2. The CvM test uses the mean squared deviation over the whole distribution, again without extra weight on the tails.
3. The Anderson-Darling test weights discrepancies in the tails more heavily, which is what risk managers care about.
4. Independence tests examine serial dependence of PITs, not the shape of their distribution.
5. The answer is Anderson-Darling.

Why A is correct: The A-D statistic places higher emphasis on tail observations, making it the goodness-of-fit test most sensitive to tail misfit.

Why the others are wrong

- **B:** CvM averages squared deviations across the entire distribution with equal weight, so it is not specifically tail-focused.
- **C:** KS looks only at the single largest gap and treats all discrepancies equally, so it can miss tail misfit.
- **D:** Independence tests check for serial correlation of PITs rather than whether their distribution is uniform, so they do not address tail fit.

Reading 8. Correlation Basics: Definitions, Applications, and Terminology

Gunter Meissner, Correlation Risk Modeling and Management, 2nd Edition (Risk Books, 2019). Chapter 1.

LO	Learning objective	Qs
8.a	describe financial correlation risk and the areas in which it appears in finance.	20
8.b	explain how correlation contributed to the global financial crisis of 2007–2009.	10
8.c	describe how correlation impacts the price of quanto options as well as other multi- asset exotic options.	17
8.d	describe the structure, uses, and payoffs of a correlation swap.	7
8.e	estimate the impact of different correlations between assets in the trading book on the VaR capital charge.	14
8.f	explain the role of correlation risk in market risk and credit risk.	7
8.g	explain how correlation risk relates to systemic and concentration risk.	13

LO 8.a

Describe financial correlation risk and the areas in which it appears in finance.

Q379 GARP book Chapter 8 end-of-chapter Q8.1

What two types of financial correlations exist?

Answer: see model answer — Static and dynamic financial correlations. Static correlations measure the relationship between assets for one specific period and do not change over time (e.g. VaR inputs, copula correlations for CDOs, the binomial default correlation model). Dynamic correlations measure the co-movement of assets over time (e.g. pairs trading, deterministic correlation approaches, stochastic correlation processes).

Step by step

1. Split financial correlations by whether they are fixed or evolve with time.
2. Static: single-period snapshot, e.g. VaR, copula correlations for CDOs, binomial default correlation.
3. Dynamic: co-movement over time, e.g. pairs trading, deterministic and stochastic correlation models.

Why this is correct: The classification depends on whether the correlation is a fixed snapshot or a time-varying process.

Q380 GARP book Chapter 8 end-of-chapter Q8.23

What do the terms ”measure of association” and ”measure of dependence” refer to in statistics?

Answer: see model answer — A measure of association quantifies the strength (and often direction) of a relationship between two variables, e.g. Pearson correlation or rank correlations such as Spearman’s rho and Kendall’s tau. A measure of dependence is a stricter notion: it should be zero only when the variables are independent, so it also captures nonlinear and non-monotonic dependence (e.g. copula-based measures). Pearson correlation can be zero for dependent variables, so it is an association measure but not a full dependence measure.

Step by step

1. Association: summary of how strongly and in which direction two variables relate (Pearson, Spearman, Kendall).
2. Dependence: full probabilistic relationship, zero only under independence.
3. Example: $Y = X^2$ with symmetric X has zero Pearson correlation yet complete dependence.

Why this is correct: Association statistics summarise a relationship; dependence is the broader concept that correlation only partly captures.

Q381 GARP book Chapter 8 end-of-chapter Q8.2

What is ”wrong-way correlation risk” or, for short, ”wrong-way risk”?

Answer: see model answer — Wrong-way risk (WWR) is the risk that exposure to a counterparty rises exactly when the counterparty’s credit quality deteriorates, because of positive correlation between the two. Example: an investor buys a CDS on French bonds from Deutsche Bank; if France and Deutsche Bank are positively correlated, the chance that both default together rises, so the protection is least reliable when it is needed. Higher correlation

therefore lowers the value of the protection (and the CDS spread).
Step by step
1. Identify the two exposures: the reference asset (e.g. France) and the protection seller (e.g. Deutsche Bank).
2. Positive correlation between them raises the joint default probability.
3. In the worst case both default and the investor loses the full exposure, because the hedge fails when it is needed.
4. Hence the CDS is worth less to the buyer as correlation rises.

Why this is correct: WWR arises because the hedge or exposure is positively correlated with the counterparty's default.

Q382 GARP book Chapter 8 end-of-chapter Q8.3

Correlations can be non-monotonous. What does this mean?

Answer: see model answer — A non-monotonic dependency means the value of a product does not move in one direction as correlation rises: it can increase over some correlation ranges and decrease over others. Example from the reading: the CDS spread on French bonds bought from Deutsche Bank may rise slightly as correlation goes from -1 to about -0.4 (negative correlation means only one of the two defaults, so replacing the hedge after a bank default is costly), and then falls as correlation increases further toward +1 (wrong-way risk).
Step by step
1. Define monotonic: output moves in one direction as the input rises.
2. Non-monotonic: direction of change reverses over the range of the input.
3. Illustration: CDS spread versus correlation between reference entity and protection seller rises over about -1 to -0.4, then falls as correlation grows.

Why this is correct: Price sensitivity to correlation can change sign across the correlation range, so 'higher correlation means lower value' need not hold everywhere.

Q383 GARP book Chapter 8 end-of-chapter Q8.4

Correlations are critical in many areas in finance. Name five.

Answer: see model answer — (1) Investments (portfolio diversification, Markowitz); (2) trading (correlation options, correlation swaps, pairs trading); (3) risk management (portfolio VaR, credit risk, counterparty and wrong-way risk); (4) global markets (co-movement across countries and asset classes, contagion in crises); (5) regulation (Basel capital rules, e.g. VaR capital charge and correlation assumptions).
Step by step
1. Recall the five areas listed in the reading: investments, trading, risk management, global markets, regulation.
2. Attach one example to each.

Why this is correct: These are the five areas the reading identifies where correlation plays an important role.

Q384 GARP book Chapter 8 end-of-chapter Q8.5

High diversification is related with low correlation. Why is this considered one of the few "free lunches" in finance?

Answer: see model answer — Portfolio return is the weighted average of asset returns, but portfolio volatility is below the weighted average of volatilities whenever correlation is below 1. Lowering correlation therefore reduces risk without sacrificing expected return, so the return/risk ratio rises at no cost. This is the Markowitz result: risk is reduced for free by combining imperfectly correlated assets.
Step by step
1. Portfolio mean: weighted average of means, independent of correlation.
2. Portfolio variance contains the term $2w_X w_Y \rho \sigma_X \sigma_Y$, which falls as ρ falls.
3. Same expected return, lower risk, so a higher return/risk ratio: a costless improvement.

Why this is correct: Correlation enters risk but not expected return, so lowering it improves the trade-off at no cost.

Q385 GARP book Chapter 8 end-of-chapter Q8.6

Create a numerical example and show why a lower correlation results in a higher return/risk ratio.

Answer: see model answer — Take two assets with expected return 10% and volatility 20% each, held 50/50. Portfolio return is 10% for any correlation. Portfolio volatility is sqrt(0.01 + 0.01 + 0.02 rho). For rho = 1: 20.0%, ratio 0.500. For rho = 0.5: 17.32%, ratio 0.577. For rho = 0: 14.14%, ratio 0.707. For rho = -0.5: 10.0%, ratio 1.000. The numerator is constant while the denominator falls as correlation falls, so the ratio rises.

Step by step

1. Assume $\mu = 10\%$, $\sigma = 20\%$ for both assets, weights 50/50, so portfolio mean = 10% regardless of ρ.
2. $\sigma_P = \sqrt{0.5^2(0.2)^2 + 0.5^2(0.2)^2 + 2(0.5)(0.5)\rho(0.2)(0.2)} = \sqrt{0.02 + 0.02\rho}$.
3. $\rho = 1$: $\sigma_P = 20.00\%$, ratio 0.500; $\rho = 0.5$: 17.32%, ratio 0.577.
4. $\rho = 0$: 14.14%, ratio 0.707; $\rho = -0.5$: 10.00%, ratio 1.000.
5. Constant return over falling risk gives a rising ratio.

Why this is correct: The covariance term in portfolio variance shrinks with correlation while expected return is unchanged.

Q386 GARP book Chapter 8 end-of-chapter Q8.8

What is ”pairs trading”?

Answer: see model answer — Pairs trading is a dynamic correlation strategy: take two historically highly correlated assets, and when their prices diverge, buy the relatively underpriced one and short the relatively overpriced one, expecting the historical relationship to reassert itself and the spread to converge. Profit comes from convergence, so the risk is that the relationship breaks down.

Step by step

1. Select two assets with a stable historical co-movement.
2. When the spread widens, go long the cheap and short the rich asset.
3. Profit if the spread reverts; loss if the relationship breaks.

Why this is correct: It is a bet that the time-varying relationship between a pair mean-reverts, which is why the reading classes it as dynamic.

Q387 GARP book Chapter 8 end-of-chapter Q8.22

Correlation risk is quite broadly defined in trading practice, referring to any co-movement of assets in time. How is the term ”correlation” defined in statistics?

Answer: see model answer — In statistics, correlation usually means the Pearson correlation coefficient: covariance of two variables divided by the product of their standard deviations. It lies between -1 and +1 and measures the strength and direction of the linear relationship between the variables only; it is not designed to capture nonlinear dependence.

Step by step

1. Covariance: $\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$.
2. Standardise: $\rho = \text{Cov}(X, Y)/(\sigma_X \sigma_Y)$.
3. Range -1 to +1; captures linear dependence only.

Why this is correct: Correlation is standardised covariance, a narrower concept than the broad ’co-movement’ used by traders.

Q388 AnalystPrep Q.1540

REBUILT — Options C and D were garbled and near-identical; rewrote the options with distinct, realistic distractors.

The original copula approach for a collateralized debt obligation (CDO) is a static financial correlation measure: it captures the default correlation of all the assets in the CDO for one fixed time period. This time period is usually equal to:

A. the reinvestment period of the CDO.
B. the maturity of the CDO.
C. the time until the first asset in the CDO defaults.
D. a one-day horizon, as in a standard VaR calculation.

Answer: B

Step by step

1. Static correlations do not change over time and measure the relationship between assets over one specific period.
2. For the copula approach to CDOs, that single period is the life of the structure, i.e. the CDO's maturity.
3. Default times of the individual assets are the random variables being correlated, not the horizon.

Why B is correct: The Gaussian copula fixes one default correlation over the whole life of the CDO, so the measurement horizon is the CDO's maturity.

Why the others are wrong

- **A:** The reinvestment period is only the phase during which proceeds are reinvested; it does not define the horizon of the default correlation.
- **C:** Confuses the horizon with the random default time; first-to-default is an outcome the model describes, not the period used.
- **D:** A one-day horizon is typical of VaR-style time-series work, not of a copula spanning the CDO's life.

Q389

AnalystPrep Q.1550

REBUILT — Option D was self-contradictory (assets rise together yet risk falls); replaced with a plausible but false statement on portfolio volatility.

A risk analyst finds that the correlation between two assets is negative. Which of the following statements is correct?

A. If one asset's value decreases, the other asset's value on average also decreases, which reduces overall risk.

B. If one asset's value decreases, the other asset's value on average increases, which reduces overall risk.

C. If one asset's value decreases, the other asset's value on average increases, which increases overall risk.

D. The standard deviation of an equally weighted portfolio of the two assets is greater than the average of the two stand-alone standard deviations.

Answer: B

Step by step

1. Negative correlation means the assets tend to move in opposite directions.
2. Losses on one asset are partly offset by gains on the other, so portfolio risk falls (diversification).
3. Portfolio volatility with $\rho < 1$ is below the weighted average of the stand-alone volatilities, and lower still when ρ is negative.

Why B is correct: Opposite co-movement means gains and losses offset, so combining the assets lowers portfolio risk.

Why the others are wrong

- **A:** Describes positive correlation (assets fall together), which would not reduce risk.
- **C:** Gets the direction right but the conclusion wrong: offsetting moves reduce, not increase, risk.
- **D:** Reverses the diversification effect; with $\rho < 1$ portfolio volatility is below the weighted average of stand-alone volatilities.

Q390

AnalystPrep Q.1559

REBUILT — Options B and C were mutually contradictory negations (giveaway); rewrote distractors to represent realistic misconceptions.

Correlation risk is an important part of risk management because different correlation values imply different risk for a portfolio. Correlation trading refers to:

A. trading assets whose prices are determined by the co-movement of two or more assets over time.

B. trading each asset only on the basis of its own stand-alone market and economic conditions.

C. trading only assets that are uncorrelated, to eliminate correlation risk.

D. trading assets whose prices depend only on the individual volatility of a single underlying asset.

Answer: A

Step by step

1. Correlation trading strategies involve assets (e.g. multi-asset options, correlation swaps) whose prices depend on how two or more assets move together.
2. This is the definition used in the reading.
3. B and D focus on single-asset drivers; C describes avoiding correlation, not trading it.

Why A is correct: Correlation trading means taking positions whose value is driven by the co-movement of assets.

Why the others are wrong

- **B:** Describes single-asset trading and ignores the co-movement that defines correlation trading.
- **C:** Avoiding correlated assets is risk avoidance, not correlation trading.
- **D:** Single-asset volatility trading is not correlation trading; the payoff must depend on joint movement.

Q391 BT answers P2.T5.502.1 · BT notes P2.T5.502.1

A colleague has computed a Pearson correlation coefficient of 0.20 between two variables but has not said what the variables are. Each of the following statements must be true EXCEPT which is false?

A. As the correlation is non-zero, we can say there is a dependency between the variables; for example, we can say "the variables are dependent".

B. If the covariance does not change, an increase in the standard deviation (volatility) of one of the variables will decrease the correlation to a value below 0.20.

C. If the correlation is a default correlation between a credit default swap (CDS) seller and the reference asset, the correlation must be monotonically increasing; i.e., the CDS spread is a continuously increasing function of a higher default correlation.

D. If the correlation is between the returns of two assets in an equally weighted portfolio, where each asset has the same daily standard deviation of 1.0%, the 10-day 99.0% value at risk (VaR) is about 5.70%.

Answer: C

Step by step

1. A: independence implies zero correlation, so non-zero correlation implies dependence. True.
2. B: $\rho = \text{cov}/(\sigma_X \sigma_Y)$; with positive covariance fixed, a larger σ lowers ρ below 0.20. True.
3. C: the CDS seller-reference default correlation relationship is not increasing; with wrong-way risk the CDS spread falls as correlation rises. False.
4. D: portfolio daily $\sigma = \sqrt{0.25 + 0.25 + 2(0.25)(0.20)} = \sqrt{0.60} = 0.7746\%$; 10-day 99% VaR $= 2.326 \times 0.7746\% \times \sqrt{10} = 5.70\%$. True.

Why C is correct: Only C is false: the CDS spread is not an increasing function of default correlation; because of wrong-way risk the spread decreases as correlation rises.

Why the others are wrong

- **A:** True statement, so it is not the exception (non-zero correlation rules out independence).
- **B:** True statement; the standard deviations are in the denominator of the correlation.
- **D:** True statement; ignoring correlation gives 5.20% and assuming $\rho = 1$ gives 7.36%, neither of which is the 5.70% result.

Q392 BT answers P2.T5.502.2 · BT notes P2.T5.502.2

Which of the following statements about the terminology surrounding "correlation" is TRUE?

A. The term "correlation" is uniquely defined.

B. Measures of association or measures of dependence refer to the linear Pearson correlation coefficient.

C. Concentration risk has the same mathematical definition as correlation risk.

D. In trading practice (e.g. correlation desks or correlation trading), the term correlation is defined broadly and can refer to any co-movement of assets over time.

Answer: D

Step by step

1. Correlation is used loosely: it may mean Pearson correlation or any measure of co-movement or dependence.
2. In trading practice it is defined broadly as co-movement of assets over time, consistent with correlation trading.
3. Hence only D is true.

Why D is correct: In practice correlation is a broad term covering any co-movement of assets over time, not one unique measure.

Why the others are wrong

- **A:** False: correlation has no single unique definition.
- **B:** False: measures of association/dependence are broader than, and typically not limited to, Pearson correlation.
- **C:** False: concentration risk and correlation risk are related (the concentration ratio moves with default correlation) but are not the same quantity.

Q393 BT notes P2.T5.22.11.3

It is important to study correlations because they are prevalent in finance. In regard to correlation risk, each of the following statements is TRUE, except which is false?

A. Wrong-way risk is a type of (special case of) correlation risk.
B. Statistical independence between two variables implies their Pearson correlation coefficient is zero.
C. Correlation risk only refers to financial variables and excludes non-financial variables (e.g. economic or geopolitical events) because they are impossible to parameterize.
D. A creditor's worst-case scenario is reduced by either a lower concentration ratio and/or a lower default correlation.

Answer: C

Step by step

1. Correlation risk is defined as loss from adverse changes in correlations between financial or nonfinancial assets (e.g. GDP, political events), so C is false.
2. A: wrong-way risk is positive correlation between exposure and counterparty default, a form of correlation risk.
3. B: independence implies zero covariance, hence zero correlation.
4. D: a lower concentration ratio and lower default correlation each lower the joint-default (worst-case) probability.

Why C is correct: The definition of correlation risk explicitly includes nonfinancial variables, so C is false.

Why the others are wrong

- **A:** True; wrong-way risk is a correlation risk.
- **B:** True; independence implies zero correlation (the converse need not hold).
- **D:** True; both a lower concentration ratio and a lower correlation reduce the worst-case joint default probability.

Q394 GARP PE1 Q55 · Mock 4 Q1

A newly hired risk analyst at a large investment bank is examining how financial correlation risk affects the bank's portfolios. The bank holds portfolios of different asset types and enters into hedging contracts with multiple counterparties. Which of the following statements would the analyst be correct to make?

A. The buyer of a CDS faces wrong-way risk when there is a positive default correlation between the reference asset and the CDS counterparty.
B. The risk-adjusted return of a portfolio typically increases when correlations of assets in the portfolio increase.
C. Dynamic correlation risk in a portfolio of pairs trades is most appropriately estimated using Gaussian copulas.
D. Correlation risk is highest during periods of relatively benign market movements when correlations are difficult to predict.

Answer: A

Step by step

1. Positive default correlation between the CDS seller and the reference asset means the seller is likely to default exactly when protection is needed: wrong-way risk. A is correct.
2. B: higher correlation reduces diversification, so the return/risk ratio falls.
3. C: Gaussian copulas are a static correlation tool; pairs trading is dynamic correlation.
4. D: correlation risk is greatest in crises when correlations jump toward one.

Why A is correct: Positive default correlation between protection seller and reference asset is the definition of wrong-way risk for the CDS buyer.

Why the others are wrong

- **B:** Reversed: higher correlations raise portfolio volatility and lower the return/risk ratio.
- **C:** A Gaussian copula is static, so it does not capture dynamic correlation in pairs trades.
- **D:** Reversed: correlation risk peaks in stressed markets (2007-09), not benign ones.

Q395 Schweser Module Quiz 8.1 Q1

Which of the following measures is most likely an example of a dynamic financial correlation measure?

A. Pairs trading.
B. Value at risk (VaR).
C. Binomial default correlation model.
D. Copula correlations for collateralized debt obligations (CDOs).

Answer: A

Step by step

1. Static financial correlations measure the relationship between assets over one specific time period.
2. Dynamic financial correlations measure the co-movement of assets over time.
3. Dynamic examples: pairs trading, deterministic correlation approaches, stochastic correlation processes.
4. Static examples: VaR, CDO copula correlations, binomial default correlation model.

Why A is correct: Pairs trading exploits how the co-movement of two assets evolves through time, so it is a dynamic correlation application.

Why the others are wrong

- **B:** VaR is a static correlation measure (correlations fixed for the chosen period); confusing a widely used risk measure with a time-evolving one.
- **C:** The binomial default correlation model is static, a one-period default dependence measure.
- **D:** CDO copula correlations are static; the copula takes a fixed correlation input.

Q396

BT answers P2.T5.501.1 · BT notes P2.T5.501.1

REBUILT — Merged two identical sources, removed the stray '(why?)' prompt, used exact means, and replaced distractors with ones produced by realistic calculation errors.

Seven paired observations of variables A and B are shown, together with their pairwise products. E(.) is the mean and StdDev(.) the population standard deviation.

n	1	2	3	4	5	6	7	E(.)	StdDev(.)
A	3.0	4.0	4.0	6.0	8.0	9.0	7.0	5.857	2.100
B	5.0	4.0	7.0	4.0	2.0	2.0	1.0	3.571	1.917
A*B	15.0	16.0	28.0	24.0	16.0	18.0	7.0	17.714	6.250

Which is NEAREST to the Pearson correlation coefficient between A and B?

A. -0.80
B. -0.51
C. -0.93
D. -0.20

Answer: A

Step by step

1. $\text{Cov}(A,B) = E(AB) - E(A)E(B) = 17.714 - 5.857 \times 3.571 = 17.714 - 20.918 = -3.204$.
2. Population standard deviations: $\sigma_A = 2.100$, $\sigma_B = 1.917$, product = 4.025.
3. $\rho = -3.204/4.025 = -0.796$, i.e. about -0.80.
4. Check: sample covariance (-3.738) over sample SDs (2.268 × 2.070) gives the same -0.796, since the n/(n-1) factors cancel.

Why A is correct: Correlation is covariance divided by the product of the two standard deviations, on a consistent (population or sample) basis; the result is -0.796.

Why the others are wrong

- **B:** Divided the covariance by $\text{StdDev}(A \times B) = 6.25$ ($-3.204/6.25 = -0.51$) instead of $\sigma_A \times \sigma_B$.
- **C:** Mixed a sample covariance (-3.738) with population standard deviations (-3.738/4.025 = -0.93); numerator and denominator must be on the same basis.
- **D:** Divided the covariance by the product of the variances (4.41 × 3.674 = 16.2), giving -0.20, instead of the product of the standard deviations.

Q397

AP Mock 3 Q2

REBUILT — Clarified that the correlation is positive and rising and that the question is about the CDS spread, and replaced the non-answer option D with a plausible distractor.

An investor holds USD 2 million of bonds issued by Issuer A and buys credit protection on Issuer A through a credit default swap (CDS) from Issuer B. The CDS spread depends on the default probability of Issuer A and on the joint default correlation of Issuer A and Issuer B (assume positive correlation). If this positive correlation increases, what is the most likely effect on the CDS spread?

A. The spread decreases, because joint default of A and B becomes more likely and the protection is less likely to pay out.
B. The spread increases, because A is more likely to default and the protection is therefore more valuable.
C. There is no effect, because the spread depends only on the default probability of the reference entity A.

D. The spread increases, because the protection seller B must be compensated for its own greater default risk.

Answer: A

Step by step

1. The buyer collects only if A defaults and B survives: P(payout) = P(A) - P(A and B).
2. Higher positive correlation raises the joint default probability P(A and B) while P(A) is unchanged.
3. P(payout) falls, so the protection is worth less and the fair CDS spread falls (wrong-way risk for the buyer).
4. The worst case for the investor is both A and B defaulting: the bond is lost and the CDS pays nothing.

Why A is correct: Positive correlation between the reference entity and the protection seller raises joint-default probability and lowers the chance of collecting, so the spread falls.

Why the others are wrong

- **B:** Correlation does not change A's own default probability; it raises the chance that B also fails, reducing the value of the protection.
- **C:** Ignores counterparty risk; the CDS spread depends on both A's default probability and the A-B joint default correlation.
- **D:** Wrong direction: the seller's default risk is borne by the buyer, who therefore pays a lower spread, not a higher one.

Q398

AnalystPrep Q.1521

REBUILT — Removed the circular wording of the stem (it stated the answer) and kept the same concept.

Which of the following best describes what a positive correlation between two pricing factors, A and B, indicates?

A. When factor A goes up in value, factor B is likely to fall in value.

B. When factor A goes down in value, factor B is likely to rise in value.

C. When factor A goes up in value, factor B tends to be unaffected.

D. When factor A goes up in value, factor B is likely to rise in value.

Answer: D

Step by step

1. Positive correlation ($\rho > 0$) means the two variables tend to move in the same direction.
2. So a rise in A is associated with a rise in B, and a fall in A with a fall in B.
3. Correlation is bounded between -1 and +1 and measures the direction and strength of co-movement, not a one-for-one change.

Why D is correct: Positive correlation means the two factors tend to move in the same direction.

Why the others are wrong

- **A:** Describes negative correlation (opposite movements).
- **B:** Also describes negative correlation, with up and down reversed.
- **C:** Describes zero correlation, i.e. no linear relationship.

LO 8.b

Explain how correlation contributed to the global financial crisis of 2007–2009.

Q399

BT answers P2.T7.507.2

REBUILT — Original relied on Malz's hedge-ratio wording; rebuilt to the Schweser 2005 GM/Ford long-equity/short-mezzanine explanation (LO 8.b) with a single clearly correct option.

In May 2005 many hedge funds were long the CDO equity tranche and short the CDO mezzanine tranche, believing losses on one would be hedged by gains on the other. After General Motors and Ford were downgraded to junk status, these funds suffered large losses. Which statement best explains the losses?

A. Equity tranche spreads fell and mezzanine spreads rose, because default correlations across all tranches increased sharply.

B. Both tranche spreads rose by a similar amount, so the long equity position lost but the short mezzanine position gained by the same amount.

C. Equity tranche spreads rose while mezzanine spreads fell, because the correlation properties across tranches were misunderstood and the tranches did not move together.

D. Recovery rates on the defaulted bonds fell to zero, which reduced the mezzanine spread but left the equity spread unchanged.

Answer: C

Step by step

1. Long the equity tranche (receiving the initial spread) loses if the market equity spread rises above the spread received.
2. Short the mezzanine tranche (paying the initial spread) loses if the market mezzanine spread falls below the spread paid.
3. After the GM/Ford downgrade the equity spread rose sharply, while correlation fell for investment-grade CDOs, lowering the mezzanine spread.
4. Both legs lost at once: the supposed hedge failed because correlation across tranches was not correctly understood.

Why C is correct: The tranche spreads moved in opposite directions, so the long equity and short mezzanine legs both lost.

Why the others are wrong

- **A:** Reverses both moves; falling equity spreads would have helped the long equity leg, and correlation fell (not rose) for investment-grade CDOs.
- **B:** Assumes the two spreads moved together, which is exactly the hedging assumption that failed; the equity spread rose while the mezzanine spread fell, so there was no offsetting gain.
- **D:** Recovery rates were not the driver; the losses came from opposite spread moves caused by changing correlation after the downgrade.

Q400 GARP book Ch.8 Q8.13

In 2005, a correlation crisis with respect to CDOs occurred that led to the default of several hedge funds. What happened?

Answer: see model answer — In May 2005 GM and Ford were downgraded to junk status. Many large hedge funds were long the CDO equity tranche and short the mezzanine tranche, believing losses on one leg would be offset by gains on the other. After the downgrades, equity tranche spreads rose sharply, causing losses on the long equity position, while correlations fell for investment-grade CDOs, so mezzanine spreads fell, causing losses on the short mezzanine position. Both legs lost because the funds misunderstood correlation across tranches, and several funds went bankrupt.

Step by step

1. May 2005: GM and Ford downgraded to junk; forced selling by institutions that cannot hold sub-investment-grade bonds.
2. Equity tranche spread widened sharply, so a long equity position (spread received below the new market spread) lost value.
3. Correlation in investment-grade CDOs fell, so the mezzanine tranche spread fell; a short mezzanine position (spread paid above the new market spread) lost value.
4. The hedge assumed the two legs offset; in fact both lost, and highly leveraged funds failed.

Why this is correct: The hedge relied on a misunderstanding of how correlation affects different tranches; a single event moved the two tranche spreads in the directions that hurt both legs.

Q401 GARP book Ch.8 Q8.14

In the global financial crisis 2007-09, many investors in the presumably safe super senior tranches got hurt. What exactly happened?

Answer: see model answer — Housing prices stagnated in 2006 and the real estate market collapsed in 2007, so mortgage default probabilities rose dramatically and default correlations across CDO tranches increased. Senior tranches were thought to be protected by the subordinated tranches below them, but with defaults widespread and highly correlated, that protection was exhausted; even AAA tranches lost about 20% of their value. Excess leverage (exposures 10 to 20 times the investment) magnified losses. Sellers of CDS protection such as AIG, which had written about $500 billion of CDSs with little reinsurance, were also overextended.

Step by step

1. Housing collapse in 2007 raised mortgage default probabilities sharply and made defaults highly correlated.
2. Higher correlation raises the chance that losses reach the senior tranches (senior tranches are short correlation); the effect on equity spreads was overshadowed by the jump in default probabilities.
3. Subordination no longer protected the senior tranches; AAA tranches lost about 20% of value.
4. Leverage of 10-20 times and concentrated CDS sellers like AIG amplified the losses.

Why this is correct: Senior tranches are safe only if defaults are limited and not highly correlated; when default probabilities and correlations jumped together, the loss buffer was exhausted.

Q402

AnalystPrep Q.1551

REBUILT — Original options were vague and D was a giveaway; rewrote stem and options so only the tranche-correlation explanation is correct.

Before and during the 2007-2009 crisis, large losses were suffered on CDO investments. According to the Schweser reading, which statement best describes the role of correlation in these losses?

A. Risk managers did not properly understand the correlation properties across the CDO tranches, even though a copula model was trusted to monitor them.

B. CDOs are inherently toxic instruments whose losses would have occurred regardless of how correlations were modeled.

C. Default correlations among mortgages and across tranches stayed low throughout the crisis, so diversification failed only because default probabilities rose.

D. Hedge funds were unaffected by correlation because their long equity and short mezzanine positions fully offset each other.

Answer: A

Step by step

1. The reading states that risk managers, institutions and investors did not understand how to properly measure correlation in structured products.
2. The copula model was trusted to monitor default correlation across tranches, yet the cross-tranche correlation properties were not correctly understood.
3. This misunderstanding, not the CDO structure itself, led to hedge fund bankruptcies in 2005 and losses on senior tranches in 2007-09.
4. Only A matches this.

Why A is correct: Losses were attributed to a failure to understand correlation properties across tranches, not to CDOs being inherently flawed.

Why the others are wrong

- **B:** Labels the CDOs themselves as toxic; the reading attributes losses to misunderstood tranche correlations.
- **C:** Default correlations rose sharply in 2007 as well as default probabilities; saying they stayed low is factually wrong.
- **D:** The 2005 hedge funds' hedge failed precisely because both legs lost money when spreads moved against them.

Q403

BT answers P2.T5.502.3 · BT notes P2.T5.502.3

About the role of correlation in the global financial crisis (GFC), Meissner shows firstly that several different types of correlation can be measured. For example, correlation between mortgages that underlie a collateralized debt obligation is different than correlation between CDO tranches. According to Meissner, each of the following is TRUE about the role of correlation in the crisis, EXCEPT which is false?

A. The Gaussian copula correlation model was widely used to value collateralized debt obligations (CDOs), but the trust in the Gaussian copula was naive; for example, low default probabilities and default correlations (between CDO assets) were often assumed as model inputs

B. Due to an increase in the default correlation among mortgages in collateralized debt obligations (CDOs), ceteris paribus, equity tranche spreads increased

C. Correlations between the tranches of the CDOs increased during the crisis, and this had a devastating effect on the super-senior tranches

D. Credit default swaps (CDS) were used as speculative instruments in ways that did not disperse risk like we expect insurance contracts to disperse risk

Answer: B

Step by step

1. The reading states that the CDO equity tranche spread typically decreases when default correlations increase (the equity tranche is long correlation).
2. So an increase in default correlation lowers the equity tranche spread and raises the tranche value, ceteris paribus; B says the opposite.
3. A, C and D are consistent with the reading (naive copula inputs, rising tranche correlation hurting senior tranches, concentrated CDS sellers like AIG).
4. B is the false statement.

Why B is correct: Higher default correlation decreases the equity tranche spread; that effect was overshadowed in 2007 by a dramatic rise in default probabilities.

Why the others are wrong

- **A:** True: the copula was trusted naively with low default probability and correlation inputs, so it is not the exception.
- **C:** True: rising correlation across tranches in 2007 eroded the protection of senior tranches, so it is not the exception.
- **D:** True: CDS sellers such as AIG concentrated rather than dispersed risk, so it is not the exception.

Q404

Schweser Module Quiz 8.2 Q3

In May 2005, several large hedge funds had speculative positions in the collateralized debt obligations (CDOs) tranches. These hedge funds were forced into bankruptcy due to a lack of understanding of correlations across tranches. Which of the following statements best describes the positions held by hedge funds at this time and the role of changing correlations? Hedge funds held a:

A. long equity tranche and short mezzanine tranche when the correlations in both tranches decreased.
B. short equity tranche and long mezzanine tranche when the correlations in both tranches increased.
C. short senior tranche and long mezzanine tranche when the correlation in the mezzanine tranche increased.
D. long mezzanine tranche and short equity tranche when the correlation in the mezzanine tranche decreased.

Answer: A

Step by step

1. The funds were long the equity tranche and short the mezzanine tranche, expecting the legs to hedge each other.
2. After the GM and Ford downgrades, the equity tranche spread rose sharply: the long equity position lost value.
3. Correlations for investment-grade CDOs fell, lowering the mezzanine spread: the short mezzanine position lost value.
4. Only A has the correct pair of positions.

Why A is correct: The funds were long equity and short mezzanine, and both legs lost as the equity spread rose and the mezzanine spread fell.

Why the others are wrong

- **B:** Reverses both positions and has correlations rising, which does not describe the actual long equity / short mezzanine book.
- **C:** Involves the senior tranche, which was not part of the hedge; the funds were not short senior.
- **D:** Reverses the positions (the funds were short mezzanine, long equity); a long mezzanine holder would actually gain when the mezzanine spread falls.

Q405

AP Mock 1 Q29

A financial institution has underwritten credit protection on the uppermost tranche of a collateralized debt obligation. All other factors constant, how would a marked reduction in the default correlation among the CDO's underlying assets, compared with the correlation assumed when the tranche was priced, affect the firm's position?

A. The impact will be ambiguous and contingent on the specific pricing model and prevailing market conditions.
B. The value of the position will significantly increase, as the likelihood of activating the credit protection diminishes.
C. There will be a notable depreciation in value, as the value of the credit protection itself escalates.
D. The position's value will remain unaffected, since only the expected default losses are relevant, and default correlation does not influence these losses.

Answer: B

Step by step

1. Underwriting protection on the senior tranche means the firm is long that tranche's credit risk (receives the

premium, pays on losses).
2. Senior tranches are short correlation: they lose only when many assets default together, which is more likely at high correlation.
3. Lower correlation makes widespread simultaneous default less likely, so the fair spread falls below the premium being received.
4. The seller's position therefore gains in value.

Why B is correct: The senior tranche suffers only from mass simultaneous default, so lower correlation reduces the chance protection is triggered and benefits the protection seller.

Why the others are wrong

- **A:** The direction is determinate under the standard model: senior tranches are short correlation.
- **C:** Gets the direction backwards; the protection becomes less valuable (not more) when correlation falls, so the seller does not depreciate.
- **D:** Correlation does not change total portfolio expected loss but redistributes it across tranches, so a single tranche's value does change.

Q406

AP Mock 1 Q30

An investment firm has sold default protection on the most senior tranche of a collateralized debt obligation. Suppose the default correlation between assets held in the CDO increases sharply, but everything else remains unchanged. How will the firm's position be affected?

A. It will neither lose nor gain value because correlation does not affect default losses.
B. It will lose significant value, since the probability of exercising the protection increases.
C. It will either increase or decrease depending on market conditions.
D. It will gain significant value because the probability of exercising the protection falls.

Answer: B

Step by step

1. The firm sold protection, so it bears the senior tranche's credit risk.
2. Senior tranches are short correlation: higher correlation raises the probability of mass simultaneous default that reaches the senior tranche.
3. The protection is therefore more likely to be called, and the premium received is now too low for the risk.
4. The position loses value (as with AIG and the monolines in 2007-08).

Why B is correct: Higher default correlation increases tail losses that hit the senior tranche, so the protection seller loses.

Why the others are wrong

- **A:** Correlation leaves total expected loss unchanged but shifts it between tranches, so the senior tranche is affected.
- **C:** The direction is determinate: senior tranches are short correlation.
- **D:** This is the answer to the opposite question (correlation falling); with rising correlation the chance of payout rises.

Q407

AnalystPrep Q.1552

REBUILT — Original asked a generic CDS definition with weak link to 8.b; rebuilt around AIG's CDS exposure in the crisis.

Before the 2007-2009 crisis the CDS market grew from about $8 trillion to $60 trillion (2004-2007), and AIG alone sold about $500 billion of CDS protection with little reinsurance. Which statement best explains why this contributed to the crisis?

A. CDS risk was concentrated in an overextended seller rather than dispersed as insurance would suggest, so rising defaults and correlation produced payouts the seller could not meet.
B. AIG had bought, not sold, most of the protection, so the failure of its counterparties left it unhedged.
C. The CDS market shrank during 2004-2007, removing the ability of institutions to hedge default risk.
D. CDS values rose as default correlation increased, so protection sellers made large gains that distorted markets.

Answer: A

Step by step

1. CDS act like insurance: the seller must be financially strong enough to pay on defaults.
2. AIG sold about $500 billion of CDS with little reinsurance, so the risk was concentrated in one overextended seller.

3. When subprime defaults and default correlations jumped in 2007, claims on protection sellers rose sharply.
4. Concentrated, under-capitalised sellers could not absorb the losses, threatening the system; only A states this.

Why A is correct: CDS are meant to transfer and spread default risk, but concentration in thinly capitalised sellers such as AIG meant the crisis losses could not be absorbed.

Why the others are wrong

- **B:** AIG was a net seller of protection, not a buyer; the problem was its sold positions.
- **C:** The market grew from $8 trillion to $60 trillion over 2004-2007, it did not shrink.
- **D:** Protection sellers lose when defaults and correlation rise; they did not gain.

Q408 BT notes P2.T5.22.12.3

REBUILT — Option C (high-PD firms have lower default correlation due to idiosyncratic factors) is not supported by the Schweser reading and the option set mixed several LOs; replaced it with a reading-based false statement and kept the CDO equity-tranche statement as the single correct answer.

Meissner argues that correlations and correlation risk are critical in investments, trading and especially risk management, and that in a systemic crisis correlations typically increase and can lead to high unexpected losses. Based on this review, which of the following statements is TRUE?

A. Because stochastic correlation is non-deterministic, stochastic correlation models should be avoided.
B. Correlation risk is less important in market risk and credit risk but more important in operational risk.
C. Default correlations between companies in the same credit-quality group are typically lower than between companies in different credit-quality groups.
D. Holding default probability constant, an increase in default correlations between mortgages in a CDO typically lowers the equity tranche spread.

Answer: D

Step by step

1. A: stochastic correlation processes are one of the main approaches to modeling correlation, so they are not to be avoided - false.
2. B: correlation risk is central to market risk (covariance matrix in VaR/ES) and credit risk (default correlation) - false.
3. C: bonds within a credit-quality level are more highly correlated than across levels - false.
4. D: higher default correlation makes equity-tranche losses less likely relative to senior losses, so the equity tranche spread typically decreases - true.

Why D is correct: Greater default correlation concentrates defaults into the same scenarios, making it less likely the equity tranche is hit alone, so its spread falls.

Why the others are wrong

- **A:** Stochastic correlation is a recognized modeling approach; non-determinism is the point, not a reason to avoid it.
- **B:** Reverses the facts: correlation is critical in both market and credit risk.
- **C:** Reverses the relationship: correlations are higher within a credit-quality level than across levels.

LO 8.c

Describe how correlation impacts the price of quanto options as well as other multi- asset exotic options.

Q409 GARP book Chapter 8 end-of-chapter Q8.7

What is ”correlation trading”?

Answer: see model answer — Correlation trading is trading assets whose prices are determined by the co-movement of one or more underlying assets over time. Examples are multi-asset (correlation) options such as exchange, quanto, rainbow and basket options, and correlation swaps, whose value is very sensitive to the correlation between underlyings. For most multi-asset options a lower correlation gives a higher option price (the worse-of option is the exception).

Step by step

1. Definition: strategies whose price depends on co-movement of assets.
2. Instruments: multi-asset options and correlation swaps.

3. Direction: lower correlation generally raises multi-asset option prices; the worse-of option is the exception.

Why this is correct: The payoff or price is driven by correlation between underlyings, so the trader takes a view on correlation.

Q410 GARP book Ch.8 Q8.9

Name three correlation options in which a lower correlation results in a higher option price.

Answer: see model answer — Any three of: the exchange option (payoff $\max(0, S_2 - S_1)$), the spread call option (payoff $\max[0, (S_2 - S_1) - K]$), the call on the maximum of two assets (payoff $\max[0, \max(S_1, S_2) - K]$), the option on the better of two assets, and the put on the minimum of two assets. Lower correlation raises the dispersion of the two prices, so the chance of a large payout rises. The one exception in this family is the option on the worse of two assets.

Step by step

1. Recall the multi-asset (rainbow) options in Schweser Figure 8.5.
2. For exchange and spread options the payoff depends on the difference S2 - S1; lower correlation increases the volatility of the difference and so the price.
3. For options on the maximum or better of two assets, lower correlation means one asset is likely high while the other is low, so the maximum is larger on average.
4. List any three of these.

Why this is correct: With lower correlation the two prices are more dispersed, which increases the probability and size of a payout for options that pay on a difference or on the best performer.

Q411 GARP book Ch.8 Q8.10

Name one correlation option in which a lower correlation results in a lower option price.

Answer: see model answer — The option on the worse of two assets (payoff = the minimum of the two asset prices, $\min(S_1, S_2)$). When correlation falls, the two prices diverge, so the minimum is more likely to be low, which reduces the option price. It is the only multi-asset option in the Schweser discussion for which a higher correlation raises the price.

Step by step

1. Recall that for nearly all multi-asset options lower correlation raises the price.
2. The exception is the option on the worse of two assets with payoff $\min(S_1, S_2)$.
3. Lower correlation spreads the assets apart, so the minimum falls on average and the price falls.

Why this is correct: The payoff is driven by the poorer performer; the more the assets diverge, the lower the minimum.

Q412 AnalystPrep Q.1546

A quanto option allows a domestic investor to convert a potential option payoff, earned in a foreign currency, back into the domestic currency at a fixed exchange rate, protecting the investor against currency risk. From a risk management standpoint, the financial institution selling a quanto option does not know which two things at the time of sale?

A. The foreign-currency amount to be converted into domestic currency, and the exchange rate at option maturity at which that foreign-currency payoff will be converted.

B. The domestic-currency amount to be converted into foreign currency, and the exchange rate at option maturity at which that domestic-currency payoff will be converted.

C. The foreign currency that is correlated with the domestic currency, and the impact of the correlation on the buying and selling of quanto options.

D. The domestic-currency amount to be converted into foreign currency, and the impact of the correlation on the buying and selling of quanto options.

Answer: A

Step by step

1. The quanto call pays off in foreign currency (the foreign asset's gain), which is converted into domestic currency.
2. The seller does not know how deep in the money the call will finish, so the foreign-currency payoff amount is

unknown.
3. The seller also does not know the spot exchange rate at exercise at which it must source the foreign currency.
4. Only option A states both unknowns in the correct direction (foreign to domestic).

Why A is correct: The payoff is in foreign currency and its size depends on the uncertain foreign asset; the FX rate needed to cover it is also unknown, and this is the risk the seller carries.

Why the others are wrong

- **B:** Reverses the conversion direction: the payoff is earned in foreign currency, not domestic.
- **C:** The currencies involved are known at sale; the correlation is an input to pricing, not an unknown amount or rate.
- **D:** Reverses the conversion direction and substitutes the correlation effect for the unknown exchange rate.

Q413

BT Mock Q46 · BT notes P2.T5.22.11.1

REBUILT — Option A (long basket option) is ambiguous because Schweser states lower correlation raises the price of nearly all multi-asset options; replaced with an unambiguous long-correlation trade (option on the worse of two).

Bertha believes that correlation between assets is going to decrease. Which of the following trades best expresses her view, i.e., gains value if correlation falls?

A. Buy an option on the worse of two assets, payoff = $\min(S_1, S_2)$.
B. Buy an option on the better of two assets, payoff = $\max(S_1, S_2)$.
C. Buy correlation in a correlation swap (pay the fixed rate, receive realized correlation).
D. Sell (short) an exchange option, payoff = $\max(0, S_2 - S_1)$.

Answer: B

Step by step

1. Bertha expects correlation to fall, so she wants a position that gains when correlation decreases (short correlation).
2. Better-of-two option: lower correlation means the assets are more dispersed, so $\max(S_1, S_2)$ is larger on average; the price rises, so owning it gains.
3. Worse-of-two: lower correlation lowers $\min(S_1, S_2)$, so a long position loses.
4. Paying fixed in a correlation swap pays off when realized correlation exceeds the fixed rate, i.e., long correlation.
5. Exchange option value rises as correlation falls; being short it loses, i.e., long correlation.

Why B is correct: The better-of-two option is worth more when correlation falls because dispersion increases the expected maximum, so a long position is short correlation.

Why the others are wrong

- **A:** The worse-of-two option is the one multi-asset option that is long correlation; its price falls as correlation falls.
- **C:** The buyer of a correlation swap pays fixed and receives realized correlation, so it profits when correlation rises, the opposite of her view.
- **D:** Exchange option price rises as correlation falls, so a short position loses; it is long correlation.

Q414

BT answers P2.T5.24.1.1

REBUILT — Source options mentioned 'yen' for a DAX/EUR quanto and had overlapping, inconsistent wording; rewrote with euros and four mutually distinct statements, keeping the Schweser result that lower correlation raises quanto prices.

A US investor buys a quanto call on the German DAX 30 index. The payoff is in euros but is converted into US dollars at a fixed exchange rate, so the investor is protected against currency risk. According to the Schweser reading, how does the correlation between the DAX 30 and the euro (relative to the dollar) affect the price of the quanto call?

A. Positive correlation increases the price, because the euro strengthens as the DAX rises and the seller needs more euros to cover the payoff.
B. Negative correlation decreases the price, because the euro strengthens as the DAX falls and the call becomes less likely to finish in the money.
C. Correlation has no effect on the price, because the conversion rate is fixed at inception.
D. Negative correlation increases the price, because the euro tends to weaken as the DAX rises, so the seller needs more euros to convert the profit into dollars.

Answer: D

Step by step

1. Lower correlation implies higher quanto prices (Schweser).
2. With negative correlation, a rising DAX comes with a weaker euro.
3. The seller must convert a euro profit into dollars at the fixed rate, so more euros are needed to meet the dollar payoff when the DAX is high.
4. Price is therefore higher for negative (lower) correlation and lower for positive correlation.

Why D is correct: Quanto price rises as correlation falls: when the index is high the foreign currency is weaker, which makes the fixed-rate conversion costlier for the seller.

Why the others are wrong

- **A:** Reverses the effect: positive correlation lowers, not raises, the price, since the euro strengthens along with the DAX.
- **B:** Gets the direction wrong: negative correlation raises the price. The correlation does not change whether the call is in the money.
- **C:** The fixed rate only removes the buyer's FX risk; the seller still bears the risk, so correlation is a pricing input.

Q415

BT answers P2.T5.501.2 · BT notes P2.T5.501.2

For which of the following multi-asset options does an increase in the correlation between S1 and S2 imply an increase in the option price? (S1 and S2 are the asset prices at maturity and K is the strike.)

A. Call on the maximum of two: payoff = $\max[0, \max(S_1, S_2) - K]$
B. Option on the worse of two: payoff = $\min(S_1, S_2)$
C. Exchange option: payoff = $\max(0, S_2 - S_1)$
D. Spread call option: payoff = $\max[0, (S_2 - S_1) - K]$

Answer: B

Step by step

1. For nearly all multi-asset options, lower correlation raises the price.
2. Call on max, exchange option and spread call benefit from dispersion between S1 and S2, so their price falls as correlation rises.
3. The worse-of-two option pays the minimum; with higher correlation the assets move together and the minimum is higher.
4. Hence only B increases with correlation.

Why B is correct: The option on the worse of two is the one exception, because high correlation keeps the poorer asset from lagging far behind the other.

Why the others are wrong

- **A:** The call on the maximum gains from dispersion, so its price falls as correlation rises.
- **C:** The exchange option is near zero as correlation approaches 1 because the spread does not move; price falls with correlation.
- **D:** The spread call depends on the volatility of S2 - S1, which shrinks as correlation rises, so price falls.

Q416

BT answers L2.T5.21.2

Let S = S&P index, Q = Nikkei index, C = cash return and K = the strike price. Each of the following gives a valid payoff for a rainbow option EXCEPT:

A. Call: $\max[0, \max(S, Q, C) - K]$
B. Call: $\max[S, Q, C, K] - K$
C. Put: $\max[0, K - \min(S, Q, C)]$
D. Put: $K - \max(S, Q, C, K)$

Answer: D

Step by step

1. A is a call on the best of three assets.
2. B equals A: $\max(S, Q, C, K) - K = \max[0, \max(S, Q, C) - K]$.
3. C is a put on the worst of three assets, valid since the holder sells the asset with the lowest price.
4. D equals $-\max[0, \max(S, Q, C) - K] \leq 0$ and can never be positive, so it is not an option payoff; the put should use the minimum: $K - \min(S, Q, C, K)$.

Why D is correct: A put pays when the asset falls below the strike, so it must be based on the minimum of the assets; using the maximum yields a non-positive value.

Why the others are wrong

- **A:** Valid rainbow call on the maximum of the three assets.
- **B:** Equivalent to A, since taking the max with K and subtracting K floors the payoff at zero.
- **C:** Valid rainbow put on the minimum of the three assets.

Q417

BT answers L2.T5.21.3a

REBUILT — Two-option true/false converted to a 4-option MCQ asking for the correct relationship between rainbow and exchange options.

Which statement best describes the relationship between rainbow options and exchange options?

A. A rainbow option is a special case of an exchange option.

B. Rainbow and exchange options are identical, since both are options on two assets.

C. An exchange option is one type of multi-asset (rainbow) option, whose payoff $\max(0, S_2 - S_1)$ depends on the difference of two assets.

D. Exchange options are single-asset options, so they are unrelated to rainbow options.

Answer: C

Step by step

1. Rainbow options (multi-asset options) are payoffs that depend on two or more assets, e.g., max or min of several assets.
2. The exchange option, with payoff $\max(0, S_2 - S_1)$, depends on two assets, so it belongs to the multi-asset family.
3. The reverse is false: a rainbow option such as a call on the max of three assets cannot be written as an exchange option.

Why C is correct: Exchange options are a member of the broader class of multi-asset options; the class is not a subset of exchange options.

Why the others are wrong

- **A:** Reverses the relationship; the rainbow class is broader.
- **B:** Not identical: rainbow payoffs use max/min of several assets, exchange payoffs use a difference.
- **D:** The exchange option depends on two assets (S1 and S2), so it is a multi-asset option.

Q418

BT answers L2.T5.21.1

REBUILT — Original 'similar to a rainbow option EXCEPT' item relied on Hull's cheapest-to-deliver analogy and had arguable options; rebuilt to test how correlation affects multi-asset option prices per Schweser.

A trader holds long positions in several two-asset options. As the correlation between the two underlying assets falls, which option would be expected to DECREASE in price?

A. An exchange option with payoff $\max(0, S_2 - S_1)$

B. A call option on the better of two assets

C. A quanto call on a foreign equity index

D. A call option on the worse of two assets (payoff based on $\min(S_1, S_2)$)

Answer: D

Step by step

1. Schweser: for the standard multi-asset correlation options (exchange, spread, better-of, etc.), a lower correlation raises the option price because the assets are more likely to diverge.
2. Exchange option: $\sigma_E = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$ rises as ρ falls, so the price rises.
3. Quanto call: lower correlation between the index and the currency means higher price.
4. The worse-of option pays on the minimum of the two assets; low correlation makes it more likely that one asset is poor, so the minimum is lower and the price falls. It is the one correlation option for which lower correlation is not desirable.

Why D is correct: The payoff of the worse-of option depends on the weaker asset, so a lower correlation increases the chance that one asset underperforms and reduces the price; all other listed options gain from dispersion.

Why the others are wrong

- **A:** Exchange option price rises as correlation falls because σ_E increases; at $\rho = 1$ it is near zero.
- **B:** Better-of (max) option benefits from dispersion, so price rises when correlation falls.
- **C:** Lower correlation raises quanto prices; reversing this sign is the common error.

Q419 BT answers L2.T5.21.3b · BT answers L2.T5.21.3c

REBUILT — Original true/false items on rainbow and basket options were generic and not covered by the Schweser text; rebuilt as a quanto-option correlation question that tests the same idea (correlation as key pricing input for multi-asset options).

A U.S. investor buys a quanto call on the Nikkei Index with a fixed USD/JPY conversion rate. The correlation between the Nikkei Index and the yen's value in dollars is expected to move from +0.4 to -0.4. All else equal, the price of the quanto call will most likely:

A. Increase, because lower correlation raises the chance the index rises while the yen falls, so the seller needs more yen per dollar to cover the payoff

B. Decrease, because lower correlation reduces the volatility of the combined payoff

C. Be unchanged, because the fixed conversion rate removes exchange-rate exposure and hence any dependence on correlation

D. Decrease, because a negative correlation means the yen strengthens when the index rises, which reduces the seller's cost

Answer: A

Step by step

1. Schweser: the seller of a quanto call does not know how deep in the money the call will be or the exchange rate at exercise.
2. Positive correlation means a rising index tends to come with a rising yen (cheaper to convert); negative correlation means a rising index tends to come with a falling yen, so each yen of gain converts to fewer dollars at market but the seller owes the fixed rate, raising the seller's cost.
3. Lower correlation therefore raises the seller's risk and the quanto price. Moving from +0.4 to -0.4 increases the price.

Why A is correct: Quanto prices rise as correlation between the underlying and the exchange rate falls, since gains in yen are then more likely to coincide with a weaker yen.

Why the others are wrong

- **B:** Treats lower correlation as risk-reducing; for these options dispersion increases the price.
- **C:** The fixed rate protects the buyer, but it leaves the seller exposed to the unknown yen level, so price still depends on correlation.
- **D:** Reverses the sign logic: with negative correlation the yen weakens when the index rises, which hurts the seller.

Q420 BT answers L2.T5.21.3c · BT answers L2.T5.21.3b

REBUILT — Original true/false item had no quantitative content; rebuilt as a numeric exchange-option volatility question showing how correlation drives price.

An exchange option pays $\max(0, S_2 - S_1)$. The assets have volatilities of 30% (S_1) and 20% (S_2), and the correlation is 0.5. The volatility used to value the option, σ_E (the volatility of S_2/S_1), is closest to:

A. 10.0%

B. 26.5%

C. 36.1%

D. 31.6%

Answer: B

Step by step

1. $\sigma_E^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$
2. $= 0.09 + 0.04 - 2(0.5)(0.30)(0.20) = 0.13 - 0.06 = 0.07$
3. $\sigma_E = \sqrt{0.07} = 26.46\% \approx 26.5\%$
4. Lower correlation raises σ_E (36.1% at $\rho = 0$) and the option price; at $\rho = 1$ it is 10%.

Why B is correct: The exchange option is priced off the volatility of the ratio, which falls as correlation rises; correlation is the key input.

Why the others are wrong

- **A:** This is the $\rho = 1$ result $(30\% - 20\%)$; it ignores the actual correlation of 0.5.
- **C:** This is the $\rho = 0$ result $\sqrt{0.13}$; it drops the correlation term.
- **D:** Subtracted only $\rho\sigma_1\sigma_2$ instead of $2\rho\sigma_1\sigma_2$: $\sqrt{0.13 - 0.03} = 31.6\%$.

Q421 BT answers L2.T5.21.3d

REBUILT — Original true/false item on worst-of/best-of options as rainbow options was a definitional giveaway; rebuilt as a payoff calculation for a worst-of call.

A call option on the worse of two assets has a strike of 90 and pays $\max(0, \min(S_1, S_2) - 90)$. At expiry $S_1 = 112$ and $S_2 = 95$. The payoff is:

A. 22
B. 5
C. 13.5
D. 17

Answer: B

Step by step

1. The worse of the two assets is $\min(112, 95) = 95$.
2. Payoff $= \max(0, 95 - 90) = 5$.

Why B is correct: A worst-of option is struck on the minimum of the assets, so the weaker asset determines the payoff.

Why the others are wrong

- **A:** Used the better asset: $112 - 90 = 22$ (best-of call).
- **C:** Used the average of the two: $103.5 - 90$.
- **D:** Used the difference $S_1 - S_2 = 17$ (spread/exchange option payoff).

Q422 BT answers L2.T5.21.3b

REBUILT — True/False item converted into a 4-option MCQ with plausible distractors so exactly one statement is correct.

Which of the following statements about rainbow options and basket options is correct?

A. Both have payoffs that depend on two or more underlying assets.
B. A basket option depends on one asset only, through the average of its prices over time.
C. A rainbow option depends on a single asset but has several possible strike prices.
D. Only rainbow options depend on multiple assets; a basket option is a single-asset option with several exercise dates.

Answer: A

Step by step

1. Rainbow options (best-of, worst-of, etc.) have payoffs driven by two or more risky assets.
2. A basket option has a payoff driven by a weighted average (portfolio) of several assets' prices.
3. Hence both are multi-asset exotic options, and their price depends on the co-movement (correlation) of the underlyings.

Why A is correct: Both are multi-asset exotics: the payoff is a function of several underlying prices, which is why correlation is a pricing input.

Why the others are wrong

- **B:** Describes an Asian (average-price) option, which averages one asset over time, not a basket.
- **C:** Confuses rainbow options with multi-strike structures; a rainbow option is defined by multiple underlyings.
- **D:** Wrongly treats a basket as single-asset; a basket is an option on a portfolio of several assets.

Q423 BT answers L2.T5.21.3c

REBUILT — True/False item converted into a 4-option MCQ with plausible distractors so exactly one statement is correct.

Which of the following statements about the effect of correlation on the price of multi-asset correlation options is correct?

A. Lower correlation raises the price of most multi-asset options (such as an exchange option), but it lowers the price of a call on the worse of two assets.
B. An exchange option is worth the most when the two asset prices are almost perfectly positively correlated.
C. Correlation affects the price of rainbow options but is irrelevant to basket options.
D. A higher correlation increases the price of every multi-asset correlation option.

Answer: A

Step by step

1. Exchange option payoff is $\max(0, S_2 - S_1)$; its volatility is $\sigma_E = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, which falls as ρ rises.

2. Check (Margrabe, $S_1 = S_2 = 100$, both vols 30%, T = 1): price = 5.35 at $\rho = 0.9$, 11.92 at $\rho = 0.5$, 16.80 at $\rho = 0$, so the price rises as correlation falls.
3. With low correlation one asset is more likely to rise while the other falls, giving a better chance of a large payout, so most multi-asset options gain value.
4. The exception is the option on the worse of two assets (payoff on the minimum), whose price falls when correlation falls.

Why A is correct: Lower correlation widens the dispersion between assets and raises the price of most multi-asset options; the worst-of option is the exception.

Why the others are wrong

- **B:** Reverses the relationship: as correlation approaches 1 the spread S2 - S1 barely changes and the exchange option price approaches zero.
- **C:** Basket value depends on portfolio variance, which depends on pairwise correlations, so correlation matters for baskets too.
- **D:** Reverses the sign: higher correlation generally lowers the price, and the worst-of option is a counterexample in the other direction.

Q424

BT answers L2.T5.21.3d

REBUILT — True/False item converted into a 4-option MCQ with plausible distractors so exactly one statement is correct.

Which of the following is an example of a rainbow option?

A. A call option whose payoff is based on the higher of two stock prices at maturity (best-of option).
B. A call on a single stock whose payoff is based on the average of that stock's daily closing prices.
C. A plain-vanilla put on a single stock.
D. A call on a single stock that is knocked out if the price touches a barrier.

Answer: A

Step by step

1. Rainbow options have payoffs depending on the best, worst or ranking of several risky assets.
2. A call on the higher of two stock prices (best-of) is therefore a rainbow option; worst-of options are also rainbow options.
3. B, C and D each depend on only one underlying asset.

Why A is correct: Best-of and worst-of options are classic rainbow options because their payoff depends on the maximum or minimum of multiple assets.

Why the others are wrong

- **B:** This is an Asian option on a single asset (path-dependent average), not multi-asset.
- **C:** A vanilla option has a single underlying, so it is not a rainbow option.
- **D:** A barrier option is path-dependent on one asset; no second asset is involved.

Q425

BT answers L2.T5.21.3e

REBUILT — True/False item converted into a 4-option MCQ with plausible distractors so exactly one statement is correct.

Asset prices S(t) and Q(t) are each lognormally distributed. A basket option pays off on the average of the two prices, [S(T) + Q(T)] / 2. Which of the following statements is correct?

A. The average price is lognormal, with volatility equal to the average of the two volatilities.
B. The average price is normally distributed, because the sum of normal variables is normal.
C. In general the average price is not lognormal, so there is no simple closed-form lognormal pricing formula and approximations or Monte Carlo simulation are used.
D. The average price is lognormal only when the correlation between S(t) and Q(t) is zero.

Answer: C

Step by step

1. $\ln S$ and $\ln Q$ are normal, so $\ln S + \ln Q = \ln(SQ)$ is normal; this is the log of the product, not of the sum.
2. The basket depends on $S + Q$, and $\ln(S + Q) \neq \ln S + \ln Q$; a sum of lognormals is not lognormal in general.
3. So the basket average has no exact lognormal distribution and is priced with approximations or Monte Carlo simulation.

Why C is correct: Normality is preserved under sums of the logs, not under sums of the prices, so the basket average is not lognormal.

Why the others are wrong

- **A:** Assumes lognormality is closed under addition and that volatilities average; neither holds.
- **B:** Applies the sum-of-normals result to prices rather than log prices; prices are lognormal, not normal.
- **D:** Zero correlation does not make a sum of lognormals lognormal; the distribution is not lognormal for any correlation except degenerate cases.

LO 8.d

Describe the structure, uses, and payoffs of a correlation swap.

Q426

AP Mock 1 Q24

The buyer of a correlation swap with three assets and a maturity of one year pays a fixed rate of 2.5%. The notional amount is USD 1 million. If the realised pairwise correlations of the daily log returns are $\rho_{2,1} = 0.2$, $\rho_{3,1} = 0.4$ and $\rho_{3,2} = 0.3$, what is the realised correlation?

A. 0.30.
B. 0.20.
C. 0.45.
D. 0.16.

Answer: A

Step by step

1. Number of pairs = $n(n-1)/2 = 3$.
2. Realised correlation $\rho_{realized} = \frac{2}{n^2-n}\sum_{i>j}\rho_{i,j} = \frac{2}{6}(0.2 + 0.4 + 0.3)$.
3. = 0.9/3 = 0.30 (the simple average of the three pairwise correlations).

Why A is correct: Realised correlation in a correlation swap is the average of all distinct pairwise correlations, which is 0.9/3 = 0.30.

Why the others are wrong

- **B:** 0.20 is just a single pair ($\rho_{2,1}$), not the average.
- **C:** 0.45 divides the sum 0.9 by 2 instead of by the number of pairs (3).
- **D:** 0.16 is a mis-normalisation (e.g., dividing by an incorrect count) and does not follow from the formula.

Q427

AnalystPrep Q.1542

A correlation swap buyer pays a fixed correlation rate of 0.28 on a notional of USD 10 million for one year, for a portfolio of three assets. The realized pairwise correlations of the daily log returns at maturity are $\rho_{2,1} = 0.7$, $\rho_{3,1} = 0.2$ and $\rho_{3,2} = 0.03$. What is the payoff to the swap buyer?

A. USD 0.28 million
B. USD 0.31 million
C. USD 0.30 million
D. USD 0.25 million

Answer: C

Step by step

1. Realised correlation $= \frac{2}{3^2-3}(0.7 + 0.2 + 0.03) = \frac{1}{3}(0.93) = 0.31$.
2. Payoff = Notional × $(\rho_{realized} - \rho_{fixed})$ = 10,000,000 × (0.31 − 0.28).
3. = USD 300,000 = USD 0.30 million.

Why C is correct: The buyer receives notional times the excess of realised over fixed correlation: 10m × 0.03 = USD 0.30m.

Why the others are wrong

- **A:** 0.28 is simply the fixed rate quoted in millions; it ignores the realised correlation.
- **B:** 0.31 is the realised correlation expressed in millions; the fixed rate was not subtracted and the notional was not applied.
- **D:** 0.25 does not correspond to any valid step; it results from an arithmetic or averaging error (e.g., mis-adding the pairwise correlations).

Q428 AnalystPrep Q.2834

Calculate the payoff to the buyer of a correlation swap on three assets. The realized pairwise correlations of the log returns at maturity are 0.24, 0.26 and 0.36. The notional is USD 12 million, the fixed correlation rate is 18%, and maturity is one year.

A. USD 0.29 million
B. USD 1.28 million
C. USD 0.24 million
D. USD 5.04 million

Answer: B

Step by step

1. Realised correlation $= \frac{2}{6}(0.24 + 0.26 + 0.36) = 0.86/3 = 0.28667$.
2. Payoff = 12,000,000 × (0.28667 − 0.18) = 12,000,000 × 0.10667.
3. = USD 1.28 million.

Why B is correct: The buyer is paid notional × (realised − fixed) = 12m × 0.10667 ≈ USD 1.28m.

Why the others are wrong

- **A:** 0.29 is the realised correlation (0.2867) written as USD millions; notional and fixed rate were ignored.
- **C:** 0.24 is just one of the pairwise correlations taken as the payoff figure; it has no basis in the formula.
- **D:** 5.04 = 12m × 0.42, which adds the fixed rate (0.18) to a pairwise correlation (0.24) instead of subtracting the fixed rate from the average realised correlation.

Q429 BT answers P2.T5.24.1.2

REBUILT — Source data were internally inconsistent (fixed rate 0.75 vs 0.15 in solution; realised 0.433 not matching the matrix) and the options were flawed; rebuilt with a consistent matrix and a seller-side payoff.

A trading firm sells (is the fixed-rate receiver in) a one-year correlation swap on three equally weighted assets. The notional is USD 20,000,000 and the fixed correlation rate is 0.35. The realised pairwise correlations at maturity are:

Correlation	Asset 1	Asset 2	Asset 3
Asset 1	1	0.3	0.2
Asset 2	0.3	1	0.4
Asset 3	0.2	0.4	1

What is the firm's payoff at maturity?

A. It receives USD 1,000,000.
B. It pays USD 1,000,000.
C. It pays USD 2,000,000.
D. It pays USD 3,666,667.

Answer: A

Step by step

1. Distinct pairs: $\rho_{2,1} = 0.3$, $\rho_{3,1} = 0.2$, $\rho_{3,2} = 0.4$.
2. Realised correlation $= \frac{2}{6}(0.3 + 0.2 + 0.4) = 0.30$.
3. The buyer's payoff is $N(\rho_{realized} - \rho_{fixed})$ = 20m × (0.30 − 0.35) = −1.0m; the seller's payoff is the opposite.
4. Seller payoff = 20,000,000 × (0.35 − 0.30) = +USD 1,000,000 (received).

Why A is correct: The seller receives the fixed rate and pays realised correlation, so it gains when realised correlation (0.30) falls below the fixed rate (0.35).

Why the others are wrong

- **B:** Sign error: this is the buyer's result (−1m) applied to the seller; the seller profits when realised < fixed.
- **C:** Uses a realised correlation of 0.45 (sum 0.9 divided by 2 rather than by 3 pairs): 20m × (0.35 − 0.45) = −2m.
- **D:** Averages the full 3×3 matrix including the diagonal ones: (3 + 2×0.9)/9 = 0.533, giving 20m × (0.35 − 0.533) = −3.67m.

Q430

GARP PE2 Q33

REBUILT — Option D (London Whale) was off-topic and arguably defensible; replaced with a clearly incorrect statement about correlation swap payoffs.

The manager of an equity trading desk presents to newly hired analysts on derivatives used to take positions on equity correlation, including rainbow options, quanto options and correlation swaps. Which of the following statements could the manager correctly make?

A. The payoff for each of these products is highest when the price movements of the underlying assets are uncorrelated.

B. Investors can take a synthetic long position in equity correlation by buying put options on an equity index and selling put options on the components of that index.

C. The payoff for the correlation buyer in a correlation swap is negative if the realized correlation of the underlying assets is higher than the fixed correlation in the swap.

D. The payoff to the buyer of a correlation swap is the notional multiplied by the difference between realised and fixed variance of the index.

Answer: B

Step by step

1. Index option implied volatility rises with correlation among constituents, while single-stock option volatilities are largely unaffected.
2. Long index options plus short options on the components therefore gains when correlation rises: a synthetic long correlation position (puts work like the calls in the Schweser example).
3. Compare A, C and D with the correlation swap payoff $N(\rho_{realized} - \rho_{fixed})$: each is inconsistent with it or with the products' payoffs.

Why B is correct: Index volatility increases with correlation, so long index options versus short constituent options profits from rising correlation.

Why the others are wrong

- **A:** Wrong: a correlation swap and many multi-asset options gain when correlation is high (or depend on sign/direction), not when assets are uncorrelated.
- **C:** Reversed: the buyer's payoff is positive when realised correlation exceeds the fixed rate.
- **D:** The correlation swap payoff is based on realised minus fixed correlation, not variance.

Q431

Mock 2 Q50

REBUILT — Distractors were all notional x a correlation level (giveaway); rebuilt options around realistic calculation mistakes.

An investor buys a correlation swap with a fixed correlation rate of 0.3 and a notional of USD 750,000 for one year on a portfolio of four assets. The pairwise correlations of the daily log returns at maturity are:

Correlation matrix	Asset 1	Asset 2	Asset 3	Asset 4
Asset 1	1	0.24	0.61	0.08
Asset 2	0.24	1	0.12	0.36
Asset 3	0.61	0.12	1	0.45
Asset 4	0.08	0.36	0.45	1

What is the payoff at maturity for the swap buyer?

A. USD 7,500.

B. USD 123,750.

C. USD 136,875.

D. USD 232,500.

Answer: A

Step by step

1. Six distinct pairs: 0.24 + 0.61 + 0.08 + 0.12 + 0.36 + 0.45 = 1.86.
2. Realised correlation $= \frac{2}{4^2-4}(1.86) = \frac{2}{12}(1.86) = 0.31$.
3. Payoff = 750,000 × (0.31 − 0.30) = USD 7,500.

Why A is correct: Payoff is notional times the small excess of realised (0.31) over fixed (0.30) correlation.

Why the others are wrong

- **B:** Divides the pair sum by n = 4 (giving 0.465) rather than by 6 pairs: 750,000 × (0.465 − 0.30) = 123,750.
- **C:** Averages all 16 matrix entries including the diagonal ones: (4 + 2×1.86)/16 = 0.4825; 750,000 × 0.1825 = 136,875.
- **D:** Multiplies the notional by the realised correlation (0.31) and forgets to subtract the fixed rate.

Q432 Schweser Module Quiz 8.2 Q1

Suppose an individual buys a correlation swap with a fixed correlation of 0.2 and a notional value of $1 million for one year. The realized pairwise correlations of the daily log returns at maturity for three assets are $\rho_{2,1} = 0.7$, $\rho_{3,1} = 0.2$, and $\rho_{3,2} = 0.3$. What is the correlation swap buyer's payoff at maturity?

A. $100,000.
B. $200,000.
C. $300,000.
D. $400,000.

Answer: B
Step by step
1. Realised correlation $= \frac{2}{3^2-3}(0.7 + 0.2 + 0.3) = \frac{1.2}{3} = 0.40$.
2. Payoff = $1,000,000 × (0.40 − 0.20) = $200,000.

Why B is correct: Realised correlation is the average pairwise correlation (0.40); the buyer is paid the excess over 0.2 times the notional.
Why the others are wrong
- **A:** Divides the sum 1.2 by 4, giving 0.30 and a payoff of 1m × 0.10.
- **C:** Uses a realised correlation of 0.5 (e.g., averaging only the extreme pairs 0.7 and 0.3), giving 1m × 0.3.
- **D:** Divides the sum by 2 instead of by 3 pairs (0.6), giving 1m × (0.6 − 0.2) = 400,000.

LO 8.e

Estimate the impact of different correlations between assets in the trading book on the VaR capital charge.

Q433 GARP book Ch.8 Q8.11

Create a numerical example of a two-asset portfolio and show that a lower correlation coefficient leads to a lower VaR number.

Answer: see model answer — Take $8 million in A (daily σ 1.5%) and $4 million in B (daily σ 2%), 10-day 99% VaR with α = 2.33. Portfolio daily σ is $\sigma_P = \sqrt{(0.12)^2 + (0.08)^2 + 2\rho(0.12)(0.08)}$ in $ millions. For ρ = 1, 0.5, 0, -0.5 this gives 0.2000, 0.1744, 0.1442, 0.1058, so the 10-day VaR $= 2.33\,\sigma_P\sqrt{10}$ is $1.474m, $1.285m, $1.063m and $0.780m. VaR falls monotonically as ρ falls because the covariance term $2\rho\sigma_1\sigma_2$ shrinks, i.e. diversification benefit grows; at ρ = 1 VaR equals the sum of stand-alone VaRs.
Step by step
1. Position volatilities: A = 8 × 0.015 = 0.12; B = 4 × 0.02 = 0.08 ($ millions per day).
2. Portfolio variance = 0.0144 + 0.0064 + 2ρ(0.12)(0.08) = 0.0208 + 0.0192ρ.
3. ρ = 1: σP = 0.2000; ρ = 0.5: 0.1744; ρ = 0: 0.1442; ρ = -0.5: 0.1058.
4. 10-day 99% VaR = 2.33 × σP × √10: 1.474, 1.285, 1.063, 0.780 ($ millions).

Why this is correct: Portfolio variance is increasing in ρ, so VaR is increasing in ρ; lower correlation means more diversification and lower VaR.

Q434 AnalystPrep Q.1549

We have calculated the VaR for a two-asset portfolio to analyze the impact of correlation between the two assets. The VaR for a 10-day horizon with a correlation coefficient of 0.7 at a 99% confidence level is $1.7486 million. What does this value imply?

A. Only once in a hundred 10-day periods will losses exceed this VaR amount ($1.7486 million).
B. We are 99% confident that we can lose more than $1.7486 million of our two-asset portfolio in the next 10 days.
C. We are 99% confident that we will not lose less than $1.7486 million of our two-asset portfolio in the next year.
D. Only once every 10,000 days will this VaR amount ($1.7486 million) be exceeded.

Answer: A
Step by step

1. A 99% VaR over a 10-day horizon is the loss exceeded with probability 1% over a 10-day period.
2. 1% of 10-day periods = 1 in 100 periods.
3. 1 in 100 ten-day periods = about 1 in 1,000 trading days, not 1 in 10,000.

Why A is correct: VaR at 99% is a loss threshold that is exceeded in only 1% of periods of the stated horizon.

Why the others are wrong

- **B:** Reverses the statement: we are 99% confident losses will NOT exceed VaR, not that we can lose more than it.
- **C:** Uses the wrong horizon (a year instead of 10 days) and treats VaR as a minimum loss.
- **D:** Mis-converts frequency: 100 ten-day periods is 1,000 days, not 10,000.

Q435

AnalystPrep Q.2639

For a portfolio having long positions in two assets, which value of correlation between the assets will yield the highest value for VaR?

A. 1
B. 0.5
C. -0.5
D. 0

Answer: A

Step by step

1. Portfolio variance = $w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\rho w_1 w_2 \sigma_1 \sigma_2$.
2. With both weights positive, variance is increasing in ρ, so it is maximised at ρ = 1.
3. VaR = α σP √t is increasing in σP, so VaR is highest at ρ = 1.

Why A is correct: With long-long positions the covariance term is positive and grows with ρ; at ρ = 1 there is no diversification and VaR equals the sum of the stand-alone VaRs.

Why the others are wrong

- **B:** ρ = 0.5 still gives some diversification benefit, so VaR is below the ρ = 1 value.
- **C:** Negative correlation acts as a partial hedge and gives a lower VaR; this would be the answer only if the question asked for the lowest.
- **D:** Zero correlation drops the covariance term entirely, giving VaR below the ρ = 1 case.

Q436

BT notes P2.T5.24.1.3 · BT answers P2.T5.24.1.3-641

REBUILT — Merged two near-identical items; replaced a giveaway/mis-worded option set (including the 'not lose anything less than' wording) with distractors from realistic calculation errors, and fixed the SD of asset 1 at 1%.

A risk manager is assessing potential losses over a 5-day period for a two-asset portfolio with the following data.

Asset	Invested amount	Daily standard deviation of returns
Asset 1	$60 million	1%
Asset 2	$7 million	4%

The correlation between the two assets is 0.6. Using the variance-covariance method with a 99% z-value of 2.326, what is the 5-day 99% VaR?

A. $1.86 million
B. $3.44 million
C. $4.16 million
D. $4.58 million

Answer: C

Step by step

1. Position volatilities: 60 × 0.01 = 0.60; 7 × 0.04 = 0.28 ($ millions).
2. Variance = $0.60^2 + 0.28^2 + 2(0.6)(0.60)(0.28) = 0.36 + 0.0784 + 0.2016 = 0.64$, so σP = $0.80 million daily.
3. 1-day 99% VaR = 2.326 × 0.80 = $1.861 million.
4. 5-day VaR = 1.861 × √5 = $4.16 million.

Why C is correct: Portfolio σ uses the full covariance matrix including the 0.6 correlation, and the horizon is scaled by the square root of 5.

Why the others are wrong

- **A:** This is the 1-day VaR; it omits the √5 scaling to 5 days.
- **B:** Sets the correlation to zero (σP = 0.662), understating risk by ignoring the positive covariance.

- **D:** Assumes perfect correlation (σP = 0.60 + 0.28 = 0.88), simply summing stand-alone volatilities and overstating VaR.

Q437

Schweser Module Quiz 8.2 Q2

REBUILT — Kept data and correct answer; replaced distractors that matched no identifiable error with realistic mistakes.

A financial institution has a two-asset portfolio with $7 million in asset A and $5 million in asset B. The correlation between the assets is 0.4, and the daily standard deviations of returns for A and B are 2% and 1%, respectively. What is the 10-day VaR of this portfolio at a 99% confidence level (α = 2.33)?

A. $0.388 million
B. $1.163 million
C. $1.226 million
D. $1.400 million

Answer: C

Step by step

1. Position volatilities: 7 × 0.02 = 0.14; 5 × 0.01 = 0.05 ($ millions).
2. Variance = $0.14^2 + 0.05^2 + 2(0.4)(0.14)(0.05) = 0.0196 + 0.0025 + 0.0056 = 0.0277$.
3. σP = $\sqrt{0.0277}$ = $0.1664 million per day.
4. 10-day VaR = 2.33 × 0.1664 × $\sqrt{10}$ = $1.226 million.

Why C is correct: The covariance term $2\rho\sigma_1\sigma_2$ adds 0.0056 to variance; scaling by α and $\sqrt{10}$ gives the 10-day 99% VaR.

Why the others are wrong

- **A:** This is the 1-day VaR (2.33 × 0.1664); the $\sqrt{10}$ horizon scaling was omitted.
- **B:** Forgot the factor of 2 on the covariance term (variance 0.0249), understating σP.
- **D:** Assumed perfect correlation (σP = 0.14 + 0.05 = 0.19), i.e. the undiversified sum of stand-alone VaRs.

Q438

AnalystPrep Q.2466

Consider the following expressions for a portfolio consisting of two assets:
I. VaR_p < VaR_1 + VaR_2
II. VaR_p = VaR_1 + VaR_2
Which of the following statements is correct?

A. When the assets are perfectly correlated, expression I is correct.
B. When the assets are perfectly correlated, expression II is correct.
C. When the assets are perfectly correlated, both expressions are correct.
D. When the assets are perfectly correlated, neither expression is correct.

Answer: B

Step by step

1. $\text{VaR}_p = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. At ρ = 1 this is $\sqrt{(\text{VaR}_1 + \text{VaR}_2)^2} = \text{VaR}_1 + \text{VaR}_2$.
3. So only expression II holds; I (strict inequality) needs ρ < 1.

Why B is correct: At ρ = 1 there is no diversification benefit, so portfolio VaR equals the sum of the stand-alone VaRs.

Why the others are wrong

- **A:** Expression I describes the diversified case (ρ < 1), not perfect correlation.
- **C:** The expressions are mutually exclusive (strict < versus =), so both cannot hold.
- **D:** Expression II does hold at ρ = 1.

Q439

AnalystPrep Q.2646 · AnalystPrep Q.3041

REBUILT — Merged two items testing the same fact (perfect correlation gives undiversified VaR); reworded and replaced weak distractors.

A risk manager computes portfolio VaR with the variance-covariance method. The portfolio VaR equals the simple sum of the individual position VaRs (the undiversified VaR) when the returns of the positions are:

A. Positively correlated, with a correlation coefficient below 1

B. Perfectly positively correlated
C. Uncorrelated
D. Perfectly negatively correlated

Answer: B
Step by step
1. For two positions, $\text{VaR}_P = \sqrt{\text{VaR}_1^2 + \text{VaR}_2^2 + 2\rho\,\text{VaR}_1\text{VaR}_2}$.
2. Set $\rho = 1$: $\text{VaR}_P = \sqrt{(\text{VaR}_1 + \text{VaR}_2)^2} = \text{VaR}_1 + \text{VaR}_2$.
3. Any $\rho < 1$ gives $\text{VaR}_P < \text{VaR}_1 + \text{VaR}_2$, so only $\rho = 1$ gives the undiversified VaR.

Why B is correct: At $\rho = 1$ the cross term is at its maximum and the square root collapses to the sum of the VaRs, so there is no diversification benefit.
Why the others are wrong
- **A:** Positive correlation below 1 still leaves VaR_P strictly below the sum; the error is treating any positive correlation as 'no diversification'.
- **C:** At $\rho = 0$ the cross term vanishes and VaR is the root of the sum of squares, well below the sum.
- **D:** At $\rho = -1$ VaR is the absolute difference of the VaRs, the minimum possible, not the maximum.

Q440 AnalystPrep Q.2648

REBUILT — Original gave 1.165 as the uncorrelated result, so rounding to 1.16 was ambiguous; changed the inputs, stated z, and rebuilt distractors.

A USD 20 million portfolio is invested in assets X and Y in the ratio 60:40. The return volatilities over the VaR horizon are 4% for X and 5% for Y. Using a 99% confidence level with z = 2.33, what are the portfolio VaRs if the assets are (i) perfectly correlated and (ii) uncorrelated?

A. Perfectly correlated: USD 2.05 million; Uncorrelated: USD 1.46 million
B. Perfectly correlated: USD 1.46 million; Uncorrelated: USD 2.05 million
C. Perfectly correlated: USD 2.05 million; Uncorrelated: USD 1.78 million
D. Perfectly correlated: USD 2.05 million; Uncorrelated: USD 0.19 million

Answer: A
Step by step
1. Positions: X = 12m, Y = 8m.
2. $\text{VaR}_X = 12 \times 0.04 \times 2.33 = 1.1184$; $\text{VaR}_Y = 8 \times 0.05 \times 2.33 = 0.9320$.
3. $\rho = 1$: 1.1184 + 0.9320 = 2.0504, about USD 2.05m.
4. $\rho = 0$: $\sqrt{1.1184^2 + 0.9320^2} = \sqrt{2.1195} = 1.4558$, about USD 1.46m.

Why A is correct: Perfect correlation adds the stand-alone VaRs; zero correlation adds them in quadrature, giving a smaller figure.
Why the others are wrong
- **B:** Swaps the two cases; diversified (uncorrelated) VaR cannot exceed the undiversified sum.
- **C:** Uses $\rho = 0.5$ (1.78) for the uncorrelated case.
- **D:** Uses $\rho = -1$ (absolute difference, 0.19) for the uncorrelated case.

Q441 AnalystPrep Q.2746

The VaR of a portfolio will be maximized when the pairwise correlations between the returns of the underlying assets are:

A. Perfectly positive
B. Perfectly negative
C. Uncorrelated
D. Either positive or negative but not perfectly negative or positive

Answer: A
Step by step
1. Two-asset VaR is increasing in ρ through the cross term $2\rho\,\text{VaR}_1\text{VaR}_2$.
2. The maximum is at $\rho = +1$: $\text{VaR}_P = \text{VaR}_1 + \text{VaR}_2$.

Why A is correct: VaR rises with correlation and is capped at the sum of the stand-alone VaRs when $\rho = 1$.
Why the others are wrong

- **B:** $\rho = -1$ gives the minimum VaR, the reverse of the relationship.
- **C:** $\rho = 0$ is an intermediate value (root of the sum of squares).
- **D:** Imperfect correlations always yield VaR below the undiversified sum, so they cannot be the maximum.

Q442 AnalystPrep Q.3146

REBUILT — Original distractors (USD 100m, 300m, 20m) were giveaways; replaced with results of realistic mistakes.

A portfolio consists of two positions with stand-alone VaRs of USD 10 million and USD 40 million. The returns of the two positions are uncorrelated. The portfolio VaR is closest to:

A. USD 30.00 million
B. USD 41.23 million
C. USD 50.00 million
D. USD 7.07 million

Answer: B

Step by step

1. $\rho = 0$ so $\text{VaR}_P = \sqrt{10^2 + 40^2}$.
2. $= \sqrt{100 + 1600} = \sqrt{1700} = 41.23$ million.

Why B is correct: Uncorrelated VaRs combine in quadrature.

Why the others are wrong

- **A:** Absolute difference 40 - 10, which is the $\rho = -1$ case.
- **C:** Simple sum, which is the $\rho = +1$ case.
- **D:** Took the square root of the sum of the VaRs, $\sqrt{50}$, instead of the root of the sum of squares.

Q443 AnalystPrep Q.3166

REBUILT — Replaced the implausible 'decrease by 3.4 million' option with a realistic mistake and tidied the stem.

A portfolio manager holds USD 60 million in BeiXin Shipping and USD 40 million in TongChi Logistics, with 95% one-day VaRs of USD 6.3 million and USD 4.1 million respectively. Returns are normally distributed and the individual VaRs stay unchanged. What is the effect on the 95% one-day portfolio VaR if the correlation between the two shares rises from zero to one?

A. The portfolio VaR will remain unchanged
B. The portfolio VaR will increase by USD 2.88 million
C. The portfolio VaR will decrease by USD 2.88 million
D. The portfolio VaR will increase by USD 4.10 million

Answer: B

Step by step

1. $\rho = 0$: $\sqrt{6.3^2 + 4.1^2} = \sqrt{56.50} = 7.517$.
2. $\rho = 1$: 6.3 + 4.1 = 10.400.
3. Change = 10.400 - 7.517 = +2.883, about USD 2.88m.

Why B is correct: Raising correlation from 0 to 1 removes the diversification benefit, so VaR rises from the quadrature value to the simple sum.

Why the others are wrong

- **A:** Ignores that the correlation term changes VaR when individual VaRs are fixed.
- **C:** Right size, wrong sign: higher correlation increases VaR.
- **D:** Compared the new VaR 10.4 with the larger single VaR 6.3 (giving 4.1) instead of the old portfolio VaR 7.517.

Q444 AnalystPrep Q.4712

REBUILT — Replaced odd distractors (1.198, 2.504) with results of realistic mistakes.

A portfolio is made up of two uncorrelated positions. The VaR of position one is USD 3 million and the VaR of position two is USD 2 million. The portfolio VaR is:

A. USD 13.000 million
B. USD 5.000 million
C. USD 3.606 million
D. USD 1.000 million

Answer: C

Step by step

1. $\rho = 0$: $\text{VaR}_P = \sqrt{3^2 + 2^2}$.
2. $= \sqrt{13} = 3.606$ million.

Why C is correct: Uncorrelated VaRs add in quadrature.

Why the others are wrong

- **A:** Added the squares and forgot the square root (13).
- **B:** Simple sum, valid only for $\rho = 1$.
- **D:** Difference, valid only for $\rho = -1$.

Q445

Mock 4 Q54

REBUILT — Replaced the arbitrary distractor 5.43 (2.33 squared) with the perfect-correlation VaR and stated z = 2.33.

A risk analyst estimates the VaR of an equally weighted two-asset portfolio to show the impact of correlation. Each asset has daily return volatility of 2%, the portfolio value is USD 20 million, and the 10-day 99% VaR is USD 2.33 million (use z = 2.33). If the correlation between the two assets doubles, the new 10-day 99% portfolio VaR is closest to:

A. USD 1.16 million
B. USD 2.55 million
C. USD 4.66 million
D. USD 2.95 million

Answer: B

Step by step

1. Equal weights and equal vols: $\sigma_P = \sigma\sqrt{0.5(1+\rho)}$.
2. Back out σ_P: $2.33/(20 \times 2.33 \times \sqrt{10}) = 0.015811$.
3. $0.02^2 \cdot 0.5(1+\rho) = 0.015811^2 = 0.00025$ gives $\rho = 0.25$.
4. Doubled: $\rho = 0.50$, $\sigma_P = 0.02\sqrt{0.75} = 0.017321$.
5. VaR $= 20 \times 0.017321 \times 2.33 \times \sqrt{10} = 2.552$ million.

Why B is correct: VaR scales with $\sqrt{0.5(1+\rho)}$, so doubling ρ from 0.25 to 0.50 raises VaR by a factor of $\sqrt{0.75/0.625} = 1.095$.

Why the others are wrong

- **A:** Halved the VaR, as if higher correlation reduced risk.
- **C:** Doubled the VaR, treating VaR as linear in correlation.
- **D:** This is the $\rho = 1$ maximum (2.95), reached only if correlation rose to 1, not doubled to 0.5.

Q446

Mock 1 Q42

A manager has a portfolio with only one position: a USD 50 million investment in Asset P, with return volatility of 7%. She is considering adding a USD 50 million position in either R or S, and wants to limit portfolio VaR to USD 13.5 million at the 95% confidence level (z = 1.645). R has volatility 11% and correlation with P of 0.6. S has volatility 14% and correlation with P of 0.2. Which addition keeps her within the risk budget?

A. Adding R only
B. Adding S only
C. Adding either R or S
D. Adding neither R nor S

Answer: A

Step by step

1. Equal weights $w = 0.5$: $\sigma_P^2 = 0.25\sigma_1^2 + 0.25\sigma_2^2 + 0.5\rho\sigma_1\sigma_2$.
2. P+R: 0.001225 + 0.003025 + 0.002310 = 0.006560; $\sigma_P = 8.099\%$; VaR = 100 x 0.08099 x 1.645 = USD 13.32m (within 13.5m).
3. P+S: 0.001225 + 0.004900 + 0.000980 = 0.007105; $\sigma_P = 8.429\%$; VaR = USD 13.87m (breach).

Why A is correct: The lower correlation of S does not offset its much higher volatility, so only R keeps VaR under USD 13.5m.

Why the others are wrong

- **B:** Assumes the lower correlation makes S the better diversifier, ignoring its higher volatility.

- **C:** S breaches the limit at USD 13.87m.
- **D:** R passes at USD 13.32m.

LO 8.f

Explain the role of correlation risk in market risk and credit risk.

Q447 GARP book Ch.8 Q8.17

How is correlation risk related to market risk and credit risk?

Answer: see model answer — Market risk (interest rate, currency, equity, commodity risk) is measured with VaR and expected shortfall, both of which depend on the covariance matrix of the assets; correlation risk is the risk that the correlations in that matrix change over time, so VaR/ES (especially tail risk) can be badly misestimated, e.g. correlations rose sharply in the 2007-2009 crisis when diversification was needed most. In credit risk, default correlation measures how likely defaults are to occur together: higher default correlation (and a higher correlation between a reference asset and a CDS seller) raises the probability of simultaneous or total losses, while lower default correlation means greater diversification of credit risk. Migration and default risk of a credit portfolio therefore cannot be assessed without correlation.

Step by step

1. Market risk: VaR and ES use the covariance matrix, so correlation is a direct input.
2. Correlation risk = changes in correlations over time change VaR/ES, especially in crises when correlations rise.
3. Credit risk: default correlation drives the chance that defaults cluster; higher correlation means less diversification and larger joint-loss risk.
4. Lower default correlation implies greater credit diversification.

Why this is correct: Correlation enters market risk through the covariance matrix behind VaR/ES and enters credit risk through default correlation, which governs joint defaults.

Q448 AnalystPrep Q.1554

Correlation risk is an important part of market risk, which is typically measured with the help of Value at Risk (VaR) concepts. Market risk is also measured using expected shortfall (ES), which characterizes tail risk. What is the purpose of expected shortfall?

A. To measure market risk for risky events, typically for the worst 0.1%, 1%, or 5% of past scenarios.
B. It measures market risk for extreme events, typically for the worst 0.01%, 0.1%, or 1% of possible future scenarios.
C. It measures market risk for extreme events, typically for the worst 0.1%, 1%, or 5% of possible future scenarios.
D. It measures market risk for risky events, typically for the worst 0.01%, 0.1%, or 1% of past scenarios.

Answer: C

Step by step

1. ES measures market risk in the extreme tail (tail risk), i.e. the average loss beyond the VaR cutoff.
2. It is forward-looking: it concerns possible future scenarios, not only past ones.
3. Typical tail levels are the worst 0.1%, 1% and 5%.
4. Only C combines 'extreme events', 'future scenarios' and 0.1%/1%/5%.

Why C is correct: ES quantifies the loss in the tail of possible future outcomes at confidence levels typically 99.9%, 99% and 95%.

Why the others are wrong

- **A:** Refers to past scenarios and 'risky' rather than extreme events; ES looks at possible future scenarios.
- **B:** Uses forward-looking scenarios but cites tail levels of 0.01%, 0.1%, 1% that are too extreme for the usual 0.1%/1%/5% levels.
- **D:** Combines both errors: past scenarios and the 0.01%/0.1%/1% levels.

Q449 AnalystPrep Q.1555

The higher the default correlation between assets, the higher the probability that investors will lose all of their investment if a single asset's price declines. Lenders can lower default risk by diversifying their portfolio. What is the best policy for lending companies to reduce default correlation risk?

A. Intersector default correlations are typically higher than intrasector default correlations, so lenders should create a sector-diversified loan portfolio.

B. Intersector default correlations are typically lower than intrasector default correlations, so lenders should create a sector-diversified loan portfolio.

C. Intrasector default correlations are typically higher than intersector default correlations, so lenders should specialize in a single-sector loan portfolio.

D. Intersector default correlations are typically higher than intrasector default correlations, so lenders should specialize in a single-sector loan portfolio.

Answer: B

Step by step

1. Historical data show default correlations are higher within an industry than across industries (systematic factors dominate).
2. If one borrower in a sector defaults, others in the same sector are more likely to default.
3. Lower default correlation means more diversified credit risk, so lenders should spread loans across sectors.

Why B is correct: Intersector default correlation is lower than intrasector, so sector diversification lowers default correlation and joint-loss risk.

Why the others are wrong

- **A:** Reverses the empirical relationship (claims intersector correlation is higher), so the diversification rationale is invalid.
- **C:** States the relationship correctly but draws the wrong conclusion: a single-sector book concentrates highly correlated exposures.
- **D:** Reverses the relationship and also recommends concentration.

Q450 AnalystPrep Q.2380

Country A is a developed economy with a vibrant financial sector that experienced a major financial crisis about 15 years ago and has since recovered, registering consistently positive growth. A survey found that the country's fund managers hold well-diversified portfolios across all sectors of the economy. They generally use the last 10 years of data and mean-variance utility to estimate correlations between assets, believing recent data is the best estimator of correlation. Suppose the country now faces a major economic crisis. Which statement is most appropriate?

A. The fund managers would not be affected as their portfolios are well diversified.

B. Portfolio diversification would most likely fail and lead to losses.

C. The returns from the fund would most likely beat other similar funds.

D. None of the above.

Answer: B

Step by step

1. The 10-year window contains no crisis, so estimated correlations are low and reflect normal conditions.
2. In a crisis correlations rise sharply (e.g. US equity correlations rose from about 27% to over 50% in 2008-09), so assets fall together.
3. Diversification benefits disappear exactly when needed, causing losses larger than the model implies.

Why B is correct: Correlations are not stable; they spike in systemic crises, so diversification based on calm-period correlations fails.

Why the others are wrong

- **A:** Assumes diversification protects in all states; ignores that correlations rise in a crisis.
- **C:** No basis; all similar funds use the same method and would suffer, so no outperformance is implied.
- **D:** Incorrect because B is correct.

Q451 Schweser Module Quiz 8.3 Q2

The relationship of correlation risk to credit risk is an important area of concern for risk managers. Which of the following statements regarding default probabilities and default correlations is incorrect?

A. Creditors benefit by diversifying exposure across industries to lower the default correlations of debtors.
B. The default term structure increases with time to maturity for most investment-grade bonds.
C. The probability of default is higher in the long-term time horizon for non-investment-grade bonds.
D. Changes in the concentration ratio are directly related to changes in default correlations.

Answer: C

Step by step

1. A: default correlations are higher within industries, so cross-industry diversification lowers them - true.
2. B: investment-grade default term structure slopes upward slightly - true.
3. C: for non-investment-grade bonds default probability is highest in the immediate horizon and falls if the issuer survives - so C is false.
4. D: concentration ratio and default correlation move together - true.

Why C is correct: Distressed issuers face their highest default risk in the near term; the term structure for junk bonds is downward sloping, not rising.

Why the others are wrong

- **A:** A true statement; the question asks for the incorrect one.
- **B:** A true statement: more time allows more adverse changes for investment-grade issuers.
- **D:** A true statement: a lower concentration ratio goes with lower default correlation.

Q452 AnalystPrep Q.2651

REBUILT — Answer was correct but two distractors (5.78%, 7.04%) were implausible; replaced with distractors from realistic formula mistakes.

A bank has issued loans to two companies, A and B. Their probabilities of default are 5% and 10%, respectively. Using the binomial default correlation model, what is the joint probability of default of the two companies if their default correlation is 0.6?

A. 3.92%
B. 4.74%
C. 4.42%
D. 0.76%

Answer: C

Step by step

1. Binomial default correlation: $\rho = \dfrac{P(A \cap B) - P_A P_B}{\sqrt{P_A(1-P_A)P_B(1-P_B)}}$, so $P(A \cap B) = \rho\sqrt{P_A(1-P_A)P_B(1-P_B)} + P_A P_B$.
2. Variance product: $0.05 \times 0.95 \times 0.10 \times 0.90 = 0.004275$; square root = 0.065383.
3. Covariance term: $0.6 \times 0.065383 = 0.039230$.
4. Independence term: $0.05 \times 0.10 = 0.005$.
5. Joint PD = 0.039230 + 0.005 = 0.044230, i.e. 4.42%.

Why C is correct: A positive default correlation adds a covariance term (rho times the product of the Bernoulli standard deviations) on top of the independent joint probability P_A*P_B.

Why the others are wrong

- **A:** 3.92% omits the independence term P_A*P_B = 0.5%; it is only the covariance component.
- **B:** 4.74% uses sqrt(P_A*P_B) instead of sqrt(P_A(1-P_A)P_B(1-P_B)) as the standard-deviation product.
- **D:** 0.76% multiplies rho by the variance product (0.004275) instead of its square root (0.6 x 0.004275 = 0.2565%), then adds 0.5%.

Q453 BT notes P2.T5.22.11.2

REBUILT — Answer was correct but distractors (zero, 0.30, 0.15) were giveaways; replaced with options from realistic formula mistakes.

Consider a loan portfolio of two loans extended to Company X and Company Y. Their default probabilities are PD(X) = 10.0% and PD(Y) = 4.0%, and the joint default probability is PD(X and Y) = 3.930%. Which is nearest to the implied (binomial) default correlation?

A. 0.35
B. 0.56

C. 0.60
D. 0.67

Answer: C
Step by step
1. $\rho = \dfrac{P(X \cap Y) - P_X P_Y}{\sqrt{P_X(1-P_X)}\sqrt{P_Y(1-P_Y)}}$.
2. Numerator: 0.03930 - 0.10*0.04 = 0.03930 - 0.00400 = 0.03530.
3. Denominator: sqrt(0.10*0.90) = 0.30000; sqrt(0.04*0.96) = 0.19596; product = 0.058788.
4. rho = 0.03530 / 0.058788 = 0.600.

Why C is correct: Default correlation is the covariance of the two default indicators (joint PD minus product of PDs) divided by the product of their Bernoulli standard deviations.
Why the others are wrong
- **A:** 0.35 divides the covariance by only 0.10 (PD of X) rather than the product of both standard deviations.
- **B:** 0.56 uses sqrt(PD_X*PD_Y) = 0.0632 in the denominator, ignoring the (1-PD) terms.
- **D:** 0.67 forgets to subtract PD_X*PD_Y: 0.0393/0.058788.

LO 8.g

Explain how correlation risk relates to systemic and concentration risk.

Q454 GARP book Ch.8 Q8.19

How can we measure the joint probability of occurrence of a binomial event such as default or no-default?

Answer: see model answer — Code default as 1 and no default as 0 (Bernoulli variables). For events A and B, $P(AB) = E(AB) = \rho_{AB}\,\sigma_A\sigma_B + P(A)P(B)$, where $\sigma_i = \sqrt{P(i)(1-P(i))}$ and ρ_{AB} is the default correlation. Setting $\rho = 0$ gives the independence case $P(A)P(B)$; $\rho = 1$ with equal default probabilities gives $P(AB) = P(A)$; higher correlation raises the joint default probability.
Step by step
1. Joint probability equals the expected value of the product of the two indicators: $P(AB) = E(AB)$.
2. From the covariance definition, $E(AB) = \mathrm{Cov}(A,B) + E(A)E(B) = \rho\,\sigma_A\sigma_B + P(A)P(B)$.
3. For a Bernoulli variable, $\sigma = \sqrt{P(1-P)}$.

Why this is correct: The covariance identity turns the correlation and the marginal default probabilities into the joint default probability.

Q455 GARP book Ch.8 Q8.20

Can two binomial events be negatively correlated but have a positive probability of joint default?

Answer: see model answer — Yes. $P(AB) = \rho\sigma_A\sigma_B + P(A)P(B)$ is positive whenever $\rho > -P(A)P(B)/(\sigma_A\sigma_B)$. For example, with P(A)=P(B)=5%, $\sigma = 0.2179$ and the threshold is about -0.0526, so a correlation of -0.03 still gives $P(AB) = -0.03(0.0475) + 0.0025 \approx 0.0011 > 0$. Only a sufficiently negative correlation (below the threshold, and bounded by the Frechet limits) drives the joint probability to zero.
Step by step
1. Joint PD: $P(AB) = \rho\sigma_A\sigma_B + P(A)P(B)$.
2. With P(A)=P(B)=5%: $\sigma_A\sigma_B = 0.05(0.95) = 0.0475$ and $P(A)P(B) = 0.0025$.
3. P(AB)>0 requires $\rho > -0.0025/0.0475 = -0.0526$.
4. E.g. $\rho = -0.03$: $P(AB) = -0.001425 + 0.0025 = 0.001075 > 0$.

Why this is correct: The product of marginals P(A)P(B) is positive, so mild negative correlation only partly offsets it.

Q456 BT notes P2.T5.501.3 · BT answers P2.T5.501.3

Commercial bank C has lent to company A and company B. Both A and B have a 9.0% default probability. If their pairwise default correlation is 0.280, which is nearest to the joint probability of default? (Use the binomial correlation measure of Lucas (1995).)

A. 0.81%
B. 3.10%
C. 6.20%
D. 8.72%

Answer: B

Step by step

1. Default indicators are Bernoulli: $\sigma = \sqrt{0.09 \times 0.91} = 0.2862$.
2. $P(AB) = \rho\, \sigma_A \sigma_B + P(A)P(B)$.
3. $= 0.280 \times 0.09 \times 0.91 + 0.09^2 = 0.022932 + 0.0081 = 0.031032$.
4. Joint default probability = 3.10%.

Why B is correct: Joint default probability is the covariance term $\rho\sigma_A\sigma_B$ added to the independence value $P(A)P(B)$.

Why the others are wrong

- **A:** 0.81% = 0.09^2 is the independence case (correlation 0); it ignores the positive correlation.
- **C:** 6.20% is roughly double the right answer (2 x 3.10%), e.g. adding the covariance term for each company; no such doubling exists in the formula.
- **D:** 8.72% is near the 9% marginal PD, which is the joint PD only when correlation is about 1; it would require a correlation of about 0.97, not 0.28.

Q457 GARP book Ch.8 Q8.12

Why do correlations typically increase in a systemic market crash?

Answer: see model answer — In a systemic crash, a common macro/liquidity shock dominates asset-specific factors, so almost all assets fall together and pairwise correlations rise toward 1. Forced selling, margin calls and flight to quality make investors liquidate across asset classes at once. Empirically, average US stock correlation rose from about 27% to over 50% between August 2008 and March 2009, and correlations with bonds and international equities also rose. Diversification therefore fails exactly when it is needed most.

Step by step

1. Systemic event: market-wide factors drive returns, idiosyncratic factors matter less.
2. Contagion and forced deleveraging cause simultaneous selling of many assets.
3. Result: correlations spike (27% to over 50% in 2008-09), so diversification benefits shrink and correlation risk is greatest in the tail.

Why this is correct: Common systematic drivers and herding in a crisis make assets move together, raising correlation and removing diversification benefit.

Q458 Mock 1 Q3

Two prime counterparties each have a marginal default probability of 5.0%. The risk committee instructs that a joint default correlation of 0.5 be applied. Which of the following is closest to their joint probability of default?

A. 2.13%.
B. 2.38%.
C. 2.63%.
D. 3.61%.

Answer: C

Step by step

1. $\sigma^2 = 0.05 \times 0.95 = 0.0475$ ($\sigma = 0.2179$).
2. Correlation term = 0.5 × 0.0475 = 0.02375.
3. Add $P_X P_Y = 0.0025$.
4. Joint PD = 0.02625 ≈ 2.63%.

Why C is correct: Joint PD = $\rho\sigma_X\sigma_Y + P_X P_Y$ adds the independent component to the correlation-driven component.

Why the others are wrong

- **A:** Subtracted $P_X P_Y$ (0.02375 − 0.0025 = 2.125%) instead of adding.
- **B:** Omitted $P_X P_Y$ and reported only the correlation term, 2.375%.
- **D:** Used $\sqrt{0.5} = 0.7071$ instead of 0.5: 0.7071 × 0.0475 + 0.0025 = 3.61%.

Q459

GARP book Ch.8 Q8.18

How is correlation risk related to systemic risk and concentration risk?

Answer: see model answer — Systemic risk (collapse of the financial system) is closely linked to correlation risk: in a crisis correlations rise sharply, so correlation risk is most severe when systemic risk is high. Concentration risk is measured by the concentration ratio (1/n for n equal loans); changes in the ratio are directly related to default correlation. A lower concentration ratio and a lower default correlation both reduce the joint probability of default (the worst-case loss), whereas with default correlation of 1 a lower ratio gives no benefit.

Step by step

1. Systemic risk: correlations jump in a crisis, so diversification fails (correlation risk is amplified).
2. Concentration risk: ratio = 1/n for equal loans; fewer, larger exposures mean a higher ratio and more concentration.
3. Joint PD = $\rho\sqrt{P_X(1-P_X)P_Y(1-P_Y)} + P_X P_Y$, so joint default falls as either the concentration ratio or ρ falls.

Why this is correct: Correlation drives both: it spikes in systemic crises, and it determines how much a lower concentration ratio actually reduces joint default risk.

Q460

AnalystPrep Q.1556

REBUILT — Original options (partially/highly dependent/independent) were vague and gave away the answer; rebuilt around the reading's 27% to 50% correlation evidence with four substantive statements.

Systemic risk is the risk of a collapse of the entire financial system. From August 2008 to March 2009, the average pairwise correlation among US stocks rose from about 27% to over 50%. Which of the following best describes the relationship between systemic risk and correlation risk?

A. They are unrelated; the rise in correlation reflected company-specific factors, so diversification benefits were unchanged.

B. In a systemic crisis correlations rise sharply, so correlation risk is greatest when diversification is most needed and diversification benefits shrink.

C. A systemic crisis lowers correlations as investors rotate across asset classes, which increases diversification benefits.

D. Correlations rose only among US stocks; correlations with bonds and international equities were unaffected.

Answer: B

Step by step

1. Systemic crisis: market-wide factors dominate asset-specific ones.
2. Observed: average stock correlation rose from 27% to over 50%, i.e. assets moved together.
3. Higher correlations mean less diversification exactly during the crisis, so correlation risk is most severe then.
4. The reading adds that correlations of equities with bonds and international equities also rose, making the problem worse.

Why B is correct: Systemic stress pushes correlations up across assets, so correlation risk is highest, and diversification weakest, in a systemic crisis.

Why the others are wrong

- **A:** Treats the increase as idiosyncratic; a market-wide rise in correlation is driven by systematic factors, not company-specific ones.
- **C:** Reverses the direction: correlations rise, not fall, in a crisis, reducing diversification.
- **D:** The reading states correlations of US equities with bonds and international equities also increased, making the effect even more severe.

Q461 AnalystPrep Q.1557

Concentration risk is the financial loss arising from exposure to a specific group of counterparties, and it is quantified with the concentration ratio. What is the correct rule of thumb for a creditor regarding the concentration ratio?

A. The lower the concentration ratio, the higher the diversification and the lower the overall portfolio default risk to the creditor.
B. The higher the concentration ratio, the higher the diversification and the lower the default risk to the creditor.
C. The lower the concentration ratio, the lower the diversification and the higher the default risk to the creditor.
D. The higher the concentration ratio, the higher the diversification and the higher the default risk to the creditor.

Answer: A

Step by step

1. For n equal-size loans to different borrowers, concentration ratio = 1/n.
2. More loans (larger n) gives a smaller ratio, e.g. 1 loan: 1.0; 100 loans: 0.01.
3. A smaller ratio means exposure is spread over more borrowers: greater diversification and lower portfolio default risk.

Why A is correct: The concentration ratio falls as exposure is spread across more counterparties, which means more diversification and lower portfolio default risk.

Why the others are wrong

- **B:** Links a high ratio to high diversification; a high ratio means concentrated exposure and less diversification.
- **C:** Links a low ratio to low diversification; this reverses the relationship.
- **D:** Links a high ratio to high diversification, which is backwards.

Q462 AnalystPrep Q.1558

MLA Commercial Bank lends Rs.10,000 to a single company X with a 10% default probability, so its concentration ratio is 1 and expected loss is Rs.1,000. If the bank instead lends the same Rs.10,000 equally among three companies, each with a 10% default probability, what is the bank's concentration ratio?

A. 0.5
B. 1
C. 0.3
D. 0.333

Answer: D

Step by step

1. Three equal loans: n = 3.
2. Concentration ratio = 1/n = 1/3 = 0.333.

Why D is correct: With equal-size loans to n different borrowers the concentration ratio is 1/n.

Why the others are wrong

- **A:** 0.5 = 1/2, the ratio for two loans.
- **B:** 1 is the ratio for a single loan, ignoring the diversification across three companies.
- **C:** 0.3 is an imprecise approximation; the ratio is exactly 1/3 = 0.333.

Q463 AnalystPrep Q.2653

What is the most likely effect of a decreasing concentration ratio on the joint probability of default, given that the default correlation between the loans is less than 1?

A. It will increase.
B. It is not possible to determine.
C. It will decrease.
D. The concentration ratio does not affect the joint probability of default.

Answer: C

Step by step

1. A lower concentration ratio means more, smaller loans (more diversification).
2. Joint PD = $\rho\sigma_X\sigma_Y + P_X P_Y$; with $\rho < 1$ defaults are not perfectly linked.
3. The probability that all loans default together therefore falls as the ratio falls.

Why C is correct: When default correlation is below 1, spreading exposure over more borrowers lowers the probability that they all default together.

Why the others are wrong

- **A:** Confuses diversification with added risk; more borrowers lowers, not raises, joint default probability.
- **B:** The reading states the direction explicitly (lower ratio, lower joint PD).
- **D:** Changes in the concentration ratio are directly related to default correlation and joint PD; it has no effect only when $\rho = 1$.

Q464

Mock 1 Q1

REBUILT — Source exhibit was reconstructed; options rounded the ratio to 0.045 and the 'no benefit' claim was loosely worded. Options restated to 0.0455 / 0.0714 and wording tightened.

Bank Q's loan book consists of bullet loans of identical size, each to a separate counterparty, in two sub-portfolios:

Sub-portfolio	Number of loans	Size per loan (USD m)	Marginal PD
Prime	14	5	5.0%
Subprime	8	5	12.0%

Which of the following is correct regarding the concentration ratio of the combined portfolio and the benefit of reducing it?

A. The ratio is 0.0455; if the pairwise default correlation across all counterparties were 1, reducing the ratio would give no reduction in joint default risk.

B. The ratio is 0.0455; reducing the ratio lowers the worst-case joint default probability irrespective of the level of default correlation.

C. The ratio is 0.0714; if the pairwise default correlation across all counterparties were 1, reducing the ratio would give no reduction in joint default risk.

D. The ratio is 0.0714; reducing the ratio lowers the worst-case joint default probability irrespective of the level of default correlation.

Answer: A

Step by step

1. Combined loans n = 14 + 8 = 22, all of equal size (USD 5m).
2. Concentration ratio = 1/22 = 0.0455.
3. Joint PD = $\rho\sigma_X\sigma_Y + P_X P_Y$. At $\rho = 1$ joint default behaves like a single exposure, so adding borrowers gives no benefit.
4. The benefit of a lower ratio therefore depends on $\rho < 1$.

Why A is correct: The ratio is 1/22 for 22 equal loans, and diversification helps only when default correlation is below 1.

Why the others are wrong

- **B:** Right ratio but overextends the rule; at $\rho = 1$ all counterparties default together and no diversification occurs.
- **C:** Uses 1/14 = 0.0714 (prime loans only) rather than all 22 loans.
- **D:** Uses 1/14 (prime only) and also ignores that the benefit vanishes at $\rho = 1$.

Q465

Schweser Module Quiz 8.3 Q1

REBUILT — Source options A/C/D did not correspond to identifiable errors; replaced with distractors from realistic mistakes (omit PX*PY, rho=0, rho=1).

A creditor makes a $4 million loan to Company X and a $4 million loan to Company Y. Each company has a 7% default probability, and the default correlation coefficient is 0.6. The expected loss for this creditor under the worst-case scenario (both default), assuming LGD of 100%, is closest to:

A. $39,200.

B. $312,480.

C. $351,680.

D. $560,000.

Answer: C

Step by step

1. $\sigma_X\sigma_Y = 0.07 \times 0.93 = 0.0651$.
2. Joint PD = 0.6 × 0.0651 + 0.07 × 0.07 = 0.03906 + 0.0049 = 0.04396.
3. Worst-case exposure = $8,000,000.

4. EL = 0.04396 × 8,000,000 = $351,680.

Why C is correct: The worst-case loss occurs when both default; weight the total $8m exposure by the joint default probability including the correlation term.

Why the others are wrong

- **A:** Set the correlation to 0: joint PD = 0.0049, EL = $39,200.
- **B:** Omitted the $P_X P_Y$ term: 0.03906 × 8m = $312,480.
- **D:** Assumed $\rho = 1$ (joint PD = 7%): 0.07 × 8m = $560,000.

Q466

Mock 1 Q2

REBUILT — Distractor B (2.38%) corresponded to a different correlation than the stem; replaced with a variance-based error. Stem made self-contained.

Bank Q holds loans to prime counterparties, each with a marginal default probability of 5.0%; default is modelled as a binomial event. The risk system reports a binomial standard deviation of 0.2179 for each. Assuming a default correlation of 0 between two prime counterparties, which is closest to their joint probability of default?

A. 0.25%.
B. 4.75%.
C. 5.00%.
D. 9.75%.

Answer: A

Step by step

1. Joint PD = $\rho\sigma_X\sigma_Y + P_X P_Y$.
2. $\rho = 0$, so the correlation term is 0 (the 0.2179 is not needed).
3. Joint PD = 0.05 × 0.05 = 0.0025 = 0.25%.

Why A is correct: With zero correlation the default events are independent, so joint PD is the product of the marginal PDs.

Why the others are wrong

- **B:** Used the variance 0.05 × 0.95 = 4.75% instead of the independent product.
- **C:** 5.00% is the marginal PD of a single counterparty (what $\rho = 1$ would give).
- **D:** 9.75% = 0.05 + 0.05 − 0.0025 is the probability that at least one defaults, not both.

Reading 9. Empirical Properties of Correlation: How Do Correlations Behave in the Real World?

Gunter Meissner, Correlation Risk Modeling and Management, 2nd Edition (Risk Books, 2019). Chapter 2.

LO	Learning objective	Qs
9.a	describe how equity correlations and correlation volatilities behave throughout various economic states.	7
9.b	calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation.	18
9.c	identify the best-fit distribution for equity, bond, and default correlations.	6

LO 9.a

Describe how equity correlations and correlation volatilities behave throughout various economic states.

Q467 GARP book Ch.9 Q9.1

In which state of the economy are equity correlations the highest?

Answer: see model answer — Equity correlations are highest in recessions. In Meissner's Dow study (1972-2017), average correlation was about 37.0% in recessions, 33.0% in normal periods and 27.5% in expansions. In bad times stocks fall together because macro factors dominate; in expansions valuations depend more on company and industry information.

Q468 GARP book Ch.9 Q9.2

In which state of the economy is equity correlation volatility high?

Answer: see model answer — Correlation volatility is high in every state (above 70%): about 80.5% in recessions, 83.0% in normal periods and 71.2% in expansions. It is highest in normal periods (not recessions), because investors are less certain about market direction then, and lowest in expansions.

Q469 GARP book Ch.9 Q9.3

REBUILT — Source stem referred to 'Questions 1 and 2'; restated the two findings so the item is self-contained.

Equity correlations are highest in recessions, and equity correlation volatility is high in all economic states. What follows from these two findings for risk management?

Answer: see model answer — Correlations rise exactly when losses are large, so diversification benefits shrink in downturns and portfolio risk is underestimated if correlations are calibrated on average or expansionary data. Because correlation volatility is high, correlation estimates are themselves unstable and must be stress-tested. Stress scenarios and risk models should therefore use higher correlation levels (and allow for high correlation volatility) for bad economic times.

Q470 AnalystPrep Q.2379

REBUILT — Original stem was verbose and did not state that the sample contains no recession, and option D ('not paramount') was a giveaway; restated the sample and made all options plausible.

A fund manager in a developed economy estimates equity correlations for portfolio risk management using only the last 10 years of monthly returns. The country has been in continuous economic expansion over this entire window; the last recession occurred 15 years ago. The manager argues that recent data is the best estimator of future correlations. Which statement is most accurate?

A. The estimated correlations will understate the correlations that prevail in a recession, so the diversification benefit in stress is overstated.

B. The estimated correlations will overstate the correlations that prevail in a recession, so the diversification benefit in stress is understated.

C. Equity correlations are independent of the state of the economy, so the choice of window does not matter.

D. The window will overstate correlation volatility, because correlation volatility is highest in expansions.

Answer: A

Step by step

1. Reading: average correlation is about 27.5% in expansions, 33.0% in normal periods and 37.0% in recessions.
2. The 10-year window contains only expansionary data, so it reflects the low-correlation state.
3. A recession would see correlations near 37% or higher, so the estimate is too low for stress conditions.
4. Hence diversification is overstated exactly when it matters: statement A.

Why A is correct: Correlations rise in recessions, so a sample with no recession gives correlation estimates below stressed levels and overstates diversification.

Why the others are wrong

- **B:** Reverses the direction: expansion correlations are the lowest, not the highest.
- **C:** Correlations are state dependent (27.5% vs 37.0%), so the window matters.
- **D:** Correlation volatility is lowest in expansions (71.2%) and highest in normal periods (83.0%), so this misstates the fact and is not the problem.

Q471 BT Mock Q41 · BT notes P2.T5.22.12.2

Meissner studied equity correlations among the 30 Dow constituents over 534 months from January 1972 to July 2017, generating 480,600 correlation values. In regard to his empirical findings on equity correlations, each of the following statements is TRUE except which is false?

A. Mean reversion should be included in equity correlation models.

B. For equity correlations, the Johnson SB distribution is better than the normal or lognormal.

C. A stressed scenario of bad economic times should incorporate higher correlation levels and higher correlation volatility.

D. Autocorrelation is a monotonically increasing function of the monthly lag; higher lags imply higher autocorrelations.

Answer: D

Step by step

1. Check each statement against the reading.
2. A: correlations show strong mean reversion (rate about 78%), so models should include it: true.
3. B: the bounded Johnson SB distribution was the best fit for equity correlations: true.
4. C: correlations are highest in recessions and correlation volatility is high, so stress should use higher levels and volatility: true.
5. D: autocorrelation is about 22% at one lag and decays to about 10% by ten months, so it falls with the lag: false.

Why D is correct: Autocorrelation decays as the lag lengthens (about 22% at one month to about 10% at ten months), the opposite of the claim in D.

Why the others are wrong

- **A:** True statement: strong mean reversion (about 78%) means it belongs in the model; not the exception.
- **B:** True statement: Johnson SB (bounded, flexible) fits equity correlations better than normal or lognormal.
- **C:** True statement: recession correlations are highest, so stress scenarios should raise level and volatility.

Q472 BT answers P2.T5.503.1 · BT notes P2.T5.503.1

REBUILT — Original statement III (positive association between correlation level and volatility) is not supported by the Schweser reading, where volatility peaks in normal periods, and the 'all correct' key was a giveaway; replaced III with a clearly false statement and adjusted options.

A junior analyst reviews the evidence on equity correlations of the 30 Dow stocks (1972 onward) and reports four findings:

I. Equity correlation levels are lowest in economic expansions and highest in recessions.

II. Equity correlation volatility is high, above 70% in each of the expansion, normal and recession states.

III. Equity correlation volatility is highest in recessions.

IV. Equity correlations are strongly mean reverting and therefore have low (positive) one-period autocorrelation.

Which of the findings are correct?

A. I only.

B. I, II and IV only.

C. I, II and III only.

D. All four are correct.

Answer: B
Key correction: BT key (D, all correct) relied on a level-volatility association not in the 2026 reading; with statement III as rewritten, correct is B.
Step by step
1. I: levels 37.0% (recession) > 33.0% (normal) > 27.5% (expansion): true.
2. II: volatilities 80.5%, 83.0%, 71.2% are all above 70%: true.
3. III: volatility is highest in normal periods (83.0%), not recessions (80.5%): false.
4. IV: mean reversion rate about 78%, so one-period autocorrelation = 1 - 0.78 = about 22%, which is low: true.
5. Correct: I, II and IV only.

Why B is correct: Correlation volatility peaks in normal periods (83.0%) rather than recessions (80.5%), so III is the only false statement.
Why the others are wrong
- **A:** Too restrictive: II and IV are also true.
- **C:** Includes III, the intuitive but wrong claim that volatility is highest in recessions, and omits IV.
- **D:** Accepts III; volatility is higher in normal periods than in recessions.

Q473 Schweser Module Quiz 9.1 Q1

A risk manager examines the correlations and correlation volatility of stocks in the Dow Jones Industrial Average for 1972 to 2017. Expansions are periods where U.S. GDP growth exceeds 3.5%, normal periods have GDP growth between 0 and 3.5%, and recessions are periods with two consecutive quarters of negative GDP growth. Which statement best characterizes correlation and correlation volatility for this sample? The risk manager will most likely find that:

A. correlations and correlation volatility are highest for recessions.
B. correlations and correlation volatility are highest for expansionary periods.
C. correlations are highest for normal periods, and correlation volatility is highest for recessions.
D. correlations are highest for recessions, and correlation volatility is highest for normal periods.

Answer: D
Step by step
1. Correlation levels: recession 37.0%, normal 33.0%, expansion 27.5%, so highest in recessions.
2. Correlation volatility: recession 80.5%, normal 83.0%, expansion 71.2%, so highest in normal periods.
3. Match: recessions for level, normal periods for volatility: D.

Why D is correct: Levels peak in recessions when stocks fall together, but correlation volatility peaks in normal periods when market direction is least certain.
Why the others are wrong
- **A:** Right on level but assumes volatility also peaks in recessions (80.5% is below 83.0%).
- **B:** Expansions have the lowest correlation and the lowest volatility.
- **C:** Swaps the two results: level is highest in recessions, volatility in normal periods.

LO 9.b

Calculate a mean reversion rate using standard regression and calculate the corresponding autocorrelation.

Q474 GARP book Ch.9 Q9.4

What is mean reversion?

Answer: see model answer — Mean reversion is the tendency of a variable (e.g. interest rates, spreads, volatility, correlations) to be pulled back toward its long-run mean over time. Statistically it is a negative relationship between the change S(t) − S(t−1) and the previous level S(t−1): when the level is above the mean the next change is expected to be negative, and vice versa. The strength of the pull is the mean reversion rate a (speed): a = 0 means none, a = 1 means complete reversion in one period.
Step by step
1. Define: $S_t - S_{t-1} = a(\mu - S_{t-1}) + \text{noise}$.
2. Level above μ implies expected decline; level below μ implies expected rise.
3. a measures the speed of reversion (fraction of the gap closed per period).

Why this is correct: Mean reversion is a negative dependence of the next change on the current deviation from the long-run mean.

Q475

AnalystPrep Q.4872

REBUILT — Options 52.12%/51.22%/20.00% were irrelevant giveaways; replaced with realistic-error values (answer 24.61% unchanged).

In December 2020 the average correlation of the Dow 30 stocks was 19.25% and its long-run mean was 27.23%. If the monthly mean reversion rate is 0.6715, what is the expected average correlation for January 2021?

A. 21.87%
B. 24.61%
C. 5.36%
D. 13.89%

Answer: B

Step by step

1. Gap = 27.23% − 19.25% = 7.98%.
2. Reversion = 0.6715 × 7.98% = 5.36%.
3. $S_t = 19.25\% + 5.36\% = 24.61\%$.

Why B is correct: The correlation closes 67.15% of the 7.98% gap in one month, rising from 19.25% to 24.61%.

Why the others are wrong

- **A:** 21.87%: used autocorrelation 1 − a = 32.85% as the reversion fraction (19.25% + 0.3285 × 7.98%).
- **C:** 5.36%: gave only the expected change, not the new level.
- **D:** 13.89%: subtracted the reversion from the current level (19.25% − 5.36%), moving away from the mean.

Q476

GARP book Ch.9 Q9.5

How can we quantify mean reversion?

Answer: see model answer — Run a standard regression of the change S(t) − S(t−1) (Y) on the prior level S(t−1) (X): Y = α + βX + ε. The slope β equals the negative of the mean reversion rate, so a = −β (e.g. β = −0.7903 implies a = 79.03%), and the long-run mean is μ = −α/β = α/a. A significantly negative β indicates mean reversion.

Step by step

1. Rewrite $S_t - S_{t-1} = a\mu - aS_{t-1}$.
2. Match to Y = α + βX: $\alpha = a\mu,\ \beta = -a$.
3. Estimate by OLS; $a = -\beta, \mu = \alpha/a$.

Why this is correct: Expanding a(μ − S(t−1)) shows that the regression slope on S(t−1) is exactly −a.

Q477

GARP book Ch.9 Q9.6

What is autocorrelation? Name two approaches for how to quantify autocorrelation.

Answer: see model answer — Autocorrelation is the correlation of a variable with its own past values, i.e. how persistent the series is (for one lag, $AC = \text{cov}(\rho_t, \rho_{t-1})/(\sigma_{\rho_t}\sigma_{\rho_{t-1}})$, the ordinary correlation coefficient applied to the series and its lag). Two approaches: (1) estimate it with an ARCH/GARCH-type model; (2) run a regression of the series on its lagged value (or compute the sample correlation between the series and its lag). For one lag it also equals 1 − a, where a is the mean reversion rate.

Step by step

1. Autocorrelation = correlation between ρt and ρt−k.
2. Approach 1: ARCH/GARCH model.
3. Approach 2: regression / sample correlation of the series with its lag.

Why this is correct: Autocorrelation is just the correlation coefficient between a series and its own lagged values.

Q478 GARP book Ch.9 Q9.7

For equity correlations, we see the typical decrease of autocorrelation with respect to time lags. What does that mean?

Answer: see model answer — Correlations are most strongly related to recent past correlations; the further back the comparison (longer lag), the weaker the relationship. In the Dow study, one-period autocorrelation was about 21%, peaked at about 26% at a two-month lag, then declined gradually to about 10% at ten months. The memory of the correlation process fades: past correlation levels have decreasing power to explain the current level as the lag grows.

Step by step

1. Compute AC for lags 1 to 10 months.
2. Values rise slightly to 26% at lag 2, then fall to about 10% at lag 10.
3. Interpretation: persistence/memory decays with time.

Why this is correct: Decay of autocorrelation with lag means the influence of past correlation on current correlation fades as the lag lengthens.

Q479 GARP book Ch.9 Q9.8

How are mean reversion and autocorrelation related?

Answer: see model answer — They are opposite properties: mean reversion is the pull back toward the long-run mean, while autocorrelation is the persistence toward recent past values. For a one-period lag, the autocorrelation equals one minus the mean reversion rate (AC = 1 − a), so the two sum to 1 (e.g. a = 79.03% gives AC of about 20.97%). High mean reversion implies low autocorrelation, as for equity correlations; bond and default correlations show lower mean reversion and higher autocorrelation.

Step by step

1. Mean reversion rate a = −β from the regression.
2. One-period autocorrelation = 1 − a.
3. Hence a + AC = 1: faster reversion means less persistence.

Why this is correct: With one lag, the share of the gap that reverts (a) and the share that persists (1 − a) must add to one.

Q480 AP Mock 1 Q9

Given the following data about a variable S:

Item	Value
S(t)	50
S(t−1)	30
Mean reversion rate	0.4

Calculate the long-run mean value for the variable.

A. 60
B. 80
C. 100
D. 75

Answer: B

Step by step

1. $S_t - S_{t-1} = a(\mu - S_{t-1})$.
2. $50 - 30 = 0.4(\mu - 30)$, so $20 = 0.4\mu - 12$.
3. $\mu = 32/0.4 = 80$.

Why B is correct: The observed change (20) is a fraction a of the gap between the mean and the previous level S(t−1), so the gap is 20/0.4 = 50 and μ = 30 + 50 = 80.

Why the others are wrong

- **A:** 60: an arbitrary value with no derivation; it ignores that the change is only 40% of the gap.
- **C:** 100: used the current level S(t) = 50 rather than S(t−1) = 30 as the base of the gap (50 + 20/0.4).
- **D:** 75: divided S(t−1) by the rate (30/0.4) and ignored the observed change.

Q481 AnalystPrep Q.2654

REBUILT — Original distractor 75 had no realistic derivation and options were weak; replaced distractors with realistic-error results (answer 80 unchanged).

Given the following data about a variable S: S(t−1) = 40, S(t) = 60, and the mean reversion rate a = 0.5. Calculate the long-run mean value for the variable.

A. 60
B. 80
C. 100
D. 50

Answer: B

Step by step

1. $S_t - S_{t-1} = a(\mu - S_{t-1})$.
2. $60 - 40 = 0.5(\mu - 40)$, so $20 = 0.5\mu - 20$.
3. $\mu = 40/0.5 = 80$.

Why B is correct: The 20-point move is half the distance to the mean, so the gap is 40 and μ = 40 + 40 = 80.

Why the others are wrong

- **A:** 60: confused the long-run mean with the current value S(t).
- **C:** 100: measured the gap from S(t) instead of S(t−1) (60 + 20/0.5 = 100).
- **D:** 50: multiplied the change by a (40 + 0.5 × 20) instead of dividing by a.

Q482 AnalystPrep Q.2835

REBUILT — Original options were mostly implausible; kept data and answer 48.78% and replaced distractors with realistic-error values.

The long-run mean of the average correlation of the Dow 30 stocks is 52.6%. The monthly average correlation observed in April 2011 was 31.26%. Given a mean reversion rate of 82.1% estimated from regression over 40 years, what is the expected average correlation one month later?

A. 17.52%
B. 35.08%
C. 48.78%
D. 13.74%

Answer: C

Step by step

1. Gap = 52.6% − 31.26% = 21.34%.
2. Reversion = 0.821 × 21.34% = 17.52%.
3. S_t = 31.26% + 17.52% = 48.78%.

Why C is correct: The correlation is expected to close 82.1% of its gap to the mean in one month, so it moves from 31.26% up to 48.78%.

Why the others are wrong

- **A:** 17.52%: gave only the expected change and forgot to add it to the starting 31.26%.
- **B:** 35.08%: used the autocorrelation (1 − a = 17.9%) instead of the mean reversion rate: 31.26% + 0.179 × 21.34%.
- **D:** 13.74%: subtracted the reversion from the current level (31.26% − 17.52%), moving away from the mean.

Q483 AnalystPrep Q.2837

REBUILT — Stem said 'exactly one year' with a rate of unknown periodicity and options 52.12%/51.22% were giveaways; fixed the period to one month and replaced distractors (answer 18.91% unchanged).

The average correlation of the Dow 30 stocks is currently 16.45%, and its long-run mean is 19.72%. Using a one-period mean reversion rate of 0.7512 (one period = one month), what is the expected average correlation one period from today?

A. 2.46%
B. 14.00%
C. 17.26%
D. 18.91%

Answer: D

Step by step

1. Gap = 19.72% − 16.45% = 3.27%.
2. Reversion = 0.7512 × 3.27% = 2.46%.
3. $S_t = 16.45\% + 2.46\% = 18.91\%$.

Why D is correct: Expected next-period correlation = current level plus a times the gap to the mean.

Why the others are wrong

- **A:** 2.46%: gave the expected change only, not the new level.
- **B:** 14.00%: subtracted the reversion from the current level (16.45% − 2.46%), moving away from the mean.
- **C:** 17.26%: used autocorrelation 1 − a = 24.88% as the reversion fraction (16.45% + 0.2488 × 3.27%).

Q484

BT answers P2.T5.503.2 · BT notes P2.T5.503.2

The long-term mean of the correlation data is 35.0%. In January 2014, the average correlation of the 30 x 30 Dow correlation matrices was 27.0%. From the regression function from 1972 to 2012, the average mean reversion rate was 77.5%. Using the simple model $s(t) - s(t-1) = a[\mu(s) - s(t-1)]$, what is the expected correlation for February 2014?

A. 27.75%
B. 28.80%
C. 33.20%
D. 37.50%

Answer: C

Step by step

1. Expected change = $a(\mu - s(t-1)) = 0.775 \times (35.0\% - 27.0\%) = 0.775 \times 8.0\% = 6.2\%$.
2. Expected February correlation = 27.0% + 6.2% = 33.2%.

Why C is correct: The model moves the correlation a fraction a of the way from its current level back to the long-run mean.

Why the others are wrong

- **A:** Barely moves January's 27.0%; it ignores the (mean minus current) gap that must be multiplied by a.
- **B:** Used 1 − a = 22.5% instead of a: 27.0% + 0.225 × 8.0% = 28.8%.
- **D:** Overshoots the long-run mean of 35%; with 0 < a < 1 the expected value must lie between 27% and 35%.

Q485

BT notes P2.T5.22.12.1

Albert has retrieved the monthly average correlation for stocks in the Dow Jones Industrial Average (DJIA) for the last ten years, i.e., each month's $\rho(t)$ is the average of the 30 x 30 pairwise correlation matrix. With this time series he computes the one-period lag autocorrelation, AC$[\rho(t), \rho(t-1)]$. If Albert's one-period lag autocorrelation is low, which of the following is most likely?

A. The regression coefficient is positive
B. The Pearson correlation coefficient is high
C. The mean reversion rate (aka, gravity) is high
D. The volatilities of correlation(t) and correlation(t−1) are low

Answer: C

Step by step

1. Mean reversion rate a = −(slope) in the regression of the change in correlation on the lagged level.
2. Autocorrelation = 1 − a, so a = 1 − autocorrelation.
3. A low autocorrelation therefore means a high mean reversion rate.

Why C is correct: Autocorrelation and the mean reversion rate sum to one, so one is high exactly when the other is low.

Why the others are wrong

- **A:** With mean reversion the slope is −a, i.e. negative; a low autocorrelation does not imply a positive slope.
- **B:** The autocorrelation is itself the Pearson correlation between $\rho(t)$ and $\rho(t-1)$; if it is low, that coefficient is low, not high.
- **D:** Volatility levels are unrelated to the autocorrelation; the link is 1 − autocorrelation = mean reversion rate.

Q486

Mock 1 Q27

REBUILT — Option A printed 3.8% (inconsistent with the computed 3.9% change) and C was an imprecise 31.0%; corrected to 3.9% and 31.1%.

In September 2021, the average monthly correlation for all the Dow stocks was 30% and the long-run mean correlation of Dow stocks was 35%. A regression of the change in correlation on the lagged level gives $Y = 0.215 - 0.78X$. What is the expected correlation for October 2021, given the mean reversion rate estimated in the regression?

A. 3.9%
B. 5.0%
C. 31.1%
D. 33.9%

Answer: D

Step by step

1. Mean reversion rate a = −slope = 0.78.
2. Expected change = 0.78 × (35% − 30%) = 3.9%.
3. Expected October correlation = 30% + 3.9% = 33.9%.

Why D is correct: The expected next-period level is the current level plus a times the gap to the long-run mean.

Why the others are wrong

- **A:** This is only the expected change, not the level; forgot to add it to September's 30%.
- **B:** This is the raw gap to the mean (35% − 30%), which applies only if a = 100%.
- **C:** Used 1 − a = 0.22 instead of a: 30% + 0.22 × 5% = 31.1%.

Q487

Mock 2 Q37

REBUILT — Replaced weak distractors (2 and 6) with ones produced by realistic errors (sign error, using 1 − a).

Suppose mean reversion exists for a variable with a value of 100 at time t − 1. The long-run mean of the variable is 120. Ignoring the stochastic term, what is the expected change in the variable for the next period if the mean reversion rate is 0.6?

A. −12
B. 8
C. 12
D. 20

Answer: C

Step by step

1. Expected change = $a(\mu - S_{t-1}) = 0.6 \times (120 - 100)$.
2. = 0.6 × 20 = +12 (the variable is expected to rise to 112).

Why C is correct: Expected change equals the mean reversion rate times the distance to the long-run mean.

Why the others are wrong

- **A:** Sign error; the variable is below its mean, so it is pulled up (+12), not down.
- **B:** Used 1 − a = 0.4 instead of a: 0.4 × 20 = 8.
- **D:** The full gap of 20 applies only if a = 1 (complete reversion in one period).

Q488

Schweser Module Quiz 9.1 Q2

Suppose mean reversion exists for a variable with a value of 30 at time period t − 1. The long-run mean of the variable is 40; ignore the stochastic term. What is the expected change in value of the variable for the next period if the mean reversion rate is 0.4?

A. −10
B. −4
C. 4
D. 10

Answer: C

Step by step

1. Expected change = $a(\mu - S_{t-1}) = 0.4 \times (40 - 30)$.
2. = +4.

Why C is correct: The variable is below its mean so it is expected to rise by a fraction a of the 10-point gap.

Why the others are wrong
- **A:** Wrong sign and ignores a (full gap, negative).
- **B:** Correct magnitude but wrong sign; a below-mean variable rises.
- **D:** Full gap of 10 applies only if a = 1; forgot to multiply by 0.4.

Q489 Mock 2 Q59

A risk analyst uses 600 months of correlation data from the S&P 500 to estimate the long-run mean correlation of common stocks and the mean reversion rate. The long-run mean correlation was 44%, and the regression gives $Y = 0.334 - 0.79X$. In July 2017 the average monthly correlation for all S&P stocks was 37%. What is the estimated one-period autocorrelation based on the mean reversion rate from the regression?

A. 21%
B. 37%
C. 56%
D. 66%

Answer: A
Step by step
1. Mean reversion rate a = −slope = 0.79.
2. Autocorrelation = 1 − a = 1 − 0.79 = 0.21 = 21%.
3. The 44% mean and 37% July level are not needed.

Why A is correct: Autocorrelation plus mean reversion rate equals one.
Why the others are wrong
- **B:** Picks up the July correlation level, which is irrelevant to the autocorrelation.
- **C:** Equals 1 − 0.44, using the long-run mean instead of the mean reversion rate.
- **D:** Roughly 1 − 0.334 = 66.6%, confusing the regression intercept with the slope.

Q490 Schweser Module Quiz 9.1 Q3

A risk manager uses 480 months of correlation data from the Dow to estimate the long-run mean correlation and the mean reversion rate. The long-run mean correlation of Dow stocks was 32%, and the regression gives $Y = 0.24 - 0.75X$. In April 20X4 the average monthly correlation for all Dow stocks was 36%. What is the expected correlation for May 20X4 using the estimated mean reversion rate?

A. 32%
B. 33%
C. 35%
D. 37%

Answer: B
Step by step
1. Mean reversion rate a = 0.75.
2. Gap = 32% − 36% = −4%; expected change = 0.75 × (−4%) = −3%.
3. Expected May correlation = 36% − 3% = 33%.

Why B is correct: Correlation reverts 75% of the way from 36% toward the 32% mean.
Why the others are wrong
- **A:** Assumes full reversion (a = 1) to the long-run mean.
- **C:** Used 1 − a = 0.25: 36% + 0.25 × (−4%) = 35%.
- **D:** Moved away from the mean: 36% + 0.25 × 4% = 37%, wrong direction and wrong rate.

Q491 Schweser Module Quiz 9.1 Q4

A risk manager uses 480 months of correlation data from the Dow to estimate the long-run mean correlation and the mean reversion rate. The long-run mean correlation of Dow stocks was 34%, and the regression gives $Y = 0.262 - 0.77X$. In April 20X4 the average monthly correlation for all Dow stocks was 33%. What is the estimated one-period autocorrelation based on the mean reversion rate from the regression?

A. 23%
B. 26%

C. 30%
D. 33%

Answer: A
Step by step
1. Mean reversion rate a = −slope = 0.77.
2. Autocorrelation = 1 − 0.77 = 0.23 = 23%.

Why A is correct: Mean reversion rate and one-period autocorrelation sum to 100%.
Why the others are wrong
- **B:** Takes the intercept 0.262 as the answer; the intercept is not the autocorrelation.
- **C:** Not the result of any valid step; the 77% rate was not subtracted from one (1 − 0.77 = 23%, not 30%).
- **D:** Picks the April correlation level, which is irrelevant.

LO 9.c

Identify the best-fit distribution for equity, bond, and default correlations.

Q492 GARP book Ch.9 Q9.9

What is the distribution of equity correlations?

Answer: see model answer — Empirical work on equity correlations found that the Johnson SB distribution (two shape parameters, a location and a scale parameter) is the best fit, based on Kolmogorov-Smirnov, Anderson-Darling and chi-squared tests, and the fit is robust across economic states. Normal, lognormal and beta distributions fit poorly. Johnson SB is bounded, which suits correlations confined to an interval.
Step by step
1. Recall the three goodness-of-fit tests (K-S, Anderson-Darling, chi-squared) applied to equity correlations.
2. Johnson SB gave the best fit and stayed robust across economic states.
3. Normal, lognormal and beta fit poorly.

Why this is correct: Johnson SB is flexible (four parameters) and bounded, matching the empirical shape of equity correlations.

Q493 GARP book Ch.9 Q9.10

When modelling stocks, bonds, commodities, exchange rates, volatilities and other financial variables, we typically assume a normal or lognormal distribution. Can we do this for equity correlations?

Answer: see model answer — No. Equity correlations are not well described by normal or lognormal distributions (nor by the beta distribution); these gave a poor fit in empirical tests. The Johnson SB distribution provides the best fit. Correlations are bounded between -1 and +1 and have a distinctive shape, so unbounded or one-sided normal/lognormal assumptions are inappropriate.
Step by step
1. Test normal, lognormal and beta distributions against empirical equity correlations.
2. All three fit poorly.
3. Johnson SB is the best fit, so the usual normal/lognormal assumption should not be carried over to equity correlations.

Why this is correct: Equity correlations are bounded and non-normal; only Johnson SB fits well.

Q494 AnalystPrep Q.2655

Which of the following distributions best fits the empirical distribution of equity correlations?
A. Chi-squared
B. Generalized extreme value
C. Pareto
D. Johnson SB

Answer: D

Step by step

1. Empirical tests (K-S, Anderson-Darling, chi-squared) on equity correlations favour one distribution.
2. The best fit is Johnson SB, robust across economic states.
3. Normal, lognormal and beta fit poorly; the other listed distributions are not used for equity correlations.

Why D is correct: Johnson SB is a flexible bounded distribution that matches equity correlations.

Why the others are wrong

- **A:** Chi-squared is defined only on positive values, but correlations can be negative; not the best fit.
- **B:** GEV is the best fit for bond correlations, not equity correlations (confusing the two assets).
- **C:** Pareto is a heavy-tailed one-sided distribution; it is not bounded and is not the fitted shape for correlations.

Q495 AnalystPrep Q.1566

Bonds and their default probabilities also have correlation distributions just like equity. Which of the following best describes the default probability correlation distribution and the bond correlation distribution?

A. Both the default probability correlation distribution and the bond correlation distribution are best modeled using the Johnson SB distribution.

B. The default probability correlation distribution shows a normal shape and is best modeled using the generalized extreme value distribution, whereas the bond correlation distribution is best modeled using the Johnson SB distribution.

C. The default probability correlation distribution is best modeled using the Johnson SB distribution, whereas the bond correlation distribution is best modeled using the generalized extreme value distribution.

D. Both the default probability correlation distribution and the bond correlation distribution are best modeled using the generalized extreme value distribution.

Answer: C

Step by step

1. Default probability correlations resemble equity correlations: best fit is Johnson SB.
2. Bond correlations: best fit is GEV (normal also fits well).
3. Statement C matches both; the others swap or merge the two.

Why C is correct: Default probability correlations behave like equity correlations (Johnson SB), whereas bond correlations are best fit by GEV.

Why the others are wrong

- **A:** Applies Johnson SB to bond correlations too; bond correlations are best fit by GEV.
- **B:** Reverses the two assignments: bond correlations, not default correlations, are GEV/normal-like.
- **D:** Applies GEV to default probability correlations; they are Johnson SB like equity.

Q496 BT answers P2.T5.503.3 · BT notes P2.T5.503.3

Consider the following four statements about the best distributional fit to equity prices, equity correlations, bond correlations, and default probability correlations:

I. Assuming a sharp volatility skew (i.e., continuously decreasing), equity prices are best fit by a lognormal distribution

II. Equity correlations can be fit by a normal or lognormal distribution

III. Bond correlations can be fit by a normal or generalized extreme value (GEV) distribution

IV. Default probability correlations can be fit by a normal or generalized extreme value (GEV) distribution

Which of the above is (are) TRUE?

A. None are true

B. III only

C. I and II only

D. All are true

Answer: B

Step by step

1. I: lognormal prices correspond to a flat volatility smile; a sharp skew implies a fatter left tail, so false.
2. II: equity correlations fit poorly with normal/lognormal (and beta); Johnson SB fits best, so false.
3. III: bond correlations are best fit by GEV and normal also fits well, so true.
4. IV: default probability correlations are best fit by Johnson SB like equity, so false.

5. Only III is true.

Why B is correct: Only bond correlations are described by normal/GEV; equity and default correlations need Johnson SB.

Why the others are wrong

- **A:** Overlooks that III is true (bond correlations fit by normal or GEV).
- **C:** Treats a skewed-vol world as lognormal and equity correlations as normal/lognormal; both are false.
- **D:** Applies the bond result to every statement, including default correlations and the skew case.

Q497 Schweser Module Quiz 9.1 Q5

In estimating correlation matrices, risk managers often assume an underlying distribution for the correlations. Which of the following statements most accurately describes the best-fit distributions for equity correlation distributions, bond correlation distributions, and default probability correlation distributions? The best-fit distributions for the equity, bond, and default probability correlation distributions, respectively, are:

A. lognormal, generalized extreme value, and normal.
B. Johnson SB, generalized extreme value, and Johnson SB.
C. beta, normal, and beta.
D. Johnson SB, normal, and beta.

Answer: B

Step by step

1. Equity correlations: Johnson SB.
2. Bond correlations: GEV (normal also a good fit).
3. Default probability correlations: Johnson SB.
4. Only B gives Johnson SB, GEV, Johnson SB.

Why B is correct: Equity and default probability correlations share the Johnson SB best fit; bond correlations are GEV.

Why the others are wrong

- **A:** Lognormal fits equity correlations poorly and normal is not the best fit for default correlations.
- **C:** Beta fits equity and default correlations poorly; normal is only a secondary fit for bonds.
- **D:** Gets equity right but uses normal for bonds (second best, not best) and beta for default correlations (poor fit).

Reading 10. Financial Correlation Modeling—Bottom-Up Approaches

Gunter Meissner, Correlation Risk Modeling and Management, 2nd Edition (Risk Books, 2019). Chapter 5, pages 126–134.

LO	Learning objective	Qs
10.a	explain the purpose of copula functions and how they are applied in finance.	8
10.b	describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets.	13
10.c	summarize the process of finding the default time of an asset correlated to all other assets in a portfolio using the Gaussian copula.	5

LO 10.a

Explain the purpose of copula functions and how they are applied in finance.

Q498

AnalystPrep Q.1588

REBUILT — Original asked only for the meaning of an inverse-function symbol (giveaway, notation damaged by OCR); rebuilt to test the roles of each component of the copula expression.

A copula function is written as $C[G_1(u_1), \dots, G_n(u_n)] = F_n[F^{-1}(G_1(u_1)), \dots, F^{-1}(G_n(u_n)); \rho_F]$, where each $G_i(u_i) \in [0, 1]$. Which of the following correctly identifies the components of this expression?

A. $G_i(u_i)$ are the marginal distributions, F_n is the joint cumulative distribution function, and ρ_F is the correlation structure of F_n.

B. $G_i(u_i)$ are the joint distributions, F_n is the marginal distribution of each asset, and ρ_F is the correlation structure of G_i.

C. $G_i(u_i)$ are the inverse distribution functions, F_n is the correlation matrix, and ρ_F is the joint cumulative distribution function.

D. $G_i(u_i)$ are the marginal distributions, F_n is the correlation structure, and ρ_F is the joint cumulative distribution function.

Answer: A

Step by step

1. Start with the inputs: each $G_i(u_i)$ is a univariate (uniform on [0,1]) marginal distribution of one variable.
2. The inverse function F^{-1} maps each marginal percentile-to-percentile onto a well-known distribution (e.g., the standard normal).
3. F_n then joins the mapped values into a single n-variate joint cumulative distribution function.
4. ρ_F is the correlation matrix that governs the dependence in F_n. So the components are: marginals G_i, joint CDF F_n, correlation structure ρ_F.

Why A is correct: A copula takes separate marginals, maps them through an inverse function to a known distribution, and joins them in a joint CDF whose dependence is set by the correlation matrix ρ_F.

Why the others are wrong

- **B:** Reverses the roles: the G_i are the individual (marginal) inputs and F_n is the joint output, not the other way round.
- **C:** Treats G_i as inverses and mixes up F_n (a distribution function) with a correlation matrix; the inverse is F^{-1}, applied to G_i.
- **D:** Gets the marginals right but swaps F_n and ρ_F; a correlation matrix is a parameter of the joint CDF, not the CDF itself.

Q499

AnalystPrep Q.1595

REBUILT — Original distractors contradicted the stem or were implausible (banning, 'not helpful'); rebuilt with plausible alternatives about why copulas were adopted and why they fell out of favor.

Flexible copula functions became popular in the financial industry around the year 2000. Which of the following best describes why they were adopted and why they later fell out of favor?

A. They offered a simple way to correlate many assets (e.g., 100+ assets in a CDO) with a single multidimensional function, but fell out of favor when the 2007–2009 financial crisis began.

B. They captured tail dependence perfectly for all asset classes, but regulators banned them from use in risk models after Basel II was implemented.

C. They removed the need to specify marginal distributions, but fell out of favor because they could only be applied

when the marginals were normal.

D. They allowed default correlation to be estimated without any assumptions, but fell out of favor after the 1998 collapse of Long-Term Capital Management.

Answer: A

Step by step

1. Recall the reading: copulas gained popularity around 2000 because they used simple methods to solve complex problems, e.g., one multidimensional function to correlate 100+ assets in a CDO.
2. Flexible copulas fell out of favor when the 2007–2009 financial crisis began.
3. Only A matches both halves; B, C and D each misstate the reason or the timing.

Why A is correct: Copulas were attractive for their simplicity in correlating very large portfolios such as CDOs, and their reputation suffered in the 2007–2009 crisis.

Why the others are wrong

- **B:** Copulas do not capture tail dependence perfectly (the Gaussian copula in particular does not) and they were never banned by regulation.
- **C:** A copula's key advantage is that marginals can be of any shape; they are mapped to a known distribution, so normal marginals are not required.
- **D:** Copulas still require assumptions, and the loss of favor is tied to the 2007–2009 crisis, not LTCM in 1998.

Q500

BT Mock Q36 · BT notes P2.T5.22.13.1

Which of the following is the BEST summary of the purpose of a copula function?

A. To utilize the Gaussian correlation matrix such that tractable pairwise correlation parameters replace intractable dependence values.

B. To specify a single multivariate distribution as a function of various (even unusual) univariate marginal distributions, by mapping them to distributions with an accessible form.

C. To translate a vector of outcomes into quantiles and map the quantiles to probabilities, such that virtually any combination of multiple outcomes can be expressed.

D. To decompose a posterior conditional distribution into a combination of the prior probability, the likelihood, and the marginal probability.

Answer: B

Step by step

1. A copula joins separate univariate marginals (of any shape) into one joint distribution.
2. It does so by mapping each marginal, percentile-to-percentile, to a known distribution such as the standard normal, where a correlation structure is easy to define.
3. Only B states this purpose; A, C and D describe something narrower or something else entirely.

Why B is correct: A copula builds a joint distribution from arbitrary marginals by mapping them to a well-behaved distribution with a defined dependence structure while preserving the marginals.

Why the others are wrong

- **A:** Describes only the Gaussian copula's use of a correlation matrix, one special case, and not the general purpose of joining marginals.
- **C:** Mentions the quantile mapping step but not its purpose (creating a joint distribution); the claim about expressing any combination of outcomes is not what a copula is for.
- **D:** This is Bayes' theorem (posterior from prior, likelihood and marginal), unrelated to copulas.

Q501

BT answers P2.T5.505.1 · BT notes P2.T5.505.1

What is a copula function?

A. A copula correlates a stock price (S) and its volatility (σ) which allows it to model equity-implied volatility skew.

B. A copula function joins (n) univariate distributions to one multivariate (n-dimensional) distribution.

C. A copula function solves for the conditional P[X|Y] as a function of the product of P[Y|X] and the ratio of marginal probabilities P[X]/P[Y].

D. A copula function is a limiting (special) case of the Pearson model, such that it only analyzes the linear relationship between variables.

Answer: B

Step by step

1. A copula maps n univariate (marginal) distributions onto a known distribution and joins them.
2. The result is one n-dimensional joint distribution that keeps the original marginals and adds a dependence structure.
3. Check the others: A is a stochastic volatility idea, C is Bayes' rule, D is a linear-correlation statement.

Why B is correct: The defining purpose of a copula is to combine n marginal distributions into a single n-dimensional joint distribution.

Why the others are wrong

- **A:** Describes a stochastic-volatility model for equity skew (e.g., Heston), not a copula.
- **C:** This is Bayes' theorem, $P[X|Y] = P[Y|X]\,P[X]/P[Y]$.
- **D:** A copula is not a special case of the Pearson model and is not limited to linear relationships; it is used precisely because Pearson correlation is limited.

Q502

BT answers P2.T6.24.26.3 · BT notes P2.T6.24.26.3

REBUILT — Original was a counterparty-risk scenario where the 'combine everything' option was a giveaway and the copula aspect was not isolated; rebuilt as a clean application of the copula approach with non-normal marginals.

A risk manager has separately estimated empirical distributions for changes in interest rates (skewed) and changes in credit spreads (heavy-tailed), and now needs a joint distribution for the two variables. Which approach best applies a copula?

A. Map each empirical marginal percentile-to-percentile to the standard normal distribution, define a correlation between the mapped variables, and thereby obtain a joint distribution that preserves both original marginals.

B. Assume that both variables are normally distributed and fit a bivariate normal distribution directly to the raw data.

C. Replace each empirical marginal with a normal distribution having the same mean and variance, and then apply the Pearson correlation to the raw data.

D. Compute the Pearson correlation of the raw data and use it as the only input, since a copula requires the marginals to be uniform on the raw scale.

Answer: A

Step by step

1. The two marginals are non-normal, so a correlation cannot be directly and sensibly defined between them.
2. A copula maps each marginal, percentile by percentile, to the standard normal, a well-behaved distribution.
3. A correlation is then specified between the mapped variables, which yields a joint distribution.
4. Because the mapping is invertible, each original marginal is preserved. Hence A.

Why A is correct: A copula indirectly defines dependence by mapping unknown marginals to a known distribution while keeping the original marginals intact.

Why the others are wrong

- **B:** Forces normal marginals and discards the skewness and heavy tails, which is exactly what a copula is designed to avoid.
- **C:** Replaces the marginals with normals, so the original shapes are lost; the joint distribution no longer preserves them.
- **D:** Misstates the method: the marginals can have any shape and are mapped via their percentiles; raw Pearson correlation alone is not a copula.

Q503

BT notes P2.T5.22.13.3

Sally works for a global financial services firm. As part of a risk aggregation project, she must build a model that aggregates potential loss distributions across several business lines. Each business has already specified its own marginal loss distribution, so her given inputs are a set of univariate marginal distributions. Sally wants to aggregate them with a copula. Each of the following is TRUE as an advantage of the copula, EXCEPT which is false?

A. The copula approach will enable her to simulate the entire distribution (i.e., in addition to the tails) of loss outcomes.

B. Given the same set of univariate marginal distributions, she will be able to test (i.e., simulate) with various one- or two-factor copula functions.

C. Several of the univariate loss distributions are skewed and/or heavy-tailed, but copula functions are flexible to this non-normal variety, as copulas place no requirements on the univariate distributions that are joined.

D. She used the linear Pearson correlation coefficient as an input to the copulas, and all major copula functions will natively ensure this parameter accurately describes the actual dependence structure, even if the dependence is dynamic.

Answer: D

Step by step

1. Statements A, B and C describe genuine advantages: copulas simulate the whole joint distribution, different copulas can be tried on the same marginals, and marginals can have any shape.
2. Statement D claims that a linear correlation input automatically captures the true dependence, including dynamic dependence.
3. This is false: Pearson correlation measures only linear dependence, and a copula does not guarantee it describes the actual (possibly non-static) dependence.
4. D is the false statement.

Why D is correct: A copula adds flexibility on the marginals but its dependence parameter is only as good as the correlation input; it does not guarantee a correct or dynamic dependence structure.

Why the others are wrong

- **A:** True: a copula yields a full joint distribution that can be simulated, not just the tail.
- **B:** True: the marginals are fixed inputs, so alternative copulas can be run on the same marginals.
- **C:** True: the marginals can be skewed or heavy-tailed because each is mapped to a known distribution.

Q504

GARP PE2 Q53

An analyst at a large investment bank is attending a seminar on copula functions and their applications in the finance industry. The analyst has been asked by the CRO to gather information on why copulas are used and which types of copulas are typically used at other financial institutions. Which of the following is correct for the analyst to report to the CRO?

A. Gaussian copulas are often used to quantify the default correlation structure underlying structured credit products.

B. A copula maps a single univariate distribution to multiple multivariate distributions.

C. A copula uses a correlation structure to transform a single multivariate distribution into a single univariate distribution.

D. Fréchet copulas incorporate a single factor and have historically been the most widely applied copulas in the finance industry.

Answer: A

Step by step

1. A copula goes from many univariate marginals to one multivariate distribution, not the reverse; this eliminates B and C.
2. The Gaussian copula is the most widely applied in finance, notably for default correlation in CDOs and other structured credit products.
3. D is wrong on both counts: Fréchet copulas are not single-factor and are not the most widely used.
4. Hence A.

Why A is correct: The Gaussian copula is the standard tool for the default correlation structure of structured credit products such as CDOs.

Why the others are wrong

- **B:** Reverses the direction: a copula maps several univariate distributions into one multivariate distribution.
- **C:** Also reverses the direction: it joins marginals into a joint distribution rather than collapsing a joint distribution to a univariate one.
- **D:** The Gaussian copula, not the Fréchet, has been the most widely applied, and the Fréchet is not a single-factor copula.

Q505

Schweser Module Quiz 10.1 Q1

REBUILT — Original equation was missing from OCR and the 'not accurately' item had a second loosely-worded option (C); rebuilt as a self-contained 'which is correct' item with exactly one right option.

A risk manager defines a copula function C that maps n marginal distributions to a joint distribution:
$C[G_1(u_1), \ldots, G_n(u_n)] = F_n[F_1^{-1}(G_1(u_1)), \ldots, F_n^{-1}(G_n(u_n));\ \rho_F]$
where $G_i(u_i)$ are the marginal distributions, F_n is the joint cumulative distribution function and ρ_F is its correlation

matrix. Which of the following statements most accurately describes this copula function?

A. The marginals $G_i(u_i)$ must themselves be standard normal distributions for the copula to be defined.

B. The marginals are mapped, percentile to percentile, to a well-behaved joint distribution F_n (such as the multivariate standard normal), and ρ_F supplies the correlation structure between them.

C. Applying the copula changes each variable's marginal distribution so that it takes on the shape of F_n.

D. ρ_F is the correlation matrix of the original, unmapped variables measured directly from their own distributions.

Answer: B

Key correction: Source asked which statement is NOT accurate and keyed A (marginals are not standard normal); that is right, but option C ('F1^-1 is the inverse of Fn') is also loosely worded, so the item was rebuilt.

Step by step

1. The marginals $G_i(u_i)$ are arbitrary (unknown-shaped) distributions, expressed as percentiles in [0,1]; they need not be normal.
2. F_i^{-1} maps each percentile to the corresponding point of a known distribution (e.g., standard normal), percentile to percentile.
3. F_n, with correlation matrix ρ_F, joins the mapped values into one multivariate distribution.
4. Because the mapping is percentile-to-percentile, the original marginals are preserved while ρ_F defines the dependence. Only B states this.

Why B is correct: A copula links arbitrary marginals to a well-defined joint distribution via percentile mapping, using ρ_F as the dependence structure while preserving each marginal.

Why the others are wrong

- **A:** Reverses the logic: the point of the copula is that marginals can be of unknown shape; only the target distribution is normal.
- **C:** Confuses mapping with transformation of the marginals; the marginals are preserved, and the mapping is only used to define the correlation.
- **D:** The unmapped marginals are the 'difficult' part where correlation cannot be directly defined; ρ_F belongs to the joint distribution F_n after mapping.

LO 10.b

Describe the Gaussian copula and explain how to use it to derive the joint probability of default of two assets.

Q506

AnalystPrep Q.1589

REBUILT — Original stem was garbled ('if Gx(ux) is uniform') and options mixed multivariate/univariate wording; restated clearly with one correct option.

In the n-variate Gaussian copula, each marginal cumulative distribution is mapped percentile to percentile to a standard normal variable, and the mapped variables are joined by the joint distribution M_n. Which statement is correct?

A. Each $N^{-1}(G_i(u_i))$ is multivariate normal, and M_n is univariate normal.

B. Each $N^{-1}(G_i(u_i))$ is a standard normal variable, and M_n is univariate standard normal.

C. Each $N^{-1}(G_i(u_i))$ is a standard normal variable, and M_n is the standard multivariate normal distribution with correlation matrix ρ_M.

D. Each $N^{-1}(G_i(u_i))$ is multivariate normal, and M_n is standard multivariate normal.

Answer: C

Step by step

1. $G_i(u_i)$ is a cumulative probability, so it is uniform on [0,1].
2. Applying the inverse standard normal CDF N^{-1} to a uniform variable gives a univariate standard normal variable (mean 0, sd 1).
3. The n mapped variables are then linked by the joint distribution M_n, the n-variate standard normal with n × n correlation matrix ρ_M.

Why C is correct: Each marginal is transformed to a univariate standard normal; the collection is joined by the multivariate standard normal M_n, which carries the correlation structure.

Why the others are wrong

- **A:** Reverses the roles: the individual mapped terms are univariate, and the joint distribution is multivariate.
- **B:** Correct on the individual terms but calls the joint distribution univariate; a joint distribution of n variables is multivariate.

- **D:** Calls each individual mapped term multivariate; each N^-1 term is a single standard normal variable.

Q507

AnalystPrep Q.1590

The Gaussian default time copula is $C_{GD}[Q_1(t), \ldots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \ldots, N^{-1}(Q_n(t)); \rho_M]$. The term N^{-1} maps each cumulative default probability $Q_i(t)$ to the univariate standard normal distribution on a percentile-to-percentile basis. Which statement is correct?

A. The 5th percentile of $Q_i(t)$ is mapped to the 5th percentile of the standard normal distribution.
B. The 4th percentile of $Q_i(t)$ is mapped to the 5th percentile of the standard normal distribution.
C. The 5th percentile of $Q_i(t)$ is mapped to the 3rd percentile of the standard normal distribution.
D. The 5th percentile of $Q_i(t)$ is mapped to the 10th percentile of the standard normal distribution.

Answer: A

Step by step

1. Percentile-to-percentile mapping means a point at the p-th percentile of the marginal maps to the p-th percentile of the standard normal.
2. So the 5th percentile of $Q_i(t)$ maps to the 5th percentile of N(0,1), i.e. $N^{-1}(0.05) = -1.645$.

Why A is correct: The mapping preserves percentile rank, so the same percentile is used on both distributions.

Why the others are wrong

- **B:** Shifts the percentile down (4th to 5th); the mapping does not change rank.
- **C:** Maps to a lower percentile; ranks are preserved, not altered.
- **D:** Maps to a higher percentile (e.g. confusing 5% one-tail with 10% two-tail); ranks are preserved.

Q508

AnalystPrep Q.1591

REBUILT — Options A (one coefficient) and B (one correlation matrix) were both defensible; rewrote so the single-parameter versus matrix distinction is unambiguous.

A risk analyst applies the Gaussian default time copula to exactly two assets, A and B. Which statement about the correlation structure is correct?

A. A single correlation coefficient ρ is required, and no correlation matrix ρ_M is needed.
B. A 2 × 2 matrix with four independent correlation parameters is required.
C. A 3 × 3 correlation matrix is required: one entry for each asset and one for the pair.
D. No correlation parameter is required because each asset is first mapped to a standard normal distribution.

Answer: A

Step by step

1. With n = 2 the joint distribution is the bivariate standard normal M_2.
2. The correlation between the two mapped variables is a single number ρ.
3. A correlation matrix ρ_M (n × n) is needed only when n > 2.

Why A is correct: For two assets the only free copula parameter is the pairwise correlation ρ, since the diagonal is 1 and the matrix is symmetric.

Why the others are wrong

- **B:** Ignores symmetry and unit diagonal; the 2 × 2 matrix has only one free parameter.
- **C:** Wrongly sizes the matrix by counting assets plus pairs; the matrix is n × n and unnecessary for n = 2.
- **D:** Confuses marginal mapping with dependence; the standard normal mapping does not specify the correlation, which must still be supplied.

Q509

AnalystPrep Q.1592

REBUILT — OCR-damaged options (Mi, Qi, M5) rewritten into a clean set where only the bivariate form is right.

Two companies, 1 and 2, are analyzed with the Gaussian default time copula. After mapping their cumulative default probabilities percentile to percentile to the standard normal distribution, which expression gives the joint distribution?

A. $M_1[N^{-1}(Q_1(t)), N^{-1}(Q_2(t)); \rho]$
B. $M_2[N^{-1}(Q_1(t)), N^{-1}(Q_2(t)); \rho]$
C. $M_2[Q_1(t), Q_2(t); \rho]$
D. $N[N^{-1}(Q_1(t)) + N^{-1}(Q_2(t))]$

Answer: B

Step by step

1. Two assets require the bivariate standard normal M_2 with a single correlation ρ.
2. Its arguments are the mapped values $N^{-1}(Q_1(t))$ and $N^{-1}(Q_2(t))$.

Why B is correct: The copula applies the bivariate standard normal CDF to the inverse-normal-transformed default probabilities.

Why the others are wrong

- **A:** Uses M_1, a univariate distribution, which cannot carry a correlation between two assets.
- **C:** Omits the N^{-1} transformation, feeding probabilities (0 to 1) into a standard normal CDF.
- **D:** Sums the transformed values into a univariate normal; this loses the joint distribution and correlation.

Q510 AnalystPrep Q.2656

A Gaussian copula maps the marginal distribution of each variable to which distribution?

A. Lognormal distribution
B. Poisson distribution
C. Standard normal distribution
D. Binomial distribution

Answer: C

Step by step

1. Each marginal is mapped percentile to percentile to the univariate standard normal (mean 0, sd 1).
2. The mapped variables are then joined by a multivariate standard normal with a chosen correlation.

Why C is correct: The Gaussian copula uses the standard normal as the well-behaved common distribution in which correlation is defined.

Why the others are wrong

- **A:** Lognormal is not the mapping target; it is a possible marginal shape.
- **B:** Poisson is a discrete count distribution, not used for the mapping.
- **D:** Binomial is discrete; the copula mapping is to a continuous standard normal.

Q511 AnalystPrep Q.6214

REBUILT — Original had a debatable qualitative key with no joint PD computed; converted to a quantitative joint default probability using the bivariate normal.

The cumulative 5-year default probabilities of Company A and Company B are 10% and 15%. A risk analyst uses a Gaussian copula with a copula correlation of 0.25 and a bivariate normal CDF (software) to estimate the joint 5-year default probability. Which value is closest to the result?

A. 1.50%
B. 2.70%
C. 4.18%
D. 10.00%

Answer: B

Step by step

1. Map each PD: $N^{-1}(0.10) = -1.2816$ and $N^{-1}(0.15) = -1.0364$.
2. Joint PD $= M_2(-1.2816, -1.0364; 0.25) = 0.02696 \approx 2.70\%$ (computed numerically).
3. Check: independence gives 10% × 15% = 1.50%; positive correlation must raise the joint PD above this, and it cannot exceed 10%.

Why B is correct: The joint probability is the bivariate standard normal CDF at the mapped quantiles; positive correlation lifts it above the independence value of 1.50%.

Why the others are wrong

- **A:** Multiplies the marginals (independence), ignoring the 0.25 correlation.
- **C:** Adds the linear-correlation covariance term, $0.015 + 0.25\sqrt{0.1 \cdot 0.9 \cdot 0.15 \cdot 0.85}$, treating the copula correlation as a default correlation.
- **D:** Takes the lower marginal PD, which corresponds to perfect dependence.

Q512 BT answers P2.T5.505.2 · BT notes P2.T5.505.2

About the Gaussian copula used in financial applications, each of the following is true EXCEPT which is not?

A. The Gaussian copula has low tail dependence, which is a weakness because dependencies (including correlations) increase in a crisis.

B. The Gaussian copula is difficult to calibrate to market prices; for example, it is difficult to calibrate CDO tranches with a single correlation model.

C. The Gaussian copula is principally static and allows only limited risk management; there is no stochastic process for the default intensity and default correlation.

D. The Gaussian copula is limited to market risk applications because it requires n × n pairwise correlation parameters, which is natural for a covariance matrix, but in credit risk there is no way to assume pairwise default correlations.

Answer: D

Step by step

1. A, B and C are recognized weaknesses (low tail dependence, calibration difficulty with a single correlation, static nature).
2. D is false: the Gaussian default time copula is designed precisely for credit risk, mapping default probabilities to a multivariate normal with correlation structure ρ_M.
3. So the EXCEPT answer is D.

Why D is correct: The Gaussian copula is widely used for default modelling, so claiming it is limited to market risk is false.

Why the others are wrong

- **A:** A true statement, so not the exception.
- **B:** A true statement, so not the exception.
- **C:** A true statement, so not the exception.

Q513 BT answers P2.T5.505.3 · BT notes P2.T5.505.3

REBUILT — Source required a spreadsheet and had OCR-damaged options; kept data, stated the software step, and made distractors realistic errors.

The marginal one-year default probabilities of asset A and asset B are 5.0% and 7.0%. The Gaussian copula correlation is 0.30. Using a bivariate standard normal CDF (software), which value is closest to the joint one-year default probability?

A. 0.105%

B. 0.350%

C. 0.930%

D. 2.018%

Answer: C

Step by step

1. $N^{-1}(0.05) = -1.6449$, $N^{-1}(0.07) = -1.4758$.
2. Joint PD $= M_2(-1.6449, -1.4758; 0.30) = 0.00930 = 0.930\%$ (computed numerically).
3. Check: independence gives 0.05 × 0.07 = 0.350%; positive correlation raises it above that, consistent with 0.930%.

Why C is correct: The joint PD is the bivariate normal CDF at the two mapped quantiles with ρ = 0.30, which exceeds the independent value of 0.350%.

Why the others are wrong

- **A:** Computes $\rho \times P_A \times P_B = 0.30 \times 0.35\%$, mistakenly scaling the independence product by the correlation.
- **B:** Multiplies the marginals (independence), ignoring ρ.
- **D:** Uses the linear default-correlation formula $0.35\% + 0.30\sqrt{0.05 \cdot 0.95 \cdot 0.07 \cdot 0.93}$, treating the copula correlation as a default correlation.

Q514 BT notes P2.T5.22.13.2

According to Meissner, the Gaussian copula is popular due to its convenient properties. In addition to convenience, what is true about the Gaussian copula?

A. The asymptotic tail dependence is high

B. The univariate marginal distributions joined are normal

C. It requires a single scalar (e.g., average) correlation because it cannot accept a correlation matrix

D. It can be utilized in the default-time copula to model the joint probability of default within various (cumulative) time periods

Answer: D

Step by step

1. Recall the Gaussian copula properties: it joins arbitrary marginals through a multivariate standard normal with correlation matrix ρ_M.
2. The marginals (e.g., cumulative default probabilities $Q_i(t)$) are mapped percentile-to-percentile via N^{-1}, so they need not be normal.
3. Because $Q_i(t)$ is defined for any horizon t, the copula gives joint default probabilities over various cumulative periods and underlies the default-time copula.
4. Its asymptotic tail dependence is zero, a known weakness in stress periods.

Why D is correct: The Gaussian copula maps any cumulative default probability term structure to a standard normal, so it can produce joint default probabilities for any horizon and drives the default-time copula.

Why the others are wrong

- **A:** Wrong: the Gaussian copula has zero asymptotic tail dependence (a weakness), not high; confuses it with a t-copula.
- **B:** Wrong: the marginals are arbitrary (unknown) distributions; only the transformed variables are standard normal.
- **C:** Wrong: it accepts a full correlation matrix ρ_M; a single average correlation is a simplification, not a requirement.

Q515

Mock 2 Q35

A portfolio strategist is training a junior analyst recently hired by her fund. The strategist would like the analyst's help with applying copulas, but she realizes this is not a concept that the analyst has worked with in the past. To educate the analyst on both correlation copulas and Gaussian copulas, the strategist is least likely to use which of the following descriptions?

A. A copula correlation model is designed to preserve the original marginal distributions while defining a correlation between them.

B. A Gaussian copula maps the marginal distribution of each variable to a normal distribution with a mean and standard deviation equal to one.

C. The purpose of a copula is to build a joint probability distribution between two or more variables and maintain those variables' individual marginal distributions.

D. A correlation copula will convert two or more unknown distributions that have distinct shapes and map them to a known distribution with well-defined properties.

Answer: B

Step by step

1. Check each statement against the definition: a Gaussian copula maps each marginal, percentile to percentile, to the standard normal distribution.
2. The standard normal has mean 0 and standard deviation 1.
3. Option B says mean and standard deviation both equal one, so it is inaccurate.
4. A, C and D correctly describe preserving marginals, building a joint distribution, and mapping to a known distribution.

Why B is correct: The standard normal target has mean 0 (not 1) and standard deviation 1, so B is the misdescription the strategist should avoid.

Why the others are wrong

- **A:** Accurate: copulas keep the marginals and add a dependence structure; not the least likely description.
- **C:** Accurate (Sklar): joint distribution built from marginals plus dependence while retaining the marginals.
- **D:** Accurate: unknown marginals of different shapes are mapped to a known distribution (standard normal) with well-defined properties.

Q516

Schweser Module Quiz 10.1 Q2

Which of the following statements best describes a Gaussian copula?

A. A major disadvantage of a Gaussian copula model is the transformation of the original marginal distributions in order to define the correlation matrix.

B. The mapping of each variable to the new distribution is done by defining a mathematical relationship between marginal and unknown distributions.
C. A Gaussian copula maps the marginal distribution of each variable to the standard normal distribution.
D. A Gaussian copula is seldom used in financial models because ordinal numbers are required.

Answer: C
Step by step
1. Definition: observations from each (unknown) marginal are mapped to the standard normal on a percentile-to-percentile basis.
2. The correlation structure is then defined on the transformed normals.
3. Only C states this; A, B, D misdescribe the method or its use.

Why C is correct: The Gaussian copula transforms each marginal, percentile by percentile, to a standard normal and then specifies correlation among the normals.
Why the others are wrong
- **A:** Wrong: the transformation is the feature that makes the correlation matrix definable, not a disadvantage.
- **B:** Wrong: the mapping is percentile-to-percentile, not a mathematical relationship between marginal and 'unknown' distributions.
- **D:** Wrong: the Gaussian copula is widely used in finance (e.g., credit portfolio default modeling); no ordinal-number requirement.

Q517 Schweser Module Quiz 10.1 Q3

A Gaussian copula is constructed to estimate the joint default probability of two assets within a one-year time period. Which of the following statements regarding this type of copula is incorrect?
A. This copula requires that the respective cumulative default probabilities are mapped to a bivariate standard normal distribution.
B. This copula defines the relationship between the variables using a default correlation matrix, ρ_M.
C. The term $N^{-1}(Q_i(t))$ maps each individual cumulative default probability for asset i for time period t on a percentile-to-percentile basis.
D. This copula is a common approach used in finance to estimate joint default probabilities.

Answer: B
Step by step
1. For two assets the copula is $C_{GD}[Q_1(t), Q_2(t)] = M_2[N^{-1}(Q_1(t)), N^{-1}(Q_2(t)); \rho]$.
2. The bivariate normal M_2 needs a single correlation coefficient ρ; the matrix ρ_M notation applies to the n-asset case.
3. So B, which cites a default correlation matrix, is the statement the reading treats as incorrect.
4. A, C, D match the construction.

Why B is correct: With only two assets the copula uses a single default correlation ρ; the matrix ρ_M belongs to the multi-asset version.
Why the others are wrong
- **A:** Correct statement (the bivariate standard normal M_2 is used), so not the answer.
- **C:** Correct: $N^{-1}(Q_i(t))$ is the percentile-to-percentile mapping to the standard normal.
- **D:** Correct: the Gaussian copula is the common industry approach to joint default probability.

Q518 Schweser Module Quiz 10.1 Q5

Suppose a risk manager owns two non-investment-grade assets B and C and has determined their individual default probabilities for the next five years. Which of the following equations best defines how a Gaussian copula is constructed by the risk manager to estimate the joint probability of these two companies defaulting within the next year, assuming a Gaussian default correlation of 0.35?
A. $C_{GD}[Q_B(t), Q_C(t)] = M_2[N^{-1}(Q_B(t)), N^{-1}(Q_C(t)); \rho]$
B. $C[G_1(u_1), \ldots, G_n(u_n)] = F_n[F_1^{-1}(G_1(u_1)), \ldots, F_n^{-1}(G_n(u_n)); \rho_F]$
C. $C_{GD}[Q_1(t), \ldots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \ldots, N^{-1}(Q_n(t)); \rho_M]$
D. $M_n(\cdot) = Q_i(\tau_i)$

Answer: A

Step by step

1. Two assets with one correlation (0.35) call for the bivariate standard normal M_2.
2. Each cumulative default probability for t = 1 year is mapped by N^{-1} to a standard normal quantile.
3. So the joint default probability is $M_2[N^{-1}(Q_B(1)), N^{-1}(Q_C(1)); 0.35]$, option A.
4. B is the generic copula definition; C is the n-asset form; D is not a copula equation.

Why A is correct: Option A applies the Gaussian default copula to exactly two assets with a single correlation coefficient.

Why the others are wrong

- **B:** Generic copula definition with generic distribution F and ρ_F; not the Gaussian default copula for these assets.
- **C:** The n-asset version with correlation matrix ρ_M; not specific to the two-asset question.
- **D:** Links a default time to a default probability ($Q_i(\tau_i)$), not a joint-probability copula equation.

LO 10.c

Summarize the process of finding the default time of an asset correlated to all other assets in a portfolio using the Gaussian copula.

Q519

AnalystPrep Q.1593

REBUILT — Original option D ('any of the components') made the item a giveaway/ill-posed; rewrote as a clear sequence-of-steps question with one correct procedure.

A risk manager wants to simulate the default time of asset i, which is correlated with the other assets in a portfolio, using the Gaussian default time copula. Which procedure is correct?

A. Draw independent uniform random numbers for each asset and convert each to a default time using its own default probability curve, ignoring the correlation matrix.

B. Use a Cholesky decomposition of the correlation matrix to draw a correlated sample $M_n(\bullet)$ from the n-variate standard normal distribution (a value in [0,1]), equate it to the cumulative default probability $Q_i(\tau)$, solve for τ, and repeat over many samples.

C. Compute the expected default time analytically in closed form by solving $Q_i(\tau) = M_n(\bullet)$ once, using the average pairwise correlation.

D. Take N^{-1} of the cumulative default probability at a single fixed horizon and record a default if one correlated draw falls below it; the horizon is the default time.

Answer: B

Step by step

1. The default time of asset i depends on its correlation with all other assets, so the draw must come from the n-variate standard normal distribution with correlation matrix ρ_M.
2. A Cholesky decomposition of ρ_M converts independent normal draws into correlated ones, giving a sample $M_n(\bullet) \in [0, 1]$.
3. Equate the sample to the market-implied cumulative default probability: $M_n(\bullet) = Q_i(\tau_i)$, and solve for τ_i (Excel or Newton-Raphson).
4. There is no closed-form solution, so repeat with many random samples (e.g., 100,000) to estimate the default time distribution.

Why B is correct: The correlated sample is generated via Cholesky from the correlation matrix and mapped to a default time through $Q_i(\tau)$; because no closed form exists, the process is repeated by simulation.

Why the others are wrong

- **A:** Independent uniforms ignore the correlation matrix, so the dependence the copula is meant to capture is lost.
- **C:** There is no closed-form solution and an average correlation does not preserve the full matrix; random sampling is required.
- **D:** This is the fixed-horizon joint-default logic (a yes/no default by a given date), not the mapping of a sampled probability to a default time τ.

Q520 AnalystPrep Q.1594

REBUILT — Original distractors ('N-variate matrix', 'average matrix') were nonsense giveaways; rewrote with plausible distractors and clearer wording.

In the Gaussian default time copula $C_{GD}[Q_1(t), \ldots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \ldots, N^{-1}(Q_n(t)); \rho_M]$, which input supplies the default dependence between asset i and the other assets?

A. The vector of cumulative default probabilities $Q_i(t)$, because higher default probabilities imply higher default correlation.

B. The vector of simulated default times τ_i, which are fed into M_n as inputs.

C. The Cholesky-decomposed matrix of hazard rates for each asset.

D. The correlation matrix ρ_M of the n-variate standard normal distribution, containing the default correlations of asset i with all n assets.

Answer: D

Step by step

1. Each $Q_i(t)$ is first mapped to a standard normal via N^{-1}, which only fixes the marginal distribution.
2. The joint dependence is introduced solely through the correlation matrix ρ_M of the n-variate standard normal M_n.
3. Default times τ_i are outputs of the simulation (via $M_n(\bullet) = Q_i(\tau_i)$), not inputs.

Why D is correct: Marginals come from $Q_i(t)$; the dependence structure of the copula is carried entirely by ρ_M.

Why the others are wrong

- **A:** $Q_i(t)$ are the marginal inputs; they set each asset's individual default probability, not the dependence.
- **B:** Default times are the result of the simulation, not an input to M_n.
- **C:** Hazard rates are not an input to the copula; Cholesky decomposition is applied to ρ_M (not hazard rates) to generate correlated samples.

Q521 AnalystPrep Q.1597

To derive the default times of a large number of correlated assets in a simulation, the sample of $M_n(\bullet)$ from the multivariate copula is found through which of the following?

A. The copula decomposition

B. The normal decomposition

C. The cumulative decomposition

D. The Cholesky decomposition

Answer: D

Step by step

1. Independent standard normal draws must be transformed so that they carry the correlation matrix ρ_M.
2. The Cholesky decomposition factors ρ_M into a lower triangular matrix A with $AA^T = \rho_M$; multiplying independent normals by A gives correlated normals.
3. Applying the normal CDF gives the sample $M_n(\bullet) \in [0, 1]$, which is mapped to default times via $M_n(\bullet) = Q_i(\tau_i)$.

Why D is correct: Cholesky decomposition of the correlation matrix is the standard technique, per the reading, for sampling $M_n(\bullet)$ when many assets are involved.

Why the others are wrong

- **A:** 'Copula decomposition' is not a sampling technique for correlated normals.
- **B:** 'Normal decomposition' is not a standard method; it would not impose the correlation matrix.
- **C:** 'Cumulative decomposition' is not a method; cumulative probabilities are what $M_n(\bullet)$ is equated to, not how it is sampled.

Q522 BT answers P2.T6.24.21.2 · BT notes P2.T6.24.21.2

You are a risk analyst tasked with simulating default scenarios for a portfolio of 10 companies over the next 5 years using the Gaussian copula model. The cumulative probabilities of default for each company over the next 1, 2, 3, 4, and 5 years are 1%, 3%, 6%, 10%, and 15%, respectively, with a default correlation of 0.2 between each pair of companies. Using the Gaussian copula model, how would you determine if a default occurs within the first year for any company?

A. Simulate a value from a multivariate normal distribution for each company. If the value is less than the inverse cumulative normal distribution value at 1% (-2.33), default occurs within the first year.

B. Calculate the cumulative probability for each company independently without using a copula. If the probability

is less than 1%, assume default occurs within the first year.

C. Use a single-factor model to compute default probability without transforming the values into a normal distribution. Default is assumed if the factor exceeds the threshold set for the first year.

D. Sum the individual default probabilities; if the total exceeds 10%, conclude that a default occurs within the portfolio's first year.

Answer: A

Step by step

1. Map the 1-year cumulative default probability to a normal threshold: $N^{-1}(1\%) = -2.326 \approx -2.33$.
2. Draw a correlated sample from the 10-variate standard normal with pairwise correlation 0.2 (via Cholesky decomposition).
3. Company j defaults within year 1 if its sampled normal value is below -2.33; compare with the 3%, 6%, 10%, 15% thresholds for later years.
4. Repeat over many simulations to estimate default frequencies and joint defaults.

Why A is correct: The Gaussian copula transforms cumulative default probabilities to normal thresholds and compares them with correlated normal draws, so default correlation is reflected.

Why the others are wrong

- **B:** Treating each company independently without a copula ignores the 0.2 default correlation, understating joint defaults.
- **C:** Skips the transformation to the normal distribution that defines the Gaussian copula; a threshold on an untransformed factor is not the copula method.
- **D:** Summing probabilities (10 × 1% = 10%) is not a default-time test for any company and ignores correlation; it has no basis in the copula framework.

Q523 Schweser Module Quiz 10.1 Q4

A risk manager is trying to estimate the default time for asset i based on the default copula correlation of asset i to n assets. Which of the following equations best defines the process that the risk manager should use to generate and map random samples to estimate the default time?

A. $C_{GD}[Q_B(t), Q_C(t)] = M_2[N^{-1}(Q_B(t)), N^{-1}(Q_C(t)); \rho]$

B. $C[G_1(u_1), \ldots, G_n(u_n)] = F_n[F_1^{-1}(G_1(u_1)), \ldots, F_n^{-1}(G_n(u_n)); \rho_F]$

C. $C_{GD}[Q_i(t), \ldots, Q_n(t)] = M_n[N^{-1}(Q_1(t)), \ldots, N^{-1}(Q_n(t)); \rho_M]$

D. $M_n(\bullet) = Q_i(\tau_i)$

Answer: D

Step by step

1. Draw a random sample $M_n(\bullet) \in [0, 1]$ from the n-variate standard normal distribution that embeds asset i's default correlations (via Cholesky).
2. Equate the sample to asset i's cumulative default probability: $M_n(\bullet) = Q_i(\tau_i)$.
3. Solve for τ_i (Excel or Newton-Raphson) and repeat many times to estimate the default time.

Why D is correct: $M_n(\bullet) = Q_i(\tau_i)$ is the equation that maps each random sample to a default time; the others define copulas, not the mapping step.

Why the others are wrong

- **A:** This is the bivariate Gaussian default time copula for two assets (B and C), giving joint default probability, not a default time for asset i.
- **B:** This is the general copula definition mapping marginals G to a joint distribution F, not the default-time mapping.
- **C:** This is the n-asset Gaussian default time copula definition; it defines the dependence structure but not how a sample is mapped to τ_i.

Reading 11. Regression Hedging and Principal Component Analysis

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 4th Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 6.

LO	Learning objective	Qs
11.a	explain the drawbacks to using a DV01-neutral hedge for a bond position.	4
11.b	describe a regression hedge and explain how it can improve a standard DV01-neutral hedge.	6
11.c	calculate the regression hedge adjustment factor, beta.	5
11.d	calculate the face value of an offsetting position needed to carry out a regression hedge.	7
11.e	calculate the face value of multiple offsetting swap positions needed to carry out a two- variable regression hedge.	3
11.f	compare and contrast level and change regressions.	5
11.g	explain why and how a regression hedge differs from a hedge based on a reverse regression.	5
11.h	describe principal component analysis and explain how it is applied to constructing a hedging portfolio.	5

LO 11.a

Explain the drawbacks to using a DV01-neutral hedge for a bond position.

Q524

AnalystPrep Q.6502

A large investment fund uses a DV01-neutral hedge to manage the interest rate exposure of its bond portfolio. Over time, market dynamics have shown this approach to be unrealistic. What is a critical drawback of relying on a DV01-neutral hedge?

A. It captures non-parallel yield curve shifts across multiple bond maturities.
B. The strategy inherently assumes that all spread-related risks are neutralized.
C. It assumes yield changes are parallel across maturities, which is rarely the case.
D. The accuracy of the hedge greatly improves with non-static historical correlations.

Answer: C

Step by step

1. A DV01-neutral hedge sizes the offsetting position so that DV01 of the hedge equals DV01 of the position.
2. This only works if the yield on the position and the yield on the hedging instrument change by the same number of basis points (a one-for-one, parallel move).
3. In practice yields at different maturities (or nominal vs. real yields) do not move one-for-one, so the hedge leaves residual risk.
4. Therefore the drawback is the assumption of parallel yield changes (C).

Why C is correct: DV01 neutrality presumes identical basis-point yield changes on the bond and the hedge; real yield changes differ across maturities, so the hedge is imprecise.

Why the others are wrong

- **A:** This is the opposite of the truth: a DV01 hedge cannot capture non-parallel shifts; that is its weakness.
- **B:** DV01 hedging addresses only rate (yield) risk; spread and other risks are not neutralized. Confuses the hedge with a full-risk hedge.
- **D:** A DV01 hedge uses no historical correlations at all; time-varying relationships are a concern for regression hedges, and 'improves' is the wrong direction anyway.

Q525

AnalystPrep Q.6503

REBUILT — Source keyed B (static historical correlations), which describes a regression hedge limitation, not a DV01-neutral hedge; rewrote the stem and options so exactly one option states the true DV01 limitation.

A trader hedges a corporate bond position with a DV01-neutral position in another fixed-income instrument but finds that the hedge performs poorly when market conditions change. Which of the following is the fundamental limitation of the DV01-neutral approach?

A. It assumes the yields on the position and the hedging instrument change by the same number of basis points, whereas the actual changes can be proportionally larger or smaller and dispersed.
B. It requires estimating a regression of historical yield changes, which may be unstable over time.
C. It accounts for convexity and so over-hedges when yields change by large amounts.
D. It can be implemented only with complex derivative instruments, which limits its use.

Answer: A

Key correction: Source key (AnalystPrep) said B (static historical correlations); that is a limitation of regression hedging. A DV01-neutral hedge uses no historical data and instead assumes a one-for-one yield change, so the original item had no clean correct option.

Step by step

1. DV01-neutral hedge: hedge DV01 equals position DV01.
2. This is exact only if both yields move by the same number of basis points (beta = 1).
3. In reality the relationship differs (e.g., nominal yield moves by more than 1 bp per 1 bp of real yield, with dispersion), leaving the position under- or over-hedged.
4. Option A states this; the others describe regression hedging, convexity or instrument type, none of which is the DV01 drawback.

Why A is correct: The DV01 hedge hard-wires a one-for-one yield change assumption; when actual yield changes differ in size, the hedge is imprecise.

Why the others are wrong

- **B:** Estimating a regression on historical data is the improvement (regression hedge), not the DV01 hedge; its reliance on past relationships is a regression-hedge caveat.
- **C:** DV01 is a first-order measure and ignores convexity; it does not include convexity adjustments.
- **D:** A DV01 hedge is simple and can be done with ordinary bonds or swaps; no complex derivatives are needed.

Q526

Schweser Module Quiz 11.1 Q1

If a trader is creating a fixed-income hedge, which hedging methodology would be least effective if the trader is concerned about the dispersion of the change in the nominal yield for a particular change in the real yield?

A. One-variable regression hedge.
B. DV01 hedge.
C. Two-variable regression hedge.
D. Principal components hedge.

Answer: B

Step by step

1. The concern is that nominal yield changes are not one-for-one with real yield changes and are dispersed.
2. A DV01 hedge assumes both yields change by the same number of basis points, so it ignores this relationship.
3. Regression hedges (one- or two-variable) estimate the empirical relationship (beta) from historical yield changes, and PCA hedges use the empirical factor structure of yield changes.
4. So the DV01 hedge is least effective (B).

Why B is correct: Only the DV01 hedge hard-wires a one-for-one yield change; the other methods are calibrated to observed yield-change relationships.

Why the others are wrong

- **A:** A one-variable regression hedge adjusts the DV01 hedge by the estimated beta, directly addressing the nominal/real relationship.
- **C:** A two-variable regression hedge uses estimated sensitivities to two instruments, so it also reflects empirical relationships.
- **D:** A principal components hedge is built from the empirical factor structure of yield changes, not a one-for-one assumption.

Q527

BT answers L2.T5.32.1

REBUILT — Original asked about key rate shifts vs DV01, which no 2026 MR LO covers; rebuilt to test the 11.a drawback of a DV01-neutral hedge.

A trader hedges a long position in a nominal Treasury bond with TIPS, using a standard DV01-neutral hedge. Which of the following is the main drawback of this hedge?

A. It assumes the nominal yield and the real yield change by the same number of basis points, whereas empirically the nominal yield tends to move by more than one basis point per basis point of real-yield change.
B. It requires the hedger to estimate the historical volatility of yield differences, which makes it harder to compute than a regression hedge.
C. It can only be applied when the bond and the hedging instrument have identical maturities.
D. It ignores the face value of the hedging instrument, so the hedge cannot be sized.

Answer: A

Step by step

1. A DV01-neutral hedge sizes the offsetting position so that DV01 of the hedge equals DV01 of the bond, implicitly assuming a one-for-one change in the two yields ($\Delta y_{hedge} = \Delta y_{bond}$, i.e. $\beta = 1$).
2. Nominal and real yields do not move one-for-one; empirically the nominal yield changes by more than 1 bp per 1 bp change in the real yield, and the relationship is noisy.
3. Hence the hedge is imprecise; a regression hedge corrects this by adjusting the DV01 hedge with the estimated β from historical yield changes.

Why A is correct: The DV01-neutral hedge presumes parallel, one-to-one yield changes between the position and the hedge; when yields have a different sensitivity (beta not equal to 1) the hedge is imprecise.

Why the others are wrong

- **B:** Reverses the comparison: the DV01 hedge needs no volatility estimation; it is the regression hedge that uses historical data.
- **C:** DV01 hedges are routinely used with different maturities and instruments (that is exactly the TIPS example); no such restriction exists.
- **D:** The face value of the hedge is precisely what the DV01 calculation solves for, so sizing is not the problem.

LO 11.b

Describe a regression hedge and explain how it can improve a standard DV01-neutral hedge.

Q528

AP Mock 2 Q15

REBUILT — Original distractors were vague and the correct option overclaimed ('ensuring'); rewrote stem and options so exactly one is clearly correct.

A fixed-income analyst currently hedges a long bond position with a DV01-neutral hedge, which implicitly assumes that the hedging instrument's yield and the bond's yield change by the same number of basis points. How does a regression hedge improve on this approach?

A. It sets the hedge amount using the historical relationship between the two yield changes (the regression slope), rather than assuming a one-for-one move.

B. It replaces the historical data with a forecast of tomorrow's yield changes, so the hedge is always exactly right.

C. It eliminates basis risk completely, because the regression residual is forced to zero.

D. It sizes the hedge from the average historical price of the two securities rather than from yield changes.

Answer: A

Step by step

1. DV01-neutral: F_hedge = F_exp × DV01_exp / DV01_hedge, which assumes Δy_exp = Δy_hedge.
2. Regression: $\Delta y_{exp} = \alpha + \beta\, \Delta y_{hedge} + \varepsilon$, estimated by least squares on historical yield changes.
3. The hedge is scaled by β: F_hedge = F_exp × (DV01_exp / DV01_hedge) × β, reflecting actual historical co-movement.
4. The R-squared and standard error indicate how much risk is removed and how much residual (basis) risk remains.

Why A is correct: The regression hedge adjusts the DV01-neutral hedge by the historical slope β of exposure yield changes on hedge yield changes, so it reflects how the yields have actually co-moved.

Why the others are wrong

- **B:** The method uses history, not forecasts, and it cannot guarantee an exactly right hedge.
- **C:** The residual (basis risk) remains; the standard error measures it. The regression only minimises its variance.
- **D:** Regression is on yield changes, not price averages; averaging prices is not the method.

Q529

AnalystPrep Q.6505

REBUILT — Original correct option was loosely worded and the distractors were giveaways; restated around beta as the risk weight with plausible alternatives.

Compared with a DV01-neutral hedge, which of the following best describes the 'risk weight' applied to the hedging instrument in a single-variable regression hedge?

A. It is fixed at 1.0 because the hedge is built to be DV01-neutral.

B. It equals the regression slope coefficient β from historical yield changes, which can differ from 1.0.

C. It equals the regression R-squared, the proportion of variance explained.

D. It equals the regression standard error, expressed in basis points.

Answer: B

Step by step

1. The DV01-neutral hedge implicitly uses a risk weight of 1 (equal yield changes).
2. In $\Delta y_{exp} = \alpha + \beta\,\Delta y_{hedge} + \varepsilon$, β is the historical bp move in the exposure yield per 1 bp move in the hedge yield.
3. The regression hedge multiplies the DV01-neutral face amount by β, so β is the risk weight (e.g., $\beta = 0.75$ cuts the hedge by 25%).

Why B is correct: The regression slope β is the empirical risk weight; the DV01-neutral hedge is the special case $\beta = 1$.

Why the others are wrong

- **A:** A weight of 1.0 describes the DV01-neutral hedge, not the regression hedge.
- **C:** R-squared measures the share of variance explained and is not used to scale the hedge.
- **D:** The standard error measures residual dispersion and drives hedged P&L volatility, not the hedge size.

Q530

BT Mock Q31 · BT notes P2.T5.22.14.1

In comparison to a yield-based DV01 hedge, which of the following is the best argument in favor of a regression hedge?

A. The regression hedge minimizes the profit and loss (P&L) variance of the hedged position.

B. The regression hedge is based on economic theory and future developments rather than empirical analysis.

C. The regression hedge does not require any out-of-sample tests, which are often unreliable if the sample is small.

D. The regression hedge solves for a risk-weight which qualifies as a valid risk-weighted asset under Basel regulations.

Answer: A

Step by step

1. Least squares chooses β to minimise the sum of squared residuals.
2. With exposure yield changes regressed on hedge yield changes, the residual is the hedged-position P&L (per unit DV01).
3. Hence the regression hedge minimises the variance of hedged P&L, while the DV01 hedge imposes a one-for-one yield move.

Why A is correct: OLS minimises residual variance, and the residual is the hedged P&L, so the regression hedge is the minimum-variance hedge given the sample.

Why the others are wrong

- **B:** The method is empirical (historical data), not theory- or forecast-based.
- **C:** Out-of-sample tests are, if anything, more important because beta may be unstable.
- **D:** The risk weight is the regression beta; it has nothing to do with Basel risk-weighted assets.

Q531

BT notes P2.T5.321.2

A trader shorts $100.0 million of nominal US Treasury 3 5/8s bonds and, employing a single-variable regression hedge, buys a face amount of 1 7/8s US TIPS. The bonds and the results of a regression of the nominal against the real yield are:

US T-Bonds	Yield	DV01
Nominal 3 5/8s	3.275%	0.0400
TIPS 1 7/8s	1.237%	0.0650

Least-Squares Regression Analysis: Δ in Nominal Yield (bps) versus Δ in Real Yield (bps)

No. of Observations	229	
R-Squared	68.0%	
Standard Error (bps)	5.300	
Regression Coeff	Value	Std Error
Constant (α)	0.0200	0.2529
Change in Real Yield (β)	1.0940	0.0595

Which is nearest to an estimate of the daily volatility of the P&L of the hedged portfolio?

A. $2,380

B. $43,760

C. $67,323

D. $212,000

Answer: D

Step by step

1. Dollar DV01 of the $100.0 million nominal position = 100,000,000 × 0.0400/100 = $40,000 per bp.
2. Hedged P&L volatility = $ DV01 × regression standard error = $40,000 × 5.300 bp = $212,000 per day.
3. Beta, R-squared and alpha size the hedge or describe fit; they are not multiplied in for the P&L volatility.

Why D is correct: The regression standard error is the daily standard deviation (in bp) of the nominal-yield change not explained by the real-yield change; multiplying by the position's dollar DV01 gives the hedged P&L volatility.

Why the others are wrong

- **A:** $40,000 × 0.0595 = $2,380 uses the standard error of beta instead of the regression standard error.
- **B:** $40,000 × 1.094 = $43,760 multiplies by beta, which sizes the hedge rather than measuring residual volatility.
- **C:** About $67.3 million (67,323 thousand) is the TIPS face amount to buy, 100 × (0.04/0.065) × 1.094; it is a hedge size, not a P&L volatility.

Q532

Schweser Module Quiz 11.1 Q3

What is a key advantage of using a regression hedge to fine-tune a DV01 hedge?

A. It assumes that term structure changes are driven by one factor.
B. The proper hedge amount may be computed for any assumed change in the term structure.
C. Bond price changes and returns can be estimated with proper measures of price sensitivity.
D. It gives an estimate of the hedged portfolio's volatility over time.

Answer: D

Step by step

1. The regression of exposure yield changes on hedge yield changes produces a residual standard error.
2. That standard error, multiplied by the position's dollar DV01, estimates the volatility of the hedged P&L.
3. So the trader can compare the expected gain with historical volatility; none of the other options is a feature of regression hedging.

Why D is correct: The regression framework automatically yields a standard error, hence an estimate of the volatility of the hedged portfolio.

Why the others are wrong

- **A:** A one-factor assumption is a limitation shared with DV01 hedging, not an advantage.
- **B:** The regression uses historical data and cannot give a proper hedge for any assumed term structure change.
- **C:** This describes duration/DV01 price-sensitivity measures in general, not a benefit specific to regression hedging.

Q533

Schweser Module Quiz 11.1 Q4

REBUILT — Original options were vaguely worded (tracks nominal rates, volatile like both); rewrote so one option states constant beta and the others are realistic misconceptions.

A trader uses a regression of changes in nominal yields on changes in real yields to set a single-variable regression hedge. Which assumption about the slope coefficient (beta) underlies the hedge?

A. Beta is constant over time, so the historical estimate is assumed to hold over the hedging horizon.
B. Beta equals 1.0, so the hedge is the same as the DV01-neutral hedge.
C. Beta is zero, so real yield changes do not affect nominal yields.
D. Beta changes each day to match the latest yield move.

Answer: A

Step by step

1. The hedge face amount is scaled by the estimated β from historical data.
2. Using that estimate for the future presumes the relationship between yield changes is stable, i.e., β is constant.
3. In practice β varies, so it should be estimated over several periods and compared.

Why A is correct: The regression hedge applies a historical β going forward, assuming that beta is stable over time.

Why the others are wrong

- **B:** Beta of 1 is the DV01-neutral special case; the regression estimate generally differs from 1.
- **C:** Beta of zero would imply no relation and no point in hedging with the TIPS.
- **D:** A daily-changing beta would make historical estimation useless; stability is the assumption.

LO 11.c

Calculate the regression hedge adjustment factor, beta.

Q534

AnalystPrep Q.6508

REBUILT — Source table implied covariance 0.4167 but stem stated 0.273 (inconsistent); replaced with new self-consistent data and asked the candidate to compute covariance and variance from the table.

A portfolio manager hedges a corporate bond portfolio with a hedging instrument using a regression hedge. Daily changes in yield (in bps) over four days are:

Day	Yp (Corporate Bond Portfolio)	Yh (Hedging Instrument)
1	1.2	2.0
2	1.8	2.5
3	0.8	1.5
4	2.4	3.0

Using sample (n − 1) statistics, what is the regression hedge adjustment factor, β, of the portfolio yield change on the hedge yield change?

A. 0.81
B. 0.92
C. 1.08
D. 1.44

Answer: C

Key correction: Source key (A, 0.66) relied on a stated covariance of 0.273 inconsistent with the table; the item was rebuilt with consistent data.

Step by step

1. Means: $\bar{Y}_h = 2.25$, $\bar{Y}_p = 1.55$.
2. Deviations of Yh: −0.25, 0.25, −0.75, 0.75; of Yp: −0.35, 0.25, −0.75, 0.85.
3. Sample covariance = (0.0875 + 0.0625 + 0.5625 + 0.6375)/3 = 1.35/3 = 0.45.
4. Sample variance of Yh = (0.0625 + 0.0625 + 0.5625 + 0.5625)/3 = 1.25/3 = 0.4167.
5. $\beta = \text{Cov}(Y_p, Y_h)/\text{Var}(Y_h) = 0.45/0.4167 = 1.08$.

Why C is correct: The regression hedge adjustment factor is the slope of the regression of the position's yield change on the hedge's yield change, $\text{Cov}(Y_p, Y_h)/\text{Var}(Y_h)$.

Why the others are wrong

- **A:** Mixes the population covariance (1.35/4 = 0.3375) with the sample variance (0.4167), giving 0.81; the n and n − 1 conventions must match.
- **B:** Reverse regression: divides the covariance by Var(Yp) = 0.49 instead of Var(Yh), giving 0.92.
- **D:** Divides the sample covariance (0.45) by the population variance of Yh (1.25/4 = 0.3125), giving 1.44; conventions are mismatched.

Q535

BT notes P2.T5.22.14.3

Erica works for a market maker who is going to hedge the purchase of Johnson & Johnson (JNJ) bonds by selling Treasury bonds. To compare a yield-based DV01 hedge to a regression hedge, Erica conducts a single-variable regression; the summary results are:

Least-Squares Regression

No. of Observations	82	
R-Squared	77.5%	
Standard Error	2.10	
Regression Coeff	Value	Std Error
Constant (α)	0.060	0.223
Change in 30y Treasury Yield (β)	0.842	0.193

Erica is interested in the advisability of a yield-based DV01 hedge, which assumes parallel shifts in the yields of the two bonds. Her hypothesis test assumes a two-sided 95.0% confidence level. Can she accept the null hypothesis of parallel shifts in yields of the two bonds?

A. She rejects the hypothesis of parallel shifts in the yields of the two bonds because the true beta is not one
B. She accepts (i.e., cannot reject) the hypothesis of parallel shifts in the yields because the true beta might be one
C. She rejects the hypothesis of parallel shifts in the yields of the two bonds because the alpha is not zero

D. From a risk-return perspective, the trade is not advised because the standard error of the regression exceeds two

Answer: B

Step by step

1. A DV01 hedge implicitly assumes a regression beta of 1, so H0: $\beta = 1$.
2. $t = (0.842 - 1)/0.193 = -0.819$.
3. Critical t (two-sided 95%, 80 df) ≈ 1.99, and $|-0.819| < 1.99$; equivalently the 95% interval $0.842 \pm 1.99 \times 0.193 = [0.458, 1.226]$ contains 1.
4. Cannot reject H0: a beta of one is statistically consistent with the data.

Why B is correct: Parallel shifts correspond to beta = 1, and 1 lies well within the confidence interval of the estimated beta, so the null cannot be rejected.

Why the others are wrong

- **A:** Concludes beta is not one just because the point estimate is 0.842; ignores sampling error (standard error 0.193).
- **C:** Tests the wrong coefficient; the intercept is insignificant (0.060/0.223 = 0.27) and in any case parallel shifts concern beta, not alpha.
- **D:** The regression standard error is a measure of fit in yield units, not a test of the beta = 1 hypothesis, and 'exceeds two' is no decision criterion.

Q536

BT notes P2.T5.321.3

A trader shorts $100.0 million of nominal US Treasury 3 5/8s bonds and hedges with a purchase of a face amount of 1 7/8s US TIPS based on the regression results of a single-variable hedge, shown here:

US T-Bonds	Yield	DV01
Nominal 3 5/8s	3.275%	0.0400
TIPS 1 7/8s	1.237%	0.0700

Least-Squares Regression Analysis: Δ in Nominal Yield (bps) versus Δ in Real Yield (bps)

No. of Observations	229	
R-Squared	68.0%	
Standard Error (bps)	2.200	
Regression Coeff	Value	Std Error
Constant (α)	0.0200	0.2529
Change in Real Yield (β)	1.3000	0.0510

Each of the following is true about this regression hedge EXCEPT for which is false?

A. The regression model estimates a correlation between changes in the nominal and real yields of about 82.5%

B. We can be 95% confident that the true value of beta is unity (1.0)

C. The daily volatility of the P&L of the hedged portfolio is about $88,000

D. The difference between a DV01-hedge and a regression hedge is about $17.14 million face amount of the TIPS

Answer: B

Step by step

1. Correlation = $\sqrt{0.680} = 0.825$: A true.
2. t for H0: $\beta = 1$ is $(1.30 - 1)/0.051 = 5.88$; 95% CI ≈ $1.30 \pm 1.97 \times 0.051 = [1.20, 1.40]$, which excludes 1: B false.
3. Nominal DV01 position = $100m × 0.0400/100 = $40,000 per bp; × residual SE 2.2 bps = $88,000: C true.
4. DV01 hedge = 100 × 0.04/0.07 = $57.14m TIPS; regression hedge = 1.3 × 57.14 = $74.29m; difference $17.14m: D true.

Why B is correct: The estimated beta of 1.30 is almost six standard errors from one, so the data reject beta = 1; the other three statements follow directly from R^2, DV01 and the standard error.

Why the others are wrong

- **A:** True statement ($\sqrt{0.68} = 0.825$), so it is not the EXCEPT answer.
- **C:** True statement ($40,000 per bp × 2.2 bp = $88,000), so it is not the EXCEPT answer.
- **D:** True statement (74.29 − 57.14 = 17.14m), so it is not the EXCEPT answer.

Q537 AP Mock 3 Q56

REBUILT — Distractors were not products of realistic errors (1.92, 1.50, 1.20); replaced with wrong-formula results (ignoring correlation, inverted ratio, dividing by correlation).

An analyst is calculating the hedge ratio for a bond portfolio using bond futures:

Input	Value
Correlation between portfolio and futures yield changes, ρ	0.60
Standard deviation of bond portfolio yield changes, σp	0.08
Standard deviation of bond futures yield changes, σh	0.05

What is the minimum-variance (regression) hedge ratio, β?

A. 0.38
B. 0.96
C. 1.60
D. 2.67

Answer: B

Step by step

1. $\beta = \rho\,\sigma_p/\sigma_h$.
2. σp/σh = 0.08/0.05 = 1.60.
3. $\beta = 0.60 \times 1.60 = 0.96$.

Why B is correct: The regression beta equals $\text{Cov}(Y_p, Y_h)/\text{Var}(Y_h) = \rho\sigma_p\sigma_h/\sigma_h^2 = \rho\sigma_p/\sigma_h$.

Why the others are wrong

- **A:** Inverted the volatility ratio: ρσh/σp = 0.60 × 0.05/0.08 = 0.375.
- **C:** Used only the volatility ratio σp/σh and ignored the correlation.
- **D:** Divided the volatility ratio by the correlation (1.60/0.60) instead of multiplying.

Q538 AnalystPrep Q.6506

A portfolio manager is hedging the price movements of a corporate bond portfolio (Yp) using a government bond futures contract (Yh). The following data is available:
The correlation between the returns of the corporate bond portfolio and the government bond futures: ρ(Yh, Yp) = 0.8
The standard deviation of the returns of the corporate bond portfolio: σ(Yp) = 0.05
The standard deviation of the returns of the government bond futures: σ(Yh) = 0.04
What is the hedge ratio (beta) that the portfolio manager should use to minimize the portfolio's exposure to interest rate risk?

A. 1.25
B. 1.00
C. 0.80
D. 0.64

Answer: B

Step by step

1. Cov(Yh,Yp) = 0.8 × 0.04 × 0.05 = 0.0016.
2. Var(Yh) = 0.04^2 = 0.0016.
3. $\beta = 0.0016/0.0016 = 1.00$; equivalently ρσp/σh = 0.8 × 1.25 = 1.00.

Why B is correct: The minimum-variance hedge ratio is the regression slope $\text{Cov}(Y_h, Y_p)/\text{Var}(Y_h)$, which here is exactly one.

Why the others are wrong

- **A:** Used the volatility ratio σp/σh = 1.25 and ignored the correlation.
- **C:** Used the correlation alone as the hedge ratio.
- **D:** Inverted the volatility ratio: ρσh/σp = 0.8 × 0.04/0.05 = 0.64.

LO 11.d

Calculate the face value of an offsetting position needed to carry out a regression hedge.

Q539

AnalystPrep Q.6509

REBUILT — Original distractors (128, 85, 110) did not correspond to any realistic error; replaced with results of omitting beta, inverting the DV01 ratio, and dividing by beta.

An asset management firm is establishing a regression hedge for its bond portfolio. The face value of the bond to be hedged is $100 million with a DV01 of 0.300 (per $100 face). The hedging instrument has a DV01 of 0.250 (per $100 face), and the regression hedge adjustment factor, $\hat{\beta}$, is 0.800. What is the face value of the offsetting position required for the regression hedge?

A. -$66.7 million
B. -$96.0 million
C. -$120.0 million
D. -$150.0 million

Answer: B

Step by step

1. Regression hedge: $F_h = -F_p \times \frac{DV01_p}{DV01_h} \times \hat{\beta}$.
2. DV01 ratio: $0.300/0.250 = 1.2$.
3. $F_h = -100 \times 1.2 \times 0.800 = -96.0$ million.

Why B is correct: The face amount that equates DV01s is scaled by the regression beta, which says how many bp the hedged yield moves per bp of the hedge yield; the negative sign means an offsetting (short) position.

Why the others are wrong

- **A:** Inverted the DV01 ratio (0.250/0.300) and then applied beta: 100 × 0.833 × 0.8 = 66.7.
- **C:** Omitted the beta adjustment (plain DV01 hedge): 100 × 1.2 = 120.
- **D:** Divided by beta instead of multiplying: 100 × 1.2 / 0.8 = 150.

Q540

AnalystPrep Q.6510

REBUILT — Original distractors (-7,500; -9,000; -12,000) were not tied to realistic errors; replaced with omit-beta, inverted-ratio and divide-by-beta results.

A bond with face value $10,000 and DV01 of 1,200 is to be hedged with a swap whose DV01 is 1,500 (both DV01s on a consistent per-face basis). The regression hedge adjustment factor, $\hat{\beta}$, is 0.75. What is the face value of the hedging swap position?

A. -$6,000
B. -$8,000
C. -$9,375
D. -$10,667

Answer: A

Step by step

1. $F_h = -F_p \times \frac{DV01_p}{DV01_h} \times \hat{\beta}$.
2. DV01 ratio: $1{,}200/1{,}500 = 0.8$.
3. $F_h = -10{,}000 \times 0.8 \times 0.75 = -6{,}000$ dollars.

Why A is correct: The DV01-neutral swap face (-$8,000) is scaled down by $\hat{\beta}$ = 0.75 because the bond's yield moves only 0.75 bp per bp move in the swap rate.

Why the others are wrong

- **B:** Omitted beta: 10,000 × 0.8 = 8,000 (plain DV01 hedge).
- **C:** Inverted the DV01 ratio before applying beta: 10,000 × 1.25 × 0.75 = 9,375.
- **D:** Divided by beta rather than multiplying: 10,000 × 0.8 / 0.75 = 10,667.

Q541

BT notes P2.T5.22.14.2

REBUILT — Distractors replaced with results of realistic errors (omit beta, inverted ratio, divide by beta); original options were not tied to specific errors.

A market maker purchases $100.0 million face of a 30-year JNJ bond and hedges the interest rate risk by selling 30-year Treasury bonds. Yields and DV01s (per $100 face):

Bond	Yield	DV01
JNJ bond	3.952%	0.3124
Treasury bond	3.764%	0.2910

The market maker regresses the JNJ yield change on the Treasury yield change and obtains intercept α = 0.0240,

slope β = 0.7150, R^2 = 73.4%. Using a single-variable regression hedge, which is nearest to the face amount of Treasury bonds sold?

A. $66.60 million
B. $76.76 million
C. $107.35 million
D. $150.15 million

Answer: B

Step by step

1. Hedge face = F_JNJ × (DV01_JNJ/DV01_T) × β.
2. DV01 ratio = 0.3124 / 0.2910 = 1.07354.
3. Plain DV01 hedge = 100 × 1.07354 = 107.35 million.
4. Regression hedge = 107.35 × 0.7150 = 76.76 million sold.

Why B is correct: Because the JNJ yield moves 0.715 bp per 1 bp move in the Treasury yield, the DV01-neutral hedge is scaled by 0.715. The intercept and R^2 are not used in the face-amount formula.

Why the others are wrong

- **A:** Inverted the DV01 ratio (0.2910/0.3124) and then multiplied by beta: 100 × 0.9315 × 0.715 = 66.60.
- **C:** Ignored beta (plain DV01 hedge): 107.35.
- **D:** Divided by beta instead of multiplying: 107.35 / 0.715 = 150.15.

Q542

BT notes P2.T5.321.1

REBUILT — Distractors replaced with results of realistic errors (omit beta, inverted ratio, divide by beta).

A trader shorts $100.0 million of nominal US Treasury 3 5/8s and hedges with a long in TIPS 1 7/8s. Yields and DV01s (per $100 face), together with a regression of nominal yield changes on real (TIPS) yield changes, are:

Bond	Yield	DV01
Nominal 3 5/8s	3.275%	0.0530
TIPS 1 7/8s	1.237%	0.0770

Regression coefficient on change in real yield (β) = 1.0940 (std error 0.0595); R-squared 56.3%; constant 0.0200. Which is nearest to the face amount of TIPS to be bought for the regression hedge?

A. $62.92 million
B. $68.83 million
C. $75.30 million
D. $158.94 million

Answer: C

Step by step

1. Hedge face = F_nominal × (DV01_nominal/DV01_TIPS) × β.
2. DV01 ratio = 0.0530 / 0.0770 = 0.68831.
3. Plain DV01 hedge = 100 × 0.68831 = 68.83 million.
4. Regression hedge = 68.83 × 1.0940 = 75.30 million of TIPS.

Why C is correct: A 1 bp change in the real yield is associated with a 1.094 bp change in the nominal yield, so more TIPS than the plain DV01 hedge is needed.

Why the others are wrong

- **A:** Divided by beta instead of multiplying: 68.83 / 1.094 = 62.92.
- **B:** Omitted beta (plain DV01 hedge): 68.83.
- **D:** Inverted the DV01 ratio and then multiplied by beta: 100 × (0.0770/0.0530) × 1.094 = 158.94.

Q543

Mock 1 Q70

REBUILT — Options lacked units and distractors (125.37, 186.95) were not tied to specific errors; restated in USD million with error-based distractors.

A hedge fund sells USD 150 million face of U.S. Treasury bonds and buys TIPS as a relative value trade. Yields and DV01s (per 100 face):

Bond	Yield	DV01
TIPS	1.475%	0.092
T-bond	3.325%	0.063

Given a yield beta of 1.012 (nominal yield change on real yield change), what is the face value of TIPS in the regression hedge?

A. USD 102.72 million
B. USD 103.95 million
C. USD 219.05 million
D. USD 221.67 million

Answer: B

Step by step

1. $F_{TIPS} = F_{Tbond} \times (DV01_{Tbond}/DV01_{TIPS}) \times \beta$.
2. DV01 ratio: $0.063/0.092 = 0.68478$.
3. Plain DV01 hedge: $150 \times 0.68478 = 102.72$ million.
4. Regression hedge: $102.72 \times 1.012 = 103.95$ million.

Why B is correct: The TIPS have the larger DV01, so less than USD 150 million face is needed; the beta of 1.012 scales the DV01 hedge up slightly.

Why the others are wrong

- **A:** Omitted beta: 150 × 0.68478 = 102.72.
- **C:** Inverted the DV01 ratio and omitted beta: 150 × 0.092/0.063 = 219.05.
- **D:** Inverted the DV01 ratio and applied beta: 150 × (0.092/0.063) × 1.012 = 221.67.

Q544

Mock 2 Q8

REBUILT — Distractors (25, 49) were not tied to realistic errors; replaced with omit-beta, inverted-ratio and omit-DV01 results.

A trader sells USD 50 million of a U.S. Treasury bond and buys TIPS as a hedge. The DV01 of the T-bond is 0.055 and the DV01 of the TIPS is 0.078 (per 100 face). The hedge adjustment factor (beta) is 1.02. Approximately what face amount of TIPS should be purchased?

A. USD 35.3 million
B. USD 36.0 million
C. USD 51.0 million
D. USD 72.3 million

Answer: B

Step by step

1. $F_{TIPS} = 50 \times (0.055/0.078) \times 1.02$.
2. $0.055/0.078 = 0.70513$, so the plain DV01 hedge is 35.26 million.
3. Regression hedge: $35.26 \times 1.02 = 35.96 \approx 36.0$ million.

Why B is correct: The TIPS have a larger DV01, so the hedge face is below USD 50 million; beta of 1.02 scales it up by 2%.

Why the others are wrong

- **A:** Omitted beta: 50 × 0.70513 = 35.26.
- **C:** Used only the beta and ignored the DV01 ratio: 50 × 1.02 = 51.0.
- **D:** Inverted the DV01 ratio: 50 × (0.078/0.055) × 1.02 = 72.3.

Q545

Schweser Module Quiz 11.1 Q2

REBUILT — Option set tidied so each wrong option reflects a specific error (divide by beta, compare with the nominal face).

A trader is shorting $100 million of a nominal Treasury and buying $89.8 million of TIPS, a position sized on DV01s (no beta adjustment). A regression shows the nominal yield changes by 1.0274 bp per 1 bp change in the real yield. Should the trader adjust the hedge, and by how much?

A. No adjustment is needed.
B. Yes, buy an additional $2.46 million of TIPS.
C. Yes, sell $2.39 million of TIPS.
D. Yes, sell $7.74 million of TIPS.

Answer: B

Step by step

1. Regression-hedged TIPS = $89.8 \times 1.0274 = 92.26$ million.

2. Adjustment: $92.26 - 89.8 = +2.46$ million, i.e. buy more TIPS.

Why B is correct: Beta above 1 means nominal yields move more than real yields, so a larger TIPS position is needed than the DV01 hedge provides.

Why the others are wrong

- **A:** Ignored the beta of 1.0274; it differs from 1, so the hedge changes.
- **C:** Divided by beta: 89.8/1.0274 = 87.41, giving a reduction of 2.39 (wrong direction).
- **D:** Compared the regression hedge ($92.26m) with the $100m nominal face instead of the existing $89.8m TIPS position.

LO 11.e

Calculate the face value of multiple offsetting swap positions needed to carry out a two- variable regression hedge.

Q546

BT notes P2.T5.22.15.1

REBUILT — Original options (41% less, 13% less, 12% more) were not tied to realistic errors; replaced with options generated by sign, DV01-neutral and averaging mistakes.

A relative value trader believes the 20-year Treasury yield is too high relative to the 10- and 30-year yields. Using a two-variable regression of daily changes in the 20-year yield on changes in the 10- and 30-year yields (Tuckman-style regression hedge), she buys USD 100.0 million face of the 20-year bond and hedges by selling the 10- and 30-year bonds.

Bond	DV01 (per 100 face)	
10-year Treasury	0.0847	
20-year Treasury	0.1446	
30-year Treasury	0.1910	
Regression coefficient	Value	Standard error
Constant (α)	0.0190	0.2111
Change in 10y yield (β10)	0.5300	0.0624
Change in 30y yield (β30)	0.7400	0.0583

How does the DV01 of her combined hedge positions (10y plus 30y) compare with the DV01 of the 20-year position being hedged?

A. 27% less
B. Equal (the hedge is DV01-neutral)
C. 27% more
D. 36.5% less

Answer: C

Step by step

1. Regression hedge face amounts: $F_{10} = 100 \times \frac{0.1446}{0.0847} \times 0.5300 = 90.48$ million; $F_{30} = 100 \times \frac{0.1446}{0.1910} \times 0.7400 = 56.02$ million.
2. DV01 of the 10y leg: $90.48 \times 0.0847/100 = 0.0766$; of the 30y leg: $56.02 \times 0.1910/100 = 0.1070$.
3. Hedge DV01 $= 0.0766 + 0.1070 = 0.1836$ versus the 20-year DV01 $100 \times 0.1446/100 = 0.1446$.
4. Equivalently, each leg's DV01 equals β_i times the 20-year DV01, so hedge/position $= \beta_{10} + \beta_{30} = 0.53 + 0.74 = 1.27$.
5. The hedge DV01 is 27% larger than the DV01 of the 20-year bond.

Why C is correct: In a regression hedge each leg carries DV01 equal to its beta times the hedged DV01, so the total hedge DV01 is $\beta_{10} + \beta_{30}$ times that of the hedged bond; 1.27 means 27% more because the betas sum to more than 1.

Why the others are wrong

- **A:** Sign error: reads the excess of the beta sum over 1 (0.27) as a shortfall.
- **B:** Assumes the regression hedge is DV01-neutral; it is DV01-neutral only when the betas sum to 1, here they sum to 1.27.
- **D:** Averages the betas ($(0.53 + 0.74)/2 = 0.635$) and treats the result as the hedge/position ratio, giving 36.5% less.

Q547

BT notes P2.T5.322.1

REBUILT — Option A (28.03) did not correspond to any realistic error; replaced with the inverted-DV01-ratio result. Restored garbled coefficient labels.

A market maker has bought USD 100.0 million face of a relatively illiquid 20-year swap (receiving fixed) and hedges by paying fixed on 10- and 30-year swaps, with weights from a two-variable regression of changes in the 20-year swap rate on changes in the 10- and 30-year swap rates.

Swap	DV01 (per 100 face)	
10-year swap	0.030	
20-year swap	0.090	
30-year swap	0.150	
Regression coefficient	Value	Standard error
Constant (α)	-0.0014	0.0040
Change in 10-year swap rate (β10)	0.2221	0.0034
Change in 30-year swap rate (β30)	0.7765	0.0037

Number of observations 1,281; R-squared 99.8%; standard error 0.140 bp.

Which is nearest to the face amount of the 30-year swap implied by the regression hedge?

A. USD 25.88 million
B. USD 46.59 million
C. USD 66.63 million
D. USD 77.65 million

Answer: B

Step by step

1. Regression hedge: $F_{30} = F_{20} \times \frac{\text{DV01}_{20}}{\text{DV01}_{30}} \times \beta_{30}$.
2. $F_{30} = 100 \times \frac{0.090}{0.150} \times 0.7765 = 100 \times 0.6 \times 0.7765 = 46.59$ million.
3. (For reference $F_{10} = 100 \times \frac{0.090}{0.030} \times 0.2221 = 66.63$ million.)

Why B is correct: The 30-year swap must supply $\beta_{30} = 77.65\%$ of the 20-year swap's DV01; since its DV01 is higher per unit of face, the face amount is scaled by 0.090/0.150.

Why the others are wrong

- **A:** Inverted the DV01 ratio (used the 10-year/20-year DV01 ratio 0.030/0.090 in place of 0.090/0.150): 100 × (0.030/0.090) × 0.7765 = 25.88.
- **C:** This is the 10-year swap face amount (100 × 0.090/0.030 × 0.2221), not the 30-year.
- **D:** Used the raw beta (0.7765 × 100) without scaling by the DV01 ratio.

Q548

GARP PE2 Q39 · Mock 4 Q40

REBUILT — Merged near-duplicates (differ only by a 65.36 vs 65.38 typo); replaced unexplained distractor A with the inverted-DV01-ratio result.

A derivatives trader has just entered a relatively illiquid 7-year USD interest-rate swap as the fixed-rate payer and wants to hedge with the more liquid 5-year and 10-year swaps. She regresses changes in the 7-year swap rate on changes in the 5- and 10-year swap rates: $\Delta y_7 = \alpha + \beta_5 \Delta y_5 + \beta_{10} \Delta y_{10} + \varepsilon$.

Regression coefficient	Value	Standard error
Constant (α)	0.0012	0.0030
Change in 5-year swap rate (β5)	0.2471	0.0025
Change in 10-year swap rate (β10)	0.6536	0.0027
Swap tenor	Swap fixed rate	DV01
5-year	2.591%	0.061
7-year	2.492%	0.084
10-year	2.475%	0.115

(Observations 1,255; R-squared 98.1%; standard error 0.12.)

What notional amounts of 5-year and 10-year swaps are needed to hedge USD 100 million notional of the 7-year swap?

A. USD 17.94 million of 5-year swaps and USD 89.48 million of 10-year swaps
B. USD 24.71 million of 5-year swaps and USD 65.36 million of 10-year swaps
C. USD 34.03 million of 5-year swaps and USD 47.74 million of 10-year swaps
D. USD 68.85 million of 5-year swaps and USD 36.52 million of 10-year swaps

Answer: C

Step by step

1. $F_5 = 100 \times \frac{0.084}{0.061} \times 0.2471 = 100 \times 1.3770 \times 0.2471 = 34.03$ million.
2. $F_{10} = 100 \times \frac{0.084}{0.115} \times 0.6536 = 100 \times 0.7304 \times 0.6536 = 47.74$ million.
3. Check: hedge DV01 $= 34.03 \times 0.061/100 + 47.74 \times 0.115/100 = 0.0757$, i.e. 0.2471 + 0.6536 = 90.1% of the 7-year DV01 of 0.084 (the betas sum to less than 1).

Why C is correct: Each hedge leg must provide β times the 7-year DV01, so its notional is $\beta \times (\text{DV01}_7/\text{DV01}_{leg}) \times$ 100 million.

Why the others are wrong

- **A:** Inverted the DV01 ratio (DV01_leg / DV01_7 instead of DV01_7 / DV01_leg).
- **B:** Used the raw betas as percentages of notional, ignoring the DV01 scaling.
- **D:** Ignored the betas and split the DV01 equally between legs: 50 × 0.084/0.061 = 68.85 and 50 × 0.084/0.115 = 36.52.

LO 11.f

Compare and contrast level and change regressions.

Q549

AnalystPrep Q.6511

REBUILT — Original options were vague (long-term trends, dampened volatility) and not grounded in the reading; rewritten around the specifications and estimator properties in LO 11.f.

An analyst hedging a bond with a regression-based hedge is choosing between two specifications for the relationship between the bond's yield y and the hedging instrument's yield x. Which statement correctly describes the level regression and how it differs from the change regression?

A. It regresses y_t on x_t (yield on yield), whereas the change regression regresses Δy_t on Δx_t.

B. It regresses Δy_t on Δx_t, whereas the change regression regresses y_t on x_t.

C. It produces biased coefficient estimates, whereas the change regression produces unbiased estimates.

D. It has serially uncorrelated error terms, whereas the change regression has serially correlated errors.

Answer: A

Step by step

1. Level-on-level regression: $y_t = \alpha + \beta x_t + \varepsilon_t$, using yield levels.
2. Change-on-change regression: $\Delta y_t = \beta \Delta x_t + \Delta \varepsilon_t$, using first differences.
3. Under both, coefficient estimates are unbiased and consistent, but errors are likely serially correlated, so estimates are not efficient.
4. Only A states the specifications correctly.

Why A is correct: The level regression uses the yield levels themselves; the change regression uses first differences of yields.

Why the others are wrong

- **B:** Swaps the definitions: first differences belong to the change regression.
- **C:** Both approaches give unbiased and consistent estimates; the issue is efficiency, not bias.
- **D:** Errors are likely serially correlated in both approaches, so this is false and also reverses the roles.

Q550

AnalystPrep Q.6512

REBUILT — Original keyed 'reduces spurious correlations' and 'noise resistance', which the reading does not support and left arguable options; rewritten around the reading's unbiased/consistent/not-efficient point.

In estimating a regression-based hedge for a bond, which of the following statements about the level and change regressions is correct?

A. Both approaches give unbiased and consistent coefficient estimates, but their error terms are likely serially correlated, so the estimates are not efficient.

B. Both approaches give efficient estimates because the errors are independent over time.

C. The change regression gives biased estimates, while the level regression gives unbiased estimates.

D. The level regression gives unbiased estimates only if the error terms follow an AR(1) process.

Answer: A

Step by step

1. Both regressions give unbiased and consistent coefficient estimates.
2. Empirically the error terms are correlated over time (serial correlation).
3. Serial correlation means the OLS estimates are not efficient.
4. Only A states this.

Why A is correct: Serially correlated errors leave OLS unbiased and consistent but inefficient, in both specifications.

Why the others are wrong

- **B:** Independent errors are the assumption, not the empirical finding; errors are serially correlated.
- **C:** Neither specification is biased; confuses efficiency with bias.
- **D:** AR(1) errors are the proposed refinement to restore efficiency, not a condition for unbiasedness.

Q551

BT notes P2.T5.22.15.2

Sam wants to regress bond Y against bond X, but he cannot decide between a level or change regression:
- Regress yield level of $Y(t)$ on yield level of $X(t)$; aka, level-on-level regression
- Regress yield change, $\Delta Y(t)$, on yield change $\Delta X(t)$; aka, change-on-change regression

Currently, the yield of bond X is constant at 5.0%, while the yield of bond Y was 1.0% yesterday. The most likely go-forward scenario is that the yield of bond Y will trend from its current value of 1.0% to approximately its "model value" of 5.0%.

Which regression is most appropriate in this circumstance?

A. Any change-on-change regression

B. Change-on-change regression but employing least-squares estimates that are BLUE

C. Level-on-level regression with autoregressive, AR(1), error dynamics, $\varepsilon(t) = \rho\,\varepsilon(t-1) + v(t)$

D. Level-on-level regression with coefficients reversed (aka, reverse regression) to smooth decay and enforce lower bound

Answer: C

Step by step

1. Bond Y's yield is expected to drift gradually toward the level implied by X (5%): the gap is persistent and mean-reverting.
2. With X constant, $\Delta X = 0$, so a change regression has no explanatory variation and cannot capture a drift toward a level.
3. A level regression with error $\varepsilon_t = \rho\varepsilon_{t-1} + v_t$, $\rho < 1$, lets the deviation decay gradually and also handles the serial correlation.
4. So C.

Why C is correct: The AR(1) error term models gradual convergence of Y toward its model value while handling serially correlated errors.

Why the others are wrong

- **A:** Changes in X are zero, so the change regression cannot capture convergence to a level.
- **B:** BLUE does not fix the issue: the change regression still ignores the level relationship, and BLUE requires uncorrelated errors.
- **D:** Reversing the regression addresses a different issue (LO 11.g) and does not model gradual decay.

Q552

BT notes P2.T5.322.2

When estimating regression-based hedges, some practitioners regress the level of yields on yields, $y_t = \alpha + \beta x_t + \varepsilon_t$, while others regress changes in yields on changes in yields, $\Delta y_t = \beta \Delta x_t + \Delta \varepsilon_t$. In regard to these two regression approaches, each of the following is true EXCEPT:

A. If the error terms are independently and identically distributed with mean zero and uncorrelated with the independent variable, then the regression coefficient estimators are unbiased, consistent, and efficient

B. Empirically, however, the error terms are likely to be correlated over time (i.e., serially correlated)

C. If the error terms are serially correlated, the regression coefficients will not be efficient, although they likely remain unbiased and consistent

D. The problem created by serial correlation cannot be overcome, and one of the two approaches must be selected, as some methodological imperfection is inevitable

Answer: D

Step by step

1. A: classical conditions give unbiased, consistent, efficient estimators: true.
2. B: errors are likely serially correlated empirically: true.
3. C: serial correlation costs efficiency only: true.
4. D: false, because a third approach with AR(1) errors, $\varepsilon_t = \rho\varepsilon_{t-1} + v_t$ with $\rho < 1$, addresses the serial correlation.

Why D is correct: The reading offers a third model with autoregressive errors that handles serial correlation, so imperfection is not unavoidable.

Why the others are wrong

- **A:** True statement (classical conditions), so not the exception.
- **B:** True statement, so not the exception.
- **C:** True statement, so not the exception.

Q553 Schweser Module Quiz 11.1 Q5

Traci York, FRM, is setting up a regression-based hedge and is trying to decide between a changes-in-yields-on-changes-in-yields approach versus a yields-on-yields approach. Which of the following is a correct statement concerning error terms in these two approaches?

A. In both cases, the error terms are completely uncorrelated.

B. With change-on-change, there is no correlation in error terms, while yield-on-yield error terms are completely correlated.

C. Error terms are correlated over time with both approaches.

D. With yield-on-yield, there is no correlation in error terms, while change-on-change error terms are completely correlated.

Answer: C

Step by step

1. Per the reading, with both specifications the error terms are unlikely to be independent: they are serially correlated.
2. So C is correct; A denies any correlation.
3. B and D assert one specification is uncorrelated and the other 'completely' correlated, which is unsupported.

Why C is correct: Both the level and change regressions have error terms correlated over time, so OLS estimates are unbiased and consistent but not efficient.

Why the others are wrong

- **A:** Ignores the serial correlation that the reading says is likely in both.
- **B:** Claims change errors are uncorrelated and level errors completely correlated; both are correlated and 'completely' is extreme.
- **D:** Reverses B and is equally extreme; neither approach is free of correlated errors.

LO 11.g

Explain why and how a regression hedge differs from a hedge based on a reverse regression.

Q554 Mock AP Mock 1 Q11 · AnalystPrep Q.6514

REBUILT — Original options were vague and partly invented (undersampling, inversely adjusts); rewritten so exactly one option accurately contrasts the two hedges.

A trader hedges a corporate bond with Treasury bonds. Which statement correctly contrasts a regression hedge with a hedge based on a reverse regression?

A. A regression hedge regresses the yield change of the Treasury on the yield change of the corporate bond, while a reverse regression does the opposite.

B. A regression hedge regresses the corporate bond's yield change on the Treasury's yield change and sets the risk weight at that slope; a reverse regression regresses the Treasury's yield change on the corporate bond's, giving a different hedge amount unless the fit is perfect.

C. Both approaches give the same hedge amount, because the reverse-regression slope is always the reciprocal of the original slope.

D. A reverse regression minimizes the variance of the corporate bond position, whereas a regression hedge min-

imizes the variance of the Treasury position.

Answer: B

Step by step

1. Regression hedge: regress the target bond's yield change (dependent) on the hedging instrument's yield change (independent); the slope β is the risk weight applied to the DV01-neutral hedge.
2. Reverse regression: swap the variables, so the hedge instrument's yield change is dependent and the target's is independent; this gives a different slope β'.
3. The two slopes are reciprocals only when the fit is perfect ($R^2 = 1$), so the hedge amounts differ in general.
4. Option B states this correctly.

Why B is correct: The two regressions have different dependent variables and hence different slopes, so they imply different risk weights and hedge amounts unless the fit is perfect.

Why the others are wrong

- **A:** Swaps the roles of the variables: the regression hedge regresses the target on the hedge instrument, not the reverse.
- **C:** The reciprocal relationship holds only when $R^2 = 1$; with imperfect fit the slopes multiply to R^2, not 1.
- **D:** Reversed objectives: the regression hedge minimizes the variance of the corporate bond (hedged) position; the reverse regression suits a trader who must hold a fixed Treasury position.

Q555

AnalystPrep Q.6513

REBUILT — Original correct option was vague and distractors were weak; rewritten with the explicit slope formulas.

An analyst must explain why a regression hedge and a reverse-regression hedge produce different hedge ratios for the same pair of assets. What is the underlying reason?

A. Regression hedges use historical data, whereas reverse-regression hedges rely on theoretical assumptions.

B. Reversing the dependent and independent variables changes the OLS slope from $\rho\,\sigma_Y/\sigma_X$ to $\rho\,\sigma_X/\sigma_Y$, and these are not reciprocals unless $|\rho| = 1$.

C. Both regressions produce the same slope because the correlation between the two series does not depend on direction.

D. The regression hedge uses cumulative yield changes, whereas the reverse regression uses only isolated daily moves.

Answer: B

Step by step

1. OLS slope of a dependent variable on an independent variable is $\rho\,\sigma_{dep}/\sigma_{indep}$.
2. Regressing Y on X gives $\rho\,\sigma_Y/\sigma_X$; regressing X on Y gives $\rho\,\sigma_X/\sigma_Y$.
3. Their product is ρ^2, so they are reciprocals only if $|\rho| = 1$.

Why B is correct: The slope depends on which variable is dependent because it scales the correlation by the dependent/independent volatility ratio.

Why the others are wrong

- **A:** Both regressions are estimated from the same historical data; neither is theoretical.
- **C:** Correlation is symmetric, but the slope also depends on the volatility ratio, which inverts when variables are swapped.
- **D:** Both use the same yield-change data; cumulative vs isolated is not the distinction.

Q556

BT answers P2.T5.25.1.1 · BT notes P2.T5.25.1.1

REBUILT — Source answer 1/β = 0.94 is wrong because the reverse slope is $R^2/\beta = 0.79$ (reciprocal only if $R^2 = 1$) and 0.79 was not offered; added 0.79 as an option and dropped 1.18.

A portfolio manager holds $100 million of corporate bonds and hedges interest rate risk with Treasury bond futures. A regression of daily corporate bond price changes (dependent) on daily Treasury futures price changes (independent) gives: slope $\beta = 1.07$, R-squared = 0.85, intercept = 0.02%. The manager then reverses the regression, making Treasury futures price changes the dependent variable and corporate bond price changes the independent variable. What is the closest estimate of the slope in the new regression?

A. 1.07

B. 0.79

C. 0.85

D. 0.94

Answer: B

Key correction: Source key (0.94) used $1/\beta$; correct is 0.79 because the reverse slope is R^2/β.

Step by step

1. Original slope $\beta = \rho\,\sigma_Y/\sigma_X = 1.07$, and $R^2 = \rho^2 = 0.85$.
2. Reverse slope $\beta' = \rho\,\sigma_X/\sigma_Y = \rho^2/\beta = R^2/\beta$.
3. $\beta' = 0.85/1.07 = 0.794 \approx 0.79$.

Why B is correct: The reverse-regression slope equals R^2 divided by the original slope; it equals the reciprocal only when $R^2 = 1$.

Why the others are wrong

- **A:** Reuses the original slope; the swap changes the slope.
- **C:** Reports R^2 itself, forgetting to divide by β.
- **D:** Takes $1/1.07 = 0.935$, which assumes a perfect fit ($R^2 = 1$); here $R^2 = 0.85$.

Q557

BT answers P2.T5.25.1.2 · BT notes P2.T5.25.1.2

A risk manager estimates a hedge ratio for an equity portfolio using futures. The correct regression has portfolio return (Y) as the dependent variable and futures return (X) as the independent variable, giving $h^* = \rho\,\sigma_Y/\sigma_X$. The manager mistakenly reverses the regression, making X dependent and Y independent. Data: portfolio return volatility σY = 12%; futures return volatility σX = 8%; correlation ρ = 0.75. What hedge ratio $h^{*\prime}$ results from the reversed regression?

A. 0.750
B. 1.125
C. 0.500
D. 0.667

Answer: C

Step by step

1. Reversed slope $h^{*\prime} = \rho\,\sigma_X/\sigma_Y$.
2. $= 0.75 \times 8/12 = 0.75 \times 0.6667$.
3. $= 0.500$.

Why C is correct: With X as the dependent variable the slope is the correlation times σX/σY, which is the inverse of the volatility ratio used in the correct regression.

Why the others are wrong

- **A:** Just the correlation; ignores the volatility ratio.
- **B:** The correct-direction ratio $0.75 \times 12/8 = 1.125$.
- **D:** Only the volatility ratio 8/12 = 0.667; forgets to multiply by ρ.

Q558

BT answers P2.T5.25.1.3 · BT notes P2.T5.25.1.3

REBUILT — Distractor with the error term was an obvious giveaway; replaced with the volatility ratio that omits rho.

A hedge ratio is estimated by regressing the asset hedged (Y) on the hedging instrument (X): $Y = \alpha + h^* X + \varepsilon$. The analyst then realizes the dependent and independent variables were mixed up. Let ρ be the correlation of X and Y and σX, σY their volatilities. If ρ < 1, which expression gives the slope $h^{*\prime}$ of the reversed regression?

A. $h^{*\prime} = \rho\,\sigma_Y/\sigma_X$
B. $h^{*\prime} = \rho\,\sigma_X/\sigma_Y$
C. $h^{*\prime} = 1/h^*$
D. $h^{*\prime} = \sigma_X/\sigma_Y$

Answer: B

Step by step

1. Slope of a regression = correlation x (σ_dependent / σ_independent).
2. Original: $h^* = \rho\,\sigma_Y/\sigma_X$.
3. Reversed (X dependent): $h^{*\prime} = \rho\,\sigma_X/\sigma_Y$.

Why B is correct: Swapping the variables swaps the volatility ratio but keeps the correlation, so the result is not 1/h* unless ρ = 1.

Why the others are wrong

- **A:** This is the original regression's slope.
- **C:** Reciprocal holds only if ρ = 1; here ρ < 1, so the product of the two slopes is $\rho^2 < 1$.

- **D:** Omits the correlation factor, which is the perfect-correlation case.

LO 11.h

Describe principal component analysis and explain how it is applied to constructing a hedging portfolio.

Q559

AP Mock 2 Q68

REBUILT — Options A and B were identical in the source and one distractor was a giveaway; replaced with distinct, plausible distractors.

An analyst is exploring principal component analysis (PCA) for constructing a hedging portfolio. What primary advantage does PCA offer when describing the risk of a portfolio of fixed-income instruments?

A. It assumes that all rates are perfectly correlated, so a single factor describes every rate change.
B. It decomposes yield curve changes into uncorrelated components, ordered by the variance they explain, so a few components capture most of the term structure's volatility.
C. It guarantees that the slope of the yield curve stays constant, so hedge ratios never need updating.
D. It requires one component per rate, so each rate is hedged separately and no simplification is needed.

Answer: B

Step by step

1. PCA turns the correlated changes in many rates (e.g., 30 swap rates) into the same number of uncorrelated principal components.
2. Each PC is chosen to have the highest possible variance given the earlier PCs; the sum of PC variances equals the sum of the rates' variances.
3. Hence the first three PCs (level, slope, short rate/curvature) approximate the total volatility, so changes in 30 rates can be expressed with three factors.
4. This matches option B.

Why B is correct: PCA gives a single empirical description of term structure behavior using a few uncorrelated, variance-ranked factors, which greatly simplifies hedging.

Why the others are wrong

- **A:** PCA estimates correlations from data and does not assume perfect correlation; if rates were perfectly correlated only one PC would exist and the analysis would be trivial.
- **C:** PCA describes how the curve moves; it guarantees nothing about a constant slope (the slope is itself a PC that varies).
- **D:** Reverses the purpose: PCA has as many PCs as rates, but only the first few are needed, which is exactly the simplification.

Q560

AnalystPrep Q.6516

REBUILT — Original distractors were giveaways (e.g., 'non-linear variances across unrelated instruments'); replaced with distractors reflecting realistic confusions with DV01 and regression hedges.

An asset manager wants to hedge a bond portfolio using principal component analysis (PCA). Which approach correctly applies PCA to construct the hedge?

A. Estimate the portfolio's sensitivity to every principal component, including the smallest ones, because each component carries the same amount of variance.
B. Compute only the portfolio DV01 and offset it with a single instrument, assuming all rates move in parallel.
C. Compute the portfolio's exposure to the first few (dominant) principal components, such as level and slope, and choose hedge positions that make the net exposure to each of them zero.
D. Regress the portfolio's yield changes on the hedging instrument's yield changes and set the hedge ratio equal to the regression beta.

Answer: C

Step by step

1. PCs are ordered by variance explained; the first three capture nearly all term structure volatility.
2. Measure how much the portfolio and each candidate hedging instrument change per one-standard-deviation move in each dominant PC.
3. Solve for hedge notionals so that the net exposure to each dominant PC is zero (one equation per PC, one hedging

instrument per equation).
4. This is option C.

Why C is correct: A PCA hedge neutralizes exposure to the few dominant, uncorrelated PCs, which hedges the bulk of term structure risk.

Why the others are wrong

- **A:** PCs do not carry equal variance; they are ranked by variance, and the small ones can be ignored.
- **B:** This is the DV01 hedge, which assumes a parallel shift (only the level factor) and ignores slope and other PCs.
- **D:** This is a regression hedge using one hedging instrument and the beta from historical yield changes, not a PCA hedge.

Q561

BT notes P2.T5.22.15.3

REBUILT — Source relied on a badly OCR-damaged PCA table that is not needed to answer; replaced with a compact self-contained stem and a clearly false option B.

A PCA of 13 USD swap rate factors (Tuckman) shows that, for the 1-year rate, the loadings in basis points are 0.23 on level, -0.16 on slope and 0.29 on short rate, and that the first three PCs explain almost all of the total variance. Which of the following statements about this PCA is FALSE?

A. The rate covariance matrix has 78 distinct covariances (plus 13 variances), whereas the PC covariance matrix is diagonal.

B. The PCs are fundamental economic factors that the analyst specifies in advance, which is why the names level, slope and short rate are assigned before the analysis is run.

C. PCA produces 13 principal components, but only a few (here three) are needed to characterize term structure fluctuations.

D. The PC volatility of the 1-year rate is about 0.40 bp, equal to $\sqrt{0.23^2 + (-0.16)^2 + 0.29^2}$, because the PCs are uncorrelated.

Answer: B

Step by step

1. A: with 13 factors, distinct covariances = $13 \times 12/2 = 78$; PCs are uncorrelated, so their covariance matrix is diagonal. True.
2. C: PCA yields as many PCs as rates (13), but the first few explain nearly all variance. True.
3. D: variance of the rate = sum of squared loadings because PCs are uncorrelated: $0.0529 + 0.0256 + 0.0841 = 0.1626$; $\sqrt{0.1626} = 0.403$ bp. True.
4. B: PCs are statistical constructs derived from the data and are named after the fact based on the shape of their loadings; they are not pre-specified fundamental factors. False.

Why B is correct: PCs are derived from the covariance structure of rates, not specified in advance; the labels level, slope and short rate are descriptive interpretations after estimation.

Why the others are wrong

- **A:** True: $13 \times 12/2 = 78$ and PCs are uncorrelated, giving a diagonal covariance matrix.
- **C:** True: only the first three PCs are needed for a good approximation.
- **D:** True: adding variances (squared loadings) is valid only because PCs are uncorrelated; $\sqrt{0.1626} = 0.40$ bp (a common slip is to forget the square root or to add loadings directly).

Q562

BT notes P2.T5.322.3

REBUILT — Option D (stability of the level PC shape around the crisis) was empirical and contestable; replaced with a definitional true statement so that only C is false.

According to Tuckman, each of the following is true about principal component analysis (PCA) of the term structure EXCEPT which is FALSE?

A. PCA provides a single empirical, multi-factor description of term structure behavior that can be applied across all bonds.

B. The first three components (for example, level, slope and short rate) tend to explain 90% or more of interest rate volatility.

C. The number of components needed grows roughly in proportion to the size of the portfolio, so for large portfolios a regression hedge becomes more practical.

D. Principal components are uncorrelated with each other, so the variance of a rate equals the sum of the squared loadings on the PCs.

Answer: C

Step by step

1. A: PCA describes all rates with the same small set of factors, unlike regression which is tied to a few bonds. True.
2. B: the first three PCs capture the vast majority (90%+) of volatility. True.
3. D: PCs are uncorrelated by construction, so variances add. True.
4. C: the number of PCs needed stays small regardless of portfolio size; this is exactly why PCA scales well to large portfolios. False.

Why C is correct: PCA's advantage is that a small, fixed number of components suffices for any number of bonds, so the required factor count does not scale with portfolio size.

Why the others are wrong

- **A:** True statement of the LO 11.h definition.
- **B:** True; the first three PCs explain most of the volatility.
- **D:** True because PCs are orthogonal by construction.

Q563

GARP PE1 Q12 · Mock 3 Q68

REBUILT — Two sources with conflicting keys (risk weights relative to the 5-year DV01 vs normalized to the hedge total) and one with notionals assigned to wrong tenors; merged, stated the risk-weight base explicitly, and rebuilt options.

A trader enters a butterfly in which the bank pays fixed on EUR 100 million notional of a 5-year swap and receives fixed on 2-year and 10-year swaps. The 2-year and 10-year notionals are chosen so the position is neutral to both the level and slope principal components. Data (change in rate in bp per 1 standard deviation move in the PC):

Swap tenor	DV01 (per EUR 100 notional)	Level PC	Slope PC
2-year	0.0285	5.06	-2.93
5-year	0.0496	5.97	-1.28
10-year	0.0731	5.43	0.02

What are the required notionals, and what are the risk weights of the 2-year and 10-year legs, defined as each leg's DV01 relative to the DV01 of the 5-year swap?

A. 2-year EUR 76.85m; 10-year EUR 46.68m; risk weights 39.1% and 60.9%.

B. 2-year EUR 46.68m; 10-year EUR 76.85m; risk weights 44.2% and 68.8%.

C. 2-year EUR 76.85m; 10-year EUR 46.68m; risk weights 44.2% and 68.8%.

D. 2-year EUR 87.02m; 10-year EUR 33.93m; risk weights 50.0% and 50.0%.

Answer: C

Key correction: Sources disagreed: GARP PE1 Q12 keyed 44.2%/68.8% (relative to the 5-year DV01) while Mock 3 Q68 keyed 39.1%/60.9% (normalized to the two-leg total). With risk weights defined relative to the 5-year DV01, as Tuckman does, 44.2%/68.8% is correct; GARP PE1's printed options also swapped the notionals between tenors.

Step by step

1. Per EUR 1m notional, PC exposure = DV01/100 × loading. The 5-year leg (EUR 100m) has level exposure 100×0.000496×5.97 = 0.29611 and slope exposure 100×0.000496×(−1.28) = −0.063488; the wings must offset these.
2. Level: 0.0014421·F2 + 0.0039693·F10 = 0.29611 ; Slope: −0.00083505·F2 + 0.0000146·F10 = −0.063488 (F in EUR m).
3. Solving the two equations: F2 = EUR 76.85m, F10 = EUR 46.68m (both received fixed, opposite to the 5-year).
4. Risk weights relative to the 5-year DV01 (100×0.0496/100 = 0.0496): 2y = 76.85×0.0285/100 / 0.0496 = 0.4416 → 44.2% ; 10y = 46.68×0.0731/100 / 0.0496 = 0.6880 → 68.8%.
5. Answer C.

Why C is correct: Solving two linear equations (level and slope neutrality) for two unknown notionals gives EUR 76.85m and EUR 46.68m, and dividing each leg's DV01 by the 5-year DV01 gives the stated risk weights.

Why the others are wrong

- **A:** Correct notionals but weights normalized to the two-leg DV01 total (39.1%/60.9%), not relative to the 5-year DV01 as the stem requires.
- **B:** Weights are right but the notionals are assigned to the wrong tenors (swapped 2-year and 10-year).
- **D:** DV01-neutral split of the 5-year DV01 equally between the wings (87.02m and 33.93m), which ignores the level and slope PC loadings.

Reading 12. Arbitrage Pricing with Term Structure Models

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 4th Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 7.

LO	Learning objective	Qs
12.a	calculate the expected discounted value of a zero-coupon security using a binomial tree.	8
12.b	construct and apply an arbitrage argument to price a call option on a zero-coupon security using replicating portfolios.	6
12.c	define risk-neutral pricing and apply it to option pricing.	11
12.d	explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift.	4
12.e	explain how the principles of arbitrage pricing of derivatives on fixed-income securities can be extended over multiple periods.	7
12.f	define option-adjusted spread (OAS) and apply it to security pricing.	5
12.g	describe the rationale behind the use of recombining trees in option pricing.	6
12.h	calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities.	4
12.i	evaluate the advantages and disadvantages of reducing the size of the time steps on the pricing of derivatives on fixed-income securities.	2
12.j	evaluate the appropriateness of the Black-Scholes-Merton model when valuing derivatives on fixed-income securities.	4

LO 12.a

Calculate the expected discounted value of a zero-coupon security using a binomial tree.

Q564

AnalystPrep Q.1617

Refer to the following binomial tree:

Date 0	Date 1
5%	5.50% (p = 1/2) or 4.50% (p = 1/2)

The six-month rate is 5% today (date 0). On date 1, six months from now, there are two possible outcomes. The 5.50% state is called the:

A. Higher state
B. Upper extreme state
C. Up-state
D. Maximum state

Answer: C

Step by step

1. In Tuckman's binomial tree the rate moves to one of two values at the next date.
2. The branch with the higher rate (5.50% > 5%) is labelled the up-state; the lower-rate branch (4.50%) is the down-state.

Why C is correct: Standard terminology labels the higher-rate branch the up-state and the lower-rate branch the down-state.

Why the others are wrong

- **A:** 'Higher state' is descriptive but not the term used in binomial-tree notation.
- **B:** 'Upper extreme state' is not standard terminology; the tree has no extreme-value concept.
- **D:** 'Maximum state' is not standard terminology; it confuses a tree branch with a maximum of a distribution.

Q565

BT notes P2.T5.300.1

REBUILT — Options were poorly designed (nearest-answer wording with arbitrary distractors); rebuilt options as realistic mistakes and asked for the exact value.

The tree below shows the true process for the one-year interest rate. The current one-year rate is 8.0%; next year it is 11.0% or 5.0% with equal probability; in year 2 it again moves by 3% up or down.

Year 0	Year 1	Year 2
8.0%	11.0%	14.0% / 8.0%
8.0%	5.0%	8.0% / 2.0%

Investors are not risk-averse, so a bond's price equals its expected discounted value under the true probabilities. What is the price of a two-year zero-coupon bond with $1,000 par?

A. $857.34
B. $858.00
C. $926.64
D. $834.17

Answer: B

Step by step

1. Year-1 values of the $1,000 payoff: up node 1000/1.11 = 900.90; down node 1000/1.05 = 952.38.
2. Expected year-1 value = 0.5(900.90) + 0.5(952.38) = 926.64.
3. Discount at today's 8%: 926.64/1.08 = 858.00.
4. The year-2 rates (14%, 8%, 2%) only matter for longer-dated bonds.

Why B is correct: Roll the payoff back one node at a time, averaging at 50% each and discounting at each node's rate.

Why the others are wrong

- **A:** $857.34 = 1000/1.08^2: discounted at the expected rate 8% twice, ignoring the convexity effect of rate volatility.
- **C:** $926.64 is the expected year-1 value; forgot the final discount at 8%.
- **D:** $834.17 = 1000/(1.08 x 1.11): followed only the up-state path.

Q566

BT answers L2.T5.40.1

REBUILT — Stem did not say the bond is a one-year zero (needed to price it from the spot rate) and the key explanation was thin; stated a one-year zero and rebuilt options as realistic mistakes.

The market six-month and one-year spot rates are 2.0% and 2.2%, respectively (semiannual compounding, per Tuckman). In a two-step binomial tree (each step is six months), the six-month rate six months from now is either 2.5% or 1.5% with equal probability. What is the market price of a one-year zero-coupon bond with face value $1,000?

A. $978.47
B. $980.30
C. $978.36
D. $957.37

Answer: C

Step by step

1. The market price of a zero is set by the one-year spot rate: two semiannual periods at 2.2%/2 = 1.1% each.
2. $P = 1000/(1.011)^2 = 1000/1.022121 = 978.36$.
3. The tree is not needed for the market price (a calibrated tree must reproduce it).

Why C is correct: A zero-coupon bond's market price is its face value discounted at the one-year spot rate with semiannual compounding.

Why the others are wrong

- **A:** $978.47 = 1000/1.022: treated 2.2% as one annual-compounded period instead of two semiannual periods at 1.1%.
- **B:** $980.30 = 1000/(1.01)^2 (also the uncalibrated tree value): used the 2.0% six-month rate for both periods rather than the one-year spot rate.
- **D:** $957.37 = 1000/(1.022)^2: used 2.2% per six-month period instead of 1.1%.

Q567

BT answers L2.T5.40.4

REBUILT — Stem relied on a call from a prior question that was not restated; stated the call (strike $990, six-month expiry on the one-year zero) and rebuilt distractors.

The six-month spot rate is 2.0%. In Tuckman's two-step tree (each step six months, semiannual compounding), six months from now the six-month rate is 2.5% or 1.5% with equal true probabilities. A European call with strike $990

expires in six months on a one-year zero-coupon bond with face value $1,000. What is the discounted expected value of the call today?

A. $1.28
B. $1.27
C. $2.53
D. $2.56

Answer: B

Step by step

1. At date 1 the bond is a six-month zero: up-rate 2.5% gives $1000/1.0125 = 987.65$; down-rate 1.5% gives $1000/1.0075 = 992.56$.
2. Call payoffs with K = 990: up = max(987.65 - 990, 0) = 0; down = max(992.56 - 990, 0) = 2.56.
3. Expected payoff = 0.5(0) + 0.5(2.5559) = 1.2779.
4. Discount at the 2.0% six-month rate: $1.2779/1.01 = 1.27$.

Why B is correct: Probability-weighted option payoff discounted one period at the current six-month rate.

Why the others are wrong

- **A:** $1.28 = 1.2779 not discounted.
- **C:** $2.53 = 2.5559/1.01: discounted the down-state payoff but forgot the 50% probability weight.
- **D:** $2.56: down-state payoff only, with neither probability weighting nor discounting.

Q568

BT answers L2.T5.41.1

In a two-step one-year binomial tree (each step six months, semiannual compounding), a one-year zero-coupon bond with face value $1,000 will in six months (when it has six months to maturity) be worth $978 in the up-state or $982.80 in the down-state. What are, respectively, the down- and up-state six-month interest rates at date 1?

A. 3.0% and 4.0%
B. 3.5% and 4.5%
C. 4.0% and 5.0%
D. 4.5% and 5.5%

Answer: B

Step by step

1. A six-month zero is worth $1000/(1 + r/2)$.
2. Up-state: $1 + r/2 = 1000/978 = 1.02249$, so $r = 2 \times 0.02249 \approx 4.5\%$.
3. Down-state: $1 + r/2 = 1000/982.80 = 1.01750$, so $r = 3.5\%$.
4. Requested order: down 3.5%, up 4.5%.

Why B is correct: Invert the six-month zero pricing formula; the lower price corresponds to the higher rate.

Why the others are wrong

- **A:** 3.0% and 4.0% would give prices of 985.22 and 980.39, not 982.80 and 978.
- **C:** 4.0% and 5.0% would give prices of 980.39 and 975.61, not 982.80 and 978.
- **D:** 4.5% and 5.5% is the wrong pair: it would give 977.995 and 973.24; the price 982.80 corresponds to 3.5%, not 4.5%.

Q569

BT answers L2.T5.41.5

REBUILT — Stem omitted the current 4% six-month rate needed for discounting and one option was OCR-damaged; stated the rate and rebuilt options as realistic mistakes.

In a two-step one-year binomial tree (each step six months, semiannual compounding) the current six-month rate is 4.0%. In six months a one-year zero-coupon bond with face value $1,000 will be worth $978 in the up-state or $982.80 in the down-state, with true probabilities of 50% each. A call option on this bond has strike $980.00 and expires in six months. What is the expected discounted value of the call?

A. $1.35
B. $1.40
C. $2.75
D. $1.37

Answer: D

Step by step

1. Up-state payoff: max(978 - 980, 0) = 0.
2. Down-state payoff: max(982.80 - 980, 0) = 2.80.
3. Expected payoff = 0.5(0) + 0.5(2.80) = 1.40.
4. Discount one period at 4.0%/2 = 2%: 1.40/1.02 = 1.37.

Why D is correct: Probability-weighted payoff discounted at the current six-month rate (2% per half-year).

Why the others are wrong

- **A:** $1.35 = 1.40/1.04: discounted by a full annual 4% instead of 2% per half-year.
- **B:** $1.40 is the undiscounted expected payoff.
- **C:** $2.75 = 2.80/1.02: discounted the down-state payoff but forgot the 50% probability weight.

Q570

BT answers L2.T5.41.2

REBUILT — Source key $967.77 contradicts its own formula (980.40/1.02 = 961.18) and no option was correct; rebuilt options around the correct value.

In a two-step one-year binomial tree (each step six months, semiannual compounding), a one-year zero-coupon bond with face value $1,000 will in six months be worth $978.00 in the up-state or $982.80 in the down-state, with 50% true probabilities. The up- and down-state six-month rates are symmetric (+0.5% and -0.5%) around today's six-month rate. What is the bond's expected discounted value today?

A. $961.18
B. $980.40
C. $942.69
D. $970.69

Answer: A

Key correction: Source key (BT answers L2.T5.41.2) said D ($967.77); correct is $961.18 because 980.40/1.02 = 961.18, and $967.77 does not follow from the source's own formula.

Step by step

1. Up-rate is about 4.5% and down-rate 3.5% (from 1000/978 and 1000/982.80), so today's six-month rate is the midpoint 4.0%.
2. Expected date-1 value = 0.5(978.00) + 0.5(982.80) = 980.40.
3. Discount one period at 4.0%/2 = 2%: 980.40/1.02 = 961.18.

Why A is correct: Average the two date-1 values and discount at today's six-month rate of 4.0% (2% per period).

Why the others are wrong

- **B:** $980.40 is the undiscounted expected date-1 value.
- **C:** $942.69 = 980.40/1.04: discounted by a full annual 4% instead of 2% per half-year.
- **D:** $970.69 = 980.40/1.01: used a 1% half-year rate (2% annual) instead of 4%/2.

Q571

Schweser Module Quiz 12.1 Q1

Which of the following statements concerning the calculation of value at a node in a fixed-income binomial interest rate tree is most accurate? The value at each node is the:

A. present value of the two possible values from the next period.
B. average of the present values of the two possible values from the next period.
C. sum of the present values of the two possible values from the next period.
D. average of the future values of the two possible values from the next period.

Answer: B

Step by step

1. At a node, take the two possible next-period values (including any coupon).
2. Weight each by its probability (50% each in the standard tree) and discount at the one-period rate at that node.
3. Equivalently: the average of the two discounted values, $V = \frac{1}{2}\left[\frac{V_u + c}{1 + r/2} + \frac{V_d + c}{1 + r/2}\right]$.

Why B is correct: Node value is the probability-weighted average of the two next-period values discounted at the node's rate.

Why the others are wrong

- **A:** There are two next-period values; a single present value ignores the averaging over states.

- **C:** Summing the two present values double-counts the cash flow; the 1/2 probabilities must be applied.
- **D:** Averaging future values without discounting ignores the time value of money.

LO 12.b

Construct and apply an arbitrage argument to price a call option on a zero-coupon security using replicating portfolios.

Q572

AnalystPrep Q.1620

REBUILT — Distractors 'call option scenario' and 'sell option scenario' were giveaways; replaced with plausible alternative state variables and tightened wording.

In arbitrage pricing of an interest rate derivative, a portfolio of underlying securities is constructed to replicate the derivative. Because the derivative's cash flows depend on the level of interest rates, the replicating portfolio must duplicate the derivative's cash flows in every possible:

A. inflation scenario
B. credit-default scenario
C. interest rate scenario
D. liquidity scenario

Answer: C

Step by step

1. The derivative's payoff is a function of the future interest rate (the node of the tree).
2. For the replica to have the same value as the derivative at every date and node, its payoff must match the derivative's payoff in each possible rate state (e.g., up and down).
3. If it matches in every interest rate scenario, no-arbitrage requires the price today to be the same.

Why C is correct: Derivative cash flows are contingent on the rate; a replica that duplicates them in every rate scenario must have the same price by no-arbitrage.

Why the others are wrong

- **A:** Inflation is not the state variable in the binomial rate model; payoffs are driven by the rate tree.
- **B:** Credit-default states are not modelled in Tuckman's default-free rate tree.
- **D:** Liquidity is not a state of the rate tree; replication is across rate states, not market-liquidity states.

Q573

AnalystPrep Q.1625

REBUILT — Original distractors (never/always/sometimes/partially) were vague and overlapping; rewrote options as distinct, plausible statements about the role of real-world probabilities.

In pricing a call option on a zero-coupon bond by replication in a binomial tree, the replica portfolio must match the option's value in both the up and the down state. What role do the real-world (true) probabilities of an up and a down move play in the arbitrage price of the option?

A. None; the arbitrage price does not depend on them
B. They weight the up- and down-state payoffs, which are then discounted at the short rate
C. They matter only if the underlying bond's expected return exceeds the risk-free rate
D. They determine the number of units of the underlying bond in the replicating portfolio

Answer: A

Step by step

1. The replica holds N units of the bond and a cash position chosen to match the option payoff in the up and down states: two equations, two unknowns.
2. N and the cash amount depend only on the state payoffs and bond prices in each state ($N = (C_{up} - C_{down})/(V_{up} - V_{down})$).
3. The option price equals the cost of the replica, computed from today's market prices of the instruments; no probabilities appear anywhere.
4. Hence the real-world probabilities are irrelevant to the arbitrage price (a notable result).

Why A is correct: The replica's composition and cost are fixed by state payoffs and market prices alone, so the true probabilities never enter the arbitrage price.

Why the others are wrong

- **B:** This describes expected discounted value with true probabilities, which is not the arbitrage price (it generally differs from the replica cost).
- **C:** Even if the bond has a risk premium, that is already embedded in its market price; the replication argument is unaffected.
- **D:** The hedge ratio is a ratio of payoff differences to price differences across states; it involves no probabilities.

Q574

BT answers L2.T5.40.3

REBUILT — Original referred to a spreadsheet only and options B/C/D were unexplained; kept the OCR-corrected data and rebuilt distractors from realistic mistakes.

The market six-month and one-year spot rates are 2.0% and 2.2% (semiannual compounding). In a two-step binomial tree with six-month steps, the six-month rate six months from now will be either 2.5% or 1.5% with equal real-world probability. A one-year zero-coupon bond has face value \$1,000. Use a replicating portfolio (of the one-year zero and the six-month zero) to price a call option that expires in six months to buy the then six-month zero with \$1,000 face at a strike of \$990. What is the arbitrage price of the call?

A. \$0.25
B. \$1.27
C. \$2.28
D. \$2.56

Answer: A

Step by step

1. Today's one-year zero price: $1000/1.011^2 = 978.358$; six-month zero: $1000/1.01 = 990.099$.
2. Bond price in six months: up (rate 2.5%) $1000/1.0125 = 987.654$; down (rate 1.5%) $1000/1.0075 = 992.556$.
3. Call payoff: up $\max(987.654 - 990, 0) = 0$; down $992.556 - 990 = 2.556$.
4. Replica: N units of the one-year zero with $N = (0 - 2.556)/(987.654 - 992.556) = 0.52144$; cash position at six months $0 - 0.52144 \times 987.654 = -515.00$ (borrowing).
5. Cost today $= 0.52144 \times 978.358 - 515.00/1.01 = 510.15 - 509.90 \approx 0.25$.
6. Check with risk-neutral p(up) = 90.06%: $(0.9006 \times 0 + 0.0994 \times 2.556)/1.01 = 0.25$.

Why A is correct: The call is replicated by 0.5214 of the one-year zero financed by borrowing; the replica's cost, \$0.25, is the arbitrage price.

Why the others are wrong

- **B:** Used the 50/50 real-world probabilities: $0.5 \times 2.556/1.01 = 1.27$; true probabilities do not give the arbitrage price.
- **C:** Applied the risk-neutral up probability (90.06%) to the down-state payoff instead of the down probability: $0.9006 \times 2.556/1.01 = 2.28$.
- **D:** Took the undiscounted down-state payoff of \$2.56 without probability weighting or discounting.

Q575

BT notes P2.T5.23.1.2

In Tuckman's illustration, the current six-month spot rate is 2.00% and the one-year spot rate is 2.15% per annum (semiannual compounding). The bond is a one-year zero with face value \$1,000. Six months from now the six-month rate will be either 2.50% or 1.50%; the real-world probabilities are 50%/50% and the risk-neutral probabilities are 80.1%/19.9%. The derivative is a call option expiring in six months on the then six-month zero (face \$1,000) with strike \$990.00.

Which of the following statements is TRUE?

A. The arbitrage price of the option is \$0.504
B. The replicating portfolio demonstrates that the expected return on all assets in the risk-neutral world is zero
C. The bond's market price of \$980.30 is given by its expected discounted value (EDV) conditional on the assumption of real-world probabilities
D. In the recombining multi-period binomial interest rate tree, risk-neutral (aka, arbitrage) pricing solves for the drift (aka, expected change) that is constant from each date to the next; i.e., the drift from date zero to date one equals the drift from date one to date two

Answer: A

Step by step

1. Bond today: $1000/1.01075^2 = 978.84$. Six months out: up 987.654, down 992.556.
2. Call payoffs: up 0; down $992.556 - 990 = 2.556$.

3. Risk-neutral p(up) solves $978.84 = [p(987.654) + (1-p)(992.556)]/1.01$, giving $p = 80.09\%$, consistent with the stated 80.1%.
4. Option price $= 0.1991 \times 2.556/1.01 = 0.504$, equal to the cost of the replicating portfolio, so A is true.

Why A is correct: The replica's cost (equivalently the risk-neutral expected discounted payoff) is $0.504.

Why the others are wrong

- **B:** In the risk-neutral world all assets earn the risk-free (short) rate, not zero.
- **C:** The EDV with 50/50 real-world probabilities is $980.30, but the bond's market price is $978.84; the risk-neutral probabilities are what reproduce the market price.
- **D:** The drift is not constant across dates: it is solved separately at each date to match market prices, so the first-step drift (about 30 bp) need not equal later drifts.

Q576

BT notes P2.T5.23.1.3

The current six-month spot rate is 4.00% and the one-year spot rate is 4.15% per annum (semiannual compounding). In a binomial tree with a 1.00% jump, the six-month rate in six months is either 3.00% or 5.00%; real-world probabilities are 50%/50% and risk-neutral probabilities are 65.2%/34.8%. The bond is a one-year zero-coupon bond with face $1,000. The derivative is a call option expiring in six months with a strike of $980.00.

Each of the following is true EXCEPT which is false?

A. The arbitrage price of the option is about $1.78

B. The cost of the portfolio that replicates the call option's cash flows (i.e., zero or $5.22) is $2.56

C. The implied drift (aka, expected change) in the six-month interest rate is about 30.4 basis points

D. The expected discounted value (EDV) of the bond using the risk-neutral probabilities returns the bond's market price of $959.76

Answer: B

Step by step

1. Bond today $1000/1.02075^2 = 959.76$. Six months out: rate 3% gives 985.22, rate 5% gives 975.61.
2. Call payoffs: 5.22 if the rate falls (bond 985.22), 0 if it rises.
3. Risk-neutral p(up) = 65.23%: $(0.6523 \times 975.61 + 0.3477 \times 985.22)/1.02 = 959.76$, so D is true.
4. Option $= 0.3477 \times 5.22/1.02 = 1.78$, equal to the replica's cost, so A is true and B (2.56) is false.
5. Drift: $0.6523 \times 5\% + 0.3477 \times 3\% - 4\% = 0.3046\% \approx 30.4$ bp, so C is true.

Why B is correct: The replica costs the arbitrage price of $1.78; $2.56 is the real-world (50/50) expected discounted payoff $0.5 \times 5.22/1.02$, which is not the arbitrage price.

Why the others are wrong

- **A:** True statement ($1.78), so it is not the exception.
- **C:** True statement (drift of about 30.4 bp), so it is not the exception.
- **D:** True statement; risk-neutral EDV reproduces $959.76, so it is not the exception.

Q577

AnalystPrep Q.1616

REBUILT — Original stem was garbled and distractors were nonsense terms ('proprietary rights/claims'); rewritten around the standard underlying vs contingent claim distinction with plausible options.

In arbitrage pricing with term structure models, the prices of securities such as zero-coupon bonds are taken as given from the market; these are called underlying securities. This distinguishes them from securities whose prices are derived by constructing a replicating portfolio of the underlying securities. These derived-price securities are called:

A. contingent claims, priced by arbitrage arguments

B. underlying claims, priced by their real-world expected discounted value

C. contingent claims, priced by their real-world expected discounted value

D. benchmark securities, priced by arbitrage arguments

Answer: A

Step by step

1. Underlying securities (e.g., zero-coupon bonds) have market prices taken as given.
2. Securities whose payoffs depend on them (e.g., options on bonds) are contingent claims.
3. Contingent claims are priced by no-arbitrage replication with the underlying securities, not by real-world expected values.

Why A is correct: Contingent claims have payoffs contingent on underlying prices and are priced by arbitrage (replication) arguments.

Why the others are wrong

- **B:** 'Underlying claims' is not the term, and real-world expected discounting does not give the arbitrage price.
- **C:** Right term but wrong method: using real-world probabilities ignores the replication argument.
- **D:** 'Benchmark securities' is not the standard term for derivatives priced by replication.

LO 12.c

Define risk-neutral pricing and apply it to option pricing.

Q578

AnalystPrep Q.1628

Risk-neutral pricing is a technique that modifies an assumed interest rate process so that any contingent claim can be priced without having to construct and price its replicating portfolio. It is an extremely efficient way to price many contingent claims under the same assumed rate process because:

A. the original interest rate process has to be modified once or more than once, and this modification only requires pricing the contingent claim(s) by arbitrage.

B. the original interest rate process has to be modified only once, and this modification only requires pricing the contingent claim(s) by arbitrage.

C. the original interest rate process does not need to be modified, and there is no need to price the contingent claim(s) by arbitrage.

D. the interest rate process is adjusted for real-world probabilities, allowing valuation of contingent claims based on expected real-world outcomes rather than arbitrage-free pricing.

Answer: B

Step by step

1. Risk-neutral probabilities are found once, by requiring the model to reproduce the market prices of the underlying securities (a pricing-by-arbitrage step).
2. With those probabilities fixed, every other contingent claim on the same rate process is valued as its expected discounted value, with no replicating portfolio needed.
3. So the process is modified only once, and that single modification uses arbitrage pricing: option B.

Why B is correct: The one-time calibration of risk-neutral probabilities to market prices is the only arbitrage step; afterwards any claim is priced by expected discounted value.

Why the others are wrong

- **A:** 'Once or more than once' is wrong: re-modifying the process for each claim would remove the efficiency; the adjustment is done once.
- **C:** Without changing true probabilities to risk-neutral ones, expected discounted values would not equal arbitrage prices; the calibration itself requires arbitrage pricing.
- **D:** Real-world probabilities do not give arbitrage-free prices because they ignore the risk penalty embedded in market prices; risk-neutral probabilities are used instead.

Q579

AnalystPrep Q.1624

REBUILT — Stem referred to an unshown example ('the one-year zero'); made self-contained.

In Tuckman's binomial example, a call option on a one-year zero-coupon bond is priced by arbitrage from the one-year zero. The risk penalty implicit in the call option price is inherited from the risk penalty of the one-year zero, that is, from the premise that the market price of the one-year zero is:

A. greater than its expected discounted value.

B. less than its expected discounted value.

C. equal to its expected discounted value.

D. not related to its expected discounted value.

Answer: B

Step by step

1. Using true probabilities, the zero's expected discounted value exceeds its market price (e.g., 961.18 vs 960.23

in Tuckman's setup).
2. The gap is the premium investors demand for rate risk, i.e., the risk penalty.
3. The option priced by arbitrage off the zero inherits this penalty, so the premise is price < EDV: option B.

Why B is correct: Risk-averse investors price the zero below its expected discounted value; arbitrage transmits that penalty to the option.

Why the others are wrong

- **A:** Reverses the sign; risk aversion lowers prices below EDV.
- **C:** If price equaled EDV there would be no risk penalty and true probabilities would already be risk neutral.
- **D:** The risk-neutral probabilities are defined precisely by the relation between the zero's market price and its EDV.

Q580

AnalystPrep Q.1629

The price of a derivative in the real economy may be computed as the discounted value under the risk-neutral probabilities. Which of the following statements about the price of an option is correct?

A. The arbitrage-free price of the option equals its expected discounted value under the risk-neutral probabilities.
B. The price of a security that is priced by arbitrage depends on investors' risk preferences.
C. Investors in the imaginary economy penalize securities for risk and do not price securities by expected discounted value.
D. The price of an option does not necessarily need to be the same in the real and imaginary economies.

Answer: A

Step by step

1. Risk-neutral pricing builds an imaginary economy in which investors are indifferent to risk and price every security at its expected discounted value.
2. The risk-neutral probabilities are chosen so the imaginary economy reproduces the market prices of the underlying securities.
3. Because the option is tied to those securities by arbitrage, its price in the imaginary economy must equal its real-economy price, and equals the expected discounted value under risk-neutral probabilities: option A.

Why A is correct: Arbitrage forces the option price to equal its expected discounted value under the risk-neutral probabilities, in both economies.

Why the others are wrong

- **B:** Arbitrage prices are independent of risk preferences; that independence is why risk-neutral pricing works.
- **C:** Reverses the logic: it is the real economy where investors penalize risk; investors in the imaginary economy are risk neutral and price by expected discounted value.
- **D:** The two economies are constructed to give identical prices for the underlying securities and hence for any derivative priced by arbitrage.

Q581

BT answers L2.T5.40.2

REBUILT — Source answered only via a spreadsheet and had weak, arbitrary distractors; options rebuilt around realistic errors.

Assume the six-month and one-year spot rates are 2.0% and 2.2%, respectively (semiannual compounding, per Tuckman). In a two-step binomial tree with six-month steps, the six-month rate six months from now will be either 2.5% (up) or 1.5% (down). Consider a zero-coupon bond with a face value of USD 1,000 and one year to maturity. What is the risk-neutral probability p of the up move (and $1 - p$ of the down move)?

A. p = 90.1% and 1-p = 9.9%
B. p = 9.9% and 1-p = 90.1%
C. p = 70.1% and 1-p = 29.9%
D. p = 50.0% and 1-p = 50.0%

Answer: A

Step by step

1. Today's price of the one-year zero: $1000/(1.011)^2 = 978.36$.
2. Prices in six months: up-rate state $1000/1.0125 = 987.65$; down-rate state $1000/1.0075 = 992.56$.
3. Require $[p(987.65) + (1 - p)(992.56)]/1.01 = 978.36$, so $p(987.65) + (1 - p)(992.56) = 988.14$.
4. $p = (992.56 - 988.14)/(992.56 - 987.65) = 0.9006 \approx 90.1\%$, and $1 - p = 9.9\%$.

Why A is correct: Risk-neutral p is the weight that makes the discounted expected six-month price equal today's market price of the zero; since the up-rate price (987.65) is the lower one, p must be large to pull the average down

to 988.14.

Why the others are wrong

- **B:** Probabilities swapped: gives the high weight to the down-rate (high-price) state, which would give an expected price above 988.14.
- **C:** Discounted the expected price at 1.1% (the one-year rate/2) instead of 1.0% (the six-month rate/2), solving $p(987.65) + (1-p)(992.56) = 978.36 \times 1.011$.
- **D:** Used the real-world 50/50 probabilities; these give an expected discounted value of 990.11/1.01 = 980.30, above the market price 978.36, so they are not risk neutral.

Q582

BT answers L2.T5.41.4

REBUILT — Stem omitted the 4% six-month rate and today's zero price (USD 960.23) needed to solve; added them and rebuilt distractors.

A one-year zero-coupon bond (face value USD 1,000) is priced on a two-step binomial tree with six-month steps (semiannual compounding, per Tuckman). The current six-month spot rate is 4.00% and the one-year zero trades at USD 960.23. In six months the bond (then a six-month zero) will be worth USD 978.00 in the up-rate state or USD 982.80 in the down-rate state; the real-world probabilities are 50% each. What are the risk-neutral up and down probabilities?

A. 70.1% and 29.9%
B. 29.9% and 70.1%
C. 50.0% and 50.0%
D. 98.2% and 1.8%

Answer: A

Step by step

1. Discount factor for the first step: $1 + 0.04/2 = 1.02$.
2. Set $[p(978.00) + (1-p)(982.80)]/1.02 = 960.23$, so $p(978.00) + (1-p)(982.80) = 979.4346$.
3. $p = (982.80 - 979.4346)/(982.80 - 978.00) = 3.3654/4.80 = 0.7011$.
4. Up probability 70.1%, down probability 29.9%.

Why A is correct: Risk-neutral probabilities make the expected discounted bond value equal the observed market price of USD 960.23, which here puts more weight on the lower (up-rate) price.

Why the others are wrong

- **B:** Swapped the probabilities: the lower price (978.00) needs the higher weight to pull the average down to 979.43.
- **C:** Used real-world 50/50 probabilities, which give an expected discounted value of 980.40/1.02 = 961.18, above the market price.
- **D:** Set $p = 960.23/978.00$, ignoring the discount factor and the down-state price.

Q583

BT answers L2.T5.41.6

REBUILT — Stem omitted the 4% rate and the zero's current price needed to get the risk-neutral probabilities; added them and rebuilt distractors (one was OCR-garbled).

A one-year zero-coupon bond (face value USD 1,000) is priced on a two-step binomial tree with six-month steps (semiannual compounding, per Tuckman). The current six-month spot rate is 4.00% and the one-year zero trades at USD 960.23. In six months the bond (then a six-month zero) will be worth USD 978.00 in the up-rate state or USD 982.80 in the down-rate state; real-world probabilities are 50% each. A European call option expiring in six months on this bond has a strike of USD 980.00. What is the option's market price (use risk-neutral pricing)?

A. USD 0.82
B. USD 0.84
C. USD 1.37
D. USD 1.92

Answer: A

Step by step

1. Risk-neutral up probability: $p = (982.80 - 960.23 \times 1.02)/(982.80 - 978.00) = 0.7011$; down probability 0.2989.
2. Option payoffs at six months: up-rate state $\max(978 - 980, 0) = 0$; down-rate state $\max(982.80 - 980, 0) = 2.80$.
3. Price $= [0.7011(0) + 0.2989(2.80)]/1.02 = 0.8369/1.02 = 0.82$.

Why A is correct: The option is priced as its expected discounted payoff under the risk-neutral probabilities, which embed the market's risk penalty.

Why the others are wrong

- **B:** Forgot to discount the expected payoff (0.2989 x 2.80 = 0.84).
- **C:** Used the real-world 50/50 probabilities: $0.5(2.80)/1.02 = 1.37$, which ignores the risk penalty.
- **D:** Applied the 70.1% probability to the wrong state (the down-rate payoff): $0.7011(2.80)/1.02 = 1.92$.

Q584

BT answers L2.T5.41.8

Which of the following best describes risk-neutral pricing?

A. Risk-neutral probabilities as inputs into expected discounted value
B. Real-world probabilities as inputs into expected discounted value
C. Risk-neutral probabilities but only in the imaginary world of risk aversion
D. Risk-neutral probabilities replace the risk-free rate to incorporate risk aversion

Answer: A

Step by step

1. Step 1: find the risk-neutral probabilities that make expected discounted values of the underlying securities equal their market prices.
2. Step 2: price the derivative as its expected discounted value using those probabilities and risk-free discounting.
3. So risk-neutral pricing = risk-neutral probabilities as inputs into expected discounted value: option A.

Why A is correct: Risk-neutral pricing keeps the expected-discounted-value form but replaces true probabilities with risk-neutral ones, so the risk penalty is carried by the probabilities.

Why the others are wrong

- **B:** Real-world probabilities ignore the risk penalty in market prices and do not reproduce arbitrage prices.
- **C:** The imaginary world is one of risk neutrality, not risk aversion; the resulting prices also hold in the real world.
- **D:** Probabilities do not replace the rate; discounting stays at the risk-free rate and the risk penalty is absorbed in the probabilities.

Q585

BT answers L2.T5.42.3

REBUILT — Original option B ('growth equals risk-free rate') is true of the risk-neutral process, making the EXCEPT ambiguous; reworded to refer to the real-world return.

In regard to the valuation of contingent claims by risk-neutral pricing, each of the following is true EXCEPT:

A. The derivative's expected discounted value under risk-neutral probabilities equals its arbitrage price.
B. The real-world expected return on the underlying security must equal the risk-free rate for the method to be valid.
C. The risk-free rate is the appropriate discount rate.
D. The derivative's price is the same in the imaginary (risk-neutral) world and in the real world.

Answer: B

Step by step

1. A, C and D are the core results: expected discounted value under risk-neutral probabilities equals the arbitrage price, discounting is at the risk-free rate, and the price is identical in both worlds.
2. Because the derivative is replicated by the underlying, its payoff is hedged and the underlying's real-world expected return does not matter.
3. The underlying earns the risk-free rate only under the risk-neutral probabilities, by construction, not as a requirement on the real world: B is false.

Why B is correct: Replication makes the real-world expected return irrelevant; it need not equal the risk-free rate.

Why the others are wrong

- **A:** True statement, so not the exception: this is the definition of risk-neutral pricing.
- **C:** True statement: after moving to risk-neutral probabilities the risk-free rate is the right discount rate.
- **D:** True statement: prices coincide because the imaginary economy is built to match the market prices of the underlying.

Q586

BT notes P2.T5.23.1.1

REBUILT — Distractors were arbitrary; rebuilt around real mistakes (real-world probability, no discounting, swapped probabilities).

An option pays either USD 6.10 or nothing in six months. The risk-free rate is 3.0% per annum with semiannual compounding. The option's expected discounted value using real-world probabilities is USD 3.00, but its market price is USD 1.80. What are the implied risk-neutral probabilities of the USD 6.10 payoff and the zero payoff?

A. 49.9% and 50.1%
B. 29.5% and 70.5%
C. 30.0% and 70.0%
D. 70.0% and 30.0%

Answer: C

Step by step

1. Six-month discount factor: $1 + 0.03/2 = 1.015$.
2. Set $[p(6.10) + (1-p)(0)]/1.015 = 1.80$.
3. $p = 1.80 \times 1.015/6.10 = 1.827/6.10 = 0.2995 \approx 30.0\%$; $1 - p = 70.0\%$.

Why C is correct: The risk-neutral p equates the discounted expected payoff to the market price, so p = price x (1 + r/2) / payoff.

Why the others are wrong

- **A:** Backed out the real-world probability from the EDV of USD 3.00 ($3.00 \times 1.015/6.10$), which ignores the market price.
- **B:** Forgot to discount: $p = 1.80/6.10 = 29.5\%$.
- **D:** Swapped the probabilities, assigning 70% to the USD 6.10 payoff.

Q587

Mock 3 Q48

An analyst is calibrating a one-period binomial interest rate tree to observed market prices. The current one-year spot rate is 4.00%. In one year the one-year rate will be either 5.50% or 3.50%. A two-year zero-coupon bond with a face value of USD 100 currently trades at USD 92.19. What is the risk-neutral probability of the upward move (in the rate)?

A. 0.35
B. 0.40
C. 0.50
D. 0.60

Answer: B

Step by step

1. Year-1 bond values: up-rate $100/1.055 = 94.7867$; down-rate $100/1.035 = 96.6184$.
2. Require $[q(94.7867) + (1-q)(96.6184)]/1.04 = 92.19$, so the numerator is $92.19 \times 1.04 = 95.8776$.
3. $q = (96.6184 - 95.8776)/(96.6184 - 94.7867) = 0.7408/1.8317 = 0.4044 \approx 0.40$.

Why B is correct: q makes the expected discounted year-1 bond value match the observed price, placing less weight on the up-rate (low-price) state than 50%.

Why the others are wrong

- **A:** Too low: with $q = 0.35$ the expected discounted value is about 92.29, not the market price 92.19.
- **C:** Assumed equal odds; the real-world 50/50 expected discounted value is 92.02, not the market price 92.19.
- **D:** The probability of the down move (1 - 0.4044 = 0.5956), mistaken for the up probability.

Q588

BT answers L2.T5.42.2

Which of the following is MOST essential to the argument that contingent claims can be valued with risk-neutral pricing?

A. The no-arbitrage price is invariant to investor risk preferences
B. The no-arbitrage price accounts for investor risk preferences
C. Investors are risk-neutral in the imaginary world
D. The discount rate impounds investor risk aversion

Answer: A

Step by step

1. A contingent claim (e.g., an option on a bond) can be replicated by a portfolio of the underlying bonds and cash, so the claim and the replicating portfolio must have the same price or an arbitrage exists.
2. The replicating (hedged) portfolio removes the claim's exposure to the risky factor, so its price does not depend on how much any investor dislikes risk.
3. Because the no-arbitrage price is the same for a highly risk-averse and a risk-seeking investor, one may compute it in any convenient world, including a fictitious one where investors are risk-neutral and discount at the risk-free rate.
4. Hence the essential premise is risk-preference invariance of the no-arbitrage price; risk-neutral investors are merely a computational device that follows from it.

Why A is correct: Risk-neutral pricing is legitimate only because replication makes the arbitrage-free price independent of investor risk preferences, which lets us price in a risk-neutral world and obtain the same answer.

Why the others are wrong

- **B:** States the opposite: if the no-arbitrage price depended on risk preferences, the price would not be common to all investors and risk-neutral valuation could not be justified.
- **C:** A true feature of the risk-neutral world, but it is a consequence/construct, not the justification; the real world does not have risk-neutral investors, and the argument works because preferences do not matter.
- **D:** Describes real-world (true-probability) valuation, where the discount rate must include a risk premium; risk-neutral pricing instead uses the risk-free rate with adjusted probabilities.

LO 12.d

Explain the difference between true and risk-neutral probabilities and apply this difference to interest rate drift.

Q589

BT answers L2.T5.42.1

Assume a binomial interest rate tree where the six-month rate either jumps up 50 basis points or jumps down 50 basis points. The real-world (true) probabilities are 50% and 50% for the up- and down-states. If the interest rate drift is +20 basis points under risk-neutral probabilities, what is the risk-neutral probability of an up-state (p)?

A. 50%
B. 60%
C. 70%
D. 80%

Answer: C

Step by step

1. Under true probabilities the expected rate change is 0.5(+50) + 0.5(-50) = 0 bp.
2. Drift = risk-neutral expected change minus true expected change = +20 bp.
3. Risk-neutral expected change: $p(50) + (1-p)(-50) = 100p - 50$ bp.
4. Set $100p - 50 = 20$, so $p = 70/100 = 70\%$.

Why C is correct: Drift is the risk-neutral expected rate change minus the true expected change; the true change is zero, so the risk-neutral expected change itself must equal +20 bp, which requires p = 70%.

Why the others are wrong

- **A:** 50% is the true probability and gives zero drift.
- **B:** 60% gives drift of only +10 bp (100(0.6) - 50).
- **D:** 80% gives drift of +30 bp (100(0.8) - 50).

Q590

BT answers L2.T5.41.3

REBUILT — Original asked for an ambiguous 'expected' price and omitted the six-month rate; added the 6-month rate and reworked options so exactly one answer (the market price) is right and distractors reflect realistic errors.

A two-step, one-year binomial tree (six-month steps) is used to price a one-year zero-coupon bond with face value $1,000. The current six-month rate is 4.0%, and in six months the bond (then a six-month zero) will be worth $978.00 in the up-state or $982.80 in the down-state. True probabilities are 50% up and 50% down. Rates use semiannual compounding. The one-year spot rate is 4.1%. What is the bond's current market price?

A. $961.18

B. $960.61
C. $960.23
D. $922.78

Answer: C
Step by step
1. The market price is the face value discounted at the one-year spot rate with semiannual compounding.
2. $P = 1000/(1 + 0.041/2)^2 = 1000/1.0205^2$
3. $P = 1000/1.04142 = 960.23$ dollars.
4. For comparison, the 50/50 true-probability discounted value is $(0.5(978) + 0.5(982.8))/1.02 = 961.18$, which does not match the market price; this gap is why risk-neutral rather than true probabilities are used to calibrate the tree.

Why C is correct: The market price is observed from the spot curve, not from true-probability expectations; the tree is calibrated to it using risk-neutral probabilities.

Why the others are wrong
- **A:** Discounted the true-probability (50/50) expected value at 4.0%/2; true probabilities do not reproduce the market price.
- **B:** Used annual compounding for one period, 1000/1.041, ignoring the semiannual convention and the two periods.
- **D:** Used 1000/1.041^2, annual compounding over two years, treating 4.1% as a per-period rate.

Q591

BT answers L2.T5.41.7

REBUILT — Original depended on data from a companion question and had giveaway options (+/-20, +/-30); made self-contained with the six-month rate and market price and replaced options with realistic error results.

A two-step, one-year binomial tree (six-month steps) prices a one-year zero-coupon bond with face value $1,000. The current six-month rate is 4.0% (semiannual compounding); in six months the rate is 4.5% (bond worth $978.00) in the up-state or 3.5% (bond worth $982.80) in the down-state. The bond's current market price is $960.23 and the true probabilities are 50% up / 50% down. Using risk-neutral probabilities calibrated to the market price, what is the implied interest rate drift (rate moves are +/-50 bp)?

A. 0 bp
B. +10 bp
C. +20 bp
D. +70 bp

Answer: C
Step by step
1. Calibrate: $960.23 = [p(978) + (1 - p)(982.8)]/1.02$, so $960.23 \times 1.02 = 979.43 = 982.8 - 4.8p$.
2. $p = (982.8 - 979.43)/4.8 \approx 70.1\%$ up, 29.9% down.
3. Risk-neutral expected change: $0.701(50) + 0.299(-50) \approx +20.2$ bp, i.e. about +20 bp.
4. True expected change: 0.5(50) + 0.5(-50) = 0 bp, so drift is about +20 bp.

Why C is correct: Drift is the risk-neutral expected rate change minus the true expected change; with p about 70% the risk-neutral change is about +20 bp versus 0 under true probabilities.

Why the others are wrong
- **A:** 0 bp is the expected change under the true 50/50 probabilities, not the risk-neutral ones.
- **B:** +10 bp comes from (p - 0.5) x 50, counting only the shift in the up-state probability and ignoring the offsetting down-state shift.
- **D:** +70 bp mistakes the risk-neutral up probability (about 70%) for the drift in basis points.

Q592

BT notes P2.T5.23.2.1

Below is a multi-period price tree for a zero-coupon bond with 1.5 years to maturity. It is similar to Tuckman's (4th edition); the only difference is that the spot rate term structure has been shifted up in parallel by 100 basis points: {3.0% at 0.5 years, 3.15% at one year, 3.25% at 1.5 years}.

Term structure	s(0.5)	s(1.0)	s(1.5)
Spot (0,T)	3.00%	3.15%	3.25%
PV of $1,000	$985.22	$969.23	$952.79
Six-month rate tree: date 0 = 3.00%; date 1 = 3.50% (up)	2.50% (down); date 2 = 4.00%	3.00%	2.00%
Risk-neutral probabilities: date 0 to 1 = 80.09% up	19.91% down; date 1 to 2 = 65.19% up	34.81% down	
Price tree: date 0 = $952.79; date 1 = $965.18 (up)	$974.73 (down); date 2 = $980.39	$985.22	$990.10; date 3 = $1

In regard to this multi-period tree, each of the following statements is true EXCEPT which is false?

A. The bond's current market price of $952.79 assumes real-world probabilities applied to its expected discounted value (EDV)

B. If the 1.5-year spot rate, s(1.5), rises from 3.25% to 3.50%, this will change at least some of the risk-neutral probabilities

C. The drifts (expected changes) in the six-month rate are +30.1 basis points (date 0 to 1) and +15.2 basis points (date 1 to 2)

D. Extending the tree to further dates (e.g., 2 years, 2.5 years) requires assumptions about future short-rate values and risk-neutral probabilities that recover a given set of bond prices

Answer: A

Step by step

1. A: The market price equals the EDV using RISK-NEUTRAL probabilities (chosen to match market prices), not real-world ones: (0.8009 × 965.18 + 0.1991 × 974.73)/1.015 = 952.79. So A is false.
2. B: probabilities are solved to fit the bond prices, so a different s(1.5) requires different probabilities. True.
3. C: Date 0 to 1: 0.8009(3.50) + 0.1991(2.50) = 3.301%, a drift of +30.1 bps. Date 1 to 2: 0.6519(4.00) + 0.3481(3.00) = 3.652% from 3.50% (and 2.652% from 2.50%), a drift of +15.2 bps. True.
4. D: Each added date needs rate assumptions and probabilities calibrated to the given bond prices. True.

Why A is correct: The tree's probabilities are risk-neutral: they are the ones that make the model price equal the observed market price, so the statement attributing the price to real-world probabilities is false.

Why the others are wrong

- **B:** True: probabilities are calibrated to the term structure, so a higher s(1.5) changes them. Not the exception.
- **C:** True: the risk-neutral expected rate change is +30.1 bps then +15.2 bps. Not the exception.
- **D:** True: extending the tree requires calibration to additional bond prices. Not the exception.

LO 12.e

Explain how the principles of arbitrage pricing of derivatives on fixed-income securities can be extended over multiple periods.

Q593

AnalystPrep Q.2662

Which of the following is the term used to describe the process of valuing a bond using a binomial interest rate tree?

A. Bootstrapping

B. Backward induction

C. Backtesting

D. Bootstrap historical simulation

Answer: B

Step by step

1. Start at the final maturity nodes where the bond value is its payoff (face value plus any coupon).
2. At each earlier node, take the average of the two next-period values (including coupons) and discount at the node's one-period rate.
3. Repeat back to date 0; this right-to-left procedure is backward induction.

Why B is correct: Valuation in a tree proceeds from the terminal nodes back to the root, which is backward induction.

Why the others are wrong

- **A:** Bootstrapping builds a spot/zero curve from par or coupon instruments; it does not roll values back through a tree.

- **C:** Backtesting compares VaR forecasts with realized losses and is unrelated to bond pricing.
- **D:** Bootstrap historical simulation resamples historical returns to estimate VaR, not to price a bond.

Q594

Mock 2 Q22

REBUILT — Options B-D were vague giveaway node-trade statements; rebuilt them as quantified tree-value conclusions that each result from a specific error.

A binomial tree for a 2-year, 5% annual-coupon bond with a market price of 99.791 (probability 50% on each branch):

Node	Bond price (ex-coupon)	Coupon	Rate
t = 0	99.791	-	4.5000%
t = 1, upper	98.384	5	6.7250%
t = 1, lower	100.179	5	4.8125%
t = 2 (all nodes)	100	5	-

Compared with the tree-implied value, what arbitrage opportunity exists?

A. None: the tree-implied value equals the market price of 99.791.
B. Sell the bond: the tree-implied value is 95.01, below the market price.
C. Buy the bond: the tree-implied value is 104.28, above the market price.
D. Sell the bond: the tree-implied value is 98.61, below the market price.

Answer: A

Step by step

1. Add the t = 1 coupon to each ex-coupon price: upper 98.384 + 5 = 103.384; lower 100.179 + 5 = 105.179.
2. Expected value = 0.5(103.384) + 0.5(105.179) = 104.2815.
3. Discount at the t = 0 rate 4.5%: 104.2815/1.045 = 99.791.
4. This equals the market price, so no arbitrage.

Why A is correct: The expected discounted value, including the coupon at t = 1, matches the market price, so the tree reprices the bond.

Why the others are wrong

- **B:** 95.01 = 0.5(98.384 + 100.179)/1.045: omitted the t = 1 coupon, creating a phantom overpricing.
- **C:** 104.28 is the expected value before discounting; forgot to divide by 1.045.
- **D:** 98.61 = 0.5(103.384/1.06725 + 105.179/1.048125): discounted each branch at its own t = 1 node rate instead of the t = 0 rate.

Q595

BT answers L2.T5.43.4

REBUILT — Source referred to a spreadsheet and had weak, unexplained distractors; stem made self-contained and options rebuilt from realistic errors.

A three-step binomial interest rate tree has six-month steps (1.5 years to maturity). Rates are annualized, semiannually compounded, so each step discounts by (1 + r/2). The six-month rate is 3.0% at date 0; 3.5% (up) or 2.5% (down) at date 1 (0.5 years); and 4.0%, 3.0% or 2.0% at date 2 (1.0 year). The risk-neutral probability is 80% for an up-move and 20% for a down-move at every node. What is the price of a $1,000 face value zero-coupon bond with 1.5 years to maturity?

A. $956.34
B. $960.59
C. $952.11
D. $964.48

Answer: C

Key correction: Source key (B, $952.11) is correct; options were rebuilt, so the letter changes to C.

Step by step

1. Date 2 values (discount $1,000 by r/2): 1000/1.02 = 980.39; 1000/1.015 = 985.22; 1000/1.01 = 990.10.
2. Date 1 up node (3.5%): (0.8 × 980.39 + 0.2 × 985.22)/1.0175 = 964.48.
3. Date 1 down node (2.5%): (0.8 × 985.22 + 0.2 × 990.10)/1.0125 = 974.02.
4. Date 0 (3.0%): (0.8 × 964.48 + 0.2 × 974.02)/1.015 = 952.11.

Why C is correct: Backward induction: at each node discount the risk-neutral expected value of the two successor prices at that node's six-month rate.

Why the others are wrong

- **A:** Uses 50/50 probabilities instead of 80/20, giving $956.34.

- **B:** Swaps the probabilities (20% up, 80% down), giving $960.59.
- **D:** Stops at date 1 and reports the up-node value ($964.48) without completing the last step.

Q596

GARP PE1 Q32

REBUILT — Tree figure was missing and the stem confused the first-step probabilities with q; reconstructed the full rate and price trees and stated that 0.70/0.30 are the first-step risk-neutral probabilities.

An analyst on a fixed-income desk is calculating risk-neutral probabilities in a zero-coupon bond price tree. Semi-annual spot rates (annualized, semiannually compounded) and prices of a bond maturing at t = 1.5 are:

t (years)	0	0.5	1.0	1.5
Rate, upper	| 4.0%	4.5%		
Rate, middle	3.5%	| 3.5%		
Rate, lower	| 3.0%	2.5%		
Price, upper	| P(1,1) = ?	978.00	1,000	
Price, middle	945.80	| 982.80	1,000	
Price, lower	| P(1,0) = ?	987.65	1,000	

The risk-neutral probability of an up-move from t = 0 is 0.70 (0.30 down). The risk-neutral probability of an up-move is the same value q at both t = 0.5 nodes. What is q?

A. 0.15
B. 0.50
C. 0.70
D. 0.85

Answer: D

Step by step

1. Node prices as functions of q: $P(1,1) = [978.00q + 982.80(1-q)]/1.02$ and $P(1,0) = [982.80q + 987.65(1-q)]/1.015$.
2. The t = 0 price must satisfy $945.80 = [0.7P(1,1) + 0.3P(1,0)]/1.0175$, so the bracket must equal 945.80 × 1.0175 = 962.35.
3. Solving numerically gives q = 0.854 (about 0.85). Check: P(1,1) = 959.51, P(1,0) = 968.98; 0.7(959.51) + 0.3(968.98) = 962.35; 962.35/1.0175 = 945.80.

Why D is correct: q is the second-step probability that makes the model price at t = 0 equal the given 945.80, which is about 0.85.

Why the others are wrong

- **A:** 1 - q: the probability of a down-move.
- **B:** Assumes risk-neutral means even odds.
- **C:** Reuses the first-step probability (0.70) for the second step; it was only the probability at t = 0.

Q597

GARP PE2 Q12 · Mock 4 Q13

REBUILT — The two sources differ in key (B vs A) because the stem never said 0.72/0.28 are first-step probabilities and q must be backed out from the 937.49 anchor; tree figure was missing. Reconstructed the full trees and unambiguous wording; one answer.

A fixed-income analyst completes a zero-coupon bond price tree. Semiannual spot rates (annualized, semiannually compounded) and prices of a bond maturing at t = 1.5 are:

t (years)	0	0.5	1.0	1.5
Rate, upper	| 4.65%	5.10%		
Rate, middle	4.20%	| 4.20%		
Rate, lower	| 3.75%	3.30%		
Price, upper	| P(1,1) = ?	975.13	1,000	
Price, middle	937.49	| 979.43	1,000	
Price, lower	| P(1,0) = ?	983.77	1,000	

The risk-neutral probabilities of an up-move/down-move from t = 0 are 0.72/0.28. The risk-neutral up-probability q from each t = 0.5 node is the same, unknown, and must be consistent with the t = 0 price. What is P(1,1)?

A. USD 954.15
B. USD 954.81
C. USD 956.00
D. USD 956.91

Answer: B

Key correction: Sources disagreed: Mock 4 Q13 keyed A (954.15), using 0.72 as q; GARP PE2 Q12 keyed B. Correct is B because q must be solved from the 937.49 anchor (q = 0.563).

Step by step

1. Write $P(1,1) = [975.13q + 979.43(1-q)]/1.02325$ and $P(1,0) = [979.43q + 983.77(1-q)]/1.01875$.
2. Require $937.49 = [0.72P(1,1) + 0.28P(1,0)]/1.021$ and solve: q = 0.563.
3. Then P(1,1) = (975.13 × 0.563 + 979.43 × 0.437)/1.02325 = 977.01/1.02325 = 954.81 (and P(1,0) = 963.27).
4. Check: (0.72 × 954.81 + 0.28 × 963.27)/1.021 = 937.49.

Why B is correct: q is not given; it is backed out so the tree reproduces the t = 0 price 937.49, giving q = 0.563 and P(1,1) = 954.81.

Why the others are wrong

- **A:** Uses 0.72 as q (treating the first-step probability as the second-step one): 954.15; this fails to reproduce 937.49.
- **C:** Uses q = 0.28 (swaps the probabilities) giving 956.00.
- **D:** Discounts at the t = 0 rate (4.2%/2) instead of the node's rate 4.65%/2: 977.01/1.021 = 956.91.

Q598

Mock 1 Q30

REBUILT — Stem omitted option expiry and exercise convention, and distractors were arbitrary; specified a European call expiring at T2 on the ex-coupon bond price and rebuilt options from realistic errors.

A derivatives trader values a European call option on a three-year, 7% annual coupon, USD 100 par bond, with strike 100, expiring at T2 (the strike is compared with the bond price immediately after the T2 coupon). Annual rates follow the tree below with a 50% probability of an up- or down-move at each node; the coupon bond's remaining cash flow at T2 is the final 107 at T3.

Node	T0	T1	T2
Upper	| 5%	8%	
Middle	2%	| 6%	
Lower	| 4%	4%	

What is the value of the option at the T1 upper node (5%)?

A. USD 0.47
B. USD 0.90
C. USD 0.45
D. USD 1.82

Answer: C

Step by step

1. Bond price at T2 (ex-coupon): 107/1.08 = 99.07; 107/1.06 = 100.94; 107/1.04 = 102.88.
2. Call payoffs max(P - 100, 0): 0, 0.943, 2.885.
3. From the T1 upper node (5%), successors are the 8% and 6% nodes: expected payoff = 0.5(0) + 0.5(0.943) = 0.4717.
4. Discount at 5%: 0.4717/1.05 = 0.449, about USD 0.45.

Why C is correct: Value at a node is the risk-neutral expected payoff of the two successors discounted at that node's rate; the 5% node leads only to the 8% and 6% states.

Why the others are wrong

- **A:** Forgot to discount at 5%: 0.47.
- **B:** Forgot the 0.5 probability weight on the 6% payoff: 0.943/1.05 = 0.90.
- **D:** Used the wrong successors (6% and 4%): 0.5(0.943 + 2.885)/1.05 = 1.82.

Q599

Schweser Module Quiz 12.2 Q1

REBUILT — OCR wrongly rendered both year-2 moves as 'rise'; corrected to rise/fall, made ex-coupon convention explicit, and rebuilt options from realistic errors.

A European put option has two years to expiration and a strike price of $101.00. The underlying asset is a 7% annual coupon bond with three years to maturity (the strike is compared with the bond price immediately after the Year 2 coupon, so the remaining cash flow is 107 at Year 3). The risk-neutral probability of an up move is 0.76 in Year 1 and 0.60 in Year 2. The current interest rate is 3.00%. At the end of Year 1 the rate will be either 5.99% (up) or 4.44% (down). If the Year 1 rate is 5.99%, the Year 2 rate will rise to 8.56% or fall to 6.34%. If the Year 1 rate is 4.44%, the Year 2 rate will rise to 6.34% or fall to 4.70%. The value of the put option today is closest to:

A. $0.52

B. $0.87
C. $1.17
D. $1.24

Answer: C
Step by step
1. Year 2 bond prices: 107/1.0856 = 98.56; 107/1.0634 = 100.62; 107/1.0470 = 102.20.
2. Put payoffs max(101 - P, 0): 2.44, 0.38, 0.
3. Year 1 up (5.99%): (0.6 × 2.437 + 0.4 × 0.379)/1.0599 = 1.52. Year 1 down (4.44%): (0.6 × 0.379 + 0.4 × 0)/1.0444 = 0.22.
4. Today: (0.76 × 1.523 + 0.24 × 0.218)/1.03 = 1.17.

Why C is correct: Backward induction on the put payoffs with node-specific discounting and the stated risk-neutral probabilities gives $1.17.
Why the others are wrong
- **A:** Uses 0.24 as the Year 1 up-probability (swaps up and down): about 0.52.
- **B:** Uses 0.4 as the Year 2 up-probability (swaps up and down): about 0.87.
- **D:** Does not discount the Year 2 payoffs back to Year 1 at the node rates (only discounts once at 3.00%): 1.24.

LO 12.f

Define option-adjusted spread (OAS) and apply it to security pricing.

Q600

Schweser Module Quiz 12.3 Q3

Which of the following statements about callable bonds compared to noncallable bonds is false?
A. They have less price volatility.
B. They have negative convexity.
C. Capital gains are capped as yields rise.
D. At low yields, reinvestment rate risk rises.

Answer: C
Step by step
1. A callable bond is a noncallable bond minus a call option held by the issuer, so its price is bounded above by the call price when yields are low.
2. Below the threshold yield y', the price-yield curve flattens (price compression), which is negative convexity (B true) and means price changes are smaller than for a noncallable bond (A true).
3. The cap on price is binding when yields FALL, so capital gains are capped as yields fall, not as yields rise; when yields are high the call is far out of the money and the bond behaves like a noncallable bond. Statement C is therefore false.
4. When yields are low the issuer is likely to call, returning the call price plus coupon that must be reinvested at lower rates, so reinvestment risk rises (D true).

Why C is correct: The issuer's call option limits upside only when yields fall; it does not cap gains when yields rise, so C misstates the direction.
Why the others are wrong
- **A:** True statement: the embedded short call dampens price moves, so selecting it means mistaking a true statement for the false one.
- **B:** True statement: price compression at low yields is negative convexity; selecting it confuses callable bonds with positively convex noncallable bonds.
- **D:** True statement: calls at low yields force reinvestment of principal at lower rates; selecting it ignores that the call price adds to the cash flow to be reinvested.

Q601

AnalystPrep Q.1632

REBUILT — Original was a fill-in-the-blank definition with a giveaway 'Cannot be determined' option; rebuilt as a numerical sign/interpretation item on the same OAS definition.

A binomial interest rate tree gives a constant-maturity Treasury swap a model value of USD 1,466.63, while its

observed market price is USD 1,464.40. The option-adjusted spread (OAS) is the constant spread that, when added to the short rates used for discounting in the tree, equates the model value to the market price. Which statement is correct?

A. The OAS is positive; the security is trading cheap relative to the model.
B. The OAS is negative; the security is trading cheap relative to the model.
C. The OAS is positive; the security is trading rich relative to the model.
D. The OAS is zero, because the model already prices the security.

Answer: A

Step by step

1. Market price (1,464.40) < model value (1,466.63): the security is trading cheap by USD 2.23.
2. Discounting at higher rates lowers present values, so to push the model value down to the market price a positive spread must be added to every discount rate.
3. Hence OAS > 0 (Schweser example: 20 bp) and the security is cheap.

Why A is correct: A market price below model value requires a larger discount rate, so the OAS is positive, which corresponds to a cheap security.

Why the others are wrong

- **B:** Reverses the sign: a negative spread would lower discount rates and raise the model value further above the market price.
- **C:** Gets the sign right but mislabels it; a market price below model value is cheap, not rich.
- **D:** An OAS of zero applies only when market price equals model value; here they differ by USD 2.23.

Q602 AnalystPrep Q.1633

The P&L of a security can be decomposed into a component due to the passage of time, a component due to changes in the factor (short rate), and a component due to changes in the option-adjusted spread (OAS). If a security is not priced in accordance with the model (its OAS is not zero), its cash flows in the tree are discounted at:

A. The short-term rate
B. The long-term rate
C. The short-term rate plus the OAS
D. The short-term rate minus the OAS

Answer: C

Step by step

1. OAS is defined as the constant spread added to the tree's short rates so that model value equals market price.
2. Therefore, when OAS is nonzero, each node's discount rate is the short rate at that node plus the OAS.
3. Cash flows themselves are not changed; only discount rates are shifted.

Why C is correct: The OAS is added to every one-period discount rate in the tree, so nonzero-OAS securities are discounted at short rate plus OAS.

Why the others are wrong

- **A:** This is the discount rate for a security priced exactly by the model (OAS = 0), so it ignores the spread.
- **B:** Tree valuation discounts node by node at the one-period short rate, not a long-term rate.
- **D:** Subtracting the spread reverses its role; the OAS is added to the short rates (it is simply negative if the market price exceeds model value).

Q603 AnalystPrep Q.1812

A fund manager holds option-embedded bonds valued with option-adjusted spreads. After a shock, all embedded options expire (are removed) while the bonds still have time to maturity. Which spread measure is then the natural substitute for the OAS for valuation purposes?

A. Credit default swap spread
B. Asset-swap spread
C. Yield spread
D. Z-spread

Answer: D

Step by step

1. OAS = spread over the tree's rates after accounting for the embedded option's value.
2. With no option, interest rate paths no longer affect cash flows, so volatility is irrelevant and OAS reduces to the zero-volatility (Z-) spread.
3. The Z-spread is the constant spread over the spot curve equating the PV of the cash flows to the price.

Why D is correct: Without optionality, the OAS collapses to the Z-spread, the constant spread to the spot curve that matches price.

Why the others are wrong

- **A:** A CDS spread measures the cost of credit protection, not a constant discounting spread over the curve.
- **B:** The asset-swap spread is the spread on a swap package against floating rate, not a spread over the spot curve.
- **C:** A yield spread compares a single yield to maturity with a benchmark yield, not a spread over the whole spot curve.

Q604

BT Mock Q26 · BT notes P2.T5.23.2.2

REBUILT — Merged two near-identical items; fixed the source typo (market price is below model value), spelled out the tree and node labels in the options; removed unverifiable recomputed values from the steps.

A stylized constant-maturity Treasury (CMT) swap has notional USD 1,000,000 and strike 2.00%, with a semiannual rate tree: date 0 = 2.00%; date 1 = 2.50% (up) or 1.50% (down); date 2 = 3.00%, 2.00%, 1.00%. Risk-neutral up-probabilities are 80.09% (date 0 to 1) and 65.20% (date 1 to 2). The expected discounted value (model value) is about USD 3,702. The observed market price is USD 3,694.81, which is USD 7.30 below the model value. This implies an OAS of about 25 basis points. Which is the best interpretation?

A. The model returns the market price if the risk-neutral probabilities are changed to 55.09% and 44.91%.

B. The model returns the market price if the cash flow at the top date-2 node is cut to USD 3,750 from USD 5,000.

C. The model returns the market price if the discount rates are 2.25% (date 0), 2.75% (date 1, up node) and 1.75% (date 1, down node), i.e., each 25 bp higher.

D. The OAS cannot be interpreted because this is a CMT swap, and an OAS requires an embedded option.

Answer: C

Step by step

1. OAS adds a constant spread to every discount rate in the tree, leaving cash flows and risk-neutral probabilities unchanged.
2. Model value exceeds market price, so heavier discounting is needed and the spread is positive: 2.00%+0.25% = 2.25%, 2.50%+0.25% = 2.75%, 1.50%+0.25% = 1.75%.
3. A uniform 25 bp upward shift in all discount rates lowers the value by a small amount, of the same order as the USD 7.30 gap between model value and market price; this is the signature of an OAS adjustment (the stem gives the 25 bp figure as the implied OAS).
4. Check A: moving the risk-neutral probabilities to 55.09% would change the node values and the model value by hundreds of USD, far more than 7.30, and probabilities are not what the OAS adjusts. Check B: cutting the top date-2 cash flow from USD 5,000 to USD 3,750 changes the swap's cash flows and would reduce the value by hundreds of USD, not 7.30.

Why C is correct: OAS is a constant spread added to all discount rates; shifting every tree rate up 25 bp reproduces the market price.

Why the others are wrong

- **A:** OAS does not alter risk-neutral probabilities; changing them to 55.09% would move the value by far more than USD 7.30.
- **B:** OAS does not alter cash flows; cutting the top cash flow by USD 1,250 would reduce the value by hundreds of USD, not 7.30.
- **D:** OAS applies to any security valued on a tree, with or without an embedded option; it is the spread that equates model to market price.

LO 12.g

Describe the rationale behind the use of recombining trees in option pricing.

Q605

BT answers L2.T5.43.2

Assume a binomial interest rate tree with six steps. At the initial node (T0), the rate is 5.0%. The probability of an up-jump is 60% (p = 0.6). The magnitude of an up-jump is +50 basis points; the magnitude of a down-jump is -30 basis points (u = +0.5%, d = -0.3%). In regard to the distribution at the end of the tree (T6), what is the mode, and what is the probability that the final rate will be the mode?

A. 5.0% mode with 27.65% probability
B. 5.0% mode with 31.10% probability
C. 5.6% mode with 27.65% probability
D. 6.4% mode with 31.10% probability

Answer: D

Step by step

1. The tree recombines, so the number of up-jumps k at T6 is binomial with $n = 6$, $p = 0.6$; the rate is $5.0\% + 0.5\%\, k - 0.3\%\, (6 - k)$.
2. Mode of the binomial: the most likely k is 4 (probabilities: k=3 is 27.65%, k=4 is 31.10%, k=5 is 18.66%).
3. Rate at k = 4: 5.0% + 4(0.5%) - 2(0.3%) = 6.4%.
4. Probability: $\binom{6}{4}(0.6)^4(0.4)^2 = 15 \times 0.1296 \times 0.16 = 0.3110$.

Why D is correct: Because the tree recombines, the terminal rate depends only on the number of up-jumps, which is binomial; the most probable count (4 ups, 2 downs) gives 6.4% with probability 31.10%.

Why the others are wrong

- **A:** Treats the starting rate as the mode and uses the k = 3 probability (27.65%); neither is correct.
- **B:** Keeps the starting rate 5.0% as the mode (ignoring the 60% up-bias) but uses the correct k = 4 probability; the rate and probability are inconsistent.
- **C:** Uses k = 3 ups (half the steps): rate 5.0% + 1.5% - 0.9% = 5.6% and probability 27.65%; k = 3 is not the most likely count when p = 0.6.

Q606

BT answers L2.T5.45.2

REBUILT — Clarified the state-numbering convention (state 0 = lowest rate, per Tuckman), which the stem left implicit.

Assume that the change to the up-state (u) = +20 bps and the change to the down-state (d) = -10 bps in a recombining binomial interest rate tree where p = 50%. The date 0 rate is 3.0%. Using the convention that state 0 is the lowest-rate node at a date (state j has j up-moves), what is the date 12, state 5 interest rate?

A. 3.3%
B. 3.6%
C. 3.9%
D. 4.2%

Answer: A

Step by step

1. State 5 at date 12 means 5 up-moves and 12 - 5 = 7 down-moves.
2. Change = 5(+20 bps) + 7(-10 bps) = +100 - 70 = +30 bps.
3. Rate = 3.0% + 0.30% = 3.3%.

Why A is correct: In a recombining tree the rate depends only on the count of up-moves, so state 5 at date 12 is 3.0% plus 5 ups and 7 downs.

Why the others are wrong

- **B:** Uses 6 ups and 6 downs (the middle state): +120 - 60 = +60 bps, giving 3.6%.
- **C:** Counts states from the top (7 ups, 5 downs): +140 - 50 = +90 bps, giving 3.9%.
- **D:** Uses 6 up-moves and ignores the down-moves: +120 bps, giving 4.2%.

Q607

BT answers L2.T5.45.3

REBUILT — Clarified the state-numbering convention and made the stem self-contained.

Assume that the change to the up-state (u) = +20 bps and the change to the down-state (d) = -10 bps in a recombining binomial interest rate tree where p = 50%. The date 0 rate is 3.0%. Using the convention that state 0 is the lowest-rate node at a date (state j has j up-moves), what is the probability that the rate evolves to the date 12, state 5 interest rate?

A. 5.37%

B. 12.08%
C. 19.34%
D. 22.56%

Answer: C
Step by step
1. Reaching date 12, state 5 requires exactly 5 up-moves in 12 steps; the number of ups is binomial(n = 12, p = 0.5).
2. $P = \binom{12}{5}(0.5)^{12} = 792/4096 = 0.1934$, i.e. 19.34%.

Why C is correct: The probability of ending in a given state of a recombining tree is the binomial probability of that number of up-moves, 792/4096.
Why the others are wrong
- **A:** Uses 3 up-moves: 220/4096 = 5.37%.
- **B:** Uses 4 up-moves: 495/4096 = 12.08%.
- **D:** Uses 6 up-moves (the most likely state): 924/4096 = 22.56%.

Q608

BT answers L2.T5.45.5

REBUILT — Source printed only three options; added a fourth plausible distractor (32) and clarified the stem to a recombining binomial tree.

A binomial interest rate tree is built over four months using weekly time steps (4 weeks per month, so 16 steps from date 0 to date 16). The tree is recombining: an up-then-down move gives the same rate as a down-then-up move. How many distinct interest rates (states) exist at the final date (date 16)?

A. 16
B. 17
C. 32
D. 65,536

Answer: B
Step by step
1. In a recombining tree, a node at date n is identified only by the number of up-moves k (0 to n), not by the path.
2. At date 16, k can be 0, 1, ..., 16, giving 16 + 1 = 17 distinct rates.
3. By contrast, a non-recombining tree has 2^16 = 65,536 distinct paths/rates at date 16.
4. Answer: 17.

Why B is correct: In a recombining tree the number of distinct states at date n grows linearly as n + 1, because only the count of up-moves matters.
Why the others are wrong
- **A:** Counts the number of steps (16) instead of the number of states; forgets that k = 0 to 16 includes 17 values.
- **C:** Uses 2n = 32, confusing linear growth with doubling per step.
- **D:** Gives 2^16, which is the number of nodes in a non-recombining tree, not a recombining one.

Q609

BT answers L2.T5.43.1

REBUILT — Original left the down-move undefined so the answer was 'not enough information'; restated with explicit constant additive moves so exactly one answer is defensible.

A multi-period binomial interest rate tree has a risk-neutral up-state probability of 60% at every node. From any node, the rate rises by 50 basis points in the up state and falls by 30 basis points in the down state, regardless of the current rate level. Is this tree recombining?

A. Yes, because the up-down and down-up paths both lead to the same rate (+20 bp from the starting rate)
B. No, because the up and down moves are of unequal size
C. No, because the up-state probability of 60% differs from the down-state probability of 40%
D. It cannot be determined without knowing the starting rate level

Answer: A
Key correction: Source key (BT answers) said D (not enough information) because the down-move was unspecified; with a constant down-move specified, the tree is recombining.
Step by step
1. Up then down: +50 bp - 30 bp = +20 bp.

2. Down then up: -30 bp + 50 bp = +20 bp.
3. Both orderings give the same rate, so the nodes merge: the tree is recombining.
4. Recombination depends on the move sizes being state-independent, not on whether they are symmetric or on the probabilities or starting level.

Why A is correct: With constant additive moves (state-independent volatility) the order of up and down moves does not matter, so the tree recombines.

Why the others are wrong

- **B:** Wrongly assumes symmetric moves are needed; only state-independence of the moves matters.
- **C:** Confuses probabilities with node values; probabilities affect pricing weights, not whether nodes merge.
- **D:** The starting level is irrelevant because moves are the same at every level.

Q610

BT answers L2.T5.43.3

Which is the best reason to employ a non-recombining interest rate tree?

A. It is analytically tractable
B. The volatility or drift of rates is state-dependent
C. It has fewer nodes
D. It allows a shorter time interval between nodes

Answer: B

Step by step

1. A non-recombining tree arises when up-then-down differs from down-then-up.
2. This happens with state-dependent volatility, e.g. rates move 50 bp above a threshold but by less below it.
3. This is economically appropriate, so it is the reason to use such a tree, despite computational burden.

Why B is correct: State-dependent volatility or drift makes order of moves matter, which forces a non-recombining tree.

Why the others are wrong

- **A:** Describes the recombining tree's advantage; non-recombining trees are hard to compute.
- **C:** Non-recombining trees have more nodes (2^n at date n versus n+1).
- **D:** Step size is independent of whether the tree recombines.

LO 12.h

Calculate the value of a constant-maturity Treasury swap, given an interest rate tree and the risk-neutral probabilities.

Q611

AnalystPrep Q.2660

REBUILT — Tree exhibit was missing and options were unverifiable; tree reconstructed in the stem and distractors replaced with realistic mistakes.

A constant-maturity Treasury (CMT) swap with face value \$1 million is struck at 6%. Every six months it pays $1{,}000{,}000 \times (y_{CMT} - 6\%)/2$, where y_{CMT} is the semiannually compounded six-month yield on the payment date. The six-month rate today is 6%. The risk-neutral probability of an up move is 45% on date 0 (to 6.5% or 5.5% at 6 months) and 70% at each 6-month node (to 7%, 6% or 5% at 1 year, recombining). The swap matures at 1 year. What is the value of the swap?

A. \$500.00
B. \$458.74
C. \$701.46
D. \$472.50

Answer: B

Step by step

1. Payments at 6 months: $1{,}000{,}000 \times (\pm 0.5\%)/2 = \pm\$2{,}500$.
2. Payments at 1 year: $+\$5{,}000$, \$0, $-\$5{,}000$ at rates 7%, 6%, 5%.
3. Upper 6-month node: $[0.7(5{,}000) + 0.3(0)]/1.0325 + 2{,}500 = \$5{,}889.83$.
4. Lower 6-month node: $[0.7(0) + 0.3(-5{,}000)]/1.0275 - 2{,}500 = -\$3{,}959.85$.
5. Today: $[0.45(5{,}889.83) + 0.55(-3{,}959.85)]/1.03 = 472.50/1.03 = \$458.74$.

Why B is correct: Backward induction: at each node, discount the risk-neutral expected next-period value at that

node's own six-month rate and add the payment due at that node.

Why the others are wrong

- **A:** $500.00 is the undiscounted risk-neutral expected value of all payments; it ignores discounting at every node.
- **C:** $701.46 leaves out the 6-month payments of ±$2,500 and values only the 1-year payments.
- **D:** $472.50 is the date-1 expected value with no final discounting by 1.03 back to today.

Q612

AnalystPrep Q.2851

REBUILT — Stem omitted the current six-month yield and the 'none of the above' option was weak; added the current yield and replaced distractors with plausible errors.

A $7 million face value stylized constant-maturity Treasury (CMT) swap is struck at 7%. It pays $7{,}000{,}000 \times (y_{CMT} - 7\%)/2$ every six months for one year, where y_{CMT} is the six-month yield. The current six-month yield is 7% and moves up or down 0.5% each six months on a recombining tree. What are the possible payoffs of the swap at 6 months and at one year?

A. 6 months: +$17,500 and -$17,500; one year: +$35,000, $0 and -$35,000
B. 6 months: +$35,000 and -$35,000; one year: +$70,000, $0 and -$70,000
C. 6 months: +$8,750 and -$8,750; one year: +$17,500, $0 and -$17,500
D. 6 months: +$17,500 and -$17,500; one year: +$17,500, $0 and -$17,500

Answer: A

Step by step

1. Payoff $= 7{,}000{,}000 \times (y - 7\%)/2$.
2. 6 months: yields 7.5% and 6.5%, so $7{,}000{,}000 \times (\pm 0.5\%)/2 = \pm\$17{,}500$.
3. 1 year: yields 8%, 7%, 6%, so $7{,}000{,}000 \times (+1\%)/2 = +\$35{,}000$, $0, and $-\$35{,}000$.

Why A is correct: Each payment is face value times the spread over the strike divided by 2 (semiannual); the spread is ±0.5% after one step and +1%, 0, -1% after two steps.

Why the others are wrong

- **B:** Forgot to divide by 2 for semiannual payments, doubling every payoff.
- **C:** Divided by 2 an extra time, halving every payoff.
- **D:** Used a single 0.5% step at one year instead of two cumulative steps (8%, 7%, 6%), so the year-one payoffs are too small.

Q613

BT answers L2.T5.44.1

REBUILT — Original options $352 and $964 could not be traced to realistic errors; replaced with computed mistake values (answer unchanged, OCR notional fixed to $1 million).

A CMT swap has notional $1 million, pays every six months notional/2 x (yield - 3%) on the six-month yield, and matures in one year. The current six-month yield is 3%; it moves +20 bp or -20 bp each six months on a recombining binomial tree. Real-world probabilities are 50%/50%; risk-neutral probabilities are 70% up / 30% down at every node. What is the price of the swap under risk-neutral pricing?

A. $0
B. $775
C. $1,170
D. $1,182

Answer: C

Step by step

1. 6-month payments: $1{,}000{,}000 \times (\pm 0.20\%)/2 = \pm\$1{,}000$. 1-year payments: +$2,000 (3.4%), $0 (3.0%), −$2,000 (2.6%).
2. Upper node: $(0.7 \times 2{,}000 + 0.3 \times 0)/1.016 + 1{,}000 = \$2{,}377.95$.
3. Lower node: $(0.7 \times 0 + 0.3 \times (-2{,}000))/1.014 - 1{,}000 = -\$1{,}591.72$.
4. Today: $(0.7 \times 2{,}377.95 + 0.3 \times (-1{,}591.72))/1.015 = 1{,}187.05/1.015 = \$1{,}169.51 \approx \$1{,}170$.

Why C is correct: Backward induction with the 70/30 risk-neutral probabilities, discounting each step at the node's own six-month rate and adding the interim payment at each node.

Why the others are wrong

- **A:** $0 assumes the symmetric ±20 bp moves cancel; with 70/30 probabilities and node-specific discount rates the value is not zero.
- **B:** $775 omits the 6-month payments of ±$1,000 and values only the 1-year payments.

- **D:** $1,182 does not discount the 1-year payoffs back from the 6-month nodes (at 3.2%/2 and 2.8%/2), discounting only once at 1.5%.

Q614 BT answers L2.T5.44.2

REBUILT — Original depended on the previous question and used a loosely named 'futures-forward effect'; made self-contained with numbers and attributed the effect to convexity (Jensen's inequality).

A CMT swap (notional $1 million, strike 3%, pays notional/2 x (six-month yield - 3%) every six months for one year) is valued on a recombining tree: six-month yield today 3%, moving ±20 bp each six months. Valued with the real-world 50/50 probabilities and node-by-node discounting, the swap's value is about -$0.96 rather than exactly zero, even though the up and down moves are symmetric. What best explains the non-zero value?

A. Convexity: discount factors are convex in the rate, so the symmetric payments are discounted at different node rates and do not offset (Jensen's inequality)
B. Mean reversion in the interest-rate process
C. The risk-neutral probabilities (70/30) differ from the 50/50 probabilities used
D. Nothing; the discounted expected value with 50/50 probabilities is exactly zero

Answer: A

Step by step

1. 6-month payments are $\pm\$1{,}000$; 1-year payments are $+\$2{,}000$, \$0, $-\$2{,}000$.
2. Upper node (3.2%): $0.5(2{,}000)/1.016 + 1{,}000 = \$1{,}984.25$; lower node (2.8%): $0.5(-2{,}000)/1.014 - 1{,}000 = -\$1{,}986.19$.
3. Today: $0.5(1{,}984.25 - 1{,}986.19)/1.015 = -\$0.96$, not zero.
4. Rates and discount factors are negatively related: the positive flows occur on up paths where discounting is heavier, and the negative flows occur on down paths where discounting is lighter, so the negative flows get more weight (a convexity-type effect of discounting node by node).

Why A is correct: Because the discount factor varies with the rate, symmetric payments are discounted at different node rates and do not exactly offset, even with equal probabilities; this is the convexity effect of discounting along the tree.

Why the others are wrong

- **B:** The tree has no mean reversion; symmetric ±20 bp steps carry no drift, so mean reversion is not what produces this value.
- **C:** The question uses the 50/50 probabilities, so the 70/30 risk-neutral probabilities play no role; they would change the price to about $1,170 rather than explain this -$0.96 residual.
- **D:** The computation gives -$0.96, small but not zero; assuming exact cancellation ignores the different discount rates at each node.

LO 12.i

Evaluate the advantages and disadvantages of reducing the size of the time steps on the pricing of derivatives on fixed-income securities.

Q615 BT answers L2.T5.45.4

According to Tuckman, we might choose time steps smaller than six months for each of the following reasons EXCEPT:

A. To ensure the security's cash flows are near a date in the tree
B. To reduce computational burden
C. To increase the granularity of potential future rates
D. To increase accuracy

Answer: B

Step by step

1. Identify the benefits of shorter time steps: cash flows fall near (or on) a tree date, the tree contains more possible future rates, and pricing accuracy improves.
2. Identify the cost: more nodes and more calculations, so computation time and complexity rise.
3. The EXCEPT item is therefore the claim that smaller steps reduce computational burden: B.

Why B is correct: Shrinking the time step multiplies the number of nodes, so computational burden increases rather than decreases; this is the trade-off for greater accuracy.

Why the others are wrong

- **A:** This is a genuine reason: cash flows rarely fall on six-month dates, so finer steps place each close to a tree date. Not the exception.
- **C:** This is a genuine reason: more steps give a finer set of possible future rates than a coarse two-state, six-month tree. Not the exception.
- **D:** This is a genuine reason: finer steps give a more accurate price. Choosing it mistakes a benefit for the exception.

Q616

Schweser Module Quiz 12.3 Q2

Which of the following regarding the use of small time steps in the binomial model is true?

A. Less realistic model.
B. More accurate model.
C. Less complicated computations.
D. Less computational expense.

Answer: B

Step by step

1. Smaller time steps let the tree approximate the continuous evolution of rates more closely, so the model is more realistic and accurate.
2. The price is a larger tree: more nodes and more backward-induction calculations, so greater complexity and computational expense.
3. Only B states a benefit; A, C and D each reverse a stated effect.

Why B is correct: Reducing the step size improves accuracy and realism at the cost of added computational complexity and expense.

Why the others are wrong

- **A:** Reverses the effect: finer steps make the model more, not less, realistic.
- **C:** Reverses the trade-off: finer steps mean more nodes and more complicated calculations.
- **D:** Reverses the trade-off: more nodes means greater, not lower, computational expense.

LO 12.j

Evaluate the appropriateness of the Black-Scholes-Merton model when valuing derivatives on fixed-income securities.

Q617

BT answers L2.T5.46.1

The application of Black-Scholes-Merton to bond valuation is difficult for all of the following reasons EXCEPT for:

A. Bonds have longer maturities
B. Bond price pulls to par
C. Bond price volatility tends to zero as maturity approaches
D. Short-term interest rate is not constant

Answer: A

Step by step

1. Recall the three BSM assumptions that fail for bonds: no upper bound on the price, constant risk-free rate, constant price volatility.
2. Pull to par (B) is the upper-bound problem: a bond's price is capped and converges to face value.
3. Volatility falling to zero at maturity (C) violates constant volatility.
4. A stochastic short rate (D) violates the constant risk-free rate assumption.
5. Longer maturity is not itself an assumption violation, so A is the EXCEPT answer.

Why A is correct: BSM fails for bonds because of the price cap/pull to par, declining volatility and a random short rate; maturity length is not one of the cited problems.

Why the others are wrong

- **B:** This is a genuine problem: the pull to par means the price has a ceiling, contradicting BSM's unbounded price.

- **C:** This is a genuine problem: BSM assumes constant volatility, but bond volatility shrinks to zero at maturity.
- **D:** This is a genuine problem: BSM assumes a constant risk-free rate, but the short rate moves and drives bond prices.

Q618 Mock 2 Q21

An asset manager is evaluating the techniques an organization uses to value derivatives on fixed-income securities and bonds with embedded options. Which of the following assumptions would NOT be correct?

A. A callable bond is assumed to have a convex price-yield curve at high yields.
B. A putable bond is assumed to have a convex price-yield curve at low yields.
C. The Black-Scholes-Merton option pricing model is one acceptable method for valuing fixed-income derivatives.
D. The price-yield curve for bonds that have no embedded options is convex.

Answer: C

Step by step

1. Check A: at high yields the call is far out of the money, so the callable bond behaves like an option-free bond and is convex. True.
2. Check B: at low yields the put is far out of the money, so the putable bond is convex. True.
3. Check D: option-free bonds have positive convexity. True.
4. Check C: BSM assumes no price cap, a constant risk-free rate and constant volatility, all violated by bonds, so it is not acceptable. This is the incorrect statement.

Why C is correct: BSM's assumptions (unbounded price, constant rate, constant volatility) do not hold for fixed-income securities, so statement C is false.

Why the others are wrong

- **A:** True statement: the call is far out of the money at high yields, so the bond is convex; choosing it confuses the low-yield region where the callable bond shows negative convexity.
- **B:** True statement: the put is far out of the money at low yields, so convexity is positive; choosing it mixes up which region the option is active in.
- **D:** True statement: option-free bonds are convex at all yields; choosing it treats convexity as absent without embedded options.

Q619 Schweser Module Quiz 12.3 Q1

The Black-Scholes-Merton option pricing model is not appropriate for valuing options on corporate bonds because corporate bonds:

A. have credit risk.
B. have an upper price bound.
C. have constant price volatility.
D. are not priced by arbitrage.

Answer: B

Step by step

1. BSM requires an unbounded underlying price, a constant risk-free rate and constant volatility.
2. A bond's price is capped (at par plus coupons when rates are zero), so there is an upper bound: B.
3. Bond volatility declines toward maturity rather than staying constant, so C is the opposite of the real problem.
4. Credit risk and arbitrage pricing are not cited as the reasons.

Why B is correct: The existence of a maximum bond price contradicts BSM's assumption of no upper limit on the underlying price.

Why the others are wrong

- **A:** Credit risk affects any bond's value but is not a cited reason BSM fails; the failure is structural (price cap, rate, volatility).
- **C:** Bond volatility is not constant, it falls as maturity approaches; the statement reverses the real violation.
- **D:** Bonds are priced by arbitrage (replication) in term structure models; this is not why BSM is unsuitable.

Q620 BT answers L2.T5.46.2

REBUILT — Original asked which assumption was 'most troubling', which is subjective (price cap, constant rate and constant volatility all fail); restated to target the price-cap assumption unambiguously.

A bond's price has a maximum value (par plus coupons, reached when interest rates are zero). Which Black-Scholes-Merton assumption does this fact most directly contradict?

A. Constant volatility of the underlying
B. No upper limit on the price of the underlying
C. Constant risk-free rate
D. Continuous, frictionless trading in the underlying

Answer: B

Key correction: Source key (BT answers) said C (constant risk-free rate) for 'most troubling'; that ranking is subjective, so the item was rebuilt around the price-cap assumption, whose answer is B.

Step by step

1. BSM lets the underlying price rise without limit (lognormal).
2. A bond's price cannot exceed the sum of its undiscounted cash flows, i.e. there is a cap.
3. This is a direct contradiction of the no-upper-limit assumption: B.
4. Declining volatility and a stochastic rate are separate violations, not the price cap.

Why B is correct: A price ceiling on bonds directly contradicts BSM's assumption that the underlying price has no upper limit.

Why the others are wrong

- **A:** Bond volatility does decline toward maturity, but that is a separate violation from the price cap described in the stem.
- **C:** The short rate is stochastic, which is a different violation; it does not follow from a maximum price.
- **D:** Frictionless continuous trading is an idealization, but it is not what the price-cap fact contradicts.

Reading 13. Expectations, Risk Premium, Convexity, and the Shape of the Term Structure

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 4th Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 8.

LO	Learning objective	Qs
13.a	explain the role of interest rate expectations in determining the shape of the term structure.	4
13.b	apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.	9
13.c	estimate the convexity effect using Jensen's inequality.	6
13.d	identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor.	7

LO 13.a

Explain the role of interest rate expectations in determining the shape of the term structure.

Q621

AnalystPrep Q.1638

REBUILT — Original distractor 0.1024 was unexplained and options mixed formats; kept the stem and replaced distractors with results of realistic mistakes.

The 2-year spot rate is 9% and the 1-year spot rate is 4% (both annually compounded). What is the 1-year forward rate one year from now?

A. 4.81%
B. 14.24%
C. 5.00%
D. 14.00%

Answer: B

Step by step

1. No-arbitrage: $(1 + s_2)^2 = (1 + s_1)(1 + f)$.
2. $f = 1.09^2/1.04 - 1 = 1.1881/1.04 - 1$.
3. $1.1881/1.04 = 1.14240$, so $f = 14.24\%$.

Why B is correct: The forward rate is the rate that makes rolling over the 1-year investment equal to the 2-year spot investment; the 2-year growth factor must be divided by the 1-year factor.

Why the others are wrong

- **A:** Divided 1.09 by 1.04 without squaring the 2-year factor (1.09/1.04 - 1 = 4.81%).
- **C:** Subtracted the spot rates (9% - 4%), ignoring compounding.
- **D:** Used the linear approximation $2s_2 - s_1 = 14\%$, ignoring compounding; the exact figure is 14.24%.

Q622

AnalystPrep Q.2853

REBUILT — Original options had two choices with the same 3-year value and an arithmetic average rounds to the same 2-year value; replaced options with distinct two-decimal answers tied to specific mistakes.

Investors value the current one-year interest rate at 11.316%. They forecast (with certainty, under pure expectations) that the one-year rate will be 13.457% for the following year and 15.658% for the year after that. What are the two-year and three-year spot rates (annual compounding)?

A. 12.38% and 13.46%
B. 26.30% and 46.07%
C. 11.32% and 12.38%
D. 12.38% and 20.86%

Answer: A

Step by step

1. Under pure expectations the spot rate is the geometric average of the one-year rates.
2. $(1 + s_2)^2 = 1.11316 \times 1.13457 = 1.26296$, so $s_2 = \sqrt{1.26296} - 1 = 12.38\%$.
3. $(1 + s_3)^3 = 1.26296 \times 1.15658 = 1.46071$, so $s_3 = 1.46071^{1/3} - 1 = 13.46\%$.

Why A is correct: With certain future rates, a multi-year zero must earn the compounded product of the one-year

rates, so the spot rate is its geometric mean.

Why the others are wrong

- **B:** Reported the cumulative multi-year returns (1.26296-1, 1.46071-1) without taking the n-th root.
- **C:** Shifted the answers by one period: gave the 1-year rate as the 2-year rate and the 2-year rate as the 3-year rate.
- **D:** Correct 2-year rate, but took the square root instead of the cube root for the 3-year product (1.46071^(1/2)-1 = 20.86%).

Q623

BT answers P2.T5.300.2 · BT notes P2.T5.300.2

Each of the following is TRUE about the shape of the term structure of spot interest rates EXCEPT for:

A. If all investors agree with certainty that the current one-year spot rate of 1.0% will increase to 2.0% next year and again to 3.0% in two years, the three-year spot rate should be almost 2.0% (with annual compounding)

B. It is unreasonable and impractical to expect pure (and strict or narrow) expectations theory to fully characterize the term structure in practice

C. Although investors' expectations about short-term rates have little or no impact on the SHAPE of the term structure, they tend to dominate its SHAPE over longer time horizons (e.g., 20 or 30 years)

D. Under an assumption of pure expectations, it is valid to hypothesize (infer) from a downward-sloping term structure that investors are forecasting a drop in future short-term (e.g., one year) interest rates

Answer: C

Step by step

1. A: $(1.01 \times 1.02 \times 1.03)^{1/3} - 1 = 1.9967\% \approx 2\%$, so A is true.
2. B: pure expectations ignores risk premium and convexity, so it cannot fully explain the curve; true.
3. C: expectations are specific and shape-determining over short horizons; over very long horizons they cannot be granular and mainly set the level (long-run short rate). The statement reverses this, so it is false.
4. D: under pure expectations, forwards equal expected future rates, so a downward-sloping curve implies expected falling rates; true.

Why C is correct: Expectations matter most for the shape at short horizons and only for the level at very long horizons, so C reverses the relationship.

Why the others are wrong

- **A:** True: the geometric average of 1%, 2%, 3% is 1.9967%, 'almost 2.0%'; choosing it means wrongly treating a true statement as the exception.
- **B:** True: the real curve also reflects risk premium and convexity, so strict expectations is not realistic.
- **D:** True: with pure expectations, forwards equal expected spot rates, so a declining curve signals expected lower short rates.

Q624

Schweser Module Quiz 13.1 Q1

REBUILT — Source equation was missing and two distractors (7.888%, 8.088%) were arbitrary; replaced with results of realistic mistakes.

An investor expects the current 1-year rate for a zero-coupon bond to remain at 6%, the 1-year rate next year to be 8%, and the 1-year rate in two years to be 10%. What is the 3-year spot rate, assuming all investors share these expectations of future 1-year rates?

A. 8.000%

B. 7.988%

C. 25.928%

D. 6.995%

Answer: B

Step by step

1. $(1 + s_3)^3 = 1.06 \times 1.08 \times 1.10 = 1.25928$.
2. $s_3 = 1.25928^{1/3} - 1$.
3. $s_3 = 1.07988 - 1 = 7.988\%$.

Why B is correct: With certain expectations the 3-year spot rate is the geometric mean of the three 1-year rates, which is slightly below the 8% arithmetic mean.

Why the others are wrong

- **A:** Used the arithmetic mean (6+8+10)/3 = 8%, ignoring compounding.
- **C:** Stopped at the 3-year cumulative return (1.25928-1) without taking the cube root.

- **D:** Averaged only the first two years: $\sqrt{1.06 \times 1.08} - 1 = 6.995\%$.

LO 13.b

Apply a risk-neutral interest rate tree to assess the effect of volatility on the shape of the term structure.

Q625 BT answers P2.T5.300.1

REBUILT — Tree graphic was missing and distractors were weak; restated the tree as a table (year-2 rates do not affect a two-year zero) and replaced options with realistic-error distractors.

The true (real-world) process for the one-year interest rate is a recombining binomial tree with equal 50% probabilities of an up or down move:

Date	Node values
Year 0	8.0%
Year 1	11.0% (up) or 5.0% (down)
Year 2	14.0%, 8.0% or 2.0%

Assume investors are risk-neutral, so a bond's price equals its expected discounted value under these probabilities with no risk premium. Which is nearest to the price of a two-year zero-coupon bond with a $1,000 par value?

A. $834.17
B. $857.34
C. $858.00
D. $881.83

Answer: C

Step by step

1. A two-year zero pays $1,000 at the end of year 2, so only the year-0 rate (8%) and the year-1 rates (11% or 5%) are used for discounting; the year-2 rates are irrelevant.
2. Year-1 values: up node $1000/1.11 = 900.90$; down node $1000/1.05 = 952.38$.
3. Expected year-1 value: $0.5(900.90 + 952.38) = 926.64$.
4. Discount one year at 8%: $926.64/1.08 = 858.00$.
5. Check: 858.00 exceeds $1000/1.08^2 = 857.34$, the price at the expected rate, because of convexity.

Why C is correct: Price equals the expected discounted value, averaging discount factors across the up and down states and then discounting at today's 8% rate. Convexity (Jensen's inequality) makes this slightly higher than the price at the expected rate.

Why the others are wrong

- **A:** Discounts only along the up path: $1000/(1.08 \times 1.11) = 834.17$, ignoring the down state.
- **B:** Discounts at the expected rate of 8% for both years, $1000/1.08^2 = 857.34$; it ignores the convexity effect of averaging prices instead of rates.
- **D:** Discounts only along the down path: $1000/(1.08 \times 1.05) = 881.83$, ignoring the up state.

Q626 BT answers P2.T5.300.3 · BT notes P2.T5.300.3

Make two assumptions: first, that investors are risk-neutral, so there is no risk premium in the term structure; second, that investors understand volatility in the sense that they agree on a binomial interest rate tree. Under these assumptions, what is the impact of higher volatility on the term structure?

A. If expected future short-term interest rates are unchanged, greater interest rate volatility has no impact
B. It is unclear, as convexity has a directionally different impact at low yields than at high yields
C. Higher volatility will increase the yield and decrease bond prices due to the convexity effect
D. Higher volatility will decrease the yield and increase bond prices due to the convexity effect

Answer: D

Step by step

1. With no risk premium, spot rates depend only on expected short rates and the convexity effect.
2. Bond prices are convex in rates, so by Jensen's inequality $E[1/(1+r)] > 1/(1+E[r])$.
3. Higher volatility widens the spread of rates around the mean, increasing the gap: the expected price rises.
4. A higher price for the same cash flow means a lower yield, so higher volatility lowers spot rates and raises bond

prices at every maturity (more so for longer maturities).

Why D is correct: Convexity means gains from falling rates exceed losses from rising rates, so average prices rise and yields fall as volatility increases, regardless of the yield level.

Why the others are wrong

- **A:** Holding expected rates fixed does not neutralise volatility; the expected price still rises through Jensen's inequality.
- **B:** Convexity is always positive for zero-coupon bonds, so the direction never changes with the level of yields.
- **C:** Reverses the sign: it confuses convexity with a positive risk premium, which would raise yields.

Q627

BT answers P2.T5.301.3 · BT notes P2.T5.301.3

Robert is asked to graph a five-year term structure of interest rates from a binomial interest rate tree. The tree recombines, with up- and down-jumps equal in size (plus or minus X basis points) and likelihood (50%), so that the expected one-year rate in each of years two through five equals today's one-year spot rate. The tree reflects significant rate volatility, and the model also includes a risk premium reflecting risk-averse investors. Which is the most plausible prediction about the term structure he produces?

A. The term structure is upward-sloping throughout
B. In the shorter end, the yield increases, but in the longer end, the yield decreases
C. In the shorter end, the yield decreases, but in the longer end, the yield increases
D. The term structure is downward-sloping throughout

Answer: B

Step by step

1. Flat expectations (each expected one-year rate equals today's rate) contribute a flat curve.
2. The risk premium raises spot rates and grows roughly linearly with maturity, so it dominates at the short end and creates an initial upward slope.
3. The convexity effect lowers spot rates and grows roughly with the square of maturity, so it eventually overwhelms the risk premium at longer maturities.
4. The curve therefore rises at first and then falls: a humped shape.

Why B is correct: The risk premium grows about linearly with maturity while convexity grows about quadratically and with high volatility, so the premium dominates short maturities and convexity dominates long ones.

Why the others are wrong

- **A:** Ignores convexity; with significant volatility convexity eventually outweighs the premium at long maturities.
- **C:** Reverses the order: convexity is negligible at short maturities, so yields cannot first fall.
- **D:** Ignores the risk premium, which dominates at the short end because convexity is tiny for short maturities.

Q628

BT notes P2.T5.23.2.3

REBUILT — Source tree/probabilities were garbled and its stated forward rates were inconsistent with the tree; rebuilt as a self-contained risk-neutral tree question on the three-year spot rate.

The one-year rate is 6.00% today, and the expected one-year rate is 6.00% in each of the next two years. The risk-neutral process is a recombining tree with equal 50% probabilities and no risk premium:

Date	One-year rate nodes
Year 0	6.00%
Year 1	8.00% (up) or 4.00% (down)
Year 2	10.00%, 6.00% or 2.00%

Which is nearest to the three-year spot rate implied by this tree?

A. 6.00%
B. 5.98%
C. 5.95%
D. 6.05%

Answer: C

Step by step

1. Date-2 values of the \$1 payoff (discounting by the year-2 rate): $1/1.10 = 0.909091$, $1/1.06 = 0.943396$, $1/1.02 = 0.980392$.
2. Date-1 up node (8%): $0.5(0.909091 + 0.943396)/1.08 = 0.857633$; down node (4%): $0.5(0.943396 +$

$0.980392)/1.04 = 0.924898$.
3. Date 0: $0.5(0.857633 + 0.924898)/1.06 = 0.840817$.
4. Spot rate: $(1/0.840817)^{1/3} - 1 = 5.95\%$.
5. This is below 6.00% (flat price $1/1.06^3 = 0.839619$) because of convexity.

Why C is correct: With expectations flat and no risk premium, the only force on the curve is convexity, which raises the expected price and so lowers the three-year spot rate below 6.00%.

Why the others are wrong

- **A:** Assumes a flat curve because expected rates are all 6%; ignores that averaging prices gives more than discounting at the average rate.
- **B:** This is the two-year spot rate (price 0.890313 gives 5.98%); it stops the roll-back one period early, understating the convexity effect that compounds with maturity.
- **D:** Gets the sign of the convexity effect wrong, adding rather than subtracting it; volatility lowers yields in a risk-neutral tree.

Q629

GARP PE2 Q74 · Mock 4 Q75

REBUILT — Interest rate tree exhibit was missing from the GARP version; used the reconstructed tree from the Mock 4 version and restated it cleanly.

An analyst on an emerging-markets fixed-income desk builds a term structure model. The 1-year spot rate is expected to be 17.00% in each of the next 3 years, and the interest rate process is a recombining risk-neutral tree with moves of plus or minus 2% each year and equal 50% probabilities:

Date	One-year rate nodes
Year 0	17.00%
Year 1	19.00% or 15.00%
Year 2	21.00%, 17.00% or 13.00%

Which correctly describes the shape of the resulting term structure?

A. Slightly downward sloping, since discounting the terminal cash flow over longer time periods when interest rates exhibit volatility results in slightly lower spot rates.

B. Flat, since the interest rate tree is based on the assumption that all expected 1-year spot rates are equal.

C. Slightly upward sloping, since there is an opportunity cost associated with long-term interest rates.

D. Steeply upward sloping, since investors will demand higher long-term interest rates as compensation for interest rate volatility.

Answer: A

Step by step

1. One-year zero: $1/1.17 = 0.854701$, spot 17.000%.
2. Two-year zero: $0.854701 \times 0.5\,(1/1.19 + 1/1.15) = 0.854701 \times 0.854951 = 0.730727$; spot $= 0.730727^{-1/2} - 1 = 16.98\%$.
3. Three-year zero (rolling back the tree with year-2 rates 21%, 17%, 13%): price $= 0.625101$; spot $= 0.625101^{-1/3} - 1 = 16.95\%$.
4. Spot rates 17.00%, 16.98%, 16.95% decline slightly with maturity, so the curve is slightly downward sloping.

Why A is correct: Expected rates are flat and the tree is risk-neutral, so the only influence is convexity: averaging discount factors gives a higher price (lower yield) than discounting at the mean rate, and the effect grows with maturity.

Why the others are wrong

- **B:** Equal expected rates would give a flat curve only if there were no volatility; ignoring convexity (Jensen's inequality) misses the small decline in spot rates.
- **C:** Has the direction reversed; an opportunity-cost or premium story does not belong in a risk-neutral tree, where convexity can only lower yields.
- **D:** Imports a risk premium for volatility, but the tree is risk-neutral (no premium), and the effect would also not be steep.

Q630

Mock 1 Q23

REBUILT — Original depended on the previous question for its setup; made self-contained with the same data.

The current one-year spot rate is 10.00%. One year from now the one-year rate will be either 18.00% or 2.00%, each with probability 0.5, so the expected one-year rate is also 10.00%. There is no risk premium. What is the implied two-year spot rate?

A. 9.42%

B. 9.71%
C. 10.00%
D. 10.29%

Answer: B
Step by step
1. Year-1 discount factor (no uncertainty): $1/1.10 = 0.909091$.
2. Expected year-2 discount factor: $0.5(1/1.18) + 0.5(1/1.02) = 0.5(0.847458 + 0.980392) = 0.913925$.
3. Two-year zero price: $0.909091 \times 0.913925 = 0.830841$.
4. Two-year spot: $(1/0.830841)^{1/2} - 1 = 9.71\%$, about 29 bp below 10%.

Why B is correct: Averaging the discount factors rather than the rates gives a price above $1/1.10^2$, so the spot rate is below the expected rate of 10% by the convexity effect.

Why the others are wrong
- **A:** This is the full one-period effect $1/0.913925 - 1 = 9.42\%$; it forgets that the first year is certain, so the effect applies to only half of the two-year horizon.
- **C:** Assumes the spot equals the expected rate, ignoring convexity.
- **D:** Gets the sign of the convexity effect wrong (10% plus 29 bp); volatility lowers yields.

Q631

Mock 1 Q24

An analyst is comparing two government bond markets. Both have a flat expected path of short-term interest rates and neither prices a risk premium. Market A exhibits substantially higher interest rate volatility than Market B. Which correctly describes the difference between the two markets' term structures?

A. Market A's long-dated spot rates will sit further above the expected short rate than Market B's, because volatility raises required returns.
B. The two term structures will be identical, since expectations and risk premiums are the same in both markets.
C. Market A's term structure will slope upward and Market B's will be flat, because volatility steepens the curve.
D. Market A's long-dated spot rates will sit further below the expected short rate than Market B's, and the gap will widen with maturity in both markets.

Answer: D
Step by step
1. Spot rate = expectations + risk premium + convexity; the first two are identical in both markets.
2. Only the convexity effect differs, and it is always negative (lowers yields).
3. Convexity is larger with higher volatility, so Market A's spot rates are further below the expected short rate.
4. In both markets the effect grows with maturity, so the gap widens with maturity.

Why D is correct: With expectations and premium equal, convexity alone shapes the difference; it lowers yields more as volatility and maturity rise.

Why the others are wrong
- **A:** Wrong sign: treats volatility as a source of a positive premium, but without a risk premium it only lowers yields.
- **B:** Ignores convexity, which depends on volatility and is not part of expectations or the risk premium.
- **C:** Wrong direction: convexity makes the curve slope downward, not upward, and Market B also has some (smaller) convexity effect.

Q632

Schweser Module Quiz 13.1 Q2

REBUILT — Original tree figure was missing and its numbers were inferred; rebuilt with a fully specified one-period-ahead tree and options built from realistic mistakes.

Investors are risk-neutral. The current 1-year spot rate is 5%. One year from now the 1-year rate will be either 9% or 3%, each with probability 0.5. What is the price today of a 2-year zero-coupon bond with face value \$1?

A. \$0.8985
B. \$0.8992
C. \$0.9070
D. \$0.9442

Answer: B

Step by step

1. Year-1 bond values: $1/1.09 = 0.91743$ (up node) and $1/1.03 = 0.97087$ (down node).
2. Expected value at year 1 (risk-neutral, 50/50): $0.5(0.91743 + 0.97087) = 0.94415$.
3. Discount one year at today's 1-year rate: $0.94415/1.05 = 0.89919 \approx 0.8992$.

Why B is correct: Under risk neutrality the bond price is the expected discounted payoff, working backward node by node; the average of the year-1 prices (not of the rates) is then discounted at today's rate.

Why the others are wrong

- **A:** Discounts at the expected rate path: $1/(1.05 \times 1.06) = 0.8985$. Using the average rate (6%) ignores convexity/Jensen's inequality and understates the price.
- **C:** $1/1.05^2 = 0.9070$: assumes the second-year rate equals today's 5% and ignores the tree.
- **D:** Averages the year-1 bond prices (0.9442) but forgets to discount back the final year at 5%.

Q633

Schweser Module Quiz 13.1 Q3

REBUILT — Original options were arbitrary values; replaced distractors with results of realistic errors (correct answer unchanged).

If investors are risk-neutral and the price of a 2-year zero-coupon bond is \$0.88035 today, what is the implied 2-year spot rate (annual compounding)?

A. 6.796%
B. 6.579%
C. 13.591%
D. 3.290%

Answer: B

Step by step

1. Price relation: $0.88035 = 1/(1 + z_2)^2$.
2. $1/0.88035 = 1.13591$.
3. $1 + z_2 = \sqrt{1.13591} = 1.06579$, so $z_2 = 6.579\%$.

Why B is correct: The 2-year spot rate is the annualized (geometric) yield on the zero: take the square root of the reciprocal price and subtract 1.

Why the others are wrong

- **A:** Divided the 2-year total return 13.591% by 2 (simple average), ignoring compounding: 6.796%.
- **C:** Forgot to take the square root: the 2-year holding return 13.591% is not an annual rate.
- **D:** Halved the correct annual rate again (6.579%/2), a confusion of per-period conversion.

LO 13.c

Estimate the convexity effect using Jensen's inequality.

Q634

BT answers P2.T5.25.2.3 · BT notes P2.T5.25.2.3

REBUILT — Stem called the Jensen price effect 'convexity' ambiguously and the sources' options (+/-1.574, 1.633, 7.633) were garbled; defined the effect explicitly and rebuilt options from realistic errors.

A two-year zero-coupon bond has a face value of \$100. The one-year spot rate is 5%. The one-year forward rate one year from now is 9% in expectation, but will actually be either 7% or 11% with equal probability. Assuming risk neutrality, the convexity effect (Jensen's inequality) is defined as the expected price of the bond minus the price computed using the expected 9% rate. Which of the following is closest to this convexity effect?

A. -0.029
B. 0.029
C. 1.604
D. 1.634

Answer: B

Step by step

1. Price if the rate is 11%: $100/(1.05 \times 1.11) = 85.8001$.
2. Price if the rate is 7%: $100/(1.05 \times 1.07) = 89.0076$.
3. Expected price: $0.5(85.8001 + 89.0076) = 87.4038$.

4. Price at the expected rate: $100/(1.05 \times 1.09) = 87.3744$.
5. Convexity effect: $87.4038 - 87.3744 = 0.0294 \approx 0.029$.

Why B is correct: Because price is a convex function of the rate, the expected price exceeds the price at the expected rate, so the effect is small and positive.

Why the others are wrong

- **A:** Sign error: Jensen's inequality for the convex price function gives expected price > price at expected rate, so the difference is positive.
- **C:** Compared the expected price with the 11% scenario price (87.4038 - 85.8001 = 1.604) instead of with the price at the expected rate.
- **D:** Compared the 7% scenario price with the price at the expected rate (89.0076 - 87.3744 = 1.633) instead of using the expected price.

Q635

BT answers P2.T5.301.2 · BT notes P2.T5.301.2

The current one-year rate is 10.0% and a binomial tree is known to investors. Next year the one-year rate will be either 14.0% or 6.0%, each with 50% probability, so the expected one-year rate in one year is 10.0%. Investors are risk-neutral and rates compound annually. If 10.0% is the two-year yield in the absence of uncertainty, what is the yield change implied by Jensen's inequality due to volatility?

A. An increase of ~25 basis points to ~10.25%
B. None
C. A decrease of ~7 basis points to 9.93%
D. A decrease of ~25 basis points to 9.75%

Answer: C

Step by step

1. Year-1 prices: $1/1.14 = 0.877193$ and $1/1.06 = 0.943396$; average $= 0.910294$.
2. Two-year price: $0.910294/1.10 = 0.827541$.
3. No-volatility price: $1/1.10^2 = 0.826446$.
4. Implied two-year yield: $(1/0.827541)^{1/2} - 1 = 9.927\%$.
5. Change: $9.927\% - 10.000\% \approx -7$ bp.

Why C is correct: Convexity makes the expected discount factor higher than the discount factor at the expected rate, so the implied yield falls below 10%.

Why the others are wrong

- **A:** Wrong direction: convexity raises prices and lowers yields.
- **B:** Ignores Jensen's inequality; the expected discount factor 0.827541 exceeds 0.826446, so the yield changes.
- **D:** Magnitude far too large; the actual price difference is only about 0.0011, i.e. about 7 bp of yield.

Q636

BT notes P2.T5.23.3.1

REBUILT — Option set had implausible distractors (-167 bp, 300 bp) and a typo in the stem; rewrote stem and options.

The one-year spot rate is 9.0% today and will be 12.0% or 6.0% next year with equal probability (expected 9.0%). The one-year discount factors next year are 0.89286 (12%) and 0.94340 (6%). The two-year discount factor today is 0.5(0.89286 + 0.94340)/1.09 = 0.84232. Assuming annual compounding and risk neutrality, what is the convexity effect (the value of convexity, measured as the size of the reduction in the two-year spot rate relative to 9.0%)?

A. -4.14 basis points
B. Zero
C. 4.14 basis points
D. 8.96 basis points

Answer: C

Step by step

1. Two-year price: $0.5(0.89286 + 0.94340)/1.09 = 0.84232$.
2. Implied two-year spot rate: $(1/0.84232)^{1/2} - 1 = 8.9586\%$.
3. Rate with no convexity: 9.00%.
4. Convexity effect: $9.0000\% - 8.9586\% = 0.0414\% = 4.14$ bp (a reduction in yield).

Why C is correct: The convexity effect (value of convexity) is the amount by which Jensen's inequality pushes the

two-year spot rate below the expected 9%, quoted as a positive magnitude.

Why the others are wrong

- **A:** Right size but negative sign; the effect is quoted as a positive magnitude of the yield reduction (value of convexity).
- **B:** Ignores Jensen's inequality; 0.84232 exceeds 1/1.09^2 = 0.84168.
- **D:** Reported the spot rate 8.96% as basis points instead of the 9.00% - 8.96% difference.

Q637

Mock 1 Q22

A fixed-income analyst is illustrating the effect of Jensen's inequality on the pricing of a zero-coupon bond. The analyst assumes that the one-year interest rate one year from now will be either 18% or 2%, each with probability 0.5, so that the expected one-year rate is 10%. The analyst compares the expected value of the one-year discount factor prevailing in one year with the discount factor evaluated at the expected 10% rate, and expresses the difference in yield terms (the expected 10% rate less the one-year rate implied by the expected discount factor). What is the implied convexity effect demonstrated by Jensen's inequality?

A. 0.48%.
B. 0.58%.
C. 1.16%.
D. 8.00%.

Answer: B

Step by step

1. Expected discount factor: $0.5(1/1.18 + 1/1.02) = 0.5(0.847458 + 0.980392) = 0.913925$.
2. Factor at expected rate: $1/1.10 = 0.909091$ (lower, per Jensen).
3. Implied yield: $1/0.913925 - 1 = 9.418\%$.
4. Convexity effect: $10.00\% - 9.418\% = 0.58\%$.

Why B is correct: Convexity of the discount function raises the expected discount factor, which lowers the implied yield by about 0.58%.

Why the others are wrong

- **A:** Reported the discount-factor difference 0.004834 as a percentage instead of converting to yield.
- **C:** Doubled the effect.
- **D:** Used 18% - 10% (the rate shock) rather than the yield difference.

Q638

Schweser Module Quiz 13.2 Q1

An investor expects the one-year rate to be 6% for the first year of a two-year zero-coupon bond, and projects a 50% probability that the one-year rate in one year is 8% and a 50% probability that it is 4%. Which inequality most accurately reflects the convexity effect for this two-year bond using Jensen's inequality? (Prices are per 1 unit of face value.)

A. 0.89031 > 0.89000.
B. 0.89000 > 0.80000.
C. 0.94340 > 0.89031.
D. 0.94373 > 0.94340.

Answer: A

Step by step

1. Expected one-year price in one year: $0.5(1/1.08 + 1/1.04) = 0.94373$.
2. Discount one more year at 6%: $0.94373/1.06 = 0.89031$.
3. Price using the expected rate: $1/1.06^2 = 0.89000$.
4. Expected price exceeds price at expected rate: 0.89031 > 0.89000.

Why A is correct: For the two-year bond, both sides of Jensen's inequality must be discounted one further year at 6%.

Why the others are wrong

- **B:** Right side 0.80000 is not a bond price from this problem; it compares nothing relevant to Jensen's inequality.
- **C:** Compares a one-year price (0.94340) with a two-year price; maturities differ, so it says nothing about convexity.
- **D:** This is the one-year-ahead inequality (0.94373 > 0.94340); it was not discounted one more year to give two-year prices.

Q639 Schweser Module Quiz 13.2 Q2

What is the impact on the bond price-yield curve if, all other factors held constant, the maturity of a zero-coupon bond increases? The pricing curve becomes:

A. less concave.
B. more concave.
C. less convex.
D. more convex.

Answer: D

Step by step

1. Convexity effect for the 2-year zero is 1.84 bp; for the 3-year it is about 5 bp.
2. The value of convexity rises with maturity.
3. So the price-yield relationship becomes more convex.

Why D is correct: Longer-maturity zeros have a larger convexity effect, so the price-yield curve is more convex.

Why the others are wrong

- **A:** Bond price-yield curves are convex, not concave.
- **B:** Same concavity error; wrong curvature direction.
- **C:** Direction is reversed: convexity increases, not decreases, with maturity.

LO 13.d

Identify the components into which the return on a bond can be decomposed, and calculate the expected return on a bond for a risk-averse investor.

Q640 BT Mock Q21 · BT notes P2.T5.23.3.2

REBUILT — Original relied on a garbled/inconsistent price-tree exhibit and vague distractors (curve shape, Jensen twist); rebuilt as a self-contained expected-return calculation with the duration-risk convention stated.

The one-year interest rate is 8.0%. Risk-averse investors require compensation of 50 basis points per year of duration risk, where the duration risk borne over a one-year holding period is that of the bond as it will exist at the end of the period (for example, a three-year zero-coupon bond becomes a two-year zero-coupon bond). What is the expected return over the next year on a three-year zero-coupon bond?

A. 8.00%
B. 8.50%
C. 9.00%
D. 9.50%

Answer: C

Step by step

1. Risk-averse expected return: $E[\text{return}] = r_0 + \lambda \times D$.
2. $r_0 = 8.00\%$ and $\lambda = 0.50\%$ per year of duration risk.
3. After one year the three-year zero is a two-year zero, so the duration risk borne is $D = 2$.
4. $E[\text{return}] = 8.00\% + 2 \times 0.50\% = 9.00\%$.

Why C is correct: The premium compensates for exposure to the rate move over the holding period; the bond's price at the end of the year is that of a two-year zero, so two units of duration risk are priced.

Why the others are wrong

- **A:** This is the risk-neutral return r_0; it omits the risk premium entirely.
- **B:** Uses only one year of duration risk (1 x 0.50%), as if the bond became a one-year zero after one year.
- **D:** Uses the current three-year maturity as the duration risk (3 x 0.50%) instead of the two-year remaining maturity at the end of the holding period.

Q641 BT answers P2.T5.25.2.1 · BT notes P2.T5.25.2.1

REBUILT — Sources differed on the short rate (OCR 1% vs 7%) and the options were nearly identical 1 bp apart; rebuilt with clean numbers and options reflecting realistic mistakes (convexity added or subtracted, duration omitted).

The short-term interest rate is 3.0%, the risk premium is 15 basis points per unit of duration, and a bond has a duration of 6 and a convexity of 50. The volatility of interest rate changes is 1.5% per year. What is the expected

return on the bond for a risk-averse investor over the next year?

A. 3.15%
B. 3.34%
C. 3.90%
D. 4.46%

Answer: C

Step by step

1. For a risk-averse investor: $E[\text{return}] = r_0 + \lambda D$.
2. $\lambda D = 0.15\% \times 6 = 0.90\%$.
3. $E[\text{return}] = 3.00\% + 0.90\% = 3.90\%$.
4. Convexity, $\frac{1}{2}C\sigma^2 = 0.5 \times 50 \times 0.015^2 = 0.56\%$, is not added or subtracted: the convexity gain in the return is offset by the lower forward rates it implies, leaving no net advantage.

Why C is correct: The expected return equals the short rate plus the risk premium times duration; convexity effects net out of the risk-averse expected return.

Why the others are wrong

- **A:** Adds the premium of 15 bp once (3.00% + 0.15%) and ignores the duration of 6.
- **B:** Subtracts the convexity term (0.56%) from 3.90%; convexity lowers forward rates, not the expected return.
- **D:** Adds the convexity term (0.56%) to 3.90%; the convexity benefit is already offset in the expected return, so adding it double counts.

Q642

BT answers P2.T5.25.2.2 · BT notes P2.T5.25.2.2

REBUILT — Source key dropped the convexity term yet its own rationale said convexity persists (internally inconsistent); rebuilt as a clean test of the zero-risk-premium expected return.

The expected one-year return on a bond can be decomposed into the forward rate, the effect of expected rate changes (proportional to duration), and a convexity effect (proportional to volatility). Suppose investors are risk neutral (risk premium $\lambda = 0$, as under the pure expectations hypothesis). How does the expected one-year return on a 10-year zero-coupon bond compare with that on a 1-year zero-coupon bond?

A. It is higher by $\lambda\times$ duration, because the 10-year bond has more interest rate risk.
B. It is higher, because the 10-year bond has greater convexity that earns extra return from rate volatility.
C. It is the same: both have an expected return equal to the short-term rate r_0.
D. It is lower, because the 10-year bond's forward rate is reduced by the convexity effect.

Answer: C

Key correction: Source key (A) said the equation simplifies to f(T)dt - D*E(dr) once the risk premium is removed; that drops the convexity term and is not a clean statement. The defensible result is that with a zero risk premium expected return equals the short rate for all maturities.

Step by step

1. For any investor: $E[\text{return}] = r_0 + \lambda \times D$.
2. Risk neutral means $\lambda = 0$, so $E[\text{return}] = r_0$ regardless of duration.
3. The 10-year zero's convexity gain is exactly offset by the lower forward rate that convexity implies, so there is no net advantage.
4. Both bonds therefore have expected return r_0.

Why C is correct: With no risk premium, expected return is independent of maturity and duration; convexity does not create an expected excess return.

Why the others are wrong

- **A:** With $\lambda = 0$ the term λD is zero, so there is no extra return for duration.
- **B:** Convexity raises the return component but lowers forward rates by the same amount; the net expected return is unchanged.
- **D:** Convexity lowers the forward rate but raises the return component; the expected return is not lowered.

Q643

BT answers P2.T5.301.1 · BT notes P2.T5.301.1

REBUILT — OCR-damaged exhibit restored; source key discounted the 90 bp premium at the date-1 nodes (giving $7.30) but the premium belongs only to the date-0 expected return (correct $7.29), so options were rebuilt.

The one-year rate is currently 7.0%. Next year it will be either 10.0% or 4.0% with equal probability (then 13.0%,

7.0% or 1.0% the year after). Ignoring risk premia, the price of a USD 1,000 par two-year zero-coupon bond is USD 874.13, the expected discounted value. Now assume investors are risk averse and require an expected return of 7.90% (a 90 basis point risk premium) on the two-year zero over the first year. Compared with the risk-neutral price of USD 874.13, what is the change in price due to the risk premium?

A. Increase of USD 7.29
B. Decrease of USD 7.29
C. Decrease of USD 7.87
D. Decrease of USD 15.20

Answer: B

Key correction: Source key (C, a $7.30 decrease) obtained its figure by adding 90 bp to the date-1 discount rates; the date-1 one-year zeros carry no further duration risk, so the premium applies only to the date-0 discount rate and the change is $7.29.

Step by step

1. Date-1 prices of the one-year zero: $1000/1.10 = 909.09$ (up) and $1000/1.04 = 961.54$ (down); expected value = 935.31.
2. Risk-neutral price: $935.31/1.07 = 874.13$.
3. Risk-averse price: discount the same expected value at the required return of 7.90%: $935.31/1.079 = 866.83$.
4. Change = $866.83 - 874.13 = -7.29$, a decrease of USD 7.29 (higher required return, lower price).

Why B is correct: A positive risk premium raises the discount rate applied to the expected payoff, which lowers the price; the 90 bp applies to the first-year return only.

Why the others are wrong

- **A:** Right magnitude but wrong sign; a higher required return reduces, not increases, the price.
- **C:** Applies 0.90% directly to the risk-neutral price (874.13 x 0.009 = 7.87) instead of re-discounting at 1.079.
- **D:** Discounts for both years at 7.90% (1000/1.079^2 = 858.93), double counting the premium.

Q644

BT notes P2.T5.23.3.3

REBUILT — Formula was OCR-garbled and the keyed trap (swapping Realized Forwards and Unchanged Term Structure) relies on material outside this reading; rebuilt on the forward-rate decomposition (expectations, risk premium, convexity) with the formula stated.

The forward rate for term T can be approximated as: $f(T) \approx r_0 + E[\Delta r] \times D + \lambda \times D -$ (convexity effect), where D is duration and the convexity effect increases with volatility. Which of the following statements about this decomposition is NOT correct?

A. The forward rate increases with the expected change in the short-term rate, scaled by the bond's duration.
B. The forward rate increases with the risk premium in proportion to duration.
C. Greater interest rate volatility lowers the forward rate through the convexity effect.
D. Because convexity raises a bond's return, it also raises the forward rate, so greater volatility steepens the forward curve.

Answer: D

Step by step

1. Read the decomposition: forward rate = short rate + expectations term + risk premium term - convexity term.
2. A: the expectations term is $E[\Delta r] \times D$, so A is true.
3. B: the risk premium term is λD, increasing in D, so B is true.
4. C and D: the convexity term enters with a negative sign, so higher volatility lowers the forward rate; C is true and D is false (convexity enhances bond returns but reduces forward rates).

Why D is correct: Convexity enhances returns but enters the forward rate with a negative sign, so greater volatility lowers, not raises, forward rates.

Why the others are wrong

- **A:** A true statement (expectations term $E[\Delta r]D$); not the exception.
- **B:** A true statement (risk premium term λD); not the exception.
- **C:** A true statement (convexity effect is subtracted and grows with volatility); not the exception.

Q645

Mock 2 Q64

A risk analyst uses a decision tree framework to estimate the price of zero-coupon bonds. The analyst assumes that investors are risk averse and require a risk premium of 50 basis points for each year of interest rate risk. What is the price that he should calculate for a 2-year zero-coupon bond with a face value of USD 100, given expected 1-year rates in one year of 10% in the up node and 6% in the down node (equal probabilities)? Assume the 1-year

spot rate is 8%.

A. USD 85.37
B. USD 92.59
C. USD 92.17
D. USD 85.76

Answer: A

Step by step

1. Date-1 prices of the 1-year zero: $100/1.10 = 90.909$ (up), $100/1.06 = 94.340$ (down).
2. Expected date-1 price: $0.5(90.909 + 94.340) = 92.625$.
3. Over the first year the 2-year zero carries one year of duration risk, so the required return is $8\% + 0.50\% = 8.50\%$.
4. Price = $92.625/1.085 = 85.37$.

Why A is correct: Risk-averse pricing discounts the expected future price at the short rate plus the risk premium for the duration risk borne.

Why the others are wrong

- **B:** Equals 100/1.08: ignores the rate tree and discounts only one period.
- **C:** Equals 100/1.085: discounts one period at the risk-adjusted rate, omitting the second period of discounting.
- **D:** Discounts the expected price at 8% (92.625/1.08 = 85.76), the risk-neutral price, ignoring the 50 bp premium.

Q646 Schweser Module Quiz 13.2 Q3

Which of the following statements represents the risk premium component in the decomposition of forward rates?

A. The return due to the passage of time.
B. The expected volatility in interest rates.
C. The compensation investors demand for convexity effects.
D. The compensation investors demand for bearing interest rate risk.

Answer: D

Step by step

1. A bond return decomposes into the forward rate (passage of time), the effect of rate changes (duration), and the effect of rate volatility (convexity).
2. Risk-neutral investors expect r_0; risk-averse investors add a risk premium, so expected return = $r_0 + \lambda \times D$.
3. The premium λ is the compensation per unit of duration for bearing interest rate risk, which is option D.

Why D is correct: Risk-averse investors require extra return proportional to duration for bearing interest rate risk; that extra return is the risk premium.

Why the others are wrong

- **A:** This describes the forward rate (return from the passage of time), not the risk premium.
- **B:** Expected rate volatility drives the convexity component, not the risk premium.
- **C:** Convexity is a separate component of forward rates (a Jensen's inequality effect from rate volatility); it is not the risk premium.

Reading 14. The Art of Term Structure Models: Drift

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 3rd Edition (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 9.

LO	Learning objective	Qs
14.a	construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift.	13
14.b	calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift.	7
14.c	describe methods for addressing the possibility of negative short-term rates in term structure models.	3
14.d	construct a short-term rate tree under the Ho-Lee Model with time-dependent drift.	4
14.e	describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices.	7
14.f	describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion.	7
14.g	calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life.	10
14.h	describe the effectiveness of the Vasicek Model.	7

LO 14.a

Construct and describe the effectiveness of a short-term interest rate tree assuming normally distributed rates, both with and without drift.

Q647

AnalystPrep Q.1654

REBUILT — Original stem wrongly implied no drift makes the curve decline for no stated reason and options were nonsensical (maturity/convexity rising by 20 bp); rewrote around the parallel-shift implication of the single-factor Model 1 with realistic distractors.

In Tuckman's Model 1 ($dr = \sigma\, dw$, zero drift, normally distributed rates), the short-term rate is the only factor. Suppose the short-term rate rises by 20 basis points. According to the model, what is the effect on the predicted term structure of spot rates?

A. The whole term structure shifts up in parallel by about 20 basis points at every maturity.
B. Short-maturity rates rise by 20 basis points but long-maturity rates rise by less because of mean reversion.
C. Only the short end rises by 20 basis points; long-maturity rates are unchanged because the drift is zero.
D. Rates rise by 20 basis points at the short end and the volatility term structure steepens.

Answer: A

Step by step

1. Model 1 has one factor, the short rate, and zero drift, so expected future rates equal the current short rate.
2. The convexity (volatility) term depends only on σ and maturity, not on the level of the short rate.
3. A 20 bp rise in the short rate therefore raises every point on the predicted curve by the same 20 bp.

Why A is correct: With a single factor and constant volatility, the predicted term structure equals the short rate minus a fixed convexity adjustment, so a change in the short rate moves all maturities equally - a parallel shift, a recognised limitation versus observed non-parallel shifts.

Why the others are wrong

- **B:** Describes mean reversion (Vasicek), which Model 1 does not have.
- **C:** Ignores that long rates are driven by expected future short rates, which are anchored to the current short rate in a zero-drift model.
- **D:** Model 1 predicts a flat volatility term structure that does not depend on the short-rate level, so nothing steepens.

Q648

AnalystPrep Q.1655

Consider Model 2 ($dr = \lambda\, dt + \sigma\, dw$) with $r_0 = 6.138\%$, $\lambda = 0.239\%$ per year and annual basis-point volatility $\sigma = 1.20\%$. Over a one-month period ($dt = 1/12$), what are the drift of the rate and the standard deviation of the rate change, respectively?

A. Drift 0.239%; standard deviation 0.18%
B. Drift 0.239%; standard deviation 0.01992%
C. Drift 0.01992%; standard deviation 0.35%
D. Drift 0.1992%; standard deviation 0.18%

Answer: C

Step by step

1. Drift over the month = $\lambda\, dt = 0.239\% \times 1/12 = 0.01992\%$.
2. Standard deviation = $\sigma\sqrt{dt} = 1.20\% \times \sqrt{1/12} = 1.20\% \times 0.28868 = 0.3464\% \approx 0.35\%$.

Why C is correct: Drift scales linearly with time ($\lambda\, dt$) while the standard deviation scales with the square root of time ($\sigma\sqrt{dt}$).

Why the others are wrong

- **A:** Quotes the annual drift without multiplying by dt, and the 0.18% standard deviation does not follow from $\sigma\sqrt{dt}$ (0.346%).
- **B:** Leaves the drift unscaled and assigns the scaled-drift number to the standard deviation.
- **D:** Decimal slip on the drift (0.1992% instead of 0.01992%) and an incorrect standard deviation of 0.18%.

Q649

AnalystPrep Q.2855

REBUILT — Stem said volatility was 207 bp 'per month' while the worked solution used it as an annual figure with dt = 1/12; made the units explicit and replaced arbitrary distractors with common-mistake values.

Using Model 2 ($dr = \lambda\, dt + \sigma\, dw$), the current short rate is 6.047%, annual basis-point volatility is 207 basis points ($\sigma = 2.07\%$) and the annual drift is $\lambda = 0.348\%$. Over one month ($dt = 1/12$) the realized value of the random variable is $dw = 0.32$. What is the change in the short rate over the month?

A. 0.6624%

B. 0.6914%

C. 1.0104%

D. 0.6266%

Answer: B

Step by step

1. Drift component = $\lambda\, dt = 0.348\%/12 = 0.0290\%$.
2. Random component = $\sigma\, dw = 2.07\% \times 0.32 = 0.6624\%$ (dw is the realization, so no further $\sqrt{dt}$).
3. $dr = 0.0290\% + 0.6624\% = 0.6914\%$.

Why B is correct: The rate change is the time-scaled drift plus the volatility times the realized shock.

Why the others are wrong

- **A:** Omits the drift term entirely (shock only).
- **C:** Adds the annual drift (0.348%) without multiplying by dt = 1/12.
- **D:** Uses $\sigma\sqrt{dt} = 0.5976\%$ in place of $\sigma\, dw$ and then adds the drift (0.0290%), ignoring the realized dw.

Q650

AnalystPrep Q.2859

REBUILT — Original stem was muddled (conflated 'consistent trend' with a constant short rate) and options were loosely worded; restated as Model 2 with r0 = 0, lambda > 0, sigma = 0 and sharper alternatives.

In Model 2 ($dr = \lambda\, dt + \sigma\, dw$) the current short rate is zero, the drift λ is positive and the basis-point volatility σ is zero. Which statement is correct?

A. The short rate will be negative about half of the time.

B. The long-term spot rate will equal the short rate at all maturities.

C. The short rate rises deterministically and never goes negative, so long-term spot rates exceed the short rate.

D. The long-term spot rate will be below the short rate because of the convexity effect.

Answer: C

Step by step

1. With $\sigma = 0$ there is no random term: $r_t = r_0 + \lambda t = \lambda t \geq 0$.
2. Spot rates are averages of expected future short rates (no convexity term when $\sigma = 0$): $\lambda T/2$, which exceeds the current short rate 0 and increases with T.

Why C is correct: Zero volatility removes both negative-rate risk and the convexity effect, leaving only the positive drift, which pushes the rate path and the spot curve upward.

Why the others are wrong

- **A:** Negative rates arise from random shocks; with zero volatility there are none.
- **B:** Equal short and long rates require zero drift as well as zero volatility (a flat curve).
- **D:** Convexity lowers long rates only when volatility is positive; here it is zero and the drift is positive.

Q651

AnalystPrep Q.4014

REBUILT — Original correct (B) but distractors such as 4.5212% and 5.5% had no traceable error; replaced with values from realistic mistakes.

Under Model 2 the current short rate is 5.26%, annual basis-point volatility is 115 basis points ($\sigma = 1.15\%$), annual drift is $\lambda = 0.25\%$, and $dt = 1/12$. After one month the random variable dw takes the value 0.25. What is the short-term rate after one month?

A. 5.7975%
B. 5.5475%
C. 5.5683%
D. 5.6128%

Answer: C

Step by step

1. Drift $= \lambda\, dt = 0.25\%/12 = 0.0208\%$.
2. Shock $= \sigma\, dw = 1.15\% \times 0.25 = 0.2875\%$.
3. $dr = 0.3083\%$; new rate $= 5.26\% + 0.3083\% = 5.5683\%$.

Why C is correct: New rate = old rate + time-scaled drift + volatility times the realized dw.

Why the others are wrong

- **A:** Added the full annual drift 0.25% instead of 0.25%/12.
- **B:** Omitted the drift: $5.26\% + 0.2875\%$.
- **D:** Replaced the realized dw by $\sqrt{dt}$: $5.26 + 0.0208 + 1.15 \times 0.2887$.

Q652

BT answers L2.T5.45.1

REBUILT — Original correct (C) but distractors were arbitrary 10 bp steps; replaced with outcomes of typical time-scaling errors.

The current short-term rate is 2.0% and annual basis-point volatility is 100 basis points (1.0 percentage point). A binomial rate tree for a zero-drift normal model has twelve monthly steps ($dt = 1/12$). What is the up-state interest rate after the first month?

A. 2.08%
B. 2.29%
C. 2.50%
D. 3.00%

Answer: B

Step by step

1. Up move $= \sigma\sqrt{dt} = 1.0\% \times \sqrt{1/12} = 0.2887\%$.
2. Up-state rate $= 2.0\% + 0.2887\% = 2.29\%$.

Why B is correct: Per-step moves in the tree are $\pm\sigma\sqrt{dt}$ around the prior rate.

Why the others are wrong

- **A:** Scaled volatility linearly ($\sigma\, dt = 0.0833\%$) instead of by the square root.
- **C:** Used $\sqrt{1/4}$ (a quarterly step) instead of $\sqrt{1/12}$.
- **D:** Added the full annual volatility with no time scaling.

Q653

BT answers P2.T5.302.1 · BT notes P2.T5.302.1

REBUILT — Merged duplicates (answers file lacked the tree); kept the reconstructed tree and replaced options with realistic-error distractors.

Analyst Greg builds a monthly interest rate tree ($dt = 1/12$) with Tuckman's Model 1 (normally distributed rates, zero drift, $dr = \sigma\, dw$). The current short rate is 4.0% and annual basis-point volatility is 180 basis points. Part of the tree:

Node	Month 0	Month 1	Month 2
Top	|	[2,2] = ?	
Up	| 4.520%		
Middle	r(0) = 4.000%	| 4.000%	
Down	| 3.480%		
Bottom	|	2.961%	

What is the missing rate at node [2,2]?

A. 4.30%
B. 4.73%

C. 5.04%
D. 4.52%

Answer: C
Step by step
1. Per-step move = $\sigma\sqrt{dt} = 1.80\% \times 0.28868 = 0.5196\%$ (matches 4.520% and 3.480% at month 1).
2. Node [2,2] is two consecutive up moves: $4.0\% + 2 \times 0.5196\% = 5.0392\% \approx 5.04\%$.

Why C is correct: The tree is recombining with zero drift, so the top node after two steps is r0 plus two up-moves of $\sigma\sqrt{dt}$.

Why the others are wrong
- **A:** Scaled volatility by dt instead of $\sqrt{dt}$: $4 + 2 \times 1.8/12$.
- **B:** Used the two-month standard deviation $\sigma\sqrt{2dt}$ (0.735%) rather than two up-steps.
- **D:** Gave the month-1 up node (one step only).

Q654

BT answers P2.T5.302.2 · BT notes P2.T5.302.2

REBUILT — Merged duplicates; the month-1 down node was illegible/missing so reconstructed it consistently (2.464%) and replaced options with realistic-error distractors.

Analyst Barbara builds a monthly interest rate tree ($dt = 1/12$) with Tuckman's Model 2 (normally distributed rates with drift, $dr = \lambda\, dt + \sigma\, dw$). The current short rate is 3.0%, annual basis-point volatility is 200 basis points and the annual drift is $\lambda = 50$ basis points. Part of the tree:

Node	Month 0	Month 1	Month 2
Top	|	4.238%	
Up	| 3.619%		
Middle	r(0) = 3.000%	| 3.083%	
Down	| 2.464%		
Bottom	|	[2,0] = ?	

What is the missing rate at node [2,0]?

A. 1.85%
B. 1.93%
C. 2.46%
D. 1.76%

Answer: B
Step by step
1. Drift per step = $\lambda\, dt = 0.50\%/12 = 0.04167\%$; shock per step = $\sigma\sqrt{dt} = 2.0\% \times 0.28868 = 0.57735\%$.
2. Two down moves: $3.0\% + 2(0.04167\%) - 2(0.57735\%)$.
3. $= 3.0 + 0.0833 - 1.1547 = 1.9286\% \approx 1.93\%$.

Why B is correct: Drift is added at every step regardless of direction, so the bottom node is r0 plus two drifts minus two volatility steps.

Why the others are wrong
- **A:** Left out the drift: $3.0 - 1.1547$.
- **C:** Gave the month-1 down node (one step only, 2.464%).
- **D:** Subtracted the drift instead of adding it: $3.0 - 0.0833 - 1.1547$.

Q655

BT answers P2.T5.302.3 · BT notes P2.T5.302.3

Tuckman's Model 1 assumes zero drift and is also called a normal model. His Model 2 adds a term for drift. Each of the following is true about these two models, according to Tuckman, EXCEPT for:

A. A weakness of Model 1 (normal model) is that the short-term rate can become negative
B. Model 1 implies a term structure that is perfectly flat at the current rate for all maturities, including the long-term rates
C. Model 2 is more capable of producing an upward-sloping term structure, which is often observed
D. Model 2 is an equilibrium model rather than an arbitrage-free model because no attempt is made to match the term structure closely

Answer: B

Step by step

1. Test each statement against the Schweser/Tuckman description of Models 1 and 2.
2. A: Model 1 has normally distributed rates, so there is always a positive probability of negative rates - true.
3. B: With zero drift, Model 1's convexity effect (larger at long maturities) pulls the predicted term structure downward, so it is downward-sloping, not perfectly flat - false.
4. C: A positive drift term can accommodate the typically observed upward-sloping curve, so Model 2 fits better - true.
5. D: Models 1 and 2 use a few constant parameters (r_0, λ, σ) and do not match the market curve, so they are equilibrium, not arbitrage-free, models - true. The EXCEPT answer is B.

Why B is correct: With no drift, the convexity effect makes Model 1's predicted term structure downward-sloping, so it is not flat at long maturities.

Why the others are wrong

- **A:** True statement: normal rates can go below zero (the main drawback of Model 1); choosing it means missing the EXCEPT wording.
- **C:** True statement: the constant drift (risk premium) lets Model 2 generate an upward slope; chosen by those who think drift does not affect the curve shape.
- **D:** True statement: Models 1 and 2 are equilibrium models because they are not calibrated to match market prices (Ho-Lee is the first arbitrage-free step).

Q656

BT answers P2.T5.306.3

REBUILT — Original five-part summary was ambiguous (statement V implied Model 3 is non-normal and Model 3 is outside this reading); replaced by a single-answer question on the effectiveness of Models 1 and 2.

Which of the following statements about Tuckman's Model 1 (no drift) and Model 2 (constant drift) is correct?

A. Model 1 predicts a hump-shaped term structure of volatility, consistent with observed volatility

B. In Model 1, a change in the short-term rate leads to a parallel shift in the predicted yield curve

C. The Model 2 tree is non-recombining because the drift is added only to the up-branches

D. Model 2's constant drift is easy to justify as an expected rate increase over very long horizons

Answer: B

Step by step

1. Model 1 has a single factor, the short rate r, and constant volatility, so a shock to r moves every point of the predicted curve by the same amount: a parallel shift. B is correct.
2. A: Model 1 predicts a flat term structure of volatility; observed volatility is hump-shaped.
3. C: In Model 2 the drift $\lambda\,dt$ is added on every branch, so up-down and down-up paths still meet; the tree recombines, with the middle node at Time 2 equal to $r_0 + 2\lambda\,dt$.
4. D: A constant drift implies expected rates keep rising (Year 10 above Year 9), which is hard to justify at long horizons.

Why B is correct: Model 1 is a one-factor model, so any shift in the short rate moves the whole predicted curve in parallel, which conflicts with observed non-parallel shifts.

Why the others are wrong

- **A:** Confuses the model with reality: Model 1 gives a flat volatility term structure, not the observed hump.
- **C:** Adding drift to all branches shifts the tree but keeps it recombining; the up-down and down-up paths still end at the same node.
- **D:** Reverses the caveat: viewing drift as expected rate change is more defensible in the short run than the long run.

Q657

Mock 4 Q57

An analyst is modelling the short-term interest rate using a model with constant drift and constant volatility: $dr = \lambda\,dt + \sigma\,dw$. The current short-term rate is 3.00%, the annual drift λ is 0.30%, and the annual basis-point volatility σ is 1.20%. What are the expected value and the standard deviation of the short-term rate two years from now?

A. Expected rate 3.30%; standard deviation 1.70%

B. Expected rate 3.60%; standard deviation 1.70%

C. Expected rate 3.60%; standard deviation 1.20%

D. Expected rate 3.60%; standard deviation 2.40%

Answer: B

Step by step

1. Expected rate: $E[r_2] = r_0 + \lambda T = 3.00\% + 0.30\% \times 2 = 3.60\%$.
2. Standard deviation: $\sigma\sqrt{T} = 1.20\% \times \sqrt{2} = 1.697\% \approx 1.70\%$.
3. Drift accumulates linearly in time; volatility scales with the square root of time.

Why B is correct: Expected change grows linearly (λT) while the standard deviation grows with $\sqrt{T}$, giving 3.60% and 1.70%.

Why the others are wrong

- **A:** Applies only one year of drift (3.00% + 0.30% = 3.30%) instead of two.
- **C:** Uses the one-year volatility without scaling by $\sqrt{2}$.
- **D:** Scales volatility linearly with time (1.20% × 2 = 2.40%) instead of by the square root of time.

Q658

Schweser Module Quiz 14.1 Q2

Using Model 2, assume a current short-term rate of 8%, an annual drift of 50 bps, and a short-term rate standard deviation of 2%. In addition, assume the ex-post realization of the dw random variable is 0.3. After constructing a 2-period interest rate tree with annual periods, what is the interest rate in the middle node at the end of Year 2?

A. 8.0%
B. 8.8%
C. 9.0%
D. 9.6%

Answer: C

Step by step

1. In the recombining Model 2 tree, the up-then-down and down-then-up paths cancel the random terms, so the middle node at Time 2 equals $r_0 + 2\lambda\, dt$.
2. $\lambda\, dt = 0.50\% \times 1 = 0.50\%$ per year.
3. Middle node = 8% + 2 × 0.50% = 9.0%. The realization $dw = 0.3$ only affects a single path, not the tree's middle node.

Why C is correct: Random shocks cancel at the middle node, leaving only two periods of drift: 8% + 2(0.5%) = 9.0%.

Why the others are wrong

- **A:** Ignores drift (the Model 1 answer, where the middle node equals r0).
- **B:** Adds only one period of drift plus the dw realization (8% + 0.5% + 0.3%).
- **D:** Adds the dw realization of 0.3 in each period on top of the two drifts (8% + 2 × (0.5% + 0.3%)).

Q659

BT answers P2.T5.303.2 · BT notes P2.T5.303.2

REBUILT — Distractors 2.77% and 4.15% did not correspond to any realistic mistake; replaced with 3.98% (full annual drift) and 5.10% (sign error), keeping stem and answer.

Analyst Barry wants to improve his short-term interest rate simulation by employing Tuckman's Model 2, which adds a drift term to Model 1: $dr = \lambda\, dt + \sigma\, dw$. The time step is one month, i.e., $dt = 1/12$. The assumptions are: the current short-term rate is 4.00%, the annual basis-point volatility is 250 bps, and the annual drift is +100 bps. In the first step of the first trial, the random uniform variable is 0.0793, which via inverse transformation gives a standard normal value of -1.410 (NORM.S.INV(7.93%) ≈ -1.410). To what level does the rate evolve in the first month, $r(1/12)$?

A. 3.98%
B. 2.98%
C. 3.07%
D. 5.10%

Answer: C

Step by step

1. Drift: $\lambda\, dt = 1.00\% \times 1/12 = 0.0833\%$.
2. Shock: $\sigma\, dw = 2.50\% \times (-1.410) \times \sqrt{1/12} = -1.0176\%$.
3. $dr = 0.0833\% - 1.0176\% = -0.9342\%$.
4. $r(1/12) = 4.00\% - 0.9342\% = 3.0658\% \approx 3.07\%$.

Why C is correct: The monthly change is the one-month drift (0.0833%) plus a shock scaled by $\sigma\sqrt{dt}$, giving 4.00%

- 0.934% = 3.07%.

Why the others are wrong

- **A:** Applies the full annual drift of 1.00% rather than 1/12 of it (4.00% + 1.00% - 1.0176%).
- **B:** Omits the drift term (4.00% - 1.0176%), i.e., uses Model 1.
- **D:** Gets the sign of the shock wrong, adding 1.0176% instead of subtracting it (4.00% + 0.0833% + 1.0176%).

LO 14.b

Calculate the short-term rate change and standard deviation of the rate change using a model with normally distributed rates and no drift.

Q660

AP Mock 1 Q13

Using a term structure model with no drift and normally distributed rates, assume that the short-term interest rate is 5%, annual volatility is 75 basis points, and after one month the realisation of dw (a normally distributed random variable with mean 0 and standard deviation √dt) is 0.2. What is the new spot rate?

A. 5.15%
B. 4.85%
C. 3.50%
D. 6.50%

Answer: A

Step by step

1. Model 1: $dr = \sigma\, dw$ with $\sigma = 0.75\%$ per year.
2. The realisation of dw (0.2) already includes the $\sqrt{dt}$ scaling, so $dr = 0.75\% \times 0.2 = 0.15\%$ = 15 bp.
3. New rate = 5.00% + 0.15% = 5.15%.

Why A is correct: With no drift the rate change is volatility times the realised dw; dw already embeds the time-step scaling.

Why the others are wrong

- **B:** Sign error: subtracts the 15 bp change instead of adding it (dw is +0.2).
- **C:** Treats 75 bp as 7.5%, giving dr = 1.5% and subtracting it.
- **D:** Treats 75 bp as 7.5%, giving dr = 1.5% and adding it.

Q661

AnalystPrep Q.1651

An FRM candidate is analyzing the movement in short-term interest rates assuming the rates are normally distributed with no drift. The current short-term rate is 7.18%, the time interval is 3 months (3/12 of a year), and volatility is 118 basis points per year. After three months, the random variable dw takes the value 0.18. What are the change in the short-term rate and the short-term rate after 3 months?

A. Change 0.2124%; rate after 3 months 7.3924%
B. Change 7.3924%; rate after 3 months 0.2124%
C. Change 1.2924%; rate after 3 months 8.4724%
D. Change 8.4724%; rate after 3 months 1.2924%

Answer: A

Step by step

1. Model 1: $dr = \sigma\, dw = 1.18\% \times 0.18 = 0.2124\%$.
2. New rate = 7.18% + 0.2124% = 7.3924%.

Why A is correct: No drift, so the change is only volatility times the realised dw, added to the starting rate.

Why the others are wrong

- **B:** Correct numbers but the labels are swapped (reports the new rate as the change).
- **C:** Multiplies the current rate (7.18%) by dw instead of the volatility: 7.18% × 0.18 = 1.2924%.
- **D:** Combines the C error with swapped labels.

Q662

AnalystPrep Q.2854

REBUILT — Original distractors (19, 16.7, 731 bp) were not tied to realistic errors; replaced with options generated by plausible mistakes and reordered.

The current short-term rate is 7.31%. Volatility is 167 basis points per year and the time interval is one month (dt = 1/12). Under the normally distributed, no-drift model (Model 1), after one month the random variable dw (mean zero, standard deviation 0.2887) takes the value 0.21. What is the change in the short-term rate?

A. 10 basis points
B. 35 basis points
C. 48 basis points
D. 3 basis points

Answer: B

Step by step

1. Model 1: $dr = \sigma\, dw$ with $\sigma = 167$ bp per year.
2. $dr = 167 \times 0.21 = 35.07$ bp, i.e. about 35 bp.
3. The 0.2887 is the standard deviation of dw, not a further multiplier; dw = 0.21 is the realised draw.

Why B is correct: The rate change is annual volatility times the realised dw; dw already contains the $\sqrt{dt}$ scaling.

Why the others are wrong

- **A:** Multiplies by the standard deviation of dw (0.2887) in addition to the realisation: 167 × 0.2887 × 0.21 ≈ 10.1 bp.
- **C:** Uses the standard deviation of dw instead of the realisation: 167 × 0.2887 ≈ 48.2 bp.
- **D:** Scales by dt = 1/12 instead of using dw: 167 × 0.21 / 12 ≈ 2.9 bp.

Q663

AnalystPrep Q.4013

REBUILT — Options 0.035 and 0.025 were giveaways and inconsistent in format; rebuilt options in % tied to realistic errors.

Under Model 1 (normally distributed rates, no drift), the current short-term rate is 5.26%, volatility is 115 basis points per year, and dt = 1/12. After one month the random variable dw, which has zero mean and standard deviation √(1/12) = 0.2887, takes the value 0.25. What is the short-term rate after one month?

A. 5.5475%
B. 5.5100%
C. 5.3430%
D. 5.5920%

Answer: A

Step by step

1. Model 1: $dr = \sigma\, dw = 1.15\% \times 0.25 = 0.2875\%$.
2. New rate = 5.26% + 0.2875% = 5.5475%.

Why A is correct: With no drift the new rate is the old rate plus volatility times the realised dw.

Why the others are wrong

- **B:** Treats dw = 0.25 as 0.25% and adds it directly, ignoring volatility (5.26% + 0.25%).
- **C:** Multiplies by the standard deviation of dw as well: 1.15% × 0.25 × 0.2887 = 0.083%.
- **D:** Uses the standard deviation of dw instead of the realisation: 1.15% × 0.2887 = 0.332%.

Q664

AnalystPrep Q.4016

REBUILT — Roman-numeral option I was internally inconsistent and options were a lopsided table; replaced with four consistent change/new-rate pairs and removed the confusing 'expected value of zero' phrasing.

Using Model 1, assume the current short-term rate is 3%, annual volatility is 100 basis points, and dw is a normally distributed random variable with mean 0 and standard deviation √dt. After one month the realisation of dw is −0.3. What are the change in the spot rate and the new spot rate?

A. Change −0.30%; new rate 2.70%
B. Change −3.00%; new rate 0.00%
C. Change +0.30%; new rate 3.30%
D. Change −0.087%; new rate 2.913%

Answer: A

Step by step

1. Model 1: $dr = \sigma\, dw = 1.00\% \times (-0.3) = -0.30\%$.
2. New rate = 3.00% − 0.30% = 2.70%.

Why A is correct: Change equals volatility times realised dw, keeping its sign; the starting rate is adjusted by that amount.

Why the others are wrong

- **B:** Misreads dw = −0.3 as −3 (decimal slip), giving −3% and a zero rate.
- **C:** Drops the negative sign of dw and adds the change.
- **D:** Multiplies by √dt = 0.2887 again: 1% × (−0.3) × 0.2887 = −0.087%.

Q665

Schweser Module Quiz 14.1 Q1

REBUILT — Source answer choices were missing in OCR (open stem); reconstructed four options from realistic errors.

Using Model 1, assume the current short-term rate is 5%, annual volatility is 80 basis points, and dw is a normally distributed random variable with mean 0 and standard deviation √dt. After one month the realisation of dw is −0.5. What are the change in the spot rate and the new spot rate?

A. Change +40 bp; new rate 5.40%

B. Change −40 bp; new rate 4.60%

C. Change −11.5 bp; new rate 4.885%

D. Change −50 bp; new rate 4.50%

Answer: B

Step by step

1. Model 1: $dr = \sigma\, dw = 0.80\% \times (-0.5) = -0.40\%$ = −40 bp.
2. New rate = 5.00% − 0.40% = 4.60%.

Why B is correct: No drift: the change is volatility times the realised dw (sign preserved).

Why the others are wrong

- **A:** Ignores the negative sign of dw.
- **C:** Applies √dt again: 0.80% × 0.5 × 0.2887 = 0.115%.
- **D:** Ignores volatility and treats dw = −0.5 as −0.5%.

Q666

BT notes P2.T5.303.1 · BT answers P2.T5.303.1

REBUILT — Two sources were identical (merged); options 3.854% and 5.393% corresponded to no identifiable error, so the distractors were replaced.

Analyst Barry simulates short-term rates with Tuckman's Model 1 (no drift): $dr = \sigma\, dw$. The time step is one month (dt = 1/12), the initial rate is 4.00%, and annual basis-point volatility is 200 bp. In the first step the random uniform draw is 0.8925, which maps via the inverse normal to a standard normal value Z = 1.240 (so $dw = Z\sqrt{dt}$). To what level does the rate evolve after the first month?

A. 3.284%

B. 4.207%

C. 4.716%

D. 6.480%

Answer: C

Step by step

1. $dw = Z\sqrt{dt} = 1.240 \times \sqrt{1/12} = 0.3579$.
2. $dr = 2.00\% \times 0.3579 = 0.7159\%$.
3. New rate = 4.00% + 0.7159% = 4.716%.

Why C is correct: The standard normal draw must be scaled by √dt to get dw, then by volatility to get the rate change.

Why the others are wrong

- **A:** Sign error: subtracts the 0.716% change.
- **B:** Scales by dt instead of √dt: 2% × 1.24 / 12 = 0.207%.
- **D:** Omits the √dt scaling: 2% × 1.24 = 2.48%, giving 6.48%.

LO 14.c

Describe methods for addressing the possibility of negative short-term rates in term structure models.

Q667

AnalystPrep Q.1653

REBUILT — Original options used undefined labels (expected/observed/shadow rates) not in the Schweser text and two options were near-identical; rebuilt around the two Schweser remedies with clear distinct statements.

A normally distributed short-rate tree allows negative rates. A common fix is to construct the tree with the desired distribution and then force every negative rate to zero. Which of the following statements about this method is correct?

A. It replaces the normal distribution with a lognormal distribution at every node, so all rates in the tree change.
B. It changes the original distribution only where rates are very low, leaving rates away from zero unchanged.
C. It introduces skewness and inappropriate volatilities across the entire distribution of rates.
D. It removes the negative rate problem only when the investment horizon is long and the current rate is high.

Answer: B

Step by step

1. The Schweser text gives two remedies for negative rates: (1) use a distribution that is always non-negative (lognormal, chi-squared); (2) force negative tree rates to zero.
2. Remedy (2) leaves every node with a positive rate as originally computed; only nodes that would be below zero are reset to 0.
3. Hence the distribution is changed only in a very low rate environment, which is why this method may be preferred to remedy (1), which reshapes the whole distribution.
4. Statement B is the only one consistent with this.

Why B is correct: Flooring at zero modifies only the nodes that would otherwise be negative, so the original distribution is altered only in a very low-rate environment.

Why the others are wrong

- **A:** Describes remedy (1) (switching to a lognormal distribution), not the zero-floor method; the tree's distribution is not replaced.
- **C:** Skewness and inappropriate volatilities are the drawbacks the text attaches to non-negative distributions such as lognormal, not to the zero floor.
- **D:** Reverses the logic: the negative-rate problem is worse for long horizons and low current rates, which is when the floor matters most, not when rates are high.

Q668

BT answers P2.T5.303.3 · BT notes P2.T5.303.3

REBUILT — Source option A ('increase the convexity effect input') is not a defined modelling input and the CIR option belongs to another reading; rebuilt as an EXCEPT question on the Schweser remedies with exactly one non-remedy.

Analyst Susan is evaluating a normally distributed short-rate model and wants to avoid negative short-term rates in the model. Each of the following approaches can help her EXCEPT:

A. Using a lognormal distribution for the short rate.
B. Using a chi-squared distribution for the short rate.
C. Adjusting the interest rate tree so that any negative rate is set to zero.
D. Increasing the number of time steps so the terminal distribution approaches a continuous normal distribution.

Answer: D

Step by step

1. Schweser's two remedies: (1) use distributions that are always non-negative, e.g. lognormal or chi-squared; (2) force negative rates in the tree to zero.
2. A and B are remedy (1); C is remedy (2).
3. D moves the terminal distribution closer to a normal distribution, which has unbounded support below zero, so a positive probability of negative rates remains (and the problem grows with horizon).
4. D is therefore the exception.

Why D is correct: A finer tree converges to a normal distribution, which still assigns positive probability to negative rates, so it is not a remedy.

Why the others are wrong

- **A:** A valid remedy: lognormal rates cannot be negative (at the cost of skewness/inappropriate volatility).

- **B:** A valid remedy: chi-squared is non-negative by construction.
- **C:** A valid remedy: flooring negative tree rates at zero changes the distribution only at very low rates.

Q669 AnalystPrep Q.1652

REBUILT — Option A (negative rates give lenders 'zero lending risk') and D were giveaway distractors; reworded all options around the economic argument in the Schweser text.

In term structure models with normally distributed rates, the short-term rate can become negative. Which of the following statements best explains why negative rates are a drawback of such models?

A. Lenders would accept negative rates because lending at a negative rate removes all credit risk.
B. Investors can instead hold cash and earn a zero rate, so lending at a materially negative rate makes little economic sense.
C. Negative rates are harmless because they always revert to positive levels within one period.
D. Negative rates occur only for short horizons, so the problem is smaller as the investment horizon lengthens.

Answer: B

Step by step

1. Lending 100 and receiving less than 100 back is dominated by holding cash at a 0% return.
2. Schweser notes a small negative rate can be rationalized only if the safety or inconvenience of holding cash is high enough.
3. So materially negative rates make little economic sense, which is the model's drawback.
4. The problem worsens with longer horizons and lower current rate levels, ruling out D.

Why B is correct: Cash offers a zero return, so no one would lend at a materially negative rate; a model that generates them is economically unrealistic.

Why the others are wrong

- **A:** Credit risk is not removed by a negative rate; lenders lose principal for certain, so they would hold cash instead.
- **C:** Nothing forces reversion to positive within one period; the model has no such mechanism.
- **D:** Reversed: negative rates become more likely as the horizon lengthens (wider terminal distribution) and when current rates are low.

LO 14.d

Construct a short-term rate tree under the Ho-Lee Model with time-dependent drift.

Q670 AnalystPrep Q.1656

REBUILT — Removed the loosely worded 'equilibrium model' preamble and tightened the stem to ask directly about the Ho-Lee time-dependent drift.

In the Ho-Lee model, the drift of the short-term rate is allowed to differ from period to period (time-dependent drift). What does the drift in a given period represent?

A. A combination of the risk premium and the expected change in the short-term rate over that period
B. Only the expected change in the short-term rate over that period
C. The expected change in the volatility of the short-term rate over that period
D. Only the risk premium demanded by investors over that period

Answer: A

Step by step

1. In Models 2 and Ho-Lee the drift term $\lambda(t)dt$ is the expected (risk-neutral) change in the short rate.
2. Per Tuckman, that drift reflects both the risk premium for bearing interest rate risk and the expected change in the short rate.
3. The two components cannot be separated, and λ is calibrated to market prices so the tree fits the term structure; volatility is a separate parameter σ.

Why A is correct: The drift is the sum of the expected rate change and the risk premium, which is why it is calibrated to market prices rather than estimated from forecasts.

Why the others are wrong

- **B:** Ignores the risk premium; the drift is risk-neutral, so it embeds compensation for risk as well as expected

changes.
- **C:** Confuses drift with volatility; volatility is the separate σ term and is constant in Ho-Lee.
- **D:** Ignores the expected-rate-change component; the drift is not the risk premium alone.

Q671

MK GARP PE1 Q48 · MK Mock 3 Q32

A risk analyst constructs a binomial interest rate tree by using the Ho-Lee model. The time step is monthly and the annualized drift is 80 bps in the first month and 120 bps in the second month. Assuming the current annualized short-term rate is 3.2% and the annual basis-point volatility is 2.1%, what is the interest rate at the lowest node after 2 months?

A. 1.82%
B. 2.15%
C. 2.76%
D. 3.03%

Answer: B

Step by step

1. dt = 1/12, so $\sqrt{dt} = 0.288675$.
2. Cumulative drift = (0.80% + 1.20%)/12 = 0.1667%.
3. Volatility step per move = $\sigma\sqrt{dt}$ = 2.1% × 0.288675 = 0.6062%.
4. Lowest node = two down moves: r = 3.2% + 0.1667% − 2(0.6062%) = 2.1543% ≈ 2.15%.

Why B is correct: In Ho-Lee, drift (scaled by dt) is added at every step regardless of direction, and the lowest node subtracts two volatility steps of $\sigma\sqrt{dt}$.

Why the others are wrong

- **A:** Subtracted the drift (3.2% − 0.1667% − 2×0.6062% = 1.82%) instead of adding it.
- **C:** Used only one down move (3.2% + 0.1667% − 0.6062% = 2.76%) instead of two.
- **D:** Subtracted the drift and applied no volatility shock (3.2% − 0.1667% = 3.03%).

Q672

BT answers P2.T5.304.1 · BT notes P2.T5.304.1

REBUILT — Source relied on a partially unreadable tree exhibit; made the stem self-contained with all inputs and replaced distractors with ones that follow from realistic errors.

The current short-term rate, r(0), is 4.00%. Under a Ho-Lee model with time-dependent drift, the time step is monthly (dt = 1/12). The annualized drifts are +100 bps in the first month and +80 bps in the second month, and the annual basis-point volatility is 200 bps. In the recombining two-month tree, what is the interest rate at the lowest node after two months?

A. 2.845%
B. 3.573%
C. 2.995%
D. 3.817%

Answer: C

Step by step

1. $\sqrt{dt} = 0.288675$; $\sigma\sqrt{dt}$ = 2.00% × 0.288675 = 0.5774%.
2. Cumulative drift = (1.00% + 0.80%) × 1/12 = 0.1500%.
3. Middle node = 4.00% + 0.15% = 4.150%; each down move subtracts 0.5774%.
4. Lowest node = 4.150% − 2(0.5774%) = 2.995%.

Why C is correct: The lowest Ho-Lee node equals r0 plus the dt-scaled cumulative drift minus two volatility steps of $\sigma\sqrt{dt}$.

Why the others are wrong

- **A:** Omitted the drift (4.00% − 1.155% = 2.845%).
- **B:** Applied only one down move (4.150% − 0.577% = 3.573%).
- **D:** Scaled volatility by dt rather than $\sqrt{dt}$ (4.150% − 2×2.00%/12 = 3.817%).

Q673 BT answers P2.T5.304.2 · BT notes P2.T5.304.2

REBUILT — Source option B (4.425%) was a rounded value; exact result is 4.427%, so options were rebuilt with an exact answer and error-based distractors.

An analyst uses the Ho-Lee model, dr = λ(t)dt + σdw, with monthly steps (dt = 1/12), current rate r(0) = 4.000%, annual basis-point volatility of 300 bps, and annualized drifts of λ(1) = 110 bps in month one and λ(2) = 70 bps in month two. In the simulation, the standard normal draw is −0.52 in month one and +0.84 in month two. What is the simulated short rate after two months, r(2/12)?

A. 4.277%
B. 4.877%
C. 4.427%
D. 5.110%

Answer: C

Key correction: Source key (BT) gave 4.425% as the answer; the exact value is 4.427% (rounding in the source option).

Step by step

1. Cumulative drift = (1.10% + 0.70%) × 1/12 = 0.1500%.
2. Sum of shocks = −0.52 + 0.84 = 0.32.
3. Volatility term = 0.32 × 3.00% × √(1/12) = 0.32 × 0.8660% = 0.2771%.
4. r(2/12) = 4.000% + 0.150% + 0.2771% = 4.427%.

Why C is correct: Ho-Lee adds the dt-scaled cumulative drift plus σ√dt times the sum of all random draws to the starting rate.

Why the others are wrong

- **A:** Left out the drift (4.000% + 0.277% = 4.277%).
- **B:** Used only the second-month shock (0.84 × 0.866% = 0.727%) with the drift: 4.877%.
- **D:** Forgot the √dt scaling on volatility (0.32 × 3.00% = 0.96%), giving 5.110%.

LO 14.e

Describe uses and benefits of the arbitrage-free models and assess the issue of fitting models to market prices.

Q674 AnalystPrep Q.1636

You are asked to start with assumptions about the interest rate process and about the risk premium demanded by the market for bearing interest rate risk, and then derive the risk-neutral process. This approach results in:

A. An arbitrage-free model
B. An equilibrium model
C. The matching of the initial term structure
D. The pricing of all fixed-income securities by arbitrage

Answer: B

Step by step

1. Identify the direction of the modeling: real-world rate process + market risk premium, then derive the risk-neutral process.
2. This is the defining procedure of an equilibrium model (e.g., Vasicek): the term structure is an output of the assumptions.
3. An arbitrage-free model works in reverse: it takes the observed prices/term structure as given and chooses the (risk-neutral) drift to reproduce them.

Why B is correct: Equilibrium models start from assumptions about the true process and the risk premium and derive the risk-neutral process and prices; arbitrage-free models start from market prices.

Why the others are wrong

- **A:** Reverses the direction: arbitrage-free models take observed prices as input and back out the risk-neutral drift.
- **C:** Matching the initial term structure is a feature of arbitrage-free models, not of an approach that derives prices from assumptions.
- **D:** Overstates what the approach does; assumption-driven models need not price every security by arbitrage and can misprice relative to the market.

Q675

AnalystPrep Q.1637

REBUILT — Original had two irrelevant distractors (marginal productivity of capital, saving propensity) and a wrong-model option; replaced with plausible distractors testing the Tuckman point about interpreting market-implied assumptions.

For an arbitrage-free interest rate model, understanding the relationship between the model's assumptions and the shape of the term structure is most important because it allows the user to:

A. separate the true expected change in rates from the risk premium embedded in the calibrated drift

B. avoid specifying any process for the short-term rate, since the market curve replaces it

C. interpret the assumptions about expectations, risk premium and volatility that the market is implicitly pricing in the observed term structure

D. confirm that observed market prices are fair, because the model reproduces them exactly

Answer: C

Step by step

1. An arbitrage-free model is calibrated (e.g., time-dependent drift in Ho-Lee) so that model prices equal market prices.
2. The calibrated drift therefore reflects what the market implies, a combination of expected rate changes and risk premium.
3. Knowing how assumptions map to curve shape lets the user judge whether those market-implied assumptions are reasonable.

Why C is correct: Because the model is fitted to the market, its parameters embody the market's implied assumptions; understanding the assumption-to-shape link is how one interprets and sanity-checks them.

Why the others are wrong

- **A:** The calibrated drift is a blend of expected change and risk premium and cannot be separated from prices alone.
- **B:** A process for the short rate is still required; calibration only fixes the drift so the tree matches prices.
- **D:** Matching prices does not show they are fair: the model may be wrong or the prices distorted (e.g., forced liquidation).

Q676

AnalystPrep Q.1657

REBUILT — Original distractors were vague/weak; rewrote options so each is a realistic misuse of arbitrage-free models while keeping the same tested point.

According to Tuckman, what is the most important use of arbitrage-free interest rate models?

A. Judging whether one security is rich or cheap relative to another

B. Forecasting the expected path of future short-term rates

C. Explaining the slope of the term structure through economic fundamentals

D. Quoting prices of illiquid or non-standard securities, and valuing derivatives, consistently with the prices of liquid securities

Answer: D

Step by step

1. Arbitrage-free models are calibrated so model prices equal observed prices of liquid securities (e.g., on-the-run Treasuries, standard swaps).
2. The calibrated model then prices securities that are not actively traded (e.g., an odd-maturity swap) and derivatives on the underlyings.
3. Hence the use is consistent quoting/market-making, not relative value or forecasting.

Why D is correct: By matching liquid market prices, arbitrage-free models let dealers price illiquid or customized instruments and derivatives consistently with the market.

Why the others are wrong

- **A:** Relative value is inappropriate because the models assume all fitted securities are fairly priced; equilibrium models suit this purpose.
- **B:** The calibrated drift mixes expected change and risk premium, so it is not a forecast of rates.
- **C:** Explaining curve shape via fundamentals is the domain of equilibrium models.

Q677

AnalystPrep Q.1658

REBUILT — Original key (economic and financial reasoning) is doubtful and the question was ambiguous; rebuilt as a clear relative-value question on when to use equilibrium rather than arbitrage-free models.

A portfolio manager wants to judge whether one bond is rich or cheap relative to another bond. According to Tuckman, which model choice is appropriate?

A. An equilibrium model, because arbitrage-free models assume both securities are fairly priced and so cannot reveal relative mispricing

B. An arbitrage-free model, because calibrating to both bond prices makes their values directly comparable

C. An arbitrage-free model, because time-dependent drift lets it fit more prices than an equilibrium model

D. Either type, because both give identical values for liquid securities

Answer: A

Step by step

1. Relative analysis asks whether the market prices of two securities are consistent with each other.
2. An arbitrage-free model is calibrated to take market prices as correct, so the fitted securities show no mispricing by construction.
3. An equilibrium model produces values from its own assumptions and so can disagree with market prices, which is the signal needed.

Why A is correct: Arbitrage-free models assume the securities being compared are properly priced, making relative-value analysis meaningless; equilibrium models are used for that purpose.

Why the others are wrong

- **B:** Calibration builds in the assumption that both prices are right, so comparing them reveals nothing.
- **C:** Better fit is a strength for pricing illiquid securities but is irrelevant to (indeed undermines) relative analysis.
- **D:** Equal values for liquid securities hold only because the arbitrage-free model was forced to fit them; equilibrium values need not match.

Q678

BT answers P2.T5.304.3 · BT notes P2.T5.304.3

Tuckman's short-term interest rate models classify into two categories: equilibrium models and arbitrage-free models. Consider the following statements about no-arbitrage interest rate models:

I. Model 1 (Normally distributed rates and no drift) and Model 2 (Constant drift and risk premium) are classified as arbitrage-free models

II. The Ho-Lee Model, due to time-dependent drift, may be used to match the observed prices of securities and, therefore, may be classified as an arbitrage-free model

III. Important uses of arbitrage-free models include quoting non-actively traded securities (based on prices of liquid securities) and to value derivatives for purposes of market-making or proprietary trading

IV. Because an arbitrage-free model matches market prices does not necessarily imply that it provides fair values and accurate hedges: the model itself could be bad, and/or market prices may not be "fair" in the context of the model

Which of the above statements are TRUE, according to Tuckman?

A. None

B. I. and II. only

C. II., III., and IV.

D. All are true

Answer: C

Step by step

1. I: Models 1 and 2 have a fixed (zero or constant) drift and cannot be fitted to a wide range of prices, so they are equilibrium models: FALSE.
2. II: Ho-Lee's time-dependent drift can be calibrated to observed prices, so it is arbitrage-free: TRUE.
3. III: Arbitrage-free models quote illiquid securities and value derivatives from liquid prices: TRUE.
4. IV: A good fit does not fix a bad model, and market prices may be distorted (supply/demand, liquidation): TRUE.
5. So II, III and IV are true.

Why C is correct: Only Models 1 and 2 are equilibrium models; Ho-Lee is arbitrage-free, and both the uses (III) and the caveats on fitting (IV) are standard Tuckman points.

Why the others are wrong

- **A:** Wrongly treats all four as false; II, III and IV are correct.
- **B:** Accepts statement I (Models 1 and 2 are equilibrium models) and rejects III and IV.

- **D:** Fails to catch that statement I misclassifies Models 1 and 2.

Q679

BT notes P2.T5.23.4.1

A trading unit wants a simple interest rate model for scenario analysis, chosen from among these candidate models in Tuckman: Model 1 (normally distributed rates, no drift), Model 2 (constant drift and risk premium), Ho-Lee (time-dependent drift), and Vasicek (mean reversion). Which of these four is easiest to fit to a wide variety of observed prices?

A. Model 1
B. Model 2
C. Ho-Lee
D. Vasicek

Answer: C

Step by step

1. Fitting to many observed prices requires a free parameter for each maturity of the curve.
2. Model 1 has none (zero drift) and Model 2 has one constant drift, so neither can match a general curve.
3. Ho-Lee's drift can differ every period ($\lambda_1, \lambda_2, \ldots$), so each λ can be solved to match the next market price.
4. Vasicek has constant parameters (k, θ, σ), so it cannot match an arbitrary curve.

Why C is correct: Ho-Lee's time-dependent drift supplies one parameter per period, so it can be calibrated to a wide variety of market prices.

Why the others are wrong

- **A:** No free drift parameter at all, so it cannot be fitted to a general set of prices.
- **B:** A single constant drift fits one slope only, not an arbitrary curve.
- **D:** Mean reversion changes curve shape, but constant parameters cannot match a general set of prices.

Q680

Schweser Module Quiz 14.2 Q3

REBUILT — Statements 1-3 were missing from the OCR; reconstructed three statements on Ho-Lee, Vasicek and arbitrage-free uses with a single defensible count.

A risk analyst makes three statements about term structure models:
Statement 1: The Ho-Lee model is more general than Model 2, since constant drift (and no drift) are special cases of Ho-Lee.
Statement 2: Like Ho-Lee, the Vasicek model, with its constant parameters, can be calibrated to match any observed term structure.
Statement 3: Arbitrage-free models are well suited to relative value analysis of two securities, since both are matched to market prices.
How many of the statements are correct?

A. 0
B. 1
C. 2
D. 3

Answer: B

Step by step

1. Statement 1: setting $\lambda_1 = \lambda_2$ gives Model 2, and $\lambda = 0$ gives Model 1, so Ho-Lee generalizes them: TRUE.
2. Statement 2: Vasicek's constant k, θ and σ do not give a free parameter per maturity, so it cannot match an arbitrary curve; Ho-Lee can: FALSE.
3. Statement 3: arbitrage-free models assume both securities are fairly priced, so relative analysis is meaningless: FALSE.
4. Exactly one statement is correct.

Why B is correct: Only Statement 1 is correct: Ho-Lee nests Models 1 and 2, but Vasicek cannot fit any curve and arbitrage-free models cannot support relative-value analysis.

Why the others are wrong

- **A:** Misses that Statement 1 is true (Models 1 and 2 are special cases of Ho-Lee).
- **C:** Credits either Statement 2 (confusing mean reversion with the flexibility of time-dependent drift) or Statement 3 (ignoring that matching prices assumes they are fair).
- **D:** Accepts both false statements; fitted prices being 'matched' does not make relative value analysis meaningful, and Vasicek has no per-period drift.

LO 14.f

Describe the process of constructing a simple and recombining tree for a short-term rate under the Vasicek Model with mean reversion.

Q681

AnalystPrep Q.6530

In the Vasicek model, a risk premium (λ) is introduced. What is the key implication of this risk premium for the observed term structure of interest rates?

A. The risk premium directly affects the volatility of the short-term rate.

B. The risk premium is indistinguishable from the long-run value of the short-term rate (θ) when observing security prices.

C. The risk premium only affects the short end of the yield curve and has no impact on long-term rates.

D. The risk premium causes the short-term rate to become negatively correlated with the long-run value (θ).

Answer: B

Step by step

1. Under the risk-neutral process the drift is $k(\theta - r) + \lambda$, which equals $k(\theta^* - r)$ with $\theta^* \approx \theta + \lambda/k$.
2. The risk premium therefore enters only as a constant shift of the long-run level used for pricing.
3. Prices depend only on θ^*, so θ and λ cannot be separately identified from security prices.

Why B is correct: The risk premium is absorbed into the risk-neutral long-run rate, so from observed prices it is indistinguishable from θ.

Why the others are wrong

- **A:** Confuses drift with volatility: λ shifts the drift only; σ is unchanged.
- **C:** The shift in the long-run level feeds into the whole term structure, not just the short end.
- **D:** No correlation between the short rate and θ is implied; θ is a constant parameter.

Q682

BT answers P2.T5.305.1 · BT notes P2.T5.305.1

REBUILT — Node label [2,1] was ambiguous because the Vasicek tree does not recombine (two distinct middle nodes) and the source used a drift-only shortcut; reframed to ask for the recombined middle node (average of the up-down and down-up nodes) with plausible distractors.

Analyst Steven builds a monthly (dt = 1/12) short-rate tree under the Vasicek model $dr = k(\theta - r)dt + \sigma dw$ with r(0) = 8.00%, θ = 3.00%, k = 0.60 and annual volatility σ = 2.00% (200 bps). After the first month the tree has r(u) = 8.3274% and r(d) = 7.1726%. The unrecombined tree is then made recombining by averaging the two middle nodes of month 2 (up-down and down-up). What is the value of the recombined middle node at month 2?

A. 7.484%

B. 7.513%

C. 7.542%

D. 7.750%

Answer: B

Step by step

1. Monthly shock: $\sigma\sqrt{dt} = 2\% \times \sqrt{1/12} = 0.5774\%$.
2. Up-down node: from 8.3274%, drift = 0.60(3 - 8.3274)/12 = -0.2664%, so 8.3274 - 0.2664 - 0.5774 = 7.4836%.
3. Down-up node: from 7.1726%, drift = 0.60(3 - 7.1726)/12 = -0.2086%, so 7.1726 - 0.2086 + 0.5774 = 7.5414%.
4. The two nodes differ (7.4836% vs 7.5414%), so the tree does not recombine; average them: (7.4836 + 7.5414)/2 = 7.5125%, about 7.513%.

Why B is correct: Mean reversion pulls the lower node up more than the upper node, so the middle nodes differ; recombining averages them to 7.5125%.

Why the others are wrong

- **A:** Used only the up-down node (7.484%) and did not average.
- **C:** Used only the down-up node (7.541%) and did not average.
- **D:** Applied one period of mean-reverting drift only (8 - 0.25 = 7.75%), i.e., stopped after the first step.

Q683

Mock 1 Q40

REBUILT — Distractors were evenly spaced offsets of the answer (giveaway pattern); replaced with results of realistic formula errors.

A risk manager is reviewing short-term interest rate assumptions by applying the Vasicek model. The first task is to estimate the long-run value of the short-term interest rate. The current interest rate is 5% and you estimate the true long-term interest rate is 5.5%, annual drift (risk premium) of 0.32%, annual volatility of 1.8%, dt = 1/12, and a mean reversion adjustment (k) of 0.04. What is the long-run value of the short-term rate, assuming risk neutrality?

A. 5.82%
B. 5.51%
C. 13.50%
D. -2.50%

Answer: C

Step by step

1. Risk-neutral long-run value: $\theta \approx r_\infty + \lambda/k$.
2. $\lambda/k = 0.32\%/0.04 = 8.0\%$.
3. $\theta = 5.5\% + 8.0\% = 13.5\%$.
4. The current rate 5%, volatility 1.8% and dt are not needed.

Why C is correct: The risk premium is capitalized into the long-run level by dividing by the reversion speed: $\theta = r_\infty + \lambda/k$.

Why the others are wrong

- **A:** Added λ directly (5.5 + 0.32) without dividing by k.
- **B:** Multiplied instead of divided: 5.5 + 0.32 x 0.04 = 5.51%.
- **D:** Subtracted the risk premium term: 5.5 - 8.0 = -2.5%.

Q684

Mock 2 Q67

REBUILT — Distractors were not traceable to realistic errors; options rebuilt from common formula mistakes keeping the same data.

Using the Vasicek model, assume a current short-term rate of 5.2% and an annual volatility of the interest rate process of 3.5%. Also assume that the long-run mean-reverting level is 14.2% with a speed of adjustment of 0.4. Within a binomial interest rate tree, what are the upper and lower node rates after the first month?

A. Upper 9.81%; lower 7.79%
B. Upper 5.91%; lower 3.89%
C. Upper 6.51%; lower 4.49%
D. Upper 9.00%; lower 2.00%

Answer: C

Step by step

1. Drift: $k(\theta - r_0)dt = 0.4 \times (14.2\% - 5.2\%) \times 1/12 = 0.30\%$.
2. Volatility: $\sigma\sqrt{dt} = 3.5\% \times \sqrt{1/12} = 1.0104\%$.
3. Upper = 5.2 + 0.30 + 1.0104 = 6.51%.
4. Lower = 5.2 + 0.30 - 1.0104 = 4.49%.

Why C is correct: Each node is r0 + monthly mean-reversion drift, plus or minus one monthly standard deviation.

Why the others are wrong

- **A:** Forgot to scale the drift by dt (3.6% instead of 0.30%): 5.2 + 3.6 ± 1.01.
- **B:** Applied the drift with the wrong sign (moved away from the mean): 5.2 - 0.30 ± 1.01.
- **D:** Used the full annual volatility 3.5% instead of $\sigma\sqrt{dt}$: 5.2 + 0.30 ± 3.5 (rounded).

Q685

Mock 3 Q26

A risk manager at a fixed-income hedge fund is evaluating methods to improve the fund's ability to model interest rate term structures. The manager would like to adopt a model that is flexible enough to incorporate both mean reversion and a risk premium and considers the Vasicek model for this purpose. Which of the following statements is correct about the Vasicek model?

A. The model incorporates mean reversion and a drift of zero.
B. The model incorporates mean reversion and models the risk premium as a component of the drift.
C. The model incorporates a drift of zero, but it cannot incorporate a risk premium.
D. The model can be used to model the time-varying risk premium, but it cannot incorporate mean reversion.

Answer: B

Step by step

1. Vasicek: $dr = k(\theta - r)dt + \sigma dw$; the drift $k(\theta - r)$ is non-zero and produces mean reversion.
2. The drift combines the expected rate change and a risk premium (λ), which shifts the risk-neutral long-run level $\theta \approx r_\infty + \lambda/k$.
3. So mean reversion and a risk premium are both embedded in the drift.

Why B is correct: Vasicek's drift is the mean-reverting term, and the risk premium enters as part of that drift.

Why the others are wrong

- **A:** Mean reversion requires a non-zero drift; zero drift describes Model 1.
- **C:** Zero drift is Model 1, and Vasicek does include a risk premium within the drift.
- **D:** Vasicek explicitly includes mean reversion (the defining feature) and uses a constant, not time-varying, premium.

Q686

Schweser Module Quiz 14.2 Q2

REBUILT — Original was an open item with answer choices and worked numbers lost to OCR; converted to a 4-option MCQ with recomputed nodes and error-based distractors.

Using the Vasicek model, assume a current short-term rate of 6.2% and an annual volatility of the interest rate process of 2.5%. Also assume that the long-run mean-reverting level is 14.2% with a speed of adjustment of 0.4. Within a binomial interest rate tree, what are the upper and lower node rates after the first month?

A. Upper 10.12%; lower 8.68%

B. Upper 8.97%; lower 3.97%

C. Upper 6.66%; lower 5.21%

D. Upper 7.19%; lower 5.74%

Answer: D

Step by step

1. Drift: $0.4 \times (14.2\% - 6.2\%) \times 1/12 = 0.2667\%$.
2. Volatility: $2.5\% \times \sqrt{1/12} = 0.7217\%$.
3. Upper = 6.2 + 0.2667 + 0.7217 = 7.19%.
4. Lower = 6.2 + 0.2667 - 0.7217 = 5.74%.

Why D is correct: Nodes are the current rate plus the one-month mean-reversion drift, plus or minus one monthly standard deviation.

Why the others are wrong

- **A:** Did not scale drift by dt (3.2% instead of 0.267%): 6.2 + 3.2 ± 0.72.
- **B:** Used the annual volatility 2.5% rather than $\sigma\sqrt{dt}$: 6.2 + 0.267 ± 2.5.
- **C:** Drift sign reversed (6.2 - 0.267 ± 0.72), moving away from the long-run level.

Q687

BT notes P2.T5.305.3

Each of the following statements is true about the Vasicek Model EXCEPT:

A. The Vasicek Model is an equilibrium model like the Cox-Ingersoll-Ross (CIR) model and unlike the Ho-Lee model, which is an arbitrage-free (aka, no-arbitrage) model

B. In the Vasicek Model, half-life is given by LN(2)/k and can be interpreted as the time it takes for the short-rate (the factor) to progress half the distance toward the long-run value (Theta)

C. The mean reversion in Vasicek dampens volatility relative to the same process without mean reversion, particularly at long horizons

D. Due to the mean-reversion, it is impossible to develop a rate tree that recombines under Vasicek

Answer: D

Step by step

1. The unadjusted Vasicek tree does not recombine because the reversion adjustment differs after up and down moves.
2. It can be made recombining by averaging the two middle nodes and solving for new probabilities (p, q) and outer nodes, so D is false.
3. A is true (equilibrium vs arbitrage-free), B is true (half-life = ln2/k), C is true (volatility term structure slopes downward).

Why D is correct: A recombining approximation of the Vasicek tree exists (average the middle nodes and re-solve

probabilities), so 'impossible' is false.

Why the others are wrong

- **A:** True statement, so not the exception: Vasicek and CIR are equilibrium models; Ho-Lee is arbitrage-free.
- **B:** True statement: half-life = ln(2)/k.
- **C:** True statement: mean reversion produces a downward-sloping volatility term structure.

LO 14.g

Calculate the Vasicek Model rate change, standard deviation of the rate change, expected rate in T years, and half-life.

Q688

AnalystPrep Q.1664

REBUILT — Units (years) were only implied and the weighted-average distractors were weak; stated units and added the implied k.

In a Vasicek model, the half-life of the short rate is 28.72 years. Which statement is correct?

A. The rate takes 28.72 years to cover half of the distance between its current value and its long-run mean-reverting level, which implies a speed of reversion $k \approx 0.0241$.

B. The rate takes 28.72 years to reach its long-run mean-reverting level, which implies $k \approx 0.0348$.

C. The rate takes 57.44 years to cover half of the distance to its long-run level, which implies $k \approx 0.0121$.

D. The rate takes 14.36 years to cover half of the distance to its long-run level, which implies $k \approx 0.0483$.

Answer: A

Step by step

1. Half-life is the time for the gap between the rate and the mean-reverting level to shrink by half: $\tau = \ln 2/k$.
2. So $k = \ln 2/28.72 = 0.6931/28.72 = 0.02413$.
3. The gap decays as e^{-kt}, so after 28.72 years it is 50% of its initial size (not zero).

Why A is correct: Half-life is the time to close half the distance to the mean-reverting level, with k = ln2/τ.

Why the others are wrong

- **B:** Treats half-life as the time to reach the mean (k = 1/28.72); the gap only halves in that time.
- **C:** Doubles the half-life; k = ln2/57.44 would be half the correct value.
- **D:** Halves the half-life; k = ln2/14.36 would be twice the correct value.

Q689

AnalystPrep Q.1665

REBUILT — Distractors were arbitrary near-misses of the key; replaced with results of realistic formula errors.

In a Vasicek model, the speed of mean reversion is $k = 0.025$ and the annual volatility is 126 basis points. The short rate in 10 years is normally distributed with an expected value of 7.4812%. What is the standard deviation of the short rate in 10 years?

A. 265 basis points

B. 353 basis points

C. 398 basis points

D. 500 basis points

Answer: B

Step by step

1. Vasicek terminal standard deviation: $\sigma_T = \sqrt{\frac{\sigma^2}{2k}(1 - e^{-2kT})}$.
2. $2k = 0.05$; $e^{-2kT} = e^{-0.5} = 0.6065$, so $1 - e^{-0.5} = 0.3935$.
3. $\sigma^2/(2k) = (1.26\%)^2/0.05 = 31.752$ (in $\%^2$).
4. $\sigma_T = \sqrt{31.752 \times 0.3935} = \sqrt{12.493} = 3.535\% \approx 353$ bp. The expected value 7.4812% is not needed.

Why B is correct: Mean reversion caps the variance of the terminal rate at $\sigma^2/2k$, so the SD is below the no-reversion value $\sigma\sqrt{T}$.

Why the others are wrong

- **A:** Used $1 - e^{-kT}$ instead of $1 - e^{-2kT}$: $\sqrt{31.752 \times 0.2212} = 2.65\%$.
- **C:** Ignored mean reversion: $\sigma\sqrt{T} = 1.26\% \times \sqrt{10} = 3.98\%$.
- **D:** Used σ^2/k instead of $\sigma^2/2k$, doubling the variance: $\sqrt{63.5 \times 0.3935} = 5.00\%$.

Q690

AnalystPrep Q.2661

REBUILT — Stem mixed ambiguous inputs (current rate, 'drift'); restated which inputs define the mean-reverting level and replaced weak options.

In a Vasicek model, the long-run true interest rate is $r_\infty = 2\%$ and the risk-premium drift term is $\lambda = 0.1\%$ per year. The mean-reverting level $\theta = r_\infty + \lambda/k$ is 4%. The current short rate is 1.5%. What is the speed of mean reversion, k?

A. 0.025
B. 0.067
C. 0.05
D. 0.04

Answer: C

Step by step

1. Schweser convention: $\theta = r_\infty + \lambda/k$.
2. $4\% = 2\% + 0.1\%/k \Rightarrow \lambda/k = 2\%$.
3. $k = 0.1\%/2\% = 0.05$. The current rate (1.5%) is not needed.

Why C is correct: The gap between θ and r∞ is exactly λ/k, so k = λ/(θ − r∞).

Why the others are wrong

- **A:** Divided λ by θ (0.1%/4%) instead of by the gap θ − r∞.
- **B:** Used the current rate in the denominator (0.1%/1.5%).
- **D:** Reported θ = 4% as a decimal, confusing the mean-reverting level with the speed parameter.

Q691

AnalystPrep Q.2856

REBUILT — Source asked for 'expected change' but its key included the random shock σdw (not an expectation); rebuilt to ask for the true expected change and the one-month standard deviation.

A Vasicek model is $dr = k\,([r_\infty + \lambda/k] - r)\,dt + \sigma\,dw$. The speed of mean reversion is $k = 0.42$, the true long-run rate is $r_\infty = 5.286\%$, the risk-premium term is $\lambda = 0.256\%$, the current short rate is 4.256% and $\sigma = 2.07\%$ per year. Over the next month ($dt = 1/12$), what are the expected change in the short rate and the standard deviation of the change?

A. 5.74 bp and 59.76 bp
B. 38.87 bp and 59.76 bp
C. 5.74 bp and 207.00 bp
D. 68.86 bp and 59.76 bp

Answer: A

Key correction: Source key (D: 38.87 and 59.76) included the random shock in the 'expected' change; the expectation excludes dw, so the expected change is 5.74 bp.

Step by step

1. Mean-reverting level: $\theta = 5.286\% + 0.256\%/0.42 = 5.8955\%$.
2. Expected change $= k(\theta - r)\,dt = 0.42 \times (5.8955\% - 4.256\%) \times 1/12 = 0.05738\% = 5.74$ bp.
3. Standard deviation $= \sigma\sqrt{dt} = 207 \times \sqrt{1/12} = 59.76$ bp.

Why A is correct: The expected change is the drift term only (E[dw]=0), and the volatility must be scaled by √dt.

Why the others are wrong

- **B:** 38.87 bp adds the realised shock $\sigma\,dw = 2.07\% \times 0.16$ (33.12 bp) to the drift; that is an actual change, not an expected one.
- **C:** Correct drift but used the annual volatility of 207 bp without scaling by $\sqrt{dt}$.
- **D:** Forgot to multiply the drift by $dt = 1/12$ (annual drift 68.86 bp).

Q692

AnalystPrep Q.2857

REBUILT — Options were arbitrary and the source working mixed 197 bp and 0.0197; replaced options with outcomes of realistic errors.

In a Vasicek model, the speed of mean reversion is 0.37 and the annual volatility is 197 basis points. The short rate is normally distributed. What is the standard deviation of the short rate after 5 years?

A. 0.0441
B. 0.0226
C. 0.0210
D. 0.0320

Answer: B

Step by step

1. $\sigma_T = \sqrt{\frac{\sigma^2}{2k}(1 - e^{-2kT})}$ with $\sigma = 0.0197, k = 0.37, T = 5$.
2. $2kT = 3.7; 1 - e^{-3.7} = 0.97530$.
3. $\sigma^2/2k = 0.00038809/0.74 = 0.00052445$.
4. $\sigma_T = \sqrt{0.00052445 \times 0.97530} = 0.02262$.

Why B is correct: With strong mean reversion the terminal SD approaches σ/√(2k) ≈ 2.29%, well under the no-reversion σ√T.

Why the others are wrong

- **A:** Ignored mean reversion: $0.0197\sqrt{5} = 0.0441$.
- **C:** Used $1 - e^{-kT}$ in place of $1 - e^{-2kT}$: $\sqrt{0.00052445 \times 0.8431} = 0.0210$.
- **D:** Used σ^2/k instead of $\sigma^2/2k$: $\sqrt{0.0010489 \times 0.9753} = 0.0320$.

Q693

AnalystPrep Q.4018

REBUILT — Option A (4.55%) matched no realistic error; replaced with the result of omitting the λ/k adjustment.

Consider a Vasicek model with speed of mean reversion $k = 0.03$, annual volatility of 200 basis points, a true long-run interest rate $r_\infty = 6\%$, a current short rate of 5.0% and an annual drift (risk-premium) term $\lambda = 0.35\%$. What is the expected short rate after 10 years?

A. 3.70%
B. 5.26%
C. 8.28%
D. 11.67%

Answer: C

Step by step

1. $\theta = r_\infty + \lambda/k = 6\% + 0.35\%/0.03 = 17.667\%$.
2. $e^{-kT} = e^{-0.3} = 0.74082$.
3. $E[r_T] = r_0 e^{-kT} + \theta(1 - e^{-kT}) = 5\%(0.74082) + 17.667\%(0.25918) = 3.704\% + 4.579\% = 8.28\%$.

Why C is correct: The expected rate is a weighted average of the current rate and θ (including λ/k), with weight e^{−kT} on the current rate.

Why the others are wrong

- **A:** Kept only the first term $r_0 e^{-kT}$, omitting the $\theta(1 - e^{-kT})$ term.
- **B:** Used $\theta = 6\%$, ignoring the λ/k adjustment: $3.704\% + 6\% \times 0.25918 = 5.26\%$.
- **D:** Reported $\lambda/k = 11.67\%$ (the adjustment alone) rather than the expected rate.

Q694

AnalystPrep Q.4019

REBUILT — Options 5.8/26.0/14.5 corresponded to no identifiable error; replaced with results of realistic formula mistakes.

Consider a Vasicek model with speed of mean reversion $k = 0.05$, annual volatility of 150 basis points, a true long-run interest rate of 6%, a current short rate of 5.0% and an annual drift term of 0.4%. What is the half-life of the model (in years)?

A. 6.93
B. 10.00
C. 13.86
D. 20.00

Answer: C

Step by step

1. Half-life $\tau = \ln 2/k$.
2. $\tau = 0.6931/0.05 = 13.86$ years. Volatility, rates and drift are irrelevant.

Why C is correct: Half-life depends only on k: the gap to θ decays as e^{−kt} and equals 50% at t = ln2/k.

Why the others are wrong

- **A:** Divided by 2k: $\ln 2/(2 \times 0.05) = 6.93$.
- **B:** Used $0.5/k$, treating the decay as linear.
- **D:** Used $1/k = 20$, the time for the gap to fall to 37% (e^{-1}), not 50%.

Q695 BT answers P2.T5.305.2 · BT notes P2.T5.305.2

Analyst Susan is employing the Vasicek model: $dr = k(\theta - r)\,dt + \sigma\,dw$. Her version is $dr = 0.40(8.00\% - r)\,dt + (3.0\%)\,dw$. The current short rate is $r(0) = 2.00\%$. Assumptions: time step one month ($dt = 1/12$); $\theta = 8.00\%$; $k = 0.40$; annual volatility 300 basis points. For the first month, the simulation uses a uniform random value of 45%, which transforms (inverse normal) into a standard normal of -0.13164. Including time scaling, $dw = -0.13164 \times \sqrt{1/12} = -0.0380$. What are the rate in the first month, $r(1/12)$, and the half-life of the process?

A. r(1/12) = 2.0860% and half-life is ~1.7 years
B. r(1/12) = 3.1920% and half-life is ~2.5 years
C. r(1/12) = 4.3333% and half-life is ~7.4 years
D. r(1/12) = 5.2136% and half-life is ~15.5 years

Answer: A

Step by step

1. Drift: $k(\theta - r)\,dt = 0.40 \times (8\% - 2\%) \times 1/12 = 0.20\%$.
2. Shock: $\sigma\,dw = 3.0\% \times (-0.0380) = -0.114\%$.
3. $dr = 0.20\% - 0.114\% = 0.086\%$, so $r(1/12) = 2.0860\%$.
4. Half-life $= \ln 2/k = 0.6931/0.40 = 1.73$ years.

Why A is correct: The rate rises by the drift less the small negative shock, and half-life is ln2/k.

Why the others are wrong

- **B:** Rate move of +1.19% is not supported by the data; half-life 2.5 years would imply k = 0.277, not 0.40.
- **C:** Rate move of +2.33% is far larger than drift (0.20%) plus shock; half-life 7.4 years would imply k = 0.094, not 0.40.
- **D:** Rate move of +3.21% is not supported by the data; half-life 15.5 years would imply k = 0.045, not 0.40.

Q696 GARP PE2 Q46 · Mock 4 Q47

A quantitative analyst on the fixed-income desk of an investment bank is applying the Vasicek model to estimate future short-term interest rates. The model is given below: $dr = k(\theta - r)\,dt + \sigma\,dw$, where dr is the change in the short-term interest rate, Î¸ is the estimated long-run value of the short-term interest rate, k is the mean reversion rate, r is the current level of the short-term interest rate, Ïƒ is the annual basis-point volatility of the short-term interest rate, dt is the time interval measured in years, and dw is a normally distributed random variable with mean zero and standard deviation equal to the square root of dt. The analyst gathers the following information:
Current short-term interest rate (r): 3.35%
Long-run value of short-term interest rate (Î¸): 4.55%
Mean reversion rate (k): 0.06
Annual basis-point volatility (Ïƒ): 120 bps
The analyst then creates an interest rate tree, determines the expected short-term interest rate after 8 years, and calculates how long it will take the short-term interest rate to revert halfway to the long-run value. Which of the following statements would be correct for the analyst to make?

A. The expected short-term interest rate is 3.81% and the half-life is 11.6 years.
B. The expected short-term interest rate is 3.81% and the half-life is 16.7 years.
C. The expected short-term interest rate is 4.09% and the half-life is 11.6 years.
D. The expected short-term interest rate is 4.09% and the half-life is 16.7 years.

Answer: A

Step by step

1. Decay factor: $e^{-kT} = e^{-0.06\times 8} = e^{-0.48} = 0.6188$.
2. Expected rate: $E[r_8] = \theta + (r_0 - \theta)e^{-kT} = 4.55\% + (3.35\% - 4.55\%)(0.6188) = 3.807\% \approx 3.81\%$. Equivalently $0.6188 \times 3.35\% + 0.3812 \times 4.55\%$.
3. Half-life: $\tau = \ln 2/k = 0.6931/0.06 = 11.55 \approx 11.6$ years.
4. The 120 bp volatility does not affect the expected rate.

Why A is correct: The expected rate is a weighted average of the current rate (weight e^{-kT}) and the long-run value (weight $1 - e^{-kT}$); the half-life is the time for the gap to halve, $\ln 2/k$.

Why the others are wrong

- **B:** Right rate, but 16.7 = 1/k is the time constant (about 63% of the gap closed), not the half-life; the half-life is ln2/k.

- **C:** Half-life is right, but the weights were swapped (0.6188 on Î¸), giving 4.09%, which overstates reversion over 8 years (only about 38% of the gap closes).
- **D:** Both errors: weights swapped (4.09%) and 1/k used instead of ln2/k (16.7).

Q697

AnalystPrep Q.1662

REBUILT — Original options were giveaways (swapped values, an odd 35.04/1.54 pair) and the shock dw was ambiguous; made the shock explicit and replaced distractors with realistic mistakes.

In a risk-neutral Vasicek model, $dr = k(\theta - r)\,dt + \sigma\,dw$ with $\theta = r_\infty + \lambda/k$. Assume k = 0.020, Ïƒ = 125 basis points per year, râˆž = 6.180%, Î» = 0.227% per year, current short rate r = 5.212%, a one-month interval (dt = 1/12), and a realized random shock of dw = 0.14 (use it directly in ÏƒÂ·dw). What are the change in the short rate over the month and the standard deviation of the monthly rate change?

A. Change of 17.66 bp; standard deviation of 36.08 bp
B. Change of 19.55 bp; standard deviation of 125.00 bp
C. Change of 19.55 bp; standard deviation of 36.08 bp
D. Change of 2.05 bp; standard deviation of 36.08 bp

Answer: C

Step by step

1. Long-run risk-neutral rate: $\theta = 6.180\% + 0.227\%/0.020 = 17.53\%$.
2. Drift: $k(\theta - r)dt = 0.020 \times (17.53\% - 5.212\%) \times (1/12) = 0.02053\% = 2.05$ bp.
3. Shock: $\sigma\,dw = 125 \times 0.14 = 17.50$ bp.
4. Rate change: $2.05 + 17.50 = 19.55$ bp.
5. Monthly standard deviation: $\sigma\sqrt{dt} = 125\sqrt{1/12} = 36.08$ bp.

Why C is correct: Vasicek's rate change is mean-reverting drift toward Î¸ (which includes the risk premium Î»/k) plus the shock; the standard deviation scales annual volatility by the square root of the time step.

Why the others are wrong

- **A:** Used râˆž = 6.18% instead of Î¸ = 17.53% in the drift (ignored the Î»/k risk-premium term), giving 0.17 + 17.50 = 17.66 bp.
- **B:** Change is right, but the annual 125 bp was quoted as the monthly standard deviation; it must be scaled by âˆš(1/12).
- **D:** Reported only the drift term (2.05 bp) and omitted ÏƒÂ·dw.

LO 14.h

Describe the effectiveness of the Vasicek Model.

Q698

AnalystPrep Q.4017

REBUILT — Original option D was a non-parallel giveaway and omitted the short-lived/high-k pairing; options rebuilt as the full 2x2 of lifespan versus mean reversion speed.

The U.S. Department of Transport has just announced an unexpected technological breakthrough that will have a major bearing on the development of self-driving autonomous vehicles. What is the most likely anticipated impact on a mean-reverting model of interest rates?

A. The economic information is long-lived with a low mean reversion parameter.
B. The economic information is short-lived with a low mean reversion parameter.
C. The economic information is long-lived with a high mean reversion parameter.
D. The economic information is short-lived with a high mean reversion parameter.

Answer: A

Step by step

1. A major technological breakthrough changes economic fundamentals, so the news is long-lived (persistent).
2. In the Vasicek model $dr = k(\theta - r)dt + \sigma dw$, k is the speed of reversion toward the long-run level.
3. Persistent shocks are expected to die out slowly, so the pull back to the long-run mean is weak: low k.
4. Long-lived information therefore pairs with a low mean reversion parameter.

Why A is correct: Long-lived economic shocks are associated with a low mean reversion parameter (slow decay),

whereas short-lived shocks are associated with a high one.

Why the others are wrong

- **B:** Pairs low k with short-lived news; a low k means slow decay, which contradicts news that fades quickly.
- **C:** Correct lifespan but wrong k: a high k implies rapid reversion, inconsistent with persistent information.
- **D:** Describes short-lived information; a technological breakthrough is structural, not transitory.

Q699

Schweser Module Quiz 14.2 Q1

The Bureau of Labor Statistics has just reported an unexpected short-term increase in high-priced luxury automobiles. What is the most likely anticipated impact on a mean-reverting model of interest rates?

A. The economic information is long-lived with a low mean reversion parameter.
B. The economic information is short-lived with a low mean reversion parameter.
C. The economic information is long-lived with a high mean reversion parameter.
D. The economic information is short-lived with a high mean reversion parameter.

Answer: D

Step by step

1. A temporary jump in one narrow sector (luxury cars) does not alter long-run fundamentals, so the information is short-lived.
2. Short-lived shocks should decay quickly toward the long-run rate θ.
3. Fast decay means a large reversion adjustment per period, i.e., a high k.
4. Short-lived information pairs with a high mean reversion parameter.

Why D is correct: Transitory news implies the rate is quickly pulled back to its long-run level, which requires a high speed-of-reversion parameter k.

Why the others are wrong

- **A:** Treats a temporary sector blip as persistent news and pairs it with slow reversion.
- **B:** Correct lifespan but wrong k: low k means slow reversion, which would make the shock last.
- **C:** Misreads the news as long-lived and then pairs it with fast reversion, which is internally inconsistent.

Q700

BT notes P2.T5.23.4.3

REBUILT — Option C contained an OCR-damaged symbol and a garbled claim, and the LO fits 14.h (model effectiveness); replaced with clear, unambiguous false statements.

Which of the following is the best interpretation of a Vasicek model with a high mean reversion parameter k?

A. Economic news affecting the short rate is short-lived
B. The half-life of a deviation from the long-run rate is long
C. A shock to the short rate has a larger effect on long-maturity rates than it would with a low k
D. The standard deviation of the short rate's terminal distribution in T years exceeds $\sigma\sqrt{T}$

Answer: A

Step by step

1. Half-life $= \ln 2/k$: a high k gives a short half-life, so deviations (shocks) decay quickly, i.e., news is short-lived (A true, B false).
2. Because shocks fade as the rate reverts, they affect long maturities less than short ones; a higher k strengthens this (C false).
3. Mean reversion pulls the rate back, so terminal variance is below the no-reversion value $\sigma^2 T$; the standard deviation is less than $\sigma\sqrt{T}$ (D false).

Why A is correct: A large k makes deviations from the long-run rate decay quickly (short half-life), so economic shocks are short-lived and long-term rates barely move.

Why the others are wrong

- **B:** Reverses the relationship: half-life ln2/k falls as k rises.
- **C:** Reverses the effect: higher k dampens the impact of shocks on long rates; the model does not imply parallel shifts.
- **D:** Mean reversion reduces dispersion; the terminal standard deviation is below Ïƒâˆšt (it equals Ïƒâˆšt only with k = 0).

Q701 AnalystPrep Q.1668

An analyst graphs the effect of a 10-basis-point increase in the short-term rate factor on the spot rate curve under the Vasicek model, which includes mean reversion. What is the effect on short-term and long-term spot rates?

A. Short-term rates decrease by about 10 bp, while longer-term rates are less affected.
B. Short-term rates increase by about 10 bp, while longer-term rates are affected by less than 10 bp.
C. Short-term rates increase by about 10 bp, while longer-term rates are affected by more than 10 bp.
D. Short-term and long-term rates all increase by 10 bp because of the increase in volatility.

Answer: B

Step by step

1. The shock is applied to the short-term rate, so short rates rise by about 10 bp.
2. With mean reversion ($k > 0$) the shocked rate is pulled back toward the long-run mean, so expected future short rates, which drive long-term spot rates, rise by less than 10 bp.
3. Hence the curve does not shift in parallel: the impact declines with maturity (in the Tuckman example the 30-year rate moves noticeably less than 10 bp).

Why B is correct: Mean reversion makes a short-rate shock dissipate over time, so long-term rates respond less than short-term rates and the shift is not parallel.

Why the others are wrong

- **A:** Wrong direction; a positive factor shock raises short rates. Only the 'longer-term less affected' part is right.
- **C:** Reverses the mean-reversion effect; Vasicek never amplifies a short-rate shock at longer maturities.
- **D:** Describes the parallel shift of Model 1 (no drift), and wrongly attributes the move to volatility rather than the factor change.

Q702 AnalystPrep Q.1669

REBUILT — Original stem was garbled and option C vague; reworded the stem and options so only the long-lived versus short-lived news distinction (and its link to k) is tested.

In a mean-reverting short-rate model, economic news is classified as long-lived or short-lived. Which statement is correct?

A. News is short-lived if it changes the market's view of the economy many years in the future, and long-lived if it changes the view only in the near future.
B. News is long-lived if it changes the market's view of the economy many years in the future, and short-lived if it changes the view only in the near future.
C. Long-lived news is associated with a high mean reversion parameter, and short-lived news with a low mean reversion parameter.
D. Both types of news are incorporated at the same speed because the mean reversion parameter is independent of the type of news.

Answer: B

Step by step

1. Long-lived news alters expectations of the economy far into the future, so the shock persists in the rate (low k).
2. Short-lived news affects only the near term, so the rate reverts quickly to its long-run mean (high k).
3. Only B states the definition correctly.

Why B is correct: Persistence of a shock defines the news type: long-horizon impact is long-lived (low k); near-term impact is short-lived (high k).

Why the others are wrong

- **A:** Reverses the definitions of long-lived and short-lived.
- **C:** Reverses the link to k: a high k means fast reversion, i.e. short-lived news.
- **D:** Ignores that a larger k dissipates news more quickly and a smaller k more slowly.

Q703 BT Mock Q16 · BT notes P2.T5.23.4.2

An analyst fits a Vasicek model when the current short-term rate is 4.60% and the long-run mean-reversion level, θ, is 13.50%. Each of the following statements is true EXCEPT which is false?

A. A key advantage of the Vasicek model is that it is a relatively simple model.
B. If the half-life is nine years, then we might expect the short rate to reach about 9.0% in nine years or so.
C. The standard deviation of the short rate in ten years is less than it would be without mean reversion.

D. Because it is a multi-factor model, the Vasicek model is popularly applied to hedge portfolios that contain short-, medium- and long-term bonds.

Answer: D

Step by step

1. Vasicek is a one-factor model (only the short rate is random), so D's premise is false.
2. Check B: after one half-life the gap to θ halves: 4.60% + 0.5 × (13.50% − 4.60%) = 9.05%, about 9.0% ($k = \ln 2/9 \approx 0.077$); true.
3. Check C: mean reversion bounds the variance, whereas without it variance grows linearly with time; true.
4. A is true: few parameters and closed-form prices.

Why D is correct: Vasicek is single-factor, so one source of risk drives all rates and it is not suited to hedging across short, medium and long maturities.

Why the others are wrong

- **A:** True statement (simplicity is an advantage), so not the exception.
- **B:** True: one half-life moves the rate halfway from 4.60% to 13.50%, i.e. 9.05%.
- **C:** True: mean reversion damps long-horizon variance relative to the no-drift model.

Q704

AnalystPrep Q.5301

REBUILT — Original option B (normal rates, constant mean reversion) is also a true Vasicek assumption, giving two defensible answers; rebuilt around the LO 14.h effectiveness points (volatility term structure, non-parallel shifts).

Which of the following best describes the effectiveness of the Vasicek model relative to a normal model with no drift (Model 1)?

A. It produces a flat term structure of volatility and parallel shifts from short-rate shocks, as in Model 1.

B. It produces a declining term structure of volatility, overstating short-term and understating long-term volatility, and a short-rate shock does not shift the curve in parallel.

C. It produces an increasing term structure of volatility, because long-term rates accumulate more shocks, and short-rate shocks move all rates equally.

D. It produces a declining term structure of volatility, but a short-rate shock still moves all rates equally.

Answer: B

Step by step

1. Model 1 (no drift) gives flat volatility across maturities and parallel shifts.
2. In Vasicek, mean reversion damps the effect of a short-rate shock at longer horizons, so the volatility term structure declines (short-term vol overstated, long-term understated).
3. For the same reason the effect of a shock falls with maturity, so shifts are not parallel. Only B has both features.

Why B is correct: Mean reversion makes shocks dissipate, giving declining volatility across maturities and non-parallel shifts, unlike Model 1.

Why the others are wrong

- **A:** Describes Model 1, not Vasicek.
- **C:** Gets the volatility slope backwards and ignores the damping of shocks at longer maturities.
- **D:** Right on declining volatility but self-contradictory: if long rates are less affected by shocks, shocks cannot move all rates equally.

Reading 15. The Art of Term Structure Models: Volatility and Distribution

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 3rd Edition (Hoboken, NJ: John Wiley & Sons, 2011). Chapter 10.

LO	Learning objective	Qs
15.a	describe the short-term rate process under a model with time-dependent volatility.	2
15.b	calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility.	4
15.c	assess the efficacy of time-dependent volatility models.	5
15.d	describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models.	5
15.e	calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models.	10
15.f	describe lognormal models with deterministic drift and mean reversion.	6

LO 15.a

Describe the short-term rate process under a model with time-dependent volatility.

Q705

AnalystPrep Q.1670

REBUILT — Original distractors were weak (circular 'volatility of Ïƒ', unrelated 'security prices'); rewrote the stem and options so each wrong option reflects a realistic confusion among Model 1, Ho-Lee, Model 3 and CIR/lognormal.

A term structure model specifies the short-term rate process as $dr = \lambda(t)\,dt + \sigma(t)\,dw$. On which factor does the volatility of the short rate depend in this model?

A. The current level of the short-term rate, as in the CIR model
B. Time only, through the function $\sigma(t)$
C. Neither the rate level nor time; volatility is a constant as in the Ho-Lee model
D. The drift $\lambda(t)$, since both are functions of time

Answer: B

Step by step

1. The diffusion term is $\sigma(t)\,dw$; the coefficient $\sigma(t)$ is a deterministic function of calendar time t.
2. Nothing in $\sigma(t)$ involves r, so basis-point volatility does not change with the rate level (that would require $\sigma\sqrt{r}$ as in CIR or σr as in lognormal models).
3. The Ho-Lee model has σ constant, so letting σ vary with t is what distinguishes the time-dependent volatility model (e.g. Model 3: $\sigma e^{-\alpha t}$).
4. Hence volatility depends on time only.

Why B is correct: The volatility coefficient $\sigma(t)$ is a deterministic function of time and contains no term in r, so the short-rate volatility varies only with time.

Why the others are wrong

- **A:** Rate-level dependence belongs to the CIR ($\sigma\sqrt{r}$) and lognormal (σr) models; here $\sigma(t)$ has no r in it.
- **C:** Describes Ho-Lee/Model 1 with constant σ; the whole point of $\sigma(t)$ is that volatility changes over time.
- **D:** Both coefficients are functions of t, but they are separate functions; volatility is not driven by the drift.

Q706

BT notes P2.T5.306.3

REBUILT — Original statement V wrongly implied Model 3 abandons normally distributed rates (it keeps them), making 'all true' defensible and the item ambiguous; rewrote V as a clearly false statement and reset options.

Consider the following summary of the interest rate models in Tuckman:
I. Model 1 is a simple equilibrium model with normally distributed rates and no drift.
II. Model 2 adds a constant drift (risk premium) to Model 1 and is also an equilibrium model with normally distributed rates.
III. The Ho-Lee model has normally distributed rates and a time-dependent drift, and can be calibrated so that it is arbitrage-free.
IV. The Vasicek model has normally distributed rates with mean reversion toward a long-run value.
V. Model 3 replaces the normal distribution of rates with a lognormal distribution while keeping volatility constant.
Which of the statements are TRUE?

A. I and II only

B. III and V only
C. I, II, III and IV only
D. All of the statements

Answer: C
Key correction: Source key (BT notes) said D (all true); that relied on a flawed statement V. In the original V, Model 3 was contrasted with models 'which assume normally distributed rates', yet Model 3 is itself normal. V has been rewritten as false.
Step by step
1. I: Model 1, $dr = \sigma\, dw$, has normal rates and no drift. True.
2. II: Model 2, $dr = \lambda\, dt + \sigma\, dw$, has constant drift and normal rates. True.
3. III: Ho-Lee, $dr = \lambda(t)\, dt + \sigma\, dw$, has time-dependent drift fitted to the market curve (arbitrage-free). True.
4. IV: Vasicek, $dr = k(\theta - r)dt + \sigma\, dw$, is normal with mean reversion. True.
5. V: Model 3, $dr = \lambda(t)dt + \sigma e^{-\alpha t} dw$, keeps normal rates and has time-dependent (not constant) volatility, and is not lognormal. False.
6. So I, II, III and IV only.

Why C is correct: Model 3 extends Ho-Lee by making volatility time-dependent while the rate distribution stays normal, so V is false and the other four statements are accurate.
Why the others are wrong
- **A:** Correct statements but incomplete: omits III (Ho-Lee) and IV (Vasicek), which are also true.
- **B:** Includes V, which is false, and omits I, II and IV, which are true.
- **D:** Accepts V as true; Model 3 differs from Ho-Lee by time-dependent volatility, not by a lognormal distribution or constant volatility.

LO 15.b

Calculate the short-term rate change and determine the behavior of the standard deviation of the rate change using a model with time-dependent volatility.

Q707 BT answers P2.T5.306.1 · BT notes P2.T5.306.1

An analyst applies Tuckman's Model 3 for the short-term rate: $dr = \lambda(t)\, dt + \sigma e^{-\alpha t}\, dw$. Assumptions:
- Time step is monthly, $dt = 1/12$ year
- Initial short rate $r(0) = 4.00\%$
- Basis-point volatility parameter $\sigma = 3.00\%$ per year
- Drift $\lambda(t)$ is constant at 1.30% per year
- $\alpha = 0.380$

In the 60th month ($t = 5.0$ years), the random uniform draw is 0.20, so the standard normal draw is NORM.S.INV(0.20) = -0.8416 and $dw = -0.8416 \times \sqrt{1/12} = -0.2430$ (already time-scaled).
Which is nearest to the change in the rate, dr, in the 60th month?
A. -1.00%
B. 0.00%
C. +1.00%
D. +2.00%

Answer: B
Step by step
1. Drift term: $\lambda\, dt = 1.30\% \times 1/12 = 0.1083\%$.
2. Volatility at t = 5: $\sigma e^{-\alpha t} = 3.00\% \times e^{-0.380 \times 5} = 3.00\% \times e^{-1.9} = 3.00\% \times 0.14957 = 0.4487\%$.
3. Shock term: $0.4487\% \times (-0.2430) = -0.1090\%$.
4. $dr = 0.1083\% - 0.1090\% = -0.0007\% \approx 0.00\%$.

Why B is correct: By month 60 the exponential decay has shrunk volatility from 3.00% to about 0.45%, so even a -0.84 standard deviation shock offsets almost exactly the 0.108% monthly drift, leaving a change near zero.
Why the others are wrong
- **A:** -1.00% is the nearest choice if volatility is not decayed and dw is applied to the full 3.00% ($3.00\% \times -0.2430 = -0.73\%$, plus drift gives about -0.62%); ignoring $e^{-\alpha t}$ overstates the shock.
- **C:** +1.00% is the nearest choice if the full annual drift of 1.30% is used without multiplying by dt, then the decayed shock of -0.109% is added (total about +1.19%); the drift is only 0.108% per month.

- **D:** +2.00% is the nearest choice if the unscaled 1.30% drift is combined with an undecayed shock whose sign is flipped (3.00% × 0.2430 = +0.73%, total about +2.03%); it ignores dt scaling, decay and the negative draw.

Q708

GARP PE1 Q54

REBUILT — Original distractors (52 bps, 178 bps) were not traceable to realistic errors; replaced with distractors from the first step only, second step only, and unscaled drift.

A bank models the short rate with $dr = \lambda(t)\,dt + \sigma(t)\,dw$, where dw has mean zero and standard deviation $\sqrt{dt}$, using a monthly binomial tree ($dt = 1/12$) with up and down moves equally likely. Inputs:

Current short rate (date 0)	2.75%
Drift $\lambda(1)$	0.0018 per year
Drift $\lambda(2)$	0.0030 per year
Annualized volatility $\sigma(1)$	0.0050
Annualized volatility $\sigma(2)$	0.0080

What is the change in the rate from date 0 to the upper node (two up-moves) at date 2?

A. 16 bps
B. 42 bps
C. 26 bps
D. 86 bps

Answer: B

Step by step

1. Up-move at date 1: $\lambda(1)dt + \sigma(1)\sqrt{dt} = 0.0018/12 + 0.0050 \times 0.28868 = 0.000150 + 0.001443 = 0.001593$ (15.9 bps).
2. Up-move at date 2: $0.0030/12 + 0.0080 \times 0.28868 = 0.000250 + 0.002309 = 0.002559$ (25.6 bps).
3. Upper node at date 2 = 2.75% + 0.1593% + 0.2559% = 3.165%.
4. Change = 0.1593% + 0.2559% = 0.4152% ≈ 42 bps.

Why B is correct: Each period's up-move equals drift $\times dt$ plus volatility $\times\sqrt{dt}$ using that date's λ and σ; the two moves add.

Why the others are wrong

- **A:** 16 bps is only the first-period move (date 0 to date 1 upper node).
- **C:** 26 bps is only the second-period move; it omits the date 1 move.
- **D:** 86 bps adds the annual drifts (18 + 30 bps) without multiplying by $dt = 1/12$, then adds the volatility terms (37.5 bps).

Q709

AnalystPrep Q.1681

REBUILT — Stem referred to an equation not shown and option A/D were poorly worded; made self-contained with the process stated and clean options.

A model of the short rate has the form $dr = \lambda(t)\,dt + \sigma(t)\,dw$, where t is measured in years. If $\sigma(1) = 1.25\%$ and $\sigma(2) = 1.28\%$, which statement is correct?

A. The volatility of the short rate is 125 bps per year at t = 1 year and 128 bps per year at t = 2 years.
B. The volatility of the short rate is 128 bps per year at t = 1 year and 125 bps per year at t = 2 years.
C. The volatility of the short rate is 125 bps per year at t = 6 months and 128 bps per year at t = 1 year.
D. The volatility of the short rate is 1.25% in the first year and 1.28% in the second year, constant within each year, so volatility is declining over time.

Answer: A

Step by step

1. $\sigma(t)$ is the annualized basis-point volatility of the short rate evaluated at date t.
2. $\sigma(1) = 1.25\%$ = 125 bps per year at t = 1 year.
3. $\sigma(2) = 1.28\%$ = 128 bps per year at t = 2 years; volatility rises slightly with time.

Why A is correct: The argument of $\sigma(\cdot)$ is the date in years, and the value is the annualized volatility at that date.

Why the others are wrong

- **B:** Swaps the two values, assigning 128 bps to year 1 and 125 bps to year 2.
- **C:** Misreads the arguments as 6 months and 1 year; the arguments are 1 and 2 years.
- **D:** Misreads $\sigma(t)$ as a step function by year and wrongly concludes volatility declines; 1.28% > 1.25%, so it increases.

Q710 AnalystPrep Q.1682

REBUILT — Original options B and D were giveaways (calling the drift lambda the volatility); replaced with plausible numeric-behaviour options.

In Model 3, $dr = \lambda(t)\,dt + \sigma e^{-\alpha t}\,dw$ with $\sigma = 1.20\%$, $\alpha = 0.25$ and $\lambda(t)$ constant. Which statement best describes the standard deviation of the short-rate change per unit of dw?

A. It starts at 1.20% per year and declines exponentially toward zero, reaching about 0.44% at t = 4 years.
B. It starts at 1.20% per year and declines exponentially toward zero, reaching about 0.73% at t = 4 years.
C. It starts at 1.20% and rises exponentially with time, reaching about 3.26% at t = 4 years.
D. It is constant at 1.20% per year; α only affects the drift.

Answer: A

Step by step

1. At t = 0, $\sigma e^{0} = 1.20\%$.
2. Since $\alpha > 0$, $e^{-\alpha t}$ decays toward zero, so volatility falls exponentially.
3. At t = 4: $1.20\% \times e^{-0.25\times 4} = 1.20\% \times e^{-1} = 0.4415\% \approx 0.44\%$.

Why A is correct: The volatility term $\sigma e^{-\alpha t}$ equals σ at t = 0 and decays exponentially at rate α toward zero.

Why the others are wrong

- **B:** 0.73% is the value at t = 2 (1.20% × e^{-0.5}), not t = 4.
- **C:** Drops the minus sign in the exponent ($1.20\% \times e^{+1} = 3.26\%$); $\alpha > 0$ gives decay, not growth.
- **D:** Confuses Model 3 with Ho-Lee/Model 1 (constant σ); α governs the decay of volatility, not drift.

LO 15.c

Assess the efficacy of time-dependent volatility models.

Q711 AnalystPrep Q.1673

REBUILT — Stem was garbled (the same sentence asked about pricing an option and about pricing and hedging securities); reworded to a clean contrast.

According to Tuckman, the purpose of the model drives the choice between a time-dependent volatility model (Model 3) and a mean-reverting model (Vasicek). Which statement is correct?

A. Time-dependent drift is best for pricing options on fixed-income securities, while time-dependent volatility is best for pricing and hedging fixed-income securities.
B. Time-dependent volatility is best for pricing options on fixed-income securities, while time-dependent drift is best for pricing and hedging fixed-income securities.
C. Time-dependent volatility is best for pricing options on fixed-income securities, while mean reversion is best for pricing and hedging fixed-income securities.
D. Mean reversion is best for pricing options on fixed-income securities, while time-dependent volatility is best for pricing and hedging fixed-income securities.

Answer: C

Step by step

1. Option pricing needs a model that can be fitted to observed option prices at several expiries: the free function $\sigma(t)$ lets the model interpolate between observed market prices.
2. Valuing or hedging securities (and options) calls for a model with sensible economics for rates over time: mean reversion provides this.
3. Match: option pricing -> time-dependent volatility; pricing/hedging securities -> mean reversion (C).

Why C is correct: Time-dependent volatility gives the flexibility to fit observed option prices, whereas mean reversion supplies the economic intuition and downward-sloping volatility term structure suited to valuing and hedging securities.

Why the others are wrong

- **A:** Reverses the roles and attributes the fitting role to drift; drift shifts the mean path, it is not the tool for matching option prices.
- **B:** Correct on the first half but time-dependent drift is not the recommended choice for hedging; the reading recommends mean reversion.
- **D:** Swaps the two purposes: mean reversion is preferred for valuing/hedging securities, time-dependent volatility for fitting option prices.

Q712 AnalystPrep Q.1683

The choice of term structure model depends on the purpose at hand. Many users prefer the mean-reversion model for valuing or hedging fixed-income securities because they dispute the time-dependent volatility model's implicit claim that the market forecasts short-term volatility far into the future. Which modification of the time-dependent volatility model addresses this objection?

A. Assuming that volatility depends on time in the near future and then settles at a constant.
B. Assuming that the short rate depends on time in the near future and then settles at a constant.
C. Assuming that volatility depends on time in the distant future and then settles at an increasing rate.
D. Assuming that volatility depends on time in the near future and then settles at a decreasing rate.

Answer: A

Step by step

1. The objection: it is implausible that the market can forecast short-rate volatility in the distant future.
2. Compromise (Tuckman): let volatility vary with time only over the near term and let it approach a constant value afterwards.
3. Only A describes volatility, not the short rate, converging to a constant.

Why A is correct: Limiting time dependence to the near future and having volatility settle to a constant removes the need for distant-horizon volatility forecasts.

Why the others are wrong

- **B:** Makes the short rate, not volatility, time-dependent; this does not address the volatility-forecast objection.
- **C:** Still requires volatility forecasts in the distant future, and an increasing rate is not a constant.
- **D:** Volatility that keeps decreasing still needs a long-horizon forecast; it does not settle at a constant.

Q713 BT answers P2.T5.306.2 · BT notes P2.T5.306.2

REBUILT — Options contained joke/giveaway wording; replaced with neutral options and kept the same two statements.

A trader prefers Model 3 (time-dependent volatility) while a colleague prefers the mean-reverting Vasicek model. The colleague makes two arguments for Vasicek:

I. Mean reversion carries economic intuition, whereas Model 3's time-dependent volatility relies on the difficult argument that the market has a forecast of short-term volatility in the distant future.

II. Mean-reverting models produce a downward-sloping term structure of volatility, which captures the behavior of rates better than the parallel shifts and flat volatility term structure of Model 3.

According to Tuckman, which of the colleague's arguments are valid?

A. Neither I nor II.
B. I only.
C. II only.
D. Both I and II.

Answer: D

Step by step

1. Argument I: a criticism of time-dependent volatility is that it implies a market forecast of short-term volatility far out into the future, which is unlikely. Valid.
2. Argument II: Model 3 shifts the yield curve in parallel; the downward-sloping volatility term structure is a point in favor of mean reversion. Valid.
3. Both valid, so D.

Why D is correct: Both points are the standard criticisms of time-dependent volatility relative to mean reversion.

Why the others are wrong

- **A:** Rejects both valid criticisms.
- **B:** Accepts I but wrongly dismisses II; the downward-sloping volatility term structure does favor mean reversion.
- **C:** Accepts II but wrongly dismisses I; the distant-horizon volatility forecast is a recognized weakness.

Q714 Schweser Module Quiz 15.1 Q1

REBUILT — Stem said 'time-dependent drift models' though the statements concern time-dependent volatility; corrected the wording.

Regarding the validity of time-dependent volatility models, which of the following statements is (are) correct?

I. They are flexible because volatility can change from period to period; however, volatility must be an increasing function of time.

II. Time-dependent volatility functions are useful for pricing interest rate caps and floors.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: B

Step by step

1. I: time-dependent volatility can be increasing, decreasing or constant between periods; no monotonicity is required. False.
2. II: a cap/floor is a series of caplets/floorlets, each paying off on a different date, so pricing depends on $\sigma(t)$ at several future dates; the flexibility helps. True.
3. Only II is correct, so B.

Why B is correct: The flexibility to choose $\sigma(t)$ per period suits multi-period caps and floors, and imposes no restriction that volatility rise over time.

Why the others are wrong

- **A:** Treats the false 'must be increasing' claim as true and rejects the valid cap/floor use.
- **C:** Accepts the false restriction in I.
- **D:** Rejects II, though caps and floors are the main application.

Q715

AnalystPrep Q.1671

REBUILT — Original only asked the caplet payoff definition; rebuilt to test why time-dependent volatility is useful for matching caplet prices.

A dealer wants to price a 5-year cap on the short rate r, made up of caplets expiring each period. Market prices of caplets at different expiries imply different volatilities. A caplet pays $\max(r - K, 0)$ on its expiry date, so its value depends on the distribution of the short rate at that date. Which model feature best allows the dealer to match all the caplet prices?

A. A single constant volatility, since all caplets share the same strike.
B. A time-dependent drift only, with constant volatility.
C. Mean reversion with constant volatility.
D. A deterministic, time-dependent volatility function $\sigma(t)$ calibrated to the caplets.

Answer: D

Step by step

1. Each caplet's value depends on the spread of the short-rate distribution at its own expiry, so a different effective volatility is needed for each expiry.
2. A deterministic $\sigma(t)$ gives enough free parameters to fit these volatilities, one expiry at a time.
3. Therefore D.

Why D is correct: A time-dependent volatility function controls the standard deviation of the short rate at each date, which is what drives each caplet's value.

Why the others are wrong

- **A:** One constant volatility cannot reproduce caplet prices that imply different volatilities by expiry.
- **B:** Drift moves the mean of the rate, not its dispersion at each date, so it cannot fit volatility differences.
- **C:** Mean reversion with constant volatility imposes a fixed volatility pattern with few parameters; it is the choice for hedging/valuation, not for exact fitting of caplet prices.

LO 15.d

Describe the short-term rate process under the Cox-Ingersoll-Ross (CIR) and lognormal models.

Q716

AP Mock 3 Q14 · GARP PE1 Q14 · Mock 3 Q66

REBUILT — Merged three near-identical items; the printed stem said CIR has 'time-dependent volatility' (wrong), so the stem and options were cleaned to describe the two models accurately.

A hedge fund's CRO asks the risk team to develop a term-structure model for fitting interest rates for the fund's options pricing. The team is comparing the Ho-Lee model (normal distribution with time-dependent drift) and the

Cox-Ingersoll-Ross (CIR) model. Which of the following statements correctly describes one of these models?

A. In the Ho-Lee model, the drift of the short rate is a constant.
B. In the Ho-Lee model, when the short rate is above its long-run equilibrium level, the drift is negative.
C. In the CIR model, the volatility of the short rate declines exponentially to a constant long-run level.
D. In the CIR model, the basis-point volatility of the short rate is proportional to the square root of the rate.

Answer: D

Step by step

1. Ho-Lee: $dr = \lambda(t)\,dt + \sigma\,dw$; the drift $\lambda(t)$ varies by period and is calibrated to fit the market curve, with no mean reversion.
2. CIR: $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$; σ is constant and the basis-point volatility $\sigma\sqrt{r}$ rises with the rate (at a decreasing rate).
3. Only statement D matches these dynamics.

Why D is correct: CIR makes basis-point volatility a function of the rate level, $\sigma\sqrt{r}$, so volatility shrinks to zero as rates approach zero.

Why the others are wrong

- **A:** Ho-Lee drift is time-dependent ($\lambda(t)$), not constant; confuses it with the simple drift model.
- **B:** Negative drift above a long-run level is mean reversion, which Ho-Lee lacks (that is Vasicek/CIR).
- **C:** CIR volatility depends on the rate level, not on time; exponentially decaying volatility is a time-dependent-volatility specification.

Q717

AnalystPrep Q.1676

REBUILT — Original options B-D were giveaways (positive rates or responsiveness to markets are not flaws); rewrote the distractors around realistic misconceptions.

In the CIR model, basis-point volatility equals $\sigma\sqrt{r}$, which is zero when the short rate is zero; together with a positive drift at $r = 0$, this prevents negative rates. What is the main weakness of models with constant basis-point volatility (such as Vasicek or Ho-Lee) that this CIR feature addresses?

A. Shocks have the same size near zero as at high rate levels, so the short rate can become negative.
B. Constant basis-point volatility makes the short rate deterministic, so no interest rate risk is modelled.
C. Constant basis-point volatility prevents the model from including mean reversion toward a long-run rate.
D. Basis-point volatility rises roughly in proportion to the rate level, overstating risk when rates are high.

Answer: A

Step by step

1. In a constant-basis-point-volatility model, the standard deviation of dr is σ regardless of the level of r.
2. When r is near zero, a normal shock of size $\sigma\sqrt{dt}$ can still push the rate below zero; a normal terminal distribution always has mass below zero, more so at long horizons.
3. CIR scales volatility by $\sqrt{r}$, so shocks vanish at $r = 0$ and a positive drift keeps rates non-negative.

Why A is correct: Volatility that does not shrink as rates approach zero lets rates go negative.

Why the others are wrong

- **B:** Constant does not mean zero; these models are stochastic with volatility σ.
- **C:** Vasicek has constant volatility and mean reversion, so the two are compatible.
- **D:** That describes the lognormal model (basis-point volatility σr), not constant-volatility models.

Q718

AnalystPrep Q.1685

REBUILT — Original stem did not identify the volatility form and option C was malformed; stem now states square-root volatility with mean reversion and distractors are plausible alternative models.

A short-rate model assumes that basis-point volatility increases with the level of the short rate, in proportion to the square root of the rate, and that the rate reverts to a long-run level θ at speed k. Which equation represents the dynamics of this model?

A. $dr = k(\theta - r)\,dt + \sigma\,dw$
B. $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$
C. $dr = k(\theta - r)\,dt + \sigma r\,dw$
D. $dr = k(r - \theta)\,dt + \sigma\sqrt{r}\,dw$

Answer: B

Step by step

1. Mean reversion toward θ: drift $k(\theta - r)\,dt$ (pulls r up when $r < \theta$).
2. Basis-point volatility proportional to $\sqrt{r}$: diffusion $\sigma\sqrt{r}\,dw$.
3. Combining gives the CIR model $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$.

Why B is correct: CIR is the mean-reverting model whose diffusion term is $\sigma\sqrt{r}$.

Why the others are wrong

- **A:** Constant diffusion σ is the Vasicek model; volatility does not depend on r.
- **C:** Diffusion σr is the lognormal model (basis-point volatility linear in r), not square-root.
- **D:** The drift sign is reversed, $k(r - \theta)$, which pushes the rate away from θ (mean aversion).

Q719

BT Mock Q11 · BT notes P2.T5.23.5.2

For his term structure model, Peter considered the Vasicek model but, on reflection, prefers the Cox-Ingersoll-Ross (CIR) model. Which of the following statements about the CIR model is TRUE?

A. A drawback of the CIR model is that it does not mean revert to a long-term rate.

B. In the CIR model, the drift is time-dependent so that the model is an arbitrage-free fit to the curve.

C. An advantage of the CIR model is that the terminal distribution of the short rate is eventually normally distributed.

D. Unlike the Vasicek model's constant volatility, the CIR model's basis-point volatility is not constant, which (with positive drift) prevents the short rate from becoming negative.

Answer: D

Step by step

1. Vasicek: $dr = k(\theta - r)\,dt + \sigma\,dw$; CIR: $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$.
2. As $r \to 0$, basis-point volatility $\sigma\sqrt{r} \to 0$ while drift $k\theta > 0$, so r cannot turn negative.
3. A, B, C contradict the CIR specification (mean reversion, constant parameters, non-negative skewed terminal distribution).

Why D is correct: State-dependent volatility $\sigma\sqrt{r}$ vanishes at zero while mean reversion pushes the rate up, so negative rates are ruled out.

Why the others are wrong

- **A:** CIR does mean revert to θ, just like Vasicek.
- **B:** CIR has constant parameters (an equilibrium model); time-dependent drift is the Ho-Lee feature.
- **C:** The CIR terminal distribution is right-skewed and non-negative, never normal; a normal distribution would allow negative rates.

Q720

Schweser Module Quiz 15.2 Q2

Which of the following statements is most likely a disadvantage of the CIR model?

A. Interest rates are always non-negative.

B. Option prices from the CIR distribution may differ significantly from those under lognormal or normal distributions.

C. Basis-point volatility increases during periods of high inflation.

D. Long-run interest rates hover around a mean-reverting level.

Answer: B

Step by step

1. Non-negative rates (A), volatility rising with the rate level (C), and mean reversion (D) are all features viewed as advantages of CIR.
2. The CIR terminal distribution is right-skewed and non-negative, so the mass in the tails differs from normal or lognormal distributions.
3. This matters particularly for out-of-the-money options, so prices may differ substantially across models, a drawback of relying on CIR.

Why B is correct: Distributional differences in the tails can cause option mispricing, which is the cited drawback; the other choices are benefits.

Why the others are wrong

- **A:** Non-negativity is a benefit, avoiding negative rates.

- **C:** Higher basis-point volatility when rates are high (e.g., inflation) is realistic, an advantage.
- **D:** Mean reversion around a long-run level is a desirable, empirically sensible feature.

LO 15.e

Calculate the short-term rate change and describe the basis point volatility using the CIR and lognormal models.

Q721

BT answers P2.T5.307.2 · BT notes P2.T5.307.2

REBUILT — Original relied on a partial tree exhibit and had OCR-damaged distractors; stem made self-contained with the lognormal dynamics and distractors rebuilt from realistic errors.

An analyst builds a binomial tree for the lognormal model without mean reversion, $dr = \lambda r\,dt + \sigma r\,dw$, with time step $dt = 1/12$ (monthly), initial rate $r(0) = 2.00\%$, constant annual drift $\lambda = 0.50\%$, and annual yield volatility $\sigma = 9.00\%$. In the tree, the up-node after one month is $r(0)\,e^{\lambda dt+\sigma\sqrt{dt}}$ and the down-node is $r(0)\,e^{\lambda dt-\sigma\sqrt{dt}}$. What is the rate at the up-node after one month, node [1,1]?

A. 1.9495%
B. 2.0535%
C. 2.0629%
D. 4.6397%

Answer: B

Step by step

1. Drift term: $\lambda\,dt = 0.005/12 = 0.000417$.
2. Volatility term: $\sigma\sqrt{dt} = 0.09 \times 0.288675 = 0.025981$.
3. Exponent (up): $0.000417 + 0.025981 = 0.026398$; $e^{0.026398} = 1.026749$.
4. Up-node: $2.00\% \times 1.026749 = 2.0535\%$.

Why B is correct: In the lognormal model the shocks are multiplicative, so the up-node scales the starting rate by $e^{\lambda dt+\sigma\sqrt{dt}}$, with drift scaled by dt and volatility by $\sqrt{dt}$.

Why the others are wrong

- **A:** This is the down-node $2.00\% \times e^{0.000417-0.025981} = 1.9495\%$; the sign of the volatility term was reversed.
- **C:** Used the full annual drift (0.50%) instead of 0.50%/12: $2.00\% \times e^{0.005+0.025981} = 2.0629\%$.
- **D:** Treated 9% as an additive basis-point volatility: $2.00\% + 0.0417\% + 9\% \times 0.2887 = 4.6397\%$, ignoring that volatility is proportional to the rate.

Q722

AP Mock 1 Q16

REBUILT — Correct answer kept (5.06%); distractors replaced with results of realistic errors and the dw convention made explicit.

Use the Cox-Ingersoll-Ross model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, with a monthly step ($dt = 1/12$), to find the short-term rate after one month.

Item	Value
Current short-term rate	5%
Long-run value of the short-term rate, θ	7.5%
Speed of mean reversion, k	0.05
Volatility, σ	0.8%
Realized dw (already scaled for the time step)	0.3

A. 5.04%
B. 5.06%
C. 5.18%
D. 5.25%

Answer: B

Step by step

1. Drift: $0.05 \times (7.5\% - 5\%) \times 1/12 = 0.0104\%$.
2. Shock: $0.8\% \times \sqrt{0.05} \times 0.3 = 0.8\% \times 0.22361 \times 0.3 = 0.0537\%$.
3. $dr = 0.0104\% + 0.0537\% = 0.0641\%$.
4. $r = 5\% + 0.0641\% = 5.064\% \approx 5.06\%$.

Why B is correct: CIR adds a mean-reverting drift scaled by dt to a shock whose size is proportional to $\sqrt{r}$; dw is given already scaled, so no further $\sqrt{dt}$ is applied.

Why the others are wrong

- **A:** Reversed the sign of $(\theta - r)$: $5\% - 0.0104\% + 0.0537\% = 5.043\%$.
- **C:** Forgot to multiply the drift by dt: $0.05 \times 2.5\% = 0.125\%$, giving $5\% + 0.125\% + 0.0537\% = 5.179\%$.
- **D:** Dropped the $\sqrt{r}$ factor (Vasicek-style constant volatility): $0.0104\% + 0.8\% \times 0.3 = 0.2504\%$, giving 5.25%.

Q723

AnalystPrep Q.1674

In the Cox-Ingersoll-Ross (CIR) model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, the basis-point volatility is not independent of the level of the short-term rate. Which of the following statements is correct regarding the CIR model?

A. The standard deviation of dr is inversely proportional to the square root of the short rate

B. The standard deviation of dr is directly proportional to the square root of dw

C. The standard deviation of dr is directly proportional to the square root of dt

D. The standard deviation of dr is directly proportional to the square root of the short rate

Answer: D

Step by step

1. The random term of CIR is $\sigma\sqrt{r}\,dw$.
2. Its standard deviation is $\sigma\sqrt{r} \times \sqrt{dt}$, so annualized basis-point volatility is $\sigma\sqrt{r}$.
3. Hence it rises with the square root of the short rate, with σ constant.

Why D is correct: Basis-point volatility equals $\sigma\sqrt{r}$, so it increases (at a decreasing rate) as the short rate rises.

Why the others are wrong

- **A:** Inverts the relationship; volatility rises, not falls, with the rate (the model fixes the high-rate/high-inflation problem).
- **B:** dw is the random shock itself, not a scale factor of volatility; its own standard deviation is $\sqrt{dt}$.
- **C:** The $\sqrt{dt}$ is a time-scaling effect common to all these models and does not describe the CIR-specific dependence on the rate level.

Q724

AnalystPrep Q.1675

REBUILT — Original option B contradicted its own explanation (σ√r vs σr) and referenced the Courtadon model outside the syllabus; recast to compare CIR and lognormal basis-point volatility.

Which of the following correctly describes yield volatility and basis-point volatility in the CIR and lognormal short-rate models (σ is the constant volatility parameter and r the short rate)?

A. In both models basis-point volatility is constant and equal to σ; only yield volatility varies with r.

B. In the CIR model basis-point volatility is $\sigma\sqrt{r}$ and in the lognormal model it is σr; in both it increases with r.

C. In the CIR model basis-point volatility is σr and in the lognormal model it is $\sigma\sqrt{r}$; in both it increases with r.

D. In both models basis-point volatility is σr and decreases as the level of the short rate rises.

Answer: B

Step by step

1. CIR: $dr = k(\theta - r)dt + \sigma\sqrt{r}\,dw$, so basis-point volatility is $\sigma\sqrt{r}$, rising at a decreasing rate.
2. Lognormal: $dr = \lambda r\,dt + \sigma r\,dw$, so basis-point volatility is σr, rising linearly.
3. Both are increasing functions of r; the parameter σ is held constant.

Why B is correct: The CIR model scales the shock by $\sqrt{r}$ and the lognormal model scales it by r, so both give basis-point volatility that increases with the rate.

Why the others are wrong

- **A:** Describes the normal (Vasicek/Ho-Lee) models with constant basis-point volatility, which CIR and lognormal were designed to replace.
- **C:** Swaps the two functional forms; σr belongs to the lognormal model and $\sigma\sqrt{r}$ to CIR.
- **D:** Gets σr for both models and wrongly claims volatility falls as rates rise; it rises with r in both.

Q725

AnalystPrep Q.2858

REBUILT — Correct answer kept (2.446%); distractors (including negative and giveaway values) replaced by realistic-error values and the volatility label corrected to the CIR sigma.

An analyst uses the CIR model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, with monthly steps ($dt = 1/12$): initial rate $r(0) = 2.11\%$, $\sigma = 3.17\%$, long-run rate $\theta = 7.64\%$, and $k = 0.57$. For the first month the realized $dw = 0.160$ (already scaled for the time step). What is the short rate after the first month?

A. 1.921%
B. 2.373%
C. 2.446%
D. 2.880%

Answer: C

Step by step

1. Drift: $0.57 \times (7.64\% - 2.11\%) \times 1/12 = 0.2627\%$.
2. Shock: $3.17\% \times \sqrt{0.0211} \times 0.160 = 3.17\% \times 0.14526 \times 0.160 = 0.0737\%$.
3. $dr = 0.2627\% + 0.0737\% = 0.3364\%$.
4. $r(1/12) = 2.11\% + 0.3364\% = 2.446\%$.

Why C is correct: The new rate is the old rate plus the mean-reverting drift (pulling toward 7.64%) plus a shock proportional to $\sqrt{r}$.

Why the others are wrong

- **A:** Reversed the sign of $(\theta - r)$: $2.11\% - 0.2627\% + 0.0737\% = 1.921\%$.
- **B:** Omitted the random shock (drift only): $2.11\% + 0.2627\% = 2.373\%$.
- **D:** Dropped $\sqrt{r}$ from the shock: $3.17\% \times 0.160 = 0.5072\%$, giving $2.11\% + 0.2627\% + 0.5072\% = 2.880\%$.

Q726

BT answers P2.T5.307.1 · BT notes P2.T5.307.1

REBUILT — OCR-damaged source; correct answer 1.390% retained but distractors (unexplained values such as 0.444%, 2.172%) replaced with realistic-error results.

An analyst uses the CIR model, $dr = 0.6(8.0\% - r)\,dt + 2.5\%\sqrt{r}\,dw$, with $dt = 1/12$, initial rate $r(0) = 1.00\%$, $\theta = 8.00\%$, $k = 0.60$, $\sigma = 2.5\%$. For the first month the random standard normal draw is 0.5534, so $dw = 0.5534\sqrt{1/12} = 0.160$. What is the short rate after one month?

A. 0.690%
B. 1.390%
C. 1.750%
D. 5.240%

Answer: B

Step by step

1. Drift: $0.60 \times (8.0\% - 1.0\%) \times 1/12 = 0.350\%$.
2. Shock: $2.5\% \times \sqrt{0.01} \times 0.160 = 2.5\% \times 0.1 \times 0.160 = 0.040\%$.
3. $dr = 0.350\% + 0.040\% = 0.390\%$.
4. $r(1/12) = 1.00\% + 0.390\% = 1.390\%$.

Why B is correct: With r far below θ the drift dominates (+35 bp) and the $\sqrt{r}$-scaled shock adds only 4 bp.

Why the others are wrong

- **A:** Reversed the sign of the drift: $1.00\% - 0.350\% + 0.040\% = 0.690\%$.
- **C:** Omitted $\sqrt{r}$ from the shock: $2.5\% \times 0.160 = 0.400\%$, giving $1.00\% + 0.350\% + 0.400\% = 1.750\%$.
- **D:** Forgot to scale the drift by dt: $0.6 \times 7\% = 4.2\%$, giving $1.00\% + 4.20\% + 0.040\% = 5.240\%$.

Q727

BT notes P2.T5.23.5.1

REBUILT — Correct answer kept (+0.149%); distractors (e.g. -2.275%, +0.003%) were unrelated to realistic errors and were replaced.

A CIR model, $dr = k(\theta - r)\,dt + \sigma\sqrt{r}\,dw$, is implemented with monthly steps ($dt = 1/12$). The current rate is 7.00%, $\sigma = 1.30\%$, $\theta = 5.00\%$, and $k = 0.30$. The random variable dw has mean zero and standard deviation $\sqrt{dt}$. If next month's shock is +2.0 standard deviations ($dw = 2.0\sqrt{1/12}$), which of the following is closest to the change in the rate?

A. -0.401%
B. +0.149%

C. +0.249%
D. +0.701%

Answer: B
Step by step
1. $dw = 2.0 \times \sqrt{1/12} = 0.57735$.
2. Drift: $0.30 \times (5\% - 7\%) \times 1/12 = -0.050\%$.
3. Shock: $1.30\% \times \sqrt{0.07} \times 0.57735 = 1.30\% \times 0.26458 \times 0.57735 = 0.1986\%$.
4. $dr = -0.050\% + 0.1986\% = +0.149\%$.

Why B is correct: Mean reversion pulls the 7% rate down toward 5% (-5 bp) but the +2σ shock (+19.9 bp) dominates.

Why the others are wrong
- **A:** Forgot to scale the drift by dt: $0.30 \times (-2\%) = -0.60\%$, giving $-0.60\% + 0.1986\% = -0.401\%$.
- **C:** Wrong sign on the drift (used $r - \theta$): $+0.050\% + 0.1986\% = +0.249\%$.
- **D:** Omitted $\sqrt{r}$ from the shock: $1.30\% \times 0.57735 = 0.7506\%$, minus 0.050% gives +0.701%.

Q728
GARP PE2 Q36

An analyst studying term structure models focuses on the Cox-Ingersoll-Ross (CIR) model and its treatment of the volatility of the short-term rate. Which of the following statements is correct regarding yield volatility and basis-point volatility in the CIR model?

A. Periods of extremely low short-term interest rates are accompanied by high basis-point volatility, which increases the possibility of negative interest rates in the CIR model.
B. In the CIR model, basis-point volatility is specified as a decreasing function of the mean reversion factor.
C. In the CIR model, yield volatility is specified as being constant while basis-point volatility is allowed to vary.
D. Basis-point volatility and yield volatility are used interchangeably to measure the same volatility in the CIR model.

Answer: C
Step by step
1. CIR keeps the volatility parameter σ constant.
2. Basis-point volatility is $\sigma\sqrt{r}$, so it varies with the level of the short rate.
3. Only C matches this; A, B and D each contradict it.

Why C is correct: σ is a constant volatility parameter, while basis-point volatility $\sigma\sqrt{r}$ changes with the rate.

Why the others are wrong
- **A:** Reverses the mechanism: low rates give low basis-point volatility, which dampens the chance of negative rates.
- **B:** Basis-point volatility increases with the short rate; mean reversion (k) governs drift, not volatility.
- **D:** The two are different measures: basis-point volatility is the absolute rate vol ($\sigma\sqrt{r}$); σ is the constant parameter.

Q729
Mock 4 Q37

A fixed-income analyst is comparing the volatility specifications of different term structure models. The analyst focuses on the Cox-Ingersoll-Ross (CIR) model, in which the short-rate process is specified as $dr = k(\theta - r)dt + \sigma\sqrt{r}\,dw$. Which of the following statements about volatility in the CIR model is correct?

A. Both basis-point volatility and yield volatility increase as the level of the short-term rate increases.
B. Both basis-point volatility and yield volatility decrease as the level of the short-term rate increases.
C. Basis-point volatility decreases and yield volatility increases as the level of the short-term rate increases.
D. Basis-point volatility increases and yield volatility decreases as the level of the short-term rate increases.

Answer: D
Step by step
1. Basis-point volatility is the coefficient on dw: $\sigma_{bp} = \sigma\sqrt{r}$, which rises with r (at a decreasing rate).
2. Yield volatility is basis-point volatility divided by the rate: $\sigma_{bp}/r = \sigma/\sqrt{r}$, which falls as r rises.
3. Check with $\sigma = 0.08$: r = 1% gives bp vol 0.80% and yield vol 80%; r = 4% gives 1.60% and 40%; r = 9% gives 2.40% and 26.7%.

4. Bp vol rises while yield vol falls, so D.

Why D is correct: In CIR, basis-point volatility grows with the square root of the rate, which is slower than proportionally, so volatility measured as a percentage of the rate declines as the rate rises.

Why the others are wrong

- **A:** Treats yield volatility as rising; that would need bp vol to grow faster than proportionally with r, whereas here it grows only like $\sqrt{r}$.
- **B:** Wrongly thinks bp volatility falls with r; the term $\sigma\sqrt{r}$ clearly increases with r.
- **C:** Reverses the two relationships: bp vol increases and yield vol decreases, not the other way round.

Q730

Schweser Module Quiz 15.2 Q1

REBUILT — Answer choices were missing from the OCR; reconstructed four options on bp and yield volatility in the lognormal model with the key (bp vol increases with r, yield vol constant).

Which of the following choices correctly characterizes basis-point volatility and yield volatility as a function of the level of the short-term rate in the lognormal model $dr = ar\,dt + \sigma r\,dw$?

A. Basis-point volatility increases with the rate; yield volatility is constant.
B. Basis-point volatility is constant; yield volatility decreases with the rate.
C. Basis-point volatility increases with the rate; yield volatility decreases with the rate.
D. Basis-point volatility is constant; yield volatility increases with the rate.

Answer: A

Step by step

1. In the lognormal model the dw coefficient is σr, so basis-point volatility $= \sigma r$, increasing linearly in r.
2. Yield volatility $= \sigma r / r = \sigma$, a constant.
3. Example with $\sigma = 20\%$: r = 2% gives bp vol 0.40% (40 bps); r = 5% gives 1.00% (100 bps); yield vol is 20% in both cases.
4. So bp vol rises with r and yield vol is constant: A.

Why A is correct: Volatility is specified as a fixed percentage of the rate, so its basis-point size scales proportionally with r while the percentage volatility stays at Ïf.

Why the others are wrong

- **B:** Describes a constant bp-vol model (Model 1/Vasicek) in which yield vol falls as σ/r, not the lognormal model.
- **C:** Describes CIR-type behavior (bp vol $\sigma\sqrt{r}$, yield vol $\sigma/\sqrt{r}$); in the lognormal model yield vol does not fall.
- **D:** Pairs constant bp vol with rising yield vol, which is not produced by any of the models in the reading; constant bp vol implies yield vol falls with r.

LO 15.f

Describe lognormal models with deterministic drift and mean reversion.

Q731

AnalystPrep Q.1679

REBUILT — Original asked what the model allows that other models do not (vague, circular, giveaway options); rebuilt as a clear comparison of the Black-Karasinski model with other models.

Which of the following statements best describes the lognormal model with mean reversion (the Black-Karasinski model)?

A. The natural log of the short rate is normally distributed and mean-reverting, rates are always positive, and the mean-reversion speed, central tendency and volatility may all be time-dependent.
B. The short rate itself is normally distributed and can become negative, with constant mean-reversion speed, central tendency and volatility.
C. Basis point volatility is proportional to the square root of the short rate, and the short rate follows a non-central chi-square distribution.
D. The model produces a naturally recombining interest rate tree, so no adjustment to time steps is required.

Answer: A

Step by step

1. The Black-Karasinski model is $d\ln r = k(t)[\ln\theta(t) - \ln r]dt + \sigma(t)dw$.

2. So $\ln r$ is normal (a time-dependent Vasicek on the log rate) and $r = e^{\ln r} > 0$.
3. The parameters k(t), θ(t), σ(t) can all depend on time, which gives the model its flexibility.

Why A is correct: Applying a time-dependent Vasicek process to ln r gives a lognormal short rate that is positive and flexible in its time dependence.

Why the others are wrong

- **B:** Describes the Vasicek model (normal rates, constant parameters); confuses the distribution of r with that of ln r.
- **C:** Describes the CIR model (volatility proportional to the square root of r).
- **D:** The tree does not recombine naturally; time steps dt must be recalibrated to force recombination.

Q732

AnalystPrep Q.1678

REBUILT — Original was a trivial definition (lognormal if log is normal) with an unrelated stem; rebuilt as a numeric node calculation for the lognormal model with deterministic drift.

In a lognormal model with deterministic drift, $\ln r_1 = \ln r_0 + \alpha_1 dt \pm \sigma\sqrt{dt}$. The current short rate is $r_0 = 5.00\%$, $\alpha_1 = 0.02$, $\sigma = 0.15$ and $dt = 1$. What is the short rate in the lower node after one period?

A. 5.93%
B. 4.39%
C. 4.30%
D. 4.35%

Answer: B

Step by step

1. Lower node: $\ln r_1 = \ln r_0 + 0.02 - 0.15$.
2. $r_1 = 5\% \times e^{0.02-0.15} = 5\% \times e^{-0.13}$.
3. $e^{-0.13} = 0.8781$, so $r_1 = 4.39\%$.

Why B is correct: Drift enters multiplicatively through the exponential, so the lower node is r0·e^{α−σ}.

Why the others are wrong

- **A:** This is the upper node, $5\% \times e^{0.17}$.
- **C:** Omitted the drift: $5\% \times e^{-0.15}$.
- **D:** Used the additive approximation $r_0(1 + \alpha - \sigma) = 5\% \times 0.87$ instead of exponentiating.

Q733

AnalystPrep Q.1680

REBUILT — Original only asked the distribution of ln r with a giveaway Bernoulli option; rebuilt as a numeric mean-reversion question in the lognormal (Black-Karasinski) model.

In a lognormal model with mean reversion, $d\ln r = k[\ln\theta - \ln r]dt + \sigma dw$, with $k = 0.25$, $\theta = 6.00\%$ and current $r = 4.00\%$. Ignoring the volatility term, what short rate corresponds to the expected $\ln r$ after $dt = 1$?

A. 4.50%
B. 4.43%
C. 4.02%
D. 4.41%

Answer: B

Step by step

1. Expected change in ln r: $0.25 \times [\ln 0.06 - \ln 0.04] = 0.25 \times 0.4055 = 0.1014$.
2. New ln r = ln(0.04) + 0.1014.
3. $r = 4\% \times e^{0.1014} = 4.43\%$.

Why B is correct: Mean reversion operates on the log of the rate, so the log rate moves toward ln θ and the rate is obtained by exponentiating.

Why the others are wrong

- **A:** Applied Vasicek in rate space: $4\% + 0.25 \times (6\% - 4\%)$.
- **C:** Used the rate-space reversion $k(\theta - r) = 0.005$ as the log change: $4\% \times e^{0.005} = 4.02\%$.
- **D:** Used the linear approximation $4\% \times (1 + 0.1014) = 4.41\%$ instead of exponentiating.

Q734 BT answers P2.T5.307.3 · BT notes P2.T5.307.3

Each of the following is true about lognormal models of the short-term interest rate process EXCEPT for:

A. In the lognormal models, the natural logarithm of the short rate is normally distributed

B. Like the Cox-Ingersoll-Ross (CIR), the lognormal models offer the advantage of not allowing negative rates

C. The lognormal model without mean reversion is similar to the Ho-Lee model but based on the natural logarithm of the short rate instead of on the short rate itself

D. The Black-Karasinski Model is similar to the Ho-Lee Model but instead is an equilibrium model with constant drift

Answer: D

Step by step

1. A: ln r is normal, so r is lognormal: true.
2. B: r = e^{ln r} > 0, as in CIR: true.
3. C: lognormal with drift is the Ho-Lee model applied to ln r: true.
4. D: Black-Karasinski is the lognormal model with mean reversion (time-dependent Vasicek on ln r), not Ho-Lee; it uses time-varying parameters and is not an equilibrium model with constant drift: false.

Why D is correct: Black-Karasinski is a mean-reverting lognormal model resembling Vasicek, so D misdescribes it.

Why the others are wrong

- **A:** True statement, not the exception.
- **B:** True statement; positivity is a shared benefit of CIR and lognormal models.
- **C:** True statement; the drift model is the Ho-Lee analogue in logs.

Q735 Schweser Module Quiz 15.2 Q3

Which of the following statements best characterizes the differences between the Ho-Lee model with drift and the lognormal model with drift?

A. In the Ho-Lee model and the lognormal model, the drift terms are multiplicative.

B. In the Ho-Lee model and the lognormal model, the drift terms are additive.

C. In the Ho-Lee model the drift terms are multiplicative, but in the lognormal model the drift terms are additive.

D. In the Ho-Lee model the drift terms are additive, but in the lognormal model the drift terms are multiplicative.

Answer: D

Step by step

1. Ho-Lee: $dr = \lambda(t)dt + \sigma dw$, so drifts add to the rate.
2. Lognormal: $d \ln r = \alpha(t)dt + \sigma dw$; rates are $r = e^{\ln r}$, so drifts add in logs and multiply in rates.
3. Hence additive for Ho-Lee, multiplicative for lognormal (which also keeps all rates positive).

Why D is correct: The lognormal tree is built in logs and then exponentiated, converting additive log drifts into multiplicative effects on rates.

Why the others are wrong

- **A:** Wrongly makes Ho-Lee multiplicative.
- **B:** Ignores that exponentiation makes lognormal drift multiplicative.
- **C:** Reverses the two models.

Q736 Schweser Module Quiz 15.2 Q4

Which of the following statements is true regarding the Black-Karasinski model?

A. The model produces an interest rate tree that is recombining by definition.

B. The model produces an interest rate tree that is recombining when the dt variable is manipulated.

C. The model is time-varying and mean-reverting with a slow speed of adjustment.

D. The model is time-varying and mean-reverting with a fast speed of adjustment.

Answer: B

Step by step

1. The model has mean reversion in ln r, so an up move followed by a down move generally does not equal a down move followed by an up move.
2. The tree recombines only if the second time step dt2 is chosen appropriately given dt1.

3. The model itself does not fix the speed of adjustment, since k(t) is a free parameter.

Why B is correct: Recombination is forced by recalibrating the time steps dt.

Why the others are wrong

- **A:** The tree does not recombine naturally.
- **C:** The speed of mean reversion is set by k(t), not inherently slow.
- **D:** Likewise not inherently fast.

Reading 16. The Vasicek and Gauss+ Models

Bruce Tuckman and Angel Serrat, Fixed Income Securities: Tools for Today's Markets, 4th Edition (Hoboken, NJ: John Wiley & Sons, 2022). Chapter 9.

LO	Learning objective	Qs
16.a	describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics.	8
16.b	compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model.	11
16.c	calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model.	6
16.d	explain how the parameters of the Gauss+ model can be estimated from empirical data.	5

LO 16.a

Describe the structure of the Gauss+ model and discuss the implications of this structure for the model's ability to replicate empirically observed interest rate dynamics.

Q737

AnalystPrep Q.6517

What structural feature of the Gauss+ model allows it to replicate the complex interest rate dynamics observed in financial markets?

A. A single mean-reverting factor capturing short-term dynamics.
B. Independent, non-mean-reverting factors that simplify modeling.
C. Three mean-reverting factors representing the short-, medium-, and long-term rate levels.
D. A linear deterministic path for rates over time.

Answer: C

Step by step

1. Gauss+ has three state variables: r_t, m_t, l_t.
2. They form a cascade: r_t reverts to m_t, m_t reverts to l_t, and l_t reverts to the fixed constant μ, each at a different speed ($\alpha_r > \alpha_m > \alpha_l$).
3. This multi-speed mean reversion lets the model match empirical rate dynamics and the volatility term structure.

Why C is correct: Layered mean reversion at different speeds across three factors gives the model the flexibility to match observed rate and volatility behavior.

Why the others are wrong

- **A:** Describes a one-factor model such as Vasicek; it lacks the medium- and long-term factors.
- **B:** Gauss+ factors are mean-reverting and linked in a cascade, not independent and non-reverting.
- **D:** Gauss+ is a stochastic model with random shocks; it does not assume a deterministic path.

Q738

AnalystPrep Q.6518

REBUILT — Original distractors were obvious giveaways and the correct option was vague; rebuilt around the volatility term structure the reading says Gauss+ replicates, with plausible alternative shapes.

Which volatility term structure can the Gauss+ model be calibrated to replicate, in line with empirically observed interest rate behavior?

A. Volatility that is highest at the short end and declines steadily with maturity.
B. Volatility that rises steadily with maturity.
C. Low volatility at the short end, higher volatility at medium maturities, and declining volatility at long maturities (hump-shaped).
D. Constant volatility across all maturities.

Answer: C

Step by step

1. The short rate r_t has no direct random shock; it only follows m_t, so short-end volatility is low.
2. The medium-term factor m_t carries its own shock σ_m, giving higher volatility at medium maturities.
3. The long-term factor reverts slowly and its influence diminishes, so volatility declines at long maturities.

4. This hump shape matches historical norms, which the single-volatility Vasicek model cannot reproduce.

Why C is correct: The cascade structure with a shock-free short rate and separately calibrated medium and long factors yields the hump-shaped volatility term structure seen in the data.

Why the others are wrong

- **A:** A monotonically declining pattern ignores the low short-end volatility caused by central-bank control; it is closer to what a one-factor model delivers.
- **B:** Volatility does not rise steadily; it declines at long maturities.
- **D:** Constant volatility is a one-parameter simplification, not an empirical feature.

Q739

AnalystPrep Q.6519

REBUILT — Original stem was vague and distractors were absurd; rebuilt as a concrete test of the correlation structure between factors under Gauss+.

In the Gauss+ model, the medium-term factor has shock $\sigma_m(\rho\, dW_t^1 + \sqrt{1-\rho^2}\, dW_t^2)$ and the long-term factor has shock $\sigma_l\, dW_t^1$, where dW_t^1 and dW_t^2 are independent. With $\rho = 0.6$, $\sigma_m = 0.01$ and $\sigma_l = 0.005$, what is the correlation between the random shocks to m_t and l_t?

A. 0.36
B. 0.60
C. 0.80
D. 0.000030

Answer: B

Step by step

1. Covariance of shocks: $\sigma_m \sigma_l \rho\, dt = 0.01 \times 0.005 \times 0.6\, dt = 0.00003\, dt$.
2. Standard deviations: $\sigma_m \sqrt{dt}$ (since $\rho^2 + (1-\rho^2) = 1$) and $\sigma_l \sqrt{dt}$.
3. Correlation $= 0.00003/(0.01 \times 0.005) = 0.6 = \rho$.

Why B is correct: Both shocks load on the same dW^1; the m-shock has weight ρ on it and unit total variance, so the correlation equals ρ.

Why the others are wrong

- **A:** Squared the correlation ($0.6^2 = 0.36$), which is the share of m-shock variance from dW^1, not the correlation.
- **C:** Used $\sqrt{1-\rho^2} = 0.8$, the weight on the independent shock dW^2.
- **D:** Reported the covariance $\rho\sigma_m\sigma_l$ without dividing by the standard deviations.

Q740

AnalystPrep Q.6520

REBUILT — Keyed answer (regime-switching) is not a feature of Gauss+ in the reading; replaced with a cascade mean-reversion question testing the same LO.

In the Gauss+ model, suppose at some date $r_t > m_t$ and $m_t > l_t > \mu$. Ignoring random shocks, which statement best describes the drift of the factors?

A. r_t drifts down toward m_t, m_t drifts down toward l_t, and l_t drifts down toward μ.
B. r_t drifts up toward m_t, while m_t and l_t drift down.
C. r_t drifts down toward μ directly, bypassing m_t.
D. r_t does not move because it has no random shock.

Answer: A

Step by step

1. Short-rate drift $-\alpha_r(r_t - m_t) < 0$ since $r_t > m_t$: it falls toward m_t.
2. Medium drift $-\alpha_m(m_t - l_t) < 0$ since $m_t > l_t$: it falls toward l_t.
3. Long drift $-\alpha_l(l_t - \mu) < 0$ since $l_t > \mu$: it falls toward μ, the only fixed target.

Why A is correct: Each factor's drift is proportional to its gap from the next factor in the cascade, so with each factor above its target all three drift down.

Why the others are wrong

- **B:** Gets the sign of the short-rate drift wrong; when $r_t > m_t$ the drift is negative.
- **C:** The short rate reverts to the moving target m_t, not directly to μ.
- **D:** Confuses having no shock with having no drift; r_t still has a drift toward m_t.

Q741 BT answers P2.T5.25.7.1

REBUILT — Original asked an ambiguous question about which adjustment factor 'has the least volatility'; rebuilt as a clear test of reversion speeds via half-lives.

In the Gauss+ model, $dr_t = -\alpha_r(r_t - m_t)dt$, $dm_t = -\alpha_m(m_t - l_t)dt + ...$, $dl_t = -\alpha_l(l_t - \mu)dt + ...$, with $\alpha_r = 0.5$, $\alpha_m = 0.2$, $\alpha_l = 0.05$. Using half-life $= \ln 2/\alpha$, which factor adjusts most slowly toward its target, and what is its half-life?

A. Short-term rate, 1.39 years
B. Medium-term factor, 3.47 years
C. Long-term factor, 13.86 years
D. Long-term factor, 20.00 years

Answer: C

Key correction: Source key (BT answers) said C for an ambiguous 'least volatility' wording; C is retained only because the item was rebuilt around the slowest-reverting factor.

Step by step

1. Half-lives: short $0.6931/0.5 = 1.39$; medium $0.6931/0.2 = 3.47$; long $0.6931/0.05 = 13.86$ years.
2. The smallest α is slowest: the long-term factor, 13.86 years.
3. This matches the reading: reversion is fastest for r and slowest for l, reflecting persistent fundamentals such as inflation and productivity.

Why C is correct: Half-life is inversely proportional to the reversion speed, so the smallest $\alpha_l = 0.05$ gives the longest half-life, 13.86 years.

Why the others are wrong

- **A:** This is the fastest-adjusting factor (largest α), the opposite of what was asked.
- **B:** The intermediate factor; α_m is not the smallest.
- **D:** Right factor but used $1/\alpha = 20$ (the mean reversion time scale) instead of $\ln 2/\alpha$.

Q742 BT notes P2.T5.23.5.3

Each of the following is true about the Gauss+ model EXCEPT which is false?

A. In the cascade form, the short-term rate mean reverts to the medium-term factor, which reverts to the long-term factor.
B. Gauss+ has three state variables r_t, m_t, l_t, denoting the short-term rate, a medium-term factor, and a long-term factor.
C. The long-term factor mean reverts to a long-term constant level and reflects long-term trends such as inflation expectations and productivity.
D. The "+" reflects an added factor that ensures a downward-sloping term structure of volatility.

Answer: D

Step by step

1. A and B restate the cascade structure and three state variables.
2. C is correct: l_t reverts to the fixed constant μ and captures long-run economic trends.
3. D is false: the model is calibrated to a hump-shaped volatility term structure (low short end, higher middle, declining long end), not a monotonically downward-sloping one.

Why D is correct: Gauss+ replicates hump-shaped, not downward-sloping, volatility term structure, so D is the false statement.

Why the others are wrong

- **A:** True; this is the cascade structure.
- **B:** True; these are the three state variables.
- **C:** True; l_t reverts to μ and represents long-term trends.

Q743 Schweser Module Quiz 16.1 Q1

Which of the following statements best describes the structure of the Gauss+ model?

A. A single-factor model with constant volatility.
B. A model without any mean reversion or volatility assumptions.
C. A two-factor model with mean reversion to a fixed long-term rate.
D. A cascade model where each factor mean reverts to the next level.

Answer: D
Step by step
1. Gauss+ has three factors: r_t, m_t, l_t.
2. r_t reverts to m_t, m_t reverts to l_t, and l_t reverts to μ: a cascade.

Why D is correct: Gauss+ is a three-factor cascade of mean-reverting factors.
Why the others are wrong
- **A:** Describes a one-factor Vasicek-type simplification.
- **B:** Gauss+ is built on mean reversion and has volatility parameters for m and l.
- **C:** It has three factors, not two; only μ is a fixed target.

Q744 Schweser Module Quiz 16.1 Q2

The role of the medium-term factor in the Gauss+ model is that the factor:
A. remains constant over time.
B. directly determines the pricing of long-term bonds.
C. reflects economic influences, such as business cycles.
D. represents the risk premium in the interest rate structure.

Answer: C
Step by step
1. m_t is the intermediate level in the cascade: r_t reverts to it and it reverts to l_t.
2. It represents broader economic influences such as business cycles and monetary policy effects.

Why C is correct: The medium-term factor captures business-cycle-type influences and acts as the moving target for the short rate.
Why the others are wrong
- **A:** m_t is dynamic, with its own drift and shock.
- **B:** m_t and l_t are not tied to specific forward rates; they influence the evolution of the short rate.
- **D:** Gauss+ factors are not a risk premium.

LO 16.b

Compare and contrast the dynamics, features, and applications of the Vasicek model and the Gauss+ model.

Q745 AP Mock 1 Q14

REBUILT — Original options were vague and two were simple reversals; rebuilt as a concrete task-to-model assignment with realistic distractors.
A bank's quant team has two tasks: (i) a training exercise illustrating the basic effect of mean reversion on a zero-coupon bond price, and (ii) pricing interest rate caps and floors consistently with the market's hump-shaped volatility term structure. Which model assignment is most appropriate?
A. Vasicek for both tasks, because its single-factor closed-form structure is sufficient for any rate derivative.
B. Vasicek for task (i) and Gauss+ for task (ii).
C. Gauss+ for task (i) and Vasicek for task (ii), because the more complex model is needed to teach mean reversion.
D. Either model for both tasks, because both are calibrated to the same volatility structure.

Answer: B
Step by step
1. Task (i) is a simple application: the Vasicek model has one factor, mean reversion, and is used for simple zero-coupon pricing, basic hedging and teaching.
2. Task (ii) needs a model that can match a hump-shaped volatility term structure (low short end, high middle, declining long end) and derivative prices such as caps and floors.
3. Vasicek has only one volatility parameter and cannot do this; Gauss+ (three factors, separate volatilities, correlation ρ) can.
4. So Vasicek for (i), Gauss+ for (ii).

Why B is correct: Vasicek's simplicity suits basic illustration and simple pricing, while Gauss+'s multifactor structure is designed for market-consistent volatility and derivative pricing.
Why the others are wrong

- **A:** Ignores that Vasicek's single volatility parameter cannot reproduce the market volatility term structure needed for caps and floors.
- **C:** Reverses the roles: mean reversion is taught most easily in the simpler model, and Vasicek cannot price caps and floors consistently.
- **D:** Wrong: the models differ precisely in their volatility structure (one volatility parameter versus hump-shaped capability).

Q746

AnalystPrep Q.1661

The risk-neutral dynamics of the Vasicek model can be written as:
$dr = k(\theta - r)dt + \sigma dw$
Here, the constant θ represents the long-term value or the central propensity of the short-term rate in the risk-neutral process, while k represents the speed of mean reversion. What will happen if the difference between r and θ increases?

A. The greater the difference between r and θ, the greater the value of the short-term rate.
B. The greater the difference between r and θ, the greater the value of k.
C. The greater the difference between r and θ, the greater the expected change in the short-term rate towards θ.
D. The greater the difference between r and θ, the greater the expected change in the short-term rate towards r.

Answer: C

Step by step

1. The expected change in r over dt is the drift $k(\theta - r)\,dt$.
2. With k fixed, a larger gap $|\theta - r|$ makes the drift larger in magnitude.
3. The sign of the drift pushes r toward θ (up if $r < \theta$, down if $r > \theta$).
4. Hence a larger gap implies a larger expected move toward θ.

Why C is correct: The drift is proportional to the gap between θ and r, so the pull toward θ strengthens as the gap widens.

Why the others are wrong

- **A:** A gap can arise from r being far above or far below θ; a larger gap does not mean a higher rate level.
- **B:** k is a fixed model parameter and does not depend on the current rate.
- **D:** The pull is toward the long-run level θ, not toward the current rate r.

Q747

AnalystPrep Q.1663

REBUILT — Original did not state whether r0 was below or above θ and had a vague option referring to market volatility; rebuilt with explicit numbers.

In a Vasicek model with $dr = k(\theta - r)dt + \sigma dw$, the current short rate is 2.0%, $\theta = 5.0\%$ and $k = 0.50$ per year. Looking at the risk-neutral distributions of the short rate at increasing horizons, which observation about the mean of the distribution is most likely?

A. The mean stays at 2.0% at every horizon because the expected value of the random shock is zero.
B. The mean jumps immediately to 5.0% for any positive horizon and stays there.
C. The mean rises without bound as the horizon lengthens because volatility accumulates over time.
D. The mean rises gradually from 2.0%, to about 3.2% at one year and about 4.8% at five years, approaching 5.0% as the horizon grows.

Answer: D

Step by step

1. Expected rate at horizon T: $E[r_T] = \theta + (r_0 - \theta)e^{-kT}$.
2. T = 1: $5\% - 3\% \times e^{-0.5} = 5\% - 3\% \times 0.6065 = 3.18\%$.
3. T = 5: $5\% - 3\% \times e^{-2.5} = 4.75\%$; T = 10: 4.98%; T = 30: about 5.00%.
4. The mean therefore climbs gradually toward the limiting value $\theta = 5\%$.

Why D is correct: Mean reversion pulls the expected rate gradually from its current value toward θ, with the mean converging to θ as the horizon increases.

Why the others are wrong

- **A:** Ignores the drift term; only the shock has zero mean, while the drift $k(\theta - r)$ moves the mean.
- **B:** Mean reversion is gradual (speed k), not instantaneous.
- **C:** Confuses the mean with the dispersion: volatility widens the distribution, but the mean converges to θ.

Q748

AnalystPrep Q.1666

REBUILT — Original options were vague and one was a non-answer; rebuilt as a clear comparison of volatility term structures with and without mean reversion.

Two short-rate models are calibrated to the same current term structure: one has no mean reversion ($k = 0$) and the other has positive mean reversion. How do their implied term structures of par-rate volatility compare?

A. Without mean reversion the volatility term structure is flat (shocks shift all rates alike); with mean reversion it declines with maturity, so the two are materially different.

B. The two are essentially identical because the current term structure fixes the volatilities.

C. With mean reversion the volatility term structure is flat; without it, volatility declines with maturity.

D. With mean reversion, long-maturity par-rate volatility is higher than short-maturity volatility.

Answer: A

Step by step

1. Without mean reversion, a shock to the short rate persists and moves all rates by roughly the same amount, so par-rate volatility is about the same at every maturity (flat).
2. With mean reversion, short rates respond to current shocks but long rates reflect expected long-run reversion, so a shock affects long rates less.
3. Volatility therefore declines with maturity; the two volatility term structures differ markedly.

Why A is correct: Mean reversion dampens the effect of shocks on long-term rates, creating a downward-sloping volatility term structure that the no-reversion model cannot produce.

Why the others are wrong

- **B:** Matching the current rate curve does not pin down volatilities; k is a separate driver of the volatility structure.
- **C:** Reverses the effects of mean reversion.
- **D:** Mean reversion lowers, not raises, long-maturity volatility relative to short-maturity volatility.

Q749

AnalystPrep Q.1667

REBUILT — Original stem was unclear ('graphed against each other') and relied on a direction not covered by the Schweser text; rebuilt around the single-volatility-parameter limitation of Vasicek.

A risk manager calibrates the Vasicek model's volatility parameter so that the model matches the market volatility at one point on the interest rate curve. What is the most likely consequence, and how does Gauss+ differ?

A. The model will automatically match market volatility at all other maturities; Gauss+ gives no improvement.

B. The model will match other maturities only if the curve is hump-shaped; Gauss+ is calibrated to a single maturity as well.

C. The model will generally mismatch market volatility at other maturities, impairing derivative pricing and hedging; Gauss+, with several volatility parameters and a correlation, can replicate a hump-shaped volatility term structure.

D. The mismatch affects zero-coupon bond prices but not derivative prices; Gauss+ is no different in this respect.

Answer: C

Step by step

1. Vasicek has only one volatility parameter, so volatility can be fitted to only one point on the curve.
2. Volatilities at other maturities are then set by the model structure and will generally differ from the market's hump-shaped pattern.
3. Mismatched volatilities impair pricing of interest rate derivatives (caps, floors) and hedging accuracy.
4. Gauss+ has volatility parameters for several factors (σm, σl) and a correlation ρ, allowing it to replicate low short-end, higher mid-curve and declining long-end volatility.

Why C is correct: One volatility parameter fits one point; a multifactor structure with several volatility parameters can fit the market volatility term structure.

Why the others are wrong

- **A:** Claims the single parameter fits everything and denies the Gauss+ improvement; both are contrary to the reading.
- **B:** Mixes up the logic: the fit depends on the number of volatility parameters, and Gauss+ is not limited to one maturity.
- **D:** Volatility errors matter most for derivatives; Gauss+ was designed to address exactly this deficiency.

Q750 AnalystPrep Q.6521

An economist is comparing the foundational assumptions of the Vasicek model and the Gauss+ model in interest rate modeling. What is a key difference in how each model addresses the dynamic nature of interest rates?

A. The Vasicek model uses a multi-factor approach to capture complex rate dynamics.
B. The Gauss+ model assumes a single-factor stochastic process with constant short-term volatility.
C. The Vasicek model focuses solely on short-term mean reversion, while the Gauss+ incorporates multi-factor dynamics for short-, medium-, and long-term rates.
D. Both models assume a single, linear deterministic path for all interest rates.

Answer: C

Step by step

1. Vasicek: one factor, the short rate, which mean-reverts to a fixed long-run level θ.
2. Gauss+: three interacting factors (short-term rt, medium-term mt, long-term lt) in a cascade.
3. So Vasicek is single-factor, Gauss+ is multifactor (statement C).

Why C is correct: Vasicek is a single-factor short-rate model, whereas Gauss+ extends it to three cascading factors.

Why the others are wrong

- **A:** Reverses the models: Vasicek is the single-factor model.
- **B:** Gauss+ is three-factor, not single-factor, and does not fix volatility at one constant value.
- **D:** Both are stochastic models with random shocks, not deterministic paths.

Q751 AnalystPrep Q.6523

REBUILT — Correct option was vaguely worded and weakly distinguished; rebuilt to name the specific feature (separate factor volatilities and correlation giving hump-shaped volatility).

Which feature of the Gauss+ model best explains why it captures observed market volatility better than the Vasicek model?

A. Its volatility parameters for the several factors and the correlation ρ allow low short-end volatility, higher mid-curve volatility and declining long-end volatility (a hump shape).
B. It fixes one volatility level across all maturities, making the volatility structure easier to calibrate.
C. It contains stochastic volatility, which the Vasicek model lacks.
D. It replaces random shocks with deterministic paths, removing volatility estimation error.

Answer: A

Step by step

1. Vasicek has one volatility parameter, so volatility is calibrated to a single point.
2. Gauss+ has three factors with separate volatilities (σm, σl) and a correlation ρ between mt and lt shocks.
3. This lets it be calibrated to a hump-shaped volatility term structure, which matches historical norms.

Why A is correct: Multiple factor volatilities plus a correlation give Gauss+ the flexibility to replicate the hump-shaped volatility structure of the market.

Why the others are wrong

- **B:** Describes the Vasicek limitation, not a Gauss+ advantage.
- **C:** Neither model has stochastic volatility; both use constant volatility parameters.
- **D:** Gauss+ remains a stochastic model; the shocks are not removed.

Q752 AnalystPrep Q.6524

REBUILT — Original correct option was vague ('nuanced interactions') and the distractors were nonsense; rebuilt around the cascading, dynamic mean-reversion targets.

Which statement correctly contrasts the mean reversion dynamics of the Vasicek and Gauss+ models?

A. Both models revert to a single fixed long-run rate; Gauss+ merely adds more noise.
B. In Vasicek the target moves dynamically, while in Gauss+ the target is fixed.
C. In Gauss+ all three factors revert to the fixed parameter μ at the same speed.
D. In Vasicek the short rate reverts to a fixed θ; in Gauss+ r reverts to m, m reverts to l, and l reverts to a fixed μ, with speeds fastest for r and slowest for l.

Answer: D

Step by step

1. Vasicek: $dr = k(\theta - r)dt + \sigma dw$, with θ a fixed target.
2. Gauss+: cascade — rt seeks mt, mt seeks lt, and lt seeks μ; only μ is fixed, the others are moving targets.
3. Reversion speeds: αr is fastest and αl slowest, so the short rate reacts quickly while the long-term factor shifts slowly.

Why D is correct: Gauss+ replaces a single fixed target with a cascade of moving targets at different speeds, giving richer long-horizon dynamics.

Why the others are wrong

- **A:** Gauss+ changes the mean-reversion structure (cascade, moving targets), not just the noise.
- **B:** Reverses the models.
- **C:** Factors revert to each other, not directly to μ, and at different speeds.

Q753

AnalystPrep Q.6525

REBUILT — Original options were vague and partly nonsensical; rebuilt around the specific Vasicek limitation affecting hedging and application.

Which limitation of the Vasicek model most directly reduces its usefulness for hedging a fixed-income portfolio across the yield curve?

A. It cannot price zero-coupon bonds in closed form.

B. It offers little flexibility in the shape of the curve, so it cannot match market rates accurately enough for effective hedging.

C. It assumes volatility is stochastic, making hedge ratios unstable.

D. It has no mean reversion, so rates drift without anchor.

Answer: B

Step by step

1. Vasicek has one factor and one volatility parameter, so the curve shapes it can produce are restricted.
2. Accurate hedging needs the model to match market rates and volatilities across maturities.
3. Vasicek's limited flexibility prevents this; Gauss+ mitigates it with three factors.

Why B is correct: A single factor with limited curve-shape flexibility cannot match market rates well enough for effective hedging.

Why the others are wrong

- **A:** Vasicek does provide closed-form zero-coupon prices; that is one of its uses.
- **C:** Vasicek assumes constant volatility, not stochastic volatility.
- **D:** Mean reversion is its defining feature.

Q754

BT answers P2.T5.25.7.2 · BT notes P2.T5.25.7.2

REBUILT — The source relied on unreadable chart images (Blue/Yellow/Green lines); replaced the charts with a table of volatility by maturity for three curves, same concept tested.

The table shows annualized rate volatility (%) by maturity for three candidate volatility term structures.

Maturity (years)	1	3	5	7	10	14
Curve X	1.81	1.48	1.21	0.99	0.74	0.49
Curve Y	1.80	2.20	2.40	2.00	1.00	0.60
Curve Z	0.80	1.20	1.50	1.80	2.10	2.40

Which assignment correctly matches the curves to the volatility term structures produced by the Vasicek and Gauss+ models?

A. Curve X is Vasicek; Curve Y is Gauss+.

B. Curve Y is Vasicek; Curve X is Gauss+.

C. Curve Z is Vasicek; Curve Y is Gauss+.

D. Curve X is Vasicek; Curve Z is Gauss+.

Answer: A

Step by step

1. Vasicek has a single volatility parameter and a single mean-reverting factor, so its volatility term structure declines monotonically with maturity (e.g. $\sigma(t) = 2\%\, e^{-0.1t}$ gives 1.81, 1.48, 1.21, 0.99, 0.74, 0.49). This matches Curve X exactly.

2. Gauss+ has short, medium and long-term factors, so it can be calibrated to a hump: low at the short end, peaking at intermediate maturities, then declining. Curve Y rises from 1.80 to 2.40 at 5 years and then falls to 0.60, which is that hump.
3. Curve Z rises with maturity throughout; neither model is described in the reading as producing this shape.
4. Correct pairing: X = Vasicek, Y = Gauss+.

Why A is correct: Vasicek's one-factor, one-volatility structure only generates a monotonically declining volatility curve, whereas Gauss+'s extra medium-term factor lets it replicate the empirically observed hump-shaped volatility term structure.

Why the others are wrong

- **B:** Reverses the models: assigns the hump to Vasicek, which cannot produce one, and the monotone decline to the richer Gauss+ model.
- **C:** Picks the steadily rising Curve Z for Vasicek; Vasicek's volatility decays with maturity, it does not rise.
- **D:** Gets Vasicek right but assigns the monotonically rising Curve Z to Gauss+, missing that Gauss+ is designed to give a hump (rise then decline), not a perpetual rise.

Q755

Schweser Module Quiz 16.1 Q3

What is the main difference or similarity in the volatility assumption between the Vasicek model and the Gauss+ model?

A. Both models assume constant volatility.
B. Both models have stochastic volatility components.
C. The Vasicek model incorporates stochastic volatility, while the Gauss+ model assumes deterministic volatility.
D. The Vasicek model assumes constant volatility, while the Gauss+ model can replicate hump-shaped volatility term structures.

Answer: D

Step by step

1. Vasicek has a single, constant volatility parameter, so volatility is calibrated to only one point on the curve and the volatility term structure cannot be hump-shaped.
2. Gauss+ adds medium-term and long-term factors, so it can be calibrated to low short-end, higher intermediate and declining long-end volatility (hump-shaped), as seen empirically.
3. Only option D states both facts correctly.

Why D is correct: Vasicek's single volatility parameter cannot match the observed hump-shaped volatility term structure; the multi-factor Gauss+ model can.

Why the others are wrong

- **A:** Wrongly extends Vasicek's constant-volatility parameter to Gauss+, ignoring that the extra factors give Gauss+ a maturity-dependent volatility structure.
- **B:** Neither model has a stochastic-volatility component; both are Gaussian with constant factor volatilities, the difference lies in the number of factors.
- **C:** Inverts the facts: Vasicek does not have stochastic volatility, and Gauss+ is more flexible, not less.

LO 16.c

Calculate changes in the short-term, medium-term, and long-term interest rate factors under the Gauss+ model.

Q756

AP Mock 2 Q71

REBUILT — Original distractors were arbitrary numbers with no realistic error behind them; replaced with options produced by sign, omitted-dt and wrong-level mistakes.

In the Gauss+ model the short-term factor follows $dr_t = a_r(m_t - r_t)\,dt$ plus a shock term that has zero expected value. Given a_r = 0.6, m_t = 0.027, r_t = 0.020 and dt = 0.01, what is the expected change in the short-term factor over the time step?

A. 0.000042
B. -0.000042
C. 0.004200
D. 0.000162

Answer: A

Step by step

1. The shock has zero mean, so only the drift matters: $E[dr_t] = a_r(m_t - r_t)\,dt$.
2. Gap: $m_t - r_t = 0.027 - 0.020 = 0.007$.
3. $E[dr_t] = 0.6 \times 0.007 \times 0.01 = 0.000042$ (0.42 bp); r is below m, so it drifts up.

Why A is correct: The short rate is pulled toward the medium-term factor in proportion to the gap, the speed and the time step; the zero-mean shock adds nothing to the expected change.

Why the others are wrong

- **B:** Sign error: used $r_t - m_t$ so the drift points away from the medium-term factor instead of toward it.
- **C:** Forgot to multiply by dt: 0.6 x 0.007 = 0.0042 is a per-year change, not a per-step change.
- **D:** Used the level of the medium-term factor (0.6 x 0.027 x 0.01) instead of the gap $m_t - r_t$.

Q757

AnalystPrep Q.6529

REBUILT — Stem asked for the 'expected' change but the key included the realized shock; reworded to ask for the change given the realized shock and rebuilt distractors.

In the Gauss+ model the long-term factor follows $dl_t = a_l(\mu - l_t)\,dt + \sigma_l\,dW_l$. Given a_l = 0.3, l_t = 0.045, μ = 0.04, σ_l = 0.025, a realized shock dW_l = 0.03 and dt = 0.01, what is the change in the long-term factor over the step?

A. 0.000735
B. -0.000015
C. 0.000750
D. 0.000765

Answer: A

Key correction: Source key (A) was numerically right but its stem asked for the 'expected' change, which would be only the drift (-0.000015); the stem is reworded to the realized change so 0.000735 is correct.

Step by step

1. Drift: $a_l(\mu - l_t)dt = 0.3 \times (0.04 - 0.045) \times 0.01 = -0.000015$.
2. Shock: $\sigma_l\,dW_l = 0.025 \times 0.03 = 0.00075$ (dW_l is already the realized shock, so no extra $\sqrt{dt}$).
3. Total: $-0.000015 + 0.00075 = 0.000735$.

Why A is correct: The change is the mean-reversion drift (negative here because l_t > μ) plus volatility times the realized shock.

Why the others are wrong

- **B:** Drift only: omitted the stochastic shock term.
- **C:** Shock only: omitted the mean-reversion drift.
- **D:** Sign error on the drift: used $l_t - \mu$, giving +0.000015 + 0.00075.

Q758

BT answers P2.T5.25.7.3 · BT notes P2.T5.25.7.3

REBUILT — Source key (2.40%) required treating 0.67 as a half-life, contradicting the stem; made the constant-m assumption explicit, used a_r = 0.67 literally and rebuilt options.

In the Gauss+ model the short-term rate adjusts deterministically toward the medium-term factor: $dr_t = -a_r(r_t - m_t)\,dt$. Assume m is held constant at 3%, r_0 = 2%, and a_r = 0.67 per year. What is the expected short-term rate after 6 months (use the exact continuous-time solution)?

A. 2.28%
B. 2.335%
C. 1.60%
D. 2.40%

Answer: A

Key correction: Source key (B, 2.40%) treated 0.67 as a half-life (a_r = ln2/0.67 = 1.034), which contradicts the stem calling 0.67 the mean reversion parameter; with a_r = 0.67 the answer is 2.28%.

Step by step

1. With m constant the ODE solves to $r_t = m + (r_0 - m)e^{-a_r t}$.
2. $e^{-0.67 \times 0.5} = e^{-0.335} = 0.7153$.
3. $r_{0.5} = 3\% + (2\% - 3\%) \times 0.7153 = 3\% - 0.7153\% = 2.28\%$.

Why A is correct: The gap to m decays exponentially at rate a_r, so after 0.5 years about 71.5% of the 1% gap remains.

Why the others are wrong

- **B:** One-step Euler approximation: 2% + 0.67 x 1% x 0.5 = 2.335%; ignores the compounding of the decay.
- **C:** Positive exponent: 3% - 1% x e^{+0.335} = 1.60%, which makes the gap grow instead of shrink.
- **D:** Treated 0.67 as a half-life (a_r = ln 2 / 0.67 = 1.034), which is not what the stem states.

Q759

BT answers P2.T5.25.8.1 · BT notes P2.T5.25.8.1

REBUILT — Stem called the fixed-shock term a 'standard deviation' (it is the realized diffusion term given dW_1 = sqrt(dt), dW_2 = 0) and the distractors were near-duplicates; reworded and rebuilt options.

The medium-term factor in the Gauss+ model follows $dm_t = -a_m(m_t - l_t)\,dt + \sigma_m(\rho\,dW_1 + \sqrt{1-\rho^2}\,dW_2)$. Given σ_m = 0.01 per year, ρ = 0.6, dt = 1/252, and a day in which dW_1 = √dt and dW_2 = 0, what is the size of the stochastic (diffusion) contribution to the daily change in m_t?

A. 0.0378%
B. 0.0630%
C. 0.0504%
D. 0.6000%

Answer: A

Key correction: BT notes keyed A (0.0374%) and BT answers keyed C (0.0378%) for the same stem; the correct value is 0.0378% (0.0374% is a miscalculation).

Step by step

1. $\sqrt{dt} = \sqrt{1/252} = 0.06299$.
2. Diffusion $= \sigma_m(\rho\,dW_1 + \sqrt{1-\rho^2}\cdot 0) = 0.01 \times 0.6 \times 0.06299$.
3. $= 0.000378 = 0.0378\%$.

Why A is correct: With dW_2 = 0 only the rho-weighted dW_1 term remains, scaled by sigma_m and the square root of dt.

Why the others are wrong

- **B:** Dropped rho: 0.01 x 0.06299 = 0.0630%.
- **C:** Used sqrt(1 - rho^2) = 0.8 instead of rho = 0.6 (the weight of the dW_2 term): 0.01 x 0.8 x 0.06299 = 0.0504%.
- **D:** Forgot the sqrt(dt) scaling: 0.01 x 0.6 = 0.6%, an annual-size shock.

Q760

BT notes P2.T5.25.7.1

REBUILT — Original asked which factor has the 'least volatility' but its answer rested on reversion speeds (half-lives), not volatility, and was ambiguous; rebuilt as a drift-change calculation using the reversion speeds.

In the Gauss+ model, $dr_t = -a_r(r_t - m_t)dt$, $dm_t = -a_m(m_t - l_t)dt + \ldots$, $dl_t = -a_l(l_t - \mu)dt + \ldots$, with a_r = 0.5, a_m = 0.2, a_l = 0.05 and μ = 5%. Currently r = 2%, m = 3%, l = 4%. Setting all random shocks to zero and using dt = 0.25 years, which factor has the smallest change over the step, and what is it?

A. Short-term factor, +0.125%
B. Medium-term factor, +0.050%
C. Long-term factor, +0.0125%
D. None; each gap is 1% so all three factors change by the same amount

Answer: C

Step by step

1. Each gap is 1% (m - r, l - m and mu - l), so the differences come only from the speeds.
2. $dr = 0.5 \times 1\% \times 0.25 = +0.125\%$.
3. $dm = 0.2 \times 1\% \times 0.25 = +0.050\%$; $dl = 0.05 \times 1\% \times 0.25 = +0.0125\%$.
4. The long-term factor, with the slowest reversion speed, moves least.

Why C is correct: With equal gaps the change is proportional to the reversion speed, and a_l is the smallest, so the long-term factor moves least (+0.0125%).

Why the others are wrong

- **A:** This is the largest change (fastest speed a_r), not the smallest.
- **B:** The medium-term speed gives +0.050%, larger than the long-term change.
- **D:** Ignores that the changes scale with the reversion speeds even though the gaps are equal.

Q761 Schweser Module Quiz 16.1 Q4

Assume the short-term rate is 5.0% and the medium-term rate is 6.0%. By using the Gauss+ model, if the corresponding mean reversion speed is 1.3 and the time step is three months, what is the change in the short-term rate?

A. -0.325%.
B. +0.325%.
C. -3.250%.
D. +3.250%.

Answer: B
Step by step
1. $dr_t = -a_r(r_t - m_t)\,dt$ with dt = 0.25.
2. $r_t - m_t = 5.0\% - 6.0\% = -1.0\%$.
3. $dr_t = -1.3 \times (-1.0\%) \times 0.25 = +0.325\%$.

Why B is correct: The short rate is below the medium-term factor so it drifts up by speed x gap x time step.
Why the others are wrong
- **A:** Sign error: treated the rate as moving away from the medium-term factor.
- **C:** Wrong sign and a decimal slip (3.250% instead of 0.325%).
- **D:** Magnitude error (ten times too large) with the correct direction.

LO 16.d

Explain how the parameters of the Gauss+ model can be estimated from empirical data.

Q762 Schweser Module Quiz 16.1 Q5

When using empirical data to estimate parameters for the Gauss+ model, which of the following parameters is calibrated by minimizing the squared errors of observed yields relative to model-predicted yields?

A. Long-term target.
B. Volatility parameter.
C. Correlation parameter.
D. Mean reversion speeds.

Answer: A
Step by step
1. The simplified Gauss+ estimation first sets the mean reversion speeds $\alpha_r, \alpha_m, \alpha_l$ from regression sensitivities.
2. Volatilities and correlation σ_m, σ_l, ρ are then fitted to the observed volatility term structure.
3. The last step finds the long-term reversion level μ by minimizing squared errors between observed and model yields.
4. So the long-term target is the parameter calibrated by least squares on yields.

Why A is correct: The long-term target μ is the only parameter pinned down by fitting model yields to observed yields across maturities.
Why the others are wrong
- **B:** Volatility parameters are fitted to the volatility term structure, not to yield levels.
- **C:** The correlation ρ is also matched to the volatility term structure.
- **D:** Mean reversion speeds are set from regression coefficients of rate changes, not from yield-level errors.

Q763 AnalystPrep Q.6526

REBUILT — Original had weak, giveaway distractors (fixed average, snapshots, median); rebuilt as a parameter-to-method matching item with plausible mismatches.

In the simplified procedure for estimating the Gauss+ model from empirical data, which of the following correctly matches a parameter with its estimation method?

A. Mean reversion speeds are found by minimizing squared errors between observed and model-implied yields.
B. The volatility and correlation parameters are found by regressing changes in the short-term rate on changes in medium- and long-term rates.
C. The long-term reversion level μ is found by minimizing squared errors between observed and model-implied

yields across maturities.

D. Mean reversion speeds are found by matching the observed volatility term structure.

Answer: C

Step by step

1. Mean reversion speeds: regression of short-rate changes on medium/long-rate changes (Steps 2-3).
2. Volatility and correlation σ_m, σ_l, ρ: replicate the volatility term structure (Steps 4-5).
3. μ: minimize squared errors of observed versus model yields (Step 6).
4. Only C matches a parameter to its correct method.

Why C is correct: μ is the last parameter calibrated, by least squares on yields over the whole term structure, so the model aligns with market expectations and risk premiums.

Why the others are wrong

- **A:** Swaps the methods: speeds come from regression sensitivities, not yield-error minimization.
- **B:** Swaps the methods: the regression gives mean reversion speeds, while volatility and correlation fit the volatility term structure.
- **D:** Mean reversion speeds do not depend on volatility, so matching the volatility term structure cannot identify them.

Q764

AnalystPrep Q.6527

REBUILT — Original asked for 'daily forward rates' as a proxy for m_t, which the reading does not support; rebuilt around the reading's regression step and why speeds can be estimated before volatilities.

In the simplified approach to estimating the Gauss+ model, mean reversion speeds are first obtained by regressing changes in a short-term rate proxy on changes in a medium-term rate and a longer-term rate. Why can the speeds be estimated at this stage without first knowing the volatility parameters?

A. The regression coefficients depend only on the volatilities and correlation, which are observed directly.

B. The regression coefficients depend only on the mean reversion parameters, not on the volatility parameters.

C. The regression coefficients depend only on the long-term reversion level μ.

D. The regression coefficients depend on both the speeds and the volatilities, so the volatilities are first set to one.

Answer: B

Step by step

1. The model's yield sensitivities to the factors are functions of the reversion speeds $\alpha_r, \alpha_m, \alpha_l$ and maturity.
2. Volatility parameters scale the size of factor moves but do not change these sensitivities.
3. Hence regression coefficients identify the speeds alone; volatilities and correlation are estimated afterward (Steps 4-5).

Why B is correct: The regression coefficients are determined only by the mean reversion speeds, so the speeds can be calibrated first and independently.

Why the others are wrong

- **A:** Reverses the dependence: volatilities do not affect the coefficients.
- **C:** μ shifts yield levels and is calibrated last by least squares; it does not drive sensitivities of changes.
- **D:** Coefficients do not depend on volatilities, and no normalization to one is used.

Q765

BT answers P2.T5.25.8.2 · BT notes P2.T5.25.8.2

In the Gauss+ model, the long-term factor l_t is assumed to follow an Ornstein-Uhlenbeck process: $dl_t = -\alpha_l(l_t - \mu)dt + \sigma_l dW_t^l$, so that deviations from its long-run target decay exponentially: $l(t) - \mu = (l(0) - \mu)e^{-\alpha_l t}$. Taking logs gives $\ln|l(t) - \mu| = \ln|l(0) - \mu| - \alpha_l t$. A regression of $\ln|l(t) - \mu|$ on time t (in years) using observed yield curve data produced a slope of -0.015 per year. How long will it take for the deviation of the long-term factor from μ to decay by half?

A. -0.015

B. 0.015

C. -4.20

D. 46.21

Answer: D

Step by step

1. Slope $= -\alpha_l = -0.015$, so $\alpha_l = 0.015$ per year.
2. Half-life: $e^{-\alpha_l h} = 0.5 \Rightarrow h = \ln 2/\alpha_l$.
3. $h = 0.6931/0.015 = 46.21$ years.

Why D is correct: The deviation decays at rate α_l, so the half-life is $\ln 2/\alpha_l$.

Why the others are wrong

- **A:** This is the regression slope, not a time.
- **B:** This is the mean reversion speed α_l; the half-life step was skipped.
- **C:** This is ln(0.015), a meaningless transformation of the speed.

Q766

BT answers P2.T5.25.8.3 · BT notes P2.T5.25.8.3

REBUILT — Original relied on an unseen scatter plot/regression exhibit and an unstated sensitivity formula, with an ambiguous 'None of the above' option; rebuilt as a self-contained calculation.

In the simplified Gauss+ estimation, a regression of weekly changes in the 1-year zero-coupon yield on weekly changes in the short-term rate gives a slope coefficient of 0.8625. Assume the model sensitivity of the yield change to the short-term rate at maturity T is $e^{-\alpha_r T}$. Which is closest to the half-life of the short-term factor, in years?

A. 0.8625
B. 4.69
C. -0.1479
D. 0.1479

Answer: B

Step by step

1. With $T = 1$: $e^{-\alpha_r} = 0.8625$, so $\alpha_r = -\ln(0.8625) = 0.1479$ per year.
2. Half-life $= \ln 2/\alpha_r = 0.6931/0.1479 = 4.69$ years.

Why B is correct: The regression slope identifies $e^{-\alpha_r}$; converting to the speed and then to $\ln 2/\alpha_r$ gives the half-life.

Why the others are wrong

- **A:** This is the raw regression slope, not a time.
- **C:** This is ln(0.8625) with the sign wrong and no half-life step.
- **D:** This is the speed α_r; forgot to compute $\ln 2/\alpha_r$.

Reading 17. Volatility Smiles and Volatility Surfaces

John C. Hull, Options, Futures, and Other Derivatives, 11th Edition (New York, NY: Pearson, 2022). Chapter 20.

LO	Learning objective	Qs
17.a	describe a volatility smile and volatility skew.	5
17.b	explain the implications of put-call parity on the implied volatility of call and put options.	16
17.c	compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset.	17
17.d	describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility.	6
17.e	describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape.	14
17.f	describe alternative ways of characterizing the volatility smile.	2
17.g	describe volatility term structures and volatility surfaces and how they may be used to price options.	12
17.h	explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures.	8
17.i	explain the impact of a single asset price jump on a volatility smile.	3

LO 17.a

Describe a volatility smile and volatility skew.

Q767 GARP book Chapter 17 Q17.1

What is a volatility smile?

Answer: see model answer — A volatility smile is the pattern obtained by plotting the Black-Scholes-Merton implied volatility of options with the same underlying and same maturity against their strike price (or strike/spot ratio). If the Black-Scholes-Merton model held exactly, implied volatility would be the same at every strike and the plot would be flat. A smile (U-shape, typical of currency options) means implied volatility is higher for deep in- and out-of-the-money options; for equity options the pattern is a downward-sloping 'smirk' or skew, with higher implied volatility at low strikes.

Step by step

1. Compute the implied volatility from market prices of options with the same maturity at different strikes.
2. Plot implied volatility against strike price (or strike divided by stock price).
3. Compare with the flat line that lognormal BSM assumptions would imply; a curved shape is the smile, a downward-sloping one the skew.

Why this is correct: The smile is the departure of implied volatility from a constant across strikes, showing that market prices are inconsistent with a lognormal distribution.

Q768 AnalystPrep Q.1706

Option traders in equity and foreign currency markets often refer to a "volatility smile." Which of the following statements gives the correct definition of a volatility smile?

A. The plot of the implied volatility of financial markets that determines option prices.
B. The plot of the implied volatility of an option with a given maturity as a function of its strike price.
C. The plot of the implied volatility of an option with an infinite life as a function of its current price.
D. The pattern of changes in historical volatility across trading days for options with different expiration dates.

Answer: B

Step by step

1. A volatility smile is built from implied volatilities, backed out of observed option prices with the BSM model.
2. Holding maturity fixed, implied volatility is computed for a range of strikes.
3. Implied volatility is then plotted against strike (or strike/stock price); a non-flat curve is the smile or skew.
4. Only B describes this construction.

Why B is correct: A smile is implied volatility as a function of strike for options of a fixed maturity.

Why the others are wrong

- **A:** Too vague: it makes implied volatility a property of the market overall rather than a function of strike for a

given maturity, so there is no curve to plot.
- **C:** A smile requires a finite, fixed maturity, and the x-axis is the strike (or strike/spot), not the option's price.
- **D:** Confuses implied with historical volatility and plots against time/expiry rather than against strike.

Q769 BT notes P2.T5.01.1

In a plot of the volatility smile, what are each of the axes, and how is the function line plotted?

Answer: see model answer — The vertical axis is the Black-Scholes-Merton implied volatility; the horizontal axis is the option's strike price, often expressed as the ratio of strike to stock price (K/S_0) as in Schweser. The line is plotted by taking options on the same underlying with the same maturity, backing out the implied volatility at each strike from market prices, and joining these points. Under constant volatility the line would be flat; in practice it is curved (currency smile) or downward-sloping (equity skew).

Step by step
1. Y-axis: implied volatility from BSM.
2. X-axis: strike price, or strike/stock price ratio.
3. For each strike, invert BSM using the market price to get the implied volatility, holding maturity fixed.
4. Connect the points to form the smile curve.

Why this is correct: The smile is a function mapping strike to implied volatility at a fixed maturity.

Q770 BT notes P2.T5.01.3

REBUILT — True/false item with no options converted to a four-option MCQ with a single defensible answer.

Two traders look at the same European option on the same underlying, with the same market price, strike and maturity. Trader 1 uses the Black-Scholes-Merton model and Trader 2 uses a different option pricing model. Which statement about the implied volatility each trader calculates, and hence the volatility smile each plots, is most accurate?

A. The implied volatilities must be identical, because they are backed out of the same market price.

B. The implied volatilities can differ, because implied volatility depends on the pricing model used to invert the market price.

C. The implied volatilities must be identical, because volatility is a property of the underlying asset alone.

D. The implied volatilities can differ only if the traders use different historical data windows.

Answer: B

Step by step
1. Implied volatility is the volatility input that makes a model price equal the market price.
2. Different models map volatility to price differently, so the same market price implies a different volatility under each model.
3. Hence the smile each trader plots is model-dependent; traders agree only if they use the same model.

Why B is correct: Implied volatility is model-dependent, so it need not be the same for all market participants.

Why the others are wrong
- **A:** Treats the common market price as sufficient; the inversion also depends on the model used.
- **C:** Confuses implied volatility with a single objective property of the asset; it is a model-specific quote.
- **D:** Mixes up implied with historical volatility; implied volatility uses no historical data window.

Q771 BT answers L2.T5.06.3 · BT notes P2.T5.05.02

REBUILT — True/false item (BT notes version had no answer) converted to a four-option MCQ with the reasoning made explicit; remapped to 17.a since it tests what a volatility smile is.

An analyst notes that a plot of option vega against the underlying stock price is hump-shaped: vega is highest for at-the-money options and lower for out-of-the-money options. Which statement about whether this plot is consistent with the volatility smile is most accurate?

A. It is consistent with the smile, because the smile also shows the largest values for at-the-money options.

B. It is not the smile: vega is a model-based sensitivity of option value to volatility, whereas the smile plots implied volatility against strike (or K/S).

C. It is consistent with the smile, because high vega at-the-money means at-the-money options must have the highest implied volatility.

D. It is not the smile, because under Black-Scholes-Merton vega would be constant across strikes, while implied volatility would be hump-shaped.

Answer: B

Key correction: Source key (BT answers L2.T5.06.3) said False, which is the same conclusion as option B; the BT notes version gave no key.

Step by step

1. The volatility smile plots implied volatility on the vertical axis against strike price or K/S on the horizontal axis.
2. The vega curve plots the partial derivative of the option value with respect to volatility against the stock price; it is computed from the model, not observed from market prices.
3. If BSM held exactly, implied volatility would be flat across strikes (no smile), yet the vega curve would still be hump-shaped.
4. Hence the hump in vega says nothing about the smile; the statement is false and B describes why.

Why B is correct: Vega is a model sensitivity and the smile is a plot of market implied volatilities; they are different objects, and a flat implied-volatility profile coexists with a hump-shaped vega.

Why the others are wrong

- **A:** Confuses the two plots by matching only the 'highest at-the-money' feature; the smile does not have that shape and is not a vega plot.
- **C:** Infers that large vega implies large implied volatility; vega measures sensitivity to volatility, not the level of implied volatility.
- **D:** Reverses the facts: under BSM implied volatility is flat while vega remains hump-shaped, not constant.

LO 17.b

Explain the implications of put-call parity on the implied volatility of call and put options.

Q772 GARP book Ch.17 Q17.8

A European call and put option have the same strike price and time to maturity. The call has an implied volatility of 30% and the put has an implied volatility of 25%. What trades would you do?

Answer: see model answer — Put-call parity requires a European call and put with the same strike and maturity to have the same implied volatility, so the call is overpriced relative to the put. Sell the call and buy the put (a short synthetic forward), and hedge by buying the underlying stock financed by borrowing the present value of the strike. At expiry the position is worth zero in every state, so the price gap is a riskless profit.

Step by step

1. Parity: $c - p = S_0 - Ke^{-rT}$ holds for market prices and for BSM prices, so $c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$.
2. Equal deviations imply equal implied volatilities for the call and the put.
3. Here 30% > 25%: the call is expensive relative to the put.
4. Trade: sell call, buy put, buy stock, borrow Ke^{-rT}; the payoff cancels at expiry and the price discrepancy is locked in.

Why this is correct: No-arbitrage parity is model-free, so differing implied volatilities for same-strike, same-maturity European options signal a mispricing that can be locked in.

Q773 GARP book Ch.17 Q17.10

The market price of a European call is \$3.00 and its price given by the Black-Scholes-Merton model with a volatility of 30% is \$3.50. The price given by this Black-Scholes-Merton model for a European put option with the same strike price and time to maturity is \$1.00. What should the market price of the put option be? Explain the reasons for your answer.

Answer: see model answer — The market put price should be \$0.50. Put-call parity holds for both market and BSM prices, so the call and put must deviate from their BSM values by the same amount: $p_{mkt} - p_{BSM} = c_{mkt} - c_{BSM} = 3.00 - 3.50 = -0.50$. Hence $p_{mkt} = 1.00 - 0.50 = 0.50$; otherwise there is an arbitrage (equivalently, call and put must share the same implied volatility).

Step by step

1. Parity for market prices: $c_{mkt} - p_{mkt} = S_0 - Ke^{-rT}$; for BSM prices the same identity holds with c_{BSM}, p_{BSM}.
2. Subtract: $c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$.
3. Call deviation $= 3.00 - 3.50 = -0.50$.
4. $p_{mkt} = 1.00 - 0.50 = 0.50$.

Why this is correct: Parity forces the pricing error in a European call and put to be identical.

Q774 GARP book Ch.17 Q17.17

A European call option on a certain stock has a strike price of $30, a time to maturity of 1 year, and an implied volatility of 30%. A European put option on the same stock has a strike price of $30, a time to maturity of 1 year, and an implied volatility of 33%. What is the arbitrage opportunity open to a trader? Does the arbitrage work only when the lognormal assumption underlying Black-Scholes-Merton holds? Explain carefully the reasons for your answer.

Answer: see model answer — The put (33%) is overpriced relative to the call (30%). Sell the put, buy the call, short the stock and lend the present value of $30 at the risk-free rate (selling the put and buying the call is a long synthetic forward, hedged by shorting the stock and lending PV(K)); this locks in a riskless profit equal to the price difference. The arbitrage does not depend on the lognormal assumption: put-call parity is a model-free no-arbitrage relation, so the implied volatilities must be equal whatever the true distribution. The lognormal assumption only matters for how a volatility maps into a price, not for the call/put relationship.

Step by step

1. Parity requires equal implied volatilities for call and put (same K, T); here 30% vs 33%.
2. Higher implied volatility means higher BSM price: the put is relatively overpriced.
3. Trade: sell put, buy call, short stock, lend $30e^{-rT}$; payoff at T is zero in all states.
4. Parity uses only payoffs at expiry ($c - p = S_T - K$), not the distribution of S_T, so the arbitrage works for any price distribution.

Why this is correct: The arbitrage rests on payoff replication, which is independent of any distributional assumption.

Q775 GARP book Ch.17 Q17.26

Consider a European call and a European put with the same strike price and time to maturity. Show that they change in value by the same amount when the volatility increases from a level Ïƒ1 to a new level Ïƒ2 within a short period of time. (Hint: Use put-call parity.)

Answer: see model answer — Parity gives $c + Ke^{-rT} = p + S_0$ (no dividends), so $c - p = S_0 - Ke^{-rT}$. The right-hand side does not depend on volatility. Therefore, when volatility moves from Ïƒ1 to Ïƒ2 (with S0, K, r, T unchanged over the short period), $c_2 - p_2 = c_1 - p_1$, i.e. $c_2 - c_1 = p_2 - p_1$: the call and put change in value by the same amount (both vegas are equal).

Step by step

1. Parity at volatility Ïƒ1: $c_1 - p_1 = S_0 - Ke^{-rT}$.
2. Parity at Ïƒ2: $c_2 - p_2 = S_0 - Ke^{-rT}$, same right-hand side because volatility does not enter it.
3. Subtract: $c_2 - c_1 = p_2 - p_1$.

Why this is correct: The parity difference c - p is independent of volatility, so any volatility-driven change in c must equal that in p.

Q776 AnalystPrep Q.1707

REBUILT — Original option D (smile most relevant near the money) was a loosely true statement, giving two defensible answers; replaced with a clearly false distractor.

Which of the following statements about the implied volatility of European options is correct?

A. For the same strike and maturity, the implied volatility of a European call equals that of a European put, so their volatility smiles are identical.

B. For the same strike and maturity, the implied volatility of a European put is higher than that of a European call whenever the underlying price distribution is not lognormal.

C. A European call with a shorter maturity has the same implied volatility as a European put with a longer maturity at the same strike.

D. For the same strike and maturity, the implied volatility of a European call is higher than that of a European put

because calls have unlimited upside.

Answer: A

Step by step

1. Put-call parity $c - p = S_0 - Ke^{-rT}$ holds for market and BSM prices alike, so $c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$.
2. The volatility that sets $c_{BSM} = c_{mkt}$ therefore also sets $p_{BSM} = p_{mkt}$.
3. Hence call and put implied volatilities coincide for the same K and T: A.

Why A is correct: Parity is a no-arbitrage identity, so same-strike, same-maturity European calls and puts must share one implied volatility, whatever the true distribution.

Why the others are wrong

- **B:** Parity is model-free and holds for non-lognormal distributions too; a gap would create an arbitrage.
- **C:** Parity links options with the same maturity; different maturities generally have different implied volatilities (term structure).
- **D:** Confuses payoff asymmetry with implied volatility; parity forces equal implied volatilities.

Q777

AnalystPrep Q.1708

REBUILT — Original distractors were not tied to specific errors (one, 0.0419, matched none); replaced with error-based options and the stem made explicit about continuous rates.

The spot value of a foreign currency is $0.55. The risk-free interest rates are 4% per year in the U.S. and 8% per year in the foreign country (continuously compounded). A European call on the currency with a 1-year maturity and a strike of $0.57 has a market price of $0.0325 (implied volatility 14.5%). Using put-call parity with the foreign risk-free rate as the continuous yield, what is the price of the European put with the same strike and maturity?

A. $0.0302
B. $0.0724
C. $0.0525
D. $0.0517

Answer: B

Step by step

1. Parity for currency options: $p = c + Ke^{-r_d T} - S_0 e^{-r_f T}$.
2. $Ke^{-r_d T} = 0.57e^{-0.04} = 0.54761$; $S_0 e^{-r_f T} = 0.55e^{-0.08} = 0.50771$.
3. $p = 0.0325 + 0.54761 - 0.50771 = 0.0724$.

Why B is correct: Parity discounts the strike at the domestic rate and the spot at the foreign rate (the yield on the currency).

Why the others are wrong

- **A:** Swapped the rates (discounted K at 8% and S at 4%).
- **C:** Ignored discounting altogether: 0.0325 + 0.57 - 0.55.
- **D:** Discounted both K and S at the domestic 4% rate, forgetting the foreign yield.

Q778

AnalystPrep Q.1710

REBUILT — Original stem was a statement scenario with a distractor on American puts and different strikes; rewritten as a direct question with clean options.

The implied volatility of a European put is found to be 25%, meaning that the BSM put price equals the market put price at that volatility. A European call has the same strike price and maturity as the put. Using put-call parity, what can be concluded about the call's implied volatility?

A. It is also 25%, because parity forces the call's pricing error to equal the put's pricing error, which is zero at 25%.
B. It is greater than 25%, because calls carry more upside risk than puts.
C. It is less than 25%, because the put premium includes a crash premium that the call lacks.
D. It cannot be determined without knowing the current underlying price.

Answer: A

Step by step

1. Parity holds for both prices: $p_{BSM} - p_{mkt} = c_{BSM} - c_{mkt}$.
2. At 25% the put's BSM price equals its market price, so the left side is zero.
3. Thus $c_{BSM} = c_{mkt}$ at 25%: the call's implied volatility is 25%.

Why A is correct: Equal pricing errors for call and put imply that the volatility removing the error in one removes it in the other.

Why the others are wrong

- **B:** Treats implied volatility as reflecting payoff asymmetry; parity forbids a different call volatility.
- **C:** Applies the equity-skew intuition across strikes; here the strike is the same, so no difference arises.
- **D:** The result follows from the parity difference alone; S0 cancels.

Q779

AnalystPrep Q.2860

REBUILT — Source data were unrealistic (call 1.35 on a 1.73 spot, giving a put of 1.10) and option B (1.098) was nearly identical to the key; rebuilt with consistent data and error-based options.

The spot exchange rate is $1.60 per unit of foreign currency. Continuously compounded risk-free rates are 5% per year in the U.S. and 3% per year in the foreign country. A 1-year European call on the currency with a strike of $1.55 trades at $0.12. What is the price of the 1-year European put with the same strike, consistent with put-call parity?

A. $0.1022
B. $0.0700
C. $0.0417
D. $0.1173

Answer: C

Step by step

1. $p = c + Ke^{-r_d T} - S_0 e^{-r_f T}$.
2. $1.55e^{-0.05} = 1.47443$; $1.60e^{-0.03} = 1.55273$.
3. $p = 0.12 + 1.47443 - 1.55273 = 0.0417$.

Why C is correct: Currency parity discounts the strike at the domestic rate and spot at the foreign rate.

Why the others are wrong

- **A:** Swapped the two rates (K at 3%, S at 5%).
- **B:** No discounting: 0.12 + 1.55 - 1.60.
- **D:** Discounted the spot at the foreign rate but left the strike undiscounted: 0.12 + 1.55 - 1.5527 gives 0.117 (1.60e^{-0.03} = 1.5527).

Q780

GARP book Ch.17 Q17.20

REBUILT — Original required DerivaGem and open-ended explanation; converted to a self-contained MCQ on the parity implication with prices supplied.

A stock trades at $40. A 6-month European call with strike $30 has an implied volatility of 35%, and a 6-month European call with strike $50 has an implied volatility of 28%. The risk-free rate is 5% (continuous) and no dividends are expected. Using BSM, the call prices are $11.155 and $0.725, so put-call parity gives put prices of $0.414 (strike 30) and $9.490 (strike 50). What are the implied volatilities of the 6-month European puts with strikes of $30 and $50?

A. 35% for the K = 30 put and 28% for the K = 50 put
B. 28% for the K = 30 put and 35% for the K = 50 put
C. 31.5% for both puts (the average of the two call volatilities)
D. Both puts have a lower implied volatility than the matching calls, since the puts are priced by parity from lower call values

Answer: A

Step by step

1. Parity: $p = c + Ke^{-rT} - S_0$; K = 30: $11.155 + 30e^{-0.025} - 40 = 0.414$; K = 50: $0.725 + 50e^{-0.025} - 40 = 9.490$.
2. Parity holds for BSM prices at any volatility, so a put priced by parity from a call at volatility Ïf has BSM-implied volatility Ïf.
3. So the put IVs are 35% (K = 30) and 28% (K = 50), equal to the calls' at each strike.

Why A is correct: Parity implies call and put with the same strike and maturity have identical implied volatility, so the skew is the same for puts and calls.

Why the others are wrong

- **B:** Swapped the strikes' volatilities.
- **C:** Averaged across strikes; implied volatility is strike-specific (skew) and parity applies strike by strike.
- **D:** Parity-derived puts are priced consistently with the calls, so their implied volatilities are equal, not lower.

Q781 BT Mock Q6 · BT notes P2.T5.23.7.2

REBUILT — Merged two copies (one misprinted K/S as 0.90); replaced the self-cancelling straddle distractors with realistic wrong-direction/wrong-hedge trades.

The risk-free rate is 5.0% (continuously compounded) and the stock price is USD 80.00. A European call and a European put on the stock both have strike USD 70.00 (K/S0 = 0.875) and maturity three months. The market-implied volatility of the call is 30.0% and that of the put is 38.0%. Which trade exploits this scenario?

A. Buy the call, sell the put, and short the stock
B. Sell the call, buy the put, and buy the stock
C. Buy the call, sell the put, and buy the stock
D. Sell the call, buy the put, and short the stock

Answer: A

Step by step

1. Put-call parity holds regardless of the price distribution, so a call and put with the same strike and maturity must have the same implied volatility. Here 30% vs 38% means parity is violated.
2. Higher implied volatility means a higher price, so the put is rich and the call is cheap relative to each other.
3. Check: BSM call at 30% = 11.845; BSM put at 38% = 1.790. Market $C - P = 10.055$.
4. Parity requires $C - P = S_0 - Ke^{-rT} = 80 - 70e^{-0.05(0.25)} = 80 - 69.130 = 10.870$.
5. Long call plus short put (a synthetic long stock) costs 10.055, less than the 10.870 it should cost, so it is cheap. Buy it and short the real stock as the offset.
6. Arbitrage profit = 10.870 - 10.055 = USD 0.815 per share, risk-free.

Why A is correct: The call is cheap and the put is expensive relative to parity, so the synthetic long stock (long call, short put) is underpriced and is hedged by shorting the actual stock.

Why the others are wrong

- **B:** The exact reverse trade: it sells the underpriced synthetic stock and buys the real one, locking in a loss of about 0.815. It comes from reading the higher put IV as a cheap put.
- **C:** Long call plus short put already gives long stock exposure; buying the stock doubles the exposure rather than hedging it, so no arbitrage is captured.
- **D:** Short synthetic stock plus short stock doubles the short exposure and takes the wrong side of the mispricing; the put is the rich leg, so it should be sold, not bought.

Q782 BT answers P2.T5.409.2 · BT notes P2.T5.409.2

REBUILT — Merged duplicates; replaced the arbitrary $5.29 distractor with the proportional-scaling error ($5.33).

An analyst estimates a stock's volatility at 34.0% per annum. The Black-Scholes-Merton (BSM) price of a European call is USD 6.95, but the call trades in the market at USD 7.23. The BSM price of a European put on the same stock with the same strike and maturity is USD 5.12. Which is NEAREST to the likely market price of the put?

A. USD 4.84
B. USD 5.12
C. USD 5.33
D. USD 5.40

Answer: D

Step by step

1. Put-call parity holds for market prices and for BSM prices with the same S, K, r, T, so $c_{BSM} - p_{BSM} = c_{mkt} - p_{mkt}$.
2. Rearranging: $c_{mkt} - c_{BSM} = p_{mkt} - p_{BSM}$; market deviations from BSM are equal for calls and puts.
3. Call deviation = 7.23 - 6.95 = 0.28.
4. $p_{mkt} = 5.12 + 0.28 = 5.40$.

Why D is correct: Parity forces the call and put to deviate from their BSM prices by the same dollar amount, so the put also trades 0.28 above its BSM price.

Why the others are wrong

- **A:** Subtracted the deviation (5.12 - 0.28) instead of adding it; the call is above its BSM price so the put must be too.
- **B:** Assumed the put trades at its BSM price, ignoring that parity transmits the call's mispricing to the put.
- **C:** Scaled the put proportionally (5.12 x 7.23/6.95 = 5.33); parity requires equal dollar, not proportional, deviations.

Q783

BT answers P2.T5.411.3 · BT notes P2.T5.411.3

REBUILT — One source lost its table and question sentence and the other had a garbled $46 row; reconstructed a consistent table (checked by BSM) with the question sentence, and replaced incoherent distractors.

A stock trades at USD 50.00 and the risk-free rate is 3.0% (continuously compounded). Options maturing in three months (T = 0.25) have these quotes:

Strike	Call price	Implied volatility
USD 48.00	7.05	59.7%
USD 46.00	7.64	54.5%
USD 50.00	6.50	63.8%

The USD 46.00-strike put is not quoted. Using a flat at-the-money volatility of 63.8%, the BSM price of the USD 46.00 put is USD 4.13. Which statement about the market price of the USD 46.00 put is correct?

A. It is USD 3.30, which is lower than the BSM price at 63.8% volatility
B. It is USD 3.30, which is higher than the BSM price at 63.8% volatility
C. It is USD 4.13, equal to the BSM price at 63.8% volatility
D. It is USD 3.64, which is lower than the BSM price at 63.8% volatility

Answer: A

Step by step

1. The market put follows from parity with the market call: $p = c + Ke^{-rT} - S_0$.
2. $Ke^{-rT} = 46e^{-0.03(0.25)} = 45.656$.
3. $p = 7.64 + 45.656 - 50.00 = 3.30$.
4. The put's implied volatility must equal the call's, 54.5%, below the ATM 63.8% (a frown: ITM/OTM options have lower IV).
5. 3.30 < 4.13, so the put trades below the BSM price at 63.8% volatility.

Why A is correct: Parity ties the put to the market call price, giving USD 3.30; this implies 54.5% volatility, below the 63.8% flat-volatility BSM price of USD 4.13.

Why the others are wrong

- **B:** Gets the price right but the comparison wrong: 3.30 is below 4.13, consistent with lower implied volatility away from the money.
- **C:** Assumes the put must be priced with the ATM volatility, ignoring that parity forces the put to share the call's 54.5% implied volatility.
- **D:** Forgot to discount the strike (7.64 + 46 - 50 = 3.64); the comparison direction is right but the price is wrong.

Q784

BT notes P2.T5.01.3 (sub 3) · BT notes P2.T5.01.3 (sub 4) · BT notes P2.T5.02.5

REBUILT — Three open-ended steps lacking answers and with doubtful inputs were combined into one self-contained MCQ (BSM call price supplied since it needs a normal table).

A non-dividend-paying stock trades at USD 15. A one-year European call with strike USD 10 has a BSM price of USD 5.50 (volatility 30%, risk-free rate 4% continuously compounded). The one-year European put with the same strike trades in the market at USD 0.60, well above its BSM price. Assuming no arbitrage, what is the nearest market price of the call?

A. USD 5.01
B. USD 5.60
C. USD 5.99
D. USD 6.10

Answer: C

Step by step

1. BSM put by parity: $p_{BSM} = c_{BSM} + Ke^{-rT} - S_0 = 5.50 + 10e^{-0.04} - 15 = 5.50 + 9.608 - 15 = 0.108$.
2. Put deviation = 0.60 - 0.108 = 0.492.
3. Parity requires the call to deviate by the same amount: $c_{mkt} = 5.50 + 0.492 = 5.99$.
4. Check: $c_{mkt} = p_{mkt} + S_0 - Ke^{-rT} = 0.60 + 15 - 9.608 = 5.99$.

Why C is correct: Market prices must satisfy parity, so the call's market price exceeds its BSM price by the same amount (0.49) as the put's.

Why the others are wrong

- **A:** Subtracted the put's deviation from the call price; both deviate in the same direction.
- **B:** Did not discount the strike (0.60 + 15 - 10 = 5.60).

- **D:** Added the whole put price to the BSM call (5.50 + 0.60) instead of only the put's deviation from its BSM value.

Q785

Mock 2 Q48

Implied volatility is used to price both call and put options. Which of the following statements best describes put-call parity for European options on a non-dividend-paying stock with the same strike and maturity in a no-arbitrage equilibrium?

A. $c + S = p + PV(X)$
B. $c + p = S - PV(X)$
C. $c - PV(X) = S + p$
D. $c - p = S - PV(X)$

Answer: D

Step by step

1. Fiduciary call: long call plus PV(X) in cash pays $\max(S_T, X)$ at expiry.
2. Protective put: long stock plus long put also pays $\max(S_T, X)$.
3. Equal payoffs imply equal values: $c + PV(X) = p + S$.
4. Rearrange: $c - p = S - PV(X)$, which is option D.
5. Because this relation is model-free, a call and put with the same strike and maturity must have the same implied volatility.

Why D is correct: Both portfolios have identical terminal payoffs, so by no-arbitrage their values are equal; rearranging gives $c - p = S - PV(X)$.

Why the others are wrong

- **A:** Pairs the call with the stock and the put with the bond: $c + S = p + PV(X)$ implies $c - p = PV(X) - S$, the wrong sign (fiduciary call and protective put mismatched).
- **B:** Adds the options ($c + p$) instead of taking their difference; the parity relation involves $c - p$.
- **C:** Moves the put to the stock side with the wrong sign: $c - PV(X) = S + p$ implies $c - p = S + PV(X)$, adding the strike's present value instead of subtracting.

Q786

Schweser Module Quiz 17.1 Q1

REBUILT — Stem was vague about which options are compared; added same strike and expiration (the condition under which parity applies) and made the options unambiguous.

A European call and a European put have the same underlying, strike price and time to expiration. Market prices of both deviate from their Black-Scholes-Merton prices (calculated with a single volatility). Put-call parity implies that the implied volatilities of the call and the put are:

A. equal, because parity holds for market prices and BSM prices, so the pricing deviations must be the same for the put and the call
B. unequal, because puts and calls respond to volatility with opposite signs
C. unequal, because the put has a higher probability of finishing in the money when the call is out of the money
D. equal only if the options are at-the-money

Answer: A

Step by step

1. Parity $c - p = S - PV(X)$ holds for market prices and for BSM prices (any volatility).
2. Subtracting: $(c_{mkt} - c_{BSM}) = (p_{mkt} - p_{BSM})$, so the dollar pricing deviation is identical for call and put.
3. Both options have positive and equal vega (same strike, maturity), so equal price deviations imply the same volatility adjustment.
4. Hence call and put implied volatilities are equal for any moneyness, not only at-the-money.

Why A is correct: Parity is model-free, so market and BSM deviations must be identical for the call and put; with equal vega this gives identical implied volatilities.

Why the others are wrong

- **B:** Confuses the opposite delta signs with vega; both calls and puts have positive vega, so volatility moves both prices the same way.
- **C:** Mixes moneyness with implied volatility; the put and call at one strike have complementary moneyness, but parity still forces equal implied volatility.
- **D:** Wrongly restricts the result to at-the-money; parity applies at every strike.

Q787 AP Mock 1 Q15

REBUILT — Distractors included an impossible value (2.5608) and nonsensical ones; replaced with results of realistic parity mistakes.

A foreign currency is valued at USD 1.80. A European call on it has market price USD 0.135, strike USD 1.70 and one year to maturity. The US risk-free rate is 5% and the foreign risk-free rate is 8% (both continuously compounded). Assuming no arbitrage, the price of a European put with strike USD 1.70 and one year to maturity is closest to:

A. USD 0.0350
B. USD 0.0905
C. USD 0.1734
D. USD 0.1795

Answer: B

Step by step

1. Parity with a yield on the underlying: $c + Ke^{-r_d T} = p + Se^{-r_f T}$.
2. $p = c + Ke^{-r_d T} - Se^{-r_f T}$.
3. $Ke^{-0.05} = 1.70 \times 0.951229 = 1.617090$; $Se^{-0.08} = 1.80 \times 0.923116 = 1.661609$.
4. $p = 0.135 + 1.617090 - 1.661609 = 0.0905$.

Why B is correct: The strike is discounted at the domestic rate and spot at the foreign rate, giving $p = 0.0905$.

Why the others are wrong

- **A:** Ignores all discounting: $0.135 + 1.70 - 1.80 = 0.035$.
- **C:** Discounts spot at the foreign rate but forgets to discount the strike: $0.135 + 1.70 - 1.661609 = 0.1734$.
- **D:** Swaps the roles of the discount factors, adding the foreign-discounted spot and subtracting the domestic-discounted strike: $c + Se^{-0.08} - Ke^{-0.05} = 0.135 + 1.661609 - 1.617090 = 0.1795$.

LO 17.c

Compare the shape of the volatility smile (or skew) to the shape of the implied distribution of the underlying asset price and to the pricing of options on the underlying asset.

Q788 GARP book Ch.17 Q17.6

What volatility smile is likely to be observed when: (a) both tails of the stock price distribution are less heavy than those of the lognormal distribution? (b) The right tail is heavier, and the left tail is less heavy, than that of a lognormal distribution?

Answer: see model answer — (a) Both tails are lighter than lognormal, so deep OTM puts and calls are worth less than BSM says and their implied volatilities are lower than the at-the-money volatility: the smile is an inverted U (a frown), with implied volatility highest at the money and falling for both low and high strikes. (b) A lighter left tail lowers the price of low-strike options (implied vol low) and a heavier right tail raises the price of high-strike options (implied vol high): implied volatility rises with strike, an upward-sloping skew.

Step by step

1. Rule: an option priced above BSM (constant vol) has a higher implied vol; below BSM, a lower implied vol.
2. Low-strike options (OTM puts) are priced off the left tail; high-strike options (OTM calls) off the right tail.
3. (a) Both tails lighter: both wings are cheaper than BSM, so implied vol is lower at both ends than at the money, giving a frown.
4. (b) Left tail lighter (low-strike vol low), right tail heavier (high-strike vol high): implied vol increases with strike.

Why this is correct: Implied volatility is the BSM volatility that reproduces the market price, so a tail that is heavier than lognormal raises the vol implied at strikes in that tail and a lighter tail lowers it.

Q789 GARP book Ch.17 Q17.9

Explain carefully why a distribution with a heavier left tail and less heavy right tail than the lognormal distribution gives rise to a downward sloping volatility smile.

Answer: see model answer — A heavier left tail makes low-strike options (OTM puts, ITM calls) more likely to pay off than under lognormal, so their market prices exceed BSM prices at the ATM vol; the BSM vol needed to

match those prices (the implied vol) is therefore high at low strikes. The lighter right tail makes high-strike options (OTM calls, ITM puts) cheaper than BSM, so their implied vol is low. Implied vol therefore falls as strike rises: a downward-sloping smile (skew), as seen in equity options. Put-call parity makes the call and put implied vols equal at each strike, so the argument holds for both.

Step by step

1. Low strikes: a put with strike K pays off only if $S_T < K$, and the heavier left tail raises $P(S_T < K)$, so the put is worth more than BSM.
2. A higher price needs a higher BSM vol, so implied vol is high at low strikes (same for the ITM call by put-call parity).
3. High strikes: a call pays off only if $S_T > K$; the lighter right tail lowers that probability, so the option is cheaper than BSM and its implied vol is low.
4. High implied vol at low K and low implied vol at high K means implied vol decreases with strike: a downward-sloping smile.

Why this is correct: Option prices in each tail are driven by that tail's probability mass, and implied vol is just the price re-expressed in BSM vol units.

Q790 GARP book Ch.17 Q17.16

Option traders sometimes refer to deep-out-of-the-money options as being options on volatility. Why do you think they do this?

Answer: see model answer — A deep OTM option pays off only if the underlying makes a very large move, and its probability of doing so, and so its price, depends almost entirely on the volatility (and the tail shape) assumed. A small change in volatility changes its value by a large percentage, while its delta is near zero so it has little exposure to small moves in the underlying. Its value therefore behaves like a bet on volatility rather than on direction.

Step by step

1. A deep OTM option has intrinsic value of zero and a very small delta, so small moves in S barely change its price.
2. Its price is the probability of a very large move times the payoff, and that probability is very sensitive to volatility and the tail assumption.
3. Hence vega relative to option value is very high: a modest rise in vol multiplies the price, so the option is effectively a bet on volatility.

Why this is correct: With near-zero delta and a payoff that exists only in the far tail, the option's value is driven by volatility rather than by the underlying's level.

Q791 GARP book Ch.17 Q17.23

A company's stock is selling for $4. The company has no outstanding debt. Analysts consider the liquidation value of the company to be at least $300,000 and there are 100,000 shares outstanding. What volatility smile would you expect to see?

Answer: see model answer — Liquidation value per share is 300,000/100,000 = $3, so the price is unlikely to fall far below $3 while there is no upper limit. The left tail is therefore lighter and the right tail heavier than lognormal. Low-strike options are cheaper than BSM (low implied vol) and high-strike options dearer (high implied vol), so implied vol increases with strike: an upward-sloping smile. Equivalently, volatility falls as the price approaches the $3 floor, i.e. volatility is positively related to price.

Step by step

1. Floor value per share = 300,000/100,000 = $3.
2. Price is $4, only 25% above the floor, so a large fall is very unlikely: lighter left tail than lognormal.
3. The upside is unbounded: right tail at least as heavy as lognormal.
4. Lighter left tail gives low implied vol at low strikes; heavier right tail gives high implied vol at high strikes: upward-sloping smile.

Why this is correct: A price floor truncates the left tail, producing the opposite skew to the leverage-driven equity skew.

Q792

GARP book Ch.17 Q17.27

REBUILT — Open-ended multi-strike computation converted to a self-contained MCQ on the butterfly-implied probability for the 1.0 strike, compared with flat 11.5% volatility.

An exchange rate is currently 1.0. The domestic and foreign risk-free rates are both 2.5% (continuously compounded). Six-month European call options with strikes 0.9, 1.0 and 1.1 have implied volatilities of 11%, 10% and 11% respectively (a smile). Using the butterfly approach (long one 0.9 call, long one 1.1 call, short two 1.0 calls, spacing $\delta = 0.1$) and the Black-Scholes-Merton (Garman-Kohlhagen) call prices, the implied probability that the exchange rate in six months lies between 0.95 and 1.05, and its comparison with the probability implied by a flat 11.5% volatility, is closest to:

A. 0.437 from the smile, which is lower than the 0.509 implied by flat volatility
B. 0.509 from the smile, which is higher than the 0.437 implied by flat volatility
C. 0.051 from the smile, which is lower than the 0.437 implied by flat volatility
D. 0.503 from the smile, which is higher than the 0.437 implied by flat volatility

Answer: B

Step by step

1. Call prices ($S = 1, r = q = 2.5\%, T = 0.5$): smile c(0.9)=0.10171, c(1.0)=0.02785, c(1.1)=0.00429.
2. Butterfly = 0.10171 + 0.00429 - 2(0.02785) = 0.05029.
3. Implied probability $\approx e^{rT} \times$ butterfly$/\delta$ = 1.01258 × 0.05029 / 0.1 = 0.509.
4. Flat 11.5%: c = 0.10226, 0.03203, 0.00500; butterfly = 0.04320; probability = 1.01258 × 0.04320 / 0.1 = 0.437.
5. The smile puts more probability near the centre (0.509 vs 0.437), so it implies a more peaked distribution with fatter tails than lognormal.

Why B is correct: The butterfly price times e^{rT}/δ approximates the risk-neutral probability of landing within δ of the middle strike; the lower ATM vol (10%) makes the 1.0 call cheap and the butterfly worth more.

Why the others are wrong

- **A:** Uses the flat-vol figure for the smile and swaps the comparison; the smile vols give the larger butterfly.
- **C:** Forgot to divide by δ (0.1), leaving the butterfly price times e^{rT} (0.051), which is not a probability on the right scale.
- **D:** Omitted the e^{rT} compounding factor (0.05029/0.1 = 0.503); the conclusion is right but the figure is not the correctly scaled risk-neutral probability.

Q793

AnalystPrep Q.1527

REBUILT — Original asked a vague 'other assumption' BSM recall question; rebuilt to test the link between BSM's constant-volatility lognormal assumption and the flat-smile/lognormal implied distribution.

In the Black-Scholes-Merton model the underlying price follows geometric Brownian motion with constant volatility. If every option price in the market were consistent with this model, which of the following would be observed?

A. Implied volatility flat across strikes, and a lognormal implied distribution of the terminal price
B. A U-shaped smile in implied volatility, and a lognormal implied distribution with fatter tails
C. A downward-sloping skew in implied volatility, and a left-skewed implied distribution
D. Implied volatility that varies with strike but a lognormal implied distribution

Answer: A

Step by step

1. Constant volatility GBM implies ln(ST) is normal, i.e. ST is lognormal.
2. One volatility prices every strike, so the volatility implied by BSM is the same at all strikes: a flat 'smile'.
3. Hence a flat implied-vol curve corresponds exactly to the lognormal distribution; any smile or skew signals a non-lognormal implied distribution.

Why A is correct: Flat implied volatility across strikes and a lognormal terminal distribution are two descriptions of the same BSM assumption.

Why the others are wrong

- **B:** Describes a currency-type smile, which arises when the implied distribution has fatter tails than lognormal; that violates, not reflects, BSM.
- **C:** Describes the equity skew, caused by a heavier left tail; the implied distribution is then not lognormal.
- **D:** Internally inconsistent: a strike-dependent implied vol means the market distribution differs from lognormal.

Q794 BT answers P2.T5.410.2 · BT notes P2.T5.410.2

Suppose the volatility is uncertain and positively correlated with the underlying asset price. Relative to prices implied by constant volatility, which of the following scenarios is MOST PLAUSIBLE?

A. Lower price for out-of-the-money (OTM) put options and higher price for in-the-money (ITM) put options
B. Lower price for out-of-the-money (OTM) call options and higher price for in-the-money (ITM) call options
C. Lower price for at-the-money (ATM) call options and higher price for at-the-money (ATM) put options
D. Lower price for in-the-money (ITM) put options and higher price for out-of-the-money (OTM) put options

Answer: A

Step by step

1. Positive correlation: volatility is high when the price is high and low when it is low.
2. The implied distribution then has a lighter left tail and a heavier right tail than lognormal (upward-sloping skew).
3. Low-strike options (OTM puts, ITM calls) depend on the left tail, so they are cheaper than constant-vol prices.
4. High-strike options (OTM calls, ITM puts) depend on the right tail, so they are dearer.
5. A matches: OTM put (low strike) cheaper, ITM put (high strike) dearer.

Why A is correct: The correlation sign fixes which tail is heavier, and low- versus high-strike options price off the left and right tails respectively.

Why the others are wrong

- **B:** Reversed for calls: OTM calls (high strike) are dearer and ITM calls (low strike) cheaper; this is the answer under negative correlation (the equity skew).
- **C:** At-the-money options are barely affected by tail shape, and put and call prices are tied together by put-call parity, so they cannot move in opposite directions.
- **D:** Reversed for puts: ITM puts have high strikes (dearer) and OTM puts low strikes (cheaper); this is the negative-correlation pattern.

Q795 BT answers P2.T5.410.3

Suppose the implied distribution of a certain option asset class exhibits, relative to the lognormal distribution, a lighter left tail and a heavier right tail. Define risk reversal (RR) as the difference in implied volatility between a 25-delta call and a -25-delta put: RR = implied volatility [25-delta call] - implied volatility [-25-delta put]. Which of the following is most likely TRUE of RR?

A. Negative risk reversal
B. Zero risk reversal
C. Positive risk reversal
D. Risk reversal approximately equal to the implied volatility of the 50-delta (ATM) call

Answer: C

Step by step

1. Lighter left tail: low-strike options are cheap, so the implied vol of the -25-delta (OTM, low-strike) put is low.
2. Heavier right tail: high-strike options are dear, so the implied vol of the 25-delta (OTM, high-strike) call is high.
3. RR = high - low > 0: positive.

Why C is correct: The 25-delta call sits in the heavy right tail and the -25-delta put in the light left tail, so call vol exceeds put vol.

Why the others are wrong

- **A:** This is the equity-style skew (heavier left tail, puts richer); here the tails are the other way round.
- **B:** Zero holds only for a symmetric smile in which both wings are equally heavy.
- **D:** RR is a difference between two implied vols, not a level; it is not tied to the ATM vol.

Q796 BT notes P2.T5.01.3

Which of the two patterns, a volatility smile or a volatility skew, is consistent with the lognormal price distribution (geometric Brownian motion) that underlies the Black-Scholes-Merton model?

Answer: see model answer — Neither. If asset prices were lognormal, as BSM assumes, implied volatility would be the same for every strike, so the plot of implied volatility against strike (or K/S) would be a horizontal line. A smile (higher implied volatility for deep ITM and deep OTM options) implies heavier tails than lognormal; a skew

(higher implied volatility at low strikes) implies a heavier left tail and lighter right tail.

Step by step

1. BSM with lognormal prices has constant volatility, so every strike backs out the same implied volatility.
2. Plotting implied volatility against strike gives a flat horizontal line.
3. A smile is U-shaped (both tails heavier than lognormal); a skew is downward sloping (left tail heavier, right tail lighter). Both depart from lognormal.

Why this is correct: Any curvature or slope in implied volatility across strikes signals an implied distribution that differs from lognormal; only a flat line matches it.

Q797 BT notes P2.T5.03.01

What is the impact on the implied volatility smile, if any, if the equity log-return distribution is platykurtic (kurtosis < 3)?

Answer: see model answer — The smile is inverted into a frown. Light tails (less probability of extreme moves than lognormal) make deep ITM and deep OTM options cheaper than BSM prices, so their implied volatilities are lower than the at-the-money implied volatility, which is the highest point on the curve.

Step by step

1. Kurtosis < 3 means thinner tails than the normal/lognormal benchmark.
2. Away-from-the-money options pay off only in the tails, so their market prices are lower than BSM prices.
3. Lower prices imply lower implied volatility at low and high strikes relative to ATM, giving an inverted-U shape: a frown.

Why this is correct: Implied volatility mirrors tail weight: heavy tails give a smile, light tails give a frown.

Q798 BT notes P2.T5.03.03

What is the impact on the implied volatility smile, if any, if the equity log-return distribution is a normal mixture distribution?

Answer: see model answer — A normal mixture has kurtosis greater than 3 (heavier tails than a single normal), so it produces a standard volatility smile: implied volatility is higher for deep ITM and deep OTM options and lower at the money. (If the mixture components are asymmetric, the smile is skewed.)

Step by step

1. A mixture of normals with different variances is leptokurtic: kurtosis > 3.
2. Heavier tails raise the market prices of deep ITM/OTM options above BSM prices.
3. Higher prices at the wings imply higher implied volatility there than ATM: a U-shaped smile.

Why this is correct: Fat tails relative to the lognormal translate into higher implied volatility for away-from-the-money strikes.

Q799 BT notes P2.T5.23.7.3

REBUILT — Original relied on unreadable graphs and ambiguous 'smirk left/right' labels; rewrote with explicit verbal definitions of each shape and verified scenario mapping.

Four archetypal shapes of implied volatility plotted against K/S are defined as follows. Smile: implied volatility is lowest at the money and higher at both low and high strikes. Frown: implied volatility is highest at the money and lower at both low and high strikes. Downward skew: implied volatility falls steadily as strike rises. Upward skew: implied volatility rises steadily as strike rises.

Consider the scenarios:

I. Asset volatility is positively correlated with the asset price.

II. The asset price distribution has a heavy left tail and a light right tail.

III. The underlying is an exchange rate that trades in a very narrow band because of central bank policy.

IV. An option expires in one week and, because of an expected announcement, the stock will jump either up or down by more than 16% within the week.

Which shapes correspond to scenarios I to IV?

A. I. Upward skew, II. Downward skew, III. Frown, IV. Frown

B. I. Downward skew, II. Upward skew, III. Frown, IV. Frown

C. I. Upward skew, II. Downward skew, III. Smile, IV. Smile

D. I. Downward skew, II. Downward skew, III. Frown, IV. Smile

Answer: A

Step by step

1. I: if volatility rises with price, high-price outcomes carry more variance, so the right tail is heavier and high-strike options have higher implied volatility: upward skew.
2. II: heavy left tail means low-strike options (OTM puts) are expensive, so implied volatility is high at low strikes and falls with strike: downward skew (typical equity pattern).
3. III: a central-bank band truncates extreme moves, so both tails are light and wing options are cheap: frown.
4. IV: a single large jump gives a bimodal-like distribution with lighter-than-lognormal wings relative to ATM: frown (Schweser LO 17.i).
5. Only A matches all four.

Why A is correct: Implied volatility is high where the implied distribution has more weight than lognormal and low where it has less: right-heavy gives upward skew, left-heavy downward skew, light tails or a single jump give a frown.

Why the others are wrong

- **B:** Reverses the direction of both skews: a heavy left tail gives high implied volatility at LOW strikes, not high strikes.
- **C:** Gets the skews right but treats light tails (III) and a single jump (IV) as heavy-tailed smiles; those cases give frowns.
- **D:** Treats positive price-volatility correlation as a downward skew (that is the negative-correlation equity case) and calls the jump a smile.

Q800

BT notes P2.T5.409.3

REBUILT — Original statements II and IV were not cleanly judgeable (II is actually true at equal strike by put-call parity; I and III ambiguous); rewrote statements to test implied distribution vs lognormal and skew pricing.

Implied volatilities computed from Google call options slope downward: implied volatility falls as the strike price rises, as is typical of equity options. An analyst draws these conclusions:

I. The implied distribution has a heavier left tail than a lognormal distribution.

II. The implied distribution has a lighter right tail than a lognormal distribution.

III. An out-of-the-money put with strike 90% of spot is priced at a higher implied volatility than an out-of-the-money call with strike 110% of spot.

IV. The implied distribution has a heavier right tail than a lognormal distribution.

Which conclusions are correct?

A. I only

B. II and III only

C. I, II and III only

D. I, II, III and IV

Answer: C

Step by step

1. Downward-sloping skew: high implied volatility at low strikes means options that pay off in large down moves are expensive, so the left tail is heavier than lognormal (I true).
2. Low implied volatility at high strikes means up-move options are cheap relative to BSM, so the right tail is lighter (II true, IV false).
3. The 90% put sits on the high-volatility side and the 110% call on the low-volatility side, so the put's implied volatility is higher (III true).
4. True statements: I, II, III.

Why C is correct: A downward-sloping skew implies a left-skewed implied distribution: heavier left tail, lighter right tail, and expensive downside options.

Why the others are wrong

- **A:** Recognises only the heavy left tail; ignores that the thin right tail and the put-versus-call volatility comparison also follow from the slope.
- **B:** Omits I; the heavy left tail is the core implication of high implied volatility at low strikes.
- **D:** Includes IV, confusing the lighter right tail with a heavier one (applying the smile rather than the skew).

Q801 GARP PE1 Q51 · Mock 3 Q29

A risk analyst at an investment bank is examining the assumptions the bank uses in its foreign exchange (FX) option pricing model. Currently, the implied volatility in a particular FX pair is relatively low if an option is at-the-money and becomes progressively higher as the option moves either more in-the-money or more out-of-the-money. How does the distribution of option prices on this FX pair implied by the Black-Scholes-Merton model compare to a lognormal distribution with the same mean and standard deviation?

A. The implied distribution has a heavier left tail and a less heavy right tail.
B. The implied distribution has a heavier left tail and a heavier right tail.
C. The implied distribution has a less heavy left tail and a heavier right tail.
D. The implied distribution has a less heavy left tail and a less heavy right tail.

Answer: B

Step by step

1. Implied volatility lowest ATM and rising in both directions is a symmetric volatility smile.
2. High implied volatility at low strikes means the market prices downside moves above lognormal: heavier left tail.
3. High implied volatility at high strikes means upside moves are priced above lognormal: heavier right tail.
4. Both tails are heavier than lognormal: B.

Why B is correct: A smile means elevated option prices (hence probability) in both tails relative to a lognormal with the same mean and standard deviation.

Why the others are wrong

- **A:** Describes the equity downward skew (high implied volatility only at low strikes); here the right wing is also elevated.
- **C:** Describes an upward (reverse) skew with only high strikes elevated; the left wing is also elevated here.
- **D:** Describes a frown (lowest implied volatility in the wings), the opposite of the smile given.

Q802 Schweser Module Quiz 17.1 Q2

An empirical distribution that exhibits a fatter right tail than that of a lognormal distribution would indicate:

A. equal implied volatilities across low and high strike prices.
B. greater implied volatilities for low strike prices.
C. greater implied volatilities for high strike prices.
D. higher implied volatilities for mid-range strike prices.

Answer: C

Key correction: Schweser's answer C is correct, but its explanation says in-the-money calls and out-of-the-money puts become expensive; high strikes are out-of-the-money calls and in-the-money puts, so those are the options that become expensive.

Step by step

1. A fatter right tail means higher probability of high underlying prices than lognormal.
2. High-strike options (OTM calls, ITM puts) pay off in that region, so their market prices exceed BSM prices.
3. Higher prices translate into higher implied volatility at high strikes: C.

Why C is correct: Implied volatility is higher at strikes where the true distribution has more probability mass than the lognormal; a fat right tail raises it at high strikes.

Why the others are wrong

- **A:** Equal implied volatilities is the flat line consistent with lognormal, which a fatter tail contradicts.
- **B:** Describes a fatter LEFT tail (equity skew); the strike side is reversed.
- **D:** Elevated mid-range volatility would imply a frown/light tails, not a fat right tail.

Q803 GARP PE1 Q56 · Mock 3 Q25

A risk committee of the board of company ABC is discussing the difference between pricing deep out-of-the-money call options on ABC stock and pricing deep out-of-the-money call options on the USD/GBP foreign exchange (FX) rate using the Black-Scholes-Merton model. The committee considers pricing each of these two options based on two distinct probability distributions of underlying asset prices at the option expiration date: a lognormal probability distribution, and an implied risk-neutral probability distribution obtained from the volatility smile for each aforementioned option of the same maturity and the same moneyness. If the implied risk-neutral probability distribution is used instead of the lognormal distribution, which of the following is correct?

A. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.
B. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.
C. The price of the option on ABC stock would be relatively low and the price of the option on USD/GBP FX rate would be relatively low compared to those computed from the lognormal counterparts.
D. The price of the option on ABC stock would be relatively high and the price of the option on USD/GBP FX rate would be relatively high compared to those computed from the lognormal counterparts.

Answer: B

Step by step

1. A deep OTM call pays off only in the right tail, so its price depends on the right-tail probability.
2. Equity: volatility skew (implied vol falls with strike) means a thinner right tail than lognormal, so the call is worth less.
3. FX: symmetric smile means both tails are heavier, so the right tail is fatter and the call is worth more.
4. Hence ABC low, USD/GBP high: B.

Why B is correct: Right-tail thickness relative to lognormal drives deep OTM call prices: thin for equities, fat for FX.

Why the others are wrong

- **A:** Reverses both: treats equity as having a fat right tail and FX as thin.
- **C:** Treats FX like equity (skew), ignoring that the FX smile raises the right tail.
- **D:** Treats equity like FX (smile), ignoring the equity skew that thins the right tail.

Q804 BT notes P2.T5.410.3

Suppose the implied distribution of an asset, relative to the lognormal distribution, has a lighter left tail and a heavier right tail. Define the risk reversal as RR = implied volatility of the 25-delta call minus implied volatility of the -25-delta put. Which of the following is most likely true of RR?

A. Negative risk reversal
B. Zero risk reversal
C. Positive risk reversal
D. Risk reversal approximately equal to the implied volatility of the 50-delta (at-the-money) call

Answer: C

Step by step

1. A heavier right tail than lognormal means out-of-the-money calls are worth more than BSM predicts, so they have higher implied volatility.
2. A lighter left tail means out-of-the-money puts are worth less than BSM predicts, so they have lower implied volatility.
3. The 25-delta call and -25-delta put are both OTM, so IV(25-delta call) > IV(-25-delta put).
4. RR = IV(call) - IV(put) > 0, so RR is positive (a forward skew, opposite to the usual equity skew).

Why C is correct: Tail weight in the implied distribution maps to implied volatility on that side, so a heavy right tail and light left tail give the call wing higher volatility and a positive RR.

Why the others are wrong

- **A:** This is the typical equity-style result (heavy left tail, light right tail); the tails are reversed here.
- **B:** RR is zero only for a symmetric smile, which requires equally heavy tails on both sides.
- **D:** RR is a difference of two volatilities, not a level; it measures skew and has no link to the ATM volatility.

LO 17.d

Describe characteristics of foreign exchange rate distributions and their implications on option prices and implied volatility.

Q805 GARP book Ch.17 Q17.15

Suppose that a central bank's policy is to allow an exchange rate to fluctuate between 0.97 and 1.03. What pattern of implied volatilities for options on the exchange rate would you expect to see?

Answer: see model answer — The terminal distribution is truncated at the band edges, so it has much thinner tails than a lognormal with the same at-the-money volatility. Options with strikes near or beyond 0.97 and 1.03 have far less chance of paying off than BSM implies, so their implied volatilities are low. Implied volatility is therefore highest at the money and falls for deep in-the-money and out-of-the-money strikes: an inverted smile (a 'frown'), the opposite of the usual FX smile.

Step by step

1. Compare the implied distribution with the lognormal: the band removes probability from both tails.
2. Thinner-than-lognormal tails mean OTM options (calls above 1.03, puts below 0.97) are worth less than BSM prices at the ATM volatility.
3. A lower price translates into a lower implied volatility at those strikes, while ATM volatility stays highest.
4. Result: implied volatility is a hump-shaped (frown) function of strike.

Why this is correct: Thin tails imply lower prices and lower implied volatility for away-from-the-money options.

Q806 GARP book Ch.17 Q17.19

An exchange rate is currently 0.8000. The volatility of the exchange rate is quoted as 12% and interest rates in the two countries are the same. Using the lognormal assumption, estimate the probability that the exchange rate in 3 months will be (a) less than 0.7000, (b) between 0.7000 and 0.7500, (c) between 0.7500 and 0.8000, (d) between 0.8000 and 0.8500, (e) between 0.8500 and 0.9000, and (f) greater than 0.9000. Based on the volatility smile usually observed in the market for exchange rates, which of these estimates would you expect to be too low and which would you expect to be too high?

Answer: see model answer — With equal interest rates the drift is zero, so $\ln S_T \sim N(\ln 0.8 - 0.5\sigma^2 T,\ \sigma\sqrt{T})$ with $\sigma\sqrt{T} = 0.06$ and mean -0.22494. Cumulative probabilities at 0.70, 0.75, 0.80, 0.85, 0.90 are 1.41%, 14.79%, 51.20%, 85.09%, 97.69%. Hence (a) 1.4%, (b) 13.4%, (c) 36.4%, (d) 33.9%, (e) 12.6%, (f) 2.3%. The market FX smile implies fatter tails than lognormal, so the tail probabilities (a) and (f) (and to a lesser degree (b) and (e)) are too low, while the central probabilities (c) and (d) are too high.

Step by step

1. Zero drift (same interest rates): mean of $\ln S_T = \ln 0.8 - 0.5(0.12)^2(0.25) = -0.22494$; sd = $0.12\sqrt{0.25} = 0.06$.
2. z-values for 0.70, 0.75, 0.80, 0.85, 0.90: -2.196, -1.046, 0.030, 1.040, 1.993.
3. Cumulative N(z): 0.0141, 0.1479, 0.5120, 0.8509, 0.9769.
4. Differences give (a) 1.41%, (b) 13.38%, (c) 36.41%, (d) 33.90%, (e) 12.59%, (f) 2.31%.
5. FX smile means fatter tails than lognormal: extreme-region probabilities are underestimated, central ones overestimated.

Why this is correct: Lognormal probabilities come from N(d) on the log-moves; the FX smile implies heavier tails so lognormal understates the tails and overstates the centre.

Q807 AnalystPrep Q.1713

In practice, exchange rates do not follow a lognormal distribution, as the exchange rate's volatility is far from constant and there are frequent jumps. Which of the following statements is true for this scenario?

A. As the maturity of the option increases, the percentage impact of non-constant volatility on prices becomes less pronounced.
B. The effects of variable volatility and jumps are independent of the maturity of options.
C. When the maturity of options increases, the volatility smile becomes less pronounced.
D. Variable volatility and jump risks have a uniform influence on options, regardless of strike or term structure.

Answer: C

Step by step

1. Non-constant volatility: with longer maturity its percentage effect on option prices grows, but its effect on implied volatility usually shrinks.
2. Jumps: the percentage effect on both prices and implied volatility shrinks as maturity increases (jumps average out).

3. Net effect: the FX smile flattens for longer-dated options, so C.

Why C is correct: Both sources of the smile weaken in implied-volatility terms with maturity, so long-dated options show a flatter smile.

Why the others are wrong

- **A:** Reverses the price effect: the percentage impact of stochastic volatility on prices becomes more, not less, pronounced with maturity.
- **B:** Both effects depend on maturity; they are strongest for short-dated options.
- **D:** The effects vary with strike (smile) and with maturity (term structure), so they are not uniform.

Q808

BT notes P2.T5.07.02

REBUILT — Original asked which currency process fits GBM but gave no key and had an ambiguous option set ('short horizon' is not a currency type); rebuilt as a clear question on when the FX smile is strongest, testing the same LO.

A risk manager compares four currency options and wants the one whose implied volatilities will deviate most from a flat line across strikes (the most pronounced volatility smile), because the BSM lognormal assumption fits it worst. Which option is it?

A. A 5-year option on a major currency pair with stable volatility

B. A 1-month option on a currency pair whose volatility frequently jumps because of central bank interventions and policy surprises

C. A 2-year option on a major currency pair with stable volatility

D. A 5-year option on an emerging-market currency

Answer: B

Step by step

1. The FX smile arises because volatility is non-constant and the rate jumps, creating fatter tails than lognormal.
2. Both effects are more pronounced in implied-volatility terms for short-dated options; long-dated options show a flatter smile.
3. Stable-volatility major pairs (A, C) deviate least; long maturity (A, D) mitigates the smile.
4. The short-dated option on a jump-prone currency (B) combines the strongest sources with the shortest maturity.

Why B is correct: The smile is most pronounced when volatility is unstable or jumpy and the option is short-dated, as for B.

Why the others are wrong

- **A:** Long maturity and stable volatility both reduce the smile, so this is among the flattest.
- **C:** Stable volatility on a major pair gives the closest fit to lognormal.
- **D:** Emerging-market currencies have jumps, but the 5-year maturity mitigates the smile, so it is less pronounced than the 1-month option in B.

Q809

BT notes P2.T5.07.02 (Q07.03)

What is the likely impact of time-varying (non-constant) volatility on the volatility smile?

Answer: see model answer — If volatility is stochastic, the terminal distribution mixes high-volatility and low-volatility states, giving heavier tails than a lognormal (excess kurtosis). Away-from-the-money options (deep ITM and OTM) are then worth more than BSM at the ATM volatility, so their implied volatilities are higher and a smile appears; for FX, where volatility has little correlation with the exchange rate, the smile is roughly symmetric. In implied-volatility terms the effect is strongest for short-dated options and becomes less pronounced as maturity increases.

Step by step

1. Stochastic volatility mixes lognormal distributions with different volatilities, producing fat tails.
2. Fat tails raise the value of OTM calls and puts relative to BSM, hence higher implied volatilities at extreme strikes.
3. With volatility uncorrelated with the rate (FX) the smile is symmetric; with maturity the smile flattens.

Why this is correct: Randomly changing volatility fattens both tails, which lifts implied volatility away from the money.

Q810 BT notes P2.T5.410.1

Suppose most market participants think that foreign currency exchange rates are lognormally distributed, such that they are comfortable using the same (constant) volatility to value options with the Black-Scholes option pricing model (BSM OPM). Risk Analyst Sally has performed an analysis, however, and she knows that the lognormal assumption is not good: rather, she expects actual distributions with heavier tails, both left and right, than the lognormal. Which trade most takes advantage of this situation?

A. Long butterfly
B. Short straddle
C. Long strangle
D. Long bull spread

Answer: C

Step by step

1. Market prices options with constant volatility, so deep OTM puts and calls are underpriced relative to Sally's heavy-tailed view.
2. She should be long the tails: buy an OTM put and an OTM call, i.e. a long strangle.
3. A long butterfly and short straddle profit when the rate stays near the centre; a bull spread is a directional bet.

Why C is correct: A long strangle buys the underpriced OTM options in both tails and gains when extreme moves occur more often than lognormal predicts.

Why the others are wrong

- **A:** A long butterfly is short the wings and profits from small moves, the opposite of a fat-tail view.
- **B:** A short straddle is short volatility and loses on large moves.
- **D:** A bull spread is a one-directional bet, but her view is symmetric and about tails, not direction.

LO 17.e

Describe the volatility smile for equity options and foreign currency options and provide possible explanations for its shape.

Q811 GARP book Chapter 17 Q17.3

What is the difference between the volatility smiles that are typically observed for equities and currencies?

Answer: see model answer — Currency options show a roughly symmetric smile: implied volatility is lowest at-the-money and rises for both deep ITM and deep OTM strikes, because exchange-rate changes have fat tails in both directions (jumps, stochastic volatility) relative to a lognormal distribution. Equity options show a downward-sloping skew (smirk): implied volatility falls as strike rises, so low-strike options (OTM puts) have the highest implied volatility. This implies a left-skewed distribution with a heavy left tail and a thin right tail, attributed to leverage and crashophobia. Both patterns are weaker for long-dated options.

Step by step

1. Currency: implied vol is U-shaped, minimum near the money, higher for both low and high strikes.
2. Reason: both tails of the exchange-rate distribution are fatter than lognormal.
3. Equity: implied vol decreases monotonically with strike (K/S), a skew or smirk.
4. Reason: heavy left tail and thin right tail, from leverage and crashophobia.

Why this is correct: Currency smile is symmetric (two fat tails); equity smile is a one-sided skew (fat left tail only).

Q812 GARP book Chapter 17 Q17.11

Explain what is meant by "crashophobia."

Answer: see model answer — Crashophobia (Mark Rubinstein's term) is the fear, prevalent since the 1987 crash, that equity markets may fall precipitously again. Traders pay a premium for deep out-of-the-money puts as crash insurance, which bids up their prices and implied volatilities. This produces the downward-sloping equity volatility skew (high implied volatility at low strikes), and the skew steepens when markets decline.

Step by step

1. Origin: the October 1987 crash led market participants to price in the possibility of another sudden large fall.

2. Demand for OTM puts as portfolio protection raises their prices.
3. Higher put prices imply higher implied volatility at low strikes.
4. Result: downward-sloping skew, more pronounced after market declines.

Why this is correct: Premium paid for protection against a crash lifts low-strike implied volatility, creating the equity skew.

Q813

GARP book Chapter 17 Q17.13

What volatility smile is likely to be observed for 6-month options when the volatility is uncertain and positively correlated with the stock price?

Answer: see model answer — When volatility is positively correlated with the stock price, rising prices come with rising volatility and falling prices with falling volatility. The right tail of the stock-price distribution is therefore heavier than lognormal and the left tail thinner. The implied volatility is an increasing function of strike (an upward-sloping skew), the reverse of the typical equity pattern: OTM calls (high strikes) have higher implied volatilities than OTM puts (low strikes).

Step by step

1. Positive correlation: stock up means volatility up, stock down means volatility down.
2. Right tail: large up moves are amplified, so the right tail is heavier than lognormal.
3. Left tail: large down moves are dampened, so the left tail is thinner than lognormal.
4. Heavy right tail raises the price and implied vol of high-strike calls; thin left tail lowers implied vol of low-strike puts.
5. Implied vol therefore rises with strike: upward-sloping skew (opposite of equities).

Why this is correct: A heavy right tail and thin left tail mean implied volatility increases with strike.

Q814

GARP book Chapter 17 Q17.25

REBUILT — Original asked the student to download index data from a website and run a test; rebuilt as a self-contained data-interpretation question on the same idea (asymmetric 3-standard-deviation moves).

An analyst studies 5,000 daily returns of an equity index. Each return is standardized by the sample mean and standard deviation. She finds 11 returns below −3 standard deviations and 5 returns above +3 standard deviations. Under a normal distribution, $P(Z < -3) = P(Z > 3) \approx 0.00135$. Which statement is the most appropriate conclusion?

A. Under normality about 6.7 observations are expected in each tail; the excess of down moves over up moves implies a left-skewed distribution, consistent with a downward-sloping implied volatility skew.

B. Both tail counts exceed what a normal distribution implies, so implied volatility should form a symmetric smile as for currency options.

C. Up moves exceed what is expected under normality, so implied volatility should rise with strike (upward-sloping skew).

D. The counts are close to the normal expectation of 6.7 per tail, so the implied volatility curve should be flat.

Answer: A

Step by step

1. Expected count per tail under normality: $5000 \times 0.00135 = 6.75 \approx 6.7$.
2. Down tail: 11 observed, well above 6.75, so the left tail is heavier than normal.
3. Up tail: 5 observed, below 6.75, so the right tail is thinner than normal.
4. Heavy left and thin right tail means a left-skewed distribution, i.e. high implied volatility for low strikes: a downward-sloping skew.

Why A is correct: Three-standard-deviation down moves occurring more often than up moves indicates a left-skewed distribution, which is the equity skew.

Why the others are wrong

- **B:** Only the down tail is above 6.75; the up tail (5) is below it, so the tails are not both fat, and a symmetric smile would be the currency pattern.
- **C:** Reverses the asymmetry; this is the pattern when volatility is positively correlated with price, not what the data show.
- **D:** Treats 11 and 5 as both close to 6.75; the down count is about 63% above expectation and the up count about 26% below.

Q815 BT notes P2.T5.02.4

REBUILT — Original open question depended on an unseen preceding exhibit and asked for a one-word answer; rebuilt with explicit numbers and answer options.

A stock trades at 60 with a risk-free rate of 5% (continuously compounded). A 6-month European put with strike 54 is priced by the Black-Scholes-Merton model using a volatility of 30%, giving a price of 1.96. The put actually trades at 2.49, which corresponds to an implied volatility of about 34%. Which explanation is most consistent with this higher implied volatility?

A. Crashophobia: investors pay a premium for out-of-the-money puts as protection against a sharp fall in equity prices.
B. A violation of put-call parity, which makes puts systematically more expensive than calls at the same strike.
C. Investors believe the left tail of the stock price distribution is thinner than lognormal.
D. Speculative demand for out-of-the-money calls raising implied volatility at low strikes.

Answer: A

Step by step

1. BSM put at $\sigma = 30\%$, $K = 54$: 1.96; market price 2.49 is higher, so implied vol (about 34%) exceeds 30%.
2. The put is out-of-the-money (K/S = 0.90), so the premium reflects demand for downside protection.
3. This is the equity skew: higher implied volatility at low strikes.
4. The standard explanation (since 1987) is crashophobia, alongside the leverage effect.

Why A is correct: Extra demand for crash protection bids up OTM put prices, raising implied volatility at low strikes.

Why the others are wrong

- **B:** Put-call parity is model-free, so a call and put with the same strike and maturity have the same implied volatility; it does not explain a skew across strikes.
- **C:** A thinner left tail would make OTM puts cheaper and lower implied volatility, the opposite of what is observed.
- **D:** Demand for calls affects high-strike options, not a low-strike put.

Q816 BT notes P2.T5.03.05

What is the impact on the implied volatility smile, if any, if end users increase demand for long positions in OTM PUT options on a company's stock for downside protection?

Answer: see model answer — Higher demand for OTM puts bids up their prices and therefore their implied volatilities. Low-strike implied volatility rises relative to at-the-money and high-strike options, so the downward-sloping equity skew becomes steeper. This is the same mechanism as crashophobia: a premium for protection against a sharp fall.

Step by step

1. Increased demand for OTM puts pushes put prices up.
2. Higher price means higher implied volatility at low strikes.
3. The left side of the curve rises, steepening the downward skew.

Why this is correct: Buying pressure in low-strike puts lifts their implied volatility, steepening the equity skew.

Q817 BT notes P2.T5.03.06

What is the impact on the implied volatility smile, if any, if option writers (market makers who take short positions) are averse to high delta call options but prefer low delta call options?

Answer: see model answer — High-delta calls are in-the-money (low strike) and low-delta calls are out-of-the-money (high strike). Writers supplying fewer low-strike calls raises their prices and implied volatilities, while ample supply of high-strike calls lowers their prices and implied volatilities. Implied volatility is therefore higher at low strikes and lower at high strikes: a downward-sloping, equity-like skew.

Step by step

1. High delta (near 1) means deep ITM call, i.e. low strike; low delta (near 0) means OTM call, i.e. high strike.
2. Less supply at low strikes raises price and implied vol there.
3. More supply at high strikes lowers price and implied vol there.
4. Net effect: implied vol falls with strike, an equity-style skew.

Why this is correct: Supply and demand imbalances across strikes push low-strike implied volatility up and high-strike down.

Q818

BT answers P2.T5.23.7.1

REBUILT — Price table was unreadable and the $75 put price was inconsistent with BSM; rebuilt with a complete table of consistent BSM and market prices.

Peter asserts that equity prices and volatility are negatively correlated. A stock trades at 60, the risk-free rate is 5% and all options have 6 months to expiry. The table gives BSM prices using a flat 30% volatility and observed market prices.

Strike	BSM call (30%)	BSM put (30%)	Market call	Market put
45	16.45	0.34	16.80	0.69
60	5.78	4.30	5.78	4.30
75	1.32	14.47	0.87	14.02

What is the shape of the implied volatility smile, and is it consistent with Peter's assertion?

A. Upward-sloping skew; consistent with Peter's assertion.

B. Downward-sloping skew; consistent with Peter's assertion.

C. Downward-sloping skew; not consistent with Peter's assertion.

D. Symmetric smile; not consistent with Peter's assertion.

Answer: B

Step by step

1. K = 45: market put 0.69 > BSM 0.34 and market call 16.80 > 16.45, so implied vol is above 30% (about 36%).
2. K = 60: market equals BSM, implied vol = 30%.
3. K = 75: market call 0.87 < 1.32 and market put 14.02 < 14.47, so implied vol is below 30% (about 26%).
4. Implied vol falls as strike rises: downward-sloping skew.
5. Negative correlation (stock falls, vol rises) fattens the left tail and thins the right tail, giving exactly this skew.

Why B is correct: Implied volatility is about 36%, 30%, 26% for strikes 45, 60, 75; this decline with strike is the pattern produced by a negative price-volatility correlation.

Why the others are wrong

- **A:** Reverses the direction: market prices exceed BSM at the low strike and fall below it at the high strike, so implied vol decreases with strike.
- **C:** Correct shape but wrong inference; a downward skew is the signature of negative correlation (leverage effect).
- **D:** A symmetric smile would need implied vol above 30% at both 45 and 75; at 75 it is below 30%.

Q819

GARP PE2 Q19 · Mock 4 Q20

An option pricing analyst at an investment bank has been asked to write a report examining the relationship between option prices and implied volatility curves. The analyst notes that the implied volatility curves of different underlying assets often have different shapes and explains the reasons why this occurs. Which of the following statements can correctly be included in the report?

A. The implied volatility smile commonly seen in equity options is due to the higher probability of a greater than three standard deviation price change than would be expected if prices are lognormally distributed.

B. The implied volatility smile commonly seen in foreign exchange rate options is due to the higher probability of a price change of between one and two standard deviations from the mean than would be expected if prices are lognormally distributed.

C. Demand for option protection against steep drops in equity prices leads to higher prices in out-of-the-money puts relative to out-of-the-money calls, which creates a downward-sloping implied volatility skew in these options.

D. Demand for option protection against the impact of unexpected central bank announcements on foreign exchange rates leads to higher prices, and higher implied volatilities, for at-the-money options relative to out-of-the-money options.

Answer: C

Step by step

1. Equity options show a downward-sloping skew, not a symmetric smile; the heavy tail is on the left only.
2. Demand for crash protection raises OTM put prices and implied volatility relative to OTM calls: statement C.
3. FX smile: implied vol is lowest at the money and higher away from the money, due to fat tails in both directions beyond lognormal.
4. Check A, B, D against these facts; each fails.

Why C is correct: Persistent demand for downside protection (crashophobia) makes low-strike puts expensive, producing the downward equity skew.

Why the others are wrong

- **A:** Describes a symmetric fat-tail effect (the currency story); equities have a heavy left tail and a thin right tail, giving a skew.
- **B:** The extra probability in FX is in the far tails; moves of one to two standard deviations are less likely than lognormal.
- **D:** FX smile has the lowest implied vol at the money; claiming ATM is highest turns the smile into a frown.

Q820

Mock 1 Q16 · Schweser Module Quiz 17.1 Q3

Which of the following is true of currency option volatility smiles?

A. Currency options generate implied volatilities that are higher for deep-in-the-money options and lower for deep-out-of-the-money options.

B. Currency options generate implied volatilities that are higher for deep-out-of-the-money options and lower for deep-in-the-money options.

C. Currency options generate implied volatilities that are higher for at-the-money options and lower for deep-out-of-the-money and deep-in-the-money options.

D. Currency options generate implied volatilities that are higher for deep-in-the-money options and deep-out-of-the-money options and lower for at-the-money options.

Answer: D

Step by step

1. Recall the shape: currency options show a roughly symmetric smile, with implied volatility lowest at the money.
2. Away-from-the-money options (both deep ITM and deep OTM) therefore have higher implied volatility than ATM options.
3. Reason: exchange rates have fatter tails than lognormal (jumps, stochastic volatility) in both directions, so both tails are priced richer than Black-Scholes-Merton implies.
4. Only D describes higher implied volatility on both wings and lowest at the money.

Why D is correct: Both tails of the FX distribution are heavier than lognormal, so implied volatility rises as strikes move away from at-the-money in either direction (a smile).

Why the others are wrong

- **A:** Describes a one-sided downward skew (equity-style); FX is symmetric, so deep OTM options do not have lower implied volatility.
- **B:** Describes a one-sided upward skew on one wing only; it ignores the high volatility of deep ITM options on the other wing.
- **C:** Describes a frown (inverted smile), which is the pattern from a price jump, not the usual currency pattern.

Q821

Mock 1 Q51

REBUILT — Option C was an obvious giveaway and A/D were trivially contradicted by the stem; rewrote distractors into plausible misconceptions while keeping the same tested idea.

During a geopolitical crisis, investors increase their demand for protection against extreme losses, and the prices of deep out-of-the-money equity put options rise significantly relative to at-the-money options. Which of the following best explains this behavior and its effect on the equity volatility skew?

A. Investors expect lower volatility during crises, so the skew flattens as deep OTM put prices rise.

B. Demand for downside tail-risk protection increases, raising implied volatility at low strikes and steepening the skew.

C. Higher demand for OTM puts raises implied volatility at high strikes, so the skew becomes an upward-sloping smirk.

D. Put-call parity forces deep OTM put prices to rise whenever call prices rise, leaving the skew unchanged.

Answer: B

Step by step

1. Deep OTM put prices rising means buyers bid up downside protection (crashophobia).
2. Option price and implied volatility are positively related, so implied volatility at low strikes rises.
3. Low-strike implied volatility rising relative to ATM and high-strike volatility makes the skew steeper.
4. Hence B.

Why B is correct: Greater demand for crash protection bids up low-strike put prices and implied volatilities, steep-

ening the left-hand skew characteristic of equity options (crashophobia).

Why the others are wrong

- **A:** Higher put prices imply higher, not lower, implied volatility; the skew steepens rather than flattens.
- **C:** Reverses the direction: the extra demand is at low strikes (puts), not high strikes, so the skew is downward-sloping.
- **D:** Put-call parity links put and call implied volatility at the same strike; it does not hold the skew fixed when demand shifts across strikes.

Q822

Mock 1 Q9

Assuming an option in the equity options portfolio has an at-the-money strike of 50, which of the following options is most likely to have the lowest level of implied volatility in real life?

A. At-the-money put.
B. In-the-money call.
C. Out-of-the-money put.
D. In-the-money put.

Answer: D

Step by step

1. Equity implied volatility falls as strike rises (downward skew/smirk).
2. Strikes: ATM put = 50; ITM call has strike below 50; OTM put has strike below 50; ITM put has strike above 50.
3. The highest strike (ITM put, K > 50) therefore has the lowest implied volatility.
4. Implied volatility depends on strike and maturity, not on whether the option is a put or call (put-call parity).

Why D is correct: With a downward equity skew, the option with the highest strike has the lowest implied volatility; the in-the-money put has a strike above 50.

Why the others are wrong

- **A:** Strike 50 gives a middle level of implied volatility, higher than for strikes above 50.
- **B:** ITM call has strike below 50, hence relatively high implied volatility; confuses ITM with high strike (true only for puts).
- **C:** OTM put has strike below 50, the region of highest implied volatility (crash protection demand).

Q823

Schweser Module Quiz 17.1 Q4

Which of the following statements regarding equity option volatility is true?

A. There is higher implied price volatility for away-from-the-money equity options.
B. ”Crashophobia” suggests actual equity volatility increases when stock prices decline.
C. Compared to the lognormal distribution, traders believe the probability of large down movements in price is similar to large up movements.
D. Increasing leverage at lower equity prices suggests increasing volatility.

Answer: D

Step by step

1. Equity options show a smirk: implied volatility is higher for low strikes and lower for high strikes.
2. Leverage explanation: when equity value falls, debt-to-equity rises, making equity riskier, so volatility rises as price falls.
3. Crashophobia is a fear of crashes (priced into low-strike options), not a statement about realized volatility.
4. Traders believe large down moves are more likely than large up moves versus lognormal.

Why D is correct: Falling equity value raises financial leverage and hence equity risk, so volatility rises as price falls, which supports higher implied volatility at low strikes.

Why the others are wrong

- **A:** That describes a symmetric smile (currency pattern); for equities only low-strike options have elevated implied volatility.
- **B:** Crashophobia is market fear priced into options, not a claim that actual volatility rises; it explains the skew via perceived crash risk.
- **C:** Traders see large down moves as more probable than large up moves (left-skewed implied distribution), which is why the skew exists.

Q824 BT notes P2.T5.03.04

If event-driven hedge funds increase demand for long positions in out-of-the-money call options on a company's stock (hoping to gain on takeover speculation), what is the effect on the company's volatility smile?

Answer: see model answer — Higher demand bids up the prices of OTM calls, which raises their implied volatilities. Because OTM calls are high-strike options, implied volatility increases on the right tail of the smile, making the skew less steep (a heavier right wing) than it would otherwise be.

Step by step

1. OTM call = strike above spot, so it sits on the right-hand side of the smile.
2. Greater buying demand raises call prices.
3. Implied volatility is the volatility that matches model price to market price, so higher price means higher implied volatility at those strikes.
4. Result: right-tail implied volatility rises.

Why this is correct: Demand-driven price increases on high-strike calls raise their implied volatility, lifting the right tail of the smile.

LO 17.f

Describe alternative ways of characterizing the volatility smile.

Q825 BT answers L2.T5.05.1 · BT notes P2.T5.05.02

A trader plots the volatility smile for vanilla European options with option delta (instead of strike) on the x-axis. Moving from left to right along the axis (low strikes to high strikes), which is the most plausible sequence of actual deltas?

A. call deltas = [0.10, 0.25, 0.50, 0.75, 0.90]
B. call deltas = [0.90, 0.75, 0.50, 0.25, 0.10]
C. call deltas = [0.10, 0.25, 0.50, 0.25, 0.10]
D. put deltas = [0.10, 0.25, 0.50, 0.25, 0.10]

Answer: B

Step by step

1. Moving left to right means the strike rises (low strike, in-the-money call, to high strike, out-of-the-money call).
2. A call's delta falls as the strike rises, from near 1 (deep ITM) to near 0 (deep OTM).
3. So call deltas must be monotonically decreasing, e.g. 0.90, 0.75, 0.50, 0.25, 0.10.
4. European put deltas lie between -1 and 0, so a positive put-delta list is impossible.
5. The familiar label '25 - 50 - 25' is only shorthand: -0.25 put, 0.50 call (about -0.50 put), 0.25 call.

Why B is correct: Call delta is a decreasing function of strike, so the x-axis runs from high call delta (low strike) to low call delta (high strike) in a monotone sequence.

Why the others are wrong

- **A:** Reverses the direction: call delta rises only as the strike falls, so this sequence would run from high strike to low strike.
- **C:** Treats the symmetric shorthand labels (10, 25, 50, 25, 10) as actual call deltas; delta is monotone in strike, not symmetric.
- **D:** Uses positive put deltas, but a European put's delta is negative (between -1 and 0), and the sequence is also non-monotone.

Q826 BT notes P2.T5.05.01

What is the advantage of using option delta on the x-axis, rather than the strike price, to plot the volatility smile?

Answer: see model answer — Delta measures how far in or out of the money an option is while implicitly reflecting its time to expiration (and volatility), so smiles for different maturities and underlying levels become comparable and more stable than when plotted against strike. Per Hull, it also lets traders apply the smile concept to options other than standard European and American calls and puts.

Why this is correct: Delta is a standardised, maturity-adjusted measure of moneyness, and it extends the smile to

non-vanilla options.

LO 17.g

Describe volatility term structures and volatility surfaces and how they may be used to price options.

Q827 GARP book Ch.17 Q17.2

What is a volatility surface?

Answer: see model answer — A volatility surface is a three-dimensional representation of implied volatility as a function of both strike price (often expressed as K/S0) and time to maturity. It combines the volatility term structure (implied volatilities across maturities for at-the-money options) with the volatility smile (implied volatilities across strikes at each maturity). Each point is the Black-Scholes-Merton implied volatility of an option with that strike and maturity.

Step by step

1. Implied volatility depends on strike (smile/skew) and on maturity (term structure).
2. Plot implied volatility against both dimensions to obtain a surface.
3. The at-the-money slice is the term structure; a fixed-maturity slice is the smile.

Why this is correct: The surface merges the smile and the term structure into one grid of implied volatilities indexed by strike and maturity.

Q828 GARP book Ch.17 Q17.5

How is a volatility surface used in option pricing?

Answer: see model answer — To price an option with a given strike and maturity, a trader reads (interpolating between grid points if needed) the implied volatility from the surface and inputs it into the Black-Scholes-Merton formula. This produces prices consistent with those of actively traded options at neighbouring strikes and maturities, and the surface can be used to check the consistency of model-generated prices with the market.

Step by step

1. Locate the option's strike (K/S0) and maturity on the surface.
2. Interpolate the implied volatility from neighbouring grid points.
3. Insert that volatility into Black-Scholes-Merton to obtain the price, which is consistent with market-quoted options.

Why this is correct: The surface supplies the strike- and maturity-specific volatility that BSM needs, so non-traded options are priced consistently with traded ones.

Q829 GARP book Ch.17 Q17.21

"The Black-Scholes-Merton model is used by traders as an interpolation tool." Discuss this view.

Answer: see model answer — The view is largely correct. Because BSM assumes constant volatility, no single volatility reproduces all market prices; traders instead quote prices as implied volatilities and use BSM only to translate between a volatility and a price. By building a volatility surface from liquid options and interpolating on it, they obtain the implied volatility (and hence BSM price) for options of other strikes and maturities that is consistent with the market. BSM thus acts as a quoting and interpolation convention rather than a literal description of asset dynamics.

Step by step

1. BSM with constant volatility is contradicted by smiles and term structures.
2. Traders therefore quote implied volatility, a one-to-one mapping to price.
3. The surface is interpolated for non-traded strikes/maturities and fed into BSM to get consistent prices.

Why this is correct: BSM is used as a mapping between prices and implied volatilities, with the volatility surface providing the interpolation across strikes and maturities.

Q830 GARP book Ch.17 Q17.22

REBUILT — Original referenced Table 17.2 that was not included and had no answer; supplied a self-contained volatility surface and converted to a numeric MCQ.

The following table shows implied volatilities (%) for a currency's options, by maturity and strike-to-spot ratio K/S0.

Maturity	K/S0 = 0.90	0.95	1.00	1.05	1.10
3 months	15.0	13.6	12.4	13.2	14.6
6 months	14.8	13.8	12.8	13.4	14.4
1 year	15.2	14.4	13.6	14.0	14.8
2 years	15.6	14.8	14.2	14.6	15.2

Using linear interpolation in both strike and maturity, what implied volatility should a trader use to price an 8-month option with K/S0 = 1.04?

A. 13.28%
B. 13.49%
C. 13.60%
D. 13.17%

Answer: B

Step by step

1. Bracket the option: maturities 6 months and 1 year; strikes 1.00 and 1.05. K/S0 = 1.04 is 0.8 of the way from 1.00 to 1.05.
2. 6-month vol at 1.04: $12.8 + 0.8(13.4 - 12.8) = 13.28\%$.
3. 1-year vol at 1.04: $13.6 + 0.8(14.0 - 13.6) = 13.92\%$.
4. 8 months is 2/6 = 1/3 of the way from 6 months to 1 year: $13.28 + (13.92 - 13.28)/3 = 13.493\%$.

Why B is correct: Interpolate in strike at each bracketing maturity, then interpolate in maturity with weight 1/3 on the longer maturity.

Why the others are wrong

- **A:** 13.28% is the 6-month vol at K/S0 = 1.04; it interpolates in strike but ignores the maturity dimension.
- **C:** 13.60% interpolates in maturity only at the K/S0 = 1.05 column, ignoring strike interpolation.
- **D:** 13.17% uses the strike weights reversed (0.2 instead of 0.8 toward 1.05) in both maturities.

Q831 GARP book Ch.17 Q17.28

REBUILT — Original referenced Table 17.2 that was not included and had no answer; supplied a self-contained volatility surface and converted to a numeric MCQ.

The following table shows implied volatilities (%) for a currency's options, by maturity and strike-to-spot ratio K/S0.

Maturity	K/S0 = 0.90	0.95	1.00	1.05	1.10
3 months	15.0	13.6	12.4	13.2	14.6
6 months	14.8	13.8	12.8	13.4	14.4
1 year	15.2	14.4	13.6	14.0	14.8
2 years	15.6	14.8	14.2	14.6	15.2

Using linear interpolation in both strike and maturity, what implied volatility should a trader use to price an 11-month option with K/S0 = 0.98?

A. 13.32%
B. 13.60%
C. 13.80%
D. 13.97%

Answer: C

Step by step

1. Bracket: maturities 6 months and 1 year; strikes 0.95 and 1.00. K/S0 = 0.98 is 0.6 of the way from 0.95 to 1.00.
2. 6-month vol: $13.8 + 0.6(12.8 - 13.8) = 13.20\%$.
3. 1-year vol: $14.4 + 0.6(13.6 - 14.4) = 13.92\%$.
4. 11 months is 5/6 of the way from 6 months to 1 year: $13.20 + (5/6)(13.92 - 13.20) = 13.80\%$.

Why C is correct: Interpolate across strike at the two bracketing maturities, then across maturity with weight 5/6 on the 1-year value.

Why the others are wrong

- **A:** 13.32% puts the maturity weight on the wrong end (1/6 instead of 5/6 toward 1 year).

- **B:** 13.60% just takes the nearest grid point (1 year, K/S0 = 1.00) with no interpolation.
- **D:** 13.97% reverses the strike weights (0.4 instead of 0.6 toward 1.00), then interpolates in maturity.

Q832

BT answers L2.T5.06.1 · BT notes P2.T5.05.02

REBUILT — Original distractor 'Independence' was a giveaway and other options were weak; reworded options so exactly one mechanism explains a downward-sloping term structure.

Which of the following is the most plausible explanation for a downward-sloping implied volatility term structure?

A. Short-dated volatility is currently high relative to its long-run level and volatility is mean reverting
B. Short-dated volatility is currently low relative to its long-run level and volatility is mean reverting
C. Returns have lighter tails than the lognormal distribution (platykurtosis)
D. Returns are independent over time, so volatility is the same at all maturities

Answer: A

Step by step

1. Implied volatility for longer maturities reflects the expected average volatility over the option's life.
2. If volatility mean reverts and is currently high, expected future volatility falls toward the long-run level, so long-dated implied vol is lower than short-dated.
3. Hence the term structure slopes downward when short-dated vol is high.

Why A is correct: Mean reversion from an elevated current volatility lowers the average expected volatility over longer horizons.

Why the others are wrong

- **B:** This gives an upward-sloping term structure, the opposite direction.
- **C:** Platykurtosis affects the shape of the smile across strikes (a frown), not the slope across maturities.
- **D:** Independence of returns with constant volatility would imply a flat term structure, not a downward slope.

Q833

BT answers L2.T5.06.2 · BT notes P2.T5.05.02

REBUILT — True/false converted to a four-option MCQ; source answer line was garbled and the second source had no answer.

If actual (observed) volatility follows a GARCH(1,1) process, which of the following describes the implied volatility term structure?

A. It is always upward sloping
B. It slopes up if current volatility is below the long-run level and down if it is above
C. It is always downward sloping
D. It is flat, because GARCH volatility is constant

Answer: B

Step by step

1. GARCH(1,1) volatility mean reverts to a long-run level.
2. Expected average volatility over longer horizons therefore moves from the current level toward the long-run level.
3. If current volatility is below the long-run level the term structure rises; if above, it falls.

Why B is correct: Mean reversion in GARCH makes the slope depend on where current volatility sits relative to the long-run level.

Why the others are wrong

- **A:** Ignores that high current volatility reverts downward, producing a downward slope.
- **C:** Ignores that low current volatility reverts upward, producing an upward slope.
- **D:** GARCH volatility is time varying, so the term structure is not flat in general.

Q834

AP Mock 2 Q46

A trader wishes to mark to market a book of plain-vanilla stock options. He is long at-the-money calls and short deep out-of-the-money calls. The trader's bonus increases as the value of his book increases, and there is a pronounced smile on these options. Which approach should the trader use to mark the book?

A. Use historical volatility so as to correct for pricing mistakes inherent in the option market.
B. For each option, use the implied volatility of the most similar option tradable on the market.
C. Use the average of the implied volatilities for the traded options, since all options should have the same implied

volatility under Black-Scholes.

D. Use the implied volatility of at-the-money options, since volatility estimation is more reliable there.

Answer: B

Step by step

1. Marking to market requires prices consistent with the market for each strike and maturity.
2. With a pronounced smile, each strike has its own implied volatility, read from the volatility surface or the nearest traded option.
3. Using the implied volatility of the most similar traded option for each position gives market-consistent values.

Why B is correct: Each option should be valued with the market implied volatility for its own strike and maturity, not a single volatility.

Why the others are wrong

- **A:** Historical volatility is not a market price and would not produce market-consistent marks.
- **C:** Averaging relies on the constant-volatility BSM assumption that the smile contradicts.
- **D:** A single at-the-money volatility ignores the smile and misstates the value of the out-of-the-money calls; it is not market-consistent and lets the trader bias the marks in his own favor.

Q835

BT answers P2.T5.411.1 · BT notes P2.T5.411.1

REBUILT — Merged two copies (one had an OCR-damaged 2-year/0.50 cell, restored to 29.8) and replaced giveaway distractors with results of realistic interpolation mistakes.

The table shows a volatility surface of Black-Scholes-Merton implied volatilities (%). Rows are maturity; columns are strike divided by current asset price, K/S(0).

Maturity	K/S(0)=0.50	0.75	1.00	1.25	1.50
1 month	36.9	28.9	20.9	14.9	15.2
3 months	35.3	28.3	21.3	15.8	16.3
6 months	32.9	26.9	20.9	16.0	16.8
1 year	32.1	27.1	22.1	17.1	17.5
2 years	29.8	25.8	21.8	16.8	18.0
5 years	30.0	27.0	24.0	18.0	19.4

The current asset price is \$30.00. A trader wants to value an out-of-the-money call with strike \$33.00 and maturity 4.0 years. Using linear interpolation in both strike and maturity, which implied volatility (%) is NEAREST to the appropriate estimate?

A. 19.80

B. 20.70

C. 21.00

D. 21.60

Answer: C

Step by step

1. K/S(0) = 33/30 = 1.10, between the 1.00 and 1.25 columns (weight 0.10/0.25 = 0.40 on the 1.25 column).
2. 2-year row at 1.10: $21.8 + 0.4(16.8 - 21.8) = 19.80$.
3. 5-year row at 1.10: $24.0 + 0.4(18.0 - 24.0) = 21.60$.
4. 4 years lies 2/3 of the way from 2 to 5 years: $19.80 + \frac{2}{3}(21.60 - 19.80) = 21.00$.

Why C is correct: A volatility surface is used by interpolating between quoted strikes and maturities; interpolating first in strike and then in maturity gives 21.00%.

Why the others are wrong

- **A:** 19.80 is the 2-year value at 1.10; the maturity interpolation toward 5 years was skipped.
- **B:** 20.70 uses a 50% weight (midpoint of 2 and 5 years) instead of 2/3 for 4 years.
- **D:** 21.60 is the 5-year value at 1.10; it ignores that 4 years is not 5 years.

Q836

BT notes P2.T5.01.2

Identify the axes in a plot of (i) the volatility term structure and (ii) the volatility surface.

Answer: see model answer — (i) The volatility term structure plots implied volatility (vertical axis) against time to expiration (horizontal axis), for at-the-money options. (ii) The volatility surface is three-dimensional: implied

volatility on the vertical axis, with time to expiration and strike price (or K/S0) on the two horizontal axes. It combines the term structure with smiles across strikes and is used to interpolate volatilities for options with any strike and maturity.

Q837 BT notes P2.T5.07.02

Which volatility smile is likely to be more pronounced: short-term or long-term maturity?

Answer: see model answer — The short-term smile is more pronounced. As maturity increases, the smile tends to flatten and the volatility surface becomes less sensitive to strike; this is visible in surfaces where the strike-related spread of implied volatilities is widest for the 1-month row and narrower for longer rows. Short-dated options are more sensitive to jump risk and skew, while over long horizons effects average out.

Q838 GARP PE2 Q3 · Mock 4 Q4

A market-maker on the FX desk must quote an FX call option expiring in 7 months with K/S0 = 1.075. The implied volatility surface (%) is:

Time to expiration	K/S0 = 0.90	0.95	1.00	1.05	1.10
1 month	9.25	8.55	8.05	8.70	9.45
3 months	9.10	8.70	8.30	8.75	9.15
6 months	9.45	9.05	8.70	9.10	9.45
1 year	9.65	9.50	9.35	9.55	9.75

What implied volatility should the market-maker use?

A. 9.18%
B. 9.28%
C. 9.34%
D. 9.65%

Answer: C

Step by step

1. Bracketing cells: 6 months and 1 year; K/S0 1.05 and 1.10. 1.075 is the midpoint of the strikes; 7 months is 1/6 of the way from 6 to 12 months.
2. Strike midpoint at 6 months: $(9.10 + 9.45)/2 = 9.275$.
3. Strike midpoint at 1 year: $(9.55 + 9.75)/2 = 9.65$.
4. Time interpolation: $9.275 + \frac{1}{6}(9.65 - 9.275) = 9.3375 \approx 9.34\%$.

Why C is correct: Interpolating linearly in both strike and maturity between the four surrounding grid points gives 9.3375%.

Why the others are wrong

- **A:** 9.18% interpolates in time only, at the 1.05 strike (9.175), without moving to 1.075.
- **B:** 9.28% interpolates in strike only at 6 months (9.275), ignoring the extra month of maturity.
- **D:** 9.65% takes the 1-year row at the strike midpoint, ignoring that 7 months is close to 6 months.

LO 17.h

Explain the impact of the volatility smile on the calculation of an option's Greek letter risk measures.

Q839 GARP book Chapter 17 Q17.4

What is a minimum variance delta?

Answer: see model answer — The minimum variance delta is the delta that accounts for how the implied volatility curve moves when the underlying price changes. Because equity prices and volatility are negatively correlated, a rise in the stock price lowers the whole smile/skew curve (and a fall raises it), and this shift tends to dominate the movement along the curve. The delta that incorporates this expected change in implied volatility is the one that minimizes the variance of the hedged position's value. For equities it is lower than the Black-Scholes-Merton delta,

which holds implied volatility fixed.

Q840

BT answers L2.T5.08.1 · BT notes P2.T5.08.01

REBUILT — BT notes version lacked the ATM setup; merged with the BT answers version and added the parameters needed to confirm the result numerically.

A call option is at-the-money with S = K = \$20 and a BSM delta of about 0.5. The implied volatility curve is flat (no volatility smile), so the Greeks are the standard BSM Greeks. Holding other inputs constant, what happens to the call's delta when the stock price falls to \$19 (slightly out-of-the-money)?

A. Delta remains 0.5
B. Delta rises above 0.5
C. Delta falls below 0.5
D. It cannot be determined without knowing the smile

Answer: C

Step by step

1. With a flat implied volatility, volatility does not change with moneyness, so there is no smile adjustment.
2. The standard BSM call delta is N(d1), which is increasing in S/K.
3. Illustration (r = 0, sigma = 20%, T = 1): at S = 20, d1 = 0.100 and N(d1) = 0.540; at S = 19, d1 = -0.156 and N(d1) = 0.438.
4. Delta falls as the option moves out-of-the-money.

Why C is correct: With flat volatility delta is N(d1), which falls as the stock falls and the call moves out-of-the-money.

Why the others are wrong

- **A:** Ignores that N(d1) changes with moneyness.
- **B:** Reverses the direction; delta rises only when the stock rises (more in-the-money).
- **D:** A smile is irrelevant here because the curve is stated to be flat; the answer is determinable.

Q841

BT answers L2.T5.08.2 · BT notes P2.T5.08.02

REBUILT — Original answer mixed a level comparison (delta vs 0.5) with the skew effect; restated so the comparison is against the flat-volatility BSM delta, with only movement along the skew considered.

An at-the-money call (S = K = \$20) is priced in an equity market with "crashophobia": implied volatility is a decreasing function of K/S (a downward-sloping skew). Consider only the movement along this fixed skew (ignore any shift of the whole curve). When the stock rises to \$21, how does the sensitivity of the option price to the stock price compare with the BSM delta computed with constant volatility?

A. It is the same, because delta depends only on N(d1)
B. It is higher, because the rise in S lowers K/S and moves the option to a point with higher implied volatility
C. It is lower, because the rise in S lowers K/S and moves the option to a point with lower implied volatility
D. It is indeterminate, because the sign of the vega term depends on the strike

Answer: B

Key correction: Source key (BT answers L2.T5.08.2) said B (delta > 0.5); the letter is unchanged but the stem now states that the comparison is with the flat-volatility BSM delta and ignores curve shifts.

Step by step

1. Total sensitivity: dC/dS = BSM delta + vega x d(sigma)/dS.
2. Stock up from 20 to 21 lowers K/S from 1.00 to 0.952.
3. The skew slopes down in K/S, so a lower K/S means higher implied volatility: d(sigma)/dS > 0.
4. Vega > 0 for a call, so vega x d(sigma)/dS > 0 and the effective sensitivity exceeds BSM delta.

Why B is correct: Moving along a downward-sloping skew as the stock rises raises implied volatility, and positive vega converts that into extra option value.

Why the others are wrong

- **A:** Treats delta as only N(d1), ignoring that implied volatility changes with the stock price.
- **C:** Gets the skew direction backwards; a lower K/S lies on the high-volatility side of the skew.
- **D:** Vega of a call is always positive, so the sign is determinate.

Q842

BT answers L2.T5.08.3 · BT notes P2.T5.08.03

REBUILT — Original key 'unclear' mixed moneyness delta with the smile effect; restated so the comparison is with the flat-volatility BSM delta, giving exactly one defensible answer.

Asset returns have a light left tail and a heavy right tail relative to the lognormal, so implied volatility is an increasing function of K/S. An at-the-money call (S = K = $20) is considered; focus only on the movement along this fixed smile (ignore shifts of the whole curve). When the stock rises to $21, how does the sensitivity of the option price to the stock price compare with the BSM delta computed with constant volatility?

A. It is the same, because delta depends only on N(d1)
B. It is higher, because the stock rise moves the option to a point with higher implied volatility
C. It is lower, because the stock rise lowers K/S and moves the option to a point with lower implied volatility
D. It is indeterminate, because moneyness and smile effects work in opposite directions

Answer: C

Key correction: Source key (BT answers L2.T5.08.3) said D (unclear); the question is now framed against the constant-volatility BSM delta, for which the correct answer is C.

Step by step

1. Total sensitivity: dC/dS = BSM delta + vega x d(sigma)/dS.
2. Stock up from 20 to 21 lowers K/S from 1.00 to 0.952.
3. Implied volatility increases with K/S, so lower K/S means lower implied volatility: d(sigma)/dS < 0.
4. Vega > 0, so vega x d(sigma)/dS < 0 and the effective sensitivity is below the BSM delta.

Why C is correct: With volatility rising in K/S, a higher stock price lowers K/S and implied volatility, so the vega term subtracts from BSM delta.

Why the others are wrong

- **A:** Ignores the change in implied volatility along the smile.
- **B:** Applies the downward-skew (equity) direction instead of the stated upward-sloping one.
- **D:** The two effects only conflict if one compares with 0.5; against the constant-volatility BSM delta the smile effect has a single sign.

Q843

Schweser Module Quiz 17.2 Q1

REBUILT — Added to cover the Schweser dominance rule and the BSM vs minimum variance delta comparison cleanly, since the BT series only tests movement along the curve.

Equity prices and implied volatilities are negatively correlated. When a stock's price rises, which statement about the delta used to hedge an equity option is most accurate?

A. The shift of the whole implied volatility curve is the smaller effect and the movement along the curve dominates
B. The minimum variance delta is higher than the BSM delta
C. The shift of the whole implied volatility curve tends to dominate the movement along it, so the minimum variance delta is lower than the BSM delta
D. The BSM delta already incorporates changes in implied volatility, so it equals the minimum variance delta

Answer: C

Step by step

1. Rule 1: a higher stock price lowers K/S and moves the option along the skew to a higher volatility.
2. Rule 2: negative correlation moves the entire curve down when the stock rises.
3. In practice rule 2 dominates, so the volatility used falls as the stock rises.
4. The delta incorporating this is the minimum variance delta, which is lower than the BSM delta.

Why C is correct: The curve shift caused by the negative price-volatility correlation outweighs the movement along the curve, so minimum variance delta is below BSM delta.

Why the others are wrong

- **A:** Reverses the dominance rule.
- **B:** Reverses the comparison; BSM delta is higher.
- **D:** BSM delta assumes implied volatility stays constant, so it ignores the change.

Q844 BT answers P2.T5.409.1 · BT notes P2.T5.409.1

In regard to the relationship, if any, between Greek vega and implied volatility, which of the following is TRUE?

A. If the implied volatility smile is a perfectly flat, horizontal line, then an increase in strike price has no effect on the option's price

B. If the implied volatility smile is a perfectly flat, horizontal line, then Greek vega is zero

C. Vega is the first derivative of the implied volatility smile; i.e., it plots the slopes of lines tangent to the implied volatility plot

D. Under the Black-Scholes-Merton model, vega is always positive and highest for an at-the-money option, but these properties do not predict any particular shape of the implied volatility smile; a skew, smirk, frown or other shape is allowed

Answer: D

Step by step

1. Vega = ∂(option price)/$\partial\sigma$. Under BSM it equals $S_0\,\phi(d_1)\sqrt{T} > 0$ for any European option, and it peaks near the money.
2. The implied volatility smile is a different object: the volatilities backed out of observed market prices across strikes. Its shape is set by supply and demand (and the true return distribution), not by vega.
3. Therefore vega's sign and ATM peak are consistent with any smile shape: statement D is true.
4. A flat smile means constant σ across strikes, which is exactly the BSM world; there the call price still falls as the strike rises, and vega is still positive.

Why D is correct: Vega is a model-based sensitivity of price to volatility, while the smile is market-implied volatility plotted against strike; the first does not dictate the shape of the second.

Why the others are wrong

- **A:** Wrong: a flat smile only means σ is the same for every strike; a higher strike still lowers a call's price (raises a put's) because moneyness changes. Confuses 'constant volatility' with 'price insensitive to strike'.
- **B:** Wrong: vega is the price sensitivity to σ and stays positive even when σ is constant across strikes. Confuses a flat smile with zero volatility sensitivity.
- **C:** Wrong: vega is the derivative of the option price with respect to volatility, not the slope of the smile (which is the derivative of implied volatility with respect to strike). Mixes up the two derivatives.

Q845 BT answers P2.T5.411.2 · BT notes P2.T5.411.2

REBUILT — Source key (A) only holds if implied volatility moves along a fixed curve; the stem did not state the empirical shift of the whole curve, so it was ambiguous and conflicted with the Schweser reading; the stem now specifies the observed behaviour and the key is B.

A one-year at-the-money equity call with strike 100 trades at 13.28, implying a volatility of 30.0% (risk-free rate 3%). The Black-Scholes-Merton (BSM) delta is N(d1) = 0.60 for the call and 0.60 − 1.0 = −0.40 for the put with the same strike and maturity. The equity option exhibits a steep implied volatility skew (heavy left tail, light right tail). Empirically, equity prices and volatility are negatively correlated, so when the stock price rises the whole implied volatility curve shifts down, and this shift dominates the move along the existing curve. Which statement best describes the true (minimum variance) deltas relative to the BSM deltas of 0.60 and −0.40?

A. True call delta is higher than 0.60 and true put delta is higher than −0.40

B. True call delta is lower than 0.60 and true put delta is lower than −0.40

C. True call delta is equal to 0.60 and true put delta is equal to −0.40

D. The relationship cannot be determined because delta is only a first-order approximation

Answer: B

Key correction: Source key (BT) said A; with the empirical downward shift of the whole volatility curve dominating (as in Schweser), the true delta is lower than the BSM delta, so B is correct.

Step by step

1. Check BSM inputs: with S=K=100, σ=30%, r=3%, T=1: d1 = (0.03+0.045)/0.30 = 0.25, N(d1)=0.599 ≈ 0.60 and call price = 13.28 (computed), so the numbers are consistent.
2. True delta = BSM delta + vega × dσ_imp/dS.
3. With a skew, a price rise lowers K/S and moves the option up the curve (σ up), but the negative price-volatility correlation shifts the whole curve down and dominates, so net dσ_imp/dS < 0.
4. Vega is positive for calls and puts alike, so the adjustment is negative for both: true call delta < 0.60.
5. Put delta = call delta − 1 (parity), so the true put delta is also lower than −0.40 (more negative).

Why B is correct: The dominant effect of a stock-price rise is a downward shift of the implied volatility curve,

which reduces the option price beyond the BSM delta effect, so the minimum variance delta is below the BSM delta for both calls and puts.

Why the others are wrong

- **A:** Considers only the move along the existing curve (σ rises as K/S falls) and ignores the downward shift of the whole curve, which dominates.
- **C:** Assumes volatility is constant when S changes, which is the BSM assumption that the smile contradicts.
- **D:** Delta being a first-order approximation does not prevent a directional comparison; the sign of the vega adjustment is determinable from the curve shift.

Q846 Schweser Module Quiz 17.2 Q1

When evaluating the impact of a volatility smile on the calculation of the option Greeks, which of the following statements is most likely correct?

A. If equity prices and volatilities are negatively correlated, equity price increases will move the option up the existing curve.

B. If equity prices and volatilities are positively correlated, equity price increases will move the entire option curve down.

C. The minimum variance delta will be higher than the Black-Scholes-Merton model delta.

D. When equity prices change, the effect from movements in the implied volatility curve up or down tends to dominate movements along the existing curve.

Answer: D

Step by step

1. Two effects follow an equity price change: movement along the existing smile (K/S changes) and a shift of the whole curve.
2. Equity prices and volatility are negatively correlated, so a price rise shifts the entire curve down (a fall shifts it up).
3. In practice the curve shift dominates movement along the curve (D).
4. The delta incorporating this shift is the minimum variance delta, which is lower than the BSM delta.

Why D is correct: Because of the negative price-volatility correlation, whole-curve shifts outweigh movement along the curve, which is why BSM delta overstates the minimum variance delta.

Why the others are wrong

- **A:** Moving up the curve is the along-the-curve effect and does not follow from negative correlation; negative correlation moves the whole curve down when prices rise.
- **B:** Reverses the correlation: the curve falls when prices rise because correlation is negative, not positive.
- **C:** Reversed: the BSM delta is higher than the minimum variance delta.

LO 17.i

Explain the impact of a single asset price jump on a volatility smile.

Q847 GARP book Chapter 17 Q17.7

What volatility smile is likely to be caused by jumps in the underlying asset price? Is the pattern likely to be more pronounced for a 2-year option than for a 3-month option?

Answer: see model answer — A single anticipated jump makes the terminal price distribution bimodal (mixture of an up-move and a down-move) with the same mean and standard deviation as the lognormal benchmark. Relative to the lognormal, there is more probability near the current price and less in the intermediate tails, so at-the-money options have a higher implied volatility than in- or out-of-the-money options: a volatility 'frown' (hump) rather than a smile. The pattern is more pronounced for the 3-month option, because over 2 years the effect of one jump is diluted by ordinary diffusion and the distribution moves closer to lognormal, so the frown flattens.

Step by step

1. Jump news makes the price distribution bimodal with the same mean and variance as the lognormal.
2. Away-from-the-money options are worth less than under BSM, so their implied volatility is lower than at-the-money: a frown.

3. A jump is a fixed one-off contribution to variance; as maturity lengthens, diffusion variance grows and the jump matters proportionally less.
4. Hence the frown is more pronounced for the 3-month option, not the 2-year option.

Why this is correct: A jump produces a bimodal distribution with the same first two moments, giving a frown (ATM implied volatility highest), and its relative impact fades as maturity lengthens.

Q848 GARP book Chapter 17 Q17.12

A stock price is currently $20. Tomorrow, news is expected to be announced that will either increase the price by $5 or decrease the price by $5. What are the problems in using Black-Scholes-Merton to value 1-month options on the stock?

Answer: see model answer — BSM assumes the price follows a continuous lognormal diffusion with constant volatility, so the 1-month terminal price is lognormal. Here the price will jump to about $25 or $15 tomorrow (roughly equal risk-neutral probabilities so the expected price stays near $20) and then diffuse, making the true terminal distribution bimodal, not lognormal. No single volatility reproduces this: a BSM value using one volatility will misprice options, and implied volatilities will vary with strike (a frown, highest near the money, lower for deep in- and out-of-the-money options). Correct valuation is a probability-weighted average of BSM values starting from $25 and $15, each with the (lower) post-news diffusion volatility.

Step by step

1. BSM requires a lognormal terminal price from continuous diffusion; an announced one-off jump violates this.
2. The jump to roughly $25 or $15 gives a bimodal terminal distribution with a mean near $20.
3. A single volatility cannot match it, so BSM misprices away-from-the-money options and implied volatilities form a frown across strikes.
4. Better approach: value as a weighted average of two BSM prices (start at $25 and $15).

Why this is correct: The jump invalidates BSM's lognormal, continuous-path assumption, producing a bimodal distribution and a volatility frown.

Q849 GARP book Chapter 17 Q17.18

REBUILT — Original open question required DerivaGem and gave no answer; converted to a self-contained MCQ with the risk-neutral probability and the computed implied-volatility pattern.

A major lawsuit ruling will be announced tomorrow. The stock price is $60. If the ruling is favorable it jumps to $75 and afterwards volatility is 25%; if unfavorable it jumps to $50 and volatility is 40%. Ignore discounting over the one day. The 6-month risk-free rate is 6% (continuously compounded), no dividends. The risk-neutral probability of a favorable ruling is p, and 6-month European call prices today are the p-weighted average of BSM prices from each post-jump price. Which statement is correct?

A. p = 0.40; implied volatility rises steadily with strike (about 40% at K = $30 up to about 48% at K = $80)
B. p = 0.40; implied volatility is hump-shaped (a frown), about 47.8% at K = $40, falling to about 40.4% at K = $80
C. p = 0.60; implied volatility is hump-shaped (a frown), about 47.8% at K = $40, falling to about 40.4% at K = $80
D. p = 0.40; implied volatility is U-shaped (a smile) with its minimum near the $60 strike

Answer: B

Step by step

1. Risk-neutral pricing: 60 = 75p + 50(1 - p), so p = 10/25 = 0.40.
2. Call price at strike K = 0.40 x BSM(S=75, vol 25%) + 0.60 x BSM(S=50, vol 40%), T = 0.5, r = 6%.
3. Prices (Python): K=30: 30.955; 40: 21.935; 50: 14.387; 60: 8.564; 70: 4.430; 80: 1.934.
4. Invert BSM with S=60 for implied volatility: 46.7%, 47.8%, 47.8%, 46.1%, 43.2%, 40.4% for K = 30, 40, 50, 60, 70, 80.
5. Implied volatility peaks at the middle strikes and is lower at the extremes: a frown, as for a price-jump distribution.

Why B is correct: The jump gives a bimodal distribution, so implied volatility is highest for strikes around $40-$50 and lower away from the money, with p = 0.40 from the martingale condition.

Why the others are wrong

- **A:** p is right, but implied volatility is not monotonically increasing; it falls as strike rises above $50 (it is a frown, not a positive skew).
- **C:** Uses p = 15/25 = 0.60, the probability of the unfavorable ruling (or computed (75-60)/25) instead of the favorable one; the expected price then is 65, not 60.

- **D:** A U-shaped smile is the pattern for fat tails (e.g. currencies), not for a single jump; the jump gives the opposite, hump-shaped pattern.

Reading 18. Fundamental Review of the Trading Book

John C. Hull, Risk Management and Financial Institutions, 6th Edition (Hoboken, NJ: John Wiley & Sons, 2023). Chapter 18.

LO	Learning objective	Qs
18.a	describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes.	13
18.b	compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons.	7
18.c	explain the FRTB revisions to Basel regulations in the following areas: Classification of positions in the trading book compared to the banking book. Backtesting, profit and loss attribution, credit risk, and securitizations.	21

LO 18.a

Describe the changes to the Basel framework for calculating market risk capital under the Fundamental Review of the Trading Book (FRTB) and the motivations for these changes.

Q850

GARP book Ch.18 Q18.1

REBUILT — Aligned to GARP 2026 Ch.18 content: GARP says the stressed period is found by searching for the 250-day period of worst market movements for the current portfolio, not self-selected.

Outline the differences between the way market risk capital is calculated in (a) Basel I, (b) Basel II.5, and (c) the FRTB.

Answer: see model answer — (a) Basel I: capital is based on a 10-day VaR at 99% confidence, estimated from recent data (typically one to four years). (b) Basel II.5: adds a stressed VaR (10-day, 99%) calibrated to the 250-day period of stressed market conditions that is worst for the current portfolio (found by searching back through history) to the current VaR, plus the incremental risk charge (IRC) for credit-dependent positions and the comprehensive risk measure (CRM) for securitizations. (c) FRTB: replaces VaR plus stressed VaR with a single expected shortfall at 97.5% confidence calibrated to a stressed period, scaled using liquidity horizons of 10, 20, 40, 60 and 120 days instead of a flat 10 days; it also introduces a stricter trading-book/banking-book boundary, VaR backtesting and P&L attribution as conditions for internal models, an incremental default risk charge (99.9% one-year VaR), and standardized treatment of securitizations.

Step by step

1. Basel I: 10-day, 99% VaR on current data.
2. Basel II.5: add stressed VaR (250-day stress window), IRC and CRM.
3. FRTB: single stressed ES at 97.5%, liquidity horizons 10-120 days, tighter book boundary, backtesting and P&L attribution, IDR charge, standardized securitizations.

Why this is correct: Each framework changed the risk measure, the calibration period and the treatment of credit and securitization risk; FRTB moves from VaR to ES and from a uniform 10-day horizon to liquidity-based horizons.

Q851

GARP book Ch.18 Q18.4

REBUILT — Aligned to GARP 2026 Ch.18 content: removed the claim about gaming, which GARP does not state.

What are the advantages of expected shortfall over value at risk?

Answer: see model answer — VaR only states a loss threshold that will not be exceeded at a given confidence level and says nothing about how bad losses can be beyond it, so it can ignore catastrophic tail outcomes (e.g., a bond portfolio with a 2% default probability has a 95% VaR of 0 but an ES of 380 million on a 950 million portfolio). ES is the expected loss conditional on exceeding the VaR level, so it captures the severity of tail losses, is more informative for fat-tailed distributions. It is also coherent (subadditive), so diversification never increases measured risk. This is why the FRTB uses 97.5% ES in place of 99% VaR.

Step by step

1. VaR ignores loss size beyond the quantile.
2. ES averages the tail losses beyond VaR, e.g. 95% ES = 40% x 950 = 380 million versus VaR 0.
3. For a normal distribution 97.5% ES ($\mu + 2.338\sigma$) is about 99% VaR ($\mu + 2.326\sigma$), but with fat tails ES is considerably larger, which motivates the FRTB switch.

Why this is correct: ES measures what happens in the tail rather than only where the tail starts, which is the

FRTB's motivation for replacing VaR.

Q852

BT answers P2.T5.405.2

REBUILT — Source relied on an off-syllabus BCBS paper with a poorly worded 'EXCEPT...LEAST true' stem and a debatable claim about ES backtesting; rebuilt around the Schweser ES-versus-VaR motivation with one clearly false statement. Aligned to GARP 2026 Ch.18 content: GARP does not tie the FRTB stressed ES to a 250-day window (searches for the worst stressed period); wording made period-neutral. Answer C unchanged (GARP: ES harder to backtest than VaR; stressed measures cannot be backtested).

Which of the following statements about the move from 99% VaR to 97.5% expected shortfall (ES) under the FRTB is INCORRECT?

A. VaR gives a loss threshold but says nothing about the size of losses beyond it, whereas ES measures the expected loss given that the threshold is exceeded.

B. For a normal distribution, 97.5% ES (mean plus 2.338 standard deviations) is almost identical to 99% VaR (mean plus 2.326 standard deviations), but the two can differ considerably when tails are fat.

C. ES is easier to backtest than VaR, which is why the FRTB requires the stressed ES itself to be backtested.

D. The FRTB bases capital on a single ES calibrated to a stressed period, rather than adding a current 10-day VaR to a stressed VaR.

Answer: C

Step by step

1. Check A: VaR ignores tail severity; ES is conditional on exceeding VaR. True (bond example: 95% VaR 0 vs ES 380 million).
2. Check B: 99% VaR = $\mu + 2.326\sigma$ and 97.5% ES = $\mu + 2.338\sigma$ under normality, so nearly equal; with fat tails ES can differ considerably. True.
3. Check C: backtesting VaR is easier than backtesting ES, and stressed ES is not backtested under the FRTB (the FRTB backtests 1-day VaR over 12 months). False.
4. Check D: FRTB replaces VaR plus stressed VaR with a single stressed ES. True.

Why C is correct: Backtesting ES is harder than backtesting VaR, and stressed ES cannot be backtested because stressed extremes need not recur with the same frequency; the FRTB therefore backtests VaR instead.

Why the others are wrong

- **A:** True statement; the central motivation for ES. Choosing it means missing that the question asks for the incorrect statement.
- **B:** True statement (2.338 vs 2.326); the equivalence under normality is why the confidence level was lowered to 97.5%.
- **D:** True statement; the Basel II.5 VaR-plus-stressed-VaR sum is replaced by one stressed ES.

Q853

Mock 2 Q36 (AP Mock 2)

REBUILT — Distractors did not correspond to identifiable errors (23,596 was not reproducible); options rebuilt from realistic mistakes while keeping the 99% row and the max-of-latest-and-multiplied-average calculation.

Prime Bank uses the VaR plus stressed VaR market risk framework (Basel II.5). The regulatory measure is a 10-day VaR at 99% confidence. Its internal models produce the following (CAD millions):

Confidence level	Latest 10-day VaR	Latest 10-day SVaR	Avg 10-day VaR, prior 60 days	Avg 10-day SVaR, prior 60 days
99.9%	1,250	2,849	1,170	3,050
99.0%	970	2,200	900	2,460
95.0%	500	1,100	561	1,235

The regulator sets the multiplication factors for both VaR and stressed VaR at 4. What is the total market risk capital charge?

A. CAD 3,170 million

B. CAD 16,880 million

C. CAD 13,440 million

D. CAD 7,184 million

Answer: C

Step by step

1. Use the 99% row: VaR leg = max(latest VaR, 4 x average VaR) = max(970, 4 x 900) = max(970, 3,600) = 3,600.
2. Stressed VaR leg = max(2,200, 4 x 2,460) = max(2,200, 9,840) = 9,840.

3. Total = 3,600 + 9,840 = CAD 13,440 million.

Why C is correct: Basel II.5 capital is the sum of the larger of the latest and multiplied-average VaR and the larger of the latest and multiplied-average stressed VaR, both at 99%.

Why the others are wrong

- **A:** 970 + 2,200: used only the latest values and ignored the multiplier-times-average floors.
- **B:** Used the 99.9% row: 4,680 + 12,200; the regulatory confidence level is 99%.
- **D:** Used the 95% row: 2,244 + 4,940; wrong confidence level.

Q854

AnalystPrep Q.2368

Which of the following presents FRTB's (Fundamental Review of the Trading Book) proposed change during the measurement of market risk capital?

A. VaR at 99% confidence
B. VaR at 99.99% confidence
C. Expected shortfall with a 97.5% confidence level
D. Expected shortfall with a 99% confidence level

Answer: C

Step by step

1. Basel I/II.5 used 10-day VaR at 99% (plus stressed VaR under II.5).
2. FRTB replaces this with expected shortfall at 97.5% confidence.
3. Answer: C.

Why C is correct: FRTB moves to 97.5% ES because it captures tail severity and, for a normal distribution, gives about the same capital as 99% VaR (2.338 vs 2.326 standard deviations).

Why the others are wrong

- **A:** This is the pre-FRTB measure (Basel I/II.5), not the change.
- **B:** No Basel market risk framework uses 99.99% VaR; the 99.9% one-year VaR applies only to the default charge.
- **D:** Confuses the ES measure with the old 99% confidence level; FRTB uses 97.5%.

Q855

BT answers P2.T7.608.1 · BT notes P2.T7.608.1

REBUILT — Option B listed the superseded horizons (10, 20, 60, 120, 250 days), making B false as well as C; corrected B to the final 10, 20, 40, 60, 120 days so exactly one option is false. Two duplicate sources merged.

The Basel Committee's Fundamental Review of the Trading Book (FRTB) revises the market risk framework. Compared to Basel I and Basel II.5, each of the following is true EXCEPT which statement is false?

A. The Basel II.5 internal models approach required the sum of current VaR and stressed VaR, whereas the FRTB requires a single stressed expected shortfall.
B. Basel I and Basel II.5 used a 10-day horizon for all risk factors, whereas the FRTB uses five liquidity horizons (10, 20, 40, 60 and 120 days) depending on the liquidity of the risk factor.
C. Basel II.5 required a default-related charge for credit-dependent positions, but the FRTB eliminates any such charge and relies on VaR and ES alone.
D. Basel I and Basel II.5 based capital on VaR at a 99% confidence level, whereas the FRTB bases capital on expected shortfall at a 97.5% confidence level.

Answer: C

Step by step

1. A: FRTB drops the VaR + stressed VaR sum for one stressed ES. True.
2. B: five liquidity horizons 10, 20, 40, 60, 120 days replace the uniform 10 days. True.
3. C: Basel II.5 introduced the IRC; the FRTB keeps default risk coverage through the incremental default risk (IDR) charge, a 99.9% one-year VaR. It does not eliminate it. False.
4. D: 99% VaR replaced by 97.5% ES. True.

Why C is correct: The FRTB replaces the IRC with the incremental default risk charge (99.9% VaR, one-year horizon), so a default charge remains.

Why the others are wrong

- **A:** True; the stressed-VaR-plus-VaR sum is replaced by one ES.
- **B:** True as stated; picking it reflects mistaking the superseded 250-day horizon list for the final one.
- **D:** True; the confidence level changes from 99% (VaR) to 97.5% (ES).

Q856 GARP PE1 Q13 · Mock 3 Q67

A regulatory analyst at a large multinational bank is examining regulatory requirements the bank must comply with under the Basel Committee's FRTB guidelines. The analyst explores how the FRTB guidelines evolved from the Basel I and Basel II.5 frameworks. Which of the following is correct regarding the FRTB?

A. While Basel I and Basel II.5 allowed market risk to be calculated at the trading desk level, FRTB requires that market risk be calculated on a firm-wide basis.

B. While Basel I and Basel II.5 emphasized the use of a standardized approach to calculating market risk, FRTB encourages each bank to develop and rely on an internal models approach.

C. FRTB standardizes the liquidity horizon for all risk factors in the market risk capital calculation as 10 days, rather than the different horizons used in Basel I and Basel II.5.

D. FRTB requires that the stressed ES measure be used in determining market risk capital, rather than the VaR and stressed VaR that were used in Basel I and Basel II.5.

Answer: D

Step by step

1. Basel I used VaR; Basel II.5 added stressed VaR.
2. FRTB replaces both with a stressed expected shortfall (97.5%).
3. Check A-C against the reading: horizons vary (10-120 days), and desk-level approval and a standardized fallback apply, so A, B, C are wrong.

Why D is correct: The defining FRTB change is a single stressed ES at 97.5% replacing VaR plus stressed VaR.

Why the others are wrong

- **A:** Reversed: FRTB introduces desk-level model approval and calculation; Basel I/II.5 were firm-wide.
- **B:** Reversed: FRTB reduces reliance on internal models, requiring the standardized approach as floor or fallback.
- **C:** Describes the old uniform 10-day horizon; FRTB uses five horizons (10, 20, 40, 60, 120 days).

Q857 Schweser Module Quiz 18.1 Q1

Which of the following statements regarding the differences between Basel I, Basel II.5, and the Fundamental Review of the Trading Book (FRTB) for market risk capital calculations is incorrect?

A. Both Basel I and Basel II.5 require calculation of VaR with a 99% confidence level.

B. FRTB requires the calculation of expected shortfall with a 97.5% confidence level.

C. FRTB requires adding a stressed VaR measure to complement the expected shortfall calculation.

D. The 10-day time horizon for market risk capital proposed under Basel I incorporates a recent period of time, which typically ranges from one to four years.

Answer: C

Step by step

1. Basel I: 10-day, 99% VaR based on recent data (one to four years) - A and D are consistent with this.
2. Basel II.5: kept the 10-day 99% VaR and added a 250-day stressed VaR.
3. FRTB: replaces both with a single 97.5% expected shortfall calibrated to a 250-day stressed period - B is correct.
4. C says FRTB adds stressed VaR to ES; it does not, so C is the incorrect statement.

Why C is correct: FRTB drops the VaR-plus-stressed-VaR combination and uses stressed-period expected shortfall alone; the stressed VaR add-on belongs to Basel II.5.

Why the others are wrong

- **A:** Correct statement: both regimes use 99% VaR, so it is not the incorrect one.
- **B:** Correct statement: FRTB uses 97.5% ES (approximately equal to 99% VaR for a normal distribution).
- **D:** Correct statement, as in the reading: the 10-day 99% VaR is a current measure because it uses recent data, typically one to four years.

Q858 Schweser Module Quiz 18.1 Q2

REBUILT — Stem did not state that a default causes a total loss of the portfolio, which the answer relies on; added that assumption.

A bank holds a bond portfolio worth $825 million. The portfolio has a 3% probability of default, in which case the entire $825 million is lost; otherwise there is no loss. What is the difference between the 95% value at risk (VaR) and the 95% expected shortfall (ES) for this portfolio?

A. Both measures will show the same result.

B. The VaR shows a loss of $495 million, while the expected shortfall shows no loss.

C. The VaR shows no loss, while the expected shortfall shows a \$495 million loss.
D. The VaR shows no loss, while the expected shortfall shows a \$395 million loss.

Answer: C
Step by step
1. 95% VaR: the loss exceeds zero only with probability 3% < 5%, so the 95% VaR is \$0.
2. The 5% tail contains the 3% default outcome (total loss) and 2% of no-loss outcomes.
3. Conditional on being in the tail, the chance of total loss is $3/5 = 60\%$.
4. $ES = 0.60 \times \$825 \text{ million} = \$495 \text{ million}$.

Why C is correct: VaR only reports the quantile (which is zero here), whereas ES averages over the whole 5% tail and so captures the 3% default loss.
Why the others are wrong
- **A:** Ignores that ES looks beyond the VaR quantile; the two measures differ sharply for this lumpy distribution.
- **B:** Reverses the two measures: VaR cannot exceed ES, and VaR is zero because the default probability is below 5%.
- **D:** VaR is right but ES is miscalculated: \$395 million is not 0.6×825; the correct tail weight of 3/5 gives \$495 million.

Q859

AnalystPrep Q.2366

REBUILT — Original asked which market variable category is not in FRTB, which is debatable because the standardized approach has vega charges and the Schweser reading does not list categories; replaced with a testable FRTB change (97.5% ES vs 99% VaR under normality).

Under FRTB, the 10-day 99% VaR is replaced by expected shortfall at a 97.5% confidence level. Suppose a trading portfolio's loss is normally distributed with mean zero and standard deviation of \$4.0 million. Using the reading's relationship that the 97.5% ES equals $\mu + 2.338\sigma$, what is the 97.5% ES (to the nearest \$0.01 million)?

A. \$7.84 million
B. \$9.30 million
C. \$9.35 million
D. \$10.66 million

Answer: C
Step by step
1. $ES_{97.5\%} = \mu + 2.338\sigma$ with $\mu = 0$ and $\sigma = 4.0$.
2. $2.338 \times 4.0 = 9.352$, so about \$9.35 million.
3. Cross-check: $\varphi(1.96)/0.025 = 2.338$.

Why C is correct: For a normal distribution the 97.5% ES multiplier is 2.338 (versus 2.326 for 99% VaR), which is why FRTB's ES calibration is roughly capital-neutral under normality.
Why the others are wrong
- **A:** Used 1.96 (the 97.5% VaR quantile) instead of the ES multiplier: $1.96 \times 4.0 = 7.84$.
- **B:** Used the 99% VaR multiplier 2.326: $2.326 \times 4.0 = 9.30$; the question asks for the 97.5% ES.
- **D:** Used the 99% ES multiplier 2.665: $2.665 \times 4.0 = 10.66$.

Q860

AnalystPrep Q.4027

Under the standardized approach, the capital requirement is the simple sum of three components: risk charges under the sensitivities-based method, a default risk charge, and a residual risk add-on. Which of the following is not a risk class as defined under the sensitivities-based method?

A. General interest rate risk
B. Foreign exchange risk
C. Commodity risk
D. Funding risk

Answer: D
Step by step
1. The sensitivities-based method defines seven risk classes: general interest rate, FX, commodity, equity, credit spread (non-securitization), credit spread (securitization non-CTP), and credit spread (securitization CTP).
2. A, B and C appear in this list.

3. Funding risk is not a defined risk class, so D is the answer.

Why D is correct: The SBM risk classes are interest rate, FX, commodity, equity and three credit spread classes; funding risk is not among them.

Why the others are wrong

- **A:** General interest rate risk is a defined SBM risk class.
- **B:** FX risk is a defined SBM risk class.
- **C:** Commodity risk is a defined SBM risk class.

Q861

AnalystPrep Q.4022

Which of the following is not a component of the standardized approach to calculating regulatory capital for banks under market risk measurement and management?

A. A risk charge calculated using a risk sensitivity approach
B. A residual risk add-on
C. A default risk charge
D. Insolvency risk charge

Answer: D

Step by step

1. The FRTB standardized capital charge is the simple sum of: sensitivities-based risk charge, default risk charge, residual risk add-on.
2. A, B and C match these three components.
3. No 'insolvency risk charge' exists, so D is the answer.

Why D is correct: The standardized approach has exactly three components; an insolvency risk charge is not one of them.

Why the others are wrong

- **A:** This is the sensitivities-based method charge, a genuine component.
- **B:** The residual risk add-on (for exotic/other risks not captured) is a genuine component.
- **C:** The default risk charge (jump-to-default) is a genuine component.

Q862

AnalystPrep Q.2365 · AnalystPrep Q.2998

REBUILT — Merged two near-duplicate items on the subset-of-variables stressed ES; rebuilt as a numerical question so the scaling ratio (and the 12-month window) is tested unambiguously.

A bank cannot find a stressed 250-day period using all market variables because of missing history for some of them. Under the FRTB it computes stressed ES on a subset of variables, giving $80 million for the stressed period. Over the most recent 12 months, ES using all variables is $40 million and ES using only the subset is $32 million. What stressed ES should the bank use?

A. $100 million
B. $64 million
C. $80 million
D. $88 million

Answer: A

Step by step

1. FRTB scales the subset stressed ES by ES(recent 12 months, all variables) / ES(recent 12 months, subset).
2. Ratio = 40 / 32 = 1.25.
3. Stressed ES = 80 x 1.25 = $100 million.

Why A is correct: The subset understates risk, so it is scaled up by the full-set-to-subset ES ratio measured over the most recent 12 months.

Why the others are wrong

- **B:** Inverted the ratio (32/40 = 0.8), scaling the figure down rather than up.
- **C:** Used the subset stressed ES without any scaling, ignoring the variables left out.
- **D:** Added the difference 40 - 32 = 8 instead of multiplying by the ratio.

LO 18.b

Compare the various liquidity horizons proposed by the FRTB for different asset classes and explain how a bank can calculate its expected shortfall using the various horizons.

Q863

GARP book Ch.18 Q18.3

REBUILT — Aligned to GARP 2026 Ch.18 content: removed the 60-day horizon overlap claim (GARP's overlapping method uses 10-day changes) and the 'about 4 non-overlapping 60-day periods' figure; stated that GARP says the lack of independence does not bias results but reduces effective sample size and makes results noisier, and that non-overlapping periods are infeasible as they need a very long history.

Explain why the use of overlapping time periods proposed by the FRTB does not give independent observations on the changes in variables.

Answer: see model answer — FRTB requires 10-day changes in risk factors from a stressed 250-day period, estimated on overlapping 10-day periods. The first simulation trial uses the changes between Day 0 and Day 10 of the stressed period, the second uses Day 1 to Day 11, and so on, so consecutive trials share 9 of their 10 daily changes. Changes in successive historical simulation trials are therefore not independent. This does not bias the results, but it reduces the effective sample size and makes the ES estimate noisier than the number of trials suggests. Non-overlapping periods (which would give independent observations) are not feasible for 10-day changes because they would require a very long historical period; the Basel Committee presumably judges that 10-day changes provide more relevant information than one-day changes despite the lack of independence.

Step by step

1. Overlapping trials shift the window by one day: trial 1 uses Day 0 to Day 10, trial 2 uses Day 1 to Day 11, and so on.
2. Adjacent windows share 9 of their 10 daily changes, so successive trials are not independent.
3. This does not bias the ES estimate but reduces the effective sample size, making results noisier; non-overlapping periods are infeasible because they would need a very long historical period.

Why this is correct: Overlap makes successive 10-day observations share almost all of their underlying daily changes, so they are not independent; this reduces the effective sample size and adds noise without biasing the estimate.

Q864

GARP book Ch.18 Q18.7

REBUILT — Original was a spreadsheet exercise needing external data (Table 12.1, author's worksheets); rebuilt as a self-contained numeric cascade liquidity-adjusted ES calculation with the Schweser horizons.

Under the FRTB internal models approach, a bank's risk factors fall into five liquidity-horizon categories: Category 1 (10 days), 2 (20 days), 3 (40 days), 4 (60 days) and 5 (120 days). ES_j denotes the 97.5% expected shortfall from a 10-day shock to risk factors in categories j through 5 only, holding categories 1 to j-1 constant. The bank estimates (in $ millions):

j	1	2	3	4	5
ES_j	10	8	6	4	2

Assuming independence of shocks across horizon buckets, what is the liquidity-adjusted ES?

A. $14.83 million
B. $17.09 million
C. $22.72 million
D. $30.00 million

Answer: B

Step by step

1. Liquidity-adjusted ES = $\sqrt{ES_1^2 + \sum_{j=2}^{5}\left(ES_j\sqrt{(LH_j - LH_{j-1})/10}\right)^2}$.
2. Scaling factors $(LH_j - LH_{j-1})/10$: j=2: 10/10 = 1; j=3: 20/10 = 2; j=4: 20/10 = 2; j=5: 60/10 = 6.
3. Terms: $10^2 = 100$; $8^2(1) = 64$; $6^2(2) = 72$; $4^2(2) = 32$; $2^2(6) = 24$.
4. Sum = 100 + 64 + 72 + 32 + 24 = 292.
5. ES = $\sqrt{292} = 17.09$ million.

Why B is correct: Each nested ES is scaled by the square root of the extra horizon length it covers relative to 10 days and the independent pieces are added in quadrature.

Why the others are wrong

- **A:** $\sqrt{100+64+36+16+4} = 14.83$: square root of the sum of squares with no horizon scaling, ignoring that longer-horizon buckets need $\sqrt{\Delta LH/10}$ scaling.
- **C:** Scaled each ES_j by $\sqrt{LH_j/10}$ (full horizon) instead of the incremental horizon $LH_j - LH_{j-1}$, double counting earlier horizons.
- **D:** Simple sum 10+8+6+4+2 = 30: assumes perfect correlation and ignores scaling and the square-root aggregation.

Q865

AnalystPrep Q.2364

REBUILT — Aligned to GARP 2026 Ch.18 content: GARP states five liquidity horizons of 10, 20, 40, 60 and 120 days (Categories 1-5); removed the unsupported 'superseded 2014-style list' claim and the Schweser attribution. Original asked a circular 'criterion' question with a weak answer; rebuilt to test the actual five FRTB liquidity horizons per Schweser.

Under the FRTB, each risk factor is assigned to one of five categories according to the time needed to exit or hedge the exposure in stressed markets. Which of the following lists the liquidity horizons (in days) of the five categories?

A. 10, 20, 40, 60, 120
B. 10, 20, 60, 120, 250
C. 1, 5, 10, 20, 60
D. 5, 10, 20, 40, 120

Answer: A

Step by step

1. FRTB categories are defined by liquidity-horizon length.
2. Category 1 = 10 days, 2 = 20, 3 = 40, 4 = 60, 5 = 120.
3. This matches list A.

Why A is correct: GARP Ch.18 specifies five liquidity horizons: 10, 20, 40, 60 and 120 days (Categories 1 to 5).

Why the others are wrong

- **B:** Includes 250, which is the length of the stressed period in days, not a liquidity horizon; GARP's longest horizon is 120 days and it has no 250-day category.
- **C:** Short horizons such as 1 and 5 days are not used; the shortest FRTB horizon is 10 days (the horizon used in Basel I and Basel II.5, which FRTB replaces with liquidity-based horizons).
- **D:** Starts at 5 days; no FRTB category has a 5-day horizon, the minimum is 10 days.

Q866

AnalystPrep Q.4028

REBUILT — Aligned to GARP 2026 Ch.18 content: GARP describes FRTB as considering changes to market variables over periods reflecting their liquidity in stressed market conditions (the liquidity horizons); steps reworded so they rely on that GARP wording rather than a Basel definition not in the chapter.

Under the Fundamental Review of the Trading Book (FRTB), liquidity risk is addressed more explicitly in the calculation of market risk capital. The concept of a liquidity horizon plays a critical role in determining how much capital banks must hold for various risk factors. Given this context, which of the following best defines a liquidity horizon under FRTB?

A. The time taken to find a willing buyer for a security or commodity
B. The time taken to successfully execute a sale transaction that guarantees a minimum level of profit
C. The time it takes to buy a security or commodity
D. The time required to sell a security in a stressed market, without materially affecting market prices

Answer: D

Step by step

1. GARP: FRTB considers changes to market variables (risk factors) that would take place in stressed market conditions over periods of time reflecting their liquidity; these periods are the liquidity horizons.
2. So a liquidity horizon concerns exiting a position in stressed markets, not buying, finding a counterparty, or earning a profit.
3. Only D describes selling in a stressed market without materially moving prices.

Why D is correct: A liquidity horizon is the period, assessed in stressed market conditions, reflecting how long it takes to exit a risk factor exposure without materially moving the market; longer horizons (10 to 120 days) imply larger capital.

Why the others are wrong

- **A:** Finding a buyer is only part of an exit; the definition requires completing the transaction, in stressed condi-

tions, without price impact.
- **B:** Confuses liquidity with profitability; the definition has no profit guarantee.
- **C:** Describes buying, whereas the horizon concerns exiting or hedging an existing exposure.

Q867

BT answers P2.T7.609.1 · BT notes P2.T5.609.1

REBUILT — Merged duplicates; the source used outdated 10/20/60/120/250 horizons and a sqrt(60/10) scaling that contradicts the Schweser formula, with vague 'five simulations per trial' wording; rewrote stem and options around the Schweser cascade (horizons 10/20/40/60/120, incremental scaling).

Under the FRTB internal models approach, risk factors are placed in five liquidity-horizon categories (10, 20, 40, 60 and 120 days). Which of the following best summarizes how the liquidity-adjusted expected shortfall is estimated?

A. ES_j is the ES from a 10-day shock to categories j through 5 only, holding categories 1 to j-1 constant. Each ES_j (j $\geq$ 2) is scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, and total ES is the square root of the sum of the squared terms, reflecting independence across horizons.

B. ES_j is the ES from a 10-day shock to categories j through 5 only, scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, but total ES is the simple sum of the five terms because variable changes are assumed perfectly correlated, the most conservative assumption.

C. ES_j is the ES from a 10-day shock to categories j through 5 only, but each is scaled by $\sqrt{LH_j/10}$ (the full horizon), and total ES is the square root of the sum of the squared terms.

D. ES is computed once from a 10-day shock to all risk factors and then multiplied by $\sqrt{120/10}$, so that every risk factor is treated as if it had the longest liquidity horizon.

Answer: A

Step by step

1. ES1 is a 10-day shock to all categories; ES2 shocks categories 2-5 (category 1 held constant), ..., ES5 shocks category 5 only.
2. Each ES_j (j = 2 to 5) is scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, i.e. squared weights of 1, 2, 2 and 6; ES1 is unscaled.
3. Shocks in different horizon buckets are assumed independent, so the pieces are combined in quadrature: $\sqrt{ES_1^2 + \sum_{j\geq 2} ES_j^2 (LH_j - LH_{j-1})/10}$.
4. Only A has all these features.

Why A is correct: The cascade removes lower-horizon categories step by step, scales each incremental horizon by the square-root rule and aggregates independent buckets in quadrature.

Why the others are wrong

- **B:** Simple sum assumes perfect correlation across buckets; the FRTB formula assumes independence and uses the square root of the sum of squares.
- **C:** Uses the full horizon LH_j instead of the incremental horizon LH_j - LH_{j-1}, double counting time already covered by lower-numbered ESs.
- **D:** Single-scaling approach that gives every factor the longest horizon; the FRTB assigns each factor its own horizon via the cascade, which is more granular and less punitive.

Q868

GARP PE2 Q7 · Mock 4 Q8

A regulatory capital analyst at a large European bank is studying the liquidity horizons specified for different risk factors in the Fundamental Review of the Trading Book (FRTB). The analyst examines how these liquidity horizons are included in and impact the bank's calculations of capital requirements. Which of the following statements is correct?

A. Under the standardized approach, the liquidity horizons of risk factors are incorporated into market risk capital calculations through the risk weights determined by the Basel Committee.

B. When a bank uses the historical simulation approach, the impact of 1-day changes in risk factors are scaled to the liquidity horizon for the risk factor by multiplying by the square root of time.

C. The standardized approach incorporates the diversification benefits between risk factors with different liquidity horizons through the risk weights set by the Basel Committee.

D. The historical simulation approach estimates a liquidity-adjusted ES by assuming a 10-day liquidity horizon for all risk factors, since their behavior will most likely be highly correlated during volatile market conditions.

Answer: A

Step by step

1. Under the standardized approach the Basel Committee sets the risk weights, calibrated to reflect each risk factor's liquidity horizon (longer horizon, larger weight).
2. Under the internal models approach, the base shock is a 10-day change, then the cascade scales ES by the square root of incremental horizons; 1-day changes are not scaled up.
3. Diversification across horizons is handled by the correlation assumptions, not risk weights, and not all factors get 10 days.

Why A is correct: The standardized approach embeds liquidity horizons directly in the Basel-set risk weights.

Why the others are wrong

- **B:** Describes Basel II.5-style square-root-of-time scaling of 1-day changes; FRTB uses 10-day shocks and a cascade scaled by incremental horizons.
- **C:** Risk weights capture liquidity horizons, not diversification; diversification in the standardized approach comes from prescribed correlations.
- **D:** The liquidity-adjusted ES uses five different horizons (10 to 120 days) in a cascade, not a uniform 10-day horizon.

Q869

Schweser Module Quiz 18.1 Q3

REBUILT — Aligned to GARP 2026 Ch.18 content: distractor A acknowledges overlapping 10-day periods over the stressed period are genuinely used by FRTB (but do not create liquidity horizons); option B reworded to GARP's Basel I/II.5 square-root-of-10 scaling of one-day changes (the original 'smaller periods extrapolated to larger' wording was partly true of GARP's cascade, which scales 10-day ES by sqrt(2) etc., so it was not a clean distractor); 'waterfall' described as GARP's 'cascade approach'. Answer D unchanged.

Which of the following statements best describes how the internal models approach (IMA) incorporates various liquidity horizons into the expected shortfall calculation?

A. A rolling 10-day approach is used over a 250-day window of time.

B. One-day changes in the risk factors are scaled up by a square root of 10 multiplier to obtain the 10-day changes used in each expected shortfall.

C. A series of weights is applied to the various liquidity horizons along with a correlation factor determined by the Basel Committee.

D. The expected shortfall is based on a waterfall of the liquidity horizon categories and is then scaled to the square root of the difference in the horizon lengths of the nested risk factors.

Answer: D

Step by step

1. Under the IMA, every risk factor is mapped to one of five liquidity horizon categories: 10, 20, 40, 60 and 120 days.
2. ES is computed over a 10-day base horizon through nested shocks: ES_1 shocks categories 1-5, ES_2 shocks categories 2-5 (category 1 held constant), and so on to ES_5 shocking only category 5.
3. These nested ES figures form the cascade approach and are combined using the extended square root rule, with each term scaled by $\sqrt{(LH_j - LH_{j-1})/10}$, i.e. the difference in horizon lengths of successive categories.
4. Hence the description matching the IMA is the waterfall scaled by the square root of the horizon-length differences: D.

Why D is correct: The IMA computes ES from a cascade of nested shocks by liquidity horizon category and combines the 10-day ES figures using the square root of the differences between successive horizon lengths, assuming behavior in each further period is independent.

Why the others are wrong

- **A:** Rolling (overlapping) 10-day changes over a 250-day stressed period are indeed how each ES is estimated, but that only sets the 10-day base horizon; it does not by itself bring in the 20, 40, 60 and 120-day liquidity horizons, which come from the cascade of nested ES calculations.
- **B:** This is the Basel I and Basel II.5 practice of deducing 10-day changes from one-day changes with a square root of 10 multiplier; FRTB instead requires banks to use actual overlapping 10-day changes from a stressed period, and handles longer liquidity horizons through the cascade of nested ES figures.
- **C:** Invents a weighting scheme and a Basel-set correlation factor; the IMA instead assumes independence across categories and uses the square root rule, with no regulatory correlation parameter.

LO 18.c

Explain the FRTB revisions to Basel regulations in the following areas: Classification of positions in the trading book compared to the banking book. Backtesting, profit and loss attribution, credit risk, and securitizations.

Q870

BT notes P2.T5.23.8.1

REBUILT — Original relied on Standardized Approach component details outside the reading and loosely worded options; rebuilt around the reading's credit risk treatment (credit spread risk, jump-to-default, IDR) and remapped to LO 18.c. Aligned to GARP 2026 Ch.18 content: option C relied on a Schweser-only IRC constant-level-of-risk/incremental-marking detail that GARP does not state; replaced with GARP's credit spread liquidity horizons (20-120 days; credit spread volatility 120 days), and removed the unsupported 'including equities' claim from the IDR step. Answer D unchanged.

Which of the following statements about the FRTB's treatment of credit-dependent positions is FALSE?

A. Credit spread risk, the risk that a change in the credit spread alters mark-to-market value, can be addressed within the expected shortfall calculation.

B. Jump-to-default risk is subject to the incremental default risk (IDR) charge, calculated as a 99.9% VaR with a one-year horizon.

C. Under the internal models approach, the liquidity horizon for credit spreads ranges from 20 to 120 days, and the liquidity horizon for credit spread volatility is 120 days.

D. The FRTB requires no market risk capital charge for jump-to-default risk, because default exposures are charged only under the credit risk rules of the banking book.

Answer: D

Step by step

1. A: true; spread changes are market risk factors in the ES process.
2. B: true; the jump-to-default capital charge under the internal models approach is a VaR with a one-year horizon and 99.9% confidence, handled like banking-book default risk.
3. C: true; credit spread liquidity horizons run from 20 to 120 days and credit spread volatility is 120 days.
4. D: false; jump-to-default is explicitly charged in the trading book through the IDR charge.

Why D is correct: The FRTB captures both credit spread risk (via ES) and jump-to-default risk (via the IDR charge) within market risk capital.

Why the others are wrong

- **A:** True statement; spread risk is part of the ES process.
- **B:** True statement; the IDR uses 99.9% VaR and one year, numbers often confused with the 97.5% ES.
- **C:** True statement; GARP Table 18.1 gives credit spread horizons of 20 to 120 days and 120 days for credit spread volatility. Choosing it means missing that the question asks for the false statement.

Q871

GARP book Ch.18 Q18.5

What is the difference between the trading book and the banking book? Why are regulators concerned about specifying whether an instrument should be one or the other in the FRTB?

Answer: see model answer — The trading book holds assets the bank intends to trade; they are marked to market and subject to market risk capital rules. The banking book holds assets intended to be held to maturity, carried at cost and subject to the (more stringent) credit risk capital rules. Because the two books attract different capital, banks could place credit-dependent assets in whichever book gave lower capital (regulatory arbitrage). FRTB therefore sets rules-based criteria (the bank must be able to trade the asset and must manage its risks on the trading desk) and restricts reclassification to extraordinary circumstances, with no capital benefit from a switch.

Step by step

1. Define trading book: traded assets, periodically marked to market, market risk capital.
2. Define banking book: held-to-maturity assets at cost, credit risk capital (more stringent).
3. Different capital rules create an incentive to choose the book with lower capital: regulatory arbitrage.
4. FRTB response: dual criteria for trading book (ability to trade; risks managed by the trading desk) and limited reclassification with no capital benefit.

Why this is correct: The concern is regulatory arbitrage: the book boundary determines which capital regime applies, so a clear, hard-to-game boundary is needed.

Q872 GARP book Ch.18 Q18.6

REBUILT — Aligned to GARP 2026 Ch.18 content: removed Schweser-only detail (IRC constant-level-of-risk/incremental marking, "IDR" label, equities in JTD charge under IMA); restated per GARP credit-spread/JTD/securitization treatment.

How are credit trades handled under the FRTB?

Answer: see model answer — Credit-dependent positions carry two risks. Credit spread risk (a change in the company's spread changes the mark-to-market value) is handled in a similar way to market risks under the internal models approach, i.e. within the expected shortfall framework, with liquidity horizons from 20 to 120 days (120 days for credit spread volatility). Jump-to-default risk (a default by the company) is handled in the same way as default risks in the banking book: under the internal models approach the charge is based on a VaR calculation with a one-year time horizon and a 99.9% confidence level (under the standardized approach it is a separate default risk charge, exposure times LGD times a default risk weight set by the Basel Committee). Securitizations must use the standardized approach because the Basel Committee found too much variation in the capital charges that different banks calculated for the same portfolio under the comprehensive risk measure.

Step by step

1. Identify the two credit risks: credit spread risk and jump-to-default risk.
2. Credit spread risk: handled like market risk in the internal models approach (ES), with liquidity horizons of 20 to 120 days.
3. Jump-to-default risk: handled like banking book default risk; internal models charge is a 99.9% VaR over a one-year horizon (standardized approach: exposure x LGD x default risk weight).
4. Securitizations: standardized approach must be used (CRM own models gave too much variation across banks).

Why this is correct: FRTB separates credit spread risk (treated like market risk, in ES) from jump-to-default risk (treated like banking book default risk, 99.9% one-year VaR) and requires the standardized approach for securitizations.

Q873 AnalystPrep Q.4032

REBUILT — Distractors C (operational-risk AMA) and D (non-existent approach) were giveaways and the stem was loose on the 12-month window; reworded and replaced with plausible distractors. Aligned to GARP 2026 Ch.18 content: removed 'static/equally weighted/calibrated' qualifiers not in GARP; GARP: one-day VaR, most recent 12 months, 99% and 97.5%, more than 12 or more than 30 exceptions -> standardized approach.

A trading desk at Exim Bank uses the internal models approach. Backtesting the desk's one-day VaR over the most recent 12 months against the desk's P&L gives 13 exceptions at the 99% confidence level and 32 exceptions at the 97.5% confidence level. Under the FRTB, which approach should be used to compute capital for this desk?

A. Internal models approach, because both exception counts are only slightly above the thresholds
B. Standardized approach
C. Internal models approach with an increased multiplier on the capital charge
D. Comprehensive risk measure (CRM) approach

Answer: B

Step by step

1. FRTB backtesting uses one-day VaR and the latest 12 months of data.
2. Thresholds: more than 12 exceptions at 99%, or more than 30 at 97.5%, forces the standardized approach.
3. Desk: 13 > 12 at 99% and 32 > 30 at 97.5%; both thresholds are exceeded.
4. So capital must be computed with the standardized approach.

Why B is correct: Exceptions exceed the FRTB limits at both confidence levels, so the desk loses internal models approval and uses the standardized approach.

Why the others are wrong

- **A:** FRTB has no tolerance band for 'slightly above'; any count beyond 12 (99%) or 30 (97.5%) fails.
- **C:** Confuses FRTB with the Basel traffic-light plus-factor multiplier; FRTB switches approach rather than scaling the multiplier.
- **D:** CRM was the Basel II.5 charge for securitizations (and FRTB removes internal models for them); it is not a fallback for desk backtest failure.

Q874 AnalystPrep Q.2367

The FRTB requires banks to backtest their internal models using a VaR measure over a specified horizon and the most recent 12 months of data. What time horizon does the FRTB specify for this VaR?

A. Thirty-day horizon

B. Seven-day horizon
C. Two-day horizon
D. One-day horizon

Answer: D
Step by step
1. FRTB suggests backtesting VaR with a one-day horizon over the latest 12 months.
2. The one-day P&L gives enough observations for a meaningful exception count (e.g. 12 at 99%, 30 at 97.5%).
3. Answer: one-day.

Why D is correct: Backtesting compares daily P&L with one-day VaR, so the horizon is one day.
Why the others are wrong
- **A:** A thirty-day horizon is not used; it would leave too few non-overlapping observations in 12 months.
- **B:** Seven days is not an FRTB horizon for backtesting (liquidity horizons are 10 to 120 days).
- **C:** Two days has no basis in the FRTB; possibly confused with the 10-day Basel I/II.5 VaR horizon.

Q875
AnalystPrep Q.2369

The FRTB defines which instruments belong in the trading book and which in the banking book. Which instruments are marked to market?
A. Instruments in the trading book
B. Instruments in the banking book
C. Instruments in both the banking and the trading books
D. Instruments in the trading book are valued using historical cost accounting

Answer: A
Step by step
1. Trading book: assets intended to be traded, periodically marked to market.
2. Banking book: assets held to maturity, carried at cost.
3. So only trading book instruments are marked to market.

Why A is correct: Trading book positions are periodically marked to market; banking book positions are held at cost.
Why the others are wrong
- **B:** Reverses the books: banking book assets are held at cost.
- **C:** Only the trading book is marked to market.
- **D:** States the opposite of the true treatment; trading book assets are not valued at historical cost.

Q876
AnalystPrep Q.2371 · AnalystPrep Q.4035

REBUILT — Merged two near-identical items (the credit risks recognized by the IRC) into one; fixed the wording so only the two Schweser risks are correct. Aligned to GARP 2026 Ch.18 content: stem attributed the two credit risks to the Basel II.5 IRC; GARP states FRTB distinguishes credit spread risk and jump-to-default risk. Stem and steps reworded; answer unchanged.

According to the FRTB reading, FRTB distinguishes two types of credit risk exposure to a company. These are:
A. Downgrade risk and jump-to-default risk
B. Downgrade risk and credit spread risk
C. Credit spread risk and jump-to-default risk
D. Credit spread risk and interest rate risk

Answer: C
Step by step
1. Credit spread risk: a change in the company's spread changes the mark-to-market value of the instrument (handled like market risk, via expected shortfall).
2. Jump-to-default risk: a default by the company (handled like banking book default risk; internal models charge is a 99.9% VaR over one year).
3. These are the two credit risks distinguished; downgrade and interest rate risk are not the named pair.

Why C is correct: The reading identifies credit spread risk and jump-to-default risk as the two types of credit risk exposure.
Why the others are wrong

- **A:** Jump-to-default is correct, but the reading pairs it with credit spread risk, not downgrade (migration) risk; the pairing is wrong.
- **B:** Pairs downgrade risk with credit spread risk; omits jump-to-default and includes a non-named risk.
- **D:** Interest rate risk is a general market risk captured in the standard ES, not a credit-specific IRC risk.

Q877

AnalystPrep Q.2999

REBUILT — Options B and C were unrelated outsourcing/legal text (giveaways); replaced with plausible but false rationales. Aligned to GARP 2026 Ch.18 content: option A and steps said 10-day ES is hard to backtest; GARP says ES is harder to backtest than VaR and a stressed measure cannot be backtested because stressed data are not expected to recur with the same frequency.

The FRTB suggests backtesting with a one-day VaR measure calculated using the most recent 12 months of data. What is the primary reason for this proposal?

A. Expected shortfall is harder to backtest than VaR, and a stressed measure cannot be backtested because stressed data are not expected to recur with the same frequency

B. VaR at the 99% level is a coherent risk measure whereas expected shortfall is not, so only VaR can be validated

C. The FRTB capital charge is itself based on one-day 99% VaR, so the same measure must be backtested

D. All risk factors under the FRTB have a one-day liquidity horizon, so one-day VaR matches the capital measure

Answer: A

Step by step

1. FRTB capital is based on stressed 97.5% expected shortfall with liquidity horizons of 10 to 120 days.
2. ES is more difficult to backtest than VaR, and a stressed measure cannot be backtested at all because stressed data are extreme and not expected to recur with the same frequency.
3. So FRTB backtests a one-day VaR over the most recent 12 months instead.

Why A is correct: Backtesting the stressed ES capital measure is impractical (ES is harder to backtest than VaR and a stressed measure cannot be backtested), so a one-day VaR is backtested.

Why the others are wrong

- **B:** Reverses the facts: VaR is not subadditive (not coherent) while ES is coherent.
- **C:** The capital charge uses expected shortfall, not one-day VaR.
- **D:** Liquidity horizons range from 10 to 120 days, not one day.

Q878

AnalystPrep Q.4029

REBUILT — Distractors (internal fraud, default risk, interest rate risk) were not plausible reasons for the book-boundary change; rewrote as plausible rationales.

Which of the following best describes the main reason the FRTB tightens the rules separating the trading book from the banking book?

A. Banks could lower capital by placing credit-dependent assets in whichever book carried the lower requirement

B. Banking book assets were being marked to market too often, making capital requirements unstable

C. Trading book assets were too illiquid to be valued, so they needed to be moved to the banking book

D. Banks held too few assets in the banking book to satisfy credit risk capital requirements

Answer: A

Step by step

1. Trading book assets face market risk capital rules; banking book assets face more stringent credit risk rules.
2. Different rules let banks hold credit-dependent assets in the trading book to relax capital: regulatory arbitrage.
3. FRTB responds with dual criteria for trading book entry and limits on reclassification.

Why A is correct: The boundary determines which capital regime applies, creating an incentive for regulatory arbitrage that FRTB aims to remove.

Why the others are wrong

- **B:** Banking book assets are held at cost, not marked to market often.
- **C:** Trading book assets are those that can be traded; illiquidity was not the motivation.
- **D:** The concern is choice of book to reduce capital, not a shortage of banking book assets.

Q879 AnalystPrep Q.6536

REBUILT — Aligned to GARP 2026 Ch.18 content: P&L attribution measures restated with GARP's notation (U = actual minus model P&L, V = actual/hypothetical P&L): Mean(U)/SD(V) within -10% to +10% and Var(U)/Var(V) below 20%; four or more breaches in 12 months force the standardized approach.

Under the FRTB profit-and-loss attribution test, U is the difference between the actual (hypothetical) and model-predicted P&L on a day and V is the actual (hypothetical) P&L. The test uses Mean(U)/SD(V), which must lie between -10% and +10%, and Var(U)/Var(V), which must be less than 20%. Which of the following readings falls outside the requirement and counts toward the four-or-more failures in 12 months that force a desk onto the standardized approach?

A. Mean(U)/SD(V) = -8%
B. Var(U)/Var(V) = 12%
C. Mean(U)/SD(V) = +9%
D. Var(U)/Var(V) = 22%

Answer: D

Step by step

1. Mean ratio Mean(U)/SD(V) must be between -10% and +10%.
2. Variance ratio Var(U)/Var(V) must be less than 20%.
3. A: -8% and C: +9% are within the band. B: 12% is below 20%.
4. D: 22% exceeds 20%, a failure.

Why D is correct: Only the variance ratio of 22% breaches its 20% limit.

Why the others are wrong

- **A:** Treats a negative sign as a breach; -8% is within the -10% to +10% band.
- **B:** Applies the ±10% band to the variance ratio; its limit is 20%.
- **C:** +9% is inside the band; near but not beyond the limit.

Q880 AnalystPrep Q.6537

A newly appointed risk officer at Delta Bank is reviewing the regulatory profit-and-loss attribution results of the firm's trading desk. Over the past 12 months, on five separate occasions the Mean(U)/SD(V) ratio exceeded +12% and the Var(U)/Var(V) ratio exceeded 25%. The bank maintains overall profitability. Under FRTB, what is the most appropriate consequence?

A. Mandatory switch to the standardized approach
B. Minor adjustments as the internal model remains valid
C. Desk receives a regulatory warning but retains model use
D. Increased capital charge but no mandatory model change

Answer: A

Step by step

1. FRTB P&L attribution thresholds: the first ratio must stay within ±10% and the second must be below 20%.
2. Each of the five days breaches both thresholds (12% > 10%, 25% > 20%).
3. The rule: four or more such occasions in a 12-month period force capital to be computed with the standardized approach.
4. Five ≥ 4, so the switch is mandatory; profitability is irrelevant.

Why A is correct: Four or more P&L attribution breaches in 12 months remove the desk's internal-model eligibility, regardless of profitability.

Why the others are wrong

- **B:** Treats the breaches as minor; the rule is a hard trigger at four, not discretionary.
- **C:** A warning-only outcome does not exist in the FRTB rule; the penalty is loss of model use.
- **D:** Confuses this with the backtesting multiplier/surcharge idea of Basel II.5; FRTB removes the model instead.

Q881 AnalystPrep Q.6538

REBUILT — Original numbers (+11%, -12%) were inconsistent with the stated reading; restated breaches explicitly against the ±10% and 20% limits and used the Schweser rule of four or more occasions.

Regulatory auditors at Zenith Bank review a trading desk's P&L attribution over the past 12 months and find: Mean(U)/SD(V) exceeded +10% on three occasions and fell below -10% on two other occasions (limit is ±10%); Var(U)/Var(V) exceeded 20% on four occasions (peaking at 28%); the desk was profitable throughout. Under

FRTB, what is the most likely regulatory outcome?

A. Immediate capital surcharge and continued internal model use
B. Mandatory switch to the standardized approach
C. Conditional approval pending further supervisory review
D. The internal model remains valid but is subjected to enhanced monitoring

Answer: B

Step by step

1. The tests: Mean ratio within ±10%, variance ratio below 20%.
2. Mean-ratio breaches: 3 + 2 = 5; variance-ratio breaches: 4. Either ratio failing counts as an occasion.
3. Four or more occasions in 12 months triggers use of the standardized approach.
4. Both counts reach the trigger, so the switch is mandatory.

Why B is correct: Reaching four or more breaches of either P&L attribution ratio in 12 months forces the standardized approach.

Why the others are wrong

- **A:** Surcharge with continued model use is not an FRTB outcome for P&L attribution failure.
- **C:** Supervisory discretion is not part of the rule; the trigger is mechanical.
- **D:** Monitoring-only ignores that the breach count is at or above the four-occasion threshold.

Q882

BT answers P2.T5.713.1 · BT notes P2.T5.713.1

REBUILT — Aligned to GARP 2026 Ch.18 content: step 3 said either the 99% or the 97.5% level may be applied; GARP requires both confidence levels to be used in the one-day VaR backtest (answer B unchanged).; corrected explanation wording on why stressed ES is not backtested (GARP: 'more difficult to back-test ES than VaR... not possible to back-test a stressed measure at all')

In comparison to Basel III, which essentially incorporated the previous quantitative backtesting requirements for the internal models approach (IMA), which of the following BEST summarizes the change(s) to backtesting under the Fundamental Review of the Trading Book (FRTB)?

A. The backtest methodology is unchanged in the FRTB
B. One-day value at risk (VaR) is backtested at both 97.5% and 99.0% confidence levels
C. There are two tests: 10-day 99.0% expected shortfall and 10-day 99.0% stressed value at risk (sVaR)
D. The 10-day horizon and 99.0% confidence level are unchanged, but these parameters are applied to expected shortfall (ES) rather than VaR

Answer: B

Step by step

1. FRTB replaces 10-day 99% VaR with 97.5% stressed ES for capital; stressed ES cannot be backtested (a stressed measure cannot be backtested at all, and ES is harder to backtest than VaR).
2. Backtesting therefore remains on one-day VaR, using the latest 12 months of data.
3. Both the 99% and the 97.5% levels are used: more than 12 exceptions at 99%, or more than 30 at 97.5%, forces the desk onto the standardized approach until neither condition continues.
4. This matches B.

Why B is correct: FRTB backtests one-day VaR at both 97.5% and 99%, because it does not backtest the stressed ES capital measure (ES is harder to backtest than VaR and a stressed measure cannot be backtested).

Why the others are wrong

- **A:** The capital metric changed to stressed ES and backtesting is done on one-day VaR at both 99% and 97.5%, so the approach is not unchanged.
- **C:** Stressed measures are explicitly not backtested, and the horizon is one day.
- **D:** Applies the old VaR parameters to ES; the stressed ES capital measure is not backtested and the backtest horizon is one day.

Q883

BT answers P2.T7.608.2 · BT notes P2.T5.608.2

REBUILT — Aligned to GARP 2026 Ch.18 content: extraordinary-circumstance examples restated per GARP (closing of trading desks; change in accounting standards)

In the past, the boundary between the regulatory banking book and the trading book created an opportunity for regulatory arbitrage. Instruments in the trading book are marked to market daily and attract market risk capital, while banking book instruments are not marked daily and attract credit risk capital. Which of the following is TRUE about the treatment of the boundary in the FRTB?

A. The FRTB collapses the banking and trading books into a single book, which solves the regulatory arbitrage problem
B. The FRTB makes no change to the boundary but subjects both books to exactly the same capital calculations
C. The FRTB establishes a more objective boundary, including assignment when a position is initiated and strict limits on subsequent movement between the books
D. The FRTB says that a sincere "intent to trade" is necessary and sufficient for trading-book inclusion; moving a position between books triggers the incremental risk charge as a penalty

Answer: C

Step by step

1. FRTB keeps two books but defines the boundary more objectively: the bank must be able to trade the asset and actively manage its risks on the desk, and price moves must affect equity.
2. Positions are assigned at inception and may be moved between books only in extraordinary circumstances (e.g., closing of trading desks, a change in accounting standards on fair value recognition).
3. Any capital benefit from reclassification is disallowed.
4. Only C reflects this.

Why C is correct: The FRTB tightens the boundary with objective criteria, assignment at inception, and strict limits on switching.

Why the others are wrong

- **A:** The two books are retained; they are not merged.
- **B:** The boundary was changed, and the books keep different capital frameworks.
- **D:** Intent to trade alone is insufficient (dual criteria are required), and the penalty is denial of any capital benefit, not the IRC.

Q884

BT answers P2.T7.609.2 · BT notes P2.T5.609.2

REBUILT — Aligned to GARP 2026 Ch.18 content: both 99% and 97.5% levels are used in the backtest (not either/or)

Each of the following statements about FRTB backtesting is true EXCEPT which is false?

A. Stressed expected shortfall measures are not backtested because extreme stressed values may not recur with the same frequency
B. Backtesting uses a one-day horizon and the latest 12 months of data
C. At the 99% level, more than 12 exceptions in the period requires the standardized approach
D. The backtest must be performed on the 97.5% stressed ES used for the capital charge, with a 10-day horizon

Answer: D

Step by step

1. FRTB does not backtest stressed ES; backtesting ES is also harder than backtesting VaR.
2. Instead it backtests VaR with a one-day horizon and the latest 12 months of data, using both 99% and 97.5% confidence levels.
3. Thresholds: more than 12 exceptions at 99%, or more than 30 at 97.5%, forces the standardized approach.
4. A, B and C state these facts; D contradicts them, so D is false.

Why D is correct: FRTB backtests one-day VaR, never the stressed ES capital measure, so D is the false statement.

Why the others are wrong

- **A:** True per the reading, so not the exception.
- **B:** True per the reading, so not the exception.
- **C:** True (12 exceptions at 99%); confusing it with the 30-exception limit at 97.5% would wrongly make it look false.

Q885

BT answers P2.T7.609.3 · BT notes P2.T5.609.3

REBUILT — Aligned to GARP 2026 Ch.18 content: removed non-GARP claim that the IRC assumed constant positions; credit spread vs jump-to-default treatment and equity inclusion restated per GARP

Basel II.5 introduced the incremental risk charge (IRC) to stop banks reducing capital by choosing the trading book over the banking book for credit-dependent instruments. How does the FRTB address credit trades?

A. The FRTB makes no changes to the IRC
B. The FRTB edits the IRC to produce a capital charge based on a 10-day, 97.5% expected shortfall (ES)
C. The FRTB replaces the IRC with a default risk charge (DRC), which recognizes jump-to-default risk and mandates the inclusion of equity products

D. The FRTB replaces the IRC with a default risk charge (DRC), which does not recognize jump-to-default risk and explicitly excludes equity products

Answer: C

Step by step

1. Basel II.5's incremental risk charge made choosing the trading book less attractive; the FRTB handles credit risk as two exposures: credit spread risk and jump-to-default risk.
2. Credit spread risk is treated like market risk (liquidity horizons 20 to 120 days within the ES process); jump-to-default risk is handled by a separate default risk charge (internal models: VaR, one-year horizon, 99.9% confidence).
3. The default risk charge applies LGD times default risk weight, and equity positions are subject to it with LGD = 100%.
4. So C is right.

Why C is correct: The FRTB replaces the IRC with a default risk charge that explicitly captures jump-to-default risk and covers equities.

Why the others are wrong

- **A:** The IRC is replaced, not retained.
- **B:** Mixes the IRC with the ES capital measure; default risk uses 99.9% VaR over one year.
- **D:** Reverses both features: jump-to-default is the point of the charge, and equities are included.

Q886

BT notes P2.T5.23.8.2

REBUILT — Original relied on vague 'probably' options and a trading/banking list not covered in the reading; replaced with a reading-based question on FRTB reclassification rules testing the same trading book vs banking book LO. Aligned to GARP 2026 Ch.18 content: examples of extraordinary circumstances changed to GARP's (closing of trading desks, change in accounting standards on fair value recognition) and removed the Schweser-style 'original method capital retained' wording; GARP only says any capital benefit is disallowed.

Under the FRTB, a bank initially assigns a position to the banking book and later wants to move it to the trading book because it expects the move to lower its capital requirement. Which statement best describes the FRTB treatment?

A. Reclassification is permitted at the bank's discretion provided it is disclosed to the supervisor.

B. Reclassification is permitted only in extraordinary circumstances, and any capital benefit from the move is disallowed.

C. Reclassification is permitted only in extraordinary circumstances, and the bank keeps the full capital benefit of the new category.

D. Reclassification is never permitted under any circumstances.

Answer: B

Step by step

1. The FRTB aims to curb regulatory arbitrage between trading book (market risk rules) and banking book (credit risk rules).
2. Transfers between the books can happen only in extraordinary circumstances (e.g., the closing of trading desks or a change in accounting standards on the recognition of fair value).
3. Even then, any capital benefit as a result of moving items between the books is disallowed.
4. Choice B states both conditions.

Why B is correct: Limiting reclassification to extraordinary events and denying any capital benefit removes the incentive to switch categories to reduce capital.

Why the others are wrong

- **A:** Treats reclassification as discretionary; this is the pre-FRTB arbitrage the rule is designed to stop.
- **C:** Gets the extraordinary-circumstances condition right but wrongly lets the bank keep the capital benefit; the benefit is disallowed.
- **D:** Overstates the rule as an absolute ban; transfers are allowed in extraordinary circumstances such as the closing of a trading desk or a change in accounting standards.

Q887

Schweser Module Quiz 18.1 Q4

REBUILT — Aligned to GARP 2026 Ch.18 content: statement I reworded from 'physically trade' to 'able to trade' as in GARP; criteria text aligned.

Which of the following statements represents a criterion for classifying an asset into the trading book?

I. The bank must be able to trade the asset.

II. The risk of the asset must be managed by the bank's trading desk.

A. I only.
B. II only.
C. Both I and II.
D. Neither I nor II.

Answer: C

Step by step

1. The FRTB says an intent to trade is no longer sufficient: the bank must be able to trade, and must be able to manage the underlying risks on a trading desk.
2. Statement I is the first criterion; statement II is the second.
3. Both are required, so the answer is both I and II.

Why C is correct: The FRTB requires more than an intent to trade: both the ability to trade and desk-level risk management must be shown.

Why the others are wrong

- **A:** Recognizes only the ability-to-trade requirement and omits management of the risks on a trading desk.
- **B:** Recognizes only desk risk management and omits the ability-to-trade requirement.
- **D:** Both statements are genuine criteria, so rejecting both is wrong.

Q888

Schweser Module Quiz 18.1 Q5

REBUILT — Source option B said 99% VaR, but the reading states the default (jump-to-default) charge uses 99.9% VaR over one year; corrected option B to 99.9%, with the other options kept as realistic wrong choices. Aligned to GARP 2026 Ch.18 content: GARP does not describe an 'incremental risk charge' as recognizing specific risks (it only notes the IRC of Basel II.5 reduced arbitrage); rebuilt on GARP's credit risk section: jump-to-default risk uses 99.9% VaR over one year, credit spread risk is handled like market risk.

Under the FRTB internal models approach, which credit-related risk has a capital charge based on a VaR calculation with a one-year horizon and a 99.9% confidence level?

A. Credit spread risk.
B. Jump-to-default risk.
C. Equity price risk.
D. Interest rate risk.

Answer: B

Step by step

1. FRTB distinguishes credit spread risk (the company's spread changes) from jump-to-default risk (the company defaults).
2. Credit spread risk is handled like market risks, using expected shortfall with liquidity horizons of 20 to 120 days.
3. Jump-to-default risk is handled like default risks in the banking book, with a capital charge based on 99.9% VaR over one year, so the answer is B.

Why B is correct: Under the internal models approach, jump-to-default risk is treated like banking book default risk and uses a 99.9% VaR with a one-year horizon.

Why the others are wrong

- **A:** Credit spread risk is handled like market risk with 97.5% stressed ES and liquidity horizons of 20 to 120 days, not 99.9% one-year VaR.
- **C:** Equity price risk is a market risk factor measured with stressed ES and 10 to 60 day liquidity horizons.
- **D:** Interest rate risk is a market risk factor measured with stressed ES and 10 to 60 day liquidity horizons.

Q889

AnalystPrep Q.2370

REBUILT — Aligned to GARP 2026 Ch.18 content: removed non-GARP 'held at cost/marked to market' accounting detail; used GARP's statement that banking book instruments are subject to credit risk capital and trading book instruments to market risk capital.

According to the Fundamental Review of the Trading Book (FRTB), the instruments in the trading book are subject to:

A. Credit risk capital
B. Market risk capital
C. Operational risk capital
D. Liquidity risk capital

Answer: B

Step by step

1. The trading book consists of instruments the bank intends to trade; the banking book of instruments expected to be held to maturity.
2. Banking book instruments are subject to credit risk capital, whereas trading book instruments are subject to market risk capital.
3. So trading book instruments are subject to market risk capital.

Why B is correct: Trading book instruments are subject to market risk capital, while banking book instruments are subject to credit risk capital.

Why the others are wrong

- **A:** Credit risk capital applies to banking book instruments, the reverse of the correct allocation.
- **C:** Operational risk capital is a firm-wide charge, not tied to trading book instruments.
- **D:** Liquidity risk is addressed by separate liquidity standards (and liquidity horizons within ES), not as the capital category for the trading book.

Q890

AP Mock 3 Q57

REBUILT — Distractors included invented or irrelevant terms (revised internal models approach, AMA); replaced with plausible market-risk alternatives. Aligned to GARP 2026 Ch.18 content: stem wording 'securitization business model' replaced by GARP's securitizations (ABS, CDOs); Basel II.5 (OCR of Basel II.5) rendered as Basel II.5 and IRC distractor tied to GARP's description.

Under the FRTB, which approach must banks use to compute capital for securitizations such as asset-backed securities and collateralized debt obligations?

A. The internal models approach (IMA) using expected shortfall.

B. The standardized approach.

C. The comprehensive risk measure (CRM) using the bank's own internal model.

D. The incremental risk charge (IRC) using the bank's own internal model.

Answer: B

Step by step

1. The comprehensive risk measure (CRM) charge introduced in Basel II.5 covered securitizations and let banks (with approval) use their own models.
2. The Basel Committee found too much variation in the capital charges different banks calculated for the same portfolio.
3. Under the FRTB the standardized approach must be used for securitizations.

Why B is correct: Model-based securitization charges varied widely across banks, so the FRTB mandates the standardized approach.

Why the others are wrong

- **A:** IMA is not permitted for securitizations under the FRTB.
- **C:** CRM was the Basel II.5 internal-model charge that the FRTB replaced for securitizations because of the variation in results across banks.
- **D:** The incremental risk charge was a Basel II.5 charge that made trading book/banking book arbitrage less attractive; it is not the securitization approach under the FRTB.